Credit_Score - Data Preperation - Correlation Strength

2024-11-10

#Opening File

```
# Set the file path for the CSV file
file_path <- "C:/Users/jivko/Documents/Data Analytics, Big Data, and Predictive Analytics/Per
sonal Project/Credit_Score_Multiclass/Credit_score_cleaned_data.csv"

# Read the CSV file into a dataframe
credit_score <- read.csv(file_path, header = TRUE)
```

```
# Print the first few rows of the dataframe
print(head(credit_score))
```

```
##   Customer_ID Age Occupation Annual_Income Monthly_Inhand_Salary
## 1  CUS_0x98ed  31 Journalist      22229.06              1672.422
## 2  CUS_0x36b4  43   Musician      17060.71              1685.725
## 3  CUS_0x105b  24 Accountant      34428.21              2575.017
## 4  CUS_0x4c4a  21 Accountant      12613.92              1155.160
## 5  CUS_0x612f  31 Journalist      34775.16              3048.930
## 6  CUS_0x2924  14   Musician      48151.86              4098.655
##   Num_Bank_Accounts Num_Credit_Card Interest_Rate Num_of_Loan
## 1                 6              4             5           3
## 2                 8              3            11           4
## 3                 2              4             8           0
## 4                 4              6            13           3
## 5                 7              9            25           5
## 6                 4              5            20           4
##   Delay_from_due_date Num_of_Delayed_Payment Changed_Credit_Limit
## 1                  10                     10                 9.44
## 2                  24                      8                 9.64
## 3                   3                     12                 8.51
## 4                  14                     17                17.51
## 5                  20                     20                17.16
## 6                  26                      9                14.94
##   Num_Credit_Inquiries Credit_Mix Outstanding_Debt Credit_Utilization_Ratio
## 1                    9   Standard          1396.14                 33.06186
## 2                    2   Standard           268.81                 36.37762
## 3                    0       Good           180.70                 39.91751
## 4                    3   Standard           569.11                 39.84149
## 5                    6        Bad          3003.45                 36.05847
## 6                    8   Standard          2083.32                 24.53459
##   Credit_History_Age Payment_of_Min_Amount Total_EMI_per_month
## 1                156                   Yes            52.13847
## 2                269                   Yes            51.93463
## 3                326                    No             0.00000
## 4                376                   Yes            16.94190
## 5                162                   Yes           126.59681
## 6                207                   Yes            80.82314
##   Amount_invested_monthly                Payment_Behaviour Monthly_Balance
## 1                44.08401 High_spent_Medium_value_payments        321.0197
## 2                39.62831   Low_spent_Small_value_payments        327.0096
## 3                84.77465  High_spent_Small_value_payments        432.7271
## 4                41.94360  Low_spent_Medium_value_payments        346.6305
## 5                67.97038 High_spent_Medium_value_payments        360.3258
## 6               180.64526  High_spent_Large_value_payments        408.3971
##   Last_Loan_9 Last_Loan_8 Last_Loan_7 Last_Loan_6 Last_Loan_5
## 1     No Loan     No Loan     No Loan     No Loan     No Loan
## 2     No Loan     No Loan     No Loan     No Loan     No Loan
## 3     No Loan     No Loan     No Loan     No Loan     No Loan
## 4     No Loan     No Loan     No Loan     No Loan     No Loan
## 5     No Loan     No Loan     No Loan     No Loan   Auto Loan
## 6     No Loan     No Loan     No Loan     No Loan     No Loan
##               Last_Loan_4             Last_Loan_3      Last_Loan_2
## 1                No Loan Debt Consolidation Loan Home Equity Loan
```

```
## 2 Debt Consolidation Loan        Home Equity Loan        Auto Loan
## 3                 No Loan                 No Loan          No Loan
## 4                 No Loan           Personal Loan     Student Loan
## 5           Mortgage Loan               Auto Loan      Payday Loan
## 6     Credit-Builder Loan               Auto Loan      Payday Loan
##               Last_Loan_1 Credit_Score
## 1 Debt Consolidation Loan     Standard
## 2               Auto Loan     Standard
## 3                 No Loan         Good
## 4        Home Equity Loan     Standard
## 5           Not Specified         Poor
## 6 Debt Consolidation Loan     Standard
```

```
# Open the dataframe in R's data viewer (use this line once)
View(credit_score)
```

#Structure of dataframe

```
str(credit_score)
```

```
## 'data.frame':    100000 obs. of  32 variables:
##  $ Customer_ID             : chr  "CUS_0x98ed" "CUS_0x36b4" "CUS_0x105b" "CUS_0x4c4a" ...
##  $ Age                     : int  31 43 24 21 31 14 36 30 17 20 ...
##  $ Occupation              : chr  "Journalist" "Musician" "Accountant" "Accountant" ...
##  $ Annual_Income           : num  22229 17061 34428 12614 34775 ...
##  $ Monthly_Inhand_Salary   : num  1672 1686 2575 1155 3049 ...
##  $ Num_Bank_Accounts       : int  6 8 2 4 7 4 8 10 9 1 ...
##  $ Num_Credit_Card         : int  4 3 4 6 9 5 4 10 5 6 ...
##  $ Interest_Rate           : num  5 11 8 13 25 20 14 21 25 7 ...
##  $ Num_of_Loan             : int  3 4 0 3 5 4 1 3 3 4 ...
##  $ Delay_from_due_date     : int  10 24 3 14 20 26 12 41 29 10 ...
##  $ Num_of_Delayed_Payment  : int  10 8 12 17 20 9 18 20 23 6 ...
##  $ Changed_Credit_Limit    : num  9.44 9.64 8.51 17.51 17.16 ...
##  $ Num_Credit_Inquiries    : int  9 2 0 3 6 8 2 11 13 2 ...
##  $ Credit_Mix              : chr  "Standard" "Standard" "Good" "Standard" ...
##  $ Outstanding_Debt        : num  1396 269 181 569 3003 ...
##  $ Credit_Utilization_Ratio: num  33.1 36.4 39.9 39.8 36.1 ...
##  $ Credit_History_Age      : int  156 269 326 376 162 207 360 139 238 362 ...
##  $ Payment_of_Min_Amount   : chr  "Yes" "Yes" "No" "Yes" ...
##  $ Total_EMI_per_month     : num  52.1 51.9 0 16.9 126.6 ...
##  $ Amount_invested_monthly : num  44.1 39.6 84.8 41.9 68 ...
##  $ Payment_Behaviour       : chr  "High_spent_Medium_value_payments" "Low_spent_Small_valu
e_payments" "High_spent_Small_value_payments" "Low_spent_Medium_value_payments" ...
##  $ Monthly_Balance         : num  321 327 433 347 360 ...
##  $ Last_Loan_9             : chr  "No Loan" "No Loan" "No Loan" "No Loan" ...
##  $ Last_Loan_8             : chr  "No Loan" "No Loan" "No Loan" "No Loan" ...
##  $ Last_Loan_7             : chr  "No Loan" "No Loan" "No Loan" "No Loan" ...
##  $ Last_Loan_6             : chr  "No Loan" "No Loan" "No Loan" "No Loan" ...
##  $ Last_Loan_5             : chr  "No Loan" "No Loan" "No Loan" "No Loan" ...
##  $ Last_Loan_4             : chr  "No Loan" "Debt Consolidation Loan" "No Loan" "No Loan"
...
##  $ Last_Loan_3             : chr  "Debt Consolidation Loan" "Home Equity Loan" "No Loan" "
Personal Loan" ...
##  $ Last_Loan_2             : chr  "Home Equity Loan" "Auto Loan" "No Loan" "Student Loan"
...
##  $ Last_Loan_1             : chr  "Debt Consolidation Loan" "Auto Loan" "No Loan" "Home Eq
uity Loan" ...
##  $ Credit_Score            : chr  "Standard" "Standard" "Good" "Standard" ...
```

#Missing Values Check

```
# Check for missing values in each column
colSums(is.na(credit_score))
```

```
##              Customer_ID                      Age               Occupation
##                        0                        0                        0
##            Annual_Income    Monthly_Inhand_Salary        Num_Bank_Accounts
##                        0                        0                        0
##          Num_Credit_Card            Interest_Rate              Num_of_Loan
##                        0                        0                        0
##      Delay_from_due_date   Num_of_Delayed_Payment     Changed_Credit_Limit
##                        0                        0                        0
##     Num_Credit_Inquiries               Credit_Mix         Outstanding_Debt
##                        0                        0                        0
## Credit_Utilization_Ratio       Credit_History_Age    Payment_of_Min_Amount
##                        0                        0                        0
##      Total_EMI_per_month  Amount_invested_monthly        Payment_Behaviour
##                        0                        0                        0
##          Monthly_Balance              Last_Loan_9              Last_Loan_8
##                        0                        0                        0
##              Last_Loan_7              Last_Loan_6              Last_Loan_5
##                        0                        0                        0
##              Last_Loan_4              Last_Loan_3              Last_Loan_2
##                        0                        0                        0
##              Last_Loan_1             Credit_Score
##                        0                        0
```

```
# Check if there are any missing values in the dataframe
any(is.na(credit_score))
```

```
## [1] FALSE
```

#Function for Numerical and Categorical Correlation - ANOVA - Eta Squared

```
# Load necessary library
if (!require("lsr")) install.packages("lsr")  # lsr package provides the etaSquared function
```

```
## Loading required package: lsr
```

```
## Warning: package 'lsr' was built under R version 4.4.2
```

```
library(lsr)

# Define function to calculate p-value, eta-squared, and show means for each level of categor
ical variable
calc_anova_eta_squared <- function(data, categorical_var, numeric_var) {

  # Convert categorical variable to factor if not already
  data[[categorical_var]] <- as.factor(data[[categorical_var]])

  # Perform ANOVA
  anova_model <- aov(data[[numeric_var]] ~ data[[categorical_var]], data = data)

  # Extract the p-value
  p_value <- summary(anova_model)[[1]][["Pr(>F)"]][1]

  # Format p-value: Display "<.0001" if p-value is very small
  formatted_p_value <- if (p_value < 0.0001) "<.0001" else format.pval(p_value, digits = 4)

  # Calculate eta-squared
  eta_squared <- etaSquared(anova_model)[1, "eta.sq"]

  # Calculate mean of numeric variable for each level of the categorical variable
  group_means <- tapply(data[[numeric_var]], data[[categorical_var]], mean)

  # Output results
  list(
    p_value = formatted_p_value,
    eta_squared = eta_squared,
    significance = ifelse(p_value < 0.05, "Significant", "Not Significant"),
    interpretation = ifelse(eta_squared < 0.06, "Weak correlation",
                            ifelse(eta_squared < 0.14, "Moderate correlation", "Strong correl
ation")),
    means = group_means  # Include the means of the numeric variable for each category
  )
}

# Example usage:
result <- calc_anova_eta_squared(credit_score, "Credit_Score", "Annual_Income")
print(result)
```

```
## $p_value
## [1] "<.0001"
##
## $eta_squared
## [1] 0.04579932
##
## $significance
## [1] "Significant"
##
## $interpretation
## [1] "Weak correlation"
##
## $means
##     Good     Poor Standard
## 65203.67 40584.52 50987.16
```

#Categorical and Numeric Attributes - ANOVA and Eta-Squared Calculation with Results Stored in a List for Categorical and Numeric Variables

```
# Load necessary libraries
if (!require("progress")) install.packages("progress")
```

```
## Loading required package: progress
```

```
## Warning: package 'progress' was built under R version 4.4.2
```

```
if (!require("lsr")) install.packages("lsr")  # lsr package for etaSquared function
library(progress)
library(lsr)

# Define function to calculate p-value, eta-squared, and show means for each level of categor
ical variable
calc_anova_eta_squared <- function(data, categorical_vars, numeric_vars) {

  # Initialize the progress bar
  total_combinations <- length(categorical_vars) * length(numeric_vars)
  pb <- progress_bar$new(
    total = total_combinations,
    format = "Processing [:bar] :percent ETA: :eta",
    clear = FALSE, width = 60
  )

  # Store results in a list
  results <- list()

  # Perform ANOVA for each combination of categorical and numeric variables
  for (cat_var in categorical_vars) {
    for (num_var in numeric_vars) {

      # Convert categorical variable to factor if not already
      data[[cat_var]] <- as.factor(data[[cat_var]])

      # Perform ANOVA
      anova_model <- aov(data[[num_var]] ~ data[[cat_var]], data = data)

      # Extract the p-value
      p_value <- summary(anova_model)[[1]][["Pr(>F)"]][1]

      # Format p-value: Display "<.0001" if p-value is very small
      formatted_p_value <- if (p_value < 0.0001) "<.0001" else format.pval(p_value, digits =
4)

      # Calculate eta-squared
      eta_squared <- etaSquared(anova_model)[1, "eta.sq"]

      # Calculate mean of numeric variable for each level of the categorical variable
      group_means <- tapply(data[[num_var]], data[[cat_var]], mean)

      # Store results
      results[[paste(cat_var, num_var, sep = "_")]] <- list(
        p_value = formatted_p_value,
        eta_squared = eta_squared,
        significance = ifelse(p_value < 0.05, "Significant", "Not Significant"),
        interpretation = ifelse(eta_squared < 0.06, "Weak correlation",
                                ifelse(eta_squared < 0.14, "Moderate correlation", "Strong co
rrelation")),
        means = group_means  # Include the means of the numeric variable for each category
```

```
      )

      # Update progress bar
      pb$tick()
    }
  }

  return(results)
}

# Example usage:
# Assuming 'credit_score' is your dataframe and you have numeric and categorical columns.
categorical_vars <- c("Credit_Mix", "Payment_of_Min_Amount", "Payment_Behaviour", "Occupatio
n",
                      "Credit_Score", "Last_Loan_1", "Last_Loan_2", "Last_Loan_3", "Last_Loan
_4",
                      "Last_Loan_5", "Last_Loan_6", "Last_Loan_7", "Last_Loan_8", "Last_Loan_
9")
numeric_vars <- c("Age", "Annual_Income", "Monthly_Inhand_Salary", "Num_Bank_Accounts", "Num_
Credit_Card",
                  "Interest_Rate", "Num_of_Loan", "Delay_from_due_date", "Num_of_Delayed_Paym
ent",
                  "Changed_Credit_Limit", "Num_Credit_Inquiries", "Outstanding_Debt", "Credit
_Utilization_Ratio",
                  "Credit_History_Age", "Total_EMI_per_month", "Amount_invested_monthly", "Mo
nthly_Balance")

# Call the function
result1 <- calc_anova_eta_squared(credit_score, categorical_vars, numeric_vars)

# Print the results (optional)
print(result1)
```

```
## $Credit_Mix_Age
## $Credit_Mix_Age$p_value
## [1] "<.0001"
##
## $Credit_Mix_Age$eta_squared
## [1] 0.05941164
##
## $Credit_Mix_Age$significance
## [1] "Significant"
##
## $Credit_Mix_Age$interpretation
## [1] "Weak correlation"
##
## $Credit_Mix_Age$means
##      Bad     Good Standard
## 29.73023 36.81734 32.83977
##
##
## $Credit_Mix_Annual_Income
## $Credit_Mix_Annual_Income$p_value
## [1] "<.0001"
##
## $Credit_Mix_Annual_Income$eta_squared
## [1] 0.1185959
##
## $Credit_Mix_Annual_Income$significance
## [1] "Significant"
##
## $Credit_Mix_Annual_Income$interpretation
## [1] "Moderate correlation"
##
## $Credit_Mix_Annual_Income$means
##      Bad     Good Standard
## 32970.45 68405.49 47732.47
##
##
## $Credit_Mix_Monthly_Inhand_Salary
## $Credit_Mix_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Credit_Mix_Monthly_Inhand_Salary$eta_squared
## [1] 0.1177831
##
## $Credit_Mix_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Credit_Mix_Monthly_Inhand_Salary$interpretation
## [1] "Moderate correlation"
##
## $Credit_Mix_Monthly_Inhand_Salary$means
##      Bad     Good Standard
```

```
## 2744.548 5683.483 3968.719
##
##
## $Credit_Mix_Num_Bank_Accounts
## $Credit_Mix_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Credit_Mix_Num_Bank_Accounts$eta_squared
## [1] 0.5260647
##
## $Credit_Mix_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Credit_Mix_Num_Bank_Accounts$interpretation
## [1] "Strong correlation"
##
## $Credit_Mix_Num_Bank_Accounts$means
##      Bad     Good Standard
## 7.967014 2.874506 5.674860
##
##
## $Credit_Mix_Num_Credit_Card
## $Credit_Mix_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Credit_Mix_Num_Credit_Card$eta_squared
## [1] 0.3161196
##
## $Credit_Mix_Num_Credit_Card$significance
## [1] "Significant"
##
## $Credit_Mix_Num_Credit_Card$interpretation
## [1] "Strong correlation"
##
## $Credit_Mix_Num_Credit_Card$means
##      Bad     Good Standard
## 7.471222 4.332313 5.325314
##
##
## $Credit_Mix_Interest_Rate
## $Credit_Mix_Interest_Rate$p_value
## [1] "<.0001"
##
## $Credit_Mix_Interest_Rate$eta_squared
## [1] 0.5687012
##
## $Credit_Mix_Interest_Rate$significance
## [1] "Significant"
##
## $Credit_Mix_Interest_Rate$interpretation
## [1] "Strong correlation"
```

```
##
## $Credit_Mix_Interest_Rate$means
##       Bad      Good  Standard
## 24.463144  6.421011 14.759030
##
##
## $Credit_Mix_Num_of_Loan
## $Credit_Mix_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Credit_Mix_Num_of_Loan$eta_squared
## [1] 0.4548029
##
## $Credit_Mix_Num_of_Loan$significance
## [1] "Significant"
##
## $Credit_Mix_Num_of_Loan$interpretation
## [1] "Strong correlation"
##
## $Credit_Mix_Num_of_Loan$means
##      Bad     Good Standard
## 6.363514 1.984992 3.091258
##
##
## $Credit_Mix_Delay_from_due_date
## $Credit_Mix_Delay_from_due_date$p_value
## [1] "<.0001"
##
## $Credit_Mix_Delay_from_due_date$eta_squared
## [1] 0.5024109
##
## $Credit_Mix_Delay_from_due_date$significance
## [1] "Significant"
##
## $Credit_Mix_Delay_from_due_date$interpretation
## [1] "Strong correlation"
##
## $Credit_Mix_Delay_from_due_date$means
##      Bad     Good Standard
## 38.52179 10.00471 19.35328
##
##
## $Credit_Mix_Num_of_Delayed_Payment
## $Credit_Mix_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Credit_Mix_Num_of_Delayed_Payment$eta_squared
## [1] 0.5753076
##
## $Credit_Mix_Num_of_Delayed_Payment$significance
## [1] "Significant"
```

```
##
## $Credit_Mix_Num_of_Delayed_Payment$interpretation
## [1] "Strong correlation"
##
## $Credit_Mix_Num_of_Delayed_Payment$means
##       Bad      Good  Standard
## 19.975976  7.074842 14.028376
##
##
## $Credit_Mix_Changed_Credit_Limit
## $Credit_Mix_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Credit_Mix_Changed_Credit_Limit$eta_squared
## [1] 0.1923268
##
## $Credit_Mix_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Credit_Mix_Changed_Credit_Limit$interpretation
## [1] "Strong correlation"
##
## $Credit_Mix_Changed_Credit_Limit$means
##      Bad     Good Standard
## 13.96101  6.16993 11.33348
##
##
## $Credit_Mix_Num_Credit_Inquiries
## $Credit_Mix_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Credit_Mix_Num_Credit_Inquiries$eta_squared
## [1] 0.4490802
##
## $Credit_Mix_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Credit_Mix_Num_Credit_Inquiries$interpretation
## [1] "Strong correlation"
##
## $Credit_Mix_Num_Credit_Inquiries$means
##      Bad     Good Standard
## 9.877440 2.803548 5.612720
##
##
## $Credit_Mix_Outstanding_Debt
## $Credit_Mix_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Credit_Mix_Outstanding_Debt$eta_squared
## [1] 0.5932736
```

```
##
## $Credit_Mix_Outstanding_Debt$significance
## [1] "Significant"
##
## $Credit_Mix_Outstanding_Debt$interpretation
## [1] "Strong correlation"
##
## $Credit_Mix_Outstanding_Debt$means
##       Bad      Good  Standard
## 3000.3726  741.3841 1064.0151
##
##
## $Credit_Mix_Credit_Utilization_Ratio
## $Credit_Mix_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Credit_Mix_Credit_Utilization_Ratio$eta_squared
## [1] 0.008200034
##
## $Credit_Mix_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Credit_Mix_Credit_Utilization_Ratio$interpretation
## [1] "Weak correlation"
##
## $Credit_Mix_Credit_Utilization_Ratio$means
##      Bad     Good Standard
## 31.65592 32.90634 32.19973
##
##
## $Credit_Mix_Credit_History_Age
## $Credit_Mix_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Credit_Mix_Credit_History_Age$eta_squared
## [1] 0.4238448
##
## $Credit_Mix_Credit_History_Age$significance
## [1] "Significant"
##
## $Credit_Mix_Credit_History_Age$interpretation
## [1] "Strong correlation"
##
## $Credit_Mix_Credit_History_Age$means
##      Bad     Good Standard
## 116.7528 293.8067 227.2440
##
##
## $Credit_Mix_Total_EMI_per_month
## $Credit_Mix_Total_EMI_per_month$p_value
## [1] "<.0001"
```

```
##
## $Credit_Mix_Total_EMI_per_month$eta_squared
## [1] 0.03119667
##
## $Credit_Mix_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Credit_Mix_Total_EMI_per_month$interpretation
## [1] "Weak correlation"
##
## $Credit_Mix_Total_EMI_per_month$means
##       Bad      Good  Standard
## 114.41221  77.99428  81.39542
##
##
## $Credit_Mix_Amount_invested_monthly
## $Credit_Mix_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Credit_Mix_Amount_invested_monthly$eta_squared
## [1] 0.04838874
##
## $Credit_Mix_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Credit_Mix_Amount_invested_monthly$interpretation
## [1] "Weak correlation"
##
## $Credit_Mix_Amount_invested_monthly$means
##      Bad     Good Standard
## 136.1722 251.5255 185.1716
##
##
## $Credit_Mix_Monthly_Balance
## $Credit_Mix_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Credit_Mix_Monthly_Balance$eta_squared
## [1] 0.1427186
##
## $Credit_Mix_Monthly_Balance$significance
## [1] "Significant"
##
## $Credit_Mix_Monthly_Balance$interpretation
## [1] "Strong correlation"
##
## $Credit_Mix_Monthly_Balance$means
##      Bad     Good Standard
## 284.0488 505.4143 397.8681
##
##
```

```
## $Payment_of_Min_Amount_Age
## $Payment_of_Min_Amount_Age$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Age$eta_squared
## [1] 0.07271744
##
## $Payment_of_Min_Amount_Age$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Age$interpretation
## [1] "Moderate correlation"
##
## $Payment_of_Min_Amount_Age$means
##       No      Yes
## 36.82282 30.91089
##
##
## $Payment_of_Min_Amount_Annual_Income
## $Payment_of_Min_Amount_Annual_Income$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Annual_Income$eta_squared
## [1] 0.09586965
##
## $Payment_of_Min_Amount_Annual_Income$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Annual_Income$interpretation
## [1] "Moderate correlation"
##
## $Payment_of_Min_Amount_Annual_Income$means
##       No      Yes
## 64858.33 40707.69
##
##
## $Payment_of_Min_Amount_Monthly_Inhand_Salary
## $Payment_of_Min_Amount_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Monthly_Inhand_Salary$eta_squared
## [1] 0.09537525
##
## $Payment_of_Min_Amount_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Monthly_Inhand_Salary$interpretation
## [1] "Moderate correlation"
##
## $Payment_of_Min_Amount_Monthly_Inhand_Salary$means
##       No      Yes
```

```
## 5390.243 3385.479
##
##
## $Payment_of_Min_Amount_Num_Bank_Accounts
## $Payment_of_Min_Amount_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Num_Bank_Accounts$eta_squared
## [1] 0.3423564
##
## $Payment_of_Min_Amount_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Num_Bank_Accounts$interpretation
## [1] "Strong correlation"
##
## $Payment_of_Min_Amount_Num_Bank_Accounts$means
##       No      Yes
## 3.532242 6.622426
##
##
## $Payment_of_Min_Amount_Num_Credit_Card
## $Payment_of_Min_Amount_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Num_Credit_Card$eta_squared
## [1] 0.1726277
##
## $Payment_of_Min_Amount_Num_Credit_Card$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Num_Credit_Card$interpretation
## [1] "Strong correlation"
##
## $Payment_of_Min_Amount_Num_Credit_Card$means
##       No      Yes
## 4.494133 6.243202
##
##
## $Payment_of_Min_Amount_Interest_Rate
## $Payment_of_Min_Amount_Interest_Rate$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Interest_Rate$eta_squared
## [1] 0.4019252
##
## $Payment_of_Min_Amount_Interest_Rate$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Interest_Rate$interpretation
## [1] "Strong correlation"
```

```
##
## $Payment_of_Min_Amount_Interest_Rate$means
##        No       Yes
##  7.824492 19.110647
##
##
## $Payment_of_Min_Amount_Num_of_Loan
## $Payment_of_Min_Amount_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Num_of_Loan$eta_squared
## [1] 0.2682129
##
## $Payment_of_Min_Amount_Num_of_Loan$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Num_of_Loan$interpretation
## [1] "Strong correlation"
##
## $Payment_of_Min_Amount_Num_of_Loan$means
##       No      Yes
## 1.999408 4.579620
##
##
## $Payment_of_Min_Amount_Delay_from_due_date
## $Payment_of_Min_Amount_Delay_from_due_date$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Delay_from_due_date$eta_squared
## [1] 0.259189
##
## $Payment_of_Min_Amount_Delay_from_due_date$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Delay_from_due_date$interpretation
## [1] "Strong correlation"
##
## $Payment_of_Min_Amount_Delay_from_due_date$means
##       No      Yes
## 11.91193 27.31921
##
##
## $Payment_of_Min_Amount_Num_of_Delayed_Payment
## $Payment_of_Min_Amount_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Num_of_Delayed_Payment$eta_squared
## [1] 0.3536356
##
## $Payment_of_Min_Amount_Num_of_Delayed_Payment$significance
## [1] "Significant"
```

```
##
## $Payment_of_Min_Amount_Num_of_Delayed_Payment$interpretation
## [1] "Strong correlation"
##
## $Payment_of_Min_Amount_Num_of_Delayed_Payment$means
##        No       Yes
##  8.816949 16.409308
##
##
## $Payment_of_Min_Amount_Changed_Credit_Limit
## $Payment_of_Min_Amount_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Changed_Credit_Limit$eta_squared
## [1] 0.2576029
##
## $Payment_of_Min_Amount_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Changed_Credit_Limit$interpretation
## [1] "Strong correlation"
##
## $Payment_of_Min_Amount_Changed_Credit_Limit$means
##        No       Yes
##  6.221857 13.233637
##
##
## $Payment_of_Min_Amount_Num_Credit_Inquiries
## $Payment_of_Min_Amount_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Num_Credit_Inquiries$eta_squared
## [1] 0.40144
##
## $Payment_of_Min_Amount_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Num_Credit_Inquiries$interpretation
## [1] "Strong correlation"
##
## $Payment_of_Min_Amount_Num_Credit_Inquiries$means
##       No      Yes
## 2.811797 7.794000
##
##
## $Payment_of_Min_Amount_Outstanding_Debt
## $Payment_of_Min_Amount_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Outstanding_Debt$eta_squared
## [1] 0.2430213
```

```
##
## $Payment_of_Min_Amount_Outstanding_Debt$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Outstanding_Debt$interpretation
## [1] "Strong correlation"
##
## $Payment_of_Min_Amount_Outstanding_Debt$means
##        No       Yes
##  736.9829 1896.6906
##
##
## $Payment_of_Min_Amount_Credit_Utilization_Ratio
## $Payment_of_Min_Amount_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Credit_Utilization_Ratio$eta_squared
## [1] 0.006952684
##
## $Payment_of_Min_Amount_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Credit_Utilization_Ratio$interpretation
## [1] "Weak correlation"
##
## $Payment_of_Min_Amount_Credit_Utilization_Ratio$means
##       No      Yes
## 32.80159 31.93267
##
##
## $Payment_of_Min_Amount_Credit_History_Age
## $Payment_of_Min_Amount_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Credit_History_Age$eta_squared
## [1] 0.3505139
##
## $Payment_of_Min_Amount_Credit_History_Age$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Credit_History_Age$interpretation
## [1] "Strong correlation"
##
## $Payment_of_Min_Amount_Credit_History_Age$means
##       No      Yes
## 292.6371 172.4490
##
##
## $Payment_of_Min_Amount_Total_EMI_per_month
## $Payment_of_Min_Amount_Total_EMI_per_month$p_value
## [1] "<.0001"
```

```
##
## $Payment_of_Min_Amount_Total_EMI_per_month$eta_squared
## [1] 0.01568684
##
## $Payment_of_Min_Amount_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Total_EMI_per_month$interpretation
## [1] "Weak correlation"
##
## $Payment_of_Min_Amount_Total_EMI_per_month$means
##       No      Yes
## 75.59000 96.82343
##
##
## $Payment_of_Min_Amount_Amount_invested_monthly
## $Payment_of_Min_Amount_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Amount_invested_monthly$eta_squared
## [1] 0.03889214
##
## $Payment_of_Min_Amount_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Amount_invested_monthly$interpretation
## [1] "Weak correlation"
##
## $Payment_of_Min_Amount_Amount_invested_monthly$means
##       No      Yes
## 240.1810 161.9493
##
##
## $Payment_of_Min_Amount_Monthly_Balance
## $Payment_of_Min_Amount_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Monthly_Balance$eta_squared
## [1] 0.1139094
##
## $Payment_of_Min_Amount_Monthly_Balance$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Monthly_Balance$interpretation
## [1] "Moderate correlation"
##
## $Payment_of_Min_Amount_Monthly_Balance$means
##       No      Yes
## 491.0850 343.7021
##
##
```

```
## $Payment_Behaviour_Age
## $Payment_Behaviour_Age$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Age$eta_squared
## [1] 0.002775788
##
## $Payment_Behaviour_Age$significance
## [1] "Significant"
##
## $Payment_Behaviour_Age$interpretation
## [1] "Weak correlation"
##
## $Payment_Behaviour_Age$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         33.99197                         33.66188
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         33.81570                         33.57642
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         33.52680                         32.58551
##
##
## $Payment_Behaviour_Annual_Income
## $Payment_Behaviour_Annual_Income$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Annual_Income$eta_squared
## [1] 0.2885956
##
## $Payment_Behaviour_Annual_Income$significance
## [1] "Significant"
##
## $Payment_Behaviour_Annual_Income$interpretation
## [1] "Strong correlation"
##
## $Payment_Behaviour_Annual_Income$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         86060.40                         58911.78
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         50723.75                         48546.96
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         62484.76                         27340.83
##
##
## $Payment_Behaviour_Monthly_Inhand_Salary
## $Payment_Behaviour_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Monthly_Inhand_Salary$eta_squared
## [1] 0.2901668
##
```

```
## $Payment_Behaviour_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Payment_Behaviour_Monthly_Inhand_Salary$interpretation
## [1] "Strong correlation"
##
## $Payment_Behaviour_Monthly_Inhand_Salary$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         7161.891                         4902.956
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         4224.426                         4041.143
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         5197.477                         2263.363
##
##
## $Payment_Behaviour_Num_Bank_Accounts
## $Payment_Behaviour_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Num_Bank_Accounts$eta_squared
## [1] 0.0234812
##
## $Payment_Behaviour_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Payment_Behaviour_Num_Bank_Accounts$interpretation
## [1] "Weak correlation"
##
## $Payment_Behaviour_Num_Bank_Accounts$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         4.713129                         5.218528
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         5.188053                         5.312483
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         5.193961                         5.837162
##
##
## $Payment_Behaviour_Num_Credit_Card
## $Payment_Behaviour_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Num_Credit_Card$eta_squared
## [1] 0.01338979
##
## $Payment_Behaviour_Num_Credit_Card$significance
## [1] "Significant"
##
## $Payment_Behaviour_Num_Credit_Card$interpretation
## [1] "Weak correlation"
##
## $Payment_Behaviour_Num_Credit_Card$means
```

```
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         5.153248                         5.451278
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         5.463831                         5.404134
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         5.412374                         5.820382
##
##
## $Payment_Behaviour_Interest_Rate
## $Payment_Behaviour_Interest_Rate$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Interest_Rate$eta_squared
## [1] 0.02957621
##
## $Payment_Behaviour_Interest_Rate$significance
## [1] "Significant"
##
## $Payment_Behaviour_Interest_Rate$interpretation
## [1] "Weak correlation"
##
## $Payment_Behaviour_Interest_Rate$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         11.91794                         14.07890
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         14.21931                         13.94506
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         13.96180                         16.25768
##
##
## $Payment_Behaviour_Num_of_Loan
## $Payment_Behaviour_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Num_of_Loan$eta_squared
## [1] 0.02333046
##
## $Payment_Behaviour_Num_of_Loan$significance
## [1] "Significant"
##
## $Payment_Behaviour_Num_of_Loan$interpretation
## [1] "Weak correlation"
##
## $Payment_Behaviour_Num_of_Loan$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         2.939180                         3.380886
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         3.432449                         3.329209
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         3.377654                         3.981446
##
```

```
##
## $Payment_Behaviour_Delay_from_due_date
## $Payment_Behaviour_Delay_from_due_date$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Delay_from_due_date$eta_squared
## [1] 0.02057324
##
## $Payment_Behaviour_Delay_from_due_date$significance
## [1] "Significant"
##
## $Payment_Behaviour_Delay_from_due_date$interpretation
## [1] "Weak correlation"
##
## $Payment_Behaviour_Delay_from_due_date$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         17.47950                         20.45214
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         20.58645                         20.39203
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         19.87208                         23.55059
##
##
## $Payment_Behaviour_Num_of_Delayed_Payment
## $Payment_Behaviour_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Num_of_Delayed_Payment$eta_squared
## [1] 0.02309311
##
## $Payment_Behaviour_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Payment_Behaviour_Num_of_Delayed_Payment$interpretation
## [1] "Weak correlation"
##
## $Payment_Behaviour_Num_of_Delayed_Payment$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         11.67699                         12.99678
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         13.09112                         13.17800
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         12.93056                         14.42303
##
##
## $Payment_Behaviour_Changed_Credit_Limit
## $Payment_Behaviour_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Changed_Credit_Limit$eta_squared
## [1] 0.008932565
```

```
##
## $Payment_Behaviour_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Payment_Behaviour_Changed_Credit_Limit$interpretation
## [1] "Weak correlation"
##
## $Payment_Behaviour_Changed_Credit_Limit$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         9.297587                        10.309594
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                        10.268622                        10.207857
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         9.886686                        11.115083
##
##
## $Payment_Behaviour_Num_Credit_Inquiries
## $Payment_Behaviour_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Num_Credit_Inquiries$eta_squared
## [1] 0.02454864
##
## $Payment_Behaviour_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Payment_Behaviour_Num_Credit_Inquiries$interpretation
## [1] "Weak correlation"
##
## $Payment_Behaviour_Num_Credit_Inquiries$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         4.754777                         5.539448
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         5.654671                         5.497552
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         5.569614                         6.479754
##
##
## $Payment_Behaviour_Outstanding_Debt
## $Payment_Behaviour_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Outstanding_Debt$eta_squared
## [1] 0.02655182
##
## $Payment_Behaviour_Outstanding_Debt$significance
## [1] "Significant"
##
## $Payment_Behaviour_Outstanding_Debt$interpretation
## [1] "Weak correlation"
##
```

```
## $Payment_Behaviour_Outstanding_Debt$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         1121.847                         1364.231
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         1367.401                         1333.896
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         1333.607                         1650.451
##
##
## $Payment_Behaviour_Credit_Utilization_Ratio
## $Payment_Behaviour_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Credit_Utilization_Ratio$eta_squared
## [1] 0.01441307
##
## $Payment_Behaviour_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Payment_Behaviour_Credit_Utilization_Ratio$interpretation
## [1] "Weak correlation"
##
## $Payment_Behaviour_Credit_Utilization_Ratio$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         33.46035                         32.59743
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         32.30471                         32.04246
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         32.24858                         31.66858
##
##
## $Payment_Behaviour_Credit_History_Age
## $Payment_Behaviour_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Credit_History_Age$eta_squared
## [1] 0.02501843
##
## $Payment_Behaviour_Credit_History_Age$significance
## [1] "Significant"
##
## $Payment_Behaviour_Credit_History_Age$interpretation
## [1] "Weak correlation"
##
## $Payment_Behaviour_Credit_History_Age$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         248.1808                         226.1191
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         225.4581                         227.9422
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         227.0204                         202.9229
```

```
##
##
## $Payment_Behaviour_Total_EMI_per_month
## $Payment_Behaviour_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Total_EMI_per_month$eta_squared
## [1] 0.07220656
##
## $Payment_Behaviour_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Payment_Behaviour_Total_EMI_per_month$interpretation
## [1] "Moderate correlation"
##
## $Payment_Behaviour_Total_EMI_per_month$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                        118.77219                        108.35951
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         93.88085                         84.98579
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         94.24831                         61.36855
##
##
## $Payment_Behaviour_Amount_invested_monthly
## $Payment_Behaviour_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Amount_invested_monthly$eta_squared
## [1] 0.05151765
##
## $Payment_Behaviour_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Payment_Behaviour_Amount_invested_monthly$interpretation
## [1] "Weak correlation"
##
## $Payment_Behaviour_Amount_invested_monthly$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         229.6089                         182.7248
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         190.6078                         219.9396
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         287.2781                         150.8571
##
##
## $Payment_Behaviour_Monthly_Balance
## $Payment_Behaviour_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Monthly_Balance$eta_squared
```

```
## [1] 0.2441823
##
## $Payment_Behaviour_Monthly_Balance$significance
## [1] "Significant"
##
## $Payment_Behaviour_Monthly_Balance$interpretation
## [1] "Strong correlation"
##
## $Payment_Behaviour_Monthly_Balance$means
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                         607.8046                         455.2748
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                         396.9412                         362.2228
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                         403.4551                         297.7876
##
##
## $Occupation_Age
## $Occupation_Age$p_value
## [1] "<.0001"
##
## $Occupation_Age$eta_squared
## [1] 0.00131741
##
## $Occupation_Age$significance
## [1] "Significant"
##
## $Occupation_Age$interpretation
## [1] "Weak correlation"
##
## $Occupation_Age$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      33.57681      33.45721      33.28378      33.06090      32.51690
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      33.34341      33.91952      33.92052      32.87951      33.51933
## Media_Manager      Musician     Scientist       Teacher        Writer
##      33.21027      33.56423      33.42957      32.66757      33.26618
##
##
## $Occupation_Annual_Income
## $Occupation_Annual_Income$p_value
## [1] "<.0001"
##
## $Occupation_Annual_Income$eta_squared
## [1] 0.000518906
##
## $Occupation_Annual_Income$significance
## [1] "Significant"
##
## $Occupation_Annual_Income$interpretation
## [1] "Weak correlation"
```

```
##
## $Occupation_Annual_Income$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      50726.12      51875.39      50128.42      49517.75      50730.65
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      50987.70      48430.09      49594.22      51466.77      50095.11
## Media_Manager      Musician     Scientist       Teacher        Writer
##      50618.44      51365.52      51196.43      49865.04      51047.66
##
##
## $Occupation_Monthly_Inhand_Salary
## $Occupation_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Occupation_Monthly_Inhand_Salary$eta_squared
## [1] 0.0004953275
##
## $Occupation_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Occupation_Monthly_Inhand_Salary$interpretation
## [1] "Weak correlation"
##
## $Occupation_Monthly_Inhand_Salary$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      4218.649      4318.224      4164.362      4121.761      4217.174
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      4238.129      4028.802      4125.812      4273.792      4167.424
## Media_Manager      Musician     Scientist       Teacher        Writer
##      4209.038      4261.559      4252.434      4147.587      4241.635
##
##
## $Occupation_Num_Bank_Accounts
## $Occupation_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Occupation_Num_Bank_Accounts$eta_squared
## [1] 0.0007162473
##
## $Occupation_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Occupation_Num_Bank_Accounts$interpretation
## [1] "Weak correlation"
##
## $Occupation_Num_Bank_Accounts$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      5.315095      5.327081      5.345387      5.386114      5.386072
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      5.378761      5.390759      5.235485      5.394900      5.429309
## Media_Manager      Musician     Scientist       Teacher        Writer
```

```
##      5.245685      5.415617      5.440243      5.349371      5.512849
##
##
## $Occupation_Num_Credit_Card
## $Occupation_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Occupation_Num_Credit_Card$eta_squared
## [1] 0.0007129517
##
## $Occupation_Num_Credit_Card$significance
## [1] "Significant"
##
## $Occupation_Num_Credit_Card$interpretation
## [1] "Weak correlation"
##
## $Occupation_Num_Credit_Card$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      5.558126      5.452521      5.619940      5.491017      5.535693
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      5.573706      5.579253      5.481821      5.475746      5.465762
## Media_Manager      Musician     Scientist       Teacher        Writer
##      5.548065      5.500157      5.509045      5.635941      5.582646
##
##
## $Occupation_Interest_Rate
## $Occupation_Interest_Rate$p_value
## [1] "<.0001"
##
## $Occupation_Interest_Rate$eta_squared
## [1] 0.0005931851
##
## $Occupation_Interest_Rate$significance
## [1] "Significant"
##
## $Occupation_Interest_Rate$interpretation
## [1] "Weak correlation"
##
## $Occupation_Interest_Rate$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      14.56702      13.98945      14.69524      14.48965      14.70746
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      14.50903      14.47368      14.50620      14.83582      14.77332
## Media_Manager      Musician     Scientist       Teacher        Writer
##      14.27381      14.61209      14.30961      14.53118      14.73985
##
##
## $Occupation_Num_of_Loan
## $Occupation_Num_of_Loan$p_value
## [1] "<.0001"
##
```

```
## $Occupation_Num_of_Loan$eta_squared
## [1] 0.001116309
##
## $Occupation_Num_of_Loan$significance
## [1] "Significant"
##
## $Occupation_Num_of_Loan$interpretation
## [1] "Weak correlation"
##
## $Occupation_Num_of_Loan$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      3.527877      3.422040      3.605952      3.697929      3.545455
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      3.641396      3.402693      3.479143      3.541045      3.608028
## Media_Manager      Musician     Scientist       Teacher        Writer
##      3.464286      3.448363      3.595492      3.483213      3.530457
##
##
## $Occupation_Delay_from_due_date
## $Occupation_Delay_from_due_date$p_value
## [1] "<.0001"
##
## $Occupation_Delay_from_due_date$eta_squared
## [1] 0.000706538
##
## $Occupation_Delay_from_due_date$significance
## [1] "Significant"
##
## $Occupation_Delay_from_due_date$interpretation
## [1] "Weak correlation"
##
## $Occupation_Delay_from_due_date$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      21.13864      20.43406      21.23527      20.82110      21.71839
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      21.12771      20.25092      20.85781      21.26912      21.26313
## Media_Manager      Musician     Scientist       Teacher        Writer
##      20.93824      20.94600      20.99318      21.81550      21.21780
##
##
## $Occupation_Num_of_Delayed_Payment
## $Occupation_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Occupation_Num_of_Delayed_Payment$eta_squared
## [1] 0.0005594529
##
## $Occupation_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Occupation_Num_of_Delayed_Payment$interpretation
```

```
## [1] "Weak correlation"
##
## $Occupation_Num_of_Delayed_Payment$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      13.47301      13.27008      13.50446      13.42677      13.28234
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      13.26173      13.36551      13.17897      13.47481      13.41160
## Media_Manager      Musician     Scientist       Teacher        Writer
##      13.13065      13.14625      13.57814      13.06640      13.37579
##
##
## $Occupation_Changed_Credit_Limit
## $Occupation_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Occupation_Changed_Credit_Limit$eta_squared
## [1] 0.001000068
##
## $Occupation_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Occupation_Changed_Credit_Limit$interpretation
## [1] "Weak correlation"
##
## $Occupation_Changed_Credit_Limit$means
##    Accountant     Architect     Developer        Doctor      Engineer
##     10.790824     10.070191     10.615238     10.402107     10.600921
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      9.980824     10.299247     10.324963     10.374876     10.330189
## Media_Manager      Musician     Scientist       Teacher        Writer
##     10.474167     10.205844     10.208304     10.585492     10.572548
##
##
## $Occupation_Num_Credit_Inquiries
## $Occupation_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Occupation_Num_Credit_Inquiries$eta_squared
## [1] 0.0009477425
##
## $Occupation_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Occupation_Num_Credit_Inquiries$interpretation
## [1] "Weak correlation"
##
## $Occupation_Num_Credit_Inquiries$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      5.826068      5.563892      5.850000      5.752741      5.870629
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      5.851835      5.745716      5.702508      5.796486      5.983323
```

```
## Media_Manager      Musician     Scientist       Teacher        Writer
##      5.628125      5.523300      5.824288      5.836331      5.831060
##
##
## $Occupation_Outstanding_Debt
## $Occupation_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Occupation_Outstanding_Debt$eta_squared
## [1] 0.001146022
##
## $Occupation_Outstanding_Debt$significance
## [1] "Significant"
##
## $Occupation_Outstanding_Debt$interpretation
## [1] "Weak correlation"
##
## $Occupation_Outstanding_Debt$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      1437.900      1343.206      1440.633      1452.149      1451.854
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      1397.060      1448.889      1351.292      1462.105      1486.373
## Media_Manager      Musician     Scientist       Teacher        Writer
##      1398.472      1404.972      1436.900      1443.942      1444.428
##
##
## $Occupation_Credit_Utilization_Ratio
## $Occupation_Credit_Utilization_Ratio$p_value
## [1] "0.08003"
##
## $Occupation_Credit_Utilization_Ratio$eta_squared
## [1] 0.0002193123
##
## $Occupation_Credit_Utilization_Ratio$significance
## [1] "Not Significant"
##
## $Occupation_Credit_Utilization_Ratio$interpretation
## [1] "Weak correlation"
##
## $Occupation_Credit_Utilization_Ratio$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      32.28438      32.35607      32.26252      32.22817      32.41186
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      32.22308      32.18259      32.20768      32.39788      32.19946
## Media_Manager      Musician     Scientist       Teacher        Writer
##      32.34060      32.33392      32.21289      32.38087      32.25891
##
##
## $Occupation_Credit_History_Age
## $Occupation_Credit_History_Age$p_value
## [1] "<.0001"
```

```
##
## $Occupation_Credit_History_Age$eta_squared
## [1] 0.0008017997
##
## $Occupation_Credit_History_Age$significance
## [1] "Significant"
##
## $Occupation_Credit_History_Age$interpretation
## [1] "Weak correlation"
##
## $Occupation_Credit_History_Age$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      220.3138      226.7216      219.0929      220.1468      217.4499
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      222.6131      225.7044      220.4944      222.2512      218.7149
## Media_Manager      Musician     Scientist       Teacher        Writer
##      222.8286      223.6071      222.9448      216.6691      218.7069
##
##
## $Occupation_Total_EMI_per_month
## $Occupation_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Occupation_Total_EMI_per_month$eta_squared
## [1] 0.0009488726
##
## $Occupation_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Occupation_Total_EMI_per_month$interpretation
## [1] "Weak correlation"
##
## $Occupation_Total_EMI_per_month$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      88.83771      89.52125      89.84890      91.92098      87.30305
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      92.46296      84.65696      84.72517      87.39131      89.01922
## Media_Manager      Musician     Scientist       Teacher        Writer
##      84.86676      87.60875      90.36341      84.48020      90.38485
##
##
## $Occupation_Amount_invested_monthly
## $Occupation_Amount_invested_monthly$p_value
## [1] "0.0004504"
##
## $Occupation_Amount_invested_monthly$eta_squared
## [1] 0.0003840157
##
## $Occupation_Amount_invested_monthly$significance
## [1] "Significant"
##
```

```
## $Occupation_Amount_invested_monthly$interpretation
## [1] "Weak correlation"
##
## $Occupation_Amount_invested_monthly$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      193.2917      197.2055      189.2158      189.6339      189.6237
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      198.5156      187.5666      189.5755      197.4318      196.5349
## Media_Manager      Musician     Scientist       Teacher        Writer
##      194.8639      197.7091      196.7895      190.3223      197.6384
##
##
## $Occupation_Monthly_Balance
## $Occupation_Monthly_Balance$p_value
## [1] "0.0003577"
##
## $Occupation_Monthly_Balance$eta_squared
## [1] 0.0003905053
##
## $Occupation_Monthly_Balance$significance
## [1] "Significant"
##
## $Occupation_Monthly_Balance$interpretation
## [1] "Weak correlation"
##
## $Occupation_Monthly_Balance$means
##    Accountant     Architect     Developer        Doctor      Engineer
##      403.5298      412.2911      403.7424      398.8748      406.3578
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##      397.7410      397.2139      402.9723      409.5640      398.4330
## Media_Manager      Musician     Scientist       Teacher        Writer
##      406.7829      407.1960      402.6366      402.9124      402.0458
##
##
## $Credit_Score_Age
## $Credit_Score_Age$p_value
## [1] "<.0001"
##
## $Credit_Score_Age$eta_squared
## [1] 0.02598354
##
## $Credit_Score_Age$significance
## [1] "Significant"
##
## $Credit_Score_Age$interpretation
## [1] "Weak correlation"
##
## $Credit_Score_Age$means
##     Good     Poor Standard
## 36.39522 31.18760 33.43160
##
```

```
##
## $Credit_Score_Annual_Income
## $Credit_Score_Annual_Income$p_value
## [1] "<.0001"
##
## $Credit_Score_Annual_Income$eta_squared
## [1] 0.04579932
##
## $Credit_Score_Annual_Income$significance
## [1] "Significant"
##
## $Credit_Score_Annual_Income$interpretation
## [1] "Weak correlation"
##
## $Credit_Score_Annual_Income$means
##     Good     Poor Standard
## 65203.67 40584.52 50987.16
##
##
## $Credit_Score_Monthly_Inhand_Salary
## $Credit_Score_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Credit_Score_Monthly_Inhand_Salary$eta_squared
## [1] 0.04461207
##
## $Credit_Score_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Credit_Score_Monthly_Inhand_Salary$interpretation
## [1] "Weak correlation"
##
## $Credit_Score_Monthly_Inhand_Salary$means
##     Good     Poor Standard
## 5403.107 3381.386 4240.741
##
##
## $Credit_Score_Num_Bank_Accounts
## $Credit_Score_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Credit_Score_Num_Bank_Accounts$eta_squared
## [1] 0.1564287
##
## $Credit_Score_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Credit_Score_Num_Bank_Accounts$interpretation
## [1] "Strong correlation"
##
## $Credit_Score_Num_Bank_Accounts$means
```

```
##     Good     Poor Standard
## 3.440206 6.526657 5.383985
##
##
## $Credit_Score_Num_Credit_Card
## $Credit_Score_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Credit_Score_Num_Credit_Card$eta_squared
## [1] 0.1635849
##
## $Credit_Score_Num_Credit_Card$significance
## [1] "Significant"
##
## $Credit_Score_Num_Credit_Card$interpretation
## [1] "Strong correlation"
##
## $Credit_Score_Num_Credit_Card$means
##     Good     Poor Standard
## 4.143314 6.625147 5.404540
##
##
## $Credit_Score_Interest_Rate
## $Credit_Score_Interest_Rate$p_value
## [1] "<.0001"
##
## $Credit_Score_Interest_Rate$eta_squared
## [1] 0.2357164
##
## $Credit_Score_Interest_Rate$significance
## [1] "Significant"
##
## $Credit_Score_Interest_Rate$interpretation
## [1] "Strong correlation"
##
## $Credit_Score_Interest_Rate$means
##      Good      Poor  Standard
##  7.662329 20.187116 13.751420
##
##
## $Credit_Score_Num_of_Loan
## $Credit_Score_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Credit_Score_Num_of_Loan$eta_squared
## [1] 0.1295394
##
## $Credit_Score_Num_of_Loan$significance
## [1] "Significant"
##
## $Credit_Score_Num_of_Loan$interpretation
```

```
## [1] "Moderate correlation"
##
## $Credit_Score_Num_of_Loan$means
##     Good     Poor Standard
## 2.202154 4.755742 3.312164
##
##
## $Credit_Score_Delay_from_due_date
## $Credit_Score_Delay_from_due_date$p_value
## [1] "<.0001"
##
## $Credit_Score_Delay_from_due_date$eta_squared
## [1] 0.1866246
##
## $Credit_Score_Delay_from_due_date$significance
## [1] "Significant"
##
## $Credit_Score_Delay_from_due_date$interpretation
## [1] "Strong correlation"
##
## $Credit_Score_Delay_from_due_date$means
##     Good     Poor Standard
## 10.89275 29.74405 19.74958
##
##
## $Credit_Score_Num_of_Delayed_Payment
## $Credit_Score_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Credit_Score_Num_of_Delayed_Payment$eta_squared
## [1] 0.1502467
##
## $Credit_Score_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Credit_Score_Num_of_Delayed_Payment$interpretation
## [1] "Strong correlation"
##
## $Credit_Score_Num_of_Delayed_Payment$means
##      Good      Poor  Standard
##  8.567422 15.845162 13.553729
##
##
## $Credit_Score_Changed_Credit_Limit
## $Credit_Score_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Credit_Score_Changed_Credit_Limit$eta_squared
## [1] 0.05093471
##
## $Credit_Score_Changed_Credit_Limit$significance
```

```
## [1] "Significant"
##
## $Credit_Score_Changed_Credit_Limit$interpretation
## [1] "Weak correlation"
##
## $Credit_Score_Changed_Credit_Limit$means
##     Good     Poor Standard
##  7.10329 11.04600 11.13252
##
##
## $Credit_Score_Num_Credit_Inquiries
## $Credit_Score_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Credit_Score_Num_Credit_Inquiries$eta_squared
## [1] 0.1924927
##
## $Credit_Score_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Credit_Score_Num_Credit_Inquiries$interpretation
## [1] "Strong correlation"
##
## $Credit_Score_Num_Credit_Inquiries$means
##     Good     Poor Standard
## 3.276587 8.154804 5.310753
##
##
## $Credit_Score_Outstanding_Debt
## $Credit_Score_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Credit_Score_Outstanding_Debt$eta_squared
## [1] 0.154202
##
## $Credit_Score_Outstanding_Debt$significance
## [1] "Significant"
##
## $Credit_Score_Outstanding_Debt$interpretation
## [1] "Strong correlation"
##
## $Credit_Score_Outstanding_Debt$means
##      Good      Poor  Standard
##  801.2563 2081.4732 1278.4196
##
##
## $Credit_Score_Credit_Utilization_Ratio
## $Credit_Score_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Credit_Score_Credit_Utilization_Ratio$eta_squared
```

```
## [1] 0.002101572
##
## $Credit_Score_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Credit_Score_Credit_Utilization_Ratio$interpretation
## [1] "Weak correlation"
##
## $Credit_Score_Credit_Utilization_Ratio$means
##     Good     Poor Standard
## 32.68611 31.98615 32.31382
##
##
## $Credit_Score_Credit_History_Age
## $Credit_Score_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Credit_Score_Credit_History_Age$eta_squared
## [1] 0.1511611
##
## $Credit_Score_Credit_History_Age$significance
## [1] "Significant"
##
## $Credit_Score_Credit_History_Age$interpretation
## [1] "Strong correlation"
##
## $Credit_Score_Credit_History_Age$means
##     Good     Poor Standard
## 284.7533 170.0483 227.8002
##
##
## $Credit_Score_Total_EMI_per_month
## $Credit_Score_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Credit_Score_Total_EMI_per_month$eta_squared
## [1] 0.00576629
##
## $Credit_Score_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Credit_Score_Total_EMI_per_month$interpretation
## [1] "Weak correlation"
##
## $Credit_Score_Total_EMI_per_month$means
##     Good     Poor Standard
## 81.73015 97.91238 85.09040
##
##
## $Credit_Score_Amount_invested_monthly
## $Credit_Score_Amount_invested_monthly$p_value
```

```
## [1] "<.0001"
##
## $Credit_Score_Amount_invested_monthly$eta_squared
## [1] 0.0246715
##
## $Credit_Score_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Credit_Score_Amount_invested_monthly$interpretation
## [1] "Weak correlation"
##
## $Credit_Score_Amount_invested_monthly$means
##     Good     Poor Standard
## 251.0849 159.0124 193.3511
##
##
## $Credit_Score_Monthly_Balance
## $Credit_Score_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Credit_Score_Monthly_Balance$eta_squared
## [1] 0.04350322
##
## $Credit_Score_Monthly_Balance$significance
## [1] "Significant"
##
## $Credit_Score_Monthly_Balance$interpretation
## [1] "Weak correlation"
##
## $Credit_Score_Monthly_Balance$means
##     Good     Poor Standard
## 478.6869 345.6344 409.8338
##
##
## $Last_Loan_1_Age
## $Last_Loan_1_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Age$eta_squared
## [1] 0.01050314
##
## $Last_Loan_1_Age$significance
## [1] "Significant"
##
## $Last_Loan_1_Age$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_1_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                33.13786                33.11551                 33.02729
##        Home Equity Loan           Mortgage Loan                  No Loan
```

```
##                33.09204                32.60876                36.32802
##           Not Specified             Payday Loan           Personal Loan
##                32.51488                33.04295                33.01600
##            Student Loan
##                32.73993
##
##
## $Last_Loan_1_Annual_Income
## $Last_Loan_1_Annual_Income$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Annual_Income$eta_squared
## [1] 0.01427276
##
## $Last_Loan_1_Annual_Income$significance
## [1] "Significant"
##
## $Last_Loan_1_Annual_Income$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_1_Annual_Income$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                48573.40                47366.74                48519.11
##        Home Equity Loan           Mortgage Loan                 No Loan
##                48780.82                50564.26                62766.88
##           Not Specified             Payday Loan           Personal Loan
##                51029.07                49667.41                46591.57
##            Student Loan
##                49221.35
##
##
## $Last_Loan_1_Monthly_Inhand_Salary
## $Last_Loan_1_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Monthly_Inhand_Salary$eta_squared
## [1] 0.01397252
##
## $Last_Loan_1_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Last_Loan_1_Monthly_Inhand_Salary$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_1_Monthly_Inhand_Salary$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                4028.291                3951.300                4025.835
##        Home Equity Loan           Mortgage Loan                 No Loan
##                4054.122                4206.215                5208.855
##           Not Specified             Payday Loan           Personal Loan
##                4244.502                4121.462                3879.665
```

```
##            Student Loan
##                4104.681
##
##
## $Last_Loan_1_Num_Bank_Accounts
## $Last_Loan_1_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Num_Bank_Accounts$eta_squared
## [1] 0.03132937
##
## $Last_Loan_1_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Last_Loan_1_Num_Bank_Accounts$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_1_Num_Bank_Accounts$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                5.557890                5.618163                5.478732
##        Home Equity Loan           Mortgage Loan                 No Loan
##                5.564694                5.504580                4.109222
##           Not Specified             Payday Loan           Personal Loan
##                5.401058                5.474367                5.701389
##            Student Loan
##                5.483154
##
##
## $Last_Loan_1_Num_Credit_Card
## $Last_Loan_1_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Num_Credit_Card$eta_squared
## [1] 0.02196558
##
## $Last_Loan_1_Num_Credit_Card$significance
## [1] "Significant"
##
## $Last_Loan_1_Num_Credit_Card$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_1_Num_Credit_Card$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                5.635935                5.663163                5.628812
##        Home Equity Loan           Mortgage Loan                 No Loan
##                5.597347                5.600229                4.685747
##           Not Specified             Payday Loan           Personal Loan
##                5.596945                5.703979                5.680455
##            Student Loan
##                5.675439
##
```

```
##
## $Last_Loan_1_Interest_Rate
## $Last_Loan_1_Interest_Rate$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Interest_Rate$eta_squared
## [1] 0.04932859
##
## $Last_Loan_1_Interest_Rate$significance
## [1] "Significant"
##
## $Last_Loan_1_Interest_Rate$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_1_Interest_Rate$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               15.424528               15.121633               15.402889
##        Home Equity Loan           Mortgage Loan                 No Loan
##               15.294694               14.952540                9.148668
##           Not Specified             Payday Loan           Personal Loan
##               14.857029               15.256532               15.510467
##            Student Loan
##               15.212121
##
##
## $Last_Loan_1_Num_of_Loan
## $Last_Loan_1_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Num_of_Loan$eta_squared
## [1] 0.2690683
##
## $Last_Loan_1_Num_of_Loan$significance
## [1] "Significant"
##
## $Last_Loan_1_Num_of_Loan$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_1_Num_of_Loan$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                3.956261                4.031020                3.941413
##        Home Equity Loan           Mortgage Loan                 No Loan
##                3.986939                3.915071                0.000000
##           Not Specified             Payday Loan           Personal Loan
##                3.950479                4.006334                4.124799
##            Student Loan
##                3.974482
##
##
## $Last_Loan_1_Delay_from_due_date
## $Last_Loan_1_Delay_from_due_date$p_value
```

```
## [1] "<.0001"
##
## $Last_Loan_1_Delay_from_due_date$eta_squared
## [1] 0.0326995
##
## $Last_Loan_1_Delay_from_due_date$significance
## [1] "Significant"
##
## $Last_Loan_1_Delay_from_due_date$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_1_Delay_from_due_date$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                21.63518                22.02000                21.84912
##        Home Equity Loan           Mortgage Loan                 No Loan
##                21.88296                21.26769                13.68636
##           Not Specified             Payday Loan           Personal Loan
##                21.81020                22.07106                22.97081
##            Student Loan
##                22.61314
##
##
## $Last_Loan_1_Num_of_Delayed_Payment
## $Last_Loan_1_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Num_of_Delayed_Payment$eta_squared
## [1] 0.03294882
##
## $Last_Loan_1_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Last_Loan_1_Num_of_Delayed_Payment$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_1_Num_of_Delayed_Payment$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                13.89687                13.92173                13.44793
##        Home Equity Loan           Mortgage Loan                 No Loan
##                13.93816                13.66799                10.18864
##           Not Specified             Payday Loan           Personal Loan
##                13.59964                13.55532                13.84179
##            Student Loan
##                13.75150
##
##
## $Last_Loan_1_Changed_Credit_Limit
## $Last_Loan_1_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Changed_Credit_Limit$eta_squared
```

```
## [1] 0.02156172
##
## $Last_Loan_1_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Last_Loan_1_Changed_Credit_Limit$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_1_Changed_Credit_Limit$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               10.671355               10.734743               10.417504
##        Home Equity Loan           Mortgage Loan                 No Loan
##               10.936894               10.574759                7.669357
##           Not Specified             Payday Loan           Personal Loan
##               10.510477               10.751081               11.093700
##            Student Loan
##               10.956992
##
##
## $Last_Loan_1_Num_Credit_Inquiries
## $Last_Loan_1_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Num_Credit_Inquiries$eta_squared
## [1] 0.05384289
##
## $Last_Loan_1_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Last_Loan_1_Num_Credit_Inquiries$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_1_Num_Credit_Inquiries$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                6.065502                6.095306                6.044242
##        Home Equity Loan           Mortgage Loan                 No Loan
##                5.992551                6.018943                3.284187
##           Not Specified             Payday Loan           Personal Loan
##                6.058806                6.121239                6.233092
##            Student Loan
##                6.203250
##
##
## $Last_Loan_1_Outstanding_Debt
## $Last_Loan_1_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Outstanding_Debt$eta_squared
## [1] 0.04277821
##
## $Last_Loan_1_Outstanding_Debt$significance
```

```
## [1] "Significant"
##
## $Last_Loan_1_Outstanding_Debt$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_1_Outstanding_Debt$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               1479.8014               1490.6415               1513.7376
##        Home Equity Loan           Mortgage Loan                 No Loan
##               1518.7150               1474.7372                766.4362
##           Not Specified             Payday Loan           Personal Loan
##               1479.9104               1523.1730               1590.2766
##            Student Loan
##               1526.2289
##
##
## $Last_Loan_1_Credit_Utilization_Ratio
## $Last_Loan_1_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Credit_Utilization_Ratio$eta_squared
## [1] 0.004325518
##
## $Last_Loan_1_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Last_Loan_1_Credit_Utilization_Ratio$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_1_Credit_Utilization_Ratio$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                32.17000                32.16559                32.17088
##        Home Equity Loan           Mortgage Loan                 No Loan
##                32.09230                32.17245                33.21509
##           Not Specified             Payday Loan           Personal Loan
##                32.21274                32.18495                32.08500
##            Student Loan
##                32.23305
##
##
## $Last_Loan_1_Credit_History_Age
## $Last_Loan_1_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Credit_History_Age$eta_squared
## [1] 0.05681776
##
## $Last_Loan_1_Credit_History_Age$significance
## [1] "Significant"
##
## $Last_Loan_1_Credit_History_Age$interpretation
```

```
## [1] "Weak correlation"
##
## $Last_Loan_1_Credit_History_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                215.3714                212.5008                216.6019
##        Home Equity Loan           Mortgage Loan                 No Loan
##                213.0363                212.7298                286.9698
##           Not Specified             Payday Loan           Personal Loan
##                215.1502                213.7621                207.8309
##            Student Loan
##                207.8238
##
##
## $Last_Loan_1_Total_EMI_per_month
## $Last_Loan_1_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Total_EMI_per_month$eta_squared
## [1] 0.1335849
##
## $Last_Loan_1_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Last_Loan_1_Total_EMI_per_month$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_1_Total_EMI_per_month$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               99.579111               98.340141               96.303114
##        Home Equity Loan           Mortgage Loan                 No Loan
##               98.433161               98.553546                3.568411
##           Not Specified             Payday Loan           Personal Loan
##              102.803388              101.660545               97.878278
##            Student Loan
##               98.360938
##
##
## $Last_Loan_1_Amount_invested_monthly
## $Last_Loan_1_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Amount_invested_monthly$eta_squared
## [1] 0.005252693
##
## $Last_Loan_1_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Last_Loan_1_Amount_invested_monthly$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_1_Amount_invested_monthly$means
```

```
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                186.2499                183.6525                185.1556
##        Home Equity Loan           Mortgage Loan                 No Loan
##                189.2747                195.3749                231.1701
##           Not Specified             Payday Loan           Personal Loan
##                196.2771                191.5321                183.0333
##            Student Loan
##                189.0806
##
##
## $Last_Loan_1_Monthly_Balance
## $Last_Loan_1_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Monthly_Balance$eta_squared
## [1] 0.06531452
##
## $Last_Loan_1_Monthly_Balance$significance
## [1] "Significant"
##
## $Last_Loan_1_Monthly_Balance$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_1_Monthly_Balance$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                383.8180                379.5978                386.2027
##        Home Equity Loan           Mortgage Loan                 No Loan
##                382.0919                390.6803                555.7131
##           Not Specified             Payday Loan           Personal Loan
##                388.3725                382.7115                374.4405
##            Student Loan
##                387.1840
##
##
## $Last_Loan_2_Age
## $Last_Loan_2_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Age$eta_squared
## [1] 0.02084425
##
## $Last_Loan_2_Age$significance
## [1] "Significant"
##
## $Last_Loan_2_Age$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_2_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                32.17257                32.79773                32.13576
##        Home Equity Loan           Mortgage Loan                 No Loan
```

```
##                32.36942                33.09943                36.14896
##           Not Specified             Payday Loan           Personal Loan
##                32.21815                32.58864                32.35764
##            Student Loan
##                32.60872
##
##
## $Last_Loan_2_Annual_Income
## $Last_Loan_2_Annual_Income$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Annual_Income$eta_squared
## [1] 0.02737054
##
## $Last_Loan_2_Annual_Income$significance
## [1] "Significant"
##
## $Last_Loan_2_Annual_Income$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_2_Annual_Income$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                47842.39                46701.37                45965.71
##        Home Equity Loan           Mortgage Loan                 No Loan
##                47927.87                46050.72                62175.79
##           Not Specified             Payday Loan           Personal Loan
##                48501.95                47317.39                46535.45
##            Student Loan
##                47038.67
##
##
## $Last_Loan_2_Monthly_Inhand_Salary
## $Last_Loan_2_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Monthly_Inhand_Salary$eta_squared
## [1] 0.02700016
##
## $Last_Loan_2_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Last_Loan_2_Monthly_Inhand_Salary$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_2_Monthly_Inhand_Salary$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                3989.816                3889.082                3826.183
##        Home Equity Loan           Mortgage Loan                 No Loan
##                3991.168                3821.250                5163.170
##           Not Specified             Payday Loan           Personal Loan
##                4026.881                3936.923                3870.103
```

```
##            Student Loan
##                3903.513
##
##
## $Last_Loan_2_Num_Bank_Accounts
## $Last_Loan_2_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Num_Bank_Accounts$eta_squared
## [1] 0.07258314
##
## $Last_Loan_2_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Last_Loan_2_Num_Bank_Accounts$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_2_Num_Bank_Accounts$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                5.731114                5.859756                5.730658
##        Home Equity Loan           Mortgage Loan                 No Loan
##                5.721855                5.816960                4.079428
##           Not Specified             Payday Loan           Personal Loan
##                5.676787                5.812500                5.752083
##            Student Loan
##                5.591612
##
##
## $Last_Loan_2_Num_Credit_Card
## $Last_Loan_2_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Num_Credit_Card$eta_squared
## [1] 0.0548874
##
## $Last_Loan_2_Num_Credit_Card$significance
## [1] "Significant"
##
## $Last_Loan_2_Num_Credit_Card$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_2_Num_Credit_Card$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                5.874882                5.807223                5.775629
##        Home Equity Loan           Mortgage Loan                 No Loan
##                5.761851                5.782707                4.639155
##           Not Specified             Payday Loan           Personal Loan
##                5.700733                5.816894                5.845602
##            Student Loan
##                5.779991
##
```

```
##
## $Last_Loan_2_Interest_Rate
## $Last_Loan_2_Interest_Rate$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Interest_Rate$eta_squared
## [1] 0.1101514
##
## $Last_Loan_2_Interest_Rate$significance
## [1] "Significant"
##
## $Last_Loan_2_Interest_Rate$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_2_Interest_Rate$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               16.308782               16.305816               15.853965
##        Home Equity Loan           Mortgage Loan                 No Loan
##               16.012762               16.353949                9.172524
##           Not Specified             Payday Loan           Personal Loan
##               15.862511               16.533392               15.662963
##            Student Loan
##               15.909091
##
##
## $Last_Loan_2_Num_of_Loan
## $Last_Loan_2_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Num_of_Loan$eta_squared
## [1] 0.4491415
##
## $Last_Loan_2_Num_of_Loan$significance
## [1] "Significant"
##
## $Last_Loan_2_Num_of_Loan$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_2_Num_of_Loan$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               4.4381492               4.4343340               4.4187621
##        Home Equity Loan           Mortgage Loan                 No Loan
##               4.4767548               4.4215033               0.4937877
##           Not Specified             Payday Loan           Personal Loan
##               4.4188818               4.4235501               4.3675926
##            Student Loan
##               4.3523899
##
##
## $Last_Loan_2_Delay_from_due_date
## $Last_Loan_2_Delay_from_due_date$p_value
```

```
## [1] "<.0001"
##
## $Last_Loan_2_Delay_from_due_date$eta_squared
## [1] 0.07499736
##
## $Last_Loan_2_Delay_from_due_date$significance
## [1] "Significant"
##
## $Last_Loan_2_Delay_from_due_date$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_2_Delay_from_due_date$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                24.35316                23.30242                22.90897
##        Home Equity Loan           Mortgage Loan                 No Loan
##                22.85848                23.77165                13.57242
##           Not Specified             Payday Loan           Personal Loan
##                22.45749                23.43464                23.16586
##            Student Loan
##                23.01722
##
##
## $Last_Loan_2_Num_of_Delayed_Payment
## $Last_Loan_2_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Num_of_Delayed_Payment$eta_squared
## [1] 0.07114403
##
## $Last_Loan_2_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Last_Loan_2_Num_of_Delayed_Payment$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_2_Num_of_Delayed_Payment$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                14.31610                14.66909                14.18919
##        Home Equity Loan           Mortgage Loan                 No Loan
##                14.22379                14.21480                10.25519
##           Not Specified             Payday Loan           Personal Loan
##                13.67862                14.30097                14.33623
##            Student Loan
##                14.08880
##
##
## $Last_Loan_2_Changed_Credit_Limit
## $Last_Loan_2_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Changed_Credit_Limit$eta_squared
```

```
## [1] 0.04513439
##
## $Last_Loan_2_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Last_Loan_2_Changed_Credit_Limit$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_2_Changed_Credit_Limit$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               11.172375               11.202690               11.211480
##        Home Equity Loan           Mortgage Loan                 No Loan
##               11.306559               10.927429                7.725006
##           Not Specified             Payday Loan           Personal Loan
##               11.181907               10.964231               11.231213
##            Student Loan
##               11.285487
##
##
## $Last_Loan_2_Num_Credit_Inquiries
## $Last_Loan_2_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Num_Credit_Inquiries$eta_squared
## [1] 0.1241025
##
## $Last_Loan_2_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Last_Loan_2_Num_Credit_Inquiries$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_2_Num_Credit_Inquiries$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                6.572356                6.509498                6.463008
##        Home Equity Loan           Mortgage Loan                 No Loan
##                6.550365                6.549239                3.256612
##           Not Specified             Payday Loan           Personal Loan
##                6.335816                6.689807                6.362847
##            Student Loan
##                6.502460
##
##
## $Last_Loan_2_Outstanding_Debt
## $Last_Loan_2_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Outstanding_Debt$eta_squared
## [1] 0.09788993
##
## $Last_Loan_2_Outstanding_Debt$significance
```

```
## [1] "Significant"
##
## $Last_Loan_2_Outstanding_Debt$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_2_Outstanding_Debt$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               1652.8526               1589.7806               1629.1380
##        Home Equity Loan           Mortgage Loan                 No Loan
##               1646.5695               1635.6748                757.8006
##           Not Specified             Payday Loan           Personal Loan
##               1560.0975               1603.7442               1635.8199
##            Student Loan
##               1634.7057
##
##
## $Last_Loan_2_Credit_Utilization_Ratio
## $Last_Loan_2_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Credit_Utilization_Ratio$eta_squared
## [1] 0.006962377
##
## $Last_Loan_2_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Last_Loan_2_Credit_Utilization_Ratio$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_2_Credit_Utilization_Ratio$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                32.10677                31.98219                32.00448
##        Home Equity Loan           Mortgage Loan                 No Loan
##                32.08737                32.10580                33.07364
##           Not Specified             Payday Loan           Personal Loan
##                32.09207                32.02588                32.03644
##            Student Loan
##                32.06111
##
##
## $Last_Loan_2_Credit_History_Age
## $Last_Loan_2_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Credit_History_Age$eta_squared
## [1] 0.1327682
##
## $Last_Loan_2_Credit_History_Age$significance
## [1] "Significant"
##
## $Last_Loan_2_Credit_History_Age$interpretation
```

```
## [1] "Moderate correlation"
##
## $Last_Loan_2_Credit_History_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                198.6010                201.0901                197.4207
##        Home Equity Loan           Mortgage Loan                 No Loan
##                204.6003                203.0557                288.4176
##           Not Specified             Payday Loan           Personal Loan
##                203.3103                200.4420                205.5833
##            Student Loan
##                200.5600
##
##
## $Last_Loan_2_Total_EMI_per_month
## $Last_Loan_2_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Total_EMI_per_month$eta_squared
## [1] 0.1869027
##
## $Last_Loan_2_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Last_Loan_2_Total_EMI_per_month$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_2_Total_EMI_per_month$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               106.10411               105.87400               108.75427
##        Home Equity Loan           Mortgage Loan                 No Loan
##               108.45752               105.21085                21.55347
##           Not Specified             Payday Loan           Personal Loan
##               109.68586               110.06839               108.69817
##            Student Loan
##               105.29569
##
##
## $Last_Loan_2_Amount_invested_monthly
## $Last_Loan_2_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Amount_invested_monthly$eta_squared
## [1] 0.01064024
##
## $Last_Loan_2_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Last_Loan_2_Amount_invested_monthly$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_2_Amount_invested_monthly$means
```

```
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                184.1584                183.1229                180.1759
##        Home Equity Loan           Mortgage Loan                 No Loan
##                185.5415                178.1354                230.7280
##           Not Specified             Payday Loan           Personal Loan
##                186.1189                182.0209                184.5349
##            Student Loan
##                182.1288
##
##
## $Last_Loan_2_Monthly_Balance
## $Last_Loan_2_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Monthly_Balance$eta_squared
## [1] 0.1081438
##
## $Last_Loan_2_Monthly_Balance$significance
## [1] "Significant"
##
## $Last_Loan_2_Monthly_Balance$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_2_Monthly_Balance$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                369.0898                363.3126                361.3916
##        Home Equity Loan           Mortgage Loan                 No Loan
##                365.3036                362.5405                534.1584
##           Not Specified             Payday Loan           Personal Loan
##                368.6342                366.3824                364.7260
##            Student Loan
##                367.6656
##
##
## $Last_Loan_3_Age
## $Last_Loan_3_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Age$eta_squared
## [1] 0.02248654
##
## $Last_Loan_3_Age$significance
## [1] "Significant"
##
## $Last_Loan_3_Age$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_3_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                32.56551                31.64959                32.13626
##        Home Equity Loan           Mortgage Loan                 No Loan
```

```
##                32.32636                32.06868                35.33863
##           Not Specified             Payday Loan           Personal Loan
##                31.65650                32.30334                31.79806
##            Student Loan
##                32.00476
##
##
## $Last_Loan_3_Annual_Income
## $Last_Loan_3_Annual_Income$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Annual_Income$eta_squared
## [1] 0.02908407
##
## $Last_Loan_3_Annual_Income$significance
## [1] "Significant"
##
## $Last_Loan_3_Annual_Income$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_3_Annual_Income$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                44218.07                45574.72                46203.36
##        Home Equity Loan           Mortgage Loan                 No Loan
##                45316.35                44918.41                58750.62
##           Not Specified             Payday Loan           Personal Loan
##                44074.34                44894.39                47072.40
##            Student Loan
##                46415.53
##
##
## $Last_Loan_3_Monthly_Inhand_Salary
## $Last_Loan_3_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Monthly_Inhand_Salary$eta_squared
## [1] 0.02904704
##
## $Last_Loan_3_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Last_Loan_3_Monthly_Inhand_Salary$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_3_Monthly_Inhand_Salary$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                3668.584                3792.997                3839.815
##        Home Equity Loan           Mortgage Loan                 No Loan
##                3769.385                3737.847                4884.270
##           Not Specified             Payday Loan           Personal Loan
##                3662.922                3726.585                3920.840
```

```
##            Student Loan
##                3857.676
##
##
## $Last_Loan_3_Num_Bank_Accounts
## $Last_Loan_3_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Num_Bank_Accounts$eta_squared
## [1] 0.09371455
##
## $Last_Loan_3_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Last_Loan_3_Num_Bank_Accounts$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_3_Num_Bank_Accounts$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                5.960047                5.947404                5.997979
##        Home Equity Loan           Mortgage Loan                 No Loan
##                5.922289                6.110594                4.363130
##           Not Specified             Payday Loan           Personal Loan
##                6.091808                5.954787                6.075218
##            Student Loan
##                5.868360
##
##
## $Last_Loan_3_Num_Credit_Card
## $Last_Loan_3_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Num_Credit_Card$eta_squared
## [1] 0.07134403
##
## $Last_Loan_3_Num_Credit_Card$significance
## [1] "Significant"
##
## $Last_Loan_3_Num_Credit_Card$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_3_Num_Credit_Card$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                5.824912                6.062158                6.053406
##        Home Equity Loan           Mortgage Loan                 No Loan
##                5.857831                6.019790                4.836880
##           Not Specified             Payday Loan           Personal Loan
##                6.022881                5.943115                5.896536
##            Student Loan
##                5.987298
##
```

```
##
## $Last_Loan_3_Interest_Rate
## $Last_Loan_3_Interest_Rate$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Interest_Rate$eta_squared
## [1] 0.1334367
##
## $Last_Loan_3_Interest_Rate$significance
## [1] "Significant"
##
## $Last_Loan_3_Interest_Rate$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_3_Interest_Rate$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                16.44536                17.21311                17.03580
##        Home Equity Loan           Mortgage Loan                 No Loan
##                17.17229                16.98021                10.48337
##           Not Specified             Payday Loan           Personal Loan
##                17.29718                16.90544                17.39950
##            Student Loan
##                16.91224
##
##
## $Last_Loan_3_Num_of_Loan
## $Last_Loan_3_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Num_of_Loan$eta_squared
## [1] 0.6064428
##
## $Last_Loan_3_Num_of_Loan$significance
## [1] "Significant"
##
## $Last_Loan_3_Num_of_Loan$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_3_Num_of_Loan$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                4.990599                5.003279                5.064665
##        Home Equity Loan           Mortgage Loan                 No Loan
##                5.034940                5.087311                1.112529
##           Not Specified             Payday Loan           Personal Loan
##                5.063277                5.050827                4.956305
##            Student Loan
##                5.032333
##
##
## $Last_Loan_3_Delay_from_due_date
## $Last_Loan_3_Delay_from_due_date$p_value
```

```
## [1] "<.0001"
##
## $Last_Loan_3_Delay_from_due_date$eta_squared
## [1] 0.1024612
##
## $Last_Loan_3_Delay_from_due_date$significance
## [1] "Significant"
##
## $Last_Loan_3_Delay_from_due_date$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_3_Delay_from_due_date$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                23.77027                24.61776                25.76155
##        Home Equity Loan           Mortgage Loan                 No Loan
##                24.73690                25.42244                15.05083
##           Not Specified             Payday Loan           Personal Loan
##                24.27726                25.10624                24.88296
##            Student Loan
##                24.61028
##
##
## $Last_Loan_3_Num_of_Delayed_Payment
## $Last_Loan_3_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Num_of_Delayed_Payment$eta_squared
## [1] 0.09580648
##
## $Last_Loan_3_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Last_Loan_3_Num_of_Delayed_Payment$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_3_Num_of_Delayed_Payment$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                14.74545                14.98811                14.76415
##        Home Equity Loan           Mortgage Loan                 No Loan
##                14.74006                14.96027                10.86540
##           Not Specified             Payday Loan           Personal Loan
##                14.78828                14.93868                14.91308
##            Student Loan
##                14.85393
##
##
## $Last_Loan_3_Changed_Credit_Limit
## $Last_Loan_3_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Changed_Credit_Limit$eta_squared
```

```
## [1] 0.06072096
##
## $Last_Loan_3_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Last_Loan_3_Changed_Credit_Limit$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_3_Changed_Credit_Limit$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               11.603079               11.939885               11.876510
##        Home Equity Loan           Mortgage Loan                 No Loan
##               11.784886               11.716950                8.271362
##           Not Specified             Payday Loan           Personal Loan
##               11.413136               11.652145               11.510643
##            Student Loan
##               11.789140
##
##
## $Last_Loan_3_Num_Credit_Inquiries
## $Last_Loan_3_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Num_Credit_Inquiries$eta_squared
## [1] 0.1475958
##
## $Last_Loan_3_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Last_Loan_3_Num_Credit_Inquiries$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_3_Num_Credit_Inquiries$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                6.794800                6.865164                7.099163
##        Home Equity Loan           Mortgage Loan                 No Loan
##                6.982530                6.996944                3.889981
##           Not Specified             Payday Loan           Personal Loan
##                6.996469                6.950059                6.870162
##            Student Loan
##                6.893620
##
##
## $Last_Loan_3_Outstanding_Debt
## $Last_Loan_3_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Outstanding_Debt$eta_squared
## [1] 0.1485868
##
## $Last_Loan_3_Outstanding_Debt$significance
```

```
## [1] "Significant"
##
## $Last_Loan_3_Outstanding_Debt$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_3_Outstanding_Debt$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               1719.6882               1808.6266               1792.7257
##        Home Equity Loan           Mortgage Loan                 No Loan
##               1761.3512               1793.1734                860.9794
##           Not Specified             Payday Loan           Personal Loan
##               1762.9195               1778.7633               1779.3268
##            Student Loan
##               1787.5842
##
##
## $Last_Loan_3_Credit_Utilization_Ratio
## $Last_Loan_3_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Credit_Utilization_Ratio$eta_squared
## [1] 0.007205305
##
## $Last_Loan_3_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Last_Loan_3_Credit_Utilization_Ratio$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_3_Credit_Utilization_Ratio$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                31.92981                31.95392                31.90889
##        Home Equity Loan           Mortgage Loan                 No Loan
##                31.99637                31.90794                32.83409
##           Not Specified             Payday Loan           Personal Loan
##                31.86404                31.97144                32.07380
##            Student Loan
##                31.91355
##
##
## $Last_Loan_3_Credit_History_Age
## $Last_Loan_3_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Credit_History_Age$eta_squared
## [1] 0.1663426
##
## $Last_Loan_3_Credit_History_Age$significance
## [1] "Significant"
##
## $Last_Loan_3_Credit_History_Age$interpretation
```

```
## [1] "Strong correlation"
##
## $Last_Loan_3_Credit_History_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                193.5517                190.4628                184.4180
##        Home Equity Loan           Mortgage Loan                 No Loan
##                190.9133                188.1461                272.7859
##           Not Specified             Payday Loan           Personal Loan
##                187.6966                187.2861                193.5237
##            Student Loan
##                187.7148
##
##
## $Last_Loan_3_Total_EMI_per_month
## $Last_Loan_3_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Total_EMI_per_month$eta_squared
## [1] 0.1981489
##
## $Last_Loan_3_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Last_Loan_3_Total_EMI_per_month$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_3_Total_EMI_per_month$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               117.55406               118.46892               118.19514
##        Home Equity Loan           Mortgage Loan                 No Loan
##               116.21988               120.06788                41.14471
##           Not Specified             Payday Loan           Personal Loan
##               115.47615               117.37859               116.50434
##            Student Loan
##               116.27163
##
##
## $Last_Loan_3_Amount_invested_monthly
## $Last_Loan_3_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Amount_invested_monthly$eta_squared
## [1] 0.01269611
##
## $Last_Loan_3_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Last_Loan_3_Amount_invested_monthly$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_3_Amount_invested_monthly$means
```

```
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                174.2157                176.3584                181.1625
##        Home Equity Loan           Mortgage Loan                 No Loan
##                173.8245                177.0571                221.4188
##           Not Specified             Payday Loan           Personal Loan
##                171.6997                175.7385                180.7216
##            Student Loan
##                178.0724
##
##
## $Last_Loan_3_Monthly_Balance
## $Last_Loan_3_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Monthly_Balance$eta_squared
## [1] 0.1192307
##
## $Last_Loan_3_Monthly_Balance$significance
## [1] "Significant"
##
## $Last_Loan_3_Monthly_Balance$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_3_Monthly_Balance$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                343.8081                344.5596                341.7081
##        Home Equity Loan           Mortgage Loan                 No Loan
##                347.6594                341.1809                497.4832
##           Not Specified             Payday Loan           Personal Loan
##                342.1382                344.1787                355.0901
##            Student Loan
##                348.0252
##
##
## $Last_Loan_4_Age
## $Last_Loan_4_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Age$eta_squared
## [1] 0.03536117
##
## $Last_Loan_4_Age$significance
## [1] "Significant"
##
## $Last_Loan_4_Age$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_4_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                30.46144                31.17549                31.13573
##        Home Equity Loan           Mortgage Loan                 No Loan
```

```
##                31.42904                30.75600                35.16826
##           Not Specified             Payday Loan           Personal Loan
##                31.11641                31.35306                31.23064
##            Student Loan
##                31.43417
##
##
## $Last_Loan_4_Annual_Income
## $Last_Loan_4_Annual_Income$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Annual_Income$eta_squared
## [1] 0.04489394
##
## $Last_Loan_4_Annual_Income$significance
## [1] "Significant"
##
## $Last_Loan_4_Annual_Income$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_4_Annual_Income$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                43283.98                42440.86                44201.97
##        Home Equity Loan           Mortgage Loan                 No Loan
##                39040.56                42609.18                57924.24
##           Not Specified             Payday Loan           Personal Loan
##                39616.81                41008.83                41024.06
##            Student Loan
##                43088.27
##
##
## $Last_Loan_4_Monthly_Inhand_Salary
## $Last_Loan_4_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Monthly_Inhand_Salary$eta_squared
## [1] 0.04473964
##
## $Last_Loan_4_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Last_Loan_4_Monthly_Inhand_Salary$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_4_Monthly_Inhand_Salary$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                3608.930                3522.238                3669.936
##        Home Equity Loan           Mortgage Loan                 No Loan
##                3249.525                3538.469                4815.413
##           Not Specified             Payday Loan           Personal Loan
##                3300.681                3408.699                3415.784
```

```
##            Student Loan
##                3573.824
##
##
## $Last_Loan_4_Num_Bank_Accounts
## $Last_Loan_4_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Num_Bank_Accounts$eta_squared
## [1] 0.1509458
##
## $Last_Loan_4_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Last_Loan_4_Num_Bank_Accounts$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_4_Num_Bank_Accounts$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                6.512288                6.438383                6.410970
##        Home Equity Loan           Mortgage Loan                 No Loan
##                6.485577                6.469040                4.439426
##           Not Specified             Payday Loan           Personal Loan
##                6.547852                6.433056                6.368616
##            Student Loan
##                6.474152
##
##
## $Last_Loan_4_Num_Credit_Card
## $Last_Loan_4_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Num_Credit_Card$eta_squared
## [1] 0.1158311
##
## $Last_Loan_4_Num_Credit_Card$significance
## [1] "Significant"
##
## $Last_Loan_4_Num_Credit_Card$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_4_Num_Credit_Card$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                6.446186                6.208157                6.261526
##        Home Equity Loan           Mortgage Loan                 No Loan
##                6.354808                6.382353                4.886593
##           Not Specified             Payday Loan           Personal Loan
##                6.335547                6.281770                6.156096
##            Student Loan
##                6.222132
##
```

```
##
## $Last_Loan_4_Interest_Rate
## $Last_Loan_4_Interest_Rate$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Interest_Rate$eta_squared
## [1] 0.2076278
##
## $Last_Loan_4_Interest_Rate$significance
## [1] "Significant"
##
## $Last_Loan_4_Interest_Rate$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_4_Interest_Rate$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                19.19831                18.49153                18.78378
##        Home Equity Loan           Mortgage Loan                 No Loan
##                19.46462                18.77864                10.86252
##           Not Specified             Payday Loan           Personal Loan
##                19.19063                18.45840                18.39209
##            Student Loan
##                18.84491
##
##
## $Last_Loan_4_Num_of_Loan
## $Last_Loan_4_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Num_of_Loan$eta_squared
## [1] 0.6860636
##
## $Last_Loan_4_Num_of_Loan$significance
## [1] "Significant"
##
## $Last_Loan_4_Num_of_Loan$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_4_Num_of_Loan$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                5.832203                5.741525                5.715421
##        Home Equity Loan           Mortgage Loan                 No Loan
##                5.792308                5.718266                1.663111
##           Not Specified             Payday Loan           Personal Loan
##                5.760937                5.706505                5.686985
##            Student Loan
##                5.596123
##
##
## $Last_Loan_4_Delay_from_due_date
## $Last_Loan_4_Delay_from_due_date$p_value
```

```
## [1] "<.0001"
##
## $Last_Loan_4_Delay_from_due_date$eta_squared
## [1] 0.1563954
##
## $Last_Loan_4_Delay_from_due_date$significance
## [1] "Significant"
##
## $Last_Loan_4_Delay_from_due_date$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_4_Delay_from_due_date$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                27.08602                26.75971                27.36328
##        Home Equity Loan           Mortgage Loan                 No Loan
##                28.99769                27.03328                15.66461
##           Not Specified             Payday Loan           Personal Loan
##                28.38496                27.38464                27.02080
##            Student Loan
##                26.66276
##
##
## $Last_Loan_4_Num_of_Delayed_Payment
## $Last_Loan_4_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Num_of_Delayed_Payment$eta_squared
## [1] 0.1515613
##
## $Last_Loan_4_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Last_Loan_4_Num_of_Delayed_Payment$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_4_Num_of_Delayed_Payment$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                16.24174                15.74823                16.03597
##        Home Equity Loan           Mortgage Loan                 No Loan
##                16.16731                16.03425                11.07913
##           Not Specified             Payday Loan           Personal Loan
##                15.89258                15.80314                15.77430
##            Student Loan
##                16.07977
##
##
## $Last_Loan_4_Changed_Credit_Limit
## $Last_Loan_4_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Changed_Credit_Limit$eta_squared
```

```
## [1] 0.09215224
##
## $Last_Loan_4_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Last_Loan_4_Changed_Credit_Limit$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_4_Changed_Credit_Limit$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               12.391873               12.612691               12.639261
##        Home Equity Loan           Mortgage Loan                 No Loan
##               12.797431               13.143522                8.493813
##           Not Specified             Payday Loan           Personal Loan
##               12.463680               12.357905               12.564852
##            Student Loan
##               12.533384
##
##
## $Last_Loan_4_Num_Credit_Inquiries
## $Last_Loan_4_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Num_Credit_Inquiries$eta_squared
## [1] 0.2198657
##
## $Last_Loan_4_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Last_Loan_4_Num_Credit_Inquiries$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_4_Num_Credit_Inquiries$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                7.858263                7.678849                7.661566
##        Home Equity Loan           Mortgage Loan                 No Loan
##                7.932885                7.706076                4.103222
##           Not Specified             Payday Loan           Personal Loan
##                7.825391                7.709153                7.557661
##            Student Loan
##                7.666195
##
##
## $Last_Loan_4_Outstanding_Debt
## $Last_Loan_4_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Outstanding_Debt$eta_squared
## [1] 0.2498042
##
## $Last_Loan_4_Outstanding_Debt$significance
```

```
## [1] "Significant"
##
## $Last_Loan_4_Outstanding_Debt$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_4_Outstanding_Debt$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               2071.5705               2045.3264               2015.9528
##        Home Equity Loan           Mortgage Loan                 No Loan
##               2080.2077               2125.2337                893.8935
##           Not Specified             Payday Loan           Personal Loan
##               2072.4878               1997.1990               2014.9273
##            Student Loan
##               2037.1306
##
##
## $Last_Loan_4_Credit_Utilization_Ratio
## $Last_Loan_4_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Credit_Utilization_Ratio$eta_squared
## [1] 0.007401874
##
## $Last_Loan_4_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Last_Loan_4_Credit_Utilization_Ratio$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_4_Credit_Utilization_Ratio$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                31.74255                31.76225                31.81897
##        Home Equity Loan           Mortgage Loan                 No Loan
##                31.83953                31.73674                32.68969
##           Not Specified             Payday Loan           Personal Loan
##                31.80963                31.84388                31.79293
##            Student Loan
##                31.94897
##
##
## $Last_Loan_4_Credit_History_Age
## $Last_Loan_4_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Credit_History_Age$eta_squared
## [1] 0.2513112
##
## $Last_Loan_4_Credit_History_Age$significance
## [1] "Significant"
##
## $Last_Loan_4_Credit_History_Age$interpretation
```

```
## [1] "Strong correlation"
##
## $Last_Loan_4_Credit_History_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                165.9102                165.8658                166.7067
##        Home Equity Loan           Mortgage Loan                 No Loan
##                164.0569                166.4737                267.3110
##           Not Specified             Payday Loan           Personal Loan
##                169.0109                166.1369                168.8707
##            Student Loan
##                171.0687
##
##
## $Last_Loan_4_Total_EMI_per_month
## $Last_Loan_4_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Total_EMI_per_month$eta_squared
## [1] 0.1402953
##
## $Last_Loan_4_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Last_Loan_4_Total_EMI_per_month$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_4_Total_EMI_per_month$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               124.42810               120.59245               130.43674
##        Home Equity Loan           Mortgage Loan                 No Loan
##               116.29632               120.84871                59.60446
##           Not Specified             Payday Loan           Personal Loan
##               121.54178               113.44266               121.17254
##            Student Loan
##               128.37856
##
##
## $Last_Loan_4_Amount_invested_monthly
## $Last_Loan_4_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Amount_invested_monthly$eta_squared
## [1] 0.01860418
##
## $Last_Loan_4_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Last_Loan_4_Amount_invested_monthly$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_4_Amount_invested_monthly$means
```

```
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                168.4357                164.7490                176.1714
##        Home Equity Loan           Mortgage Loan                 No Loan
##                155.7614                169.1947                217.8956
##           Not Specified             Payday Loan           Personal Loan
##                158.5334                160.2934                163.4035
##            Student Loan
##                171.7394
##
##
## $Last_Loan_4_Monthly_Balance
## $Last_Loan_4_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Monthly_Balance$eta_squared
## [1] 0.1338757
##
## $Last_Loan_4_Monthly_Balance$significance
## [1] "Significant"
##
## $Last_Loan_4_Monthly_Balance$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_4_Monthly_Balance$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                320.0375                319.4613                322.3169
##        Home Equity Loan           Mortgage Loan                 No Loan
##                312.0249                320.3249                475.8668
##           Not Specified             Payday Loan           Personal Loan
##                311.7967                320.1772                319.5899
##            Student Loan
##                321.3112
##
##
## $Last_Loan_5_Age
## $Last_Loan_5_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Age$eta_squared
## [1] 0.05060869
##
## $Last_Loan_5_Age$significance
## [1] "Significant"
##
## $Last_Loan_5_Age$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_5_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                29.68341                30.04319                29.23441
##        Home Equity Loan           Mortgage Loan                 No Loan
```

```
##                29.65647                29.89012                34.90869
##           Not Specified             Payday Loan           Personal Loan
##                30.48227                29.23953                29.24915
##            Student Loan
##                29.60705
##
##
## $Last_Loan_5_Annual_Income
## $Last_Loan_5_Annual_Income$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Annual_Income$eta_squared
## [1] 0.07146011
##
## $Last_Loan_5_Annual_Income$significance
## [1] "Significant"
##
## $Last_Loan_5_Annual_Income$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_5_Annual_Income$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                34272.44                34238.57                32716.29
##        Home Equity Loan           Mortgage Loan                 No Loan
##                37164.91                36359.39                57279.51
##           Not Specified             Payday Loan           Personal Loan
##                35252.83                35368.97                35033.90
##            Student Loan
##                35660.15
##
##
## $Last_Loan_5_Monthly_Inhand_Salary
## $Last_Loan_5_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Monthly_Inhand_Salary$eta_squared
## [1] 0.07090903
##
## $Last_Loan_5_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Last_Loan_5_Monthly_Inhand_Salary$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_5_Monthly_Inhand_Salary$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                2859.512                2851.844                2710.034
##        Home Equity Loan           Mortgage Loan                 No Loan
##                3087.949                3034.855                4760.321
##           Not Specified             Payday Loan           Personal Loan
##                2934.703                2937.456                2918.302
```

```
##            Student Loan
##                2976.206
##
##
## $Last_Loan_5_Num_Bank_Accounts
## $Last_Loan_5_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Num_Bank_Accounts$eta_squared
## [1] 0.2465806
##
## $Last_Loan_5_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Last_Loan_5_Num_Bank_Accounts$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_5_Num_Bank_Accounts$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                7.379089                7.453089                7.352608
##        Home Equity Loan           Mortgage Loan                 No Loan
##                7.362059                7.181395                4.515780
##           Not Specified             Payday Loan           Personal Loan
##                7.126222                7.242524                7.235795
##            Student Loan
##                7.445192
##
##
## $Last_Loan_5_Num_Credit_Card
## $Last_Loan_5_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Num_Credit_Card$eta_squared
## [1] 0.1991373
##
## $Last_Loan_5_Num_Credit_Card$significance
## [1] "Significant"
##
## $Last_Loan_5_Num_Credit_Card$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_5_Num_Credit_Card$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                7.023949                6.860412                6.917234
##        Home Equity Loan           Mortgage Loan                 No Loan
##                6.976176                6.753488                4.922656
##           Not Specified             Payday Loan           Personal Loan
##                6.994193                6.874103                6.992614
##            Student Loan
##                6.916987
##
```

```
##
## $Last_Loan_5_Interest_Rate
## $Last_Loan_5_Interest_Rate$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Interest_Rate$eta_squared
## [1] 0.3506369
##
## $Last_Loan_5_Interest_Rate$significance
## [1] "Significant"
##
## $Last_Loan_5_Interest_Rate$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_5_Interest_Rate$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                22.19626                22.60183                22.42630
##        Home Equity Loan           Mortgage Loan                 No Loan
##                22.58118                22.12093                11.10124
##           Not Specified             Payday Loan           Personal Loan
##                21.66259                22.47608                22.26818
##            Student Loan
##                22.66923
##
##
## $Last_Loan_5_Num_of_Loan
## $Last_Loan_5_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Num_of_Loan$eta_squared
## [1] 0.6923063
##
## $Last_Loan_5_Num_of_Loan$significance
## [1] "Significant"
##
## $Last_Loan_5_Num_of_Loan$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_5_Num_of_Loan$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                6.633178                6.643021                6.557823
##        Home Equity Loan           Mortgage Loan                 No Loan
##                6.632941                6.567442                2.183138
##           Not Specified             Payday Loan           Personal Loan
##                6.606357                6.590909                6.540909
##            Student Loan
##                6.653846
##
##
## $Last_Loan_5_Delay_from_due_date
## $Last_Loan_5_Delay_from_due_date$p_value
```

```
## [1] "<.0001"
##
## $Last_Loan_5_Delay_from_due_date$eta_squared
## [1] 0.2752315
##
## $Last_Loan_5_Delay_from_due_date$significance
## [1] "Significant"
##
## $Last_Loan_5_Delay_from_due_date$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_5_Delay_from_due_date$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                33.09755                31.92363                34.01219
##        Home Equity Loan           Mortgage Loan                 No Loan
##                34.25971                31.58750                15.90846
##           Not Specified             Payday Loan           Personal Loan
##                33.07641                32.24910                32.44830
##            Student Loan
##                32.55929
##
##
## $Last_Loan_5_Num_of_Delayed_Payment
## $Last_Loan_5_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Num_of_Delayed_Payment$eta_squared
## [1] 0.2443192
##
## $Last_Loan_5_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Last_Loan_5_Num_of_Delayed_Payment$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_5_Num_of_Delayed_Payment$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                17.89194                17.97054                18.28713
##        Home Equity Loan           Mortgage Loan                 No Loan
##                18.27794                17.84390                11.27563
##           Not Specified             Payday Loan           Personal Loan
##                18.07304                17.76794                17.73977
##            Student Loan
##                18.15288
##
##
## $Last_Loan_5_Changed_Credit_Limit
## $Last_Loan_5_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Changed_Credit_Limit$eta_squared
```

```
## [1] 0.1582061
##
## $Last_Loan_5_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Last_Loan_5_Changed_Credit_Limit$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_5_Changed_Credit_Limit$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               14.492605               13.983879               14.562596
##        Home Equity Loan           Mortgage Loan                 No Loan
##               14.191518               14.558093                8.604043
##           Not Specified             Payday Loan           Personal Loan
##               14.912213               13.903194               14.859205
##            Student Loan
##               14.583269
##
##
## $Last_Loan_5_Num_Credit_Inquiries
## $Last_Loan_5_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Num_Credit_Inquiries$eta_squared
## [1] 0.3624486
##
## $Last_Loan_5_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Last_Loan_5_Num_Credit_Inquiries$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_5_Num_Credit_Inquiries$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                9.289720                9.410469                9.487245
##        Home Equity Loan           Mortgage Loan                 No Loan
##                9.243529                9.111919                4.232046
##           Not Specified             Payday Loan           Personal Loan
##                9.133557                9.143242                9.244318
##            Student Loan
##                9.419872
##
##
## $Last_Loan_5_Outstanding_Debt
## $Last_Loan_5_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Outstanding_Debt$eta_squared
## [1] 0.4570059
##
## $Last_Loan_5_Outstanding_Debt$significance
```

```
## [1] "Significant"
##
## $Last_Loan_5_Outstanding_Debt$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_5_Outstanding_Debt$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               2710.0483               2599.6439               2604.6561
##        Home Equity Loan           Mortgage Loan                 No Loan
##               2638.0373               2564.0933                908.7115
##           Not Specified             Payday Loan           Personal Loan
##               2508.5072               2591.3780               2607.0624
##            Student Loan
##               2599.2678
##
##
## $Last_Loan_5_Credit_Utilization_Ratio
## $Last_Loan_5_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Credit_Utilization_Ratio$eta_squared
## [1] 0.006801796
##
## $Last_Loan_5_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Last_Loan_5_Credit_Utilization_Ratio$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_5_Credit_Utilization_Ratio$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                31.74913                31.76971                31.62940
##        Home Equity Loan           Mortgage Loan                 No Loan
##                31.62663                31.67893                32.56335
##           Not Specified             Payday Loan           Personal Loan
##                31.66152                31.65129                31.46420
##            Student Loan
##                31.64558
##
##
## $Last_Loan_5_Credit_History_Age
## $Last_Loan_5_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Credit_History_Age$eta_squared
## [1] 0.4105915
##
## $Last_Loan_5_Credit_History_Age$significance
## [1] "Significant"
##
## $Last_Loan_5_Credit_History_Age$interpretation
```

```
## [1] "Strong correlation"
##
## $Last_Loan_5_Credit_History_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                124.6659                122.9165                123.7608
##        Home Equity Loan           Mortgage Loan                 No Loan
##                126.5024                126.2140                263.5619
##           Not Specified             Payday Loan           Personal Loan
##                124.4071                125.6555                124.7568
##            Student Loan
##                125.2744
##
##
## $Last_Loan_5_Total_EMI_per_month
## $Last_Loan_5_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Total_EMI_per_month$eta_squared
## [1] 0.07531979
##
## $Last_Loan_5_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Last_Loan_5_Total_EMI_per_month$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_5_Total_EMI_per_month$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               126.50971               119.45313               112.58197
##        Home Equity Loan           Mortgage Loan                 No Loan
##               122.90178               122.81392                73.15498
##           Not Specified             Payday Loan           Personal Loan
##               124.06827               124.04082               120.50282
##            Student Loan
##               130.34175
##
##
## $Last_Loan_5_Amount_invested_monthly
## $Last_Loan_5_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Amount_invested_monthly$eta_squared
## [1] 0.03015825
##
## $Last_Loan_5_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Last_Loan_5_Amount_invested_monthly$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_5_Amount_invested_monthly$means
```

```
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                142.9689                137.7935                134.9922
##        Home Equity Loan           Mortgage Loan                 No Loan
##                151.7536                146.4393                216.0470
##           Not Specified             Payday Loan           Personal Loan
##                141.5543                142.2680                140.6856
##            Student Loan
##                147.9263
##
##
## $Last_Loan_5_Monthly_Balance
## $Last_Loan_5_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Monthly_Balance$eta_squared
## [1] 0.1374797
##
## $Last_Loan_5_Monthly_Balance$significance
## [1] "Significant"
##
## $Last_Loan_5_Monthly_Balance$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_5_Monthly_Balance$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                281.6461                281.1275                281.0784
##        Home Equity Loan           Mortgage Loan                 No Loan
##                280.0708                289.7132                456.2017
##           Not Specified             Payday Loan           Personal Loan
##                285.4053                285.4200                287.1862
##            Student Loan
##                280.9067
##
##
## $Last_Loan_6_Age
## $Last_Loan_6_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Age$eta_squared
## [1] 0.0327336
##
## $Last_Loan_6_Age$significance
## [1] "Significant"
##
## $Last_Loan_6_Age$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_6_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                29.65858                30.21319                30.10674
##        Home Equity Loan           Mortgage Loan                 No Loan
```

```
##                29.91274                29.08863                34.36796
##           Not Specified             Payday Loan           Personal Loan
##                29.57546                29.59003                29.24415
##            Student Loan
##                30.53154
##
##
## $Last_Loan_6_Annual_Income
## $Last_Loan_6_Annual_Income$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Annual_Income$eta_squared
## [1] 0.05053067
##
## $Last_Loan_6_Annual_Income$significance
## [1] "Significant"
##
## $Last_Loan_6_Annual_Income$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_6_Annual_Income$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                36272.44                33695.10                32627.86
##        Home Equity Loan           Mortgage Loan                 No Loan
##                36114.72                34678.16                55200.29
##           Not Specified             Payday Loan           Personal Loan
##                36520.14                34105.38                35535.76
##            Student Loan
##                33632.74
##
##
## $Last_Loan_6_Monthly_Inhand_Salary
## $Last_Loan_6_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Monthly_Inhand_Salary$eta_squared
## [1] 0.05004667
##
## $Last_Loan_6_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Last_Loan_6_Monthly_Inhand_Salary$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_6_Monthly_Inhand_Salary$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                3015.651                2802.055                2713.587
##        Home Equity Loan           Mortgage Loan                 No Loan
##                2992.693                2905.571                4587.582
##           Not Specified             Payday Loan           Personal Loan
##                3060.698                2823.079                2964.453
```

```
##            Student Loan
##                2805.098
##
##
## $Last_Loan_6_Num_Bank_Accounts
## $Last_Loan_6_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Num_Bank_Accounts$eta_squared
## [1] 0.1810463
##
## $Last_Loan_6_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Last_Loan_6_Num_Bank_Accounts$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_6_Num_Bank_Accounts$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                7.319579                7.351227                7.506899
##        Home Equity Loan           Mortgage Loan                 No Loan
##                7.295860                7.450251                4.765912
##           Not Specified             Payday Loan           Personal Loan
##                7.373476                7.545015                7.289839
##            Student Loan
##                7.338785
##
##
## $Last_Loan_6_Num_Credit_Card
## $Last_Loan_6_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Num_Credit_Card$eta_squared
## [1] 0.1451678
##
## $Last_Loan_6_Num_Credit_Card$significance
## [1] "Significant"
##
## $Last_Loan_6_Num_Credit_Card$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_6_Num_Credit_Card$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                6.940939                7.041411                6.812094
##        Home Equity Loan           Mortgage Loan                 No Loan
##                7.026380                7.165970                5.103697
##           Not Specified             Payday Loan           Personal Loan
##                6.961890                6.976562                6.944810
##            Student Loan
##                6.883956
##
```

```
##
## $Last_Loan_6_Interest_Rate
## $Last_Loan_6_Interest_Rate$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Interest_Rate$eta_squared
## [1] 0.2543723
##
## $Last_Loan_6_Interest_Rate$significance
## [1] "Significant"
##
## $Last_Loan_6_Interest_Rate$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_6_Interest_Rate$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                22.50809                22.64417                23.01299
##        Home Equity Loan           Mortgage Loan                 No Loan
##                22.47403                22.82274                12.12210
##           Not Specified             Payday Loan           Personal Loan
##                22.57012                22.24702                22.45614
##            Student Loan
##                22.65109
##
##
## $Last_Loan_6_Num_of_Loan
## $Last_Loan_6_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Num_of_Loan$eta_squared
## [1] 0.6451307
##
## $Last_Loan_6_Num_of_Loan$significance
## [1] "Significant"
##
## $Last_Loan_6_Num_of_Loan$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_6_Num_of_Loan$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                7.061489                7.064417                7.103896
##        Home Equity Loan           Mortgage Loan                 No Loan
##                7.188312                7.234114                2.458589
##           Not Specified             Payday Loan           Personal Loan
##                7.140244                7.110119                7.114035
##            Student Loan
##                7.127726
##
##
## $Last_Loan_6_Delay_from_due_date
## $Last_Loan_6_Delay_from_due_date$p_value
```

```
## [1] "<.0001"
##
## $Last_Loan_6_Delay_from_due_date$eta_squared
## [1] 0.2070276
##
## $Last_Loan_6_Delay_from_due_date$significance
## [1] "Significant"
##
## $Last_Loan_6_Delay_from_due_date$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_6_Delay_from_due_date$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                32.50566                33.84586                35.15422
##        Home Equity Loan           Mortgage Loan                 No Loan
##                33.86810                34.62667                17.37998
##           Not Specified             Payday Loan           Personal Loan
##                32.74390                33.03385                32.90826
##            Student Loan
##                32.17445
##
##
## $Last_Loan_6_Num_of_Delayed_Payment
## $Last_Loan_6_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Num_of_Delayed_Payment$eta_squared
## [1] 0.178268
##
## $Last_Loan_6_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Last_Loan_6_Num_of_Delayed_Payment$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_6_Num_of_Delayed_Payment$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                17.95307                18.33014                18.52110
##        Home Equity Loan           Mortgage Loan                 No Loan
##                17.95657                18.36789                11.88482
##           Not Specified             Payday Loan           Personal Loan
##                18.21341                18.68899                17.50731
##            Student Loan
##                17.93341
##
##
## $Last_Loan_6_Changed_Credit_Limit
## $Last_Loan_6_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Changed_Credit_Limit$eta_squared
```

```
## [1] 0.1123997
##
## $Last_Loan_6_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Last_Loan_6_Changed_Credit_Limit$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_6_Changed_Credit_Limit$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               15.547362               15.205890               14.725471
##        Home Equity Loan           Mortgage Loan                 No Loan
##               14.199935               13.699381                9.154552
##           Not Specified             Payday Loan           Personal Loan
##               14.496296               14.599003               14.429664
##            Student Loan
##               13.732336
##
##
## $Last_Loan_6_Num_Credit_Inquiries
## $Last_Loan_6_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Num_Credit_Inquiries$eta_squared
## [1] 0.2527083
##
## $Last_Loan_6_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Last_Loan_6_Num_Credit_Inquiries$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_6_Num_Credit_Inquiries$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                9.343042                9.421396                9.369318
##        Home Equity Loan           Mortgage Loan                 No Loan
##                9.182630                9.353679                4.711966
##           Not Specified             Payday Loan           Personal Loan
##                9.137195                9.540179                9.277778
##            Student Loan
##                9.259735
##
##
## $Last_Loan_6_Outstanding_Debt
## $Last_Loan_6_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Outstanding_Debt$eta_squared
## [1] 0.3570039
##
## $Last_Loan_6_Outstanding_Debt$significance
```

```
## [1] "Significant"
##
## $Last_Loan_6_Outstanding_Debt$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_6_Outstanding_Debt$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                2766.225                2723.828                2729.374
##        Home Equity Loan           Mortgage Loan                 No Loan
##                2722.335                2663.455                1049.143
##           Not Specified             Payday Loan           Personal Loan
##                2621.487                2726.136                2578.654
##            Student Loan
##                2666.333
##
##
## $Last_Loan_6_Credit_Utilization_Ratio
## $Last_Loan_6_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Credit_Utilization_Ratio$eta_squared
## [1] 0.005342324
##
## $Last_Loan_6_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Last_Loan_6_Credit_Utilization_Ratio$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_6_Credit_Utilization_Ratio$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                31.50816                31.75633                31.54198
##        Home Equity Loan           Mortgage Loan                 No Loan
##                31.59067                31.70229                32.48801
##           Not Specified             Payday Loan           Personal Loan
##                31.44808                31.60212                31.73012
##            Student Loan
##                31.57360
##
##
## $Last_Loan_6_Credit_History_Age
## $Last_Loan_6_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Credit_History_Age$eta_squared
## [1] 0.2943054
##
## $Last_Loan_6_Credit_History_Age$significance
## [1] "Significant"
##
## $Last_Loan_6_Credit_History_Age$interpretation
```

```
## [1] "Strong correlation"
##
## $Last_Loan_6_Credit_History_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                117.0049                124.3160                120.3636
##        Home Equity Loan           Mortgage Loan                 No Loan
##                120.3019                123.6371                250.7673
##           Not Specified             Payday Loan           Personal Loan
##                122.5671                122.0238                124.0380
##            Student Loan
##                126.3411
##
##
## $Last_Loan_6_Total_EMI_per_month
## $Last_Loan_6_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Total_EMI_per_month$eta_squared
## [1] 0.06203297
##
## $Last_Loan_6_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Last_Loan_6_Total_EMI_per_month$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_6_Total_EMI_per_month$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               132.39909               126.03133               125.72991
##        Home Equity Loan           Mortgage Loan                 No Loan
##               132.18615               119.60629                76.94729
##           Not Specified             Payday Loan           Personal Loan
##               121.66686               119.30900               131.79044
##            Student Loan
##               124.26634
##
##
## $Last_Loan_6_Amount_invested_monthly
## $Last_Loan_6_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Amount_invested_monthly$eta_squared
## [1] 0.02109988
##
## $Last_Loan_6_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Last_Loan_6_Amount_invested_monthly$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_6_Amount_invested_monthly$means
```

```
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                148.6404                137.8537                135.2227
##        Home Equity Loan           Mortgage Loan                 No Loan
##                145.8809                140.6618                209.1076
##           Not Specified             Payday Loan           Personal Loan
##                149.1007                140.9892                143.4179
##            Student Loan
##                137.0811
##
##
## $Last_Loan_6_Monthly_Balance
## $Last_Loan_6_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Monthly_Balance$eta_squared
## [1] 0.1101719
##
## $Last_Loan_6_Monthly_Balance$significance
## [1] "Significant"
##
## $Last_Loan_6_Monthly_Balance$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_6_Monthly_Balance$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                272.9729                269.9184                272.2192
##        Home Equity Loan           Mortgage Loan                 No Loan
##                274.3158                278.5738                442.4020
##           Not Specified             Payday Loan           Personal Loan
##                276.0804                272.0587                270.6325
##            Student Loan
##                273.9254
##
##
## $Last_Loan_7_Age
## $Last_Loan_7_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Age$eta_squared
## [1] 0.01937737
##
## $Last_Loan_7_Age$significance
## [1] "Significant"
##
## $Last_Loan_7_Age$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_7_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                30.44875                29.02500                30.18296
##        Home Equity Loan           Mortgage Loan                 No Loan
```

```
##                29.00785                29.84710                33.93022
##           Not Specified             Payday Loan           Personal Loan
##                29.50560                29.74792                29.60545
##            Student Loan
##                30.49495
##
##
## $Last_Loan_7_Annual_Income
## $Last_Loan_7_Annual_Income$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Annual_Income$eta_squared
## [1] 0.03290375
##
## $Last_Loan_7_Annual_Income$significance
## [1] "Significant"
##
## $Last_Loan_7_Annual_Income$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_7_Annual_Income$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                33529.29                31402.80                31665.53
##        Home Equity Loan           Mortgage Loan                 No Loan
##                37280.25                36133.40                53394.38
##           Not Specified             Payday Loan           Personal Loan
##                34290.73                33755.78                32788.06
##            Student Loan
##                34809.91
##
##
## $Last_Loan_7_Monthly_Inhand_Salary
## $Last_Loan_7_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Monthly_Inhand_Salary$eta_squared
## [1] 0.03252882
##
## $Last_Loan_7_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Last_Loan_7_Monthly_Inhand_Salary$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_7_Monthly_Inhand_Salary$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                2792.963                2612.908                2624.066
##        Home Equity Loan           Mortgage Loan                 No Loan
##                3106.792                3003.225                4437.836
##           Not Specified             Payday Loan           Personal Loan
##                2856.115                2822.125                2747.455
```

```
##            Student Loan
##                2900.693
##
##
## $Last_Loan_7_Num_Bank_Accounts
## $Last_Loan_7_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Num_Bank_Accounts$eta_squared
## [1] 0.1258934
##
## $Last_Loan_7_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Last_Loan_7_Num_Bank_Accounts$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_7_Num_Bank_Accounts$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                7.563750                7.527907                7.519553
##        Home Equity Loan           Mortgage Loan                 No Loan
##                7.541885                7.669085                4.984376
##           Not Specified             Payday Loan           Personal Loan
##                7.674129                7.613542                7.545024
##            Student Loan
##                7.443813
##
##
## $Last_Loan_7_Num_Credit_Card
## $Last_Loan_7_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Num_Credit_Card$eta_squared
## [1] 0.09951085
##
## $Last_Loan_7_Num_Credit_Card$significance
## [1] "Significant"
##
## $Last_Loan_7_Num_Credit_Card$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_7_Num_Credit_Card$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                7.083750                7.076744                7.234637
##        Home Equity Loan           Mortgage Loan                 No Loan
##                7.337696                6.994978                5.261747
##           Not Specified             Payday Loan           Personal Loan
##                6.902985                7.096875                7.028436
##            Student Loan
##                7.095328
##
```

```
##
## $Last_Loan_7_Interest_Rate
## $Last_Loan_7_Interest_Rate$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Interest_Rate$eta_squared
## [1] 0.1671163
##
## $Last_Loan_7_Interest_Rate$significance
## [1] "Significant"
##
## $Last_Loan_7_Interest_Rate$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_7_Interest_Rate$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                23.57000                23.17674                23.24581
##        Home Equity Loan           Mortgage Loan                 No Loan
##                22.86387                22.94643                13.03928
##           Not Specified             Payday Loan           Personal Loan
##                22.45274                23.32500                22.90047
##            Student Loan
##                23.19192
##
##
## $Last_Loan_7_Num_of_Loan
## $Last_Loan_7_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Num_of_Loan$eta_squared
## [1] 0.5174891
##
## $Last_Loan_7_Num_of_Loan$significance
## [1] "Significant"
##
## $Last_Loan_7_Num_of_Loan$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_7_Num_of_Loan$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                7.750000                7.693023                7.653631
##        Home Equity Loan           Mortgage Loan                 No Loan
##                7.706806                7.816964                2.797387
##           Not Specified             Payday Loan           Personal Loan
##                7.701493                7.795833                7.701422
##            Student Loan
##                7.843434
##
##
## $Last_Loan_7_Delay_from_due_date
## $Last_Loan_7_Delay_from_due_date$p_value
```

```
## [1] "<.0001"
##
## $Last_Loan_7_Delay_from_due_date$eta_squared
## [1] 0.1551894
##
## $Last_Loan_7_Delay_from_due_date$significance
## [1] "Significant"
##
## $Last_Loan_7_Delay_from_due_date$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_7_Delay_from_due_date$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                35.34063                34.80988                36.49022
##        Home Equity Loan           Mortgage Loan                 No Loan
##                35.50196                34.34487                18.62414
##           Not Specified             Payday Loan           Personal Loan
##                34.10572                34.82135                35.15936
##            Student Loan
##                35.30871
##
##
## $Last_Loan_7_Num_of_Delayed_Payment
## $Last_Loan_7_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Num_of_Delayed_Payment$eta_squared
## [1] 0.1255078
##
## $Last_Loan_7_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Last_Loan_7_Num_of_Delayed_Payment$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_7_Num_of_Delayed_Payment$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                18.81000                18.93953                18.78142
##        Home Equity Loan           Mortgage Loan                 No Loan
##                18.25785                19.19978                12.40265
##           Not Specified             Payday Loan           Personal Loan
##                18.18781                18.45365                18.19491
##            Student Loan
##                18.84470
##
##
## $Last_Loan_7_Changed_Credit_Limit
## $Last_Loan_7_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Changed_Credit_Limit$eta_squared
```

```
## [1] 0.07145013
##
## $Last_Loan_7_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Last_Loan_7_Changed_Credit_Limit$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_7_Changed_Credit_Limit$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               14.831550               14.923930               14.391564
##        Home Equity Loan           Mortgage Loan                 No Loan
##               15.381597               14.017433                9.636303
##           Not Specified             Payday Loan           Personal Loan
##               15.455149               14.645208               13.748033
##            Student Loan
##               15.014242
##
##
## $Last_Loan_7_Num_Credit_Inquiries
## $Last_Loan_7_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Num_Credit_Inquiries$eta_squared
## [1] 0.156056
##
## $Last_Loan_7_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Last_Loan_7_Num_Credit_Inquiries$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_7_Num_Credit_Inquiries$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                9.486875                9.165116                9.474860
##        Home Equity Loan           Mortgage Loan                 No Loan
##                9.451571                9.244420                5.135772
##           Not Specified             Payday Loan           Personal Loan
##                9.426617                9.329688                9.550948
##            Student Loan
##                9.704545
##
##
## $Last_Loan_7_Outstanding_Debt
## $Last_Loan_7_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Outstanding_Debt$eta_squared
## [1] 0.2741405
##
## $Last_Loan_7_Outstanding_Debt$significance
```

```
## [1] "Significant"
##
## $Last_Loan_7_Outstanding_Debt$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_7_Outstanding_Debt$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                2867.083                2914.899                2693.295
##        Home Equity Loan           Mortgage Loan                 No Loan
##                2906.711                2923.897                1173.832
##           Not Specified             Payday Loan           Personal Loan
##                2930.106                2923.128                2711.349
##            Student Loan
##                2939.662
##
##
## $Last_Loan_7_Credit_Utilization_Ratio
## $Last_Loan_7_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Credit_Utilization_Ratio$eta_squared
## [1] 0.003609967
##
## $Last_Loan_7_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Last_Loan_7_Credit_Utilization_Ratio$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_7_Credit_Utilization_Ratio$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                31.77552                31.61125                31.45559
##        Home Equity Loan           Mortgage Loan                 No Loan
##                31.40869                31.49702                32.41210
##           Not Specified             Payday Loan           Personal Loan
##                31.76153                31.44735                31.56792
##            Student Loan
##                31.50912
##
##
## $Last_Loan_7_Credit_History_Age
## $Last_Loan_7_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Credit_History_Age$eta_squared
## [1] 0.1921504
##
## $Last_Loan_7_Credit_History_Age$significance
## [1] "Significant"
##
## $Last_Loan_7_Credit_History_Age$interpretation
```

```
## [1] "Strong correlation"
##
## $Last_Loan_7_Credit_History_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                110.8300                118.4442                118.3994
##        Home Equity Loan           Mortgage Loan                 No Loan
##                120.7251                116.2009                239.4642
##           Not Specified             Payday Loan           Personal Loan
##                114.7139                114.2625                119.8981
##            Student Loan
##                117.4545
##
##
## $Last_Loan_7_Total_EMI_per_month
## $Last_Loan_7_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Total_EMI_per_month$eta_squared
## [1] 0.04181449
##
## $Last_Loan_7_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Last_Loan_7_Total_EMI_per_month$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_7_Total_EMI_per_month$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               132.52002               125.54421               128.57868
##        Home Equity Loan           Mortgage Loan                 No Loan
##               137.52329               132.70483                81.20234
##           Not Specified             Payday Loan           Personal Loan
##               131.79962               116.91652               115.48870
##            Student Loan
##               136.96867
##
##
## $Last_Loan_7_Amount_invested_monthly
## $Last_Loan_7_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Amount_invested_monthly$eta_squared
## [1] 0.0139466
##
## $Last_Loan_7_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Last_Loan_7_Amount_invested_monthly$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_7_Amount_invested_monthly$means
```

```
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                133.8331                130.8571                130.3457
##        Home Equity Loan           Mortgage Loan                 No Loan
##                153.3046                143.6344                203.2400
##           Not Specified             Payday Loan           Personal Loan
##                142.0229                136.5670                137.7657
##            Student Loan
##                143.0450
##
##
## $Last_Loan_7_Monthly_Balance
## $Last_Loan_7_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Monthly_Balance$eta_squared
## [1] 0.07782853
##
## $Last_Loan_7_Monthly_Balance$significance
## [1] "Significant"
##
## $Last_Loan_7_Monthly_Balance$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_7_Monthly_Balance$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                263.1549                255.1622                258.7676
##        Home Equity Loan           Mortgage Loan                 No Loan
##                266.0050                260.4806                428.4900
##           Not Specified             Payday Loan           Personal Loan
##                260.6182                259.1844                263.5241
##            Student Loan
##                257.2521
##
##
## $Last_Loan_8_Age
## $Last_Loan_8_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Age$eta_squared
## [1] 0.008954794
##
## $Last_Loan_8_Age$significance
## [1] "Significant"
##
## $Last_Loan_8_Age$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_8_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                29.33750                29.59748                27.72895
##        Home Equity Loan           Mortgage Loan                 No Loan
```

```
##                29.63851                29.83967                33.58234
##           Not Specified             Payday Loan           Personal Loan
##                29.33737                30.16892                31.98750
##            Student Loan
##                30.41551
##
##
## $Last_Loan_8_Annual_Income
## $Last_Loan_8_Annual_Income$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Annual_Income$eta_squared
## [1] 0.01727158
##
## $Last_Loan_8_Annual_Income$significance
## [1] "Significant"
##
## $Last_Loan_8_Annual_Income$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_8_Annual_Income$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                33455.98                30164.14                33385.71
##        Home Equity Loan           Mortgage Loan                 No Loan
##                29572.34                32681.41                51901.17
##           Not Specified             Payday Loan           Personal Loan
##                32455.22                33026.74                33800.15
##            Student Loan
##                34466.96
##
##
## $Last_Loan_8_Monthly_Inhand_Salary
## $Last_Loan_8_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Monthly_Inhand_Salary$eta_squared
## [1] 0.01702978
##
## $Last_Loan_8_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Last_Loan_8_Monthly_Inhand_Salary$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_8_Monthly_Inhand_Salary$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                2821.347                2483.277                2791.611
##        Home Equity Loan           Mortgage Loan                 No Loan
##                2503.088                2729.244                4314.139
##           Not Specified             Payday Loan           Personal Loan
##                2733.295                2701.933                2830.924
```

```
##            Student Loan
##                2858.238
##
##
## $Last_Loan_8_Num_Bank_Accounts
## $Last_Loan_8_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Num_Bank_Accounts$eta_squared
## [1] 0.08168034
##
## $Last_Loan_8_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Last_Loan_8_Num_Bank_Accounts$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_8_Num_Bank_Accounts$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                8.011111                8.126147                7.994737
##        Home Equity Loan           Mortgage Loan                 No Loan
##                8.161036                8.351902                5.162809
##           Not Specified             Payday Loan           Personal Loan
##                7.873656                7.792793                7.983333
##            Student Loan
##                7.962963
##
##
## $Last_Loan_8_Num_Credit_Card
## $Last_Loan_8_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Num_Credit_Card$eta_squared
## [1] 0.06842971
##
## $Last_Loan_8_Num_Credit_Card$significance
## [1] "Significant"
##
## $Last_Loan_8_Num_Credit_Card$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_8_Num_Credit_Card$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                7.433333                7.272936                7.510526
##        Home Equity Loan           Mortgage Loan                 No Loan
##                7.251126                7.395380                5.383523
##           Not Specified             Payday Loan           Personal Loan
##                7.658602                7.477477                7.672222
##            Student Loan
##                7.620370
##
```

```
##
## $Last_Loan_8_Interest_Rate
## $Last_Loan_8_Interest_Rate$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Interest_Rate$eta_squared
## [1] 0.09994461
##
## $Last_Loan_8_Interest_Rate$significance
## [1] "Significant"
##
## $Last_Loan_8_Interest_Rate$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_8_Interest_Rate$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                24.77778                24.45872                22.57895
##        Home Equity Loan           Mortgage Loan                 No Loan
##                24.27027                24.98913                13.76476
##           Not Specified             Payday Loan           Personal Loan
##                25.00000                24.54054                23.97778
##            Student Loan
##                25.23148
##
##
## $Last_Loan_8_Num_of_Loan
## $Last_Loan_8_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Num_of_Loan$eta_squared
## [1] 0.3241683
##
## $Last_Loan_8_Num_of_Loan$significance
## [1] "Significant"
##
## $Last_Loan_8_Num_of_Loan$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_8_Num_of_Loan$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                8.566667                8.513761                8.494737
##        Home Equity Loan           Mortgage Loan                 No Loan
##                8.612613                8.597826                3.145160
##           Not Specified             Payday Loan           Personal Loan
##                8.537634                8.504505                8.533333
##            Student Loan
##                8.472222
##
##
## $Last_Loan_8_Delay_from_due_date
## $Last_Loan_8_Delay_from_due_date$p_value
```

```
## [1] "<.0001"
##
## $Last_Loan_8_Delay_from_due_date$eta_squared
## [1] 0.1111827
##
## $Last_Loan_8_Delay_from_due_date$significance
## [1] "Significant"
##
## $Last_Loan_8_Delay_from_due_date$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_8_Delay_from_due_date$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                39.59722                37.84060                38.55263
##        Home Equity Loan           Mortgage Loan                 No Loan
##                38.97523                38.74049                19.69037
##           Not Specified             Payday Loan           Personal Loan
##                38.65457                38.11824                39.30833
##            Student Loan
##                40.06250
##
##
## $Last_Loan_8_Num_of_Delayed_Payment
## $Last_Loan_8_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Num_of_Delayed_Payment$eta_squared
## [1] 0.08800093
##
## $Last_Loan_8_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Last_Loan_8_Num_of_Delayed_Payment$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_8_Num_of_Delayed_Payment$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                20.51944                19.91972                19.89211
##        Home Equity Loan           Mortgage Loan                 No Loan
##                20.02815                20.20380                12.81175
##           Not Specified             Payday Loan           Personal Loan
##                19.87097                19.89640                19.86111
##            Student Loan
##                19.93287
##
##
## $Last_Loan_8_Changed_Credit_Limit
## $Last_Loan_8_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Changed_Credit_Limit$eta_squared
```

```
## [1] 0.048037
##
## $Last_Loan_8_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Last_Loan_8_Changed_Credit_Limit$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_8_Changed_Credit_Limit$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               14.110000               16.560092               16.254053
##        Home Equity Loan           Mortgage Loan                 No Loan
##               15.634640               14.624565                9.982087
##           Not Specified             Payday Loan           Personal Loan
##               16.843065               16.482703               13.945833
##            Student Loan
##               15.837361
##
##
## $Last_Loan_8_Num_Credit_Inquiries
## $Last_Loan_8_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Num_Credit_Inquiries$eta_squared
## [1] 0.08594714
##
## $Last_Loan_8_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Last_Loan_8_Num_Credit_Inquiries$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_8_Num_Credit_Inquiries$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                9.684722                9.498853                9.598684
##        Home Equity Loan           Mortgage Loan                 No Loan
##                9.557432               10.369565                5.458527
##           Not Specified             Payday Loan           Personal Loan
##               10.034946               10.141892                9.520833
##            Student Loan
##               10.057870
##
##
## $Last_Loan_8_Outstanding_Debt
## $Last_Loan_8_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Outstanding_Debt$eta_squared
## [1] 0.2135934
##
## $Last_Loan_8_Outstanding_Debt$significance
```

```
## [1] "Significant"
##
## $Last_Loan_8_Outstanding_Debt$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_8_Outstanding_Debt$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                3167.563                3390.913                3393.613
##        Home Equity Loan           Mortgage Loan                 No Loan
##                3277.374                3296.153                1277.755
##           Not Specified             Payday Loan           Personal Loan
##                3369.756                3499.012                3287.527
##            Student Loan
##                3358.800
##
##
## $Last_Loan_8_Credit_Utilization_Ratio
## $Last_Loan_8_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Credit_Utilization_Ratio$eta_squared
## [1] 0.001917061
##
## $Last_Loan_8_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Last_Loan_8_Credit_Utilization_Ratio$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_8_Credit_Utilization_Ratio$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                31.67080                31.12253                31.20835
##        Home Equity Loan           Mortgage Loan                 No Loan
##                31.61924                31.47520                32.34573
##           Not Specified             Payday Loan           Personal Loan
##                31.67320                31.70507                31.57774
##            Student Loan
##                31.49994
##
##
## $Last_Loan_8_Credit_History_Age
## $Last_Loan_8_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Credit_History_Age$eta_squared
## [1] 0.1035773
##
## $Last_Loan_8_Credit_History_Age$significance
## [1] "Significant"
##
## $Last_Loan_8_Credit_History_Age$interpretation
```

```
## [1] "Moderate correlation"
##
## $Last_Loan_8_Credit_History_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                105.1556                101.4817                109.8158
##        Home Equity Loan           Mortgage Loan                 No Loan
##                105.7523                105.0000                230.1312
##           Not Specified             Payday Loan           Personal Loan
##                104.5323                100.4459                106.7556
##            Student Loan
##                115.7222
##
##
## $Last_Loan_8_Total_EMI_per_month
## $Last_Loan_8_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Total_EMI_per_month$eta_squared
## [1] 0.02149842
##
## $Last_Loan_8_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Last_Loan_8_Total_EMI_per_month$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_8_Total_EMI_per_month$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               119.98035               128.39454               133.19025
##        Home Equity Loan           Mortgage Loan                 No Loan
##               126.00783               122.93488                84.89239
##           Not Specified             Payday Loan           Personal Loan
##               121.27701               134.30999               136.62359
##            Student Loan
##               153.28785
##
##
## $Last_Loan_8_Amount_invested_monthly
## $Last_Loan_8_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Amount_invested_monthly$eta_squared
## [1] 0.007436446
##
## $Last_Loan_8_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Last_Loan_8_Amount_invested_monthly$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_8_Amount_invested_monthly$means
```

```
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                131.0848                118.5934                145.1908
##        Home Equity Loan           Mortgage Loan                 No Loan
##                126.3364                136.9462                198.3250
##           Not Specified             Payday Loan           Personal Loan
##                136.9737                131.4996                139.9571
##            Student Loan
##                141.1097
##
##
## $Last_Loan_8_Monthly_Balance
## $Last_Loan_8_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Monthly_Balance$eta_squared
## [1] 0.04240605
##
## $Last_Loan_8_Monthly_Balance$significance
## [1] "Significant"
##
## $Last_Loan_8_Monthly_Balance$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_8_Monthly_Balance$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                257.5707                248.7529                237.2248
##        Home Equity Loan           Mortgage Loan                 No Loan
##                243.2331                241.9413                415.7782
##           Not Specified             Payday Loan           Personal Loan
##                243.3819                244.9393                242.0737
##            Student Loan
##                245.4941
##
##
## $Last_Loan_9_Age
## $Last_Loan_9_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Age$eta_squared
## [1] 0.005666897
##
## $Last_Loan_9_Age$significance
## [1] "Significant"
##
## $Last_Loan_9_Age$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_9_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                28.81027                30.14583                30.38776
##        Home Equity Loan           Mortgage Loan                 No Loan
```

```
##                30.38235                29.20192                33.46570
##           Not Specified             Payday Loan           Personal Loan
##                30.82075                27.41164                28.04167
##            Student Loan
##                29.73958
##
##
## $Last_Loan_9_Annual_Income
## $Last_Loan_9_Annual_Income$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Annual_Income$eta_squared
## [1] 0.008808132
##
## $Last_Loan_9_Annual_Income$significance
## [1] "Significant"
##
## $Last_Loan_9_Annual_Income$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_9_Annual_Income$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                38236.04                33574.68                25793.44
##        Home Equity Loan           Mortgage Loan                 No Loan
##                34021.66                30431.75                51212.64
##           Not Specified             Payday Loan           Personal Loan
##                32117.67                32301.90                37136.62
##            Student Loan
##                32311.03
##
##
## $Last_Loan_9_Monthly_Inhand_Salary
## $Last_Loan_9_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Monthly_Inhand_Salary$eta_squared
## [1] 0.00867235
##
## $Last_Loan_9_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Last_Loan_9_Monthly_Inhand_Salary$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_9_Monthly_Inhand_Salary$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                3231.157                2789.234                2156.672
##        Home Equity Loan           Mortgage Loan                 No Loan
##                2852.357                2552.556                4257.118
##           Not Specified             Payday Loan           Personal Loan
##                2664.080                2686.973                3105.175
```

```
##            Student Loan
##                2664.321
##
##
## $Last_Loan_9_Num_Bank_Accounts
## $Last_Loan_9_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Num_Bank_Accounts$eta_squared
## [1] 0.0415347
##
## $Last_Loan_9_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Last_Loan_9_Num_Bank_Accounts$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_9_Num_Bank_Accounts$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                8.049107                8.201389                8.007653
##        Home Equity Loan           Mortgage Loan                 No Loan
##                8.022059                7.880769                5.263085
##           Not Specified             Payday Loan           Personal Loan
##                7.849057                8.209052                7.973958
##            Student Loan
##                7.833333
##
##
## $Last_Loan_9_Num_Credit_Card
## $Last_Loan_9_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Num_Credit_Card$eta_squared
## [1] 0.0342679
##
## $Last_Loan_9_Num_Credit_Card$significance
## [1] "Significant"
##
## $Last_Loan_9_Num_Credit_Card$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_9_Num_Credit_Card$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                7.388393                7.701389                7.334184
##        Home Equity Loan           Mortgage Loan                 No Loan
##                7.375000                7.711538                5.457501
##           Not Specified             Payday Loan           Personal Loan
##                7.037736                7.640086                7.390625
##            Student Loan
##                7.187500
##
```

```
##
## $Last_Loan_9_Interest_Rate
## $Last_Loan_9_Interest_Rate$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Interest_Rate$eta_squared
## [1] 0.0521583
##
## $Last_Loan_9_Interest_Rate$significance
## [1] "Significant"
##
## $Last_Loan_9_Interest_Rate$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_9_Interest_Rate$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                24.92857                25.51852                24.28571
##        Home Equity Loan           Mortgage Loan                 No Loan
##                24.82353                23.86154                14.13280
##           Not Specified             Payday Loan           Personal Loan
##                24.01887                24.44828                24.64583
##            Student Loan
##                23.91667
##
##
## $Last_Loan_9_Num_of_Loan
## $Last_Loan_9_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Num_of_Loan$eta_squared
## [1] 0.2003074
##
## $Last_Loan_9_Num_of_Loan$significance
## [1] "Significant"
##
## $Last_Loan_9_Num_of_Loan$interpretation
## [1] "Strong correlation"
##
## $Last_Loan_9_Num_of_Loan$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                9.000000                9.000000                9.000000
##        Home Equity Loan           Mortgage Loan                 No Loan
##                9.000000                9.000000                3.313613
##           Not Specified             Payday Loan           Personal Loan
##                9.000000                9.000000                9.000000
##            Student Loan
##                9.000000
##
##
## $Last_Loan_9_Delay_from_due_date
## $Last_Loan_9_Delay_from_due_date$p_value
```

```
## [1] "<.0001"
##
## $Last_Loan_9_Delay_from_due_date$eta_squared
## [1] 0.05785833
##
## $Last_Loan_9_Delay_from_due_date$significance
## [1] "Significant"
##
## $Last_Loan_9_Delay_from_due_date$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_9_Delay_from_due_date$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                39.57812                41.06713                38.92602
##        Home Equity Loan           Mortgage Loan                 No Loan
##                36.58578                40.42115                20.35556
##           Not Specified             Payday Loan           Personal Loan
##                36.13208                38.05388                39.21615
##            Student Loan
##                39.32292
##
##
## $Last_Loan_9_Num_of_Delayed_Payment
## $Last_Loan_9_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Num_of_Delayed_Payment$eta_squared
## [1] 0.04775206
##
## $Last_Loan_9_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Last_Loan_9_Num_of_Delayed_Payment$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_9_Num_of_Delayed_Payment$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                20.81920                19.73611                19.96429
##        Home Equity Loan           Mortgage Loan                 No Loan
##                19.05147                20.59615                13.05571
##           Not Specified             Payday Loan           Personal Loan
##                20.58019                20.52586                20.13281
##            Student Loan
##                19.66927
##
##
## $Last_Loan_9_Changed_Credit_Limit
## $Last_Loan_9_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Changed_Credit_Limit$eta_squared
```

```
## [1] 0.02247617
##
## $Last_Loan_9_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Last_Loan_9_Changed_Credit_Limit$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_9_Changed_Credit_Limit$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                15.67491                16.98620                15.24286
##        Home Equity Loan           Mortgage Loan                 No Loan
##                15.70569                15.90662                10.18785
##           Not Specified             Payday Loan           Personal Loan
##                14.00311                14.68319                14.96094
##            Student Loan
##                15.36198
##
##
## $Last_Loan_9_Num_Credit_Inquiries
## $Last_Loan_9_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Num_Credit_Inquiries$eta_squared
## [1] 0.04307318
##
## $Last_Loan_9_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Last_Loan_9_Num_Credit_Inquiries$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_9_Num_Credit_Inquiries$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                9.926339                9.916667                9.548469
##        Home Equity Loan           Mortgage Loan                 No Loan
##                9.708333                9.684615                5.612540
##           Not Specified             Payday Loan           Personal Loan
##                9.818396               10.058190                9.333333
##            Student Loan
##                9.856771
##
##
## $Last_Loan_9_Outstanding_Debt
## $Last_Loan_9_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Outstanding_Debt$eta_squared
## [1] 0.1078362
##
## $Last_Loan_9_Outstanding_Debt$significance
```

```
## [1] "Significant"
##
## $Last_Loan_9_Outstanding_Debt$interpretation
## [1] "Moderate correlation"
##
## $Last_Loan_9_Outstanding_Debt$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                3195.326                3489.891                3320.270
##        Home Equity Loan           Mortgage Loan                 No Loan
##                3238.044                3349.249                1350.326
##           Not Specified             Payday Loan           Personal Loan
##                3270.814                3396.232                3305.347
##            Student Loan
##                3283.802
##
##
## $Last_Loan_9_Credit_Utilization_Ratio
## $Last_Loan_9_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Credit_Utilization_Ratio$eta_squared
## [1] 0.00107605
##
## $Last_Loan_9_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Last_Loan_9_Credit_Utilization_Ratio$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_9_Credit_Utilization_Ratio$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                31.43484                31.35203                31.60947
##        Home Equity Loan           Mortgage Loan                 No Loan
##                31.26565                31.51689                32.31830
##           Not Specified             Payday Loan           Personal Loan
##                31.41720                31.28580                31.54838
##            Student Loan
##                31.74897
##
##
## $Last_Loan_9_Credit_History_Age
## $Last_Loan_9_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Credit_History_Age$eta_squared
## [1] 0.05354044
##
## $Last_Loan_9_Credit_History_Age$significance
## [1] "Significant"
##
## $Last_Loan_9_Credit_History_Age$interpretation
```

```
## [1] "Weak correlation"
##
## $Last_Loan_9_Credit_History_Age$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               116.37500               110.92593               105.54082
##        Home Equity Loan           Mortgage Loan                 No Loan
##               103.83333               102.34615               225.82019
##           Not Specified             Payday Loan           Personal Loan
##               113.72642                97.62069               103.27083
##            Student Loan
##               102.20833
##
##
## $Last_Loan_9_Total_EMI_per_month
## $Last_Loan_9_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Total_EMI_per_month$eta_squared
## [1] 0.01204687
##
## $Last_Loan_9_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Last_Loan_9_Total_EMI_per_month$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_9_Total_EMI_per_month$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##               136.08191               131.13983               119.60177
##        Home Equity Loan           Mortgage Loan                 No Loan
##               137.69141               144.67145                86.44943
##           Not Specified             Payday Loan           Personal Loan
##               128.77163               120.36359               156.61423
##            Student Loan
##               111.60208
##
##
## $Last_Loan_9_Amount_invested_monthly
## $Last_Loan_9_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Amount_invested_monthly$eta_squared
## [1] 0.003584309
##
## $Last_Loan_9_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Last_Loan_9_Amount_invested_monthly$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_9_Amount_invested_monthly$means
```

```
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                151.6015                136.8925                109.8221
##        Home Equity Loan           Mortgage Loan                 No Loan
##                146.2397                138.7579                195.9791
##           Not Specified             Payday Loan           Personal Loan
##                128.6830                136.6235                145.9338
##            Student Loan
##                131.5123
##
##
## $Last_Loan_9_Monthly_Balance
## $Last_Loan_9_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Last_Loan_9_Monthly_Balance$eta_squared
## [1] 0.02432217
##
## $Last_Loan_9_Monthly_Balance$significance
## [1] "Significant"
##
## $Last_Loan_9_Monthly_Balance$interpretation
## [1] "Weak correlation"
##
## $Last_Loan_9_Monthly_Balance$means
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                252.2877                223.0924                247.5236
##        Home Equity Loan           Mortgage Loan                 No Loan
##                238.8564                229.5969                410.1805
##           Not Specified             Payday Loan           Personal Loan
##                235.8453                236.6636                240.1519
##            Student Loan
##                227.9868
```

Numeric and Numeric Correlations - RValue/ Pearson Correlation

```
# Load necessary library
if (!require("Hmisc")) install.packages("Hmisc")  # Hmisc package for rcorr function
```

```
## Loading required package: Hmisc
```

```
## Warning: package 'Hmisc' was built under R version 4.4.2
```

```
##
## Attaching package: 'Hmisc'
```

```
## The following objects are masked from 'package:base':
##
##     format.pval, units
```

```
library(Hmisc)

# Select only the relevant numeric attributes
numeric_data <- credit_score[, c("Age", "Annual_Income", "Monthly_Inhand_Salary",
                                 "Num_Bank_Accounts", "Num_Credit_Card", "Interest_Rate",
                                 "Num_of_Loan", "Delay_from_due_date", "Num_of_Delayed_Paymen
t",
                                 "Changed_Credit_Limit", "Num_Credit_Inquiries",
                                 "Outstanding_Debt", "Credit_Utilization_Ratio",
                                 "Credit_History_Age", "Total_EMI_per_month",
                                 "Amount_invested_monthly", "Monthly_Balance")]

# Calculate Pearson correlation coefficients and p-values
cor_results <- rcorr(as.matrix(numeric_data), type = "pearson")

# Extract correlation coefficients and p-values
cor_matrix <- cor_results$r  # Pearson correlation coefficients
p_matrix <- cor_results$P    # p-values

# Interpret and store results
results2 <- list()  # Rename output list to "results2"

for (i in 1:(ncol(cor_matrix) - 1)) {
  for (j in (i + 1):ncol(cor_matrix)) {
    cor_val <- cor_matrix[i, j]
    p_val <- p_matrix[i, j]

    # Format p-value: show "<.0001" for very small values, otherwise use four decimals
    formatted_p_val <- if (p_val < 0.0001) "<.0001" else formatC(p_val, format = "f", digits
= 4)

    # Interpretation based on specified thresholds
    interpretation <- if (cor_val == 1) {
      "Perfect positive correlation"
    } else if (cor_val == -1) {
      "Perfect negative correlation"
    } else if (cor_val >= 0.7) {
      "Strong positive correlation"
    } else if (cor_val >= 0.3) {
      "Moderate positive correlation"
    } else if (cor_val > 0) {
      "Weak positive correlation"
    } else if (cor_val > -0.3) {
      "Weak negative correlation"
    } else if (cor_val >= -0.7) {
      "Moderate negative correlation"
    } else {
      "Strong negative correlation"
    }

    # Significance check based on numeric p-value
```

```
    significance <- ifelse(p_val < 0.05, "Significant", "Not Significant")

    # If p-value is formatted as "<.0001", set significance to "Significant"
    if (formatted_p_val == "<.0001") {
      significance <- "Significant"
    }

    # Store results in the list
    results2[[paste(names(numeric_data)[i], names(numeric_data)[j], sep = "_")]] <- list(
      correlation_coefficient = cor_val,
      p_value = formatted_p_val,
      significance = significance,
      interpretation = interpretation
    )
  }
}

# Print the results (optional)
print(results2)
```

```
## $Age_Annual_Income
## $Age_Annual_Income$correlation_coefficient
## [1] 0.0914981
##
## $Age_Annual_Income$p_value
## [1] "<.0001"
##
## $Age_Annual_Income$significance
## [1] "Significant"
##
## $Age_Annual_Income$interpretation
## [1] "Weak positive correlation"
##
##
## $Age_Monthly_Inhand_Salary
## $Age_Monthly_Inhand_Salary$correlation_coefficient
## [1] 0.09059984
##
## $Age_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Age_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Age_Monthly_Inhand_Salary$interpretation
## [1] "Weak positive correlation"
##
##
## $Age_Num_Bank_Accounts
## $Age_Num_Bank_Accounts$correlation_coefficient
## [1] -0.1903769
##
## $Age_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Age_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Age_Num_Bank_Accounts$interpretation
## [1] "Weak negative correlation"
##
##
## $Age_Num_Credit_Card
## $Age_Num_Credit_Card$correlation_coefficient
## [1] -0.1485107
##
## $Age_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Age_Num_Credit_Card$significance
## [1] "Significant"
```

```
##
## $Age_Num_Credit_Card$interpretation
## [1] "Weak negative correlation"
##
##
## $Age_Interest_Rate
## $Age_Interest_Rate$correlation_coefficient
## [1] -0.2175194
##
## $Age_Interest_Rate$p_value
## [1] "<.0001"
##
## $Age_Interest_Rate$significance
## [1] "Significant"
##
## $Age_Interest_Rate$interpretation
## [1] "Weak negative correlation"
##
##
## $Age_Num_of_Loan
## $Age_Num_of_Loan$correlation_coefficient
## [1] -0.2135159
##
## $Age_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Age_Num_of_Loan$significance
## [1] "Significant"
##
## $Age_Num_of_Loan$interpretation
## [1] "Weak negative correlation"
##
##
## $Age_Delay_from_due_date
## $Age_Delay_from_due_date$correlation_coefficient
## [1] -0.174132
##
## $Age_Delay_from_due_date$p_value
## [1] "<.0001"
##
## $Age_Delay_from_due_date$significance
## [1] "Significant"
##
## $Age_Delay_from_due_date$interpretation
## [1] "Weak negative correlation"
##
##
## $Age_Num_of_Delayed_Payment
## $Age_Num_of_Delayed_Payment$correlation_coefficient
## [1] -0.1842916
##
```

```
## $Age_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Age_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Age_Num_of_Delayed_Payment$interpretation
## [1] "Weak negative correlation"
##
##
## $Age_Changed_Credit_Limit
## $Age_Changed_Credit_Limit$correlation_coefficient
## [1] -0.1566206
##
## $Age_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Age_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Age_Changed_Credit_Limit$interpretation
## [1] "Weak negative correlation"
##
##
## $Age_Num_Credit_Inquiries
## $Age_Num_Credit_Inquiries$correlation_coefficient
## [1] -0.2509299
##
## $Age_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Age_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Age_Num_Credit_Inquiries$interpretation
## [1] "Weak negative correlation"
##
##
## $Age_Outstanding_Debt
## $Age_Outstanding_Debt$correlation_coefficient
## [1] -0.2023667
##
## $Age_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Age_Outstanding_Debt$significance
## [1] "Significant"
##
## $Age_Outstanding_Debt$interpretation
## [1] "Weak negative correlation"
##
```

```
##
## $Age_Credit_Utilization_Ratio
## $Age_Credit_Utilization_Ratio$correlation_coefficient
## [1] 0.02549872
##
## $Age_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Age_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Age_Credit_Utilization_Ratio$interpretation
## [1] "Weak positive correlation"
##
##
## $Age_Credit_History_Age
## $Age_Credit_History_Age$correlation_coefficient
## [1] 0.2346224
##
## $Age_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Age_Credit_History_Age$significance
## [1] "Significant"
##
## $Age_Credit_History_Age$interpretation
## [1] "Weak positive correlation"
##
##
## $Age_Total_EMI_per_month
## $Age_Total_EMI_per_month$correlation_coefficient
## [1] -0.06925344
##
## $Age_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Age_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Age_Total_EMI_per_month$interpretation
## [1] "Weak negative correlation"
##
##
## $Age_Amount_invested_monthly
## $Age_Amount_invested_monthly$correlation_coefficient
## [1] 0.05736039
##
## $Age_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Age_Amount_invested_monthly$significance
```

```
## [1] "Significant"
##
## $Age_Amount_invested_monthly$interpretation
## [1] "Weak positive correlation"
##
##
## $Age_Monthly_Balance
## $Age_Monthly_Balance$correlation_coefficient
## [1] 0.1165916
##
## $Age_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Age_Monthly_Balance$significance
## [1] "Significant"
##
## $Age_Monthly_Balance$interpretation
## [1] "Weak positive correlation"
##
##
## $Annual_Income_Monthly_Inhand_Salary
## $Annual_Income_Monthly_Inhand_Salary$correlation_coefficient
## [1] 0.9982013
##
## $Annual_Income_Monthly_Inhand_Salary$p_value
## [1] "<.0001"
##
## $Annual_Income_Monthly_Inhand_Salary$significance
## [1] "Significant"
##
## $Annual_Income_Monthly_Inhand_Salary$interpretation
## [1] "Strong positive correlation"
##
##
## $Annual_Income_Num_Bank_Accounts
## $Annual_Income_Num_Bank_Accounts$correlation_coefficient
## [1] -0.2840476
##
## $Annual_Income_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Annual_Income_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Annual_Income_Num_Bank_Accounts$interpretation
## [1] "Weak negative correlation"
##
##
## $Annual_Income_Num_Credit_Card
## $Annual_Income_Num_Credit_Card$correlation_coefficient
## [1] -0.2178769
```

```
##
## $Annual_Income_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Annual_Income_Num_Credit_Card$significance
## [1] "Significant"
##
## $Annual_Income_Num_Credit_Card$interpretation
## [1] "Weak negative correlation"
##
##
## $Annual_Income_Interest_Rate
## $Annual_Income_Interest_Rate$correlation_coefficient
## [1] -0.3023524
##
## $Annual_Income_Interest_Rate$p_value
## [1] "<.0001"
##
## $Annual_Income_Interest_Rate$significance
## [1] "Significant"
##
## $Annual_Income_Interest_Rate$interpretation
## [1] "Moderate negative correlation"
##
##
## $Annual_Income_Num_of_Loan
## $Annual_Income_Num_of_Loan$correlation_coefficient
## [1] -0.2555596
##
## $Annual_Income_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Annual_Income_Num_of_Loan$significance
## [1] "Significant"
##
## $Annual_Income_Num_of_Loan$interpretation
## [1] "Weak negative correlation"
##
##
## $Annual_Income_Delay_from_due_date
## $Annual_Income_Delay_from_due_date$correlation_coefficient
## [1] -0.2501626
##
## $Annual_Income_Delay_from_due_date$p_value
## [1] "<.0001"
##
## $Annual_Income_Delay_from_due_date$significance
## [1] "Significant"
##
## $Annual_Income_Delay_from_due_date$interpretation
## [1] "Weak negative correlation"
```

```
##
##
## $Annual_Income_Num_of_Delayed_Payment
## $Annual_Income_Num_of_Delayed_Payment$correlation_coefficient
## [1] -0.2860564
##
## $Annual_Income_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Annual_Income_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Annual_Income_Num_of_Delayed_Payment$interpretation
## [1] "Weak negative correlation"
##
##
## $Annual_Income_Changed_Credit_Limit
## $Annual_Income_Changed_Credit_Limit$correlation_coefficient
## [1] -0.1753545
##
## $Annual_Income_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Annual_Income_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Annual_Income_Changed_Credit_Limit$interpretation
## [1] "Weak negative correlation"
##
##
## $Annual_Income_Num_Credit_Inquiries
## $Annual_Income_Num_Credit_Inquiries$correlation_coefficient
## [1] -0.2808249
##
## $Annual_Income_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Annual_Income_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Annual_Income_Num_Credit_Inquiries$interpretation
## [1] "Weak negative correlation"
##
##
## $Annual_Income_Outstanding_Debt
## $Annual_Income_Outstanding_Debt$correlation_coefficient
## [1] -0.2693193
##
## $Annual_Income_Outstanding_Debt$p_value
## [1] "<.0001"
##
```

```
## $Annual_Income_Outstanding_Debt$significance
## [1] "Significant"
##
## $Annual_Income_Outstanding_Debt$interpretation
## [1] "Weak negative correlation"
##
##
## $Annual_Income_Credit_Utilization_Ratio
## $Annual_Income_Credit_Utilization_Ratio$correlation_coefficient
## [1] 0.1759301
##
## $Annual_Income_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Annual_Income_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Annual_Income_Credit_Utilization_Ratio$interpretation
## [1] "Weak positive correlation"
##
##
## $Annual_Income_Credit_History_Age
## $Annual_Income_Credit_History_Age$correlation_coefficient
## [1] 0.2723774
##
## $Annual_Income_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Annual_Income_Credit_History_Age$significance
## [1] "Significant"
##
## $Annual_Income_Credit_History_Age$interpretation
## [1] "Weak positive correlation"
##
##
## $Annual_Income_Total_EMI_per_month
## $Annual_Income_Total_EMI_per_month$correlation_coefficient
## [1] 0.3898632
##
## $Annual_Income_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Annual_Income_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Annual_Income_Total_EMI_per_month$interpretation
## [1] "Moderate positive correlation"
##
##
## $Annual_Income_Amount_invested_monthly
## $Annual_Income_Amount_invested_monthly$correlation_coefficient
```

```
## [1] 0.6274419
##
## $Annual_Income_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Annual_Income_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Annual_Income_Amount_invested_monthly$interpretation
## [1] "Moderate positive correlation"
##
##
## $Annual_Income_Monthly_Balance
## $Annual_Income_Monthly_Balance$correlation_coefficient
## [1] 0.7055278
##
## $Annual_Income_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Annual_Income_Monthly_Balance$significance
## [1] "Significant"
##
## $Annual_Income_Monthly_Balance$interpretation
## [1] "Strong positive correlation"
##
##
## $Monthly_Inhand_Salary_Num_Bank_Accounts
## $Monthly_Inhand_Salary_Num_Bank_Accounts$correlation_coefficient
## [1] -0.2832543
##
## $Monthly_Inhand_Salary_Num_Bank_Accounts$p_value
## [1] "<.0001"
##
## $Monthly_Inhand_Salary_Num_Bank_Accounts$significance
## [1] "Significant"
##
## $Monthly_Inhand_Salary_Num_Bank_Accounts$interpretation
## [1] "Weak negative correlation"
##
##
## $Monthly_Inhand_Salary_Num_Credit_Card
## $Monthly_Inhand_Salary_Num_Credit_Card$correlation_coefficient
## [1] -0.2169451
##
## $Monthly_Inhand_Salary_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Monthly_Inhand_Salary_Num_Credit_Card$significance
## [1] "Significant"
##
## $Monthly_Inhand_Salary_Num_Credit_Card$interpretation
```

```
## [1] "Weak negative correlation"
##
##
## $Monthly_Inhand_Salary_Interest_Rate
## $Monthly_Inhand_Salary_Interest_Rate$correlation_coefficient
## [1] -0.301907
##
## $Monthly_Inhand_Salary_Interest_Rate$p_value
## [1] "<.0001"
##
## $Monthly_Inhand_Salary_Interest_Rate$significance
## [1] "Significant"
##
## $Monthly_Inhand_Salary_Interest_Rate$interpretation
## [1] "Moderate negative correlation"
##
##
## $Monthly_Inhand_Salary_Num_of_Loan
## $Monthly_Inhand_Salary_Num_of_Loan$correlation_coefficient
## [1] -0.2544056
##
## $Monthly_Inhand_Salary_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Monthly_Inhand_Salary_Num_of_Loan$significance
## [1] "Significant"
##
## $Monthly_Inhand_Salary_Num_of_Loan$interpretation
## [1] "Weak negative correlation"
##
##
## $Monthly_Inhand_Salary_Delay_from_due_date
## $Monthly_Inhand_Salary_Delay_from_due_date$correlation_coefficient
## [1] -0.2495546
##
## $Monthly_Inhand_Salary_Delay_from_due_date$p_value
## [1] "<.0001"
##
## $Monthly_Inhand_Salary_Delay_from_due_date$significance
## [1] "Significant"
##
## $Monthly_Inhand_Salary_Delay_from_due_date$interpretation
## [1] "Weak negative correlation"
##
##
## $Monthly_Inhand_Salary_Num_of_Delayed_Payment
## $Monthly_Inhand_Salary_Num_of_Delayed_Payment$correlation_coefficient
## [1] -0.2851434
##
## $Monthly_Inhand_Salary_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
```

```
##
## $Monthly_Inhand_Salary_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Monthly_Inhand_Salary_Num_of_Delayed_Payment$interpretation
## [1] "Weak negative correlation"
##
##
## $Monthly_Inhand_Salary_Changed_Credit_Limit
## $Monthly_Inhand_Salary_Changed_Credit_Limit$correlation_coefficient
## [1] -0.175308
##
## $Monthly_Inhand_Salary_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Monthly_Inhand_Salary_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Monthly_Inhand_Salary_Changed_Credit_Limit$interpretation
## [1] "Weak negative correlation"
##
##
## $Monthly_Inhand_Salary_Num_Credit_Inquiries
## $Monthly_Inhand_Salary_Num_Credit_Inquiries$correlation_coefficient
## [1] -0.2804296
##
## $Monthly_Inhand_Salary_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Monthly_Inhand_Salary_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Monthly_Inhand_Salary_Num_Credit_Inquiries$interpretation
## [1] "Weak negative correlation"
##
##
## $Monthly_Inhand_Salary_Outstanding_Debt
## $Monthly_Inhand_Salary_Outstanding_Debt$correlation_coefficient
## [1] -0.2690784
##
## $Monthly_Inhand_Salary_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Monthly_Inhand_Salary_Outstanding_Debt$significance
## [1] "Significant"
##
## $Monthly_Inhand_Salary_Outstanding_Debt$interpretation
## [1] "Weak negative correlation"
##
##
## $Monthly_Inhand_Salary_Credit_Utilization_Ratio
```

```
## $Monthly_Inhand_Salary_Credit_Utilization_Ratio$correlation_coefficient
## [1] 0.1760806
##
## $Monthly_Inhand_Salary_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Monthly_Inhand_Salary_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Monthly_Inhand_Salary_Credit_Utilization_Ratio$interpretation
## [1] "Weak positive correlation"
##
##
## $Monthly_Inhand_Salary_Credit_History_Age
## $Monthly_Inhand_Salary_Credit_History_Age$correlation_coefficient
## [1] 0.2715092
##
## $Monthly_Inhand_Salary_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Monthly_Inhand_Salary_Credit_History_Age$significance
## [1] "Significant"
##
## $Monthly_Inhand_Salary_Credit_History_Age$interpretation
## [1] "Weak positive correlation"
##
##
## $Monthly_Inhand_Salary_Total_EMI_per_month
## $Monthly_Inhand_Salary_Total_EMI_per_month$correlation_coefficient
## [1] 0.3883769
##
## $Monthly_Inhand_Salary_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Monthly_Inhand_Salary_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Monthly_Inhand_Salary_Total_EMI_per_month$interpretation
## [1] "Moderate positive correlation"
##
##
## $Monthly_Inhand_Salary_Amount_invested_monthly
## $Monthly_Inhand_Salary_Amount_invested_monthly$correlation_coefficient
## [1] 0.6281073
##
## $Monthly_Inhand_Salary_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Monthly_Inhand_Salary_Amount_invested_monthly$significance
## [1] "Significant"
##
```

```
## $Monthly_Inhand_Salary_Amount_invested_monthly$interpretation
## [1] "Moderate positive correlation"
##
##
## $Monthly_Inhand_Salary_Monthly_Balance
## $Monthly_Inhand_Salary_Monthly_Balance$correlation_coefficient
## [1] 0.7067414
##
## $Monthly_Inhand_Salary_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Monthly_Inhand_Salary_Monthly_Balance$significance
## [1] "Significant"
##
## $Monthly_Inhand_Salary_Monthly_Balance$interpretation
## [1] "Strong positive correlation"
##
##
## $Num_Bank_Accounts_Num_Credit_Card
## $Num_Bank_Accounts_Num_Credit_Card$correlation_coefficient
## [1] 0.4426824
##
## $Num_Bank_Accounts_Num_Credit_Card$p_value
## [1] "<.0001"
##
## $Num_Bank_Accounts_Num_Credit_Card$significance
## [1] "Significant"
##
## $Num_Bank_Accounts_Num_Credit_Card$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_Bank_Accounts_Interest_Rate
## $Num_Bank_Accounts_Interest_Rate$correlation_coefficient
## [1] 0.5842968
##
## $Num_Bank_Accounts_Interest_Rate$p_value
## [1] "<.0001"
##
## $Num_Bank_Accounts_Interest_Rate$significance
## [1] "Significant"
##
## $Num_Bank_Accounts_Interest_Rate$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_Bank_Accounts_Num_of_Loan
## $Num_Bank_Accounts_Num_of_Loan$correlation_coefficient
## [1] 0.4724834
##
## $Num_Bank_Accounts_Num_of_Loan$p_value
```

```
## [1] "<.0001"
##
## $Num_Bank_Accounts_Num_of_Loan$significance
## [1] "Significant"
##
## $Num_Bank_Accounts_Num_of_Loan$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_Bank_Accounts_Delay_from_due_date
## $Num_Bank_Accounts_Delay_from_due_date$correlation_coefficient
## [1] 0.5602764
##
## $Num_Bank_Accounts_Delay_from_due_date$p_value
## [1] "<.0001"
##
## $Num_Bank_Accounts_Delay_from_due_date$significance
## [1] "Significant"
##
## $Num_Bank_Accounts_Delay_from_due_date$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_Bank_Accounts_Num_of_Delayed_Payment
## $Num_Bank_Accounts_Num_of_Delayed_Payment$correlation_coefficient
## [1] 0.6013749
##
## $Num_Bank_Accounts_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Num_Bank_Accounts_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Num_Bank_Accounts_Num_of_Delayed_Payment$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_Bank_Accounts_Changed_Credit_Limit
## $Num_Bank_Accounts_Changed_Credit_Limit$correlation_coefficient
## [1] 0.3313534
##
## $Num_Bank_Accounts_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Num_Bank_Accounts_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Num_Bank_Accounts_Changed_Credit_Limit$interpretation
## [1] "Moderate positive correlation"
##
##
```

```
## $Num_Bank_Accounts_Num_Credit_Inquiries
## $Num_Bank_Accounts_Num_Credit_Inquiries$correlation_coefficient
## [1] 0.5193313
##
## $Num_Bank_Accounts_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Num_Bank_Accounts_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Num_Bank_Accounts_Num_Credit_Inquiries$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_Bank_Accounts_Outstanding_Debt
## $Num_Bank_Accounts_Outstanding_Debt$correlation_coefficient
## [1] 0.507045
##
## $Num_Bank_Accounts_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Num_Bank_Accounts_Outstanding_Debt$significance
## [1] "Significant"
##
## $Num_Bank_Accounts_Outstanding_Debt$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_Bank_Accounts_Credit_Utilization_Ratio
## $Num_Bank_Accounts_Credit_Utilization_Ratio$correlation_coefficient
## [1] -0.07177803
##
## $Num_Bank_Accounts_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Num_Bank_Accounts_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Num_Bank_Accounts_Credit_Utilization_Ratio$interpretation
## [1] "Weak negative correlation"
##
##
## $Num_Bank_Accounts_Credit_History_Age
## $Num_Bank_Accounts_Credit_History_Age$correlation_coefficient
## [1] -0.4853322
##
## $Num_Bank_Accounts_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Num_Bank_Accounts_Credit_History_Age$significance
## [1] "Significant"
```

```
##
## $Num_Bank_Accounts_Credit_History_Age$interpretation
## [1] "Moderate negative correlation"
##
##
## $Num_Bank_Accounts_Total_EMI_per_month
## $Num_Bank_Accounts_Total_EMI_per_month$correlation_coefficient
## [1] 0.09751977
##
## $Num_Bank_Accounts_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Num_Bank_Accounts_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Num_Bank_Accounts_Total_EMI_per_month$interpretation
## [1] "Weak positive correlation"
##
##
## $Num_Bank_Accounts_Amount_invested_monthly
## $Num_Bank_Accounts_Amount_invested_monthly$correlation_coefficient
## [1] -0.1822093
##
## $Num_Bank_Accounts_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Num_Bank_Accounts_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Num_Bank_Accounts_Amount_invested_monthly$interpretation
## [1] "Weak negative correlation"
##
##
## $Num_Bank_Accounts_Monthly_Balance
## $Num_Bank_Accounts_Monthly_Balance$correlation_coefficient
## [1] -0.3004481
##
## $Num_Bank_Accounts_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Num_Bank_Accounts_Monthly_Balance$significance
## [1] "Significant"
##
## $Num_Bank_Accounts_Monthly_Balance$interpretation
## [1] "Moderate negative correlation"
##
##
## $Num_Credit_Card_Interest_Rate
## $Num_Credit_Card_Interest_Rate$correlation_coefficient
## [1] 0.4979123
##
```

```
## $Num_Credit_Card_Interest_Rate$p_value
## [1] "<.0001"
##
## $Num_Credit_Card_Interest_Rate$significance
## [1] "Significant"
##
## $Num_Credit_Card_Interest_Rate$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_Credit_Card_Num_of_Loan
## $Num_Credit_Card_Num_of_Loan$correlation_coefficient
## [1] 0.4179652
##
## $Num_Credit_Card_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Num_Credit_Card_Num_of_Loan$significance
## [1] "Significant"
##
## $Num_Credit_Card_Num_of_Loan$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_Credit_Card_Delay_from_due_date
## $Num_Credit_Card_Delay_from_due_date$correlation_coefficient
## [1] 0.4795745
##
## $Num_Credit_Card_Delay_from_due_date$p_value
## [1] "<.0001"
##
## $Num_Credit_Card_Delay_from_due_date$significance
## [1] "Significant"
##
## $Num_Credit_Card_Delay_from_due_date$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_Credit_Card_Num_of_Delayed_Payment
## $Num_Credit_Card_Num_of_Delayed_Payment$correlation_coefficient
## [1] 0.4235605
##
## $Num_Credit_Card_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Num_Credit_Card_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Num_Credit_Card_Num_of_Delayed_Payment$interpretation
## [1] "Moderate positive correlation"
##
```

```
##
## $Num_Credit_Card_Changed_Credit_Limit
## $Num_Credit_Card_Changed_Credit_Limit$correlation_coefficient
## [1] 0.2532154
##
## $Num_Credit_Card_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Num_Credit_Card_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Num_Credit_Card_Changed_Credit_Limit$interpretation
## [1] "Weak positive correlation"
##
##
## $Num_Credit_Card_Num_Credit_Inquiries
## $Num_Credit_Card_Num_Credit_Inquiries$correlation_coefficient
## [1] 0.4602084
##
## $Num_Credit_Card_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Num_Credit_Card_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Num_Credit_Card_Num_Credit_Inquiries$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_Credit_Card_Outstanding_Debt
## $Num_Credit_Card_Outstanding_Debt$correlation_coefficient
## [1] 0.4903615
##
## $Num_Credit_Card_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Num_Credit_Card_Outstanding_Debt$significance
## [1] "Significant"
##
## $Num_Credit_Card_Outstanding_Debt$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_Credit_Card_Credit_Utilization_Ratio
## $Num_Credit_Card_Credit_Utilization_Ratio$correlation_coefficient
## [1] -0.05524781
##
## $Num_Credit_Card_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Num_Credit_Card_Credit_Utilization_Ratio$significance
```

```
## [1] "Significant"
##
## $Num_Credit_Card_Credit_Utilization_Ratio$interpretation
## [1] "Weak negative correlation"
##
##
## $Num_Credit_Card_Credit_History_Age
## $Num_Credit_Card_Credit_History_Age$correlation_coefficient
## [1] -0.4172234
##
## $Num_Credit_Card_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Num_Credit_Card_Credit_History_Age$significance
## [1] "Significant"
##
## $Num_Credit_Card_Credit_History_Age$interpretation
## [1] "Moderate negative correlation"
##
##
## $Num_Credit_Card_Total_EMI_per_month
## $Num_Credit_Card_Total_EMI_per_month$correlation_coefficient
## [1] 0.1051108
##
## $Num_Credit_Card_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Num_Credit_Card_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Num_Credit_Card_Total_EMI_per_month$interpretation
## [1] "Weak positive correlation"
##
##
## $Num_Credit_Card_Amount_invested_monthly
## $Num_Credit_Card_Amount_invested_monthly$correlation_coefficient
## [1] -0.1429999
##
## $Num_Credit_Card_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Num_Credit_Card_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Num_Credit_Card_Amount_invested_monthly$interpretation
## [1] "Weak negative correlation"
##
##
## $Num_Credit_Card_Monthly_Balance
## $Num_Credit_Card_Monthly_Balance$correlation_coefficient
## [1] -0.238718
```

```
##
## $Num_Credit_Card_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Num_Credit_Card_Monthly_Balance$significance
## [1] "Significant"
##
## $Num_Credit_Card_Monthly_Balance$interpretation
## [1] "Weak negative correlation"
##
##
## $Interest_Rate_Num_of_Loan
## $Interest_Rate_Num_of_Loan$correlation_coefficient
## [1] 0.5591562
##
## $Interest_Rate_Num_of_Loan$p_value
## [1] "<.0001"
##
## $Interest_Rate_Num_of_Loan$significance
## [1] "Significant"
##
## $Interest_Rate_Num_of_Loan$interpretation
## [1] "Moderate positive correlation"
##
##
## $Interest_Rate_Delay_from_due_date
## $Interest_Rate_Delay_from_due_date$correlation_coefficient
## [1] 0.5890878
##
## $Interest_Rate_Delay_from_due_date$p_value
## [1] "<.0001"
##
## $Interest_Rate_Delay_from_due_date$significance
## [1] "Significant"
##
## $Interest_Rate_Delay_from_due_date$interpretation
## [1] "Moderate positive correlation"
##
##
## $Interest_Rate_Num_of_Delayed_Payment
## $Interest_Rate_Num_of_Delayed_Payment$correlation_coefficient
## [1] 0.5714163
##
## $Interest_Rate_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Interest_Rate_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Interest_Rate_Num_of_Delayed_Payment$interpretation
## [1] "Moderate positive correlation"
```

```
##
##
## $Interest_Rate_Changed_Credit_Limit
## $Interest_Rate_Changed_Credit_Limit$correlation_coefficient
## [1] 0.3669429
##
## $Interest_Rate_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Interest_Rate_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Interest_Rate_Changed_Credit_Limit$interpretation
## [1] "Moderate positive correlation"
##
##
## $Interest_Rate_Num_Credit_Inquiries
## $Interest_Rate_Num_Credit_Inquiries$correlation_coefficient
## [1] 0.6335194
##
## $Interest_Rate_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Interest_Rate_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Interest_Rate_Num_Credit_Inquiries$interpretation
## [1] "Moderate positive correlation"
##
##
## $Interest_Rate_Outstanding_Debt
## $Interest_Rate_Outstanding_Debt$correlation_coefficient
## [1] 0.6294144
##
## $Interest_Rate_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Interest_Rate_Outstanding_Debt$significance
## [1] "Significant"
##
## $Interest_Rate_Outstanding_Debt$interpretation
## [1] "Moderate positive correlation"
##
##
## $Interest_Rate_Credit_Utilization_Ratio
## $Interest_Rate_Credit_Utilization_Ratio$correlation_coefficient
## [1] -0.07569978
##
## $Interest_Rate_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
```

```
## $Interest_Rate_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Interest_Rate_Credit_Utilization_Ratio$interpretation
## [1] "Weak negative correlation"
##
##
## $Interest_Rate_Credit_History_Age
## $Interest_Rate_Credit_History_Age$correlation_coefficient
## [1] -0.5762261
##
## $Interest_Rate_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Interest_Rate_Credit_History_Age$significance
## [1] "Significant"
##
## $Interest_Rate_Credit_History_Age$interpretation
## [1] "Moderate negative correlation"
##
##
## $Interest_Rate_Total_EMI_per_month
## $Interest_Rate_Total_EMI_per_month$correlation_coefficient
## [1] 0.1340717
##
## $Interest_Rate_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Interest_Rate_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Interest_Rate_Total_EMI_per_month$interpretation
## [1] "Weak positive correlation"
##
##
## $Interest_Rate_Amount_invested_monthly
## $Interest_Rate_Amount_invested_monthly$correlation_coefficient
## [1] -0.1949567
##
## $Interest_Rate_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Interest_Rate_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Interest_Rate_Amount_invested_monthly$interpretation
## [1] "Weak negative correlation"
##
##
## $Interest_Rate_Monthly_Balance
## $Interest_Rate_Monthly_Balance$correlation_coefficient
```

```
## [1] -0.3288446
##
## $Interest_Rate_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Interest_Rate_Monthly_Balance$significance
## [1] "Significant"
##
## $Interest_Rate_Monthly_Balance$interpretation
## [1] "Moderate negative correlation"
##
##
## $Num_of_Loan_Delay_from_due_date
## $Num_of_Loan_Delay_from_due_date$correlation_coefficient
## [1] 0.5012433
##
## $Num_of_Loan_Delay_from_due_date$p_value
## [1] "<.0001"
##
## $Num_of_Loan_Delay_from_due_date$significance
## [1] "Significant"
##
## $Num_of_Loan_Delay_from_due_date$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_of_Loan_Num_of_Delayed_Payment
## $Num_of_Loan_Num_of_Delayed_Payment$correlation_coefficient
## [1] 0.474618
##
## $Num_of_Loan_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Num_of_Loan_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Num_of_Loan_Num_of_Delayed_Payment$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_of_Loan_Changed_Credit_Limit
## $Num_of_Loan_Changed_Credit_Limit$correlation_coefficient
## [1] 0.3703803
##
## $Num_of_Loan_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Num_of_Loan_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Num_of_Loan_Changed_Credit_Limit$interpretation
```

```
## [1] "Moderate positive correlation"
##
##
## $Num_of_Loan_Num_Credit_Inquiries
## $Num_of_Loan_Num_Credit_Inquiries$correlation_coefficient
## [1] 0.566149
##
## $Num_of_Loan_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Num_of_Loan_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Num_of_Loan_Num_Credit_Inquiries$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_of_Loan_Outstanding_Debt
## $Num_of_Loan_Outstanding_Debt$correlation_coefficient
## [1] 0.6387126
##
## $Num_of_Loan_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Num_of_Loan_Outstanding_Debt$significance
## [1] "Significant"
##
## $Num_of_Loan_Outstanding_Debt$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_of_Loan_Credit_Utilization_Ratio
## $Num_of_Loan_Credit_Utilization_Ratio$correlation_coefficient
## [1] -0.1004692
##
## $Num_of_Loan_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Num_of_Loan_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Num_of_Loan_Credit_Utilization_Ratio$interpretation
## [1] "Weak negative correlation"
##
##
## $Num_of_Loan_Credit_History_Age
## $Num_of_Loan_Credit_History_Age$correlation_coefficient
## [1] -0.6057249
##
## $Num_of_Loan_Credit_History_Age$p_value
## [1] "<.0001"
```

```
##
## $Num_of_Loan_Credit_History_Age$significance
## [1] "Significant"
##
## $Num_of_Loan_Credit_History_Age$interpretation
## [1] "Moderate negative correlation"
##
##
## $Num_of_Loan_Total_EMI_per_month
## $Num_of_Loan_Total_EMI_per_month$correlation_coefficient
## [1] 0.4320645
##
## $Num_of_Loan_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Num_of_Loan_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Num_of_Loan_Total_EMI_per_month$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_of_Loan_Amount_invested_monthly
## $Num_of_Loan_Amount_invested_monthly$correlation_coefficient
## [1] -0.1644227
##
## $Num_of_Loan_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Num_of_Loan_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Num_of_Loan_Amount_invested_monthly$interpretation
## [1] "Weak negative correlation"
##
##
## $Num_of_Loan_Monthly_Balance
## $Num_of_Loan_Monthly_Balance$correlation_coefficient
## [1] -0.4337439
##
## $Num_of_Loan_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Num_of_Loan_Monthly_Balance$significance
## [1] "Significant"
##
## $Num_of_Loan_Monthly_Balance$interpretation
## [1] "Moderate negative correlation"
##
##
## $Delay_from_due_date_Num_of_Delayed_Payment
```

```
## $Delay_from_due_date_Num_of_Delayed_Payment$correlation_coefficient
## [1] 0.5422376
##
## $Delay_from_due_date_Num_of_Delayed_Payment$p_value
## [1] "<.0001"
##
## $Delay_from_due_date_Num_of_Delayed_Payment$significance
## [1] "Significant"
##
## $Delay_from_due_date_Num_of_Delayed_Payment$interpretation
## [1] "Moderate positive correlation"
##
##
## $Delay_from_due_date_Changed_Credit_Limit
## $Delay_from_due_date_Changed_Credit_Limit$correlation_coefficient
## [1] 0.293719
##
## $Delay_from_due_date_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Delay_from_due_date_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Delay_from_due_date_Changed_Credit_Limit$interpretation
## [1] "Weak positive correlation"
##
##
## $Delay_from_due_date_Num_Credit_Inquiries
## $Delay_from_due_date_Num_Credit_Inquiries$correlation_coefficient
## [1] 0.5407855
##
## $Delay_from_due_date_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Delay_from_due_date_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Delay_from_due_date_Num_Credit_Inquiries$interpretation
## [1] "Moderate positive correlation"
##
##
## $Delay_from_due_date_Outstanding_Debt
## $Delay_from_due_date_Outstanding_Debt$correlation_coefficient
## [1] 0.5717128
##
## $Delay_from_due_date_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Delay_from_due_date_Outstanding_Debt$significance
## [1] "Significant"
##
```

```
## $Delay_from_due_date_Outstanding_Debt$interpretation
## [1] "Moderate positive correlation"
##
##
## $Delay_from_due_date_Credit_Utilization_Ratio
## $Delay_from_due_date_Credit_Utilization_Ratio$correlation_coefficient
## [1] -0.06379612
##
## $Delay_from_due_date_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Delay_from_due_date_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Delay_from_due_date_Credit_Utilization_Ratio$interpretation
## [1] "Weak negative correlation"
##
##
## $Delay_from_due_date_Credit_History_Age
## $Delay_from_due_date_Credit_History_Age$correlation_coefficient
## [1] -0.4911832
##
## $Delay_from_due_date_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Delay_from_due_date_Credit_History_Age$significance
## [1] "Significant"
##
## $Delay_from_due_date_Credit_History_Age$interpretation
## [1] "Moderate negative correlation"
##
##
## $Delay_from_due_date_Total_EMI_per_month
## $Delay_from_due_date_Total_EMI_per_month$correlation_coefficient
## [1] 0.1385045
##
## $Delay_from_due_date_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Delay_from_due_date_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Delay_from_due_date_Total_EMI_per_month$interpretation
## [1] "Weak positive correlation"
##
##
## $Delay_from_due_date_Amount_invested_monthly
## $Delay_from_due_date_Amount_invested_monthly$correlation_coefficient
## [1] -0.1625426
##
## $Delay_from_due_date_Amount_invested_monthly$p_value
```

```
## [1] "<.0001"
##
## $Delay_from_due_date_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Delay_from_due_date_Amount_invested_monthly$interpretation
## [1] "Weak negative correlation"
##
##
## $Delay_from_due_date_Monthly_Balance
## $Delay_from_due_date_Monthly_Balance$correlation_coefficient
## [1] -0.281933
##
## $Delay_from_due_date_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Delay_from_due_date_Monthly_Balance$significance
## [1] "Significant"
##
## $Delay_from_due_date_Monthly_Balance$interpretation
## [1] "Weak negative correlation"
##
##
## $Num_of_Delayed_Payment_Changed_Credit_Limit
## $Num_of_Delayed_Payment_Changed_Credit_Limit$correlation_coefficient
## [1] 0.3261093
##
## $Num_of_Delayed_Payment_Changed_Credit_Limit$p_value
## [1] "<.0001"
##
## $Num_of_Delayed_Payment_Changed_Credit_Limit$significance
## [1] "Significant"
##
## $Num_of_Delayed_Payment_Changed_Credit_Limit$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_of_Delayed_Payment_Num_Credit_Inquiries
## $Num_of_Delayed_Payment_Num_Credit_Inquiries$correlation_coefficient
## [1] 0.5007269
##
## $Num_of_Delayed_Payment_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Num_of_Delayed_Payment_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Num_of_Delayed_Payment_Num_Credit_Inquiries$interpretation
## [1] "Moderate positive correlation"
##
##
```

```
## $Num_of_Delayed_Payment_Outstanding_Debt
## $Num_of_Delayed_Payment_Outstanding_Debt$correlation_coefficient
## [1] 0.5048743
##
## $Num_of_Delayed_Payment_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Num_of_Delayed_Payment_Outstanding_Debt$significance
## [1] "Significant"
##
## $Num_of_Delayed_Payment_Outstanding_Debt$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_of_Delayed_Payment_Credit_Utilization_Ratio
## $Num_of_Delayed_Payment_Credit_Utilization_Ratio$correlation_coefficient
## [1] -0.07375247
##
## $Num_of_Delayed_Payment_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Num_of_Delayed_Payment_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Num_of_Delayed_Payment_Credit_Utilization_Ratio$interpretation
## [1] "Weak negative correlation"
##
##
## $Num_of_Delayed_Payment_Credit_History_Age
## $Num_of_Delayed_Payment_Credit_History_Age$correlation_coefficient
## [1] -0.4821971
##
## $Num_of_Delayed_Payment_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Num_of_Delayed_Payment_Credit_History_Age$significance
## [1] "Significant"
##
## $Num_of_Delayed_Payment_Credit_History_Age$interpretation
## [1] "Moderate negative correlation"
##
##
## $Num_of_Delayed_Payment_Total_EMI_per_month
## $Num_of_Delayed_Payment_Total_EMI_per_month$correlation_coefficient
## [1] 0.1105219
##
## $Num_of_Delayed_Payment_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Num_of_Delayed_Payment_Total_EMI_per_month$significance
## [1] "Significant"
```

```
##
## $Num_of_Delayed_Payment_Total_EMI_per_month$interpretation
## [1] "Weak positive correlation"
##
##
## $Num_of_Delayed_Payment_Amount_invested_monthly
## $Num_of_Delayed_Payment_Amount_invested_monthly$correlation_coefficient
## [1] -0.1826502
##
## $Num_of_Delayed_Payment_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Num_of_Delayed_Payment_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Num_of_Delayed_Payment_Amount_invested_monthly$interpretation
## [1] "Weak negative correlation"
##
##
## $Num_of_Delayed_Payment_Monthly_Balance
## $Num_of_Delayed_Payment_Monthly_Balance$correlation_coefficient
## [1] -0.3052632
##
## $Num_of_Delayed_Payment_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Num_of_Delayed_Payment_Monthly_Balance$significance
## [1] "Significant"
##
## $Num_of_Delayed_Payment_Monthly_Balance$interpretation
## [1] "Moderate negative correlation"
##
##
## $Changed_Credit_Limit_Num_Credit_Inquiries
## $Changed_Credit_Limit_Num_Credit_Inquiries$correlation_coefficient
## [1] 0.3802539
##
## $Changed_Credit_Limit_Num_Credit_Inquiries$p_value
## [1] "<.0001"
##
## $Changed_Credit_Limit_Num_Credit_Inquiries$significance
## [1] "Significant"
##
## $Changed_Credit_Limit_Num_Credit_Inquiries$interpretation
## [1] "Moderate positive correlation"
##
##
## $Changed_Credit_Limit_Outstanding_Debt
## $Changed_Credit_Limit_Outstanding_Debt$correlation_coefficient
## [1] 0.458872
##
```

```
## $Changed_Credit_Limit_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Changed_Credit_Limit_Outstanding_Debt$significance
## [1] "Significant"
##
## $Changed_Credit_Limit_Outstanding_Debt$interpretation
## [1] "Moderate positive correlation"
##
##
## $Changed_Credit_Limit_Credit_Utilization_Ratio
## $Changed_Credit_Limit_Credit_Utilization_Ratio$correlation_coefficient
## [1] -0.0479998
##
## $Changed_Credit_Limit_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Changed_Credit_Limit_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Changed_Credit_Limit_Credit_Utilization_Ratio$interpretation
## [1] "Weak negative correlation"
##
##
## $Changed_Credit_Limit_Credit_History_Age
## $Changed_Credit_Limit_Credit_History_Age$correlation_coefficient
## [1] -0.4251643
##
## $Changed_Credit_Limit_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Changed_Credit_Limit_Credit_History_Age$significance
## [1] "Significant"
##
## $Changed_Credit_Limit_Credit_History_Age$interpretation
## [1] "Moderate negative correlation"
##
##
## $Changed_Credit_Limit_Total_EMI_per_month
## $Changed_Credit_Limit_Total_EMI_per_month$correlation_coefficient
## [1] 0.1019709
##
## $Changed_Credit_Limit_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Changed_Credit_Limit_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Changed_Credit_Limit_Total_EMI_per_month$interpretation
## [1] "Weak positive correlation"
##
```

```
##
## $Changed_Credit_Limit_Amount_invested_monthly
## $Changed_Credit_Limit_Amount_invested_monthly$correlation_coefficient
## [1] -0.1151546
##
## $Changed_Credit_Limit_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Changed_Credit_Limit_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Changed_Credit_Limit_Amount_invested_monthly$interpretation
## [1] "Weak negative correlation"
##
##
## $Changed_Credit_Limit_Monthly_Balance
## $Changed_Credit_Limit_Monthly_Balance$correlation_coefficient
## [1] -0.2053662
##
## $Changed_Credit_Limit_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Changed_Credit_Limit_Monthly_Balance$significance
## [1] "Significant"
##
## $Changed_Credit_Limit_Monthly_Balance$interpretation
## [1] "Weak negative correlation"
##
##
## $Num_Credit_Inquiries_Outstanding_Debt
## $Num_Credit_Inquiries_Outstanding_Debt$correlation_coefficient
## [1] 0.5980139
##
## $Num_Credit_Inquiries_Outstanding_Debt$p_value
## [1] "<.0001"
##
## $Num_Credit_Inquiries_Outstanding_Debt$significance
## [1] "Significant"
##
## $Num_Credit_Inquiries_Outstanding_Debt$interpretation
## [1] "Moderate positive correlation"
##
##
## $Num_Credit_Inquiries_Credit_Utilization_Ratio
## $Num_Credit_Inquiries_Credit_Utilization_Ratio$correlation_coefficient
## [1] -0.07728014
##
## $Num_Credit_Inquiries_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Num_Credit_Inquiries_Credit_Utilization_Ratio$significance
```

```
## [1] "Significant"
##
## $Num_Credit_Inquiries_Credit_Utilization_Ratio$interpretation
## [1] "Weak negative correlation"
##
##
## $Num_Credit_Inquiries_Credit_History_Age
## $Num_Credit_Inquiries_Credit_History_Age$correlation_coefficient
## [1] -0.610655
##
## $Num_Credit_Inquiries_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Num_Credit_Inquiries_Credit_History_Age$significance
## [1] "Significant"
##
## $Num_Credit_Inquiries_Credit_History_Age$interpretation
## [1] "Moderate negative correlation"
##
##
## $Num_Credit_Inquiries_Total_EMI_per_month
## $Num_Credit_Inquiries_Total_EMI_per_month$correlation_coefficient
## [1] 0.1545127
##
## $Num_Credit_Inquiries_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Num_Credit_Inquiries_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Num_Credit_Inquiries_Total_EMI_per_month$interpretation
## [1] "Weak positive correlation"
##
##
## $Num_Credit_Inquiries_Amount_invested_monthly
## $Num_Credit_Inquiries_Amount_invested_monthly$correlation_coefficient
## [1] -0.1764706
##
## $Num_Credit_Inquiries_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Num_Credit_Inquiries_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Num_Credit_Inquiries_Amount_invested_monthly$interpretation
## [1] "Weak negative correlation"
##
##
## $Num_Credit_Inquiries_Monthly_Balance
## $Num_Credit_Inquiries_Monthly_Balance$correlation_coefficient
## [1] -0.3222718
```

```
##
## $Num_Credit_Inquiries_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Num_Credit_Inquiries_Monthly_Balance$significance
## [1] "Significant"
##
## $Num_Credit_Inquiries_Monthly_Balance$interpretation
## [1] "Moderate negative correlation"
##
##
## $Outstanding_Debt_Credit_Utilization_Ratio
## $Outstanding_Debt_Credit_Utilization_Ratio$correlation_coefficient
## [1] -0.07116478
##
## $Outstanding_Debt_Credit_Utilization_Ratio$p_value
## [1] "<.0001"
##
## $Outstanding_Debt_Credit_Utilization_Ratio$significance
## [1] "Significant"
##
## $Outstanding_Debt_Credit_Utilization_Ratio$interpretation
## [1] "Weak negative correlation"
##
##
## $Outstanding_Debt_Credit_History_Age
## $Outstanding_Debt_Credit_History_Age$correlation_coefficient
## [1] -0.6293173
##
## $Outstanding_Debt_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Outstanding_Debt_Credit_History_Age$significance
## [1] "Significant"
##
## $Outstanding_Debt_Credit_History_Age$interpretation
## [1] "Moderate negative correlation"
##
##
## $Outstanding_Debt_Total_EMI_per_month
## $Outstanding_Debt_Total_EMI_per_month$correlation_coefficient
## [1] 0.1655062
##
## $Outstanding_Debt_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Outstanding_Debt_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Outstanding_Debt_Total_EMI_per_month$interpretation
## [1] "Weak positive correlation"
```

```
##
##
## $Outstanding_Debt_Amount_invested_monthly
## $Outstanding_Debt_Amount_invested_monthly$correlation_coefficient
## [1] -0.173813
##
## $Outstanding_Debt_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Outstanding_Debt_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Outstanding_Debt_Amount_invested_monthly$interpretation
## [1] "Weak negative correlation"
##
##
## $Outstanding_Debt_Monthly_Balance
## $Outstanding_Debt_Monthly_Balance$correlation_coefficient
## [1] -0.317174
##
## $Outstanding_Debt_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Outstanding_Debt_Monthly_Balance$significance
## [1] "Significant"
##
## $Outstanding_Debt_Monthly_Balance$interpretation
## [1] "Moderate negative correlation"
##
##
## $Credit_Utilization_Ratio_Credit_History_Age
## $Credit_Utilization_Ratio_Credit_History_Age$correlation_coefficient
## [1] 0.07323533
##
## $Credit_Utilization_Ratio_Credit_History_Age$p_value
## [1] "<.0001"
##
## $Credit_Utilization_Ratio_Credit_History_Age$significance
## [1] "Significant"
##
## $Credit_Utilization_Ratio_Credit_History_Age$interpretation
## [1] "Weak positive correlation"
##
##
## $Credit_Utilization_Ratio_Total_EMI_per_month
## $Credit_Utilization_Ratio_Total_EMI_per_month$correlation_coefficient
## [1] 0.01326283
##
## $Credit_Utilization_Ratio_Total_EMI_per_month$p_value
## [1] "<.0001"
##
```

```
## $Credit_Utilization_Ratio_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Credit_Utilization_Ratio_Total_EMI_per_month$interpretation
## [1] "Weak positive correlation"
##
##
## $Credit_Utilization_Ratio_Amount_invested_monthly
## $Credit_Utilization_Ratio_Amount_invested_monthly$correlation_coefficient
## [1] -0.005790128
##
## $Credit_Utilization_Ratio_Amount_invested_monthly$p_value
## [1] "0.0671"
##
## $Credit_Utilization_Ratio_Amount_invested_monthly$significance
## [1] "Not Significant"
##
## $Credit_Utilization_Ratio_Amount_invested_monthly$interpretation
## [1] "Weak negative correlation"
##
##
## $Credit_Utilization_Ratio_Monthly_Balance
## $Credit_Utilization_Ratio_Monthly_Balance$correlation_coefficient
## [1] 0.2451642
##
## $Credit_Utilization_Ratio_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Credit_Utilization_Ratio_Monthly_Balance$significance
## [1] "Significant"
##
## $Credit_Utilization_Ratio_Monthly_Balance$interpretation
## [1] "Weak positive correlation"
##
##
## $Credit_History_Age_Total_EMI_per_month
## $Credit_History_Age_Total_EMI_per_month$correlation_coefficient
## [1] -0.1718104
##
## $Credit_History_Age_Total_EMI_per_month$p_value
## [1] "<.0001"
##
## $Credit_History_Age_Total_EMI_per_month$significance
## [1] "Significant"
##
## $Credit_History_Age_Total_EMI_per_month$interpretation
## [1] "Weak negative correlation"
##
##
## $Credit_History_Age_Amount_invested_monthly
## $Credit_History_Age_Amount_invested_monthly$correlation_coefficient
```

```
## [1] 0.1757446
##
## $Credit_History_Age_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Credit_History_Age_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Credit_History_Age_Amount_invested_monthly$interpretation
## [1] "Weak positive correlation"
##
##
## $Credit_History_Age_Monthly_Balance
## $Credit_History_Age_Monthly_Balance$correlation_coefficient
## [1] 0.3257627
##
## $Credit_History_Age_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Credit_History_Age_Monthly_Balance$significance
## [1] "Significant"
##
## $Credit_History_Age_Monthly_Balance$interpretation
## [1] "Moderate positive correlation"
##
##
## $Total_EMI_per_month_Amount_invested_monthly
## $Total_EMI_per_month_Amount_invested_monthly$correlation_coefficient
## [1] 0.2419131
##
## $Total_EMI_per_month_Amount_invested_monthly$p_value
## [1] "<.0001"
##
## $Total_EMI_per_month_Amount_invested_monthly$significance
## [1] "Significant"
##
## $Total_EMI_per_month_Amount_invested_monthly$interpretation
## [1] "Weak positive correlation"
##
##
## $Total_EMI_per_month_Monthly_Balance
## $Total_EMI_per_month_Monthly_Balance$correlation_coefficient
## [1] 0.05122924
##
## $Total_EMI_per_month_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Total_EMI_per_month_Monthly_Balance$significance
## [1] "Significant"
##
## $Total_EMI_per_month_Monthly_Balance$interpretation
```

```
## [1] "Weak positive correlation"
##
##
## $Amount_invested_monthly_Monthly_Balance
## $Amount_invested_monthly_Monthly_Balance$correlation_coefficient
## [1] 0.04678245
##
## $Amount_invested_monthly_Monthly_Balance$p_value
## [1] "<.0001"
##
## $Amount_invested_monthly_Monthly_Balance$significance
## [1] "Significant"
##
## $Amount_invested_monthly_Monthly_Balance$interpretation
## [1] "Weak positive correlation"
```

Categorical and Categorical Correlations - Phi Coefficient/Cramer’s V

```
# Load necessary libraries
if (!require("vcd")) install.packages("vcd")  # vcd package for Cramer's V
```

```
## Loading required package: vcd
```

```
## Warning: package 'vcd' was built under R version 4.4.2
```

```
## Loading required package: grid
```

```
if (!require("stats")) install.packages("stats")  # For Chi-squared test
if (!require("lsr")) install.packages("lsr")  # For Cramer's V calculation
library(vcd)
library(lsr)

# Define function to calculate p-value, Cramer's V, and significance for categorical variable
s
calc_chisq_cramers_v <- function(data, categorical_vars) {

  # Initialize the progress bar
  total_combinations <- length(categorical_vars) * (length(categorical_vars) - 1) / 2
  pb <- progress_bar$new(
    total = total_combinations,
    format = "Processing [:bar] :percent ETA: :eta",
    clear = FALSE, width = 60
  )

  # Store results in a list
  results3 <- list()

  # Perform Chi-Square Test and Cramer's V for each combination of categorical variables
  for (i in 1:(length(categorical_vars) - 1)) {
    for (j in (i + 1):length(categorical_vars)) {

      cat_var1 <- categorical_vars[i]
      cat_var2 <- categorical_vars[j]

      # Create contingency table
      contingency_table <- table(data[[cat_var1]], data[[cat_var2]])

      # Perform Chi-Square Test
      chi_sq_test <- chisq.test(contingency_table)

      # Extract p-value
      p_value <- chi_sq_test$p.value

      # Format p-value: Display "<.0001" if p-value is very small
      formatted_p_value <- if (p_value < 0.0001) "<.0001" else format.pval(p_value, digits =
4)

      # Calculate Cramer's V
      cramers_v <- assocstats(contingency_table)$cramer
      association_strength <- if (cramers_v < 0.1) {
        "Weak association"
      } else if (cramers_v >= 0.1 && cramers_v < 0.3) {
        "Moderate association"
      } else if (cramers_v >= 0.3 && cramers_v < 0.5) {
        "Strong association"
      } else {
        "Very strong association"
      }
```

```
      # Store results in the list without chi-square statistic
      results3[[paste(cat_var1, cat_var2, sep = "_")]] <- list(
        p_value = formatted_p_value,
        significance = ifelse(p_value < 0.05, "Significant", "Not Significant"),
        cramers_v = cramers_v,
        association_strength = association_strength
      )

      # Update progress bar
      pb$tick()
    }
  }

  return(results3)
}

# Example usage:
# Assuming 'credit_score' is your dataframe with categorical variables
categorical_vars <- c("Credit_Mix", "Payment_of_Min_Amount", "Payment_Behaviour", "Occupatio
n",
                      "Credit_Score", "Last_Loan_1", "Last_Loan_2", "Last_Loan_3", "Last_Loan
_4",
                      "Last_Loan_5", "Last_Loan_6", "Last_Loan_7", "Last_Loan_8", "Last_Loan_
9")

# Call the function
results3 <- calc_chisq_cramers_v(credit_score, categorical_vars)
```

```
## Warning in chisq.test(contingency_table): Chi-squared approximation may be
## incorrect
```

```
# Print the results (optional)
print(results3)
```

```
## $Credit_Mix_Payment_of_Min_Amount
## $Credit_Mix_Payment_of_Min_Amount$p_value
## [1] "<.0001"
##
## $Credit_Mix_Payment_of_Min_Amount$significance
## [1] "Significant"
##
## $Credit_Mix_Payment_of_Min_Amount$cramers_v
## [1] 0.8194101
##
## $Credit_Mix_Payment_of_Min_Amount$association_strength
## [1] "Very strong association"
##
##
## $Credit_Mix_Payment_Behaviour
## $Credit_Mix_Payment_Behaviour$p_value
## [1] "<.0001"
##
## $Credit_Mix_Payment_Behaviour$significance
## [1] "Significant"
##
## $Credit_Mix_Payment_Behaviour$cramers_v
## [1] 0.1358269
##
## $Credit_Mix_Payment_Behaviour$association_strength
## [1] "Moderate association"
##
##
## $Credit_Mix_Occupation
## $Credit_Mix_Occupation$p_value
## [1] "<.0001"
##
## $Credit_Mix_Occupation$significance
## [1] "Significant"
##
## $Credit_Mix_Occupation$cramers_v
## [1] 0.03347124
##
## $Credit_Mix_Occupation$association_strength
## [1] "Weak association"
##
##
## $Credit_Mix_Credit_Score
## $Credit_Mix_Credit_Score$p_value
## [1] "<.0001"
##
## $Credit_Mix_Credit_Score$significance
## [1] "Significant"
##
## $Credit_Mix_Credit_Score$cramers_v
## [1] 0.4499412
```

```
##
## $Credit_Mix_Credit_Score$association_strength
## [1] "Strong association"
##
##
## $Credit_Mix_Last_Loan_1
## $Credit_Mix_Last_Loan_1$p_value
## [1] "<.0001"
##
## $Credit_Mix_Last_Loan_1$significance
## [1] "Significant"
##
## $Credit_Mix_Last_Loan_1$cramers_v
## [1] 0.1705401
##
## $Credit_Mix_Last_Loan_1$association_strength
## [1] "Moderate association"
##
##
## $Credit_Mix_Last_Loan_2
## $Credit_Mix_Last_Loan_2$p_value
## [1] "<.0001"
##
## $Credit_Mix_Last_Loan_2$significance
## [1] "Significant"
##
## $Credit_Mix_Last_Loan_2$cramers_v
## [1] 0.2494943
##
## $Credit_Mix_Last_Loan_2$association_strength
## [1] "Moderate association"
##
##
## $Credit_Mix_Last_Loan_3
## $Credit_Mix_Last_Loan_3$p_value
## [1] "<.0001"
##
## $Credit_Mix_Last_Loan_3$significance
## [1] "Significant"
##
## $Credit_Mix_Last_Loan_3$cramers_v
## [1] 0.2944384
##
## $Credit_Mix_Last_Loan_3$association_strength
## [1] "Moderate association"
##
##
## $Credit_Mix_Last_Loan_4
## $Credit_Mix_Last_Loan_4$p_value
## [1] "<.0001"
##
```

```
## $Credit_Mix_Last_Loan_4$significance
## [1] "Significant"
##
## $Credit_Mix_Last_Loan_4$cramers_v
## [1] 0.3780629
##
## $Credit_Mix_Last_Loan_4$association_strength
## [1] "Strong association"
##
##
## $Credit_Mix_Last_Loan_5
## $Credit_Mix_Last_Loan_5$p_value
## [1] "<.0001"
##
## $Credit_Mix_Last_Loan_5$significance
## [1] "Significant"
##
## $Credit_Mix_Last_Loan_5$cramers_v
## [1] 0.5016568
##
## $Credit_Mix_Last_Loan_5$association_strength
## [1] "Very strong association"
##
##
## $Credit_Mix_Last_Loan_6
## $Credit_Mix_Last_Loan_6$p_value
## [1] "<.0001"
##
## $Credit_Mix_Last_Loan_6$significance
## [1] "Significant"
##
## $Credit_Mix_Last_Loan_6$cramers_v
## [1] 0.4411703
##
## $Credit_Mix_Last_Loan_6$association_strength
## [1] "Strong association"
##
##
## $Credit_Mix_Last_Loan_7
## $Credit_Mix_Last_Loan_7$p_value
## [1] "<.0001"
##
## $Credit_Mix_Last_Loan_7$significance
## [1] "Significant"
##
## $Credit_Mix_Last_Loan_7$cramers_v
## [1] 0.3827456
##
## $Credit_Mix_Last_Loan_7$association_strength
## [1] "Strong association"
##
```

```
##
## $Credit_Mix_Last_Loan_8
## $Credit_Mix_Last_Loan_8$p_value
## [1] "<.0001"
##
## $Credit_Mix_Last_Loan_8$significance
## [1] "Significant"
##
## $Credit_Mix_Last_Loan_8$cramers_v
## [1] 0.3525246
##
## $Credit_Mix_Last_Loan_8$association_strength
## [1] "Strong association"
##
##
## $Credit_Mix_Last_Loan_9
## $Credit_Mix_Last_Loan_9$p_value
## [1] "<.0001"
##
## $Credit_Mix_Last_Loan_9$significance
## [1] "Significant"
##
## $Credit_Mix_Last_Loan_9$cramers_v
## [1] 0.2536091
##
## $Credit_Mix_Last_Loan_9$association_strength
## [1] "Moderate association"
##
##
## $Payment_of_Min_Amount_Payment_Behaviour
## $Payment_of_Min_Amount_Payment_Behaviour$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Payment_Behaviour$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Payment_Behaviour$cramers_v
## [1] 0.1635161
##
## $Payment_of_Min_Amount_Payment_Behaviour$association_strength
## [1] "Moderate association"
##
##
## $Payment_of_Min_Amount_Occupation
## $Payment_of_Min_Amount_Occupation$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Occupation$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Occupation$cramers_v
```

```
## [1] 0.02870354
##
## $Payment_of_Min_Amount_Occupation$association_strength
## [1] "Weak association"
##
##
## $Payment_of_Min_Amount_Credit_Score
## $Payment_of_Min_Amount_Credit_Score$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Credit_Score$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Credit_Score$cramers_v
## [1] 0.4735023
##
## $Payment_of_Min_Amount_Credit_Score$association_strength
## [1] "Strong association"
##
##
## $Payment_of_Min_Amount_Last_Loan_1
## $Payment_of_Min_Amount_Last_Loan_1$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Last_Loan_1$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Last_Loan_1$cramers_v
## [1] 0.2318237
##
## $Payment_of_Min_Amount_Last_Loan_1$association_strength
## [1] "Moderate association"
##
##
## $Payment_of_Min_Amount_Last_Loan_2
## $Payment_of_Min_Amount_Last_Loan_2$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Last_Loan_2$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Last_Loan_2$cramers_v
## [1] 0.3445669
##
## $Payment_of_Min_Amount_Last_Loan_2$association_strength
## [1] "Strong association"
##
##
## $Payment_of_Min_Amount_Last_Loan_3
## $Payment_of_Min_Amount_Last_Loan_3$p_value
## [1] "<.0001"
```

```
##
## $Payment_of_Min_Amount_Last_Loan_3$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Last_Loan_3$cramers_v
## [1] 0.3625512
##
## $Payment_of_Min_Amount_Last_Loan_3$association_strength
## [1] "Strong association"
##
##
## $Payment_of_Min_Amount_Last_Loan_4
## $Payment_of_Min_Amount_Last_Loan_4$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Last_Loan_4$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Last_Loan_4$cramers_v
## [1] 0.4356745
##
## $Payment_of_Min_Amount_Last_Loan_4$association_strength
## [1] "Strong association"
##
##
## $Payment_of_Min_Amount_Last_Loan_5
## $Payment_of_Min_Amount_Last_Loan_5$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Last_Loan_5$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Last_Loan_5$cramers_v
## [1] 0.5478854
##
## $Payment_of_Min_Amount_Last_Loan_5$association_strength
## [1] "Very strong association"
##
##
## $Payment_of_Min_Amount_Last_Loan_6
## $Payment_of_Min_Amount_Last_Loan_6$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Last_Loan_6$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Last_Loan_6$cramers_v
## [1] 0.4517481
##
## $Payment_of_Min_Amount_Last_Loan_6$association_strength
## [1] "Strong association"
```

```
##
##
## $Payment_of_Min_Amount_Last_Loan_7
## $Payment_of_Min_Amount_Last_Loan_7$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Last_Loan_7$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Last_Loan_7$cramers_v
## [1] 0.3453267
##
## $Payment_of_Min_Amount_Last_Loan_7$association_strength
## [1] "Strong association"
##
##
## $Payment_of_Min_Amount_Last_Loan_8
## $Payment_of_Min_Amount_Last_Loan_8$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Last_Loan_8$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Last_Loan_8$cramers_v
## [1] 0.2299925
##
## $Payment_of_Min_Amount_Last_Loan_8$association_strength
## [1] "Moderate association"
##
##
## $Payment_of_Min_Amount_Last_Loan_9
## $Payment_of_Min_Amount_Last_Loan_9$p_value
## [1] "<.0001"
##
## $Payment_of_Min_Amount_Last_Loan_9$significance
## [1] "Significant"
##
## $Payment_of_Min_Amount_Last_Loan_9$cramers_v
## [1] 0.1654585
##
## $Payment_of_Min_Amount_Last_Loan_9$association_strength
## [1] "Moderate association"
##
##
## $Payment_Behaviour_Occupation
## $Payment_Behaviour_Occupation$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Occupation$significance
## [1] "Significant"
##
```

```
## $Payment_Behaviour_Occupation$cramers_v
## [1] 0.02595969
##
## $Payment_Behaviour_Occupation$association_strength
## [1] "Weak association"
##
##
## $Payment_Behaviour_Credit_Score
## $Payment_Behaviour_Credit_Score$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Credit_Score$significance
## [1] "Significant"
##
## $Payment_Behaviour_Credit_Score$cramers_v
## [1] 0.09071711
##
## $Payment_Behaviour_Credit_Score$association_strength
## [1] "Weak association"
##
##
## $Payment_Behaviour_Last_Loan_1
## $Payment_Behaviour_Last_Loan_1$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Last_Loan_1$significance
## [1] "Significant"
##
## $Payment_Behaviour_Last_Loan_1$cramers_v
## [1] 0.04031992
##
## $Payment_Behaviour_Last_Loan_1$association_strength
## [1] "Weak association"
##
##
## $Payment_Behaviour_Last_Loan_2
## $Payment_Behaviour_Last_Loan_2$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Last_Loan_2$significance
## [1] "Significant"
##
## $Payment_Behaviour_Last_Loan_2$cramers_v
## [1] 0.04936506
##
## $Payment_Behaviour_Last_Loan_2$association_strength
## [1] "Weak association"
##
##
## $Payment_Behaviour_Last_Loan_3
## $Payment_Behaviour_Last_Loan_3$p_value
```

```
## [1] "<.0001"
##
## $Payment_Behaviour_Last_Loan_3$significance
## [1] "Significant"
##
## $Payment_Behaviour_Last_Loan_3$cramers_v
## [1] 0.04747032
##
## $Payment_Behaviour_Last_Loan_3$association_strength
## [1] "Weak association"
##
##
## $Payment_Behaviour_Last_Loan_4
## $Payment_Behaviour_Last_Loan_4$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Last_Loan_4$significance
## [1] "Significant"
##
## $Payment_Behaviour_Last_Loan_4$cramers_v
## [1] 0.05909902
##
## $Payment_Behaviour_Last_Loan_4$association_strength
## [1] "Weak association"
##
##
## $Payment_Behaviour_Last_Loan_5
## $Payment_Behaviour_Last_Loan_5$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Last_Loan_5$significance
## [1] "Significant"
##
## $Payment_Behaviour_Last_Loan_5$cramers_v
## [1] 0.0706799
##
## $Payment_Behaviour_Last_Loan_5$association_strength
## [1] "Weak association"
##
##
## $Payment_Behaviour_Last_Loan_6
## $Payment_Behaviour_Last_Loan_6$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Last_Loan_6$significance
## [1] "Significant"
##
## $Payment_Behaviour_Last_Loan_6$cramers_v
## [1] 0.06080149
##
## $Payment_Behaviour_Last_Loan_6$association_strength
```

```
## [1] "Weak association"
##
##
## $Payment_Behaviour_Last_Loan_7
## $Payment_Behaviour_Last_Loan_7$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Last_Loan_7$significance
## [1] "Significant"
##
## $Payment_Behaviour_Last_Loan_7$cramers_v
## [1] 0.05280862
##
## $Payment_Behaviour_Last_Loan_7$association_strength
## [1] "Weak association"
##
##
## $Payment_Behaviour_Last_Loan_8
## $Payment_Behaviour_Last_Loan_8$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Last_Loan_8$significance
## [1] "Significant"
##
## $Payment_Behaviour_Last_Loan_8$cramers_v
## [1] 0.04182599
##
## $Payment_Behaviour_Last_Loan_8$association_strength
## [1] "Weak association"
##
##
## $Payment_Behaviour_Last_Loan_9
## $Payment_Behaviour_Last_Loan_9$p_value
## [1] "<.0001"
##
## $Payment_Behaviour_Last_Loan_9$significance
## [1] "Significant"
##
## $Payment_Behaviour_Last_Loan_9$cramers_v
## [1] 0.03421204
##
## $Payment_Behaviour_Last_Loan_9$association_strength
## [1] "Weak association"
##
##
## $Occupation_Credit_Score
## $Occupation_Credit_Score$p_value
## [1] "<.0001"
##
## $Occupation_Credit_Score$significance
## [1] "Significant"
```

```
##
## $Occupation_Credit_Score$cramers_v
## [1] 0.0300571
##
## $Occupation_Credit_Score$association_strength
## [1] "Weak association"
##
##
## $Occupation_Last_Loan_1
## $Occupation_Last_Loan_1$p_value
## [1] "<.0001"
##
## $Occupation_Last_Loan_1$significance
## [1] "Significant"
##
## $Occupation_Last_Loan_1$cramers_v
## [1] 0.03119128
##
## $Occupation_Last_Loan_1$association_strength
## [1] "Weak association"
##
##
## $Occupation_Last_Loan_2
## $Occupation_Last_Loan_2$p_value
## [1] "<.0001"
##
## $Occupation_Last_Loan_2$significance
## [1] "Significant"
##
## $Occupation_Last_Loan_2$cramers_v
## [1] 0.03375707
##
## $Occupation_Last_Loan_2$association_strength
## [1] "Weak association"
##
##
## $Occupation_Last_Loan_3
## $Occupation_Last_Loan_3$p_value
## [1] "<.0001"
##
## $Occupation_Last_Loan_3$significance
## [1] "Significant"
##
## $Occupation_Last_Loan_3$cramers_v
## [1] 0.03860316
##
## $Occupation_Last_Loan_3$association_strength
## [1] "Weak association"
##
##
## $Occupation_Last_Loan_4
```

```
## $Occupation_Last_Loan_4$p_value
## [1] "<.0001"
##
## $Occupation_Last_Loan_4$significance
## [1] "Significant"
##
## $Occupation_Last_Loan_4$cramers_v
## [1] 0.03372037
##
## $Occupation_Last_Loan_4$association_strength
## [1] "Weak association"
##
##
## $Occupation_Last_Loan_5
## $Occupation_Last_Loan_5$p_value
## [1] "<.0001"
##
## $Occupation_Last_Loan_5$significance
## [1] "Significant"
##
## $Occupation_Last_Loan_5$cramers_v
## [1] 0.03159487
##
## $Occupation_Last_Loan_5$association_strength
## [1] "Weak association"
##
##
## $Occupation_Last_Loan_6
## $Occupation_Last_Loan_6$p_value
## [1] "<.0001"
##
## $Occupation_Last_Loan_6$significance
## [1] "Significant"
##
## $Occupation_Last_Loan_6$cramers_v
## [1] 0.03027574
##
## $Occupation_Last_Loan_6$association_strength
## [1] "Weak association"
##
##
## $Occupation_Last_Loan_7
## $Occupation_Last_Loan_7$p_value
## [1] "<.0001"
##
## $Occupation_Last_Loan_7$significance
## [1] "Significant"
##
## $Occupation_Last_Loan_7$cramers_v
## [1] 0.03040452
##
```

```
## $Occupation_Last_Loan_7$association_strength
## [1] "Weak association"
##
##
## $Occupation_Last_Loan_8
## $Occupation_Last_Loan_8$p_value
## [1] "<.0001"
##
## $Occupation_Last_Loan_8$significance
## [1] "Significant"
##
## $Occupation_Last_Loan_8$cramers_v
## [1] 0.0336853
##
## $Occupation_Last_Loan_8$association_strength
## [1] "Weak association"
##
##
## $Occupation_Last_Loan_9
## $Occupation_Last_Loan_9$p_value
## [1] "<.0001"
##
## $Occupation_Last_Loan_9$significance
## [1] "Significant"
##
## $Occupation_Last_Loan_9$cramers_v
## [1] 0.03254446
##
## $Occupation_Last_Loan_9$association_strength
## [1] "Weak association"
##
##
## $Credit_Score_Last_Loan_1
## $Credit_Score_Last_Loan_1$p_value
## [1] "<.0001"
##
## $Credit_Score_Last_Loan_1$significance
## [1] "Significant"
##
## $Credit_Score_Last_Loan_1$cramers_v
## [1] 0.1131661
##
## $Credit_Score_Last_Loan_1$association_strength
## [1] "Moderate association"
##
##
## $Credit_Score_Last_Loan_2
## $Credit_Score_Last_Loan_2$p_value
## [1] "<.0001"
##
## $Credit_Score_Last_Loan_2$significance
```

```
## [1] "Significant"
##
## $Credit_Score_Last_Loan_2$cramers_v
## [1] 0.1666405
##
## $Credit_Score_Last_Loan_2$association_strength
## [1] "Moderate association"
##
##
## $Credit_Score_Last_Loan_3
## $Credit_Score_Last_Loan_3$p_value
## [1] "<.0001"
##
## $Credit_Score_Last_Loan_3$significance
## [1] "Significant"
##
## $Credit_Score_Last_Loan_3$cramers_v
## [1] 0.1740602
##
## $Credit_Score_Last_Loan_3$association_strength
## [1] "Moderate association"
##
##
## $Credit_Score_Last_Loan_4
## $Credit_Score_Last_Loan_4$p_value
## [1] "<.0001"
##
## $Credit_Score_Last_Loan_4$significance
## [1] "Significant"
##
## $Credit_Score_Last_Loan_4$cramers_v
## [1] 0.2102532
##
## $Credit_Score_Last_Loan_4$association_strength
## [1] "Moderate association"
##
##
## $Credit_Score_Last_Loan_5
## $Credit_Score_Last_Loan_5$p_value
## [1] "<.0001"
##
## $Credit_Score_Last_Loan_5$significance
## [1] "Significant"
##
## $Credit_Score_Last_Loan_5$cramers_v
## [1] 0.2712832
##
## $Credit_Score_Last_Loan_5$association_strength
## [1] "Moderate association"
##
##
```

```
## $Credit_Score_Last_Loan_6
## $Credit_Score_Last_Loan_6$p_value
## [1] "<.0001"
##
## $Credit_Score_Last_Loan_6$significance
## [1] "Significant"
##
## $Credit_Score_Last_Loan_6$cramers_v
## [1] 0.2250358
##
## $Credit_Score_Last_Loan_6$association_strength
## [1] "Moderate association"
##
##
## $Credit_Score_Last_Loan_7
## $Credit_Score_Last_Loan_7$p_value
## [1] "<.0001"
##
## $Credit_Score_Last_Loan_7$significance
## [1] "Significant"
##
## $Credit_Score_Last_Loan_7$cramers_v
## [1] 0.1767788
##
## $Credit_Score_Last_Loan_7$association_strength
## [1] "Moderate association"
##
##
## $Credit_Score_Last_Loan_8
## $Credit_Score_Last_Loan_8$p_value
## [1] "<.0001"
##
## $Credit_Score_Last_Loan_8$significance
## [1] "Significant"
##
## $Credit_Score_Last_Loan_8$cramers_v
## [1] 0.1222828
##
## $Credit_Score_Last_Loan_8$association_strength
## [1] "Moderate association"
##
##
## $Credit_Score_Last_Loan_9
## $Credit_Score_Last_Loan_9$p_value
## [1] "<.0001"
##
## $Credit_Score_Last_Loan_9$significance
## [1] "Significant"
##
## $Credit_Score_Last_Loan_9$cramers_v
## [1] 0.08719994
```

```
##
## $Credit_Score_Last_Loan_9$association_strength
## [1] "Weak association"
##
##
## $Last_Loan_1_Last_Loan_2
## $Last_Loan_1_Last_Loan_2$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Last_Loan_2$significance
## [1] "Significant"
##
## $Last_Loan_1_Last_Loan_2$cramers_v
## [1] 0.2230195
##
## $Last_Loan_1_Last_Loan_2$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_1_Last_Loan_3
## $Last_Loan_1_Last_Loan_3$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Last_Loan_3$significance
## [1] "Significant"
##
## $Last_Loan_1_Last_Loan_3$cramers_v
## [1] 0.1536396
##
## $Last_Loan_1_Last_Loan_3$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_1_Last_Loan_4
## $Last_Loan_1_Last_Loan_4$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Last_Loan_4$significance
## [1] "Significant"
##
## $Last_Loan_1_Last_Loan_4$cramers_v
## [1] 0.1141948
##
## $Last_Loan_1_Last_Loan_4$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_1_Last_Loan_5
## $Last_Loan_1_Last_Loan_5$p_value
## [1] "<.0001"
##
```

```
## $Last_Loan_1_Last_Loan_5$significance
## [1] "Significant"
##
## $Last_Loan_1_Last_Loan_5$cramers_v
## [1] 0.08278096
##
## $Last_Loan_1_Last_Loan_5$association_strength
## [1] "Weak association"
##
##
## $Last_Loan_1_Last_Loan_6
## $Last_Loan_1_Last_Loan_6$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Last_Loan_6$significance
## [1] "Significant"
##
## $Last_Loan_1_Last_Loan_6$cramers_v
## [1] 0.07072193
##
## $Last_Loan_1_Last_Loan_6$association_strength
## [1] "Weak association"
##
##
## $Last_Loan_1_Last_Loan_7
## $Last_Loan_1_Last_Loan_7$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Last_Loan_7$significance
## [1] "Significant"
##
## $Last_Loan_1_Last_Loan_7$cramers_v
## [1] 0.05757405
##
## $Last_Loan_1_Last_Loan_7$association_strength
## [1] "Weak association"
##
##
## $Last_Loan_1_Last_Loan_8
## $Last_Loan_1_Last_Loan_8$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Last_Loan_8$significance
## [1] "Significant"
##
## $Last_Loan_1_Last_Loan_8$cramers_v
## [1] 0.04386797
##
## $Last_Loan_1_Last_Loan_8$association_strength
## [1] "Weak association"
##
```

```
##
## $Last_Loan_1_Last_Loan_9
## $Last_Loan_1_Last_Loan_9$p_value
## [1] "<.0001"
##
## $Last_Loan_1_Last_Loan_9$significance
## [1] "Significant"
##
## $Last_Loan_1_Last_Loan_9$cramers_v
## [1] 0.03608927
##
## $Last_Loan_1_Last_Loan_9$association_strength
## [1] "Weak association"
##
##
## $Last_Loan_2_Last_Loan_3
## $Last_Loan_2_Last_Loan_3$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Last_Loan_3$significance
## [1] "Significant"
##
## $Last_Loan_2_Last_Loan_3$cramers_v
## [1] 0.2301789
##
## $Last_Loan_2_Last_Loan_3$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_2_Last_Loan_4
## $Last_Loan_2_Last_Loan_4$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Last_Loan_4$significance
## [1] "Significant"
##
## $Last_Loan_2_Last_Loan_4$cramers_v
## [1] 0.1679801
##
## $Last_Loan_2_Last_Loan_4$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_2_Last_Loan_5
## $Last_Loan_2_Last_Loan_5$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Last_Loan_5$significance
## [1] "Significant"
##
## $Last_Loan_2_Last_Loan_5$cramers_v
```

```
## [1] 0.1223087
##
## $Last_Loan_2_Last_Loan_5$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_2_Last_Loan_6
## $Last_Loan_2_Last_Loan_6$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Last_Loan_6$significance
## [1] "Significant"
##
## $Last_Loan_2_Last_Loan_6$cramers_v
## [1] 0.1019006
##
## $Last_Loan_2_Last_Loan_6$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_2_Last_Loan_7
## $Last_Loan_2_Last_Loan_7$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Last_Loan_7$significance
## [1] "Significant"
##
## $Last_Loan_2_Last_Loan_7$cramers_v
## [1] 0.08017032
##
## $Last_Loan_2_Last_Loan_7$association_strength
## [1] "Weak association"
##
##
## $Last_Loan_2_Last_Loan_8
## $Last_Loan_2_Last_Loan_8$p_value
## [1] "<.0001"
##
## $Last_Loan_2_Last_Loan_8$significance
## [1] "Significant"
##
## $Last_Loan_2_Last_Loan_8$cramers_v
## [1] 0.05928495
##
## $Last_Loan_2_Last_Loan_8$association_strength
## [1] "Weak association"
##
##
## $Last_Loan_2_Last_Loan_9
## $Last_Loan_2_Last_Loan_9$p_value
## [1] "<.0001"
```

```
##
## $Last_Loan_2_Last_Loan_9$significance
## [1] "Significant"
##
## $Last_Loan_2_Last_Loan_9$cramers_v
## [1] 0.0437891
##
## $Last_Loan_2_Last_Loan_9$association_strength
## [1] "Weak association"
##
##
## $Last_Loan_3_Last_Loan_4
## $Last_Loan_3_Last_Loan_4$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Last_Loan_4$significance
## [1] "Significant"
##
## $Last_Loan_3_Last_Loan_4$cramers_v
## [1] 0.2442995
##
## $Last_Loan_3_Last_Loan_4$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_3_Last_Loan_5
## $Last_Loan_3_Last_Loan_5$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Last_Loan_5$significance
## [1] "Significant"
##
## $Last_Loan_3_Last_Loan_5$cramers_v
## [1] 0.1766684
##
## $Last_Loan_3_Last_Loan_5$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_3_Last_Loan_6
## $Last_Loan_3_Last_Loan_6$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Last_Loan_6$significance
## [1] "Significant"
##
## $Last_Loan_3_Last_Loan_6$cramers_v
## [1] 0.1465826
##
## $Last_Loan_3_Last_Loan_6$association_strength
## [1] "Moderate association"
```

```
##
##
## $Last_Loan_3_Last_Loan_7
## $Last_Loan_3_Last_Loan_7$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Last_Loan_7$significance
## [1] "Significant"
##
## $Last_Loan_3_Last_Loan_7$cramers_v
## [1] 0.1134939
##
## $Last_Loan_3_Last_Loan_7$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_3_Last_Loan_8
## $Last_Loan_3_Last_Loan_8$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Last_Loan_8$significance
## [1] "Significant"
##
## $Last_Loan_3_Last_Loan_8$cramers_v
## [1] 0.07815494
##
## $Last_Loan_3_Last_Loan_8$association_strength
## [1] "Weak association"
##
##
## $Last_Loan_3_Last_Loan_9
## $Last_Loan_3_Last_Loan_9$p_value
## [1] "<.0001"
##
## $Last_Loan_3_Last_Loan_9$significance
## [1] "Significant"
##
## $Last_Loan_3_Last_Loan_9$cramers_v
## [1] 0.06408399
##
## $Last_Loan_3_Last_Loan_9$association_strength
## [1] "Weak association"
##
##
## $Last_Loan_4_Last_Loan_5
## $Last_Loan_4_Last_Loan_5$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Last_Loan_5$significance
## [1] "Significant"
##
```

```
## $Last_Loan_4_Last_Loan_5$cramers_v
## [1] 0.2418975
##
## $Last_Loan_4_Last_Loan_5$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_4_Last_Loan_6
## $Last_Loan_4_Last_Loan_6$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Last_Loan_6$significance
## [1] "Significant"
##
## $Last_Loan_4_Last_Loan_6$cramers_v
## [1] 0.200791
##
## $Last_Loan_4_Last_Loan_6$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_4_Last_Loan_7
## $Last_Loan_4_Last_Loan_7$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Last_Loan_7$significance
## [1] "Significant"
##
## $Last_Loan_4_Last_Loan_7$cramers_v
## [1] 0.1551323
##
## $Last_Loan_4_Last_Loan_7$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_4_Last_Loan_8
## $Last_Loan_4_Last_Loan_8$p_value
## [1] "<.0001"
##
## $Last_Loan_4_Last_Loan_8$significance
## [1] "Significant"
##
## $Last_Loan_4_Last_Loan_8$cramers_v
## [1] 0.1081874
##
## $Last_Loan_4_Last_Loan_8$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_4_Last_Loan_9
## $Last_Loan_4_Last_Loan_9$p_value
```

```
## [1] "<.0001"
##
## $Last_Loan_4_Last_Loan_9$significance
## [1] "Significant"
##
## $Last_Loan_4_Last_Loan_9$cramers_v
## [1] 0.08098319
##
## $Last_Loan_4_Last_Loan_9$association_strength
## [1] "Weak association"
##
##
## $Last_Loan_5_Last_Loan_6
## $Last_Loan_5_Last_Loan_6$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Last_Loan_6$significance
## [1] "Significant"
##
## $Last_Loan_5_Last_Loan_6$cramers_v
## [1] 0.2794955
##
## $Last_Loan_5_Last_Loan_6$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_5_Last_Loan_7
## $Last_Loan_5_Last_Loan_7$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Last_Loan_7$significance
## [1] "Significant"
##
## $Last_Loan_5_Last_Loan_7$cramers_v
## [1] 0.2135997
##
## $Last_Loan_5_Last_Loan_7$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_5_Last_Loan_8
## $Last_Loan_5_Last_Loan_8$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Last_Loan_8$significance
## [1] "Significant"
##
## $Last_Loan_5_Last_Loan_8$cramers_v
## [1] 0.1453474
##
## $Last_Loan_5_Last_Loan_8$association_strength
```

```
## [1] "Moderate association"
##
##
## $Last_Loan_5_Last_Loan_9
## $Last_Loan_5_Last_Loan_9$p_value
## [1] "<.0001"
##
## $Last_Loan_5_Last_Loan_9$significance
## [1] "Significant"
##
## $Last_Loan_5_Last_Loan_9$cramers_v
## [1] 0.1111073
##
## $Last_Loan_5_Last_Loan_9$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_6_Last_Loan_7
## $Last_Loan_6_Last_Loan_7$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Last_Loan_7$significance
## [1] "Significant"
##
## $Last_Loan_6_Last_Loan_7$cramers_v
## [1] 0.2596141
##
## $Last_Loan_6_Last_Loan_7$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_6_Last_Loan_8
## $Last_Loan_6_Last_Loan_8$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Last_Loan_8$significance
## [1] "Significant"
##
## $Last_Loan_6_Last_Loan_8$cramers_v
## [1] 0.1763347
##
## $Last_Loan_6_Last_Loan_8$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_6_Last_Loan_9
## $Last_Loan_6_Last_Loan_9$p_value
## [1] "<.0001"
##
## $Last_Loan_6_Last_Loan_9$significance
## [1] "Significant"
```

```
##
## $Last_Loan_6_Last_Loan_9$cramers_v
## [1] 0.130096
##
## $Last_Loan_6_Last_Loan_9$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_7_Last_Loan_8
## $Last_Loan_7_Last_Loan_8$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Last_Loan_8$significance
## [1] "Significant"
##
## $Last_Loan_7_Last_Loan_8$cramers_v
## [1] 0.2305511
##
## $Last_Loan_7_Last_Loan_8$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_7_Last_Loan_9
## $Last_Loan_7_Last_Loan_9$p_value
## [1] "<.0001"
##
## $Last_Loan_7_Last_Loan_9$significance
## [1] "Significant"
##
## $Last_Loan_7_Last_Loan_9$cramers_v
## [1] 0.1712199
##
## $Last_Loan_7_Last_Loan_9$association_strength
## [1] "Moderate association"
##
##
## $Last_Loan_8_Last_Loan_9
## $Last_Loan_8_Last_Loan_9$p_value
## [1] "<.0001"
##
## $Last_Loan_8_Last_Loan_9$significance
## [1] "Significant"
##
## $Last_Loan_8_Last_Loan_9$cramers_v
## [1] 0.2590623
##
## $Last_Loan_8_Last_Loan_9$association_strength
## [1] "Moderate association"
```

#Contigency Table for Categorical and Categorical Attributes

```
# Load necessary library
if (!require("progress")) install.packages("progress")  # For progress bar
library(progress)

# Define function to calculate contingency tables with totals and percentages
get_contingency_tables_with_totals <- function(data, categorical_vars) {

  # Initialize the progress bar
  total_combinations <- length(categorical_vars) * (length(categorical_vars) - 1) / 2
  pb <- progress_bar$new(
    total = total_combinations,
    format = "Processing [:bar] :percent ETA: :eta",
    clear = FALSE, width = 60
  )

  # Store results in a list
  contingency_tables_with_totals <- list()

  # Generate contingency tables for each combination of categorical variables
  for (i in 1:(length(categorical_vars) - 1)) {
    for (j in (i + 1):length(categorical_vars)) {

      cat_var1 <- categorical_vars[i]
      cat_var2 <- categorical_vars[j]

      # Create contingency table
      contingency_table <- table(data[[cat_var1]], data[[cat_var2]])

      # Calculate totals
      row_totals <- margin.table(contingency_table, 1)  # Row totals
      col_totals <- margin.table(contingency_table, 2)  # Column totals
      grand_total <- sum(contingency_table)             # Overall total

      # Calculate percentages
      cell_percentage <- (contingency_table / grand_total) * 100
      row_percentage <- prop.table(contingency_table, 1) * 100
      col_percentage <- prop.table(contingency_table, 2) * 100

      # Create a list to store the table with totals and percentages
      contingency_tables_with_totals[[paste(cat_var1, cat_var2, sep = "_")]] <- list(
        contingency_table = contingency_table,
        row_totals = row_totals,
        col_totals = col_totals,
        grand_total = grand_total,
        cell_percentage = cell_percentage,
        row_percentage = row_percentage,
        col_percentage = col_percentage
      )

      # Update progress bar
      pb$tick()
```

```
    }
  }

  return(contingency_tables_with_totals)
}

# Example usage:
# Assuming 'credit_score' is your dataframe with categorical variables
categorical_vars <- c("Credit_Mix", "Payment_of_Min_Amount", "Payment_Behaviour", "Occupatio
n",
                      "Credit_Score", "Last_Loan_1", "Last_Loan_2", "Last_Loan_3", "Last_Loan
_4",
                      "Last_Loan_5", "Last_Loan_6", "Last_Loan_7", "Last_Loan_8", "Last_Loan_
9")

# Call the function
contingency_tables_with_totals <- get_contingency_tables_with_totals(credit_score, categorica
l_vars)

# Print the contingency tables with totals and percentages (optional)
print(contingency_tables_with_totals)
```

```
## $Credit_Mix_Payment_of_Min_Amount
## $Credit_Mix_Payment_of_Min_Amount$contingency_table
##
##               No   Yes
##   Bad          0 23768
##   Good     30384     0
##   Standard 10184 35664
##
## $Credit_Mix_Payment_of_Min_Amount$row_totals
##
##      Bad     Good Standard
##    23768    30384    45848
##
## $Credit_Mix_Payment_of_Min_Amount$col_totals
##
##    No   Yes
## 40568 59432
##
## $Credit_Mix_Payment_of_Min_Amount$grand_total
## [1] 100000
##
## $Credit_Mix_Payment_of_Min_Amount$cell_percentage
##
##                No    Yes
##   Bad       0.000 23.768
##   Good     30.384  0.000
##   Standard 10.184 35.664
##
## $Credit_Mix_Payment_of_Min_Amount$row_percentage
##
##                   No       Yes
##   Bad        0.00000 100.00000
##   Good     100.00000   0.00000
##   Standard  22.21253  77.78747
##
## $Credit_Mix_Payment_of_Min_Amount$col_percentage
##
##                  No      Yes
##   Bad       0.00000 39.99192
##   Good     74.89647  0.00000
##   Standard 25.10353 60.00808
##
##
## $Credit_Mix_Payment_Behaviour
## $Credit_Mix_Payment_Behaviour$contingency_table
##
##            High_spent_Large_value_payments High_spent_Medium_value_payments
##   Bad                                 1971                             4464
##   Good                                6635                             6849
##   Standard                            6833                             9499
##
```

```
##            High_spent_Small_value_payments Low_spent_Large_value_payments
##   Bad                                 1746                           1522
##   Good                                2557                           2253
##   Standard                            3950                           3579
##
##            Low_spent_Medium_value_payments Low_spent_Small_value_payments
##   Bad                                 2463                          11602
##   Good                                4020                           8070
##   Standard                            5009                          16978
##
## $Credit_Mix_Payment_Behaviour$row_totals
##
##      Bad     Good Standard
##    23768    30384    45848
##
## $Credit_Mix_Payment_Behaviour$col_totals
##
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                            15439                           20812
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                             8253                            7354
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                            11492                           36650
##
## $Credit_Mix_Payment_Behaviour$grand_total
## [1] 100000
##
## $Credit_Mix_Payment_Behaviour$cell_percentage
##
##            High_spent_Large_value_payments High_spent_Medium_value_payments
##   Bad                                1.971                           4.464
##   Good                               6.635                           6.849
##   Standard                           6.833                           9.499
##
##            High_spent_Small_value_payments Low_spent_Large_value_payments
##   Bad                                1.746                           1.522
##   Good                               2.557                           2.253
##   Standard                           3.950                           3.579
##
##            Low_spent_Medium_value_payments Low_spent_Small_value_payments
##   Bad                                2.463                          11.602
##   Good                               4.020                           8.070
##   Standard                           5.009                          16.978
##
## $Credit_Mix_Payment_Behaviour$row_percentage
##
##            High_spent_Large_value_payments High_spent_Medium_value_payments
##   Bad                             8.292662                       18.781555
##   Good                           21.837151                       22.541469
##   Standard                       14.903594                       20.718461
##
```

```
##            High_spent_Small_value_payments Low_spent_Large_value_payments
##   Bad                             7.346011                       6.403568
##   Good                            8.415613                       7.415087
##   Standard                        8.615425                       7.806229
##
##            Low_spent_Medium_value_payments Low_spent_Small_value_payments
##   Bad                            10.362673                      48.813531
##   Good                           13.230648                      26.560032
##   Standard                       10.925231                      37.031059
##
## $Credit_Mix_Payment_Behaviour$col_percentage
##
##            High_spent_Large_value_payments High_spent_Medium_value_payments
##   Bad                             12.76637                         21.44916
##   Good                            42.97558                         32.90890
##   Standard                        44.25805                         45.64194
##
##            High_spent_Small_value_payments Low_spent_Large_value_payments
##   Bad                             21.15594                       20.69622
##   Good                            30.98267                       30.63639
##   Standard                        47.86138                       48.66739
##
##            Low_spent_Medium_value_payments Low_spent_Small_value_payments
##   Bad                             21.43230                       31.65621
##   Good                            34.98086                       22.01910
##   Standard                        43.58684                       46.32469
##
##
## $Credit_Mix_Occupation
## $Credit_Mix_Occupation$contingency_table
##
##            Accountant Architect Developer Doctor Engineer Entrepreneur
##   Bad            1616      1512      1648   1672     1792         1568
##   Good           2096      2064      2008   1896     2088         2144
##   Standard       3032      3248      3064   3000     2984         2936
##
##            Journalist Lawyer Manager Mechanic Media_Manager Musician Scientist
##   Bad            1448   1560    1624     1704          1384     1480      1688
##   Good           1968   2176    2008     2048          2072     1992      2096
##   Standard       3120   3360    2800     3024          3264     2880      2960
##
##            Teacher Writer
##   Bad         1632   1440
##   Good        2032   1696
##   Standard    3008   3168
##
## $Credit_Mix_Occupation$row_totals
##
##      Bad     Good Standard
##    23768    30384    45848
##
```

```
## $Credit_Mix_Occupation$col_totals
##
##    Accountant     Architect     Developer        Doctor      Engineer
##          6744          6824          6720          6568          6864
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##          6648          6536          7096          6432          6776
## Media_Manager      Musician     Scientist       Teacher        Writer
##          6720          6352          6744          6672          6304
##
## $Credit_Mix_Occupation$grand_total
## [1] 100000
##
## $Credit_Mix_Occupation$cell_percentage
##
##            Accountant Architect Developer Doctor Engineer Entrepreneur
##   Bad           1.616     1.512     1.648  1.672    1.792        1.568
##   Good          2.096     2.064     2.008  1.896    2.088        2.144
##   Standard      3.032     3.248     3.064  3.000    2.984        2.936
##
##            Journalist Lawyer Manager Mechanic Media_Manager Musician Scientist
##   Bad           1.448  1.560   1.624    1.704         1.384    1.480     1.688
##   Good          1.968  2.176   2.008    2.048         2.072    1.992     2.096
##   Standard      3.120  3.360   2.800    3.024         3.264    2.880     2.960
##
##            Teacher Writer
##   Bad        1.632  1.440
##   Good       2.032  1.696
##   Standard   3.008  3.168
##
## $Credit_Mix_Occupation$row_percentage
##
##            Accountant Architect Developer   Doctor Engineer Entrepreneur
##   Bad        6.799058  6.361494  6.933692 7.034668 7.539549     6.597105
##   Good       6.898368  6.793049  6.608741 6.240126 6.872038     7.056345
##   Standard   6.613157  7.084278  6.682952 6.543361 6.508463     6.403769
##
##            Journalist   Lawyer  Manager Mechanic Media_Manager Musician
##   Bad        6.092225 6.563447 6.832716 7.169303      5.822955 6.226860
##   Good       6.477093 7.161664 6.608741 6.740390      6.819379 6.556082
##   Standard   6.805095 7.328564 6.107137 6.595708      7.119176 6.281626
##
##            Scientist  Teacher   Writer
##   Bad       7.101986 6.866375 6.058566
##   Good      6.898368 6.687730 5.581885
##   Standard  6.456116 6.560810 6.909789
##
## $Credit_Mix_Occupation$col_percentage
##
##            Accountant Architect Developer   Doctor Engineer Entrepreneur
##   Bad        23.96204  22.15709  24.52381 25.45676 26.10723     23.58604
##   Good       31.07948  30.24619  29.88095 28.86724 30.41958     32.25030
```

```
##   Standard   44.95848  47.59672  45.59524 45.67600 43.47319     44.16366
##
##            Journalist   Lawyer  Manager Mechanic Media_Manager Musician
##   Bad        22.15422 21.98422 25.24876 25.14758      20.59524 23.29975
##   Good       30.11016 30.66516 31.21891 30.22432      30.83333 31.36020
##   Standard   47.73562 47.35062 43.53234 44.62810      48.57143 45.34005
##
##            Scientist  Teacher   Writer
##   Bad       25.02966 24.46043 22.84264
##   Good      31.07948 30.45564 26.90355
##   Standard  43.89087 45.08393 50.25381
##
##
## $Credit_Mix_Credit_Score
## $Credit_Mix_Credit_Score$contingency_table
##
##             Good  Poor Standard
##   Bad        351 14289     9128
##   Good     14848  4851    10685
##   Standard  2629  9858    33361
##
## $Credit_Mix_Credit_Score$row_totals
##
##      Bad     Good Standard
##    23768    30384    45848
##
## $Credit_Mix_Credit_Score$col_totals
##
##     Good     Poor Standard
##    17828    28998    53174
##
## $Credit_Mix_Credit_Score$grand_total
## [1] 100000
##
## $Credit_Mix_Credit_Score$cell_percentage
##
##              Good   Poor Standard
##   Bad       0.351 14.289    9.128
##   Good     14.848  4.851   10.685
##   Standard  2.629  9.858   33.361
##
## $Credit_Mix_Credit_Score$row_percentage
##
##                 Good      Poor  Standard
##   Bad       1.476775 60.118647 38.404578
##   Good     48.867825 15.965640 35.166535
##   Standard  5.734165 21.501483 72.764352
##
## $Credit_Mix_Credit_Score$col_percentage
##
##                 Good      Poor  Standard
```

```
##   Bad       1.968813 49.275812 17.166284
##   Good     83.284721 16.728740 20.094407
##   Standard 14.746466 33.995448 62.739309
##
##
## $Credit_Mix_Last_Loan_1
## $Credit_Mix_Last_Loan_1$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad           2400                2776                    2640
##   Good          2416                2664                    2752
##   Standard      4512                4360                    4576
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                  2632          2368       0          2520        2744
##   Good                 2536          2656    6296          2824        2856
##   Standard             4632          4584    5112          4672        4504
##
##            Personal Loan Student Loan
##   Bad               2944         2744
##   Good              2680         2704
##   Standard          4312         4584
##
## $Credit_Mix_Last_Loan_1$row_totals
##
##      Bad     Good Standard
##    23768    30384    45848
##
## $Credit_Mix_Last_Loan_1$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    9328                    9800                    9968
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    9800                    9608                   11408
##           Not Specified             Payday Loan           Personal Loan
##                   10016                   10104                    9936
##            Student Loan
##                   10032
##
## $Credit_Mix_Last_Loan_1$grand_total
## [1] 100000
##
## $Credit_Mix_Last_Loan_1$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad          2.400               2.776                   2.640
##   Good         2.416               2.664                   2.752
##   Standard     4.512               4.360                   4.576
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                 2.632         2.368   0.000         2.520       2.744
```

```
##   Good                2.536         2.656   6.296         2.824       2.856
##   Standard            4.632         4.584   5.112         4.672       4.504
##
##            Personal Loan Student Loan
##   Bad              2.944        2.744
##   Good             2.680        2.704
##   Standard         4.312        4.584
##
## $Credit_Mix_Last_Loan_1$row_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad      10.097610           11.679569               11.107371
##   Good      7.951553            8.767773                9.057399
##   Standard  9.841214            9.509684                9.980806
##
##            Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Bad             11.073713      9.962975  0.000000     10.602491   11.544934
##   Good             8.346498      8.741443 20.721432      9.294365    9.399684
##   Standard        10.102949      9.998255 11.149887     10.190194    9.823765
##
##            Personal Loan Student Loan
##   Bad          12.386402    11.544934
##   Good          8.820432     8.899421
##   Standard      9.404990     9.998255
##
## $Credit_Mix_Last_Loan_1$col_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad       25.72899            28.32653                26.48475
##   Good      25.90051            27.18367                27.60835
##   Standard  48.37050            44.48980                45.90690
##
##            Home Equity Loan Mortgage Loan  No Loan Not Specified Payday Loan
##   Bad              26.85714      24.64613  0.00000      25.15974    27.15756
##   Good             25.87755      27.64363 55.18934      28.19489    28.26603
##   Standard         47.26531      47.71024 44.81066      46.64537    44.57641
##
##            Personal Loan Student Loan
##   Bad           29.62963     27.35247
##   Good          26.97262     26.95375
##   Standard      43.39775     45.69378
##
##
## $Credit_Mix_Last_Loan_2
## $Credit_Mix_Last_Loan_2$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad           2704                2568                    2472
##   Good          1984                1768                    1928
##   Standard      3784                4192                    3872
##
```

```
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                  2784          2712       0          2440        2824
##   Good                 2192          1976   12192          2224        2016
##   Standard             3800          3720   10344          4064        4264
##
##            Personal Loan Student Loan
##   Bad               2624         2640
##   Good              1992         2112
##   Standard          4024         3784
##
## $Credit_Mix_Last_Loan_2$row_totals
##
##      Bad     Good Standard
##    23768    30384    45848
##
## $Credit_Mix_Last_Loan_2$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    8472                    8528                    8272
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    8776                    8408                   22536
##           Not Specified             Payday Loan           Personal Loan
##                    8728                    9104                    8640
##            Student Loan
##                    8536
##
## $Credit_Mix_Last_Loan_2$grand_total
## [1] 100000
##
## $Credit_Mix_Last_Loan_2$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad          2.704               2.568                   2.472
##   Good         1.984               1.768                   1.928
##   Standard     3.784               4.192                   3.872
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                 2.784         2.712   0.000         2.440       2.824
##   Good                2.192         1.976  12.192         2.224       2.016
##   Standard            3.800         3.720  10.344         4.064       4.264
##
##            Personal Loan Student Loan
##   Bad              2.624        2.640
##   Good             1.992        2.112
##   Standard         4.024        3.784
##
## $Credit_Mix_Last_Loan_2$row_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad      11.376641           10.804443               10.400539
##   Good      6.529753            5.818852                6.345445
```

```
##   Standard  8.253359            9.143256                8.445298
##
##            Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Bad             11.713228     11.410300  0.000000     10.265904   11.881521
##   Good             7.214323      6.503423 40.126382      7.319642    6.635071
##   Standard         8.288257      8.113767 22.561508      8.864073    9.300297
##
##            Personal Loan Student Loan
##   Bad          11.040054    11.107371
##   Good          6.556082     6.951027
##   Standard      8.776828     8.253359
##
## $Credit_Mix_Last_Loan_2$col_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad       31.91690            30.11257                29.88395
##   Good      23.41832            20.73171                23.30754
##   Standard  44.66478            49.15572                46.80851
##
##            Home Equity Loan Mortgage Loan  No Loan Not Specified Payday Loan
##   Bad              31.72288      32.25500  0.00000      27.95600    31.01933
##   Good             24.97721      23.50143 54.10011      25.48121    22.14411
##   Standard         43.29991      44.24358 45.89989      46.56279    46.83656
##
##            Personal Loan Student Loan
##   Bad           30.37037     30.92784
##   Good          23.05556     24.74227
##   Standard      46.57407     44.32990
##
##
## $Credit_Mix_Last_Loan_3
## $Credit_Mix_Last_Loan_3$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad           2296                2720                     2560
##   Good          1416                1432                     1320
##   Standard      3096                3168                     3048
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                  2384          2584    1200          2672        2536
##   Good                 1304          1224   18192          1408        1344
##   Standard             2952          3064   18856          3000        2888
##
##            Personal Loan Student Loan
##   Bad               2272         2544
##   Good              1288         1456
##   Standard          2848         2928
##
## $Credit_Mix_Last_Loan_3$row_totals
##
##      Bad     Good Standard
```

```
##    23768    30384    45848
##
## $Credit_Mix_Last_Loan_3$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    6808                    7320                    6928
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    6640                    6872                   38248
##           Not Specified             Payday Loan           Personal Loan
##                    7080                    6768                    6408
##            Student Loan
##                    6928
##
## $Credit_Mix_Last_Loan_3$grand_total
## [1] 100000
##
## $Credit_Mix_Last_Loan_3$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad          2.296               2.720                   2.560
##   Good         1.416               1.432                   1.320
##   Standard     3.096               3.168                   3.048
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                 2.384         2.584   1.200         2.672       2.536
##   Good                1.304         1.224  18.192         1.408       1.344
##   Standard            2.952         3.064  18.856         3.000       2.888
##
##            Personal Loan Student Loan
##   Bad              2.272        2.544
##   Good             1.288        1.456
##   Standard         2.848        2.928
##
## $Credit_Mix_Last_Loan_3$row_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad       9.660047           11.443958               10.770784
##   Good      4.660348            4.713007                4.344392
##   Standard  6.752748            6.909789                6.648054
##
##            Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Bad             10.030293     10.871760  5.048805     11.242006   10.669808
##   Good             4.291732      4.028436 59.873618      4.634018    4.423381
##   Standard         6.438667      6.682952 41.127203      6.543361    6.299075
##
##            Personal Loan Student Loan
##   Bad           9.559071    10.703467
##   Good          4.239073     4.791996
##   Standard      6.211830     6.386320
##
## $Credit_Mix_Last_Loan_3$col_percentage
```

```
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad      33.725029           37.158470               36.951501
##   Good     20.799060           19.562842               19.053118
##   Standard 45.475911           43.278689               43.995381
##
##            Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Bad             35.903614     37.601863  3.137419     37.740113   37.470449
##   Good            19.638554     17.811409 47.563271     19.887006   19.858156
##   Standard        44.457831     44.586729 49.299310     42.372881   42.671395
##
##            Personal Loan Student Loan
##   Bad          35.455680    36.720554
##   Good         20.099875    21.016166
##   Standard     44.444444    42.263279
##
##
## $Credit_Mix_Last_Loan_4
## $Credit_Mix_Last_Loan_4$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad           2320                2568                    2352
##   Good           560                 752                     632
##   Standard      1840                2344                    2048
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                  2488          2472    2192          2496        2448
##   Good                  616           712   24544           528         744
##   Standard             2096          1984   27264          2096        2096
##
##            Personal Loan Student Loan
##   Bad               2200         2232
##   Good               696          600
##   Standard          1960         2120
##
## $Credit_Mix_Last_Loan_4$row_totals
##
##      Bad     Good Standard
##    23768    30384    45848
##
## $Credit_Mix_Last_Loan_4$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    4720                    5664                    5032
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    5200                    5168                   54000
##           Not Specified             Payday Loan           Personal Loan
##                    5120                    5288                    4856
##            Student Loan
##                    4952
##
```

```
## $Credit_Mix_Last_Loan_4$grand_total
## [1] 100000
##
## $Credit_Mix_Last_Loan_4$cell_percentage
##
##           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad          2.320               2.568                   2.352
##   Good         0.560               0.752                   0.632
##   Standard     1.840               2.344                   2.048
##
##           Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                 2.488         2.472   2.192         2.496       2.448
##   Good                0.616         0.712  24.544         0.528       0.744
##   Standard            2.096         1.984  27.264         2.096       2.096
##
##           Personal Loan Student Loan
##   Bad              2.200        2.232
##   Good             0.696        0.600
##   Standard         1.960        2.120
##
## $Credit_Mix_Last_Loan_4$row_percentage
##
##           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad       9.761023           10.804443                9.895658
##   Good      1.843075            2.474987                2.080042
##   Standard  4.013261            5.112546                4.466934
##
##           Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Bad             10.467856     10.400539  9.222484     10.501515   10.299562
##   Good             2.027383      2.343339 80.779358      1.737757    2.448657
##   Standard         4.571628      4.327343 59.466062      4.571628    4.571628
##
##           Personal Loan Student Loan
##   Bad           9.256143     9.390778
##   Good          2.290679     1.974724
##   Standard      4.274996     4.623975
##
## $Credit_Mix_Last_Loan_4$col_percentage
##
##           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad      49.152542           45.338983               46.740859
##   Good     11.864407           13.276836               12.559618
##   Standard 38.983051           41.384181               40.699523
##
##           Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Bad             47.846154     47.832817  4.059259     48.750000   46.293495
##   Good            11.846154     13.777090 45.451852     10.312500   14.069592
##   Standard        40.307692     38.390093 50.488889     40.937500   39.636914
##
##           Personal Loan Student Loan
##   Bad          45.304778    45.072698
```

```
##   Good         14.332784    12.116317
##   Standard     40.362438    42.810985
##
##
## $Credit_Mix_Last_Loan_5
## $Credit_Mix_Last_Loan_5$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad           2408                2392                    2480
##   Good             0                   0                       0
##   Standard      1016                1104                    1048
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                  2344          2248    3192          2128        2112
##   Good                    0             0   30384             0           0
##   Standard             1056          1192   35880          1144        1232
##
##            Personal Loan Student Loan
##   Bad               2288         2176
##   Good                 0            0
##   Standard          1232          944
##
## $Credit_Mix_Last_Loan_5$row_totals
##
##      Bad     Good Standard
##    23768    30384    45848
##
## $Credit_Mix_Last_Loan_5$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    3424                    3496                    3528
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    3400                    3440                   69456
##           Not Specified             Payday Loan           Personal Loan
##                    3272                    3344                    3520
##            Student Loan
##                    3120
##
## $Credit_Mix_Last_Loan_5$grand_total
## [1] 100000
##
## $Credit_Mix_Last_Loan_5$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad          2.408               2.392                   2.480
##   Good         0.000               0.000                   0.000
##   Standard     1.016               1.104                   1.048
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                 2.344         2.248   3.192         2.128       2.112
##   Good                0.000         0.000  30.384         0.000       0.000
```

```
##   Standard            1.056         1.192  35.880         1.144       1.232
##
##            Personal Loan Student Loan
##   Bad              2.288        2.176
##   Good             0.000        0.000
##   Standard         1.232        0.944
##
## $Credit_Mix_Last_Loan_5$row_percentage
##
##             Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad       10.131269           10.063952               10.434197
##   Good       0.000000            0.000000                0.000000
##   Standard   2.216018            2.407957                2.285814
##
##            Home Equity Loan Mortgage Loan    No Loan Not Specified Payday Loan
##   Bad              9.861999      9.458095  13.429822      8.953214    8.885897
##   Good             0.000000      0.000000 100.000000      0.000000    0.000000
##   Standard         2.303263      2.599895  78.258594      2.495202    2.687140
##
##            Personal Loan Student Loan
##   Bad           9.626388     9.155167
##   Good          0.000000     0.000000
##   Standard      2.687140     2.058977
##
## $Credit_Mix_Last_Loan_5$col_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad      70.327103           68.421053               70.294785
##   Good      0.000000            0.000000                0.000000
##   Standard 29.672897           31.578947               29.705215
##
##            Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Bad             68.941176     65.348837  4.595715     65.036675   63.157895
##   Good             0.000000      0.000000 43.745681      0.000000    0.000000
##   Standard        31.058824     34.651163 51.658604     34.963325   36.842105
##
##            Personal Loan Student Loan
##   Bad          65.000000    69.743590
##   Good          0.000000     0.000000
##   Standard     35.000000    30.256410
##
##
## $Credit_Mix_Last_Loan_6
## $Credit_Mix_Last_Loan_6$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad           1768                1832                    1784
##   Good             0                   0                       0
##   Standard       704                 776                     680
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
```

```
##   Bad                 1744          1760    7448          1872        1968
##   Good                   0             0   30384             0           0
##   Standard             720           632   39152           752         720
##
##            Personal Loan Student Loan
##   Bad               1840         1752
##   Good                 0            0
##   Standard           896          816
##
## $Credit_Mix_Last_Loan_6$row_totals
##
##      Bad     Good Standard
##    23768    30384    45848
##
## $Credit_Mix_Last_Loan_6$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    2472                    2608                    2464
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    2464                    2392                   76984
##           Not Specified             Payday Loan           Personal Loan
##                    2624                    2688                    2736
##            Student Loan
##                    2568
##
## $Credit_Mix_Last_Loan_6$grand_total
## [1] 100000
##
## $Credit_Mix_Last_Loan_6$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad          1.768               1.832                   1.784
##   Good         0.000               0.000                   0.000
##   Standard     0.704               0.776                   0.680
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                 1.744         1.760   7.448         1.872       1.968
##   Good                0.000         0.000  30.384         0.000       0.000
##   Standard            0.720         0.632  39.152         0.752       0.720
##
##            Personal Loan Student Loan
##   Bad              1.840        1.752
##   Good             0.000        0.000
##   Standard         0.896        0.816
##
## $Credit_Mix_Last_Loan_6$row_percentage
##
##             Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad        7.438573            7.707842                7.505890
##   Good       0.000000            0.000000                0.000000
##   Standard   1.535509            1.692549                1.483162
```

```
##
##            Home Equity Loan Mortgage Loan    No Loan Not Specified Payday Loan
##   Bad              7.337597      7.404914  31.336250      7.876136    8.280040
##   Good             0.000000      0.000000 100.000000      0.000000    0.000000
##   Standard         1.570407      1.378468  85.395219      1.640202    1.570407
##
##            Personal Loan Student Loan
##   Bad           7.741501     7.371255
##   Good          0.000000     0.000000
##   Standard      1.954284     1.779794
##
## $Credit_Mix_Last_Loan_6$col_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad      71.521036           70.245399               72.402597
##   Good      0.000000            0.000000                0.000000
##   Standard 28.478964           29.754601               27.597403
##
##            Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Bad             70.779221     73.578595  9.674738     71.341463   73.214286
##   Good             0.000000      0.000000 39.467941      0.000000    0.000000
##   Standard        29.220779     26.421405 50.857321     28.658537   26.785714
##
##            Personal Loan Student Loan
##   Bad          67.251462    68.224299
##   Good          0.000000     0.000000
##   Standard     32.748538    31.775701
##
##
## $Credit_Mix_Last_Loan_7
## $Credit_Mix_Last_Loan_7$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad           1248                1368                    1072
##   Good             0                   0                       0
##   Standard       352                 352                     360
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                  1184          1464   12144          1232        1512
##   Good                    0             0   30384             0           0
##   Standard              344           328   42600           376         408
##
##            Personal Loan Student Loan
##   Bad               1264         1280
##   Good                 0            0
##   Standard           424          304
##
## $Credit_Mix_Last_Loan_7$row_totals
##
##      Bad     Good Standard
##    23768    30384    45848
```

```
##
## $Credit_Mix_Last_Loan_7$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    1600                    1720                    1432
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    1528                    1792                   85128
##           Not Specified             Payday Loan           Personal Loan
##                    1608                    1920                    1688
##            Student Loan
##                    1584
##
## $Credit_Mix_Last_Loan_7$grand_total
## [1] 100000
##
## $Credit_Mix_Last_Loan_7$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad          1.248               1.368                   1.072
##   Good         0.000               0.000                   0.000
##   Standard     0.352               0.352                   0.360
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                 1.184         1.464  12.144         1.232       1.512
##   Good                0.000         0.000  30.384         0.000       0.000
##   Standard            0.344         0.328  42.600         0.376       0.408
##
##            Personal Loan Student Loan
##   Bad              1.264        1.280
##   Good             0.000        0.000
##   Standard         0.424        0.304
##
## $Credit_Mix_Last_Loan_7$row_percentage
##
##              Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad        5.2507573           5.7556378               4.5102659
##   Good       0.0000000           0.0000000               0.0000000
##   Standard   0.7677543           0.7677543               0.7852033
##
##            Home Equity Loan Mortgage Loan     No Loan Not Specified Payday Loan
##   Bad             4.9814877     6.1595422  51.0939078     5.1834399   6.3614944
##   Good            0.0000000     0.0000000 100.0000000     0.0000000   0.0000000
##   Standard        0.7503054     0.7154074  92.9157215     0.8201012   0.8898971
##
##            Personal Loan Student Loan
##   Bad          5.3180747    5.3853921
##   Good         0.0000000    0.0000000
##   Standard     0.9247950    0.6630605
##
## $Credit_Mix_Last_Loan_7$col_percentage
##
```

```
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad       78.00000            79.53488                74.86034
##   Good       0.00000             0.00000                 0.00000
##   Standard  22.00000            20.46512                25.13966
##
##            Home Equity Loan Mortgage Loan  No Loan Not Specified Payday Loan
##   Bad              77.48691      81.69643 14.26558      76.61692    78.75000
##   Good              0.00000       0.00000 35.69213       0.00000     0.00000
##   Standard         22.51309      18.30357 50.04229      23.38308    21.25000
##
##            Personal Loan Student Loan
##   Bad           74.88152     80.80808
##   Good           0.00000      0.00000
##   Standard      25.11848     19.19192
##
##
## $Credit_Mix_Last_Loan_8
## $Credit_Mix_Last_Loan_8$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad            720                 872                     760
##   Good             0                   0                       0
##   Standard         0                   0                       0
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                  888           736   16576           744         888
##   Good                   0             0   30384             0           0
##   Standard               0             0   45848             0           0
##
##            Personal Loan Student Loan
##   Bad                720          864
##   Good                 0            0
##   Standard             0            0
##
## $Credit_Mix_Last_Loan_8$row_totals
##
##      Bad     Good Standard
##    23768    30384    45848
##
## $Credit_Mix_Last_Loan_8$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     720                     872                     760
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     888                     736                   92808
##           Not Specified             Payday Loan           Personal Loan
##                     744                     888                     720
##            Student Loan
##                     864
##
## $Credit_Mix_Last_Loan_8$grand_total
```

```
## [1] 100000
##
## $Credit_Mix_Last_Loan_8$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad          0.720               0.872                   0.760
##   Good         0.000               0.000                   0.000
##   Standard     0.000               0.000                   0.000
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                 0.888         0.736  16.576         0.744       0.888
##   Good                0.000         0.000  30.384         0.000       0.000
##   Standard            0.000         0.000  45.848         0.000       0.000
##
##            Personal Loan Student Loan
##   Bad              0.720        0.864
##   Good             0.000        0.000
##   Standard         0.000        0.000
##
## $Credit_Mix_Last_Loan_8$row_percentage
##
##             Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad        3.029283            3.668798                3.197577
##   Good       0.000000            0.000000                0.000000
##   Standard   0.000000            0.000000                0.000000
##
##            Home Equity Loan Mortgage Loan    No Loan Not Specified Payday Loan
##   Bad              3.736116      3.096600  69.740828      3.130259    3.736116
##   Good             0.000000      0.000000 100.000000      0.000000    0.000000
##   Standard         0.000000      0.000000 100.000000      0.000000    0.000000
##
##            Personal Loan Student Loan
##   Bad           3.029283     3.635140
##   Good          0.000000     0.000000
##   Standard      0.000000     0.000000
##
## $Credit_Mix_Last_Loan_8$col_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad      100.00000           100.00000               100.00000
##   Good       0.00000             0.00000                 0.00000
##   Standard   0.00000             0.00000                 0.00000
##
##            Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Bad             100.00000     100.00000  17.86053     100.00000   100.00000
##   Good              0.00000       0.00000  32.73856       0.00000     0.00000
##   Standard          0.00000       0.00000  49.40091       0.00000     0.00000
##
##            Personal Loan Student Loan
##   Bad          100.00000    100.00000
##   Good           0.00000      0.00000
```

```
##   Standard       0.00000      0.00000
##
##
## $Credit_Mix_Last_Loan_9
## $Credit_Mix_Last_Loan_9$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad            448                 432                     392
##   Good             0                   0                       0
##   Standard         0                   0                       0
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                   408           520   19912           424         464
##   Good                    0             0   30384             0           0
##   Standard                0             0   45848             0           0
##
##            Personal Loan Student Loan
##   Bad                384          384
##   Good                 0            0
##   Standard             0            0
##
## $Credit_Mix_Last_Loan_9$row_totals
##
##      Bad     Good Standard
##    23768    30384    45848
##
## $Credit_Mix_Last_Loan_9$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     448                     432                     392
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     408                     520                   96144
##           Not Specified             Payday Loan           Personal Loan
##                     424                     464                     384
##            Student Loan
##                     384
##
## $Credit_Mix_Last_Loan_9$grand_total
## [1] 100000
##
## $Credit_Mix_Last_Loan_9$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad          0.448               0.432                   0.392
##   Good         0.000               0.000                   0.000
##   Standard     0.000               0.000                   0.000
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Bad                 0.408         0.520  19.912         0.424       0.464
##   Good                0.000         0.000  30.384         0.000       0.000
##   Standard            0.000         0.000  45.848         0.000       0.000
```

```
##
##            Personal Loan Student Loan
##   Bad              0.384        0.384
##   Good             0.000        0.000
##   Standard         0.000        0.000
##
## $Credit_Mix_Last_Loan_9$row_percentage
##
##             Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad        1.884887            1.817570                1.649276
##   Good       0.000000            0.000000                0.000000
##   Standard   0.000000            0.000000                0.000000
##
##            Home Equity Loan Mortgage Loan    No Loan Not Specified Payday Loan
##   Bad              1.716594      2.187816  83.776506      1.783911    1.952205
##   Good             0.000000      0.000000 100.000000      0.000000    0.000000
##   Standard         0.000000      0.000000 100.000000      0.000000    0.000000
##
##            Personal Loan Student Loan
##   Bad           1.615618     1.615618
##   Good          0.000000     0.000000
##   Standard      0.000000     0.000000
##
## $Credit_Mix_Last_Loan_9$col_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Bad       100.0000            100.0000                100.0000
##   Good        0.0000              0.0000                  0.0000
##   Standard    0.0000              0.0000                  0.0000
##
##            Home Equity Loan Mortgage Loan  No Loan Not Specified Payday Loan
##   Bad              100.0000      100.0000  20.7106      100.0000    100.0000
##   Good               0.0000        0.0000  31.6026        0.0000      0.0000
##   Standard           0.0000        0.0000  47.6868        0.0000      0.0000
##
##            Personal Loan Student Loan
##   Bad           100.0000     100.0000
##   Good            0.0000       0.0000
##   Standard        0.0000       0.0000
##
##
## $Payment_of_Min_Amount_Payment_Behaviour
## $Payment_of_Min_Amount_Payment_Behaviour$contingency_table
##
##       High_spent_Large_value_payments High_spent_Medium_value_payments
##   No                             8426                             9041
##   Yes                            7013                            11771
##
##       High_spent_Small_value_payments Low_spent_Large_value_payments
##   No                             3477                           3080
##   Yes                            4776                           4274
```

```
##
##       Low_spent_Medium_value_payments Low_spent_Small_value_payments
##   No                             5040                          11504
##   Yes                            6452                          25146
##
## $Payment_of_Min_Amount_Payment_Behaviour$row_totals
##
##    No   Yes
## 40568 59432
##
## $Payment_of_Min_Amount_Payment_Behaviour$col_totals
##
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                            15439                           20812
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                             8253                            7354
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                            11492                           36650
##
## $Payment_of_Min_Amount_Payment_Behaviour$grand_total
## [1] 100000
##
## $Payment_of_Min_Amount_Payment_Behaviour$cell_percentage
##
##       High_spent_Large_value_payments High_spent_Medium_value_payments
##   No                           8.426                           9.041
##   Yes                          7.013                          11.771
##
##       High_spent_Small_value_payments Low_spent_Large_value_payments
##   No                           3.477                          3.080
##   Yes                          4.776                          4.274
##
##       Low_spent_Medium_value_payments Low_spent_Small_value_payments
##   No                           5.040                         11.504
##   Yes                          6.452                         25.146
##
## $Payment_of_Min_Amount_Payment_Behaviour$row_percentage
##
##       High_spent_Large_value_payments High_spent_Medium_value_payments
##   No                       20.770065                       22.286038
##   Yes                      11.800040                       19.805829
##
##       High_spent_Small_value_payments Low_spent_Large_value_payments
##   No                        8.570795                       7.592191
##   Yes                       8.036075                       7.191412
##
##       Low_spent_Medium_value_payments Low_spent_Small_value_payments
##   No                       12.423585                      28.357326
##   Yes                      10.856104                      42.310540
##
## $Payment_of_Min_Amount_Payment_Behaviour$col_percentage
```

```
##
##       High_spent_Large_value_payments High_spent_Medium_value_payments
##   No                         54.57607                         43.44128
##   Yes                        45.42393                         56.55872
##
##       High_spent_Small_value_payments Low_spent_Large_value_payments
##   No                         42.13013                       41.88197
##   Yes                        57.86987                       58.11803
##
##       Low_spent_Medium_value_payments Low_spent_Small_value_payments
##   No                         43.85660                       31.38881
##   Yes                        56.14340                       68.61119
##
##
## $Payment_of_Min_Amount_Occupation
## $Payment_of_Min_Amount_Occupation$contingency_table
##
##       Accountant Architect Developer Doctor Engineer Entrepreneur Journalist
##   No        2720      2784      2680   2600     2648         2816       2592
##   Yes       4024      4040      4040   3968     4216         3832       3944
##
##       Lawyer Manager Mechanic Media_Manager Musician Scientist Teacher Writer
##   No    2904    2808     2752          2712     2728      2720    2696   2408
##   Yes   4192    3624     4024          4008     3624      4024    3976   3896
##
## $Payment_of_Min_Amount_Occupation$row_totals
##
##    No   Yes
## 40568 59432
##
## $Payment_of_Min_Amount_Occupation$col_totals
##
##    Accountant     Architect     Developer        Doctor      Engineer
##          6744          6824          6720          6568          6864
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##          6648          6536          7096          6432          6776
## Media_Manager      Musician     Scientist       Teacher        Writer
##          6720          6352          6744          6672          6304
##
## $Payment_of_Min_Amount_Occupation$grand_total
## [1] 100000
##
## $Payment_of_Min_Amount_Occupation$cell_percentage
##
##       Accountant Architect Developer Doctor Engineer Entrepreneur Journalist
##   No       2.720     2.784     2.680  2.600    2.648        2.816      2.592
##   Yes      4.024     4.040     4.040  3.968    4.216        3.832      3.944
##
##       Lawyer Manager Mechanic Media_Manager Musician Scientist Teacher Writer
##   No   2.904   2.808    2.752         2.712    2.728     2.720   2.696  2.408
##   Yes  4.192   3.624    4.024         4.008    3.624     4.024   3.976  3.896
```

```
##
## $Payment_of_Min_Amount_Occupation$row_percentage
##
##       Accountant Architect Developer   Doctor Engineer Entrepreneur Journalist
##   No    6.704792  6.862552  6.606192 6.408992 6.527312     6.941432   6.389272
##   Yes   6.770763  6.797685  6.797685 6.676538 7.093822     6.447705   6.636156
##
##         Lawyer  Manager Mechanic Media_Manager Musician Scientist  Teacher
##   No  7.158351 6.921712 6.783672      6.685072 6.724512  6.704792 6.645632
##   Yes 7.053439 6.097725 6.770763      6.743842 6.097725  6.770763 6.689999
##
##         Writer
##   No  5.935713
##   Yes 6.555391
##
## $Payment_of_Min_Amount_Occupation$col_percentage
##
##       Accountant Architect Developer   Doctor Engineer Entrepreneur Journalist
##   No    40.33215  40.79719  39.88095 39.58587 38.57809     42.35860   39.65728
##   Yes   59.66785  59.20281  60.11905 60.41413 61.42191     57.64140   60.34272
##
##         Lawyer  Manager Mechanic Media_Manager Musician Scientist  Teacher
##   No  40.92446 43.65672 40.61393      40.35714 42.94710  40.33215 40.40767
##   Yes 59.07554 56.34328 59.38607      59.64286 57.05290  59.66785 59.59233
##
##         Writer
##   No  38.19797
##   Yes 61.80203
##
##
## $Payment_of_Min_Amount_Credit_Score
## $Payment_of_Min_Amount_Credit_Score$contingency_table
##
##        Good  Poor Standard
##   No  15620  5439    19509
##   Yes  2208 23559    33665
##
## $Payment_of_Min_Amount_Credit_Score$row_totals
##
##    No   Yes
## 40568 59432
##
## $Payment_of_Min_Amount_Credit_Score$col_totals
##
##     Good     Poor Standard
##    17828    28998    53174
##
## $Payment_of_Min_Amount_Credit_Score$grand_total
## [1] 100000
##
## $Payment_of_Min_Amount_Credit_Score$cell_percentage
```

```
##
##         Good   Poor Standard
##   No  15.620  5.439   19.509
##   Yes  2.208 23.559   33.665
##
## $Payment_of_Min_Amount_Credit_Score$row_percentage
##
##           Good     Poor Standard
##   No  38.50325 13.40712 48.08963
##   Yes  3.71517 39.64026 56.64457
##
## $Payment_of_Min_Amount_Credit_Score$col_percentage
##
##           Good     Poor Standard
##   No  87.61499 18.75647 36.68898
##   Yes 12.38501 81.24353 63.31102
##
##
## $Payment_of_Min_Amount_Last_Loan_1
## $Payment_of_Min_Amount_Last_Loan_1$contingency_table
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No       3384                3560                    3752             3488
##   Yes      5944                6240                    6216             6312
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No           3576    8240          3744        3624          3528
##   Yes          6032    3168          6272        6480          6408
##
##       Student Loan
##   No          3672
##   Yes         6360
##
## $Payment_of_Min_Amount_Last_Loan_1$row_totals
##
##    No   Yes
## 40568 59432
##
## $Payment_of_Min_Amount_Last_Loan_1$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    9328                    9800                    9968
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    9800                    9608                   11408
##           Not Specified             Payday Loan           Personal Loan
##                   10016                   10104                    9936
##            Student Loan
##                   10032
##
## $Payment_of_Min_Amount_Last_Loan_1$grand_total
## [1] 100000
```

```
##
## $Payment_of_Min_Amount_Last_Loan_1$cell_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No      3.384               3.560                   3.752            3.488
##   Yes     5.944               6.240                   6.216            6.312
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No          3.576   8.240         3.744       3.624         3.528
##   Yes         6.032   3.168         6.272       6.480         6.408
##
##       Student Loan
##   No         3.672
##   Yes        6.360
##
## $Payment_of_Min_Amount_Last_Loan_1$row_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No   8.341550            8.775389                9.248669         8.597910
##   Yes 10.001346           10.499394               10.459012        10.620541
##
##       Mortgage Loan   No Loan Not Specified Payday Loan Personal Loan
##   No       8.814829 20.311576      9.228949    8.933149      8.696510
##   Yes     10.149414  5.330462     10.553237   10.903217     10.782070
##
##       Student Loan
##   No      9.051469
##   Yes    10.701306
##
## $Payment_of_Min_Amount_Last_Loan_1$col_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No   36.27787            36.32653                37.64045         35.59184
##   Yes  63.72213            63.67347                62.35955         64.40816
##
##       Mortgage Loan  No Loan Not Specified Payday Loan Personal Loan
##   No       37.21898 72.23001      37.38019    35.86698      35.50725
##   Yes      62.78102 27.76999      62.61981    64.13302      64.49275
##
##       Student Loan
##   No      36.60287
##   Yes     63.39713
##
##
## $Payment_of_Min_Amount_Last_Loan_2
## $Payment_of_Min_Amount_Last_Loan_2$contingency_table
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No       2632                2664                    2592             2768
##   Yes      5840                5864                    5680             6008
##
```

```
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No           2608   16200          2952        2688          2704
##   Yes          5800    6336          5776        6416          5936
##
##       Student Loan
##   No          2760
##   Yes         5776
##
## $Payment_of_Min_Amount_Last_Loan_2$row_totals
##
##    No   Yes
## 40568 59432
##
## $Payment_of_Min_Amount_Last_Loan_2$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    8472                    8528                    8272
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    8776                    8408                   22536
##           Not Specified             Payday Loan           Personal Loan
##                    8728                    9104                    8640
##            Student Loan
##                    8536
##
## $Payment_of_Min_Amount_Last_Loan_2$grand_total
## [1] 100000
##
## $Payment_of_Min_Amount_Last_Loan_2$cell_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No      2.632               2.664                   2.592            2.768
##   Yes     5.840               5.864                   5.680            6.008
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No          2.608  16.200         2.952       2.688         2.704
##   Yes         5.800   6.336         5.776       6.416         5.936
##
##       Student Loan
##   No         2.760
##   Yes        5.776
##
## $Payment_of_Min_Amount_Last_Loan_2$row_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No   6.487872            6.566752                6.389272         6.823112
##   Yes  9.826356            9.866738                9.557141        10.109032
##
##       Mortgage Loan   No Loan Not Specified Payday Loan Personal Loan
##   No       6.428712 39.932952      7.276671    6.625912      6.665352
##   Yes      9.759052 10.660923      9.718670   10.795531      9.987885
##
```

```
##       Student Loan
##   No      6.803392
##   Yes     9.718670
##
## $Payment_of_Min_Amount_Last_Loan_2$col_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No   31.06704            31.23827                31.33462         31.54057
##   Yes  68.93296            68.76173                68.66538         68.45943
##
##       Mortgage Loan  No Loan Not Specified Payday Loan Personal Loan
##   No       31.01808 71.88498      33.82218    29.52548      31.29630
##   Yes      68.98192 28.11502      66.17782    70.47452      68.70370
##
##       Student Loan
##   No      32.33365
##   Yes     67.66635
##
##
## $Payment_of_Min_Amount_Last_Loan_3
## $Payment_of_Min_Amount_Last_Loan_3$contingency_table
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No       1928                1880                    1760             1816
##   Yes      4880                5440                    5168             4824
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No           1776   24160          1824        1768          1744
##   Yes          5096   14088          5256        5000          4664
##
##       Student Loan
##   No          1912
##   Yes         5016
##
## $Payment_of_Min_Amount_Last_Loan_3$row_totals
##
##    No   Yes
## 40568 59432
##
## $Payment_of_Min_Amount_Last_Loan_3$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    6808                    7320                    6928
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    6640                    6872                   38248
##           Not Specified             Payday Loan           Personal Loan
##                    7080                    6768                    6408
##            Student Loan
##                    6928
##
## $Payment_of_Min_Amount_Last_Loan_3$grand_total
```

```
## [1] 100000
##
## $Payment_of_Min_Amount_Last_Loan_3$cell_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No      1.928               1.880                   1.760            1.816
##   Yes     4.880               5.440                   5.168            4.824
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No          1.776  24.160         1.824       1.768         1.744
##   Yes         5.096  14.088         5.256       5.000         4.664
##
##       Student Loan
##   No         1.912
##   Yes        5.016
##
## $Payment_of_Min_Amount_Last_Loan_3$row_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No   4.752514            4.634194                4.338395         4.476435
##   Yes  8.211065            9.153318                8.695652         8.116839
##
##       Mortgage Loan   No Loan Not Specified Payday Loan Personal Loan
##   No       4.377835 59.554329      4.496155    4.358115      4.298955
##   Yes      8.574505 23.704402      8.843721    8.412976      7.847624
##
##       Student Loan
##   No      4.713074
##   Yes     8.439898
##
## $Payment_of_Min_Amount_Last_Loan_3$col_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No   28.31962            25.68306                25.40416         27.34940
##   Yes  71.68038            74.31694                74.59584         72.65060
##
##       Mortgage Loan  No Loan Not Specified Payday Loan Personal Loan
##   No       25.84400 63.16670      25.76271    26.12293      27.21598
##   Yes      74.15600 36.83330      74.23729    73.87707      72.78402
##
##       Student Loan
##   No      27.59815
##   Yes     72.40185
##
##
## $Payment_of_Min_Amount_Last_Loan_4
## $Payment_of_Min_Amount_Last_Loan_4$contingency_table
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No        784                1016                     848              792
##   Yes      3936                4648                    4184             4408
```

```
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No           1008   32560           824         944           920
##   Yes          4160   21440          4296        4344          3936
##
##       Student Loan
##   No           872
##   Yes         4080
##
## $Payment_of_Min_Amount_Last_Loan_4$row_totals
##
##    No   Yes
## 40568 59432
##
## $Payment_of_Min_Amount_Last_Loan_4$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    4720                    5664                    5032
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    5200                    5168                   54000
##           Not Specified             Payday Loan           Personal Loan
##                    5120                    5288                    4856
##            Student Loan
##                    4952
##
## $Payment_of_Min_Amount_Last_Loan_4$grand_total
## [1] 100000
##
## $Payment_of_Min_Amount_Last_Loan_4$cell_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No      0.784               1.016                   0.848            0.792
##   Yes     3.936               4.648                   4.184            4.408
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No          1.008  32.560         0.824       0.944         0.920
##   Yes         4.160  21.440         4.296       4.344         3.936
##
##       Student Loan
##   No         0.872
##   Yes        4.080
##
## $Payment_of_Min_Amount_Last_Loan_4$row_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No   1.932558            2.504437                2.090317         1.952278
##   Yes  6.622695            7.820703                7.039978         7.416880
##
##       Mortgage Loan   No Loan Not Specified Payday Loan Personal Loan
##   No       2.484717 80.260304      2.031158    2.326957      2.267797
##   Yes      6.999596 36.074842      7.228429    7.309194      6.622695
```

```
##
##       Student Loan
##   No      2.149477
##   Yes     6.864989
##
## $Payment_of_Min_Amount_Last_Loan_4$col_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No   16.61017            17.93785                16.85215         15.23077
##   Yes  83.38983            82.06215                83.14785         84.76923
##
##       Mortgage Loan  No Loan Not Specified Payday Loan Personal Loan
##   No       19.50464 60.29630      16.09375    17.85174      18.94563
##   Yes      80.49536 39.70370      83.90625    82.14826      81.05437
##
##       Student Loan
##   No      17.60905
##   Yes     82.39095
##
##
## $Payment_of_Min_Amount_Last_Loan_5
## $Payment_of_Min_Amount_Last_Loan_5$contingency_table
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No          0                   0                       0                0
##   Yes      3424                3496                    3528             3400
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No              0   40568             0           0             0
##   Yes          3440   28888          3272        3344          3520
##
##       Student Loan
##   No             0
##   Yes         3120
##
## $Payment_of_Min_Amount_Last_Loan_5$row_totals
##
##    No   Yes
## 40568 59432
##
## $Payment_of_Min_Amount_Last_Loan_5$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    3424                    3496                    3528
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    3400                    3440                   69456
##           Not Specified             Payday Loan           Personal Loan
##                    3272                    3344                    3520
##            Student Loan
##                    3120
##
```

```
## $Payment_of_Min_Amount_Last_Loan_5$grand_total
## [1] 100000
##
## $Payment_of_Min_Amount_Last_Loan_5$cell_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No      0.000               0.000                   0.000            0.000
##   Yes     3.424               3.496                   3.528            3.400
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No          0.000  40.568         0.000       0.000         0.000
##   Yes         3.440  28.888         3.272       3.344         3.520
##
##       Student Loan
##   No         0.000
##   Yes        3.120
##
## $Payment_of_Min_Amount_Last_Loan_5$row_percentage
##
##        Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No    0.000000            0.000000                0.000000         0.000000
##   Yes   5.761206            5.882353                5.936196         5.720824
##
##       Mortgage Loan    No Loan Not Specified Payday Loan Personal Loan
##   No       0.000000 100.000000      0.000000    0.000000      0.000000
##   Yes      5.788128  48.606811      5.505452    5.626598      5.922735
##
##       Student Loan
##   No      0.000000
##   Yes     5.249697
##
## $Payment_of_Min_Amount_Last_Loan_5$col_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No     0.0000              0.0000                  0.0000           0.0000
##   Yes  100.0000            100.0000                100.0000         100.0000
##
##       Mortgage Loan  No Loan Not Specified Payday Loan Personal Loan
##   No         0.0000  58.4082        0.0000      0.0000        0.0000
##   Yes      100.0000  41.5918      100.0000    100.0000      100.0000
##
##       Student Loan
##   No        0.0000
##   Yes     100.0000
##
##
## $Payment_of_Min_Amount_Last_Loan_6
## $Payment_of_Min_Amount_Last_Loan_6$contingency_table
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No          0                   0                       0                0
```

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##   Yes      2472                2608                    2464             2464
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No              0   40568             0           0             0
##   Yes          2392   36416          2624        2688          2736
##
##       Student Loan
##   No             0
##   Yes         2568
##
## $Payment_of_Min_Amount_Last_Loan_6$row_totals
##
##    No   Yes
## 40568 59432
##
## $Payment_of_Min_Amount_Last_Loan_6$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    2472                    2608                    2464
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    2464                    2392                   76984
##           Not Specified             Payday Loan           Personal Loan
##                    2624                    2688                    2736
##            Student Loan
##                    2568
##
## $Payment_of_Min_Amount_Last_Loan_6$grand_total
## [1] 100000
##
## $Payment_of_Min_Amount_Last_Loan_6$cell_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No      0.000               0.000                   0.000            0.000
##   Yes     2.472               2.608                   2.464            2.464
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No          0.000  40.568         0.000       0.000         0.000
##   Yes         2.392  36.416         2.624       2.688         2.736
##
##       Student Loan
##   No         0.000
##   Yes        2.568
##
## $Payment_of_Min_Amount_Last_Loan_6$row_percentage
##
##        Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No    0.000000            0.000000                0.000000         0.000000
##   Yes   4.159375            4.388208                4.145915         4.145915
##
##       Mortgage Loan    No Loan Not Specified Payday Loan Personal Loan
##   No       0.000000 100.000000      0.000000    0.000000      0.000000
```

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##   Yes      4.024768  61.273388      4.415130    4.522816      4.603581
##
##       Student Loan
##   No      0.000000
##   Yes     4.320905
##
## $Payment_of_Min_Amount_Last_Loan_6$col_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No    0.00000             0.00000                 0.00000          0.00000
##   Yes 100.00000           100.00000               100.00000        100.00000
##
##       Mortgage Loan   No Loan Not Specified Payday Loan Personal Loan
##   No        0.00000  52.69666       0.00000     0.00000       0.00000
##   Yes     100.00000  47.30334     100.00000   100.00000     100.00000
##
##       Student Loan
##   No       0.00000
##   Yes    100.00000
##
##
## $Payment_of_Min_Amount_Last_Loan_7
## $Payment_of_Min_Amount_Last_Loan_7$contingency_table
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No          0                   0                       0                0
##   Yes      1600                1720                    1432             1528
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No              0   40568             0           0             0
##   Yes          1792   44560          1608        1920          1688
##
##       Student Loan
##   No             0
##   Yes         1584
##
## $Payment_of_Min_Amount_Last_Loan_7$row_totals
##
##    No   Yes
## 40568 59432
##
## $Payment_of_Min_Amount_Last_Loan_7$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    1600                    1720                    1432
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    1528                    1792                   85128
##           Not Specified             Payday Loan           Personal Loan
##                    1608                    1920                    1688
##            Student Loan
##                    1584
```

```
##
## $Payment_of_Min_Amount_Last_Loan_7$grand_total
## [1] 100000
##
## $Payment_of_Min_Amount_Last_Loan_7$cell_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No      0.000               0.000                   0.000            0.000
##   Yes     1.600               1.720                   1.432            1.528
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No          0.000  40.568         0.000       0.000         0.000
##   Yes         1.792  44.560         1.608       1.920         1.688
##
##       Student Loan
##   No         0.000
##   Yes        1.584
##
## $Payment_of_Min_Amount_Last_Loan_7$row_percentage
##
##        Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No    0.000000            0.000000                0.000000         0.000000
##   Yes   2.692152            2.894064                2.409476         2.571006
##
##       Mortgage Loan    No Loan Not Specified Payday Loan Personal Loan
##   No       0.000000 100.000000      0.000000    0.000000      0.000000
##   Yes      3.015211  74.976444      2.705613    3.230583      2.840221
##
##       Student Loan
##   No      0.000000
##   Yes     2.665231
##
## $Payment_of_Min_Amount_Last_Loan_7$col_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No     0.0000              0.0000                  0.0000           0.0000
##   Yes  100.0000            100.0000                100.0000         100.0000
##
##       Mortgage Loan  No Loan Not Specified Payday Loan Personal Loan
##   No         0.0000  47.6553        0.0000      0.0000        0.0000
##   Yes      100.0000  52.3447      100.0000    100.0000      100.0000
##
##       Student Loan
##   No        0.0000
##   Yes     100.0000
##
##
## $Payment_of_Min_Amount_Last_Loan_8
## $Payment_of_Min_Amount_Last_Loan_8$contingency_table
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
```

```
##   No          0                   0                       0                0
##   Yes       720                 872                     760              888
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No              0   40568             0           0             0
##   Yes           736   52240           744         888           720
##
##       Student Loan
##   No             0
##   Yes          864
##
## $Payment_of_Min_Amount_Last_Loan_8$row_totals
##
##    No   Yes
## 40568 59432
##
## $Payment_of_Min_Amount_Last_Loan_8$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     720                     872                     760
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     888                     736                   92808
##           Not Specified             Payday Loan           Personal Loan
##                     744                     888                     720
##            Student Loan
##                     864
##
## $Payment_of_Min_Amount_Last_Loan_8$grand_total
## [1] 100000
##
## $Payment_of_Min_Amount_Last_Loan_8$cell_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No      0.000               0.000                   0.000            0.000
##   Yes     0.720               0.872                   0.760            0.888
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No          0.000  40.568         0.000       0.000         0.000
##   Yes         0.736  52.240         0.744       0.888         0.720
##
##       Student Loan
##   No         0.000
##   Yes        0.864
##
## $Payment_of_Min_Amount_Last_Loan_8$row_percentage
##
##        Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No    0.000000            0.000000                0.000000         0.000000
##   Yes   1.211469            1.467223                1.278772         1.494145
##
##       Mortgage Loan    No Loan Not Specified Payday Loan Personal Loan
```

```
##   No       0.000000 100.000000      0.000000    0.000000      0.000000
##   Yes      1.238390  87.898775      1.251851    1.494145      1.211469
##
##       Student Loan
##   No      0.000000
##   Yes     1.453762
##
## $Payment_of_Min_Amount_Last_Loan_8$col_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No    0.00000             0.00000                 0.00000          0.00000
##   Yes 100.00000           100.00000               100.00000        100.00000
##
##       Mortgage Loan   No Loan Not Specified Payday Loan Personal Loan
##   No        0.00000  43.71175       0.00000     0.00000       0.00000
##   Yes     100.00000  56.28825     100.00000   100.00000     100.00000
##
##       Student Loan
##   No       0.00000
##   Yes    100.00000
##
##
## $Payment_of_Min_Amount_Last_Loan_9
## $Payment_of_Min_Amount_Last_Loan_9$contingency_table
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No          0                   0                       0                0
##   Yes       448                 432                     392              408
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No              0   40568             0           0             0
##   Yes           520   55576           424         464           384
##
##       Student Loan
##   No             0
##   Yes          384
##
## $Payment_of_Min_Amount_Last_Loan_9$row_totals
##
##    No   Yes
## 40568 59432
##
## $Payment_of_Min_Amount_Last_Loan_9$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     448                     432                     392
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     408                     520                   96144
##           Not Specified             Payday Loan           Personal Loan
##                     424                     464                     384
##            Student Loan
```

```
##                     384
##
## $Payment_of_Min_Amount_Last_Loan_9$grand_total
## [1] 100000
##
## $Payment_of_Min_Amount_Last_Loan_9$cell_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No      0.000               0.000                   0.000            0.000
##   Yes     0.448               0.432                   0.392            0.408
##
##       Mortgage Loan No Loan Not Specified Payday Loan Personal Loan
##   No          0.000  40.568         0.000       0.000         0.000
##   Yes         0.520  55.576         0.424       0.464         0.384
##
##       Student Loan
##   No         0.000
##   Yes        0.384
##
## $Payment_of_Min_Amount_Last_Loan_9$row_percentage
##
##         Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No    0.0000000           0.0000000               0.0000000        0.0000000
##   Yes   0.7538027           0.7268811               0.6595773        0.6864989
##
##       Mortgage Loan     No Loan Not Specified Payday Loan Personal Loan
##   No      0.0000000 100.0000000     0.0000000   0.0000000     0.0000000
##   Yes     0.8749495  93.5119128     0.7134204   0.7807242     0.6461166
##
##       Student Loan
##   No     0.0000000
##   Yes    0.6461166
##
## $Payment_of_Min_Amount_Last_Loan_9$col_percentage
##
##       Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan
##   No    0.00000             0.00000                 0.00000          0.00000
##   Yes 100.00000           100.00000               100.00000        100.00000
##
##       Mortgage Loan   No Loan Not Specified Payday Loan Personal Loan
##   No        0.00000  42.19504       0.00000     0.00000       0.00000
##   Yes     100.00000  57.80496     100.00000   100.00000     100.00000
##
##       Student Loan
##   No       0.00000
##   Yes    100.00000
##
##
## $Payment_Behaviour_Occupation
## $Payment_Behaviour_Occupation$contingency_table
##
```

```
##                                   Accountant Architect Developer Doctor
##   High_spent_Large_value_payments         957      1101      1049   1020
##   High_spent_Medium_value_payments       1539      1504      1458   1306
##   High_spent_Small_value_payments         567       569       598    498
##   Low_spent_Large_value_payments          532       507       468    430
##   Low_spent_Medium_value_payments         690       790       749    786
##   Low_spent_Small_value_payments         2459      2353      2398   2528
##
##                                   Engineer Entrepreneur Journalist Lawyer
##   High_spent_Large_value_payments      1050         1000        912   1130
##   High_spent_Medium_value_payments     1612         1389       1358   1367
##   High_spent_Small_value_payments       600          615        576    594
##   Low_spent_Large_value_payments        447          545        452    559
##   Low_spent_Medium_value_payments       741          725        788    791
##   Low_spent_Small_value_payments       2414         2374       2450   2655
##
##                                   Manager Mechanic Media_Manager Musician
##   High_spent_Large_value_payments     1077      912          1105     1038
##   High_spent_Medium_value_payments    1391     1374          1328     1218
##   High_spent_Small_value_payments      496      489           547      488
##   Low_spent_Large_value_payments       530      523           483      502
##   Low_spent_Medium_value_payments      648      806           840      789
##   Low_spent_Small_value_payments      2290     2672          2417     2317
##
##                                   Scientist Teacher Writer
##   High_spent_Large_value_payments       1056    1010   1022
##   High_spent_Medium_value_payments      1334    1333   1301
##   High_spent_Small_value_payments        623     593    400
##   Low_spent_Large_value_payments         461     462    453
##   Low_spent_Medium_value_payments        765     812    772
##   Low_spent_Small_value_payments        2505    2462   2356
##
## $Payment_Behaviour_Occupation$row_totals
##
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                            15439                            20812
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                             8253                             7354
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                            11492                            36650
##
## $Payment_Behaviour_Occupation$col_totals
##
##    Accountant     Architect     Developer        Doctor      Engineer
##          6744          6824          6720          6568          6864
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##          6648          6536          7096          6432          6776
## Media_Manager      Musician     Scientist       Teacher        Writer
##          6720          6352          6744          6672          6304
##
## $Payment_Behaviour_Occupation$grand_total
```

```
## [1] 100000
##
## $Payment_Behaviour_Occupation$cell_percentage
##
##                                    Accountant Architect Developer Doctor
##   High_spent_Large_value_payments       0.957     1.101     1.049  1.020
##   High_spent_Medium_value_payments      1.539     1.504     1.458  1.306
##   High_spent_Small_value_payments       0.567     0.569     0.598  0.498
##   Low_spent_Large_value_payments        0.532     0.507     0.468  0.430
##   Low_spent_Medium_value_payments       0.690     0.790     0.749  0.786
##   Low_spent_Small_value_payments        2.459     2.353     2.398  2.528
##
##                                    Engineer Entrepreneur Journalist Lawyer
##   High_spent_Large_value_payments     1.050        1.000      0.912  1.130
##   High_spent_Medium_value_payments    1.612        1.389      1.358  1.367
##   High_spent_Small_value_payments     0.600        0.615      0.576  0.594
##   Low_spent_Large_value_payments      0.447        0.545      0.452  0.559
##   Low_spent_Medium_value_payments     0.741        0.725      0.788  0.791
##   Low_spent_Small_value_payments      2.414        2.374      2.450  2.655
##
##                                    Manager Mechanic Media_Manager Musician
##   High_spent_Large_value_payments    1.077    0.912         1.105    1.038
##   High_spent_Medium_value_payments   1.391    1.374         1.328    1.218
##   High_spent_Small_value_payments    0.496    0.489         0.547    0.488
##   Low_spent_Large_value_payments     0.530    0.523         0.483    0.502
##   Low_spent_Medium_value_payments    0.648    0.806         0.840    0.789
##   Low_spent_Small_value_payments     2.290    2.672         2.417    2.317
##
##                                    Scientist Teacher Writer
##   High_spent_Large_value_payments      1.056   1.010  1.022
##   High_spent_Medium_value_payments     1.334   1.333  1.301
##   High_spent_Small_value_payments      0.623   0.593  0.400
##   Low_spent_Large_value_payments       0.461   0.462  0.453
##   Low_spent_Medium_value_payments      0.765   0.812  0.772
##   Low_spent_Small_value_payments       2.505   2.462  2.356
##
## $Payment_Behaviour_Occupation$row_percentage
##
##                                    Accountant Architect Developer   Doctor
##   High_spent_Large_value_payments    6.198588  7.131291  6.794482 6.606646
##   High_spent_Medium_value_payments   7.394772  7.226600  7.005574 6.275226
##   High_spent_Small_value_payments    6.870229  6.894463  7.245850 6.034169
##   Low_spent_Large_value_payments     7.234158  6.894207  6.363884 5.847158
##   Low_spent_Medium_value_payments    6.004177  6.874347  6.517577 6.839541
##   Low_spent_Small_value_payments     6.709413  6.420191  6.542974 6.897681
##
##                                    Engineer Entrepreneur Journalist   Lawyer
##   High_spent_Large_value_payments  6.800959     6.477103   5.907118 7.319127
##   High_spent_Medium_value_payments 7.745531     6.674034   6.525082 6.568326
##   High_spent_Small_value_payments  7.270084     7.451836   6.979280 7.197383
##   Low_spent_Large_value_payments   6.078325     7.410933   6.146315 7.601305
```

```
##   Low_spent_Medium_value_payments  6.447964     6.308737   6.856944 6.883049
##   Low_spent_Small_value_payments   6.586630     6.477490   6.684857 7.244202
##
##                                     Manager Mechanic Media_Manager Musician
##   High_spent_Large_value_payments  6.975840 5.907118      7.157199 6.723233
##   High_spent_Medium_value_payments 6.683644 6.601960      6.380934 5.852393
##   High_spent_Small_value_payments  6.009936 5.925118      6.627893 5.913001
##   Low_spent_Large_value_payments   7.206962 7.111776      6.567854 6.826217
##   Low_spent_Medium_value_payments  5.638705 7.013575      7.309433 6.865646
##   Low_spent_Small_value_payments   6.248295 7.290587      6.594816 6.321965
##
##                                    Scientist  Teacher   Writer
##   High_spent_Large_value_payments   6.839821 6.541874 6.619600
##   High_spent_Medium_value_payments  6.409764 6.404959 6.251201
##   High_spent_Small_value_payments   7.548770 7.185266 4.846722
##   Low_spent_Large_value_payments    6.268697 6.282295 6.159913
##   Low_spent_Medium_value_payments   6.656805 7.065785 6.717717
##   Low_spent_Small_value_payments    6.834925 6.717599 6.428377
##
## $Payment_Behaviour_Occupation$col_percentage
##
##                                    Accountant Architect Developer    Doctor
##   High_spent_Large_value_payments   14.190391 16.134232 15.610119 15.529842
##   High_spent_Medium_value_payments  22.820285 22.039859 21.696429 19.884287
##   High_spent_Small_value_payments    8.407473  8.338218  8.898810  7.582217
##   Low_spent_Large_value_payments     7.888493  7.429660  6.964286  6.546894
##   Low_spent_Medium_value_payments   10.231317 11.576788 11.145833 11.967113
##   Low_spent_Small_value_payments    36.462040 34.481243 35.684524 38.489647
##
##                                     Engineer Entrepreneur Journalist    Lawyer
##   High_spent_Large_value_payments  15.297203    15.042118  13.953488 15.924464
##   High_spent_Medium_value_payments 23.484848    20.893502  20.777234 19.264374
##   High_spent_Small_value_payments   8.741259     9.250903   8.812729  8.370913
##   Low_spent_Large_value_payments    6.512238     8.197954   6.915545  7.877678
##   Low_spent_Medium_value_payments  10.795455    10.905535  12.056304 11.147125
##   Low_spent_Small_value_payments   35.168998    35.709988  37.484700 37.415445
##
##                                      Manager  Mechanic Media_Manager  Musician
##   High_spent_Large_value_payments  16.744403 13.459268     16.443452 16.341310
##   High_spent_Medium_value_payments 21.626244 20.277450     19.761905 19.175063
##   High_spent_Small_value_payments   7.711443  7.216647      8.139881  7.682620
##   Low_spent_Large_value_payments    8.240050  7.718418      7.187500  7.903023
##   Low_spent_Medium_value_payments  10.074627 11.894923     12.500000 12.421285
##   Low_spent_Small_value_payments   35.603234 39.433294     35.967262 36.476700
##
##                                    Scientist   Teacher    Writer
##   High_spent_Large_value_payments  15.658363 15.137890 16.211929
##   High_spent_Medium_value_payments 19.780546 19.979017 20.637690
##   High_spent_Small_value_payments   9.237841  8.887890  6.345178
##   Low_spent_Large_value_payments    6.835706  6.924460  7.185914
##   Low_spent_Medium_value_payments  11.343416 12.170264 12.246193
```

```
##   Low_spent_Small_value_payments   37.144128 36.900480 37.373096
##
##
## $Payment_Behaviour_Credit_Score
## $Payment_Behaviour_Credit_Score$contingency_table
##
##                                     Good  Poor Standard
##   High_spent_Large_value_payments   3734  3127     8578
##   High_spent_Medium_value_payments  3973  5701    11138
##   High_spent_Small_value_payments   1459  2258     4536
##   Low_spent_Large_value_payments    1355  1920     4079
##   Low_spent_Medium_value_payments   2316  3396     5780
##   Low_spent_Small_value_payments    4991 12596    19063
##
## $Payment_Behaviour_Credit_Score$row_totals
##
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                            15439                            20812
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                             8253                             7354
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                            11492                            36650
##
## $Payment_Behaviour_Credit_Score$col_totals
##
##     Good     Poor Standard
##    17828    28998    53174
##
## $Payment_Behaviour_Credit_Score$grand_total
## [1] 100000
##
## $Payment_Behaviour_Credit_Score$cell_percentage
##
##                                      Good   Poor Standard
##   High_spent_Large_value_payments   3.734  3.127    8.578
##   High_spent_Medium_value_payments  3.973  5.701   11.138
##   High_spent_Small_value_payments   1.459  2.258    4.536
##   Low_spent_Large_value_payments    1.355  1.920    4.079
##   Low_spent_Medium_value_payments   2.316  3.396    5.780
##   Low_spent_Small_value_payments    4.991 12.596   19.063
##
## $Payment_Behaviour_Credit_Score$row_percentage
##
##                                         Good     Poor Standard
##   High_spent_Large_value_payments  24.18550 20.25390 55.56059
##   High_spent_Medium_value_payments 19.08995 27.39285 53.51720
##   High_spent_Small_value_payments  17.67842 27.35975 54.96183
##   Low_spent_Large_value_payments   18.42535 26.10824 55.46641
##   Low_spent_Medium_value_payments  20.15315 29.55099 50.29586
##   Low_spent_Small_value_payments   13.61801 34.36835 52.01364
##
```

```
## $Payment_Behaviour_Credit_Score$col_percentage
##
##                                          Good      Poor  Standard
##   High_spent_Large_value_payments  20.944582 10.783502 16.131944
##   High_spent_Medium_value_payments 22.285169 19.659977 20.946327
##   High_spent_Small_value_payments   8.183756  7.786744  8.530485
##   Low_spent_Large_value_payments    7.600404  6.621146  7.671042
##   Low_spent_Medium_value_payments  12.990801 11.711152 10.869974
##   Low_spent_Small_value_payments   27.995288 43.437478 35.850228
##
##
## $Payment_Behaviour_Last_Loan_1
## $Payment_Behaviour_Last_Loan_1$contingency_table
##
##                                    Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments       1399                1493
##   High_spent_Medium_value_payments      1813                2055
##   High_spent_Small_value_payments        814                 761
##   Low_spent_Large_value_payments         638                 775
##   Low_spent_Medium_value_payments       1016                1053
##   Low_spent_Small_value_payments        3648                3663
##
##                                    Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                     1458             1381
##   High_spent_Medium_value_payments                    2036             2076
##   High_spent_Small_value_payments                      769              742
##   Low_spent_Large_value_payments                       740              782
##   Low_spent_Medium_value_payments                     1232              998
##   Low_spent_Small_value_payments                      3733             3821
##
##                                    Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments           1365    2351          1498
##   High_spent_Medium_value_payments          1970    2578          2048
##   High_spent_Small_value_payments            794     987           890
##   Low_spent_Large_value_payments             686     910           825
##   Low_spent_Medium_value_payments           1201    1489          1082
##   Low_spent_Small_value_payments            3592    3093          3673
##
##                                    Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments         1555          1314         1625
##   High_spent_Medium_value_payments        2080          1944         2212
##   High_spent_Small_value_payments          876           853          767
##   Low_spent_Large_value_payments           723           658          617
##   Low_spent_Medium_value_payments         1145          1166         1110
##   Low_spent_Small_value_payments          3725          4001         3701
##
## $Payment_Behaviour_Last_Loan_1$row_totals
##
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                            15439                            20812
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
```

```
##                             8253                             7354
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                            11492                            36650
##
## $Payment_Behaviour_Last_Loan_1$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    9328                    9800                    9968
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    9800                    9608                   11408
##           Not Specified             Payday Loan           Personal Loan
##                   10016                   10104                    9936
##            Student Loan
##                   10032
##
## $Payment_Behaviour_Last_Loan_1$grand_total
## [1] 100000
##
## $Payment_Behaviour_Last_Loan_1$cell_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments      1.399               1.493
##   High_spent_Medium_value_payments     1.813               2.055
##   High_spent_Small_value_payments      0.814               0.761
##   Low_spent_Large_value_payments       0.638               0.775
##   Low_spent_Medium_value_payments      1.016               1.053
##   Low_spent_Small_value_payments       3.648               3.663
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                    1.458            1.381
##   High_spent_Medium_value_payments                   2.036            2.076
##   High_spent_Small_value_payments                    0.769            0.742
##   Low_spent_Large_value_payments                     0.740            0.782
##   Low_spent_Medium_value_payments                    1.232            0.998
##   Low_spent_Small_value_payments                     3.733            3.821
##
##                                   Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments         1.365   2.351         1.498
##   High_spent_Medium_value_payments        1.970   2.578         2.048
##   High_spent_Small_value_payments         0.794   0.987         0.890
##   Low_spent_Large_value_payments          0.686   0.910         0.825
##   Low_spent_Medium_value_payments         1.201   1.489         1.082
##   Low_spent_Small_value_payments          3.592   3.093         3.673
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments       1.555         1.314        1.625
##   High_spent_Medium_value_payments      2.080         1.944        2.212
##   High_spent_Small_value_payments       0.876         0.853        0.767
##   Low_spent_Large_value_payments        0.723         0.658        0.617
##   Low_spent_Medium_value_payments       1.145         1.166        1.110
##   Low_spent_Small_value_payments        3.725         4.001        3.701
```

```
##
## $Payment_Behaviour_Last_Loan_1$row_percentage
##
##                                    Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments   9.061468            9.670315
##   High_spent_Medium_value_payments  8.711320            9.874111
##   High_spent_Small_value_payments   9.863080            9.220889
##   Low_spent_Large_value_payments    8.675551           10.538482
##   Low_spent_Medium_value_payments   8.840933            9.162896
##   Low_spent_Small_value_payments    9.953615            9.994543
##
##                                    Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                 9.443617         8.944880
##   High_spent_Medium_value_payments                9.782818         9.975014
##   High_spent_Small_value_payments                 9.317824         8.990670
##   Low_spent_Large_value_payments                 10.062551        10.633669
##   Low_spent_Medium_value_payments                10.720501         8.684302
##   Low_spent_Small_value_payments                 10.185539        10.425648
##
##                                    Mortgage Loan   No Loan Not Specified
##   High_spent_Large_value_payments       8.841246 15.227670      9.702701
##   High_spent_Medium_value_payments      9.465693 12.387084      9.840477
##   High_spent_Small_value_payments       9.620744 11.959288     10.783957
##   Low_spent_Large_value_payments        9.328257 12.374218     11.218385
##   Low_spent_Medium_value_payments      10.450748 12.956840      9.415245
##   Low_spent_Small_value_payments        9.800819  8.439291     10.021828
##
##                                    Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments    10.071896      8.510914    10.525293
##   High_spent_Medium_value_payments    9.994234      9.340765    10.628484
##   High_spent_Small_value_payments    10.614322     10.335636     9.293590
##   Low_spent_Large_value_payments      9.831384      8.947512     8.389992
##   Low_spent_Medium_value_payments     9.963453     10.146189     9.658893
##   Low_spent_Small_value_payments     10.163711     10.916780    10.098226
##
## $Payment_Behaviour_Last_Loan_1$col_percentage
##
##                                    Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments  14.997856           15.234694
##   High_spent_Medium_value_payments 19.436106           20.969388
##   High_spent_Small_value_payments   8.726415            7.765306
##   Low_spent_Large_value_payments    6.839623            7.908163
##   Low_spent_Medium_value_payments  10.891938           10.744898
##   Low_spent_Small_value_payments   39.108062           37.377551
##
##                                    Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                14.626806        14.091837
##   High_spent_Medium_value_payments               20.425361        21.183673
##   High_spent_Small_value_payments                 7.714687         7.571429
##   Low_spent_Large_value_payments                  7.423756         7.979592
##   Low_spent_Medium_value_payments                12.359551        10.183673
```

```
##   Low_spent_Small_value_payments                 37.449839        38.989796
##
##                                     Mortgage Loan   No Loan Not Specified
##   High_spent_Large_value_payments      14.206911 20.608345     14.956070
##   High_spent_Medium_value_payments     20.503747 22.598177     20.447284
##   High_spent_Small_value_payments       8.263947  8.651823      8.885783
##   Low_spent_Large_value_payments        7.139883  7.976858      8.236821
##   Low_spent_Medium_value_payments      12.500000 13.052244     10.802716
##   Low_spent_Small_value_payments       37.385512 27.112553     36.671326
##
##                                     Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments    15.389945     13.224638    16.198166
##   High_spent_Medium_value_payments   20.585907     19.565217    22.049442
##   High_spent_Small_value_payments     8.669834      8.584944     7.645534
##   Low_spent_Large_value_payments      7.155582      6.622383     6.150319
##   Low_spent_Medium_value_payments    11.332146     11.735105    11.064593
##   Low_spent_Small_value_payments     36.866587     40.267713    36.891946
##
##
## $Payment_Behaviour_Last_Loan_2
## $Payment_Behaviour_Last_Loan_2$contingency_table
##
##                                     Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments       1272                1136
##   High_spent_Medium_value_payments      1741                1687
##   High_spent_Small_value_payments        607                 678
##   Low_spent_Large_value_payments         585                 600
##   Low_spent_Medium_value_payments        906                1017
##   Low_spent_Small_value_payments        3361                3410
##
##                                     Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                     1080             1279
##   High_spent_Medium_value_payments                    1706             1702
##   High_spent_Small_value_payments                      739              778
##   Low_spent_Large_value_payments                       584              632
##   Low_spent_Medium_value_payments                      922              910
##   Low_spent_Small_value_payments                      3241             3475
##
##                                     Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments           1136    4588          1251
##   High_spent_Medium_value_payments          1791    4995          1771
##   High_spent_Small_value_payments            718    1917           662
##   Low_spent_Large_value_payments             562    1824           679
##   Low_spent_Medium_value_payments            943    2799           993
##   Low_spent_Small_value_payments            3258    6413          3372
##
##                                     Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments         1287          1151         1259
##   High_spent_Medium_value_payments        1871          1827         1721
##   High_spent_Small_value_payments          801           623          730
##   Low_spent_Large_value_payments           666           553          669
```

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##   Low_spent_Medium_value_payments         1118           908          976
##   Low_spent_Small_value_payments          3361          3578         3181
##
## $Payment_Behaviour_Last_Loan_2$row_totals
##
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                            15439                            20812
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                             8253                             7354
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                            11492                            36650
##
## $Payment_Behaviour_Last_Loan_2$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    8472                    8528                    8272
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    8776                    8408                   22536
##           Not Specified             Payday Loan           Personal Loan
##                    8728                    9104                    8640
##            Student Loan
##                    8536
##
## $Payment_Behaviour_Last_Loan_2$grand_total
## [1] 100000
##
## $Payment_Behaviour_Last_Loan_2$cell_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments      1.272               1.136
##   High_spent_Medium_value_payments     1.741               1.687
##   High_spent_Small_value_payments      0.607               0.678
##   Low_spent_Large_value_payments       0.585               0.600
##   Low_spent_Medium_value_payments      0.906               1.017
##   Low_spent_Small_value_payments       3.361               3.410
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                    1.080            1.279
##   High_spent_Medium_value_payments                   1.706            1.702
##   High_spent_Small_value_payments                    0.739            0.778
##   Low_spent_Large_value_payments                     0.584            0.632
##   Low_spent_Medium_value_payments                    0.922            0.910
##   Low_spent_Small_value_payments                     3.241            3.475
##
##                                   Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments          1.136   4.588         1.251
##   High_spent_Medium_value_payments         1.791   4.995         1.771
##   High_spent_Small_value_payments          0.718   1.917         0.662
##   Low_spent_Large_value_payments           0.562   1.824         0.679
##   Low_spent_Medium_value_payments          0.943   2.799         0.993
##   Low_spent_Small_value_payments           3.258   6.413         3.372
```

```
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments        1.287         1.151        1.259
##   High_spent_Medium_value_payments       1.871         1.827        1.721
##   High_spent_Small_value_payments        0.801         0.623        0.730
##   Low_spent_Large_value_payments         0.666         0.553        0.669
##   Low_spent_Medium_value_payments        1.118         0.908        0.976
##   Low_spent_Small_value_payments         3.361         3.578        3.181
##
## $Payment_Behaviour_Last_Loan_2$row_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments   8.238876            7.357990
##   High_spent_Medium_value_payments  8.365366            8.105900
##   High_spent_Small_value_payments   7.354901            8.215194
##   Low_spent_Large_value_payments    7.954855            8.158825
##   Low_spent_Medium_value_payments   7.883745            8.849635
##   Low_spent_Small_value_payments    9.170532            9.304229
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                 6.995272         8.284215
##   High_spent_Medium_value_payments                8.197194         8.177974
##   High_spent_Small_value_payments                 8.954320         9.426875
##   Low_spent_Large_value_payments                  7.941256         8.593962
##   Low_spent_Medium_value_payments                 8.022973         7.918552
##   Low_spent_Small_value_payments                  8.843111         9.481583
##
##                                   Mortgage Loan   No Loan Not Specified
##   High_spent_Large_value_payments       7.357990 29.716951      8.102856
##   High_spent_Medium_value_payments      8.605612 24.000577      8.509514
##   High_spent_Small_value_payments       8.699867 23.227917      8.021326
##   Low_spent_Large_value_payments        7.642100 24.802828      9.233070
##   Low_spent_Medium_value_payments       8.205708 24.356074      8.640794
##   Low_spent_Small_value_payments        8.889495 17.497954      9.200546
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments     8.336032      7.455146     8.154673
##   High_spent_Medium_value_payments    8.990006      8.778589     8.269268
##   High_spent_Small_value_payments     9.705562      7.548770     8.845268
##   Low_spent_Large_value_payments      9.056296      7.519717     9.097090
##   Low_spent_Medium_value_payments     9.728507      7.901149     8.492865
##   Low_spent_Small_value_payments      9.170532      9.762619     8.679400
##
## $Payment_Behaviour_Last_Loan_2$col_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments  15.014164           13.320826
##   High_spent_Medium_value_payments 20.550047           19.781895
##   High_spent_Small_value_payments   7.164778            7.950281
##   Low_spent_Large_value_payments    6.905099            7.035647
##   Low_spent_Medium_value_payments  10.694051           11.925422
```

```
##   Low_spent_Small_value_payments   39.671860           39.985929
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                13.056093        14.573838
##   High_spent_Medium_value_payments               20.623791        19.393801
##   High_spent_Small_value_payments                 8.933752         8.865087
##   Low_spent_Large_value_payments                  7.059961         7.201459
##   Low_spent_Medium_value_payments                11.146035        10.369189
##   Low_spent_Small_value_payments                 39.180368        39.596627
##
##                                   Mortgage Loan   No Loan Not Specified
##   High_spent_Large_value_payments      13.510942 20.358537     14.333181
##   High_spent_Medium_value_payments     21.301142 22.164537     20.291017
##   High_spent_Small_value_payments       8.539486  8.506390      7.584785
##   Low_spent_Large_value_payments        6.684110  8.093717      7.779560
##   Low_spent_Medium_value_payments      11.215509 12.420128     11.377177
##   Low_spent_Small_value_payments       38.748811 28.456692     38.634280
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments    14.136643     13.321759    14.749297
##   High_spent_Medium_value_payments   20.551406     21.145833    20.161668
##   High_spent_Small_value_payments     8.798330      7.210648     8.552015
##   Low_spent_Large_value_payments      7.315466      6.400463     7.837395
##   Low_spent_Medium_value_payments    12.280316     10.509259    11.433927
##   Low_spent_Small_value_payments     36.917838     41.412037    37.265698
##
##
## $Payment_Behaviour_Last_Loan_3
## $Payment_Behaviour_Last_Loan_3$contingency_table
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments        895                 987
##   High_spent_Medium_value_payments      1381                1573
##   High_spent_Small_value_payments        517                 568
##   Low_spent_Large_value_payments         481                 447
##   Low_spent_Medium_value_payments        790                 753
##   Low_spent_Small_value_payments        2744                2992
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                     843              923
##   High_spent_Medium_value_payments                   1493             1445
##   High_spent_Small_value_payments                     540              539
##   Low_spent_Large_value_payments                      546              483
##   Low_spent_Medium_value_payments                     783              639
##   Low_spent_Small_value_payments                     2723             2611
##
##                                   Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments            927    6996           973
##   High_spent_Medium_value_payments          1335    8256          1322
##   High_spent_Small_value_payments            532    3294           625
##   Low_spent_Large_value_payments             462    3035           470
```

```
##   Low_spent_Medium_value_payments            748    4766           708
##   Low_spent_Small_value_payments            2868   11901          2982
##
##                                     Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments          884           953         1058
##   High_spent_Medium_value_payments        1359          1351         1297
##   High_spent_Small_value_payments          542           498          598
##   Low_spent_Large_value_payments           502           378          550
##   Low_spent_Medium_value_payments          823           713          769
##   Low_spent_Small_value_payments          2658          2515         2656
##
## $Payment_Behaviour_Last_Loan_3$row_totals
##
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                            15439                            20812
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                             8253                             7354
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                            11492                            36650
##
## $Payment_Behaviour_Last_Loan_3$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    6808                    7320                    6928
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    6640                    6872                   38248
##           Not Specified             Payday Loan           Personal Loan
##                    7080                    6768                    6408
##            Student Loan
##                    6928
##
## $Payment_Behaviour_Last_Loan_3$grand_total
## [1] 100000
##
## $Payment_Behaviour_Last_Loan_3$cell_percentage
##
##                                     Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments      0.895               0.987
##   High_spent_Medium_value_payments     1.381               1.573
##   High_spent_Small_value_payments      0.517               0.568
##   Low_spent_Large_value_payments       0.481               0.447
##   Low_spent_Medium_value_payments      0.790               0.753
##   Low_spent_Small_value_payments       2.744               2.992
##
##                                     Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                    0.843            0.923
##   High_spent_Medium_value_payments                   1.493            1.445
##   High_spent_Small_value_payments                    0.540            0.539
##   Low_spent_Large_value_payments                     0.546            0.483
##   Low_spent_Medium_value_payments                    0.783            0.639
##   Low_spent_Small_value_payments                     2.723            2.611
```

```
##
##                                   Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments          0.927   6.996         0.973
##   High_spent_Medium_value_payments         1.335   8.256         1.322
##   High_spent_Small_value_payments          0.532   3.294         0.625
##   Low_spent_Large_value_payments           0.462   3.035         0.470
##   Low_spent_Medium_value_payments          0.748   4.766         0.708
##   Low_spent_Small_value_payments           2.868  11.901         2.982
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments        0.884         0.953        1.058
##   High_spent_Medium_value_payments       1.359         1.351        1.297
##   High_spent_Small_value_payments        0.542         0.498        0.598
##   Low_spent_Large_value_payments         0.502         0.378        0.550
##   Low_spent_Medium_value_payments        0.823         0.713        0.769
##   Low_spent_Small_value_payments         2.658         2.515        2.656
##
## $Payment_Behaviour_Last_Loan_3$row_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments   5.797008            6.392901
##   High_spent_Medium_value_payments  6.635595            7.558140
##   High_spent_Small_value_payments   6.264389            6.882346
##   Low_spent_Large_value_payments    6.540658            6.078325
##   Low_spent_Medium_value_payments   6.874347            6.552384
##   Low_spent_Small_value_payments    7.487040            8.163711
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                 5.460198         5.978366
##   High_spent_Medium_value_payments                7.173746         6.943110
##   High_spent_Small_value_payments                 6.543075         6.530958
##   Low_spent_Large_value_payments                  7.424531         6.567854
##   Low_spent_Medium_value_payments                 6.813435         5.560390
##   Low_spent_Small_value_payments                  7.429741         7.124147
##
##                                   Mortgage Loan   No Loan Not Specified
##   High_spent_Large_value_payments       6.004275 45.313816      6.302222
##   High_spent_Medium_value_payments      6.414569 39.669421      6.352105
##   High_spent_Small_value_payments       6.446141 39.912759      7.573004
##   Low_spent_Large_value_payments        6.282295 41.270057      6.391080
##   Low_spent_Medium_value_payments       6.508876 41.472329      6.160808
##   Low_spent_Small_value_payments        7.825375 32.472033      8.136426
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments     5.725759      6.172680     6.852775
##   High_spent_Medium_value_payments    6.529887      6.491447     6.231982
##   High_spent_Small_value_payments     6.567309      6.034169     7.245850
##   Low_spent_Large_value_payments      6.826217      5.140060     7.478923
##   Low_spent_Medium_value_payments     7.161504      6.204316     6.691612
##   Low_spent_Small_value_payments      7.252387      6.862210     7.246930
##
```

```
## $Payment_Behaviour_Last_Loan_3$col_percentage
##
##                                    Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments  13.146298           13.483607
##   High_spent_Medium_value_payments 20.284959           21.489071
##   High_spent_Small_value_payments   7.594007            7.759563
##   Low_spent_Large_value_payments    7.065217            6.106557
##   Low_spent_Medium_value_payments  11.603995           10.286885
##   Low_spent_Small_value_payments   40.305523           40.874317
##
##                                    Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                12.168014        13.900602
##   High_spent_Medium_value_payments               21.550231        21.762048
##   High_spent_Small_value_payments                 7.794457         8.117470
##   Low_spent_Large_value_payments                  7.881062         7.274096
##   Low_spent_Medium_value_payments                11.301963         9.623494
##   Low_spent_Small_value_payments                 39.304273        39.322289
##
##                                    Mortgage Loan   No Loan Not Specified
##   High_spent_Large_value_payments      13.489523 18.291152     13.742938
##   High_spent_Medium_value_payments     19.426659 21.585442     18.672316
##   High_spent_Small_value_payments       7.741560  8.612215      8.827684
##   Low_spent_Large_value_payments        6.722934  7.935055      6.638418
##   Low_spent_Medium_value_payments      10.884750 12.460782     10.000000
##   Low_spent_Small_value_payments       41.734575 31.115352     42.118644
##
##                                    Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments    13.061466     14.872035    15.271363
##   High_spent_Medium_value_payments   20.079787     21.083021    18.721132
##   High_spent_Small_value_payments     8.008274      7.771536     8.631640
##   Low_spent_Large_value_payments      7.417258      5.898876     7.938799
##   Low_spent_Medium_value_payments    12.160165     11.126717    11.099885
##   Low_spent_Small_value_payments     39.273050     39.247815    38.337182
##
##
## $Payment_Behaviour_Last_Loan_4
## $Payment_Behaviour_Last_Loan_4$contingency_table
##
##                                    Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments        560                 741
##   High_spent_Medium_value_payments      1035                1120
##   High_spent_Small_value_payments        263                 476
##   Low_spent_Large_value_payments         338                 430
##   Low_spent_Medium_value_payments        502                 628
##   Low_spent_Small_value_payments        2022                2269
##
##                                    Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                      614              579
##   High_spent_Medium_value_payments                    1062             1054
##   High_spent_Small_value_payments                      441              461
##   Low_spent_Large_value_payments                       338              360
```

```
##   Low_spent_Medium_value_payments                      516              604
##   Low_spent_Small_value_payments                      2061             2142
##
##                                   Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments            643    9804           544
##   High_spent_Medium_value_payments          1095   11769           995
##   High_spent_Small_value_payments            484    4551           440
##   Low_spent_Large_value_payments             261    4260           322
##   Low_spent_Medium_value_payments            453    6512           601
##   Low_spent_Small_value_payments            2232   17104          2218
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments          633           676          645
##   High_spent_Medium_value_payments         981           882          819
##   High_spent_Small_value_payments          354           405          378
##   Low_spent_Large_value_payments           367           323          355
##   Low_spent_Medium_value_payments          530           559          587
##   Low_spent_Small_value_payments          2423          2011         2168
##
## $Payment_Behaviour_Last_Loan_4$row_totals
##
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                            15439                            20812
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                             8253                             7354
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                            11492                            36650
##
## $Payment_Behaviour_Last_Loan_4$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    4720                    5664                    5032
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    5200                    5168                   54000
##           Not Specified             Payday Loan           Personal Loan
##                    5120                    5288                    4856
##            Student Loan
##                    4952
##
## $Payment_Behaviour_Last_Loan_4$grand_total
## [1] 100000
##
## $Payment_Behaviour_Last_Loan_4$cell_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments      0.560               0.741
##   High_spent_Medium_value_payments     1.035               1.120
##   High_spent_Small_value_payments      0.263               0.476
##   Low_spent_Large_value_payments       0.338               0.430
##   Low_spent_Medium_value_payments      0.502               0.628
##   Low_spent_Small_value_payments       2.022               2.269
```

```
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                    0.614            0.579
##   High_spent_Medium_value_payments                   1.062            1.054
##   High_spent_Small_value_payments                    0.441            0.461
##   Low_spent_Large_value_payments                     0.338            0.360
##   Low_spent_Medium_value_payments                    0.516            0.604
##   Low_spent_Small_value_payments                     2.061            2.142
##
##                                   Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments         0.643   9.804         0.544
##   High_spent_Medium_value_payments        1.095  11.769         0.995
##   High_spent_Small_value_payments         0.484   4.551         0.440
##   Low_spent_Large_value_payments          0.261   4.260         0.322
##   Low_spent_Medium_value_payments         0.453   6.512         0.601
##   Low_spent_Small_value_payments          2.232  17.104         2.218
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments       0.633         0.676        0.645
##   High_spent_Medium_value_payments      0.981         0.882        0.819
##   High_spent_Small_value_payments       0.354         0.405        0.378
##   Low_spent_Large_value_payments        0.367         0.323        0.355
##   Low_spent_Medium_value_payments       0.530         0.559        0.587
##   Low_spent_Small_value_payments        2.423         2.011        2.168
##
## $Payment_Behaviour_Last_Loan_4$row_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments   3.627178            4.799534
##   High_spent_Medium_value_payments  4.973092            5.381511
##   High_spent_Small_value_payments   3.186720            5.767600
##   Low_spent_Large_value_payments    4.596138            5.847158
##   Low_spent_Medium_value_payments   4.368256            5.464671
##   Low_spent_Small_value_payments    5.517053            6.190996
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                3.976942         3.750243
##   High_spent_Medium_value_payments               5.102825         5.064386
##   High_spent_Small_value_payments                5.343511         5.585848
##   Low_spent_Large_value_payments                 4.596138         4.895295
##   Low_spent_Medium_value_payments                4.490080         5.255830
##   Low_spent_Small_value_payments                 5.623465         5.844475
##
##                                   Mortgage Loan   No Loan Not Specified
##   High_spent_Large_value_payments       4.164778 63.501522      3.523544
##   High_spent_Medium_value_payments      5.261388 56.549106      4.780896
##   High_spent_Small_value_payments       5.864534 55.143584      5.331395
##   Low_spent_Large_value_payments        3.549089 57.927658      4.378569
##   Low_spent_Medium_value_payments       3.941873 56.665506      5.229725
##   Low_spent_Small_value_payments        6.090041 46.668486      6.051842
##
```

```
##                                    Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments     4.100006      4.378522     4.177732
##   High_spent_Medium_value_payments    4.713627      4.237940     3.935230
##   High_spent_Small_value_payments     4.289349      4.907306     4.580153
##   Low_spent_Large_value_payments      4.990481      4.392168     4.827305
##   Low_spent_Medium_value_payments     4.611904      4.864253     5.107901
##   Low_spent_Small_value_payments      6.611187      5.487040     5.915416
##
## $Payment_Behaviour_Last_Loan_4$col_percentage
##
##                                    Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments  11.864407           13.082627
##   High_spent_Medium_value_payments 21.927966           19.774011
##   High_spent_Small_value_payments   5.572034            8.403955
##   Low_spent_Large_value_payments    7.161017            7.591808
##   Low_spent_Medium_value_payments  10.635593           11.087571
##   Low_spent_Small_value_payments   42.838983           40.060028
##
##                                    Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                12.201908        11.134615
##   High_spent_Medium_value_payments               21.104928        20.269231
##   High_spent_Small_value_payments                 8.763911         8.865385
##   Low_spent_Large_value_payments                  6.717011         6.923077
##   Low_spent_Medium_value_payments                10.254372        11.615385
##   Low_spent_Small_value_payments                 40.957870        41.192308
##
##                                    Mortgage Loan   No Loan Not Specified
##   High_spent_Large_value_payments      12.441950 18.155556     10.625000
##   High_spent_Medium_value_payments     21.188080 21.794444     19.433594
##   High_spent_Small_value_payments       9.365325  8.427778      8.593750
##   Low_spent_Large_value_payments        5.050310  7.888889      6.289063
##   Low_spent_Medium_value_payments       8.765480 12.059259     11.738281
##   Low_spent_Small_value_payments       43.188854 31.674074     43.320312
##
##                                    Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments    11.970499     13.920923    13.025040
##   High_spent_Medium_value_payments   18.551437     18.163097    16.538772
##   High_spent_Small_value_payments     6.694402      8.340198     7.633279
##   Low_spent_Large_value_payments      6.940242      6.651565     7.168821
##   Low_spent_Medium_value_payments    10.022693     11.511532    11.853796
##   Low_spent_Small_value_payments     45.820726     41.412685    43.780291
##
##
## $Payment_Behaviour_Last_Loan_5
## $Payment_Behaviour_Last_Loan_5$contingency_table
##
##                                    Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments        277                 359
##   High_spent_Medium_value_payments       674                 596
##   High_spent_Small_value_payments        267                 212
##   Low_spent_Large_value_payments         194                 253
```

```
##   Low_spent_Medium_value_payments        357                 392
##   Low_spent_Small_value_payments        1655                1684
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                     287              375
##   High_spent_Medium_value_payments                    597              588
##   High_spent_Small_value_payments                     219              247
##   Low_spent_Large_value_payments                      271              205
##   Low_spent_Medium_value_payments                     475              397
##   Low_spent_Small_value_payments                     1679             1588
##
##                                   Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments           302   12549           312
##   High_spent_Medium_value_payments          679   15031           673
##   High_spent_Small_value_payments           317    5885           238
##   Low_spent_Large_value_payments            233    5342           198
##   Low_spent_Medium_value_payments           397    8178           324
##   Low_spent_Small_value_payments           1512   22471          1527
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments         320           367          291
##   High_spent_Medium_value_payments        666           628          680
##   High_spent_Small_value_payments         263           328          277
##   Low_spent_Large_value_payments          195           244          219
##   Low_spent_Medium_value_payments         295           359          318
##   Low_spent_Small_value_payments         1605          1594         1335
##
## $Payment_Behaviour_Last_Loan_5$row_totals
##
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                            15439                            20812
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                             8253                             7354
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                            11492                            36650
##
## $Payment_Behaviour_Last_Loan_5$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    3424                    3496                    3528
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    3400                    3440                   69456
##           Not Specified             Payday Loan           Personal Loan
##                    3272                    3344                    3520
##            Student Loan
##                    3120
##
## $Payment_Behaviour_Last_Loan_5$grand_total
## [1] 100000
##
## $Payment_Behaviour_Last_Loan_5$cell_percentage
```

```
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments      0.277               0.359
##   High_spent_Medium_value_payments     0.674               0.596
##   High_spent_Small_value_payments      0.267               0.212
##   Low_spent_Large_value_payments       0.194               0.253
##   Low_spent_Medium_value_payments      0.357               0.392
##   Low_spent_Small_value_payments       1.655               1.684
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                    0.287            0.375
##   High_spent_Medium_value_payments                   0.597            0.588
##   High_spent_Small_value_payments                    0.219            0.247
##   Low_spent_Large_value_payments                     0.271            0.205
##   Low_spent_Medium_value_payments                    0.475            0.397
##   Low_spent_Small_value_payments                     1.679            1.588
##
##                                   Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments         0.302  12.549         0.312
##   High_spent_Medium_value_payments        0.679  15.031         0.673
##   High_spent_Small_value_payments         0.317   5.885         0.238
##   Low_spent_Large_value_payments          0.233   5.342         0.198
##   Low_spent_Medium_value_payments         0.397   8.178         0.324
##   Low_spent_Small_value_payments          1.512  22.471         1.527
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments       0.320         0.367        0.291
##   High_spent_Medium_value_payments      0.666         0.628        0.680
##   High_spent_Small_value_payments       0.263         0.328        0.277
##   Low_spent_Large_value_payments        0.195         0.244        0.219
##   Low_spent_Medium_value_payments       0.295         0.359        0.318
##   Low_spent_Small_value_payments        1.605         1.594        1.335
##
## $Payment_Behaviour_Last_Loan_5$row_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments   1.794158            2.325280
##   High_spent_Medium_value_payments  3.238516            2.863732
##   High_spent_Small_value_payments   3.235187            2.568763
##   Low_spent_Large_value_payments    2.638020            3.440305
##   Low_spent_Medium_value_payments   3.106509            3.411069
##   Low_spent_Small_value_payments    4.515689            4.594816
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                1.858929         2.428914
##   High_spent_Medium_value_payments               2.868537         2.825293
##   High_spent_Small_value_payments                2.653581         2.992851
##   Low_spent_Large_value_payments                 3.685069         2.787599
##   Low_spent_Medium_value_payments                4.133310         3.454577
##   Low_spent_Small_value_payments                 4.581173         4.332879
##
```

```
##                                   Mortgage Loan   No Loan Not Specified
##   High_spent_Large_value_payments       1.956085 81.281171      2.020856
##   High_spent_Medium_value_payments      3.262541 72.222756      3.233711
##   High_spent_Small_value_payments       3.841028 71.307403      2.883800
##   Low_spent_Large_value_payments        3.168344 72.640740      2.692412
##   Low_spent_Medium_value_payments       3.454577 71.162548      2.819353
##   Low_spent_Small_value_payments        4.125512 61.312415      4.166439
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments     2.072673      2.377097     1.884837
##   High_spent_Medium_value_payments    3.200077      3.017490     3.267346
##   High_spent_Small_value_payments     3.186720      3.974312     3.356355
##   Low_spent_Large_value_payments      2.651618      3.317922     2.977971
##   Low_spent_Medium_value_payments     2.567003      3.123912     2.767142
##   Low_spent_Small_value_payments      4.379263      4.349250     3.642565
##
## $Payment_Behaviour_Last_Loan_5$col_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments   8.089953           10.268879
##   High_spent_Medium_value_payments 19.684579           17.048055
##   High_spent_Small_value_payments   7.797897            6.064073
##   Low_spent_Large_value_payments    5.665888            7.236842
##   Low_spent_Medium_value_payments  10.426402           11.212815
##   Low_spent_Small_value_payments   48.335280           48.169336
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                 8.134921        11.029412
##   High_spent_Medium_value_payments               16.921769        17.294118
##   High_spent_Small_value_payments                 6.207483         7.264706
##   Low_spent_Large_value_payments                  7.681406         6.029412
##   Low_spent_Medium_value_payments                13.463719        11.676471
##   Low_spent_Small_value_payments                 47.590703        46.705882
##
##                                   Mortgage Loan   No Loan Not Specified
##   High_spent_Large_value_payments       8.779070 18.067554      9.535452
##   High_spent_Medium_value_payments     19.738372 21.641039     20.568460
##   High_spent_Small_value_payments       9.215116  8.472990      7.273839
##   Low_spent_Large_value_payments        6.773256  7.691200      6.051345
##   Low_spent_Medium_value_payments      11.540698 11.774361      9.902200
##   Low_spent_Small_value_payments       43.953488 32.352856     46.668704
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments     9.569378     10.426136     9.326923
##   High_spent_Medium_value_payments   19.916268     17.840909    21.794872
##   High_spent_Small_value_payments     7.864833      9.318182     8.878205
##   Low_spent_Large_value_payments      5.831340      6.931818     7.019231
##   Low_spent_Medium_value_payments     8.821770     10.198864    10.192308
##   Low_spent_Small_value_payments     47.996411     45.284091    42.788462
##
##
```

```
## $Payment_Behaviour_Last_Loan_6
## $Payment_Behaviour_Last_Loan_6$contingency_table
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments        262                 239
##   High_spent_Medium_value_payments       429                 461
##   High_spent_Small_value_payments        254                 174
##   Low_spent_Large_value_payments         188                 169
##   Low_spent_Medium_value_payments        253                 281
##   Low_spent_Small_value_payments        1086                1284
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                      224              248
##   High_spent_Medium_value_payments                     477              475
##   High_spent_Small_value_payments                      156              197
##   Low_spent_Large_value_payments                       132              153
##   Low_spent_Medium_value_payments                      245              250
##   Low_spent_Small_value_payments                      1230             1141
##
##                                   Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments           245   13291           257
##   High_spent_Medium_value_payments          442   16504           471
##   High_spent_Small_value_payments           159    6478           207
##   Low_spent_Large_value_payments            130    5895           156
##   Low_spent_Medium_value_payments           276    8998           271
##   Low_spent_Small_value_payments           1140   25818          1262
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments         242           239          192
##   High_spent_Medium_value_payments        485           558          510
##   High_spent_Small_value_payments         176           239          213
##   Low_spent_Large_value_payments          203           182          146
##   Low_spent_Medium_value_payments         326           302          290
##   Low_spent_Small_value_payments         1256          1216         1217
##
## $Payment_Behaviour_Last_Loan_6$row_totals
##
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                            15439                            20812
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                             8253                             7354
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                            11492                            36650
##
## $Payment_Behaviour_Last_Loan_6$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    2472                    2608                    2464
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    2464                    2392                   76984
##           Not Specified             Payday Loan           Personal Loan
```

```
##                    2624                    2688                    2736
##            Student Loan
##                    2568
##
## $Payment_Behaviour_Last_Loan_6$grand_total
## [1] 100000
##
## $Payment_Behaviour_Last_Loan_6$cell_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments      0.262               0.239
##   High_spent_Medium_value_payments     0.429               0.461
##   High_spent_Small_value_payments      0.254               0.174
##   Low_spent_Large_value_payments       0.188               0.169
##   Low_spent_Medium_value_payments      0.253               0.281
##   Low_spent_Small_value_payments       1.086               1.284
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                    0.224            0.248
##   High_spent_Medium_value_payments                   0.477            0.475
##   High_spent_Small_value_payments                    0.156            0.197
##   Low_spent_Large_value_payments                     0.132            0.153
##   Low_spent_Medium_value_payments                    0.245            0.250
##   Low_spent_Small_value_payments                     1.230            1.141
##
##                                   Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments         0.245  13.291         0.257
##   High_spent_Medium_value_payments        0.442  16.504         0.471
##   High_spent_Small_value_payments         0.159   6.478         0.207
##   Low_spent_Large_value_payments          0.130   5.895         0.156
##   Low_spent_Medium_value_payments         0.276   8.998         0.271
##   Low_spent_Small_value_payments          1.140  25.818         1.262
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments       0.242         0.239        0.192
##   High_spent_Medium_value_payments      0.485         0.558        0.510
##   High_spent_Small_value_payments       0.176         0.239        0.213
##   Low_spent_Large_value_payments        0.203         0.182        0.146
##   Low_spent_Medium_value_payments       0.326         0.302        0.290
##   Low_spent_Small_value_payments        1.256         1.216        1.217
##
## $Payment_Behaviour_Last_Loan_6$row_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments   1.697001            1.548028
##   High_spent_Medium_value_payments  2.061311            2.215068
##   High_spent_Small_value_payments   3.077669            2.108324
##   Low_spent_Large_value_payments    2.556432            2.298069
##   Low_spent_Medium_value_payments   2.201532            2.445179
##   Low_spent_Small_value_payments    2.963165            3.503411
##
```

```
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                 1.450871         1.606322
##   High_spent_Medium_value_payments                2.291947         2.282337
##   High_spent_Small_value_payments                 1.890222         2.387011
##   Low_spent_Large_value_payments                  1.794942         2.080500
##   Low_spent_Medium_value_payments                 2.131918         2.175426
##   Low_spent_Small_value_payments                  3.356071         3.113233
##
##                                   Mortgage Loan   No Loan Not Specified
##   High_spent_Large_value_payments       1.586890 86.087182      1.664616
##   High_spent_Medium_value_payments      2.123775 79.300404      2.263117
##   High_spent_Small_value_payments       1.926572 78.492669      2.508179
##   Low_spent_Large_value_payments        1.767745 80.160457      2.121295
##   Low_spent_Medium_value_payments       2.401671 78.297946      2.358162
##   Low_spent_Small_value_payments        3.110505 70.444748      3.443383
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments     1.567459      1.548028     1.243604
##   High_spent_Medium_value_payments    2.330386      2.681145     2.450509
##   High_spent_Small_value_payments     2.132558      2.895917     2.580880
##   Low_spent_Large_value_payments      2.760403      2.474844     1.985314
##   Low_spent_Medium_value_payments     2.836756      2.627915     2.523495
##   Low_spent_Small_value_payments      3.427012      3.317872     3.320600
##
## $Payment_Behaviour_Last_Loan_6$col_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments  10.598706            9.164110
##   High_spent_Medium_value_payments 17.354369           17.676380
##   High_spent_Small_value_payments  10.275081            6.671779
##   Low_spent_Large_value_payments    7.605178            6.480061
##   Low_spent_Medium_value_payments  10.234628           10.774540
##   Low_spent_Small_value_payments   43.932039           49.233129
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                 9.090909        10.064935
##   High_spent_Medium_value_payments               19.358766        19.277597
##   High_spent_Small_value_payments                 6.331169         7.995130
##   Low_spent_Large_value_payments                  5.357143         6.209416
##   Low_spent_Medium_value_payments                 9.943182        10.146104
##   Low_spent_Small_value_payments                 49.918831        46.306818
##
##                                   Mortgage Loan   No Loan Not Specified
##   High_spent_Large_value_payments      10.242475 17.264626      9.794207
##   High_spent_Medium_value_payments     18.478261 21.438221     17.949695
##   High_spent_Small_value_payments       6.647157  8.414736      7.888720
##   Low_spent_Large_value_payments        5.434783  7.657435      5.945122
##   Low_spent_Medium_value_payments      11.538462 11.688143     10.327744
##   Low_spent_Small_value_payments       47.658863 33.536839     48.094512
##
##                                   Payday Loan Personal Loan Student Loan
```

```
##   High_spent_Large_value_payments     9.002976      8.735380     7.476636
##   High_spent_Medium_value_payments   18.043155     20.394737    19.859813
##   High_spent_Small_value_payments     6.547619      8.735380     8.294393
##   Low_spent_Large_value_payments      7.552083      6.652047     5.685358
##   Low_spent_Medium_value_payments    12.127976     11.038012    11.292835
##   Low_spent_Small_value_payments     46.726190     44.444444    47.390966
##
##
## $Payment_Behaviour_Last_Loan_7
## $Payment_Behaviour_Last_Loan_7$contingency_table
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments        135                 133
##   High_spent_Medium_value_payments       348                 275
##   High_spent_Small_value_payments         99                 171
##   Low_spent_Large_value_payments          96                 119
##   Low_spent_Medium_value_payments        182                 182
##   Low_spent_Small_value_payments         740                 840
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                     124              132
##   High_spent_Medium_value_payments                    245              271
##   High_spent_Small_value_payments                     131              131
##   Low_spent_Large_value_payments                       98               86
##   Low_spent_Medium_value_payments                     150              160
##   Low_spent_Small_value_payments                      684              748
##
##                                   Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments            210   14087           127
##   High_spent_Medium_value_payments           330   18143           263
##   High_spent_Small_value_payments            130    7087           176
##   Low_spent_Large_value_payments             110    6425           119
##   Low_spent_Medium_value_payments            176    9969           173
##   Low_spent_Small_value_payments             836   29417           750
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments          177           185          129
##   High_spent_Medium_value_payments         337           272          328
##   High_spent_Small_value_payments          121           114           93
##   Low_spent_Large_value_payments           126            83           92
##   Low_spent_Medium_value_payments          188           181          131
##   Low_spent_Small_value_payments           971           853          811
##
## $Payment_Behaviour_Last_Loan_7$row_totals
##
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                            15439                            20812
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                             8253                             7354
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                            11492                            36650
```

```
##
## $Payment_Behaviour_Last_Loan_7$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    1600                    1720                    1432
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    1528                    1792                   85128
##           Not Specified             Payday Loan           Personal Loan
##                    1608                    1920                    1688
##            Student Loan
##                    1584
##
## $Payment_Behaviour_Last_Loan_7$grand_total
## [1] 100000
##
## $Payment_Behaviour_Last_Loan_7$cell_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments      0.135               0.133
##   High_spent_Medium_value_payments     0.348               0.275
##   High_spent_Small_value_payments      0.099               0.171
##   Low_spent_Large_value_payments       0.096               0.119
##   Low_spent_Medium_value_payments      0.182               0.182
##   Low_spent_Small_value_payments       0.740               0.840
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                    0.124            0.132
##   High_spent_Medium_value_payments                   0.245            0.271
##   High_spent_Small_value_payments                    0.131            0.131
##   Low_spent_Large_value_payments                     0.098            0.086
##   Low_spent_Medium_value_payments                    0.150            0.160
##   Low_spent_Small_value_payments                     0.684            0.748
##
##                                   Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments          0.210  14.087         0.127
##   High_spent_Medium_value_payments         0.330  18.143         0.263
##   High_spent_Small_value_payments          0.130   7.087         0.176
##   Low_spent_Large_value_payments           0.110   6.425         0.119
##   Low_spent_Medium_value_payments          0.176   9.969         0.173
##   Low_spent_Small_value_payments           0.836  29.417         0.750
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments        0.177         0.185        0.129
##   High_spent_Medium_value_payments       0.337         0.272        0.328
##   High_spent_Small_value_payments        0.121         0.114        0.093
##   Low_spent_Large_value_payments         0.126         0.083        0.092
##   Low_spent_Medium_value_payments        0.188         0.181        0.131
##   Low_spent_Small_value_payments         0.971         0.853        0.811
##
## $Payment_Behaviour_Last_Loan_7$row_percentage
##
```

```
##                                    Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments   0.8744090           0.8614548
##   High_spent_Medium_value_payments  1.6721122           1.3213531
##   High_spent_Small_value_payments   1.1995638           2.0719738
##   Low_spent_Large_value_payments    1.3054120           1.6181670
##   Low_spent_Medium_value_payments   1.5837104           1.5837104
##   Low_spent_Small_value_payments    2.0190996           2.2919509
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                0.8031608        0.8549777
##   High_spent_Medium_value_payments               1.1772055        1.3021334
##   High_spent_Small_value_payments                1.5873016        1.5873016
##   Low_spent_Large_value_payments                 1.3326081        1.1694316
##   Low_spent_Medium_value_payments                1.3052558        1.3922729
##   Low_spent_Small_value_payments                 1.8663029        2.0409277
##
##                                   Mortgage Loan    No Loan Not Specified
##   High_spent_Large_value_payments      1.3601917 91.2429562     0.8225921
##   High_spent_Medium_value_payments     1.5856237 87.1756679     1.2636940
##   High_spent_Small_value_payments      1.5751848 85.8718042     2.1325579
##   Low_spent_Large_value_payments       1.4957846 87.3674191     1.6181670
##   Low_spent_Medium_value_payments      1.5315002 86.7473025     1.5053951
##   Low_spent_Small_value_payments       2.2810368 80.2646658     2.0463847
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments    1.1464473     1.1982641    0.8355463
##   High_spent_Medium_value_payments   1.6192581     1.3069383    1.5760138
##   High_spent_Small_value_payments    1.4661335     1.3813159    1.1268630
##   Low_spent_Large_value_payments     1.7133533     1.1286375    1.2510199
##   Low_spent_Medium_value_payments    1.6359206     1.5750087    1.1399234
##   Low_spent_Small_value_payments     2.6493861     2.3274216    2.2128240
##
## $Payment_Behaviour_Last_Loan_7$col_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments   8.437500            7.732558
##   High_spent_Medium_value_payments 21.750000           15.988372
##   High_spent_Small_value_payments   6.187500            9.941860
##   Low_spent_Large_value_payments    6.000000            6.918605
##   Low_spent_Medium_value_payments  11.375000           10.581395
##   Low_spent_Small_value_payments   46.250000           48.837209
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                 8.659218         8.638743
##   High_spent_Medium_value_payments               17.108939        17.735602
##   High_spent_Small_value_payments                 9.148045         8.573298
##   Low_spent_Large_value_payments                  6.843575         5.628272
##   Low_spent_Medium_value_payments                10.474860        10.471204
##   Low_spent_Small_value_payments                 47.765363        48.952880
##
##                                   Mortgage Loan   No Loan Not Specified
```

```
##   High_spent_Large_value_payments      11.718750 16.548022      7.898010
##   High_spent_Medium_value_payments     18.415179 21.312612     16.355721
##   High_spent_Small_value_payments       7.254464  8.325110     10.945274
##   Low_spent_Large_value_payments        6.138393  7.547458      7.400498
##   Low_spent_Medium_value_payments       9.821429 11.710601     10.758706
##   Low_spent_Small_value_payments       46.651786 34.556198     46.641791
##
##                                    Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments     9.218750     10.959716     8.143939
##   High_spent_Medium_value_payments   17.552083     16.113744    20.707071
##   High_spent_Small_value_payments     6.302083      6.753555     5.871212
##   Low_spent_Large_value_payments      6.562500      4.917062     5.808081
##   Low_spent_Medium_value_payments     9.791667     10.722749     8.270202
##   Low_spent_Small_value_payments     50.572917     50.533175    51.199495
##
##
## $Payment_Behaviour_Last_Loan_8
## $Payment_Behaviour_Last_Loan_8$contingency_table
##
##                                    Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments         76                  68
##   High_spent_Medium_value_payments       132                 139
##   High_spent_Small_value_payments         43                  64
##   Low_spent_Large_value_payments          48                  71
##   Low_spent_Medium_value_payments         93                  90
##   Low_spent_Small_value_payments         328                 440
##
##                                    Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                       47               50
##   High_spent_Medium_value_payments                     144              180
##   High_spent_Small_value_payments                       57               59
##   Low_spent_Large_value_payments                        55               52
##   Low_spent_Medium_value_payments                       48              117
##   Low_spent_Small_value_payments                       409              430
##
##                                    Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments             78   14835            67
##   High_spent_Medium_value_payments            87   19517           140
##   High_spent_Small_value_payments             45    7763            43
##   Low_spent_Large_value_payments              47    6887            56
##   Low_spent_Medium_value_payments             71   10800            91
##   Low_spent_Small_value_payments             408   33006           347
##
##                                    Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments           99            51           68
##   High_spent_Medium_value_payments         185           131          157
##   High_spent_Small_value_payments           38            56           85
##   Low_spent_Large_value_payments            50            39           49
##   Low_spent_Medium_value_payments           48            58           76
##   Low_spent_Small_value_payments           468           385          429
##
```

```
## $Payment_Behaviour_Last_Loan_8$row_totals
##
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                            15439                           20812
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                             8253                            7354
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                            11492                           36650
##
## $Payment_Behaviour_Last_Loan_8$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     720                     872                     760
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     888                     736                   92808
##           Not Specified             Payday Loan           Personal Loan
##                     744                     888                     720
##            Student Loan
##                     864
##
## $Payment_Behaviour_Last_Loan_8$grand_total
## [1] 100000
##
## $Payment_Behaviour_Last_Loan_8$cell_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments      0.076               0.068
##   High_spent_Medium_value_payments     0.132               0.139
##   High_spent_Small_value_payments      0.043               0.064
##   Low_spent_Large_value_payments       0.048               0.071
##   Low_spent_Medium_value_payments      0.093               0.090
##   Low_spent_Small_value_payments       0.328               0.440
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                    0.047            0.050
##   High_spent_Medium_value_payments                   0.144            0.180
##   High_spent_Small_value_payments                    0.057            0.059
##   Low_spent_Large_value_payments                     0.055            0.052
##   Low_spent_Medium_value_payments                    0.048            0.117
##   Low_spent_Small_value_payments                     0.409            0.430
##
##                                   Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments          0.078  14.835         0.067
##   High_spent_Medium_value_payments         0.087  19.517         0.140
##   High_spent_Small_value_payments          0.045   7.763         0.043
##   Low_spent_Large_value_payments           0.047   6.887         0.056
##   Low_spent_Medium_value_payments          0.071  10.800         0.091
##   Low_spent_Small_value_payments           0.408  33.006         0.347
##
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments        0.099         0.051        0.068
```

```
##   High_spent_Medium_value_payments       0.185         0.131        0.157
##   High_spent_Small_value_payments        0.038         0.056        0.085
##   Low_spent_Large_value_payments         0.050         0.039        0.049
##   Low_spent_Medium_value_payments        0.048         0.058        0.076
##   Low_spent_Small_value_payments         0.468         0.385        0.429
##
## $Payment_Behaviour_Last_Loan_8$row_percentage
##
##                                     Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments   0.4922599           0.4404430
##   High_spent_Medium_value_payments  0.6342495           0.6678839
##   High_spent_Small_value_payments   0.5210227           0.7754756
##   Low_spent_Large_value_payments    0.6527060           0.9654610
##   Low_spent_Medium_value_payments   0.8092586           0.7831535
##   Low_spent_Small_value_payments    0.8949523           1.2005457
##
##                                    Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                0.3044239        0.3238552
##   High_spent_Medium_value_payments               0.6919085        0.8648856
##   High_spent_Small_value_payments                0.6906579        0.7148916
##   Low_spent_Large_value_payments                 0.7478923        0.7070982
##   Low_spent_Medium_value_payments                0.4176819        1.0180995
##   Low_spent_Small_value_payments                 1.1159618        1.1732606
##
##                                    Mortgage Loan    No Loan Not Specified
##   High_spent_Large_value_payments      0.5052141 96.0878295     0.4339659
##   High_spent_Medium_value_payments     0.4180281 93.7776283     0.6726888
##   High_spent_Small_value_payments      0.5452563 94.0627651     0.5210227
##   Low_spent_Large_value_payments       0.6391080 93.6497144     0.7614903
##   Low_spent_Medium_value_payments      0.6178211 93.9784198     0.7918552
##   Low_spent_Small_value_payments       1.1132333 90.0572988     0.9467940
##
##                                    Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments    0.6412332     0.3303323    0.4404430
##   High_spent_Medium_value_payments   0.8889102     0.6294446    0.7543725
##   High_spent_Small_value_payments    0.4604386     0.6785411    1.0299285
##   Low_spent_Large_value_payments     0.6799021     0.5303236    0.6663041
##   Low_spent_Medium_value_payments    0.4176819     0.5046989    0.6613296
##   Low_spent_Small_value_payments     1.2769441     1.0504775    1.1705321
##
## $Payment_Behaviour_Last_Loan_8$col_percentage
##
##                                    Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments  10.555556            7.798165
##   High_spent_Medium_value_payments 18.333333           15.940367
##   High_spent_Small_value_payments   5.972222            7.339450
##   Low_spent_Large_value_payments    6.666667            8.142202
##   Low_spent_Medium_value_payments  12.916667           10.321101
##   Low_spent_Small_value_payments   45.555556           50.458716
##
##                                    Debt Consolidation Loan Home Equity Loan
```

```
##   High_spent_Large_value_payments                 6.184211         5.630631
##   High_spent_Medium_value_payments               18.947368        20.270270
##   High_spent_Small_value_payments                 7.500000         6.644144
##   Low_spent_Large_value_payments                  7.236842         5.855856
##   Low_spent_Medium_value_payments                 6.315789        13.175676
##   Low_spent_Small_value_payments                 53.815789        48.423423
##
##                                     Mortgage Loan   No Loan Not Specified
##   High_spent_Large_value_payments      10.597826 15.984613      9.005376
##   High_spent_Medium_value_payments     11.820652 21.029437     18.817204
##   High_spent_Small_value_payments       6.114130  8.364581      5.779570
##   Low_spent_Large_value_payments        6.385870  7.420696      7.526882
##   Low_spent_Medium_value_payments       9.646739 11.636928     12.231183
##   Low_spent_Small_value_payments       55.434783 35.563745     46.639785
##
##                                     Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments    11.148649      7.083333     7.870370
##   High_spent_Medium_value_payments   20.833333     18.194444    18.171296
##   High_spent_Small_value_payments     4.279279      7.777778     9.837963
##   Low_spent_Large_value_payments      5.630631      5.416667     5.671296
##   Low_spent_Medium_value_payments     5.405405      8.055556     8.796296
##   Low_spent_Small_value_payments     52.702703     53.472222    49.652778
##
##
## $Payment_Behaviour_Last_Loan_9
## $Payment_Behaviour_Last_Loan_9$contingency_table
##
##                                     Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments         71                  28
##   High_spent_Medium_value_payments        81                  98
##   High_spent_Small_value_payments         21                  44
##   Low_spent_Large_value_payments          28                  20
##   Low_spent_Medium_value_payments         85                  27
##   Low_spent_Small_value_payments         162                 215
##
##                                     Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                       27               36
##   High_spent_Medium_value_payments                      32               58
##   High_spent_Small_value_payments                       30               33
##   Low_spent_Large_value_payments                        11               11
##   Low_spent_Medium_value_payments                       45               57
##   Low_spent_Small_value_payments                       247              213
##
##                                     Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments             32   15072            65
##   High_spent_Medium_value_payments            68   20152            60
##   High_spent_Small_value_payments             29    7987            29
##   Low_spent_Large_value_payments              50    7125            20
##   Low_spent_Medium_value_payments             48   11101            28
##   Low_spent_Small_value_payments             293   34707           222
##
```

```
##                                   Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments           20            53           35
##   High_spent_Medium_value_payments          93            98           72
##   High_spent_Small_value_payments           27            27           26
##   Low_spent_Large_value_payments            41            23           25
##   Low_spent_Medium_value_payments           45            22           34
##   Low_spent_Small_value_payments           238           161          192
##
## $Payment_Behaviour_Last_Loan_9$row_totals
##
##  High_spent_Large_value_payments High_spent_Medium_value_payments
##                            15439                            20812
##  High_spent_Small_value_payments   Low_spent_Large_value_payments
##                             8253                             7354
##  Low_spent_Medium_value_payments   Low_spent_Small_value_payments
##                            11492                            36650
##
## $Payment_Behaviour_Last_Loan_9$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     448                     432                     392
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     408                     520                   96144
##           Not Specified             Payday Loan           Personal Loan
##                     424                     464                     384
##            Student Loan
##                     384
##
## $Payment_Behaviour_Last_Loan_9$grand_total
## [1] 100000
##
## $Payment_Behaviour_Last_Loan_9$cell_percentage
##
##                                   Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments      0.071               0.028
##   High_spent_Medium_value_payments     0.081               0.098
##   High_spent_Small_value_payments      0.021               0.044
##   Low_spent_Large_value_payments       0.028               0.020
##   Low_spent_Medium_value_payments      0.085               0.027
##   Low_spent_Small_value_payments       0.162               0.215
##
##                                   Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                    0.027            0.036
##   High_spent_Medium_value_payments                   0.032            0.058
##   High_spent_Small_value_payments                    0.030            0.033
##   Low_spent_Large_value_payments                     0.011            0.011
##   Low_spent_Medium_value_payments                    0.045            0.057
##   Low_spent_Small_value_payments                     0.247            0.213
##
##                                   Mortgage Loan No Loan Not Specified
##   High_spent_Large_value_payments          0.032  15.072         0.065
```

```
##   High_spent_Medium_value_payments         0.068  20.152         0.060
##   High_spent_Small_value_payments          0.029   7.987         0.029
##   Low_spent_Large_value_payments           0.050   7.125         0.020
##   Low_spent_Medium_value_payments          0.048  11.101         0.028
##   Low_spent_Small_value_payments           0.293  34.707         0.222
##
##                                     Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments        0.020         0.053        0.035
##   High_spent_Medium_value_payments       0.093         0.098        0.072
##   High_spent_Small_value_payments        0.027         0.027        0.026
##   Low_spent_Large_value_payments         0.041         0.023        0.025
##   Low_spent_Medium_value_payments        0.045         0.022        0.034
##   Low_spent_Small_value_payments         0.238         0.161        0.192
##
## $Payment_Behaviour_Last_Loan_9$row_percentage
##
##                                      Auto Loan Credit-Builder Loan
##   High_spent_Large_value_payments   0.4598743           0.1813589
##   High_spent_Medium_value_payments  0.3891985           0.4708822
##   High_spent_Small_value_payments   0.2544529           0.5331395
##   Low_spent_Large_value_payments    0.3807452           0.2719608
##   Low_spent_Medium_value_payments   0.7396450           0.2349460
##   Low_spent_Small_value_payments    0.4420191           0.5866303
##
##                                     Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                0.1748818        0.2331757
##   High_spent_Medium_value_payments               0.1537574        0.2786854
##   High_spent_Small_value_payments                0.3635042        0.3998546
##   Low_spent_Large_value_payments                 0.1495785        0.1495785
##   Low_spent_Medium_value_payments                0.3915767        0.4959972
##   Low_spent_Small_value_payments                 0.6739427        0.5811733
##
##                                     Mortgage Loan    No Loan Not Specified
##   High_spent_Large_value_payments      0.2072673 97.6229030     0.4210117
##   High_spent_Medium_value_payments     0.3267346 96.8287526     0.2882952
##   High_spent_Small_value_payments      0.3513874 96.7769296     0.3513874
##   Low_spent_Large_value_payments       0.6799021 96.8860484     0.2719608
##   Low_spent_Medium_value_payments      0.4176819 96.5976331     0.2436478
##   Low_spent_Small_value_payments       0.7994543 94.6984993     0.6057299
##
##                                     Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments    0.1295421     0.3432865    0.2266986
##   High_spent_Medium_value_payments   0.4468576     0.4708822    0.3459543
##   High_spent_Small_value_payments    0.3271538     0.3271538    0.3150370
##   Low_spent_Large_value_payments     0.5575197     0.3127550    0.3399510
##   Low_spent_Medium_value_payments    0.3915767     0.1914375    0.2958580
##   Low_spent_Small_value_payments     0.6493861     0.4392906    0.5238745
##
## $Payment_Behaviour_Last_Loan_9$col_percentage
##
##                                     Auto Loan Credit-Builder Loan
```

```
##   High_spent_Large_value_payments  15.848214            6.481481
##   High_spent_Medium_value_payments 18.080357           22.685185
##   High_spent_Small_value_payments   4.687500           10.185185
##   Low_spent_Large_value_payments    6.250000            4.629630
##   Low_spent_Medium_value_payments  18.973214            6.250000
##   Low_spent_Small_value_payments   36.160714           49.768519
##
##                                    Debt Consolidation Loan Home Equity Loan
##   High_spent_Large_value_payments                 6.887755         8.823529
##   High_spent_Medium_value_payments                8.163265        14.215686
##   High_spent_Small_value_payments                 7.653061         8.088235
##   Low_spent_Large_value_payments                  2.806122         2.696078
##   Low_spent_Medium_value_payments                11.479592        13.970588
##   Low_spent_Small_value_payments                 63.010204        52.205882
##
##                                    Mortgage Loan   No Loan Not Specified
##   High_spent_Large_value_payments       6.153846 15.676485     15.330189
##   High_spent_Medium_value_payments     13.076923 20.960226     14.150943
##   High_spent_Small_value_payments       5.576923  8.307331      6.839623
##   Low_spent_Large_value_payments        9.615385  7.410759      4.716981
##   Low_spent_Medium_value_payments       9.230769 11.546222      6.603774
##   Low_spent_Small_value_payments       56.346154 36.098977     52.358491
##
##                                    Payday Loan Personal Loan Student Loan
##   High_spent_Large_value_payments     4.310345     13.802083     9.114583
##   High_spent_Medium_value_payments   20.043103     25.520833    18.750000
##   High_spent_Small_value_payments     5.818966      7.031250     6.770833
##   Low_spent_Large_value_payments      8.836207      5.989583     6.510417
##   Low_spent_Medium_value_payments     9.698276      5.729167     8.854167
##   Low_spent_Small_value_payments     51.293103     41.927083    50.000000
##
##
## $Occupation_Credit_Score
## $Occupation_Credit_Score$contingency_table
##
##                 Good Poor Standard
##   Accountant    1205 2037     3502
##   Architect     1233 1855     3736
##   Developer     1210 1924     3586
##   Doctor        1164 1804     3600
##   Engineer      1264 2094     3506
##   Entrepreneur  1185 2033     3430
##   Journalist    1254 1840     3442
##   Lawyer        1259 1965     3872
##   Manager       1190 1856     3386
##   Mechanic      1107 2044     3625
##   Media_Manager 1244 1764     3712
##   Musician      1192 1786     3374
##   Scientist     1201 2062     3481
##   Teacher       1224 2027     3421
##   Writer         896 1907     3501
```

```
##
## $Occupation_Credit_Score$row_totals
##
##    Accountant     Architect     Developer        Doctor      Engineer
##          6744          6824          6720          6568          6864
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##          6648          6536          7096          6432          6776
## Media_Manager      Musician     Scientist       Teacher        Writer
##          6720          6352          6744          6672          6304
##
## $Occupation_Credit_Score$col_totals
##
##     Good     Poor Standard
##    17828    28998    53174
##
## $Occupation_Credit_Score$grand_total
## [1] 100000
##
## $Occupation_Credit_Score$cell_percentage
##
##                  Good  Poor Standard
##   Accountant    1.205 2.037    3.502
##   Architect     1.233 1.855    3.736
##   Developer     1.210 1.924    3.586
##   Doctor        1.164 1.804    3.600
##   Engineer      1.264 2.094    3.506
##   Entrepreneur  1.185 2.033    3.430
##   Journalist    1.254 1.840    3.442
##   Lawyer        1.259 1.965    3.872
##   Manager       1.190 1.856    3.386
##   Mechanic      1.107 2.044    3.625
##   Media_Manager 1.244 1.764    3.712
##   Musician      1.192 1.786    3.374
##   Scientist     1.201 2.062    3.481
##   Teacher       1.224 2.027    3.421
##   Writer        0.896 1.907    3.501
##
## $Occupation_Credit_Score$row_percentage
##
##                     Good     Poor Standard
##   Accountant    17.86773 30.20463 51.92764
##   Architect     18.06858 27.18347 54.74795
##   Developer     18.00595 28.63095 53.36310
##   Doctor        17.72229 27.46650 54.81121
##   Engineer      18.41492 30.50699 51.07809
##   Entrepreneur  17.82491 30.58063 51.59446
##   Journalist    19.18605 28.15177 52.66218
##   Lawyer        17.74239 27.69166 54.56595
##   Manager       18.50124 28.85572 52.64303
##   Mechanic      16.33707 30.16529 53.49764
##   Media_Manager 18.51190 26.25000 55.23810
```

```
##   Musician      18.76574 28.11713 53.11713
##   Scientist     17.80842 30.57533 51.61625
##   Teacher       18.34532 30.38070 51.27398
##   Writer        14.21320 30.25063 55.53617
##
## $Occupation_Credit_Score$col_percentage
##
##                     Good     Poor Standard
##   Accountant    6.759031 7.024622 6.585925
##   Architect     6.916087 6.396993 7.025990
##   Developer     6.787077 6.634940 6.743897
##   Doctor        6.529055 6.221119 6.770226
##   Engineer      7.089971 7.221188 6.593448
##   Entrepreneur  6.646848 7.010828 6.450521
##   Journalist    7.033879 6.345265 6.473088
##   Lawyer        7.061925 6.776329 7.281754
##   Manager       6.674893 6.400441 6.367774
##   Mechanic      6.209334 7.048762 6.817242
##   Media_Manager 6.977788 6.083178 6.980855
##   Musician      6.686112 6.159045 6.345206
##   Scientist     6.736594 7.110835 6.546432
##   Teacher       6.865605 6.990137 6.433595
##   Writer        5.025802 6.576316 6.584045
##
##
## $Occupation_Last_Loan_1
## $Occupation_Last_Loan_1$contingency_table
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant          552                 712                      656
##   Architect           720                 640                      656
##   Developer           560                 656                      680
##   Doctor              608                 592                      520
##   Engineer            656                 704                      824
##   Entrepreneur        584                 816                      688
##   Journalist          576                 544                      632
##   Lawyer              672                 696                      704
##   Manager             528                 592                      624
##   Mechanic            680                 600                      672
##   Media_Manager       680                 616                      640
##   Musician            600                 648                      608
##   Scientist           608                 736                      736
##   Teacher             608                 648                      648
##   Writer              696                 600                      680
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant                592           616     840           624
##   Architect                 624           608     768           696
##   Developer                 624           744     816           592
##   Doctor                    736           728     584           656
##   Engineer                  632           592     824           688
```

```
##   Entrepreneur               696           584     656           808
##   Journalist                 616           624     928           608
##   Lawyer                     712           776     688           696
##   Manager                    672           680     800           600
##   Mechanic                   664           576     672           656
##   Media_Manager              696           704     800           696
##   Musician                   672           512     784           712
##   Scientist                  656           560     768           680
##   Teacher                    680           632     784           632
##   Writer                     528           672     696           672
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant            672           800          680
##   Architect             680           768          664
##   Developer             736           656          656
##   Doctor                696           648          800
##   Engineer              672           632          640
##   Entrepreneur          624           584          608
##   Journalist            688           680          640
##   Lawyer                744           712          696
##   Manager               632           696          608
##   Mechanic              848           736          672
##   Media_Manager         680           656          552
##   Musician              584           528          704
##   Scientist             688           640          672
##   Teacher               616           648          776
##   Writer                544           552          664
##
## $Occupation_Last_Loan_1$row_totals
##
##    Accountant     Architect     Developer        Doctor      Engineer
##          6744          6824          6720          6568          6864
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##          6648          6536          7096          6432          6776
## Media_Manager      Musician     Scientist       Teacher        Writer
##          6720          6352          6744          6672          6304
##
## $Occupation_Last_Loan_1$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    9328                    9800                    9968
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    9800                    9608                   11408
##           Not Specified             Payday Loan           Personal Loan
##                   10016                   10104                    9936
##            Student Loan
##                   10032
##
## $Occupation_Last_Loan_1$grand_total
## [1] 100000
##
```

```
## $Occupation_Last_Loan_1$cell_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant        0.552               0.712                   0.656
##   Architect         0.720               0.640                   0.656
##   Developer         0.560               0.656                   0.680
##   Doctor            0.608               0.592                   0.520
##   Engineer          0.656               0.704                   0.824
##   Entrepreneur      0.584               0.816                   0.688
##   Journalist        0.576               0.544                   0.632
##   Lawyer            0.672               0.696                   0.704
##   Manager           0.528               0.592                   0.624
##   Mechanic          0.680               0.600                   0.672
##   Media_Manager     0.680               0.616                   0.640
##   Musician          0.600               0.648                   0.608
##   Scientist         0.608               0.736                   0.736
##   Teacher           0.608               0.648                   0.648
##   Writer            0.696               0.600                   0.680
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant               0.592         0.616   0.840         0.624
##   Architect                0.624         0.608   0.768         0.696
##   Developer                0.624         0.744   0.816         0.592
##   Doctor                   0.736         0.728   0.584         0.656
##   Engineer                 0.632         0.592   0.824         0.688
##   Entrepreneur             0.696         0.584   0.656         0.808
##   Journalist               0.616         0.624   0.928         0.608
##   Lawyer                   0.712         0.776   0.688         0.696
##   Manager                  0.672         0.680   0.800         0.600
##   Mechanic                 0.664         0.576   0.672         0.656
##   Media_Manager            0.696         0.704   0.800         0.696
##   Musician                 0.672         0.512   0.784         0.712
##   Scientist                0.656         0.560   0.768         0.680
##   Teacher                  0.680         0.632   0.784         0.632
##   Writer                   0.528         0.672   0.696         0.672
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant          0.672         0.800        0.680
##   Architect           0.680         0.768        0.664
##   Developer           0.736         0.656        0.656
##   Doctor              0.696         0.648        0.800
##   Engineer            0.672         0.632        0.640
##   Entrepreneur        0.624         0.584        0.608
##   Journalist          0.688         0.680        0.640
##   Lawyer              0.744         0.712        0.696
##   Manager             0.632         0.696        0.608
##   Mechanic            0.848         0.736        0.672
##   Media_Manager       0.680         0.656        0.552
##   Musician            0.584         0.528        0.704
##   Scientist           0.688         0.640        0.672
##   Teacher             0.616         0.648        0.776
```

```
##   Writer              0.544         0.552        0.664
##
## $Occupation_Last_Loan_1$row_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     8.185053           10.557533                9.727165
##   Architect     10.550996            9.378664                9.613130
##   Developer      8.333333            9.761905               10.119048
##   Doctor         9.257004            9.013398                7.917174
##   Engineer       9.557110           10.256410               12.004662
##   Entrepreneur   8.784597           12.274368               10.348977
##   Journalist     8.812729            8.323133                9.669523
##   Lawyer         9.470124            9.808343                9.921082
##   Manager        8.208955            9.203980                9.701493
##   Mechanic      10.035419            8.854782                9.917355
##   Media_Manager 10.119048            9.166667                9.523810
##   Musician       9.445844           10.201511                9.571788
##   Scientist      9.015421           10.913405               10.913405
##   Teacher        9.112710            9.712230                9.712230
##   Writer        11.040609            9.517766               10.786802
##
##                 Home Equity Loan Mortgage Loan   No Loan Not Specified
##   Accountant            8.778173      9.134045 12.455516      9.252669
##   Architect             9.144197      8.909730 11.254396     10.199297
##   Developer             9.285714     11.071429 12.142857      8.809524
##   Doctor               11.205847     11.084044  8.891596      9.987820
##   Engineer              9.207459      8.624709 12.004662     10.023310
##   Entrepreneur         10.469314      8.784597  9.867629     12.154031
##   Journalist            9.424725      9.547124 14.198286      9.302326
##   Lawyer               10.033822     10.935738  9.695603      9.808343
##   Manager              10.447761     10.572139 12.437811      9.328358
##   Mechanic              9.799292      8.500590  9.917355      9.681228
##   Media_Manager        10.357143     10.476190 11.904762     10.357143
##   Musician             10.579345      8.060453 12.342569     11.209068
##   Scientist             9.727165      8.303677 11.387900     10.083037
##   Teacher              10.191847      9.472422 11.750600      9.472422
##   Writer                8.375635     10.659898 11.040609     10.659898
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant       9.964413     11.862396    10.083037
##   Architect        9.964830     11.254396     9.730363
##   Developer       10.952381      9.761905     9.761905
##   Doctor          10.596833      9.866017    12.180268
##   Engineer         9.790210      9.207459     9.324009
##   Entrepreneur     9.386282      8.784597     9.145608
##   Journalist      10.526316     10.403917     9.791922
##   Lawyer          10.484780     10.033822     9.808343
##   Manager          9.825871     10.820896     9.452736
##   Mechanic        12.514758     10.861865     9.917355
##   Media_Manager   10.119048      9.761905     8.214286
##   Musician         9.193955      8.312343    11.083123
```

```
##   Scientist       10.201661      9.489917     9.964413
##   Teacher          9.232614      9.712230    11.630695
##   Writer           8.629442      8.756345    10.532995
##
## $Occupation_Last_Loan_1$col_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     5.917667            7.265306                6.581059
##   Architect      7.718696            6.530612                6.581059
##   Developer      6.003431            6.693878                6.821830
##   Doctor         6.518010            6.040816                5.216693
##   Engineer       7.032590            7.183673                8.266453
##   Entrepreneur   6.260720            8.326531                6.902087
##   Journalist     6.174957            5.551020                6.340289
##   Lawyer         7.204117            7.102041                7.062600
##   Manager        5.660377            6.040816                6.260032
##   Mechanic       7.289880            6.122449                6.741573
##   Media_Manager  7.289880            6.285714                6.420546
##   Musician       6.432247            6.612245                6.099518
##   Scientist      6.518010            7.510204                7.383628
##   Teacher        6.518010            6.612245                6.500803
##   Writer         7.461407            6.122449                6.821830
##
##                 Home Equity Loan Mortgage Loan  No Loan Not Specified
##   Accountant            6.040816      6.411324 7.363254      6.230032
##   Architect             6.367347      6.328060 6.732118      6.948882
##   Developer             6.367347      7.743547 7.152875      5.910543
##   Doctor                7.510204      7.577019 5.119215      6.549521
##   Engineer              6.448980      6.161532 7.223001      6.869010
##   Entrepreneur          7.102041      6.078268 5.750351      8.067093
##   Journalist            6.285714      6.494588 8.134642      6.070288
##   Lawyer                7.265306      8.076603 6.030856      6.948882
##   Manager               6.857143      7.077435 7.012623      5.990415
##   Mechanic              6.775510      5.995004 5.890603      6.549521
##   Media_Manager         7.102041      7.327227 7.012623      6.948882
##   Musician              6.857143      5.328893 6.872370      7.108626
##   Scientist             6.693878      5.828476 6.732118      6.789137
##   Teacher               6.938776      6.577852 6.872370      6.309904
##   Writer                5.387755      6.994172 6.100982      6.709265
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant       6.650831      8.051530     6.778309
##   Architect        6.730008      7.729469     6.618820
##   Developer        7.284244      6.602254     6.539075
##   Doctor           6.888361      6.521739     7.974482
##   Engineer         6.650831      6.360709     6.379585
##   Entrepreneur     6.175772      5.877617     6.060606
##   Journalist       6.809184      6.843800     6.379585
##   Lawyer           7.363420      7.165862     6.937799
##   Manager          6.254949      7.004831     6.060606
##   Mechanic         8.392716      7.407407     6.698565
```

```
##   Media_Manager    6.730008      6.602254     5.502392
##   Musician         5.779889      5.314010     7.017544
##   Scientist        6.809184      6.441224     6.698565
##   Teacher          6.096595      6.521739     7.735247
##   Writer           5.384006      5.555556     6.618820
##
##
## $Occupation_Last_Loan_2
## $Occupation_Last_Loan_2$contingency_table
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant          592                 528                     496
##   Architect           536                 600                     448
##   Developer           536                 664                     528
##   Doctor              472                 472                     632
##   Engineer            688                 584                     488
##   Entrepreneur        592                 528                     568
##   Journalist          528                 600                     552
##   Lawyer              584                 608                     720
##   Manager             464                 536                     496
##   Mechanic            560                 656                     632
##   Media_Manager       600                 520                     456
##   Musician            592                 432                     560
##   Scientist           712                 656                     584
##   Teacher             576                 496                     472
##   Writer              440                 648                     640
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant                608           672    1560           600
##   Architect                 600           560    1536           680
##   Developer                 560           592    1464           584
##   Doctor                    712           600    1368           632
##   Engineer                  576           512    1584           600
##   Entrepreneur              672           640    1400           600
##   Journalist                488           520    1520           552
##   Lawyer                    480           544    1616           616
##   Manager                   560           632    1552           568
##   Mechanic                  616           448    1416           576
##   Media_Manager             600           616    1688           584
##   Musician                  608           424    1544           536
##   Scientist                 496           512    1432           560
##   Teacher                   648           648    1512           576
##   Writer                    552           488    1344           464
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant            472           592          624
##   Architect             696           552          616
##   Developer             640           568          584
##   Doctor                640           496          544
##   Engineer              656           624          552
##   Entrepreneur          552           488          608
```

```
##   Journalist            552           512          712
##   Lawyer                688           704          536
##   Manager               552           528          544
##   Mechanic              608           632          632
##   Media_Manager         608           528          520
##   Musician              624           584          448
##   Scientist             600           560          632
##   Teacher               576           672          496
##   Writer                640           600          488
##
## $Occupation_Last_Loan_2$row_totals
##
##    Accountant     Architect     Developer        Doctor      Engineer
##          6744          6824          6720          6568          6864
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##          6648          6536          7096          6432          6776
## Media_Manager      Musician     Scientist       Teacher        Writer
##          6720          6352          6744          6672          6304
##
## $Occupation_Last_Loan_2$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    8472                    8528                    8272
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    8776                    8408                   22536
##           Not Specified             Payday Loan           Personal Loan
##                    8728                    9104                    8640
##            Student Loan
##                    8536
##
## $Occupation_Last_Loan_2$grand_total
## [1] 100000
##
## $Occupation_Last_Loan_2$cell_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant        0.592               0.528                   0.496
##   Architect         0.536               0.600                   0.448
##   Developer         0.536               0.664                   0.528
##   Doctor            0.472               0.472                   0.632
##   Engineer          0.688               0.584                   0.488
##   Entrepreneur      0.592               0.528                   0.568
##   Journalist        0.528               0.600                   0.552
##   Lawyer            0.584               0.608                   0.720
##   Manager           0.464               0.536                   0.496
##   Mechanic          0.560               0.656                   0.632
##   Media_Manager     0.600               0.520                   0.456
##   Musician          0.592               0.432                   0.560
##   Scientist         0.712               0.656                   0.584
##   Teacher           0.576               0.496                   0.472
##   Writer            0.440               0.648                   0.640
```

```
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant               0.608         0.672   1.560         0.600
##   Architect                0.600         0.560   1.536         0.680
##   Developer                0.560         0.592   1.464         0.584
##   Doctor                   0.712         0.600   1.368         0.632
##   Engineer                 0.576         0.512   1.584         0.600
##   Entrepreneur             0.672         0.640   1.400         0.600
##   Journalist               0.488         0.520   1.520         0.552
##   Lawyer                   0.480         0.544   1.616         0.616
##   Manager                  0.560         0.632   1.552         0.568
##   Mechanic                 0.616         0.448   1.416         0.576
##   Media_Manager            0.600         0.616   1.688         0.584
##   Musician                 0.608         0.424   1.544         0.536
##   Scientist                0.496         0.512   1.432         0.560
##   Teacher                  0.648         0.648   1.512         0.576
##   Writer                   0.552         0.488   1.344         0.464
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant          0.472         0.592        0.624
##   Architect           0.696         0.552        0.616
##   Developer           0.640         0.568        0.584
##   Doctor              0.640         0.496        0.544
##   Engineer            0.656         0.624        0.552
##   Entrepreneur        0.552         0.488        0.608
##   Journalist          0.552         0.512        0.712
##   Lawyer              0.688         0.704        0.536
##   Manager             0.552         0.528        0.544
##   Mechanic            0.608         0.632        0.632
##   Media_Manager       0.608         0.528        0.520
##   Musician            0.624         0.584        0.448
##   Scientist           0.600         0.560        0.632
##   Teacher             0.576         0.672        0.496
##   Writer              0.640         0.600        0.488
##
## $Occupation_Last_Loan_2$row_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     8.778173            7.829181                7.354686
##   Architect      7.854631            8.792497                6.565064
##   Developer      7.976190            9.880952                7.857143
##   Doctor         7.186358            7.186358                9.622412
##   Engineer      10.023310            8.508159                7.109557
##   Entrepreneur   8.904934            7.942238                8.543923
##   Journalist     8.078335            9.179927                8.445532
##   Lawyer         8.229989            8.568207               10.146561
##   Manager        7.213930            8.333333                7.711443
##   Mechanic       8.264463            9.681228                9.327037
##   Media_Manager  8.928571            7.738095                6.785714
##   Musician       9.319899            6.801008                8.816121
##   Scientist     10.557533            9.727165                8.659549
```

```
##   Teacher        8.633094            7.434053                 7.074341
##   Writer         6.979695           10.279188                10.152284
##
##                 Home Equity Loan Mortgage Loan   No Loan Not Specified
##   Accountant            9.015421      9.964413 23.131673      8.896797
##   Architect             8.792497      8.206331 22.508792      9.964830
##   Developer             8.333333      8.809524 21.785714      8.690476
##   Doctor               10.840438      9.135201 20.828258      9.622412
##   Engineer              8.391608      7.459207 23.076923      8.741259
##   Entrepreneur         10.108303      9.626955 21.058965      9.025271
##   Journalist            7.466340      7.955936 23.255814      8.445532
##   Lawyer                6.764374      7.666291 22.773393      8.680947
##   Manager               8.706468      9.825871 24.129353      8.830846
##   Mechanic              9.090909      6.611570 20.897285      8.500590
##   Media_Manager         8.928571      9.166667 25.119048      8.690476
##   Musician              9.571788      6.675063 24.307305      8.438287
##   Scientist             7.354686      7.591934 21.233689      8.303677
##   Teacher               9.712230      9.712230 22.661871      8.633094
##   Writer                8.756345      7.741117 21.319797      7.360406
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant       6.998814      8.778173     9.252669
##   Architect       10.199297      8.089097     9.026964
##   Developer        9.523810      8.452381     8.690476
##   Doctor           9.744214      7.551766     8.282582
##   Engineer         9.557110      9.090909     8.041958
##   Entrepreneur     8.303249      7.340554     9.145608
##   Journalist       8.445532      7.833537    10.893513
##   Lawyer           9.695603      9.921082     7.553551
##   Manager          8.582090      8.208955     8.457711
##   Mechanic         8.972845      9.327037     9.327037
##   Media_Manager    9.047619      7.857143     7.738095
##   Musician         9.823678      9.193955     7.052897
##   Scientist        8.896797      8.303677     9.371293
##   Teacher          8.633094     10.071942     7.434053
##   Writer          10.152284      9.517766     7.741117
##
## $Occupation_Last_Loan_2$col_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     6.987724            6.191370                5.996132
##   Architect      6.326723            7.035647                5.415861
##   Developer      6.326723            7.786116                6.382979
##   Doctor         5.571294            5.534709                7.640232
##   Engineer       8.120869            6.848030                5.899420
##   Entrepreneur   6.987724            6.191370                6.866538
##   Journalist     6.232295            7.035647                6.673114
##   Lawyer         6.893296            7.129456                8.704062
##   Manager        5.476865            6.285178                5.996132
##   Mechanic       6.610009            7.692308                7.640232
##   Media_Manager  7.082153            6.097561                5.512573
```

```
##   Musician       6.987724            5.065666                6.769826
##   Scientist      8.404155            7.692308                7.059961
##   Teacher        6.798867            5.816135                5.705996
##   Writer         5.193579            7.598499                7.736944
##
##                 Home Equity Loan Mortgage Loan  No Loan Not Specified
##   Accountant            6.927985      7.992388 6.922258      6.874427
##   Architect             6.836828      6.660324 6.815761      7.791017
##   Developer             6.381039      7.040913 6.496273      6.691109
##   Doctor                8.113036      7.136061 6.070288      7.241063
##   Engineer              6.563355      6.089439 7.028754      6.874427
##   Entrepreneur          7.657247      7.611798 6.212283      6.874427
##   Journalist            5.560620      6.184586 6.744764      6.324473
##   Lawyer                5.469462      6.470029 7.170749      7.057745
##   Manager               6.381039      7.516651 6.886759      6.507791
##   Mechanic              7.019143      5.328259 6.283280      6.599450
##   Media_Manager         6.836828      7.326356 7.490238      6.691109
##   Musician              6.927985      5.042816 6.851260      6.141155
##   Scientist             5.651778      6.089439 6.354278      6.416132
##   Teacher               7.383774      7.706946 6.709265      6.599450
##   Writer                6.289881      5.803996 5.963791      5.316224
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant       5.184534      6.851852     7.310216
##   Architect        7.644991      6.388889     7.216495
##   Developer        7.029877      6.574074     6.841612
##   Doctor           7.029877      5.740741     6.373008
##   Engineer         7.205624      7.222222     6.466729
##   Entrepreneur     6.063269      5.648148     7.122774
##   Journalist       6.063269      5.925926     8.341143
##   Lawyer           7.557118      8.148148     6.279288
##   Manager          6.063269      6.111111     6.373008
##   Mechanic         6.678383      7.314815     7.403936
##   Media_Manager    6.678383      6.111111     6.091846
##   Musician         6.854130      6.759259     5.248360
##   Scientist        6.590510      6.481481     7.403936
##   Teacher          6.326889      7.777778     5.810684
##   Writer           7.029877      6.944444     5.716963
##
##
## $Occupation_Last_Loan_3
## $Occupation_Last_Loan_3$contingency_table
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant          368                 512                      560
##   Architect           472                 568                      512
##   Developer           408                 464                      480
##   Doctor              496                 344                      392
##   Engineer            472                 560                      512
##   Entrepreneur        472                 464                      544
##   Journalist          424                 472                      448
```

```
##   Lawyer              480                 336                      392
##   Manager             480                 496                      488
##   Mechanic            472                 496                      400
##   Media_Manager       432                 544                      680
##   Musician            416                 448                      440
##   Scientist           512                 448                      344
##   Teacher             448                 616                      384
##   Writer              456                 552                      352
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant                392           520    2568           328
##   Architect                 336           424    2752           512
##   Developer                 472           560    2528           464
##   Doctor                    432           448    2352           552
##   Engineer                  504           424    2672           536
##   Entrepreneur              504           584    2336           472
##   Journalist                480           336    2536           448
##   Lawyer                    504           376    2912           552
##   Manager                   408           432    2456           480
##   Mechanic                  568           464    2472           552
##   Media_Manager             312           344    2720           424
##   Musician                  416           480    2536           432
##   Scientist                 520           560    2504           544
##   Teacher                   408           488    2512           408
##   Writer                    384           432    2392           376
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant            496           480          520
##   Architect             488           328          432
##   Developer             408           448          488
##   Doctor                544           504          504
##   Engineer              400           392          392
##   Entrepreneur          416           488          368
##   Journalist            472           352          568
##   Lawyer                472           576          496
##   Manager               424           368          400
##   Mechanic              416           472          464
##   Media_Manager         440           360          464
##   Musician              368           424          392
##   Scientist             440           400          472
##   Teacher               512           344          552
##   Writer                472           472          416
##
## $Occupation_Last_Loan_3$row_totals
##
##    Accountant     Architect     Developer        Doctor      Engineer
##          6744          6824          6720          6568          6864
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##          6648          6536          7096          6432          6776
## Media_Manager      Musician     Scientist       Teacher         Writer
##          6720          6352          6744          6672          6304
```

```
##
## $Occupation_Last_Loan_3$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    6808                    7320                    6928
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    6640                    6872                   38248
##           Not Specified             Payday Loan           Personal Loan
##                    7080                    6768                    6408
##            Student Loan
##                    6928
##
## $Occupation_Last_Loan_3$grand_total
## [1] 100000
##
## $Occupation_Last_Loan_3$cell_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant        0.368               0.512                   0.560
##   Architect         0.472               0.568                   0.512
##   Developer         0.408               0.464                   0.480
##   Doctor            0.496               0.344                   0.392
##   Engineer          0.472               0.560                   0.512
##   Entrepreneur      0.472               0.464                   0.544
##   Journalist        0.424               0.472                   0.448
##   Lawyer            0.480               0.336                   0.392
##   Manager           0.480               0.496                   0.488
##   Mechanic          0.472               0.496                   0.400
##   Media_Manager     0.432               0.544                   0.680
##   Musician          0.416               0.448                   0.440
##   Scientist         0.512               0.448                   0.344
##   Teacher           0.448               0.616                   0.384
##   Writer            0.456               0.552                   0.352
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant               0.392         0.520   2.568         0.328
##   Architect                0.336         0.424   2.752         0.512
##   Developer                0.472         0.560   2.528         0.464
##   Doctor                   0.432         0.448   2.352         0.552
##   Engineer                 0.504         0.424   2.672         0.536
##   Entrepreneur             0.504         0.584   2.336         0.472
##   Journalist               0.480         0.336   2.536         0.448
##   Lawyer                   0.504         0.376   2.912         0.552
##   Manager                  0.408         0.432   2.456         0.480
##   Mechanic                 0.568         0.464   2.472         0.552
##   Media_Manager            0.312         0.344   2.720         0.424
##   Musician                 0.416         0.480   2.536         0.432
##   Scientist                0.520         0.560   2.504         0.544
##   Teacher                  0.408         0.488   2.512         0.408
##   Writer                   0.384         0.432   2.392         0.376
##
```

```
##                 Payday Loan Personal Loan Student Loan
##   Accountant          0.496         0.480        0.520
##   Architect           0.488         0.328        0.432
##   Developer           0.408         0.448        0.488
##   Doctor              0.544         0.504        0.504
##   Engineer            0.400         0.392        0.392
##   Entrepreneur        0.416         0.488        0.368
##   Journalist          0.472         0.352        0.568
##   Lawyer              0.472         0.576        0.496
##   Manager             0.424         0.368        0.400
##   Mechanic            0.416         0.472        0.464
##   Media_Manager       0.440         0.360        0.464
##   Musician            0.368         0.424        0.392
##   Scientist           0.440         0.400        0.472
##   Teacher             0.512         0.344        0.552
##   Writer              0.472         0.472        0.416
##
## $Occupation_Last_Loan_3$row_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     5.456702            7.591934                8.303677
##   Architect      6.916764            8.323564                7.502931
##   Developer      6.071429            6.904762                7.142857
##   Doctor         7.551766            5.237515                5.968331
##   Engineer       6.876457            8.158508                7.459207
##   Entrepreneur   7.099880            6.979543                8.182912
##   Journalist     6.487148            7.221542                6.854345
##   Lawyer         6.764374            4.735062                5.524239
##   Manager        7.462687            7.711443                7.587065
##   Mechanic       6.965762            7.319953                5.903188
##   Media_Manager  6.428571            8.095238               10.119048
##   Musician       6.549118            7.052897                6.926952
##   Scientist      7.591934            6.642942                5.100830
##   Teacher        6.714628            9.232614                5.755396
##   Writer         7.233503            8.756345                5.583756
##
##                 Home Equity Loan Mortgage Loan   No Loan Not Specified
##   Accountant            5.812574      7.710558 38.078292      4.863582
##   Architect             4.923798      6.213365 40.328253      7.502931
##   Developer             7.023810      8.333333 37.619048      6.904762
##   Doctor                6.577345      6.820950 35.809988      8.404385
##   Engineer              7.342657      6.177156 38.927739      7.808858
##   Entrepreneur          7.581227      8.784597 35.138387      7.099880
##   Journalist            7.343941      5.140759 38.800490      6.854345
##   Lawyer                7.102593      5.298760 41.037204      7.779030
##   Manager               6.343284      6.716418 38.184080      7.462687
##   Mechanic              8.382527      6.847698 36.481700      8.146399
##   Media_Manager         4.642857      5.119048 40.476190      6.309524
##   Musician              6.549118      7.556675 39.924433      6.801008
##   Scientist             7.710558      8.303677 37.129300      8.066429
##   Teacher               6.115108      7.314149 37.649880      6.115108
```

```
##   Writer                6.091371      6.852792 37.944162      5.964467
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant       7.354686      7.117438     7.710558
##   Architect        7.151231      4.806565     6.330598
##   Developer        6.071429      6.666667     7.261905
##   Doctor           8.282582      7.673569     7.673569
##   Engineer         5.827506      5.710956     5.710956
##   Entrepreneur     6.257521      7.340554     5.535499
##   Journalist       7.221542      5.385557     8.690330
##   Lawyer           6.651635      8.117249     6.989853
##   Manager          6.592040      5.721393     6.218905
##   Mechanic         6.139315      6.965762     6.847698
##   Media_Manager    6.547619      5.357143     6.904762
##   Musician         5.793451      6.675063     6.171285
##   Scientist        6.524318      5.931198     6.998814
##   Teacher          7.673861      5.155875     8.273381
##   Writer           7.487310      7.487310     6.598985
##
## $Occupation_Last_Loan_3$col_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     5.405405            6.994536                8.083141
##   Architect      6.933020            7.759563                7.390300
##   Developer      5.992949            6.338798                6.928406
##   Doctor         7.285546            4.699454                5.658199
##   Engineer       6.933020            7.650273                7.390300
##   Entrepreneur   6.933020            6.338798                7.852194
##   Journalist     6.227967            6.448087                6.466513
##   Lawyer         7.050529            4.590164                5.658199
##   Manager        7.050529            6.775956                7.043880
##   Mechanic       6.933020            6.775956                5.773672
##   Media_Manager  6.345476            7.431694                9.815242
##   Musician       6.110458            6.120219                6.351039
##   Scientist      7.520564            6.120219                4.965358
##   Teacher        6.580494            8.415301                5.542725
##   Writer         6.698002            7.540984                5.080831
##
##                 Home Equity Loan Mortgage Loan  No Loan Not Specified
##   Accountant            5.903614      7.566938 6.714077      4.632768
##   Architect             5.060241      6.169965 7.195147      7.231638
##   Developer             7.108434      8.149010 6.609496      6.553672
##   Doctor                6.506024      6.519208 6.149341      7.796610
##   Engineer              7.590361      6.169965 6.985986      7.570621
##   Entrepreneur          7.590361      8.498254 6.107509      6.666667
##   Journalist            7.228916      4.889406 6.630412      6.327684
##   Lawyer                7.590361      5.471478 7.613470      7.796610
##   Manager               6.144578      6.286380 6.421251      6.779661
##   Mechanic              8.554217      6.752037 6.463083      7.796610
##   Media_Manager         4.698795      5.005821 7.111483      5.988701
##   Musician              6.265060      6.984866 6.630412      6.101695
```

```
##   Scientist             7.831325      8.149010 6.546748      7.683616
##   Teacher               6.144578      7.101281 6.567664      5.762712
##   Writer                5.783133      6.286380 6.253922      5.310734
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant       7.328605      7.490637     7.505774
##   Architect        7.210402      5.118602     6.235566
##   Developer        6.028369      6.991261     7.043880
##   Doctor           8.037825      7.865169     7.274827
##   Engineer         5.910165      6.117353     5.658199
##   Entrepreneur     6.146572      7.615481     5.311778
##   Journalist       6.973995      5.493134     8.198614
##   Lawyer           6.973995      8.988764     7.159353
##   Manager          6.264775      5.742821     5.773672
##   Mechanic         6.146572      7.365793     6.697460
##   Media_Manager    6.501182      5.617978     6.697460
##   Musician         5.437352      6.616729     5.658199
##   Scientist        6.501182      6.242197     6.812933
##   Teacher          7.565012      5.368290     7.967667
##   Writer           6.973995      7.365793     6.004619
##
##
## $Occupation_Last_Loan_4
## $Occupation_Last_Loan_4$contingency_table
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant          264                 432                      392
##   Architect           272                 328                      368
##   Developer           352                 368                      240
##   Doctor              288                 368                      312
##   Engineer            392                 408                      376
##   Entrepreneur        312                 344                      400
##   Journalist          336                 344                      312
##   Lawyer              336                 544                      344
##   Manager             296                 408                      376
##   Mechanic            304                 352                      304
##   Media_Manager       280                 384                      304
##   Musician            344                 336                      304
##   Scientist           312                 312                      312
##   Teacher             360                 416                      392
##   Writer              272                 320                      296
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant                328           400    3648           296
##   Architect                 368           288    3896           304
##   Developer                 400           368    3608           432
##   Doctor                    344           424    3456           368
##   Engineer                  312           336    3672           272
##   Entrepreneur              320           352    3352           408
##   Journalist                288           360    3672           264
##   Lawyer                    392           336    3760           368
```

```
##   Manager                    408           328    3488           248
##   Mechanic                   448           328    3576           440
##   Media_Manager              352           312    3720           296
##   Musician                   368           192    3560           344
##   Scientist                  224           416    3536           408
##   Teacher                    264           392    3616           344
##   Writer                     384           336    3440           328
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant            352           360          272
##   Architect             400           288          312
##   Developer             264           344          344
##   Doctor                328           344          336
##   Engineer              352           392          352
##   Entrepreneur          400           312          448
##   Journalist            376           296          288
##   Lawyer                320           344          352
##   Manager               304           312          264
##   Mechanic              384           264          376
##   Media_Manager         408           296          368
##   Musician              320           360          224
##   Scientist             392           384          448
##   Teacher               368           240          280
##   Writer                320           320          288
##
## $Occupation_Last_Loan_4$row_totals
##
##    Accountant     Architect     Developer        Doctor      Engineer
##          6744          6824          6720          6568          6864
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##          6648          6536          7096          6432          6776
## Media_Manager      Musician     Scientist       Teacher        Writer
##          6720          6352          6744          6672          6304
##
## $Occupation_Last_Loan_4$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    4720                    5664                    5032
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    5200                    5168                   54000
##           Not Specified             Payday Loan           Personal Loan
##                    5120                    5288                    4856
##            Student Loan
##                    4952
##
## $Occupation_Last_Loan_4$grand_total
## [1] 100000
##
## $Occupation_Last_Loan_4$cell_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
```

```
##   Accountant        0.264              0.432                   0.392
##   Architect         0.272              0.328                   0.368
##   Developer         0.352              0.368                   0.240
##   Doctor            0.288              0.368                   0.312
##   Engineer          0.392              0.408                   0.376
##   Entrepreneur      0.312              0.344                   0.400
##   Journalist        0.336              0.344                   0.312
##   Lawyer            0.336              0.544                   0.344
##   Manager           0.296              0.408                   0.376
##   Mechanic          0.304              0.352                   0.304
##   Media_Manager     0.280              0.384                   0.304
##   Musician          0.344              0.336                   0.304
##   Scientist         0.312              0.312                   0.312
##   Teacher           0.360              0.416                   0.392
##   Writer            0.272              0.320                   0.296
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant               0.328         0.400   3.648         0.296
##   Architect                0.368         0.288   3.896         0.304
##   Developer                0.400         0.368   3.608         0.432
##   Doctor                   0.344         0.424   3.456         0.368
##   Engineer                 0.312         0.336   3.672         0.272
##   Entrepreneur             0.320         0.352   3.352         0.408
##   Journalist               0.288         0.360   3.672         0.264
##   Lawyer                   0.392         0.336   3.760         0.368
##   Manager                  0.408         0.328   3.488         0.248
##   Mechanic                 0.448         0.328   3.576         0.440
##   Media_Manager            0.352         0.312   3.720         0.296
##   Musician                 0.368         0.192   3.560         0.344
##   Scientist                0.224         0.416   3.536         0.408
##   Teacher                  0.264         0.392   3.616         0.344
##   Writer                   0.384         0.336   3.440         0.328
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant          0.352         0.360        0.272
##   Architect           0.400         0.288        0.312
##   Developer           0.264         0.344        0.344
##   Doctor              0.328         0.344        0.336
##   Engineer            0.352         0.392        0.352
##   Entrepreneur        0.400         0.312        0.448
##   Journalist          0.376         0.296        0.288
##   Lawyer              0.320         0.344        0.352
##   Manager             0.304         0.312        0.264
##   Mechanic            0.384         0.264        0.376
##   Media_Manager       0.408         0.296        0.368
##   Musician            0.320         0.360        0.224
##   Scientist           0.392         0.384        0.448
##   Teacher             0.368         0.240        0.280
##   Writer              0.320         0.320        0.288
##
## $Occupation_Last_Loan_4$row_percentage
```

```
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     3.914591            6.405694                5.812574
##   Architect      3.985932            4.806565                5.392732
##   Developer      5.238095            5.476190                3.571429
##   Doctor         4.384896            5.602923                4.750305
##   Engineer       5.710956            5.944056                5.477855
##   Entrepreneur   4.693141            5.174489                6.016847
##   Journalist     5.140759            5.263158                4.773562
##   Lawyer         4.735062            7.666291                4.847802
##   Manager        4.601990            6.343284                5.845771
##   Mechanic       4.486423            5.194805                4.486423
##   Media_Manager  4.166667            5.714286                4.523810
##   Musician       5.415617            5.289673                4.785894
##   Scientist      4.626335            4.626335                4.626335
##   Teacher        5.395683            6.235012                5.875300
##   Writer         4.314721            5.076142                4.695431
##
##                 Home Equity Loan Mortgage Loan   No Loan Not Specified
##   Accountant            4.863582      5.931198 54.092527      4.389087
##   Architect             5.392732      4.220399 57.092614      4.454865
##   Developer             5.952381      5.476190 53.690476      6.428571
##   Doctor                5.237515      6.455542 52.618758      5.602923
##   Engineer              4.545455      4.895105 53.496503      3.962704
##   Entrepreneur          4.813478      5.294826 50.421179      6.137184
##   Journalist            4.406365      5.507956 56.181151      4.039168
##   Lawyer                5.524239      4.735062 52.987599      5.186020
##   Manager               6.343284      5.099502 54.228856      3.855721
##   Mechanic              6.611570      4.840614 52.774498      6.493506
##   Media_Manager         5.238095      4.642857 55.357143      4.404762
##   Musician              5.793451      3.022670 56.045340      5.415617
##   Scientist             3.321471      6.168446 52.431791      6.049822
##   Teacher               3.956835      5.875300 54.196643      5.155875
##   Writer                6.091371      5.329949 54.568528      5.203046
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant       5.219454      5.338078     4.033215
##   Architect        5.861665      4.220399     4.572098
##   Developer        3.928571      5.119048     5.119048
##   Doctor           4.993910      5.237515     5.115713
##   Engineer         5.128205      5.710956     5.128205
##   Entrepreneur     6.016847      4.693141     6.738869
##   Journalist       5.752754      4.528764     4.406365
##   Lawyer           4.509583      4.847802     4.960541
##   Manager          4.726368      4.850746     4.104478
##   Mechanic         5.667060      3.896104     5.548996
##   Media_Manager    6.071429      4.404762     5.476190
##   Musician         5.037783      5.667506     3.526448
##   Scientist        5.812574      5.693950     6.642942
##   Teacher          5.515588      3.597122     4.196643
##   Writer           5.076142      5.076142     4.568528
```

```
##
## $Occupation_Last_Loan_4$col_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     5.593220            7.627119                7.790143
##   Architect      5.762712            5.790960                7.313196
##   Developer      7.457627            6.497175                4.769475
##   Doctor         6.101695            6.497175                6.200318
##   Engineer       8.305085            7.203390                7.472178
##   Entrepreneur   6.610169            6.073446                7.949126
##   Journalist     7.118644            6.073446                6.200318
##   Lawyer         7.118644            9.604520                6.836248
##   Manager        6.271186            7.203390                7.472178
##   Mechanic       6.440678            6.214689                6.041335
##   Media_Manager  5.932203            6.779661                6.041335
##   Musician       7.288136            5.932203                6.041335
##   Scientist      6.610169            5.508475                6.200318
##   Teacher        7.627119            7.344633                7.790143
##   Writer         5.762712            5.649718                5.882353
##
##                 Home Equity Loan Mortgage Loan  No Loan Not Specified
##   Accountant            6.307692      7.739938 6.755556      5.781250
##   Architect             7.076923      5.572755 7.214815      5.937500
##   Developer             7.692308      7.120743 6.681481      8.437500
##   Doctor                6.615385      8.204334 6.400000      7.187500
##   Engineer              6.000000      6.501548 6.800000      5.312500
##   Entrepreneur          6.153846      6.811146 6.207407      7.968750
##   Journalist            5.538462      6.965944 6.800000      5.156250
##   Lawyer                7.538462      6.501548 6.962963      7.187500
##   Manager               7.846154      6.346749 6.459259      4.843750
##   Mechanic              8.615385      6.346749 6.622222      8.593750
##   Media_Manager         6.769231      6.037152 6.888889      5.781250
##   Musician              7.076923      3.715170 6.592593      6.718750
##   Scientist             4.307692      8.049536 6.548148      7.968750
##   Teacher               5.076923      7.585139 6.696296      6.718750
##   Writer                7.384615      6.501548 6.370370      6.406250
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant       6.656581      7.413509     5.492730
##   Architect        7.564297      5.930807     6.300485
##   Developer        4.992436      7.084020     6.946688
##   Doctor           6.202723      7.084020     6.785137
##   Engineer         6.656581      8.072488     7.108239
##   Entrepreneur     7.564297      6.425041     9.046850
##   Journalist       7.110439      6.095552     5.815832
##   Lawyer           6.051437      7.084020     7.108239
##   Manager          5.748865      6.425041     5.331179
##   Mechanic         7.261725      5.436573     7.592892
##   Media_Manager    7.715582      6.095552     7.431341
##   Musician         6.051437      7.413509     4.523425
##   Scientist        7.413011      7.907743     9.046850
```

```
##   Teacher          6.959153      4.942339     5.654281
##   Writer           6.051437      6.589786     5.815832
##
##
## $Occupation_Last_Loan_5
## $Occupation_Last_Loan_5$contingency_table
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant          232                 224                      224
##   Architect           216                 208                      152
##   Developer           176                 304                      280
##   Doctor              256                 224                      296
##   Engineer            344                 264                      240
##   Entrepreneur        280                 224                      224
##   Journalist          248                 192                      208
##   Lawyer              240                 320                      208
##   Manager             184                 160                      232
##   Mechanic            184                 296                      224
##   Media_Manager       248                 248                      328
##   Musician            168                 216                      216
##   Scientist           248                 280                      264
##   Teacher             224                 168                      224
##   Writer              176                 168                      208
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant                 280           248    4624           200
##   Architect                  272           216    4920           216
##   Developer                  232           256    4632           288
##   Doctor                     184           264    4424           224
##   Engineer                   240           160    4696           200
##   Entrepreneur               248           264    4528           240
##   Journalist                 224           272    4640           176
##   Lawyer                     200           256    4888           224
##   Manager                    208           264    4488           224
##   Mechanic                   256           264    4672           232
##   Media_Manager              240           224    4672           200
##   Musician                   192           136    4584           256
##   Scientist                  240           208    4584           136
##   Teacher                    232           224    4680           176
##   Writer                     152           184    4424           280
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant            248           248          216
##   Architect             200           256          168
##   Developer             176           248          128
##   Doctor                232           216          248
##   Engineer              216           280          224
##   Entrepreneur          232           224          184
##   Journalist            232           200          144
##   Lawyer                232           288          240
##   Manager               200           224          248
```

```
##   Mechanic              232           200          216
##   Media_Manager         200           168          192
##   Musician              216           208          160
##   Scientist             240           264          280
##   Teacher               288           240          216
##   Writer                200           256          256
##
## $Occupation_Last_Loan_5$row_totals
##
##    Accountant     Architect     Developer        Doctor      Engineer
##          6744          6824          6720          6568          6864
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##          6648          6536          7096          6432          6776
## Media_Manager      Musician     Scientist       Teacher        Writer
##          6720          6352          6744          6672          6304
##
## $Occupation_Last_Loan_5$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    3424                    3496                    3528
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    3400                    3440                   69456
##           Not Specified             Payday Loan           Personal Loan
##                    3272                    3344                    3520
##            Student Loan
##                    3120
##
## $Occupation_Last_Loan_5$grand_total
## [1] 100000
##
## $Occupation_Last_Loan_5$cell_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant        0.232               0.224                   0.224
##   Architect         0.216               0.208                   0.152
##   Developer         0.176               0.304                   0.280
##   Doctor            0.256               0.224                   0.296
##   Engineer          0.344               0.264                   0.240
##   Entrepreneur      0.280               0.224                   0.224
##   Journalist        0.248               0.192                   0.208
##   Lawyer            0.240               0.320                   0.208
##   Manager           0.184               0.160                   0.232
##   Mechanic          0.184               0.296                   0.224
##   Media_Manager     0.248               0.248                   0.328
##   Musician          0.168               0.216                   0.216
##   Scientist         0.248               0.280                   0.264
##   Teacher           0.224               0.168                   0.224
##   Writer            0.176               0.168                   0.208
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant               0.280         0.248   4.624         0.200
```

```
##   Architect                0.272         0.216   4.920         0.216
##   Developer                0.232         0.256   4.632         0.288
##   Doctor                   0.184         0.264   4.424         0.224
##   Engineer                 0.240         0.160   4.696         0.200
##   Entrepreneur             0.248         0.264   4.528         0.240
##   Journalist               0.224         0.272   4.640         0.176
##   Lawyer                   0.200         0.256   4.888         0.224
##   Manager                  0.208         0.264   4.488         0.224
##   Mechanic                 0.256         0.264   4.672         0.232
##   Media_Manager            0.240         0.224   4.672         0.200
##   Musician                 0.192         0.136   4.584         0.256
##   Scientist                0.240         0.208   4.584         0.136
##   Teacher                  0.232         0.224   4.680         0.176
##   Writer                   0.152         0.184   4.424         0.280
##
##                  Payday Loan Personal Loan Student Loan
##   Accountant          0.248         0.248        0.216
##   Architect           0.200         0.256        0.168
##   Developer           0.176         0.248        0.128
##   Doctor              0.232         0.216        0.248
##   Engineer            0.216         0.280        0.224
##   Entrepreneur        0.232         0.224        0.184
##   Journalist          0.232         0.200        0.144
##   Lawyer              0.232         0.288        0.240
##   Manager             0.200         0.224        0.248
##   Mechanic            0.232         0.200        0.216
##   Media_Manager       0.200         0.168        0.192
##   Musician            0.216         0.208        0.160
##   Scientist           0.240         0.264        0.280
##   Teacher             0.288         0.240        0.216
##   Writer              0.200         0.256        0.256
##
## $Occupation_Last_Loan_5$row_percentage
##
##                  Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant      3.440095            3.321471                3.321471
##   Architect       3.165299            3.048066                2.227433
##   Developer       2.619048            4.523810                4.166667
##   Doctor          3.897686            3.410475                4.506699
##   Engineer        5.011655            3.846154                3.496503
##   Entrepreneur    4.211793            3.369434                3.369434
##   Journalist      3.794370            2.937576                3.182375
##   Lawyer          3.382187            4.509583                2.931229
##   Manager         2.860697            2.487562                3.606965
##   Mechanic        2.715466            4.368359                3.305785
##   Media_Manager   3.690476            3.690476                4.880952
##   Musician        2.644836            3.400504                3.400504
##   Scientist       3.677343            4.151839                3.914591
##   Teacher         3.357314            2.517986                3.357314
##   Writer          2.791878            2.664975                3.299492
##
```

```
##                 Home Equity Loan Mortgage Loan   No Loan Not Specified
##   Accountant            4.151839      3.677343 68.564650      2.965599
##   Architect             3.985932      3.165299 72.098476      3.165299
##   Developer             3.452381      3.809524 68.928571      4.285714
##   Doctor                2.801462      4.019488 67.356882      3.410475
##   Engineer              3.496503      2.331002 68.414918      2.913753
##   Entrepreneur          3.730445      3.971119 68.110710      3.610108
##   Journalist            3.427173      4.161567 70.991432      2.692778
##   Lawyer                2.818489      3.607666 68.883878      3.156708
##   Manager               3.233831      4.104478 69.776119      3.482587
##   Mechanic              3.778040      3.896104 68.949233      3.423849
##   Media_Manager         3.571429      3.333333 69.523810      2.976190
##   Musician              3.022670      2.141058 72.166247      4.030227
##   Scientist             3.558719      3.084223 67.971530      2.016607
##   Teacher               3.477218      3.357314 70.143885      2.637890
##   Writer                2.411168      2.918782 70.177665      4.441624
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant       3.677343      3.677343     3.202847
##   Architect        2.930832      3.751465     2.461899
##   Developer        2.619048      3.690476     1.904762
##   Doctor           3.532278      3.288672     3.775883
##   Engineer         3.146853      4.079254     3.263403
##   Entrepreneur     3.489771      3.369434     2.767750
##   Journalist       3.549572      3.059976     2.203182
##   Lawyer           3.269448      4.058625     3.382187
##   Manager          3.109453      3.482587     3.855721
##   Mechanic         3.423849      2.951594     3.187721
##   Media_Manager    2.976190      2.500000     2.857143
##   Musician         3.400504      3.274559     2.518892
##   Scientist        3.558719      3.914591     4.151839
##   Teacher          4.316547      3.597122     3.237410
##   Writer           3.172589      4.060914     4.060914
##
## $Occupation_Last_Loan_5$col_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     6.775701            6.407323                6.349206
##   Architect      6.308411            5.949657                4.308390
##   Developer      5.140187            8.695652                7.936508
##   Doctor         7.476636            6.407323                8.390023
##   Engineer      10.046729            7.551487                6.802721
##   Entrepreneur   8.177570            6.407323                6.349206
##   Journalist     7.242991            5.491991                5.895692
##   Lawyer         7.009346            9.153318                5.895692
##   Manager        5.373832            4.576659                6.575964
##   Mechanic       5.373832            8.466819                6.349206
##   Media_Manager  7.242991            7.093822                9.297052
##   Musician       4.906542            6.178490                6.122449
##   Scientist      7.242991            8.009153                7.482993
##   Teacher        6.542056            4.805492                6.349206
```

```
##   Writer         5.140187            4.805492                5.895692
##
##                 Home Equity Loan Mortgage Loan   No Loan Not Specified
##   Accountant            8.235294      7.209302  6.657452      6.112469
##   Architect             8.000000      6.279070  7.083621      6.601467
##   Developer             6.823529      7.441860  6.668970      8.801956
##   Doctor                5.411765      7.674419  6.369500      6.845966
##   Engineer              7.058824      4.651163  6.761115      6.112469
##   Entrepreneur          7.294118      7.674419  6.519235      7.334963
##   Journalist            6.588235      7.906977  6.680488      5.378973
##   Lawyer                5.882353      7.441860  7.037549      6.845966
##   Manager               6.117647      7.674419  6.461645      6.845966
##   Mechanic              7.529412      7.674419  6.726561      7.090465
##   Media_Manager         7.058824      6.511628  6.726561      6.112469
##   Musician              5.647059      3.953488  6.599862      7.823961
##   Scientist             7.058824      6.046512  6.599862      4.156479
##   Teacher               6.823529      6.511628  6.738079      5.378973
##   Writer                4.470588      5.348837  6.369500      8.557457
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant       7.416268      7.045455     6.923077
##   Architect        5.980861      7.272727     5.384615
##   Developer        5.263158      7.045455     4.102564
##   Doctor           6.937799      6.136364     7.948718
##   Engineer         6.459330      7.954545     7.179487
##   Entrepreneur     6.937799      6.363636     5.897436
##   Journalist       6.937799      5.681818     4.615385
##   Lawyer           6.937799      8.181818     7.692308
##   Manager          5.980861      6.363636     7.948718
##   Mechanic         6.937799      5.681818     6.923077
##   Media_Manager    5.980861      4.772727     6.153846
##   Musician         6.459330      5.909091     5.128205
##   Scientist        7.177033      7.500000     8.974359
##   Teacher          8.612440      6.818182     6.923077
##   Writer           5.980861      7.272727     8.205128
##
##
## $Occupation_Last_Loan_6
## $Occupation_Last_Loan_6$contingency_table
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant          160                 192                      200
##   Architect           144                 152                      152
##   Developer           136                 136                      200
##   Doctor              144                 208                      208
##   Engineer            208                 224                      176
##   Entrepreneur        200                 160                      192
##   Journalist          152                 192                      112
##   Lawyer              176                 144                      128
##   Manager             200                 216                      120
##   Mechanic            168                 120                      208
```

```
##   Media_Manager       144                 200                      152
##   Musician            136                 192                      112
##   Scientist           176                 168                      184
##   Teacher             168                 184                      144
##   Writer              160                 120                      176
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant                160           168    5144           160
##   Architect                 168           112    5376           200
##   Developer                 168           216    5064           200
##   Doctor                    160           240    4856           176
##   Engineer                  128           152    5232           248
##   Entrepreneur              128           160    5112           168
##   Journalist                216           144    5152           136
##   Lawyer                    192           136    5632           136
##   Manager                   168           168    4928           144
##   Mechanic                  176           120    5280           136
##   Media_Manager             192           128    5088           224
##   Musician                  136           192    4968           216
##   Scientist                 152           136    5160           152
##   Teacher                   160           152    5152           200
##   Writer                    160           168    4840           128
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant            192           216          152
##   Architect             144           176          200
##   Developer             248           184          168
##   Doctor                232           200          144
##   Engineer              160           216          120
##   Entrepreneur          144           176          208
##   Journalist            120           160          152
##   Lawyer                176           216          160
##   Manager               144           152          192
##   Mechanic              208           200          160
##   Media_Manager         200           168          224
##   Musician              120           112          168
##   Scientist             240           232          144
##   Teacher               176           176          160
##   Writer                184           152          216
##
## $Occupation_Last_Loan_6$row_totals
##
##    Accountant     Architect     Developer        Doctor      Engineer
##          6744          6824          6720          6568          6864
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##          6648          6536          7096          6432          6776
## Media_Manager      Musician     Scientist       Teacher        Writer
##          6720          6352          6744          6672          6304
##
## $Occupation_Last_Loan_6$col_totals
##
```

```
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    2472                    2608                    2464
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    2464                    2392                   76984
##           Not Specified             Payday Loan           Personal Loan
##                    2624                    2688                    2736
##            Student Loan
##                    2568
##
## $Occupation_Last_Loan_6$grand_total
## [1] 100000
##
## $Occupation_Last_Loan_6$cell_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant        0.160               0.192                   0.200
##   Architect         0.144               0.152                   0.152
##   Developer         0.136               0.136                   0.200
##   Doctor            0.144               0.208                   0.208
##   Engineer          0.208               0.224                   0.176
##   Entrepreneur      0.200               0.160                   0.192
##   Journalist        0.152               0.192                   0.112
##   Lawyer            0.176               0.144                   0.128
##   Manager           0.200               0.216                   0.120
##   Mechanic          0.168               0.120                   0.208
##   Media_Manager     0.144               0.200                   0.152
##   Musician          0.136               0.192                   0.112
##   Scientist         0.176               0.168                   0.184
##   Teacher           0.168               0.184                   0.144
##   Writer            0.160               0.120                   0.176
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant               0.160         0.168   5.144         0.160
##   Architect                0.168         0.112   5.376         0.200
##   Developer                0.168         0.216   5.064         0.200
##   Doctor                   0.160         0.240   4.856         0.176
##   Engineer                 0.128         0.152   5.232         0.248
##   Entrepreneur             0.128         0.160   5.112         0.168
##   Journalist               0.216         0.144   5.152         0.136
##   Lawyer                   0.192         0.136   5.632         0.136
##   Manager                  0.168         0.168   4.928         0.144
##   Mechanic                 0.176         0.120   5.280         0.136
##   Media_Manager            0.192         0.128   5.088         0.224
##   Musician                 0.136         0.192   4.968         0.216
##   Scientist                0.152         0.136   5.160         0.152
##   Teacher                  0.160         0.152   5.152         0.200
##   Writer                   0.160         0.168   4.840         0.128
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant          0.192         0.216        0.152
##   Architect           0.144         0.176        0.200
```

```
##   Developer           0.248         0.184        0.168
##   Doctor              0.232         0.200        0.144
##   Engineer            0.160         0.216        0.120
##   Entrepreneur        0.144         0.176        0.208
##   Journalist          0.120         0.160        0.152
##   Lawyer              0.176         0.216        0.160
##   Manager             0.144         0.152        0.192
##   Mechanic            0.208         0.200        0.160
##   Media_Manager       0.200         0.168        0.224
##   Musician            0.120         0.112        0.168
##   Scientist           0.240         0.232        0.144
##   Teacher             0.176         0.176        0.160
##   Writer              0.184         0.152        0.216
##
## $Occupation_Last_Loan_6$row_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     2.372479            2.846975                2.965599
##   Architect      2.110199            2.227433                2.227433
##   Developer      2.023810            2.023810                2.976190
##   Doctor         2.192448            3.166870                3.166870
##   Engineer       3.030303            3.263403                2.564103
##   Entrepreneur   3.008424            2.406739                2.888087
##   Journalist     2.325581            2.937576                1.713586
##   Lawyer         2.480271            2.029312                1.803833
##   Manager        3.109453            3.358209                1.865672
##   Mechanic       2.479339            1.770956                3.069658
##   Media_Manager  2.142857            2.976190                2.261905
##   Musician       2.141058            3.022670                1.763224
##   Scientist      2.609727            2.491103                2.728351
##   Teacher        2.517986            2.757794                2.158273
##   Writer         2.538071            1.903553                2.791878
##
##                 Home Equity Loan Mortgage Loan   No Loan Not Specified
##   Accountant            2.372479      2.491103 76.275208      2.372479
##   Architect             2.461899      1.641266 78.780774      2.930832
##   Developer             2.500000      3.214286 75.357143      2.976190
##   Doctor                2.436054      3.654080 73.934227      2.679659
##   Engineer              1.864802      2.214452 76.223776      3.613054
##   Entrepreneur          1.925391      2.406739 76.895307      2.527076
##   Journalist            3.304774      2.203182 78.824969      2.080783
##   Lawyer                2.705750      1.916573 79.368658      1.916573
##   Manager               2.611940      2.611940 76.616915      2.238806
##   Mechanic              2.597403      1.770956 77.922078      2.007084
##   Media_Manager         2.857143      1.904762 75.714286      3.333333
##   Musician              2.141058      3.022670 78.211587      3.400504
##   Scientist             2.253855      2.016607 76.512456      2.253855
##   Teacher               2.398082      2.278177 77.218225      2.997602
##   Writer                2.538071      2.664975 76.776650      2.030457
##
##                 Payday Loan Personal Loan Student Loan
```

```
##   Accountant       2.846975      3.202847     2.253855
##   Architect        2.110199      2.579132     2.930832
##   Developer        3.690476      2.738095     2.500000
##   Doctor           3.532278      3.045067     2.192448
##   Engineer         2.331002      3.146853     1.748252
##   Entrepreneur     2.166065      2.647413     3.128761
##   Journalist       1.835985      2.447980     2.325581
##   Lawyer           2.480271      3.043968     2.254791
##   Manager          2.238806      2.363184     2.985075
##   Mechanic         3.069658      2.951594     2.361275
##   Media_Manager    2.976190      2.500000     3.333333
##   Musician         1.889169      1.763224     2.644836
##   Scientist        3.558719      3.440095     2.135231
##   Teacher          2.637890      2.637890     2.398082
##   Writer           2.918782      2.411168     3.426396
##
## $Occupation_Last_Loan_6$col_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     6.472492            7.361963                8.116883
##   Architect      5.825243            5.828221                6.168831
##   Developer      5.501618            5.214724                8.116883
##   Doctor         5.825243            7.975460                8.441558
##   Engineer       8.414239            8.588957                7.142857
##   Entrepreneur   8.090615            6.134969                7.792208
##   Journalist     6.148867            7.361963                4.545455
##   Lawyer         7.119741            5.521472                5.194805
##   Manager        8.090615            8.282209                4.870130
##   Mechanic       6.796117            4.601227                8.441558
##   Media_Manager  5.825243            7.668712                6.168831
##   Musician       5.501618            7.361963                4.545455
##   Scientist      7.119741            6.441718                7.467532
##   Teacher        6.796117            7.055215                5.844156
##   Writer         6.472492            4.601227                7.142857
##
##                 Home Equity Loan Mortgage Loan   No Loan Not Specified
##   Accountant            6.493506      7.023411  6.681908      6.097561
##   Architect             6.818182      4.682274  6.983269      7.621951
##   Developer             6.818182      9.030100  6.577990      7.621951
##   Doctor                6.493506     10.033445  6.307804      6.707317
##   Engineer              5.194805      6.354515  6.796217      9.451220
##   Entrepreneur          5.194805      6.688963  6.640341      6.402439
##   Journalist            8.766234      6.020067  6.692300      5.182927
##   Lawyer                7.792208      5.685619  7.315806      5.182927
##   Manager               6.818182      7.023411  6.401330      5.487805
##   Mechanic              7.142857      5.016722  6.858568      5.182927
##   Media_Manager         7.792208      5.351171  6.609166      8.536585
##   Musician              5.519481      8.026756  6.453289      8.231707
##   Scientist             6.168831      5.685619  6.702691      5.792683
##   Teacher               6.493506      6.354515  6.692300      7.621951
##   Writer                6.493506      7.023411  6.287021      4.878049
```

```
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant       7.142857      7.894737     5.919003
##   Architect        5.357143      6.432749     7.788162
##   Developer        9.226190      6.725146     6.542056
##   Doctor           8.630952      7.309942     5.607477
##   Engineer         5.952381      7.894737     4.672897
##   Entrepreneur     5.357143      6.432749     8.099688
##   Journalist       4.464286      5.847953     5.919003
##   Lawyer           6.547619      7.894737     6.230530
##   Manager          5.357143      5.555556     7.476636
##   Mechanic         7.738095      7.309942     6.230530
##   Media_Manager    7.440476      6.140351     8.722741
##   Musician         4.464286      4.093567     6.542056
##   Scientist        8.928571      8.479532     5.607477
##   Teacher          6.547619      6.432749     6.230530
##   Writer           6.845238      5.555556     8.411215
##
##
## $Occupation_Last_Loan_7
## $Occupation_Last_Loan_7$contingency_table
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant          104                 112                      128
##   Architect           136                  88                       72
##   Developer           112                 112                      144
##   Doctor               88                 144                      104
##   Engineer            112                  88                      120
##   Entrepreneur         96                 152                       72
##   Journalist          112                  88                       64
##   Lawyer               96                 144                      104
##   Manager             120                 128                      104
##   Mechanic            104                  80                       48
##   Media_Manager        64                 120                      128
##   Musician            112                 160                      104
##   Scientist           128                 136                       72
##   Teacher             104                  96                       80
##   Writer              112                  72                       88
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant                 128           136    5712            96
##   Architect                  128           120    5896            96
##   Developer                  104           128    5600            96
##   Doctor                     136           128    5448           128
##   Engineer                   112           136    5768           160
##   Entrepreneur                96           152    5616           112
##   Journalist                 120            88    5704           112
##   Lawyer                     112           144    6112            88
##   Manager                     96           120    5416            80
##   Mechanic                   112           128    5768           136
##   Media_Manager               56           128    5736            96
```

```
##   Musician                    64            40    5392           128
##   Scientist                   72           128    5752           112
##   Teacher                     88            96    5784            88
##   Writer                     104           120    5424            80
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant            112            96          120
##   Architect             112            96           80
##   Developer             200            96          128
##   Doctor                128           136          128
##   Engineer              152            88          128
##   Entrepreneur          144           112           96
##   Journalist             96            96           56
##   Lawyer                 80           120           96
##   Manager               128           112          128
##   Mechanic              160           104          136
##   Media_Manager          88           160          144
##   Musician              168            72          112
##   Scientist              96           112          136
##   Teacher               136           152           48
##   Writer                120           136           48
##
## $Occupation_Last_Loan_7$row_totals
##
##    Accountant     Architect     Developer        Doctor      Engineer
##          6744          6824          6720          6568          6864
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##          6648          6536          7096          6432          6776
## Media_Manager      Musician     Scientist       Teacher        Writer
##          6720          6352          6744          6672          6304
##
## $Occupation_Last_Loan_7$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    1600                    1720                    1432
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    1528                    1792                   85128
##           Not Specified             Payday Loan           Personal Loan
##                    1608                    1920                    1688
##            Student Loan
##                    1584
##
## $Occupation_Last_Loan_7$grand_total
## [1] 100000
##
## $Occupation_Last_Loan_7$cell_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant        0.104               0.112                   0.128
##   Architect         0.136               0.088                   0.072
##   Developer         0.112               0.112                   0.144
```

```
##   Doctor            0.088               0.144                    0.104
##   Engineer          0.112               0.088                    0.120
##   Entrepreneur      0.096               0.152                    0.072
##   Journalist        0.112               0.088                    0.064
##   Lawyer            0.096               0.144                    0.104
##   Manager           0.120               0.128                    0.104
##   Mechanic          0.104               0.080                    0.048
##   Media_Manager     0.064               0.120                    0.128
##   Musician          0.112               0.160                    0.104
##   Scientist         0.128               0.136                    0.072
##   Teacher           0.104               0.096                    0.080
##   Writer            0.112               0.072                    0.088
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant               0.128         0.136   5.712         0.096
##   Architect                0.128         0.120   5.896         0.096
##   Developer                0.104         0.128   5.600         0.096
##   Doctor                   0.136         0.128   5.448         0.128
##   Engineer                 0.112         0.136   5.768         0.160
##   Entrepreneur             0.096         0.152   5.616         0.112
##   Journalist               0.120         0.088   5.704         0.112
##   Lawyer                   0.112         0.144   6.112         0.088
##   Manager                  0.096         0.120   5.416         0.080
##   Mechanic                 0.112         0.128   5.768         0.136
##   Media_Manager            0.056         0.128   5.736         0.096
##   Musician                 0.064         0.040   5.392         0.128
##   Scientist                0.072         0.128   5.752         0.112
##   Teacher                  0.088         0.096   5.784         0.088
##   Writer                   0.104         0.120   5.424         0.080
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant          0.112         0.096        0.120
##   Architect           0.112         0.096        0.080
##   Developer           0.200         0.096        0.128
##   Doctor              0.128         0.136        0.128
##   Engineer            0.152         0.088        0.128
##   Entrepreneur        0.144         0.112        0.096
##   Journalist          0.096         0.096        0.056
##   Lawyer              0.080         0.120        0.096
##   Manager             0.128         0.112        0.128
##   Mechanic            0.160         0.104        0.136
##   Media_Manager       0.088         0.160        0.144
##   Musician            0.168         0.072        0.112
##   Scientist           0.096         0.112        0.136
##   Teacher             0.136         0.152        0.048
##   Writer              0.120         0.136        0.048
##
## $Occupation_Last_Loan_7$row_percentage
##
##                  Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     1.5421115           1.6607355                1.8979834
```

```
##   Architect      1.9929660           1.2895662                1.0550996
##   Developer      1.6666667           1.6666667                2.1428571
##   Doctor         1.3398295           2.1924482                1.5834348
##   Engineer       1.6317016           1.2820513                1.7482517
##   Entrepreneur   1.4440433           2.2864019                1.0830325
##   Journalist     1.7135863           1.3463892                0.9791922
##   Lawyer         1.3528749           2.0293123                1.4656144
##   Manager        1.8656716           1.9900498                1.6169154
##   Mechanic       1.5348288           1.1806375                0.7083825
##   Media_Manager  0.9523810           1.7857143                1.9047619
##   Musician       1.7632242           2.5188917                1.6372796
##   Scientist      1.8979834           2.0166074                1.0676157
##   Teacher        1.5587530           1.4388489                1.1990408
##   Writer         1.7766497           1.1421320                1.3959391
##
##                 Home Equity Loan Mortgage Loan    No Loan Not Specified
##   Accountant           1.8979834     2.0166074 84.6975089     1.4234875
##   Architect            1.8757327     1.7584994 86.4009379     1.4067995
##   Developer            1.5476190     1.9047619 83.3333333     1.4285714
##   Doctor               2.0706456     1.9488429 82.9476248     1.9488429
##   Engineer             1.6317016     1.9813520 84.0326340     2.3310023
##   Entrepreneur         1.4440433     2.2864019 84.4765343     1.6847172
##   Journalist           1.8359853     1.3463892 87.2705018     1.7135863
##   Lawyer               1.5783540     2.0293123 86.1330327     1.2401353
##   Manager              1.4925373     1.8656716 84.2039801     1.2437811
##   Mechanic             1.6528926     1.8890201 85.1239669     2.0070838
##   Media_Manager        0.8333333     1.9047619 85.3571429     1.4285714
##   Musician             1.0075567     0.6297229 84.8866499     2.0151134
##   Scientist            1.0676157     1.8979834 85.2906287     1.6607355
##   Teacher              1.3189448     1.4388489 86.6906475     1.3189448
##   Writer               1.6497462     1.9035533 86.0406091     1.2690355
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant      1.6607355     1.4234875    1.7793594
##   Architect       1.6412661     1.4067995    1.1723329
##   Developer       2.9761905     1.4285714    1.9047619
##   Doctor          1.9488429     2.0706456    1.9488429
##   Engineer        2.2144522     1.2820513    1.8648019
##   Entrepreneur    2.1660650     1.6847172    1.4440433
##   Journalist      1.4687882     1.4687882    0.8567931
##   Lawyer          1.1273957     1.6910936    1.3528749
##   Manager         1.9900498     1.7412935    1.9900498
##   Mechanic        2.3612751     1.5348288    2.0070838
##   Media_Manager   1.3095238     2.3809524    2.1428571
##   Musician        2.6448363     1.1335013    1.7632242
##   Scientist       1.4234875     1.6607355    2.0166074
##   Teacher         2.0383693     2.2781775    0.7194245
##   Writer          1.9035533     2.1573604    0.7614213
##
## $Occupation_Last_Loan_7$col_percentage
##
```

```
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     6.500000            6.511628                8.938547
##   Architect      8.500000            5.116279                5.027933
##   Developer      7.000000            6.511628               10.055866
##   Doctor         5.500000            8.372093                7.262570
##   Engineer       7.000000            5.116279                8.379888
##   Entrepreneur   6.000000            8.837209                5.027933
##   Journalist     7.000000            5.116279                4.469274
##   Lawyer         6.000000            8.372093                7.262570
##   Manager        7.500000            7.441860                7.262570
##   Mechanic       6.500000            4.651163                3.351955
##   Media_Manager  4.000000            6.976744                8.938547
##   Musician       7.000000            9.302326                7.262570
##   Scientist      8.000000            7.906977                5.027933
##   Teacher        6.500000            5.581395                5.586592
##   Writer         7.000000            4.186047                6.145251
##
##                 Home Equity Loan Mortgage Loan   No Loan Not Specified
##   Accountant            8.376963      7.589286  6.709896      5.970149
##   Architect             8.376963      6.696429  6.926041      5.970149
##   Developer             6.806283      7.142857  6.578329      5.970149
##   Doctor                8.900524      7.142857  6.399774      7.960199
##   Engineer              7.329843      7.589286  6.775679      9.950249
##   Entrepreneur          6.282723      8.482143  6.597124      6.965174
##   Journalist            7.853403      4.910714  6.700498      6.965174
##   Lawyer                7.329843      8.035714  7.179776      5.472637
##   Manager               6.282723      6.696429  6.362184      4.975124
##   Mechanic              7.329843      7.142857  6.775679      8.457711
##   Media_Manager         3.664921      7.142857  6.738089      5.970149
##   Musician              4.188482      2.232143  6.333991      7.960199
##   Scientist             4.712042      7.142857  6.756884      6.965174
##   Teacher               5.759162      5.357143  6.794474      5.472637
##   Writer                6.806283      6.696429  6.371582      4.975124
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant       5.833333      5.687204     7.575758
##   Architect        5.833333      5.687204     5.050505
##   Developer       10.416667      5.687204     8.080808
##   Doctor           6.666667      8.056872     8.080808
##   Engineer         7.916667      5.213270     8.080808
##   Entrepreneur     7.500000      6.635071     6.060606
##   Journalist       5.000000      5.687204     3.535354
##   Lawyer           4.166667      7.109005     6.060606
##   Manager          6.666667      6.635071     8.080808
##   Mechanic         8.333333      6.161137     8.585859
##   Media_Manager    4.583333      9.478673     9.090909
##   Musician         8.750000      4.265403     7.070707
##   Scientist        5.000000      6.635071     8.585859
##   Teacher          7.083333      9.004739     3.030303
##   Writer           6.250000      8.056872     3.030303
##
```

```
##
## $Occupation_Last_Loan_8
## $Occupation_Last_Loan_8$contingency_table
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant           40                  72                      56
##   Architect            48                  96                      40
##   Developer            48                  48                      48
##   Doctor               40                  56                      24
##   Engineer             24                  64                      56
##   Entrepreneur         56                  32                      24
##   Journalist           56                  72                      32
##   Lawyer               80                  24                      40
##   Manager              64                  56                      56
##   Mechanic             40                  48                      88
##   Media_Manager        32                  56                      88
##   Musician             40                  40                      72
##   Scientist            32                 112                      64
##   Teacher              56                  48                      48
##   Writer               64                  48                      24
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant                  64            32    6296            64
##   Architect                   72            24    6352            48
##   Developer                   64            56    6136            80
##   Doctor                      64            72    6016            64
##   Engineer                    80            56    6392            48
##   Entrepreneur                40            56    6216            72
##   Journalist                  64            24    6144            40
##   Lawyer                      72            40    6688            40
##   Manager                     48            72    5880            64
##   Mechanic                    48            32    6224            40
##   Media_Manager               64            48    6288            56
##   Musician                    72            48    5800            40
##   Scientist                   32            64    6224            32
##   Teacher                     32            32    6296            32
##   Writer                      72            80    5856            24
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant             40            24           56
##   Architect              56            56           32
##   Developer              64            72          104
##   Doctor                120            64           48
##   Engineer               24            56           64
##   Entrepreneur           64            40           48
##   Journalist             32            48           24
##   Lawyer                 16            32           64
##   Manager                72            56           64
##   Mechanic              120            72           64
##   Media_Manager          24            40           24
##   Musician               72            56          112
```

```
##   Scientist              64            48           72
##   Teacher                64            24           40
##   Writer                 56            32           48
##
## $Occupation_Last_Loan_8$row_totals
##
##    Accountant     Architect     Developer        Doctor      Engineer
##          6744          6824          6720          6568          6864
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##          6648          6536          7096          6432          6776
## Media_Manager      Musician     Scientist       Teacher        Writer
##          6720          6352          6744          6672          6304
##
## $Occupation_Last_Loan_8$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     720                     872                     760
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     888                     736                   92808
##           Not Specified             Payday Loan           Personal Loan
##                     744                     888                     720
##            Student Loan
##                     864
##
## $Occupation_Last_Loan_8$grand_total
## [1] 100000
##
## $Occupation_Last_Loan_8$cell_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant        0.040               0.072                   0.056
##   Architect         0.048               0.096                   0.040
##   Developer         0.048               0.048                   0.048
##   Doctor            0.040               0.056                   0.024
##   Engineer          0.024               0.064                   0.056
##   Entrepreneur      0.056               0.032                   0.024
##   Journalist        0.056               0.072                   0.032
##   Lawyer            0.080               0.024                   0.040
##   Manager           0.064               0.056                   0.056
##   Mechanic          0.040               0.048                   0.088
##   Media_Manager     0.032               0.056                   0.088
##   Musician          0.040               0.040                   0.072
##   Scientist         0.032               0.112                   0.064
##   Teacher           0.056               0.048                   0.048
##   Writer            0.064               0.048                   0.024
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant               0.064         0.032   6.296         0.064
##   Architect                0.072         0.024   6.352         0.048
##   Developer                0.064         0.056   6.136         0.080
##   Doctor                   0.064         0.072   6.016         0.064
```

```
##   Engineer                 0.080         0.056   6.392         0.048
##   Entrepreneur             0.040         0.056   6.216         0.072
##   Journalist               0.064         0.024   6.144         0.040
##   Lawyer                   0.072         0.040   6.688         0.040
##   Manager                  0.048         0.072   5.880         0.064
##   Mechanic                 0.048         0.032   6.224         0.040
##   Media_Manager            0.064         0.048   6.288         0.056
##   Musician                 0.072         0.048   5.800         0.040
##   Scientist                0.032         0.064   6.224         0.032
##   Teacher                  0.032         0.032   6.296         0.032
##   Writer                   0.072         0.080   5.856         0.024
##
##                  Payday Loan Personal Loan Student Loan
##   Accountant          0.040         0.024        0.056
##   Architect           0.056         0.056        0.032
##   Developer           0.064         0.072        0.104
##   Doctor              0.120         0.064        0.048
##   Engineer            0.024         0.056        0.064
##   Entrepreneur        0.064         0.040        0.048
##   Journalist          0.032         0.048        0.024
##   Lawyer              0.016         0.032        0.064
##   Manager             0.072         0.056        0.064
##   Mechanic            0.120         0.072        0.064
##   Media_Manager       0.024         0.040        0.024
##   Musician            0.072         0.056        0.112
##   Scientist           0.064         0.048        0.072
##   Teacher             0.064         0.024        0.040
##   Writer              0.056         0.032        0.048
##
## $Occupation_Last_Loan_8$row_percentage
##
##                  Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     0.5931198           1.0676157               0.8303677
##   Architect      0.7033998           1.4067995               0.5861665
##   Developer      0.7142857           0.7142857               0.7142857
##   Doctor         0.6090134           0.8526188               0.3654080
##   Engineer       0.3496503           0.9324009               0.8158508
##   Entrepreneur   0.8423586           0.4813478               0.3610108
##   Journalist     0.8567931           1.1015912               0.4895961
##   Lawyer         1.1273957           0.3382187               0.5636979
##   Manager        0.9950249           0.8706468               0.8706468
##   Mechanic       0.5903188           0.7083825               1.2987013
##   Media_Manager  0.4761905           0.8333333               1.3095238
##   Musician       0.6297229           0.6297229               1.1335013
##   Scientist      0.4744958           1.6607355               0.9489917
##   Teacher        0.8393285           0.7194245               0.7194245
##   Writer         1.0152284           0.7614213               0.3807107
##
##                  Home Equity Loan Mortgage Loan    No Loan Not Specified
##   Accountant            0.9489917     0.4744958 93.3570581     0.9489917
##   Architect             1.0550996     0.3516999 93.0832356     0.7033998
```

```
##   Developer            0.9523810     0.8333333 91.3095238     1.1904762
##   Doctor               0.9744214     1.0962241 91.5956151     0.9744214
##   Engineer             1.1655012     0.8158508 93.1235431     0.6993007
##   Entrepreneur         0.6016847     0.8423586 93.5018051     1.0830325
##   Journalist           0.9791922     0.3671971 94.0024480     0.6119951
##   Lawyer               1.0146561     0.5636979 94.2502818     0.5636979
##   Manager              0.7462687     1.1194030 91.4179104     0.9950249
##   Mechanic             0.7083825     0.4722550 91.8536009     0.5903188
##   Media_Manager        0.9523810     0.7142857 93.5714286     0.8333333
##   Musician             1.1335013     0.7556675 91.3098237     0.6297229
##   Scientist            0.4744958     0.9489917 92.2894425     0.4744958
##   Teacher              0.4796163     0.4796163 94.3645084     0.4796163
##   Writer               1.1421320     1.2690355 92.8934010     0.3807107
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant      0.5931198     0.3558719    0.8303677
##   Architect       0.8206331     0.8206331    0.4689332
##   Developer       0.9523810     1.0714286    1.5476190
##   Doctor          1.8270402     0.9744214    0.7308161
##   Engineer        0.3496503     0.8158508    0.9324009
##   Entrepreneur    0.9626955     0.6016847    0.7220217
##   Journalist      0.4895961     0.7343941    0.3671971
##   Lawyer          0.2254791     0.4509583    0.9019166
##   Manager         1.1194030     0.8706468    0.9950249
##   Mechanic        1.7709563     1.0625738    0.9445100
##   Media_Manager   0.3571429     0.5952381    0.3571429
##   Musician        1.1335013     0.8816121    1.7632242
##   Scientist       0.9489917     0.7117438    1.0676157
##   Teacher         0.9592326     0.3597122    0.5995204
##   Writer          0.8883249     0.5076142    0.7614213
##
## $Occupation_Last_Loan_8$col_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     5.555556            8.256881                7.368421
##   Architect      6.666667           11.009174                5.263158
##   Developer      6.666667            5.504587                6.315789
##   Doctor         5.555556            6.422018                3.157895
##   Engineer       3.333333            7.339450                7.368421
##   Entrepreneur   7.777778            3.669725                3.157895
##   Journalist     7.777778            8.256881                4.210526
##   Lawyer        11.111111            2.752294                5.263158
##   Manager        8.888889            6.422018                7.368421
##   Mechanic       5.555556            5.504587               11.578947
##   Media_Manager  4.444444            6.422018               11.578947
##   Musician       5.555556            4.587156                9.473684
##   Scientist      4.444444           12.844037                8.421053
##   Teacher        7.777778            5.504587                6.315789
##   Writer         8.888889            5.504587                3.157895
##
##                 Home Equity Loan Mortgage Loan   No Loan Not Specified
```

```
##   Accountant            7.207207      4.347826  6.783898      8.602151
##   Architect             8.108108      3.260870  6.844238      6.451613
##   Developer             7.207207      7.608696  6.611499     10.752688
##   Doctor                7.207207      9.782609  6.482200      8.602151
##   Engineer              9.009009      7.608696  6.887337      6.451613
##   Entrepreneur          4.504505      7.608696  6.697698      9.677419
##   Journalist            7.207207      3.260870  6.620119      5.376344
##   Lawyer                8.108108      5.434783  7.206275      5.376344
##   Manager               5.405405      9.782609  6.335661      8.602151
##   Mechanic              5.405405      4.347826  6.706318      5.376344
##   Media_Manager         7.207207      6.521739  6.775278      7.526882
##   Musician              8.108108      6.521739  6.249461      5.376344
##   Scientist             3.603604      8.695652  6.706318      4.301075
##   Teacher               3.603604      4.347826  6.783898      4.301075
##   Writer                8.108108     10.869565  6.309801      3.225806
##
##                  Payday Loan Personal Loan Student Loan
##   Accountant       4.504505      3.333333     6.481481
##   Architect        6.306306      7.777778     3.703704
##   Developer        7.207207     10.000000    12.037037
##   Doctor          13.513514      8.888889     5.555556
##   Engineer         2.702703      7.777778     7.407407
##   Entrepreneur     7.207207      5.555556     5.555556
##   Journalist       3.603604      6.666667     2.777778
##   Lawyer           1.801802      4.444444     7.407407
##   Manager          8.108108      7.777778     7.407407
##   Mechanic        13.513514     10.000000     7.407407
##   Media_Manager    2.702703      5.555556     2.777778
##   Musician         8.108108      7.777778    12.962963
##   Scientist        7.207207      6.666667     8.333333
##   Teacher          7.207207      3.333333     4.629630
##   Writer           6.306306      4.444444     5.555556
##
##
## $Occupation_Last_Loan_9
## $Occupation_Last_Loan_9$contingency_table
##
##                  Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant            32                  32                      16
##   Architect             32                  24                       0
##   Developer             32                  48                      40
##   Doctor                32                  16                      16
##   Engineer              16                  48                      40
##   Entrepreneur          32                  24                       0
##   Journalist            24                  16                      16
##   Lawyer                24                  24                      24
##   Manager               16                  32                      24
##   Mechanic              64                  40                      32
##   Media_Manager         32                  16                      48
##   Musician              24                  32                      64
##   Scientist             32                  24                      32
```

```
##   Teacher              40                  16                       8
##   Writer               16                  40                      32
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant                  40            16    6512            40
##   Architect                   40            32    6568            24
##   Developer                   48            32    6400            48
##   Doctor                      16            56    6320            32
##   Engineer                    16            40    6600            32
##   Entrepreneur                56            64    6408             8
##   Journalist                  32            64    6288            16
##   Lawyer                      16             8    6880            32
##   Manager                     32            56    6104            32
##   Mechanic                    32            16    6456            40
##   Media_Manager               32            32    6488             8
##   Musician                     8            32    6096            32
##   Scientist                    0            24    6488            48
##   Teacher                      8            24    6472            24
##   Writer                      32            24    6064             8
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant             24            24            8
##   Architect              24            64           16
##   Developer              24            32           16
##   Doctor                 32            16           32
##   Engineer               48            24            0
##   Entrepreneur            0            32           24
##   Journalist             32             8           40
##   Lawyer                 24            32           32
##   Manager                64            32           40
##   Mechanic               32            16           48
##   Media_Manager          16            16           32
##   Musician               24             8           32
##   Scientist              48            32           16
##   Teacher                32            24           24
##   Writer                 40            24           24
##
## $Occupation_Last_Loan_9$row_totals
##
##    Accountant     Architect     Developer        Doctor      Engineer
##          6744          6824          6720          6568          6864
##  Entrepreneur    Journalist        Lawyer       Manager      Mechanic
##          6648          6536          7096          6432          6776
## Media_Manager      Musician     Scientist       Teacher        Writer
##          6720          6352          6744          6672          6304
##
## $Occupation_Last_Loan_9$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     448                     432                     392
##        Home Equity Loan           Mortgage Loan                 No Loan
```

```
##                     408                     520                   96144
##           Not Specified             Payday Loan           Personal Loan
##                     424                     464                     384
##            Student Loan
##                     384
##
## $Occupation_Last_Loan_9$grand_total
## [1] 100000
##
## $Occupation_Last_Loan_9$cell_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant        0.032               0.032                   0.016
##   Architect         0.032               0.024                   0.000
##   Developer         0.032               0.048                   0.040
##   Doctor            0.032               0.016                   0.016
##   Engineer          0.016               0.048                   0.040
##   Entrepreneur      0.032               0.024                   0.000
##   Journalist        0.024               0.016                   0.016
##   Lawyer            0.024               0.024                   0.024
##   Manager           0.016               0.032                   0.024
##   Mechanic          0.064               0.040                   0.032
##   Media_Manager     0.032               0.016                   0.048
##   Musician          0.024               0.032                   0.064
##   Scientist         0.032               0.024                   0.032
##   Teacher           0.040               0.016                   0.008
##   Writer            0.016               0.040                   0.032
##
##                 Home Equity Loan Mortgage Loan No Loan Not Specified
##   Accountant               0.040         0.016   6.512         0.040
##   Architect                0.040         0.032   6.568         0.024
##   Developer                0.048         0.032   6.400         0.048
##   Doctor                   0.016         0.056   6.320         0.032
##   Engineer                 0.016         0.040   6.600         0.032
##   Entrepreneur             0.056         0.064   6.408         0.008
##   Journalist               0.032         0.064   6.288         0.016
##   Lawyer                   0.016         0.008   6.880         0.032
##   Manager                  0.032         0.056   6.104         0.032
##   Mechanic                 0.032         0.016   6.456         0.040
##   Media_Manager            0.032         0.032   6.488         0.008
##   Musician                 0.008         0.032   6.096         0.032
##   Scientist                0.000         0.024   6.488         0.048
##   Teacher                  0.008         0.024   6.472         0.024
##   Writer                   0.032         0.024   6.064         0.008
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant          0.024         0.024        0.008
##   Architect           0.024         0.064        0.016
##   Developer           0.024         0.032        0.016
##   Doctor              0.032         0.016        0.032
##   Engineer            0.048         0.024        0.000
```

```
##   Entrepreneur        0.000         0.032        0.024
##   Journalist          0.032         0.008        0.040
##   Lawyer              0.024         0.032        0.032
##   Manager             0.064         0.032        0.040
##   Mechanic            0.032         0.016        0.048
##   Media_Manager       0.016         0.016        0.032
##   Musician            0.024         0.008        0.032
##   Scientist           0.048         0.032        0.016
##   Teacher             0.032         0.024        0.024
##   Writer              0.040         0.024        0.024
##
## $Occupation_Last_Loan_9$row_percentage
##
##                  Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     0.4744958           0.4744958               0.2372479
##   Architect      0.4689332           0.3516999               0.0000000
##   Developer      0.4761905           0.7142857               0.5952381
##   Doctor         0.4872107           0.2436054               0.2436054
##   Engineer       0.2331002           0.6993007               0.5827506
##   Entrepreneur   0.4813478           0.3610108               0.0000000
##   Journalist     0.3671971           0.2447980               0.2447980
##   Lawyer         0.3382187           0.3382187               0.3382187
##   Manager        0.2487562           0.4975124               0.3731343
##   Mechanic       0.9445100           0.5903188               0.4722550
##   Media_Manager  0.4761905           0.2380952               0.7142857
##   Musician       0.3778338           0.5037783               1.0075567
##   Scientist      0.4744958           0.3558719               0.4744958
##   Teacher        0.5995204           0.2398082               0.1199041
##   Writer         0.2538071           0.6345178               0.5076142
##
##                 Home Equity Loan Mortgage Loan    No Loan Not Specified
##   Accountant           0.5931198     0.2372479 96.5599051     0.5931198
##   Architect            0.5861665     0.4689332 96.2485346     0.3516999
##   Developer            0.7142857     0.4761905 95.2380952     0.7142857
##   Doctor               0.2436054     0.8526188 96.2241169     0.4872107
##   Engineer             0.2331002     0.5827506 96.1538462     0.4662005
##   Entrepreneur         0.8423586     0.9626955 96.3898917     0.1203369
##   Journalist           0.4895961     0.9791922 96.2056304     0.2447980
##   Lawyer               0.2254791     0.1127396 96.9560316     0.4509583
##   Manager              0.4975124     0.8706468 94.9004975     0.4975124
##   Mechanic             0.4722550     0.2361275 95.2774498     0.5903188
##   Media_Manager        0.4761905     0.4761905 96.5476190     0.1190476
##   Musician             0.1259446     0.5037783 95.9697733     0.5037783
##   Scientist            0.0000000     0.3558719 96.2040332     0.7117438
##   Teacher              0.1199041     0.3597122 97.0023981     0.3597122
##   Writer               0.5076142     0.3807107 96.1928934     0.1269036
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant      0.3558719     0.3558719    0.1186240
##   Architect       0.3516999     0.9378664    0.2344666
##   Developer       0.3571429     0.4761905    0.2380952
```

```
##   Doctor          0.4872107     0.2436054    0.4872107
##   Engineer        0.6993007     0.3496503    0.0000000
##   Entrepreneur    0.0000000     0.4813478    0.3610108
##   Journalist      0.4895961     0.1223990    0.6119951
##   Lawyer          0.3382187     0.4509583    0.4509583
##   Manager         0.9950249     0.4975124    0.6218905
##   Mechanic        0.4722550     0.2361275    0.7083825
##   Media_Manager   0.2380952     0.2380952    0.4761905
##   Musician        0.3778338     0.1259446    0.5037783
##   Scientist       0.7117438     0.4744958    0.2372479
##   Teacher         0.4796163     0.3597122    0.3597122
##   Writer          0.6345178     0.3807107    0.3807107
##
## $Occupation_Last_Loan_9$col_percentage
##
##                 Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Accountant     7.142857            7.407407                4.081633
##   Architect      7.142857            5.555556                0.000000
##   Developer      7.142857           11.111111               10.204082
##   Doctor         7.142857            3.703704                4.081633
##   Engineer       3.571429           11.111111               10.204082
##   Entrepreneur   7.142857            5.555556                0.000000
##   Journalist     5.357143            3.703704                4.081633
##   Lawyer         5.357143            5.555556                6.122449
##   Manager        3.571429            7.407407                6.122449
##   Mechanic      14.285714            9.259259                8.163265
##   Media_Manager  7.142857            3.703704               12.244898
##   Musician       5.357143            7.407407               16.326531
##   Scientist      7.142857            5.555556                8.163265
##   Teacher        8.928571            3.703704                2.040816
##   Writer         3.571429            9.259259                8.163265
##
##                 Home Equity Loan Mortgage Loan   No Loan Not Specified
##   Accountant            9.803922      3.076923  6.773174      9.433962
##   Architect             9.803922      6.153846  6.831420      5.660377
##   Developer            11.764706      6.153846  6.656682     11.320755
##   Doctor                3.921569     10.769231  6.573473      7.547170
##   Engineer              3.921569      7.692308  6.864703      7.547170
##   Entrepreneur         13.725490     12.307692  6.665002      1.886792
##   Journalist            7.843137     12.307692  6.540190      3.773585
##   Lawyer                3.921569      1.538462  7.155933      7.547170
##   Manager               7.843137     10.769231  6.348810      7.547170
##   Mechanic              7.843137      3.076923  6.714928      9.433962
##   Media_Manager         7.843137      6.153846  6.748211      1.886792
##   Musician              1.960784      6.153846  6.340489      7.547170
##   Scientist             0.000000      4.615385  6.748211     11.320755
##   Teacher               1.960784      4.615385  6.731569      5.660377
##   Writer                7.843137      4.615385  6.307206      1.886792
##
##                 Payday Loan Personal Loan Student Loan
##   Accountant       5.172414      6.250000     2.083333
```

```
##   Architect        5.172414     16.666667     4.166667
##   Developer        5.172414      8.333333     4.166667
##   Doctor           6.896552      4.166667     8.333333
##   Engineer        10.344828      6.250000     0.000000
##   Entrepreneur     0.000000      8.333333     6.250000
##   Journalist       6.896552      2.083333    10.416667
##   Lawyer           5.172414      8.333333     8.333333
##   Manager         13.793103      8.333333    10.416667
##   Mechanic         6.896552      4.166667    12.500000
##   Media_Manager    3.448276      4.166667     8.333333
##   Musician         5.172414      2.083333     8.333333
##   Scientist       10.344828      8.333333     4.166667
##   Teacher          6.896552      6.250000     6.250000
##   Writer           8.620690      6.250000     6.250000
##
##
## $Credit_Score_Last_Loan_1
## $Credit_Score_Last_Loan_1$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good          1442                1646                    1578
##   Poor          2883                3089                    3105
##   Standard      5003                5065                    5285
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                 1537          1632    3444          1780        1643
##   Poor                 2957          3022    1336          2982        3142
##   Standard             5306          4954    6628          5254        5319
##
##            Personal Loan Student Loan
##   Good              1472         1654
##   Poor              3237         3245
##   Standard          5227         5133
##
## $Credit_Score_Last_Loan_1$row_totals
##
##     Good     Poor Standard
##    17828    28998    53174
##
## $Credit_Score_Last_Loan_1$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    9328                    9800                    9968
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    9800                    9608                   11408
##           Not Specified             Payday Loan           Personal Loan
##                   10016                   10104                    9936
##            Student Loan
##                   10032
##
## $Credit_Score_Last_Loan_1$grand_total
```

```
## [1] 100000
##
## $Credit_Score_Last_Loan_1$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good         1.442               1.646                   1.578
##   Poor         2.883               3.089                   3.105
##   Standard     5.003               5.065                   5.285
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                1.537         1.632   3.444         1.780       1.643
##   Poor                2.957         3.022   1.336         2.982       3.142
##   Standard            5.306         4.954   6.628         5.254       5.319
##
##            Personal Loan Student Loan
##   Good             1.472        1.654
##   Poor             3.237        3.245
##   Standard         5.227        5.133
##
## $Credit_Score_Last_Loan_1$row_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      8.088400            9.232668                8.851245
##   Poor      9.942065           10.652459               10.707635
##   Standard  9.408734            9.525332                9.939068
##
##            Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Good             8.621270      9.154140 19.317927      9.984294    9.215840
##   Poor            10.197255     10.421408  4.607214     10.283468   10.835230
##   Standard         9.978561      9.316583 12.464738      9.880769   10.003009
##
##            Personal Loan Student Loan
##   Good          8.256675     9.277541
##   Poor         11.162839    11.190427
##   Standard      9.829992     9.653214
##
## $Credit_Score_Last_Loan_1$col_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      15.45883            16.79592                15.83066
##   Poor      30.90695            31.52041                31.14968
##   Standard  53.63422            51.68367                53.01966
##
##            Home Equity Loan Mortgage Loan  No Loan Not Specified Payday Loan
##   Good             15.68367      16.98585 30.18934      17.77157    16.26089
##   Poor             30.17347      31.45296 11.71108      29.77236    31.09660
##   Standard         54.14286      51.56120 58.09958      52.45607    52.64252
##
##            Personal Loan Student Loan
##   Good          14.81481     16.48724
##   Poor          32.57850     32.34649
```

```
##   Standard      52.60668     51.16627
##
##
## $Credit_Score_Last_Loan_2
## $Credit_Score_Last_Loan_2$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good          1189                1170                    1079
##   Poor          3032                2881                    2834
##   Standard      4251                4477                    4359
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                 1407          1225    6614          1371        1226
##   Poor                 2905          2836    2569          2882        3263
##   Standard             4464          4347   13353          4475        4615
##
##            Personal Loan Student Loan
##   Good              1235         1312
##   Poor              2891         2905
##   Standard          4514         4319
##
## $Credit_Score_Last_Loan_2$row_totals
##
##     Good     Poor Standard
##    17828    28998    53174
##
## $Credit_Score_Last_Loan_2$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    8472                    8528                    8272
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    8776                    8408                   22536
##           Not Specified             Payday Loan           Personal Loan
##                    8728                    9104                    8640
##            Student Loan
##                    8536
##
## $Credit_Score_Last_Loan_2$grand_total
## [1] 100000
##
## $Credit_Score_Last_Loan_2$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good         1.189               1.170                   1.079
##   Poor         3.032               2.881                   2.834
##   Standard     4.251               4.477                   4.359
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                1.407         1.225   6.614         1.371       1.226
##   Poor                2.905         2.836   2.569         2.882       3.263
##   Standard            4.464         4.347  13.353         4.475       4.615
```

```
##
##            Personal Loan Student Loan
##   Good             1.235        1.312
##   Poor             2.891        2.905
##   Standard         4.514        4.319
##
## $Credit_Score_Last_Loan_2$row_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      6.669284            6.562710                6.052277
##   Poor     10.455894            9.935168                9.773088
##   Standard  7.994509            8.419528                8.197615
##
##            Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Good             7.892080      6.871214 37.098945      7.690150    6.876823
##   Poor            10.017932      9.779985  8.859232      9.938616   11.252500
##   Standard         8.395080      8.175048 25.111897      8.415767    8.679054
##
##            Personal Loan Student Loan
##   Good          6.927305     7.359210
##   Poor          9.969653    10.017932
##   Standard      8.489111     8.122391
##
## $Credit_Score_Last_Loan_2$col_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      14.03447            13.71951                13.04400
##   Poor      35.78848            33.78283                34.26015
##   Standard  50.17705            52.49765                52.69584
##
##            Home Equity Loan Mortgage Loan  No Loan Not Specified Payday Loan
##   Good             16.03236      14.56946 29.34860      15.70807    13.46661
##   Poor             33.10164      33.72978 11.39954      33.02016    35.84139
##   Standard         50.86600      51.70076 59.25186      51.27177    50.69200
##
##            Personal Loan Student Loan
##   Good          14.29398     15.37020
##   Poor          33.46065     34.03233
##   Standard      52.24537     50.59747
##
##
## $Credit_Score_Last_Loan_3
## $Credit_Score_Last_Loan_3$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good           871                 950                     807
##   Poor          2354                2639                    2732
##   Standard      3583                3731                    3389
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                  828           814   10055           859         869
```

```
##   Poor                 2371          2507    6326          2690        2577
##   Standard             3441          3551   21867          3531        3322
##
##            Personal Loan Student Loan
##   Good               844          931
##   Poor              2331         2471
##   Standard          3233         3526
##
## $Credit_Score_Last_Loan_3$row_totals
##
##     Good     Poor Standard
##    17828    28998    53174
##
## $Credit_Score_Last_Loan_3$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    6808                    7320                    6928
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    6640                    6872                   38248
##           Not Specified             Payday Loan           Personal Loan
##                    7080                    6768                    6408
##            Student Loan
##                    6928
##
## $Credit_Score_Last_Loan_3$grand_total
## [1] 100000
##
## $Credit_Score_Last_Loan_3$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good         0.871               0.950                   0.807
##   Poor         2.354               2.639                   2.732
##   Standard     3.583               3.731                   3.389
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                0.828         0.814  10.055         0.859       0.869
##   Poor                2.371         2.507   6.326         2.690       2.577
##   Standard            3.441         3.551  21.867         3.531       3.322
##
##            Personal Loan Student Loan
##   Good             0.844        0.931
##   Poor             2.331        2.471
##   Standard         3.233        3.526
##
## $Credit_Score_Last_Loan_3$row_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      4.885573            5.328696                4.526587
##   Poor      8.117801            9.100628                9.421339
##   Standard  6.738256            7.016587                6.373416
##
```

```
##            Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Good             4.644380      4.565851 56.400045      4.818263    4.874355
##   Poor             8.176426      8.645424 21.815298      9.276502    8.886820
##   Standard         6.471208      6.678076 41.123481      6.640463    6.247414
##
##            Personal Loan Student Loan
##   Good          4.734126     5.222123
##   Poor          8.038485     8.521277
##   Standard      6.080039     6.631060
##
## $Credit_Score_Last_Loan_3$col_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      12.79377            12.97814                11.64838
##   Poor      34.57697            36.05191                39.43418
##   Standard  52.62926            50.96995                48.91744
##
##            Home Equity Loan Mortgage Loan  No Loan Not Specified Payday Loan
##   Good             12.46988      11.84517 26.28896      12.13277    12.83983
##   Poor             35.70783      36.48137 16.53943      37.99435    38.07624
##   Standard         51.82229      51.67346 57.17162      49.87288    49.08392
##
##            Personal Loan Student Loan
##   Good          13.17104     13.43822
##   Poor          36.37640     35.66686
##   Standard      50.45256     50.89492
##
##
## $Credit_Score_Last_Loan_4
## $Credit_Score_Last_Loan_4$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good           393                 565                     496
##   Poor          2057                2250                    2026
##   Standard      2270                2849                    2510
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                  382           519   13657           412         508
##   Poor                 2264          2139    9763          2266        2124
##   Standard             2554          2510   30580          2442        2656
##
##            Personal Loan Student Loan
##   Good               442          454
##   Poor              1944         2165
##   Standard          2470         2333
##
## $Credit_Score_Last_Loan_4$row_totals
##
##     Good     Poor Standard
##    17828    28998    53174
##
```

```
## $Credit_Score_Last_Loan_4$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    4720                    5664                    5032
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    5200                    5168                   54000
##           Not Specified             Payday Loan           Personal Loan
##                    5120                    5288                    4856
##            Student Loan
##                    4952
##
## $Credit_Score_Last_Loan_4$grand_total
## [1] 100000
##
## $Credit_Score_Last_Loan_4$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good         0.393               0.565                   0.496
##   Poor         2.057               2.250                   2.026
##   Standard     2.270               2.849                   2.510
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                0.382         0.519  13.657         0.412       0.508
##   Poor                2.264         2.139   9.763         2.266       2.124
##   Standard            2.554         2.510  30.580         2.442       2.656
##
##            Personal Loan Student Loan
##   Good             0.442        0.454
##   Poor             1.944        2.165
##   Standard         2.470        2.333
##
## $Credit_Score_Last_Loan_4$row_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      2.204398            3.169172                2.782140
##   Poor      7.093593            7.759156                6.986689
##   Standard  4.269004            5.357882                4.720352
##
##            Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Good             2.142697      2.911151 76.604218      2.310972    2.849450
##   Poor             7.807435      7.376371 33.667839      7.814332    7.324643
##   Standard         4.803099      4.720352 57.509309      4.592470    4.994922
##
##            Personal Loan Student Loan
##   Good          2.479246     2.546556
##   Poor          6.703911     7.466032
##   Standard      4.645127     4.387483
##
## $Credit_Score_Last_Loan_4$col_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
```

```
##   Good      8.326271            9.975282                9.856916
##   Poor     43.580508           39.724576               40.262321
##   Standard 48.093220           50.300141               49.880763
##
##            Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Good             7.346154     10.042570 25.290741      8.046875    9.606657
##   Poor            43.538462     41.389319 18.079630     44.257812   40.166415
##   Standard        49.115385     48.568111 56.629630     47.695312   50.226929
##
##            Personal Loan Student Loan
##   Good          9.102142     9.168013
##   Poor         40.032949    43.719709
##   Standard     50.864909    47.112278
##
##
## $Credit_Score_Last_Loan_5
## $Credit_Score_Last_Loan_5$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good            74                  90                      85
##   Poor          1908                1850                    1796
##   Standard      1442                1556                    1647
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                  71            89   17130            91          76
##   Poor                1822          1822   13047          1708        1755
##   Standard            1507          1529   39279          1473        1513
##
##            Personal Loan Student Loan
##   Good               69           53
##   Poor             1645         1645
##   Standard         1806         1422
##
## $Credit_Score_Last_Loan_5$row_totals
##
##     Good     Poor Standard
##    17828    28998    53174
##
## $Credit_Score_Last_Loan_5$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    3424                    3496                    3528
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    3400                    3440                   69456
##           Not Specified             Payday Loan           Personal Loan
##                    3272                    3344                    3520
##            Student Loan
##                    3120
##
## $Credit_Score_Last_Loan_5$grand_total
## [1] 100000
```

```
##
## $Credit_Score_Last_Loan_5$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good         0.074               0.090                   0.085
##   Poor         1.908               1.850                   1.796
##   Standard     1.442               1.556                   1.647
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                0.071         0.089  17.130         0.091       0.076
##   Poor                1.822         1.822  13.047         1.708       1.755
##   Standard            1.507         1.529  39.279         1.473       1.513
##
##            Personal Loan Student Loan
##   Good             0.069        0.053
##   Poor             1.645        1.645
##   Standard         1.806        1.422
##
## $Credit_Score_Last_Loan_5$row_percentage
##
##             Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      0.4150774           0.5048239               0.4767781
##   Poor      6.5797641           6.3797503               6.1935306
##   Standard  2.7118517           2.9262421               3.0973784
##
##            Home Equity Loan Mortgage Loan    No Loan Not Specified Payday Loan
##   Good            0.3982499     0.4992147 96.0848104     0.5104330   0.4262957
##   Poor            6.2831919     6.2831919 44.9927581     5.8900614   6.0521415
##   Standard        2.8340918     2.8754655 73.8688081     2.7701508   2.8453756
##
##            Personal Loan Student Loan
##   Good         0.3870316    0.2972852
##   Poor         5.6728050    5.6728050
##   Standard     3.3963967    2.6742393
##
## $Credit_Score_Last_Loan_5$col_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      2.161215            2.574371                2.409297
##   Poor     55.724299           52.917620               50.907029
##   Standard 42.114486           44.508009               46.683673
##
##            Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Good             2.088235      2.587209 24.663096      2.781174    2.272727
##   Poor            53.588235     52.965116 18.784554     52.200489   52.482057
##   Standard        44.323529     44.447674 56.552350     45.018337   45.245215
##
##            Personal Loan Student Loan
##   Good          1.960227     1.698718
##   Poor         46.732955    52.724359
##   Standard     51.306818    45.576923
```

```
##
##
## $Credit_Score_Last_Loan_6
## $Credit_Score_Last_Loan_6$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good            58                  51                      48
##   Poor          1281                1331                    1326
##   Standard      1133                1226                    1090
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                   67            28   17324            58          68
##   Poor                 1339          1302   16938          1310        1353
##   Standard             1058          1062   42722          1256        1267
##
##            Personal Loan Student Loan
##   Good                56           70
##   Poor              1410         1408
##   Standard          1270         1090
##
## $Credit_Score_Last_Loan_6$row_totals
##
##     Good     Poor Standard
##    17828    28998    53174
##
## $Credit_Score_Last_Loan_6$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    2472                    2608                    2464
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    2464                    2392                   76984
##           Not Specified             Payday Loan           Personal Loan
##                    2624                    2688                    2736
##            Student Loan
##                    2568
##
## $Credit_Score_Last_Loan_6$grand_total
## [1] 100000
##
## $Credit_Score_Last_Loan_6$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good         0.058               0.051                   0.048
##   Poor         1.281               1.331                   1.326
##   Standard     1.133               1.226                   1.090
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                0.067         0.028  17.324         0.058       0.068
##   Poor                1.339         1.302  16.938         1.310       1.353
##   Standard            1.058         1.062  42.722         1.256       1.267
##
```

```
##            Personal Loan Student Loan
##   Good             0.056        0.070
##   Poor             1.410        1.408
##   Standard         1.270        1.090
##
## $Credit_Score_Last_Loan_6$row_percentage
##
##             Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      0.3253309           0.2860669               0.2692394
##   Poor      4.4175460           4.5899717               4.5727292
##   Standard  2.1307406           2.3056381               2.0498740
##
##            Home Equity Loan Mortgage Loan    No Loan Not Specified Payday Loan
##   Good            0.3758133     0.1570563 97.1729863     0.3253309   0.3814225
##   Poor            4.6175598     4.4899648 58.4109249     4.5175529   4.6658390
##   Standard        1.9896942     1.9972167 80.3437770     2.3620566   2.3827434
##
##            Personal Loan Student Loan
##   Good         0.3141126    0.3926408
##   Poor         4.8624043    4.8555073
##   Standard     2.3883853    2.0498740
##
## $Credit_Score_Last_Loan_6$col_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      2.346278            1.955521                1.948052
##   Poor     51.820388           51.035276               53.814935
##   Standard 45.833333           47.009202               44.237013
##
##            Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Good             2.719156      1.170569 22.503377      2.210366    2.529762
##   Poor            54.342532     54.431438 22.001974     49.923780   50.334821
##   Standard        42.938312     44.397993 55.494648     47.865854   47.135417
##
##            Personal Loan Student Loan
##   Good          2.046784     2.725857
##   Poor         51.535088    54.828660
##   Standard     46.418129    42.445483
##
##
## $Credit_Score_Last_Loan_7
## $Credit_Score_Last_Loan_7$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good            44                  44                      22
##   Poor           858                 927                     806
##   Standard       698                 749                     604
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                   34            28   17513            26          66
##   Poor                  815           965   21070           811         970
```

```
##   Standard              679           799   46545           771         884
##
##            Personal Loan Student Loan
##   Good                25           26
##   Poor               898          878
##   Standard           765          680
##
## $Credit_Score_Last_Loan_7$row_totals
##
##     Good     Poor Standard
##    17828    28998    53174
##
## $Credit_Score_Last_Loan_7$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    1600                    1720                    1432
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    1528                    1792                   85128
##           Not Specified             Payday Loan           Personal Loan
##                    1608                    1920                    1688
##            Student Loan
##                    1584
##
## $Credit_Score_Last_Loan_7$grand_total
## [1] 100000
##
## $Credit_Score_Last_Loan_7$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good         0.044               0.044                   0.022
##   Poor         0.858               0.927                   0.806
##   Standard     0.698               0.749                   0.604
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                0.034         0.028  17.513         0.026       0.066
##   Poor                0.815         0.965  21.070         0.811       0.970
##   Standard            0.679         0.799  46.545         0.771       0.884
##
##            Personal Loan Student Loan
##   Good             0.025        0.026
##   Poor             0.898        0.878
##   Standard         0.765        0.680
##
## $Credit_Score_Last_Loan_7$row_percentage
##
##             Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      0.2468028           0.2468028               0.1234014
##   Poor      2.9588247           3.1967722               2.7795020
##   Standard  1.3126716           1.4085831               1.1358935
##
##            Home Equity Loan Mortgage Loan    No Loan Not Specified Payday Loan
```

```
##   Good            0.1907112     0.1570563 98.2331164     0.1458380   0.3702042
##   Poor            2.8105387     3.3278157 72.6601835     2.7967446   3.3450583
##   Standard        1.2769399     1.5026141 87.5333810     1.4499567   1.6624666
##
##            Personal Loan Student Loan
##   Good         0.1402289    0.1458380
##   Poor         3.0967653    3.0277950
##   Standard     1.4386730    1.2788205
##
## $Credit_Score_Last_Loan_7$col_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      2.750000            2.558140                1.536313
##   Poor     53.625000           53.895349               56.284916
##   Standard 43.625000           43.546512               42.178771
##
##            Home Equity Loan Mortgage Loan   No Loan Not Specified Payday Loan
##   Good             2.225131      1.562500 20.572550      1.616915    3.437500
##   Poor            53.337696     53.850446 24.750963     50.435323   50.520833
##   Standard        44.437173     44.587054 54.676487     47.947761   46.041667
##
##            Personal Loan Student Loan
##   Good          1.481043     1.641414
##   Poor         53.199052    55.429293
##   Standard     45.319905    42.929293
##
##
## $Credit_Score_Last_Loan_8
## $Credit_Score_Last_Loan_8$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good             5                  24                      14
##   Poor           379                 465                     395
##   Standard       336                 383                     351
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                   18            19   17696            17          15
##   Poor                  474           371   25095           435         454
##   Standard              396           346   50017           292         419
##
##            Personal Loan Student Loan
##   Good                 6           14
##   Poor               448          482
##   Standard           266          368
##
## $Credit_Score_Last_Loan_8$row_totals
##
##     Good     Poor Standard
##    17828    28998    53174
##
## $Credit_Score_Last_Loan_8$col_totals
```

```
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     720                     872                     760
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     888                     736                   92808
##           Not Specified             Payday Loan           Personal Loan
##                     744                     888                     720
##            Student Loan
##                     864
##
## $Credit_Score_Last_Loan_8$grand_total
## [1] 100000
##
## $Credit_Score_Last_Loan_8$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good         0.005               0.024                   0.014
##   Poor         0.379               0.465                   0.395
##   Standard     0.336               0.383                   0.351
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                0.018         0.019  17.696         0.017       0.015
##   Poor                0.474         0.371  25.095         0.435       0.454
##   Standard            0.396         0.346  50.017         0.292       0.419
##
##            Personal Loan Student Loan
##   Good             0.006        0.014
##   Poor             0.448        0.482
##   Standard         0.266        0.368
##
## $Credit_Score_Last_Loan_8$row_percentage
##
##              Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      0.02804577          0.13461970              0.07852816
##   Poor      1.30698669          1.60355887              1.36216291
##   Standard  0.63188776          0.72027683              0.66009704
##
##            Home Equity Loan Mortgage Loan     No Loan Not Specified Payday Loan
##   Good           0.10096477    0.10657393 99.25959165    0.09535562  0.08413731
##   Poor           1.63459549    1.27939858 86.54045107    1.50010346  1.56562522
##   Standard       0.74472487    0.65069395 94.06288788    0.54914056  0.78797909
##
##            Personal Loan Student Loan
##   Good        0.03365492   0.07852816
##   Poor        1.54493413   1.66218360
##   Standard    0.50024448   0.69206755
##
## $Credit_Score_Last_Loan_8$col_percentage
##
##             Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      0.6944444           2.7522936               1.8421053
```

```
##   Poor     52.6388889          53.3256881              51.9736842
##   Standard 46.6666667          43.9220183              46.1842105
##
##            Home Equity Loan Mortgage Loan    No Loan Not Specified Payday Loan
##   Good            2.0270270     2.5815217 19.0673218     2.2849462   1.6891892
##   Poor           53.3783784    50.4076087 27.0396949    58.4677419  51.1261261
##   Standard       44.5945946    47.0108696 53.8929834    39.2473118  47.1846847
##
##            Personal Loan Student Loan
##   Good         0.8333333    1.6203704
##   Poor        62.2222222   55.7870370
##   Standard    36.9444444   42.5925926
##
##
## $Credit_Score_Last_Loan_9
## $Credit_Score_Last_Loan_9$contingency_table
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good            13                   3                       6
##   Poor           218                 233                     224
##   Standard       217                 196                     162
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                    4            11   17759             0           7
##   Poor                  207           249   26931           262         282
##   Standard              197           260   51454           162         175
##
##            Personal Loan Student Loan
##   Good                16            9
##   Poor               199          193
##   Standard           169          182
##
## $Credit_Score_Last_Loan_9$row_totals
##
##     Good     Poor Standard
##    17828    28998    53174
##
## $Credit_Score_Last_Loan_9$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     448                     432                     392
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     408                     520                   96144
##           Not Specified             Payday Loan           Personal Loan
##                     424                     464                     384
##            Student Loan
##                     384
##
## $Credit_Score_Last_Loan_9$grand_total
## [1] 100000
##
```

```
## $Credit_Score_Last_Loan_9$cell_percentage
##
##            Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good         0.013               0.003                   0.006
##   Poor         0.218               0.233                   0.224
##   Standard     0.217               0.196                   0.162
##
##            Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
##   Good                0.004         0.011  17.759         0.000       0.007
##   Poor                0.207         0.249  26.931         0.262       0.282
##   Standard            0.197         0.260  51.454         0.162       0.175
##
##            Personal Loan Student Loan
##   Good             0.016        0.009
##   Poor             0.199        0.193
##   Standard         0.169        0.182
##
## $Credit_Score_Last_Loan_9$row_percentage
##
##              Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      0.07291900          0.01682746              0.03365492
##   Poor      0.75177598          0.80350369              0.77246707
##   Standard  0.40809418          0.36860120              0.30466017
##
##            Home Equity Loan Mortgage Loan     No Loan Not Specified Payday Loan
##   Good           0.02243662    0.06170070 99.61296836    0.00000000  0.03926408
##   Poor           0.71384233    0.85867991 92.87192220    0.90351059  0.97248086
##   Standard       0.37048181    0.48896077 96.76533644    0.30466017  0.32910821
##
##            Personal Loan Student Loan
##   Good        0.08974647   0.05048239
##   Poor        0.68625422   0.66556314
##   Standard    0.31782450   0.34227254
##
## $Credit_Score_Last_Loan_9$col_percentage
##
##             Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Good      2.9017857           0.6944444               1.5306122
##   Poor     48.6607143          53.9351852              57.1428571
##   Standard 48.4375000          45.3703704              41.3265306
##
##            Home Equity Loan Mortgage Loan    No Loan Not Specified Payday Loan
##   Good            0.9803922     2.1153846 18.4712515     0.0000000   1.5086207
##   Poor           50.7352941    47.8846154 28.0111083    61.7924528  60.7758621
##   Standard       48.2843137    50.0000000 53.5176402    38.2075472  37.7155172
##
##            Personal Loan Student Loan
##   Good         4.1666667    2.3437500
##   Poor        51.8229167   50.2604167
##   Standard    44.0104167   47.3958333
##
```

```
##
## $Last_Loan_1_Last_Loan_2
## $Last_Loan_1_Last_Loan_2$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                    968                1032                     872
##   Credit-Builder Loan          888                1056                     744
##   Debt Consolidation Loan      848                 872                     904
##   Home Equity Loan             984                1080                     864
##   Mortgage Loan                832                 896                     936
##   No Loan                        0                   0                       0
##   Not Specified               1040                 896                     936
##   Payday Loan                 1064                 896                    1080
##   Personal Loan               1016                 832                    1008
##   Student Loan                 832                 968                     928
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            776           920    1152           888
##   Credit-Builder Loan                 1040           984    1280           864
##   Debt Consolidation Loan              992           976    1264          1080
##   Home Equity Loan                    1008           928    1136           920
##   Mortgage Loan                       1040           800    1176          1048
##   No Loan                                0             0   11408             0
##   Not Specified                        952           976    1408           920
##   Payday Loan                         1008           896    1200          1024
##   Personal Loan                        920           936    1272          1088
##   Student Loan                        1040           992    1240           896
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                      944           888          888
##   Credit-Builder Loan            984           960         1000
##   Debt Consolidation Loan       1136           936          960
##   Home Equity Loan              1080           968          832
##   Mortgage Loan                 1000           952          928
##   No Loan                          0             0            0
##   Not Specified                  960           976          952
##   Payday Loan                    976           976          984
##   Personal Loan                  992           984          888
##   Student Loan                  1032          1000         1104
##
## $Last_Loan_1_Last_Loan_2$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    9328                    9800                    9968
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    9800                    9608                   11408
##           Not Specified             Payday Loan           Personal Loan
##                   10016                   10104                    9936
##            Student Loan
##                   10032
##
```

```
## $Last_Loan_1_Last_Loan_2$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    8472                    8528                    8272
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    8776                    8408                   22536
##           Not Specified             Payday Loan           Personal Loan
##                    8728                    9104                    8640
##            Student Loan
##                    8536
##
## $Last_Loan_1_Last_Loan_2$grand_total
## [1] 100000
##
## $Last_Loan_1_Last_Loan_2$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.968               1.032                   0.872
##   Credit-Builder Loan        0.888               1.056                   0.744
##   Debt Consolidation Loan    0.848               0.872                   0.904
##   Home Equity Loan           0.984               1.080                   0.864
##   Mortgage Loan              0.832               0.896                   0.936
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              1.040               0.896                   0.936
##   Payday Loan                1.064               0.896                   1.080
##   Personal Loan              1.016               0.832                   1.008
##   Student Loan               0.832               0.968                   0.928
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                         0.776         0.920   1.152         0.888
##   Credit-Builder Loan               1.040         0.984   1.280         0.864
##   Debt Consolidation Loan           0.992         0.976   1.264         1.080
##   Home Equity Loan                  1.008         0.928   1.136         0.920
##   Mortgage Loan                     1.040         0.800   1.176         1.048
##   No Loan                           0.000         0.000  11.408         0.000
##   Not Specified                     0.952         0.976   1.408         0.920
##   Payday Loan                       1.008         0.896   1.200         1.024
##   Personal Loan                     0.920         0.936   1.272         1.088
##   Student Loan                      1.040         0.992   1.240         0.896
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.944         0.888        0.888
##   Credit-Builder Loan          0.984         0.960        1.000
##   Debt Consolidation Loan      1.136         0.936        0.960
##   Home Equity Loan             1.080         0.968        0.832
##   Mortgage Loan                1.000         0.952        0.928
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.960         0.976        0.952
##   Payday Loan                  0.976         0.976        0.984
##   Personal Loan                0.992         0.984        0.888
##   Student Loan                 1.032         1.000        1.104
```

```
##
## $Last_Loan_1_Last_Loan_2$row_percentage
##
##                           Auto Loan Credit-Builder Loan
##   Auto Loan               10.377358           11.063465
##   Credit-Builder Loan      9.061224           10.775510
##   Debt Consolidation Loan  8.507223            8.747994
##   Home Equity Loan        10.040816           11.020408
##   Mortgage Loan            8.659450            9.325562
##   No Loan                  0.000000            0.000000
##   Not Specified           10.383387            8.945687
##   Payday Loan             10.530483            8.867775
##   Personal Loan           10.225443            8.373591
##   Student Loan             8.293461            9.649123
##
##                          Debt Consolidation Loan Home Equity Loan
##   Auto Loan                               9.348199         8.319039
##   Credit-Builder Loan                     7.591837        10.612245
##   Debt Consolidation Loan                 9.069021         9.951846
##   Home Equity Loan                        8.816327        10.285714
##   Mortgage Loan                           9.741882        10.824313
##   No Loan                                 0.000000         0.000000
##   Not Specified                           9.345048         9.504792
##   Payday Loan                            10.688836         9.976247
##   Personal Loan                          10.144928         9.259259
##   Student Loan                            9.250399        10.366826
##
##                          Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                    9.862779  12.349914      9.519726   10.120069
##   Credit-Builder Loan         10.040816  13.061224      8.816327   10.040816
##   Debt Consolidation Loan      9.791332  12.680578     10.834671   11.396469
##   Home Equity Loan             9.469388  11.591837      9.387755   11.020408
##   Mortgage Loan                8.326395  12.239800     10.907577   10.407993
##   No Loan                      0.000000 100.000000      0.000000    0.000000
##   Not Specified                9.744409  14.057508      9.185304    9.584665
##   Payday Loan                  8.867775  11.876485     10.134600    9.659541
##   Personal Loan                9.420290  12.801932     10.950081    9.983897
##   Student Loan                 9.888357  12.360447      8.931419   10.287081
##
##                          Personal Loan Student Loan
##   Auto Loan                    9.519726     9.519726
##   Credit-Builder Loan          9.795918    10.204082
##   Debt Consolidation Loan      9.390048     9.630819
##   Home Equity Loan             9.877551     8.489796
##   Mortgage Loan                9.908410     9.658618
##   No Loan                      0.000000     0.000000
##   Not Specified                9.744409     9.504792
##   Payday Loan                  9.659541     9.738717
##   Personal Loan                9.903382     8.937198
##   Student Loan                 9.968102    11.004785
##
```

```
## $Last_Loan_1_Last_Loan_2$col_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan              11.425873           12.101313               10.541586
##   Credit-Builder Loan    10.481586           12.382739                8.994197
##   Debt Consolidation Loan 10.009443           10.225141               10.928433
##   Home Equity Loan       11.614731           12.664165               10.444874
##   Mortgage Loan           9.820585           10.506567               11.315280
##   No Loan                 0.000000            0.000000                0.000000
##   Not Specified          12.275732           10.506567               11.315280
##   Payday Loan            12.559018           10.506567               13.056093
##   Personal Loan          11.992446            9.756098               12.185687
##   Student Loan            9.820585           11.350844               11.218569
##
##                          Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                       8.842297     10.941960  5.111821
##   Credit-Builder Loan            11.850501     11.703140  5.679801
##   Debt Consolidation Loan        11.303555     11.607992  5.608804
##   Home Equity Loan               11.485871     11.037108  5.040824
##   Mortgage Loan                  11.850501      9.514748  5.218317
##   No Loan                         0.000000      0.000000 50.621228
##   Not Specified                  10.847767     11.607992  6.247781
##   Payday Loan                    11.485871     10.656518  5.324814
##   Personal Loan                  10.483136     11.132255  5.644302
##   Student Loan                   11.850501     11.798287  5.502307
##
##                          Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                   10.174152   10.369069     10.277778    10.402999
##   Credit-Builder Loan          9.899175   10.808436     11.111111    11.715089
##   Debt Consolidation Loan     12.373969   12.478032     10.833333    11.246485
##   Home Equity Loan            10.540788   11.862917     11.203704     9.746954
##   Mortgage Loan               12.007333   10.984183     11.018519    10.871603
##   No Loan                      0.000000    0.000000      0.000000     0.000000
##   Not Specified               10.540788   10.544815     11.296296    11.152765
##   Payday Loan                 11.732356   10.720562     11.296296    11.527648
##   Personal Loan               12.465628   10.896309     11.388889    10.402999
##   Student Loan                10.265811   11.335677     11.574074    12.933458
##
##
## $Last_Loan_1_Last_Loan_3
## $Last_Loan_1_Last_Loan_3$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                    672                 848                     744
##   Credit-Builder Loan          736                 832                     720
##   Debt Consolidation Loan      792                 640                     704
##   Home Equity Loan             728                 800                     912
##   Mortgage Loan                768                 880                     784
##   No Loan                        0                   0                       0
##   Not Specified                840                 856                     752
##   Payday Loan                  824                 872                     720
```

```
##   Personal Loan                 720                 800                     824
##   Student Loan                  728                 792                     768
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                           744           728    2776           808
##   Credit-Builder Loan                 848           736    3040           744
##   Debt Consolidation Loan             776           800    3088           800
##   Home Equity Loan                    600           712    2904           800
##   Mortgage Loan                       648           776    2944           712
##   No Loan                               0             0   11408             0
##   Not Specified                       688           792    3176           784
##   Payday Loan                         832           712    3024           728
##   Personal Loan                       752           824    2864           864
##   Student Loan                        752           792    3024           840
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                      752           624          632
##   Credit-Builder Loan            728           728          688
##   Debt Consolidation Loan        792           832          744
##   Home Equity Loan               768           720          856
##   Mortgage Loan                  616           744          736
##   No Loan                          0             0            0
##   Not Specified                  744           600          784
##   Payday Loan                    800           800          792
##   Personal Loan                  768           728          792
##   Student Loan                   800           632          904
##
## $Last_Loan_1_Last_Loan_3$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    9328                    9800                    9968
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    9800                    9608                   11408
##           Not Specified             Payday Loan           Personal Loan
##                   10016                   10104                    9936
##            Student Loan
##                   10032
##
## $Last_Loan_1_Last_Loan_3$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    6808                    7320                    6928
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    6640                    6872                   38248
##           Not Specified             Payday Loan           Personal Loan
##                    7080                    6768                    6408
##            Student Loan
##                    6928
##
## $Last_Loan_1_Last_Loan_3$grand_total
## [1] 100000
```

```
##
## $Last_Loan_1_Last_Loan_3$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.672               0.848                   0.744
##   Credit-Builder Loan        0.736               0.832                   0.720
##   Debt Consolidation Loan    0.792               0.640                   0.704
##   Home Equity Loan           0.728               0.800                   0.912
##   Mortgage Loan              0.768               0.880                   0.784
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.840               0.856                   0.752
##   Payday Loan                0.824               0.872                   0.720
##   Personal Loan              0.720               0.800                   0.824
##   Student Loan               0.728               0.792                   0.768
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.744         0.728   2.776         0.808
##   Credit-Builder Loan                0.848         0.736   3.040         0.744
##   Debt Consolidation Loan            0.776         0.800   3.088         0.800
##   Home Equity Loan                   0.600         0.712   2.904         0.800
##   Mortgage Loan                      0.648         0.776   2.944         0.712
##   No Loan                            0.000         0.000  11.408         0.000
##   Not Specified                      0.688         0.792   3.176         0.784
##   Payday Loan                        0.832         0.712   3.024         0.728
##   Personal Loan                      0.752         0.824   2.864         0.864
##   Student Loan                       0.752         0.792   3.024         0.840
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.752         0.624        0.632
##   Credit-Builder Loan          0.728         0.728        0.688
##   Debt Consolidation Loan      0.792         0.832        0.744
##   Home Equity Loan             0.768         0.720        0.856
##   Mortgage Loan                0.616         0.744        0.736
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.744         0.600        0.784
##   Payday Loan                  0.800         0.800        0.792
##   Personal Loan                0.768         0.728        0.792
##   Student Loan                 0.800         0.632        0.904
##
## $Last_Loan_1_Last_Loan_3$row_percentage
##
##                           Auto Loan Credit-Builder Loan
##   Auto Loan                7.204117            9.090909
##   Credit-Builder Loan      7.510204            8.489796
##   Debt Consolidation Loan  7.945425            6.420546
##   Home Equity Loan         7.428571            8.163265
##   Mortgage Loan            7.993339            9.159034
##   No Loan                  0.000000            0.000000
##   Not Specified            8.386581            8.546326
##   Payday Loan              8.155186            8.630245
##   Personal Loan            7.246377            8.051530
```

```
##   Student Loan              7.256778            7.894737
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                7.975986         7.975986
##   Credit-Builder Loan                      7.346939         8.653061
##   Debt Consolidation Loan                  7.062600         7.784912
##   Home Equity Loan                         9.306122         6.122449
##   Mortgage Loan                            8.159867         6.744380
##   No Loan                                  0.000000         0.000000
##   Not Specified                            7.507987         6.869010
##   Payday Loan                              7.125891         8.234363
##   Personal Loan                            8.293076         7.568438
##   Student Loan                             7.655502         7.496013
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                     7.804460  29.759863      8.662093    8.061750
##   Credit-Builder Loan           7.510204  31.020408      7.591837    7.428571
##   Debt Consolidation Loan       8.025682  30.979133      8.025682    7.945425
##   Home Equity Loan              7.265306  29.632653      8.163265    7.836735
##   Mortgage Loan                 8.076603  30.641132      7.410491    6.411324
##   No Loan                       0.000000 100.000000      0.000000    0.000000
##   Not Specified                 7.907348  31.709265      7.827476    7.428115
##   Payday Loan                   7.046714  29.928741      7.205067    7.917656
##   Personal Loan                 8.293076  28.824477      8.695652    7.729469
##   Student Loan                  7.894737  30.143541      8.373206    7.974482
##
##                           Personal Loan Student Loan
##   Auto Loan                     6.689537     6.775300
##   Credit-Builder Loan           7.428571     7.020408
##   Debt Consolidation Loan       8.346709     7.463884
##   Home Equity Loan              7.346939     8.734694
##   Mortgage Loan                 7.743547     7.660283
##   No Loan                       0.000000     0.000000
##   Not Specified                 5.990415     7.827476
##   Payday Loan                   7.917656     7.838480
##   Personal Loan                 7.326892     7.971014
##   Student Loan                  6.299841     9.011164
##
## $Last_Loan_1_Last_Loan_3$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                 9.870740           11.584699                10.739030
##   Credit-Builder Loan      10.810811           11.366120                10.392610
##   Debt Consolidation Loan 11.633373            8.743169                10.161663
##   Home Equity Loan         10.693302           10.928962                13.163972
##   Mortgage Loan            11.280846           12.021858                11.316397
##   No Loan                   0.000000            0.000000                 0.000000
##   Not Specified            12.338425           11.693989                10.854503
##   Payday Loan              12.103408           11.912568                10.392610
##   Personal Loan            10.575793           10.928962                11.893764
##   Student Loan             10.693302           10.819672                11.085450
```

```
##
##                          Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                     11.204819     10.593714  7.257896
##   Credit-Builder Loan           12.771084     10.710128  7.948128
##   Debt Consolidation Loan       11.686747     11.641444  8.073625
##   Home Equity Loan               9.036145     10.360885  7.592554
##   Mortgage Loan                  9.759036     11.292200  7.697134
##   No Loan                        0.000000      0.000000 29.826396
##   Not Specified                 10.361446     11.525029  8.303702
##   Payday Loan                   12.530120     10.360885  7.906296
##   Personal Loan                 11.325301     11.990687  7.487973
##   Student Loan                  11.325301     11.525029  7.906296
##
##                          Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                   11.412429   11.111111      9.737828     9.122402
##   Credit-Builder Loan         10.508475   10.756501     11.360799     9.930716
##   Debt Consolidation Loan     11.299435   11.702128     12.983770    10.739030
##   Home Equity Loan            11.299435   11.347518     11.235955    12.355658
##   Mortgage Loan               10.056497    9.101655     11.610487    10.623557
##   No Loan                      0.000000    0.000000      0.000000     0.000000
##   Not Specified               11.073446   10.992908      9.363296    11.316397
##   Payday Loan                 10.282486   11.820331     12.484395    11.431871
##   Personal Loan               12.203390   11.347518     11.360799    11.431871
##   Student Loan                11.864407   11.820331      9.862672    13.048499
##
##
## $Last_Loan_1_Last_Loan_4
## $Last_Loan_1_Last_Loan_4$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                    512                 592                     560
##   Credit-Builder Loan          648                 560                     584
##   Debt Consolidation Loan      504                 624                     536
##   Home Equity Loan             552                 608                     592
##   Mortgage Loan                488                 728                     496
##   No Loan                        0                   0                       0
##   Not Specified                464                 696                     464
##   Payday Loan                  544                 728                     680
##   Personal Loan                472                 664                     672
##   Student Loan                 536                 464                     448
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            560           472    4496           520
##   Credit-Builder Loan                  648           704    4480           600
##   Debt Consolidation Loan              600           512    4968           592
##   Home Equity Loan                     576           520    4680           560
##   Mortgage Loan                        504           504    4792           536
##   No Loan                                0             0   11408             0
##   Not Specified                        576           680    5024           472
##   Payday Loan                          576           504    4760           664
##   Personal Loan                        648           592    4504           488
```

```
##   Student Loan                        512           680    4888           688
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                       504           584          528
##   Credit-Builder Loan             528           520          528
##   Debt Consolidation Loan         616           536          480
##   Home Equity Loan                544           560          608
##   Mortgage Loan                   648           512          400
##   No Loan                           0             0            0
##   Not Specified                   600           440          600
##   Payday Loan                     608           528          512
##   Personal Loan                   664           648          584
##   Student Loan                    576           528          712
##
## $Last_Loan_1_Last_Loan_4$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    9328                    9800                    9968
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    9800                    9608                   11408
##           Not Specified             Payday Loan           Personal Loan
##                   10016                   10104                    9936
##            Student Loan
##                   10032
##
## $Last_Loan_1_Last_Loan_4$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    4720                    5664                    5032
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    5200                    5168                   54000
##           Not Specified             Payday Loan           Personal Loan
##                    5120                    5288                    4856
##            Student Loan
##                    4952
##
## $Last_Loan_1_Last_Loan_4$grand_total
## [1] 100000
##
## $Last_Loan_1_Last_Loan_4$cell_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                   0.512               0.592                   0.560
##   Credit-Builder Loan         0.648               0.560                   0.584
##   Debt Consolidation Loan     0.504               0.624                   0.536
##   Home Equity Loan            0.552               0.608                   0.592
##   Mortgage Loan               0.488               0.728                   0.496
##   No Loan                     0.000               0.000                   0.000
##   Not Specified               0.464               0.696                   0.464
##   Payday Loan                 0.544               0.728                   0.680
##   Personal Loan               0.472               0.664                   0.672
```

```
##   Student Loan                0.536               0.464                   0.448
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.560         0.472   4.496         0.520
##   Credit-Builder Loan                0.648         0.704   4.480         0.600
##   Debt Consolidation Loan            0.600         0.512   4.968         0.592
##   Home Equity Loan                   0.576         0.520   4.680         0.560
##   Mortgage Loan                      0.504         0.504   4.792         0.536
##   No Loan                            0.000         0.000  11.408         0.000
##   Not Specified                      0.576         0.680   5.024         0.472
##   Payday Loan                        0.576         0.504   4.760         0.664
##   Personal Loan                      0.648         0.592   4.504         0.488
##   Student Loan                       0.512         0.680   4.888         0.688
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                     0.504         0.584        0.528
##   Credit-Builder Loan           0.528         0.520        0.528
##   Debt Consolidation Loan       0.616         0.536        0.480
##   Home Equity Loan              0.544         0.560        0.608
##   Mortgage Loan                 0.648         0.512        0.400
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.600         0.440        0.600
##   Payday Loan                   0.608         0.528        0.512
##   Personal Loan                 0.664         0.648        0.584
##   Student Loan                  0.576         0.528        0.712
##
## $Last_Loan_1_Last_Loan_4$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 5.488851            6.346484
##   Credit-Builder Loan       6.612245            5.714286
##   Debt Consolidation Loan   5.056180            6.260032
##   Home Equity Loan          5.632653            6.204082
##   Mortgage Loan             5.079101            7.577019
##   No Loan                   0.000000            0.000000
##   Not Specified             4.632588            6.948882
##   Payday Loan               5.384006            7.205067
##   Personal Loan             4.750403            6.682770
##   Student Loan              5.342903            4.625199
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                6.003431         6.003431
##   Credit-Builder Loan                      5.959184         6.612245
##   Debt Consolidation Loan                  5.377207         6.019262
##   Home Equity Loan                         6.040816         5.877551
##   Mortgage Loan                            5.162365         5.245629
##   No Loan                                  0.000000         0.000000
##   Not Specified                            4.632588         5.750799
##   Payday Loan                              6.730008         5.700713
##   Personal Loan                            6.763285         6.521739
##   Student Loan                             4.465710         5.103668
```

```
##
##                          Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                   5.060034  48.198971      5.574614    5.403087
##   Credit-Builder Loan         7.183673  45.714286      6.122449    5.387755
##   Debt Consolidation Loan     5.136437  49.839486      5.939005    6.179775
##   Home Equity Loan            5.306122  47.755102      5.714286    5.551020
##   Mortgage Loan               5.245629  49.875104      5.578684    6.744380
##   No Loan                     0.000000 100.000000      0.000000    0.000000
##   Not Specified               6.789137  50.159744      4.712460    5.990415
##   Payday Loan                 4.988124  47.110055      6.571655    6.017419
##   Personal Loan               5.958132  45.330113      4.911433    6.682770
##   Student Loan                6.778309  48.724083      6.858054    5.741627
##
##                          Personal Loan Student Loan
##   Auto Loan                   6.260720     5.660377
##   Credit-Builder Loan         5.306122     5.387755
##   Debt Consolidation Loan     5.377207     4.815409
##   Home Equity Loan            5.714286     6.204082
##   Mortgage Loan               5.328893     4.163197
##   No Loan                     0.000000     0.000000
##   Not Specified               4.392971     5.990415
##   Payday Loan                 5.225653     5.067300
##   Personal Loan               6.521739     5.877617
##   Student Loan                5.263158     7.097289
##
## $Last_Loan_1_Last_Loan_4$col_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               10.847458           10.451977               11.128776
##   Credit-Builder Loan     13.728814            9.887006               11.605723
##   Debt Consolidation Loan 10.677966           11.016949               10.651828
##   Home Equity Loan        11.694915           10.734463               11.764706
##   Mortgage Loan           10.338983           12.853107                9.856916
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified            9.830508           12.288136                9.220986
##   Payday Loan             11.525424           12.853107               13.513514
##   Personal Loan           10.000000           11.723164               13.354531
##   Student Loan            11.355932            8.192090                8.903021
##
##                          Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                       10.769231      9.133127  8.325926
##   Credit-Builder Loan             12.461538     13.622291  8.296296
##   Debt Consolidation Loan         11.538462      9.907121  9.200000
##   Home Equity Loan                11.076923     10.061920  8.666667
##   Mortgage Loan                    9.692308      9.752322  8.874074
##   No Loan                          0.000000      0.000000 21.125926
##   Not Specified                   11.076923     13.157895  9.303704
##   Payday Loan                     11.076923      9.752322  8.814815
##   Personal Loan                   12.461538     11.455108  8.340741
##   Student Loan                     9.846154     13.157895  9.051852
##
```

```
##                          Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                   10.156250    9.531014     12.026359    10.662359
##   Credit-Builder Loan         11.718750    9.984871     10.708402    10.662359
##   Debt Consolidation Loan     11.562500   11.649017     11.037891     9.693053
##   Home Equity Loan            10.937500   10.287443     11.532125    12.277868
##   Mortgage Loan               10.468750   12.254160     10.543657     8.077544
##   No Loan                      0.000000    0.000000      0.000000     0.000000
##   Not Specified                9.218750   11.346445      9.060956    12.116317
##   Payday Loan                 12.968750   11.497731     10.873147    10.339257
##   Personal Loan                9.531250   12.556732     13.344316    11.793215
##   Student Loan                13.437500   10.892587     10.873147    14.378029
##
##
## $Last_Loan_1_Last_Loan_5
## $Last_Loan_1_Last_Loan_5$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                    288                 376                     384
##   Credit-Builder Loan          528                 400                     432
##   Debt Consolidation Loan      384                 384                     368
##   Home Equity Loan             416                 400                     344
##   Mortgage Loan                328                 352                     352
##   No Loan                        0                   0                       0
##   Not Specified                400                 392                     408
##   Payday Loan                  328                 392                     376
##   Personal Loan                384                 392                     488
##   Student Loan                 368                 408                     376
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            232           384    6224           296
##   Credit-Builder Loan                  368           360    6296           384
##   Debt Consolidation Loan              320           360    6624           368
##   Home Equity Loan                     296           384    6480           328
##   Mortgage Loan                        432           336    6376           400
##   No Loan                                0             0   11408             0
##   Not Specified                        336           392    6648           304
##   Payday Loan                          464           400    6536           360
##   Personal Loan                        496           424    6288           416
##   Student Loan                         456           400    6576           416
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                      360           416          368
##   Credit-Builder Loan            344           344          344
##   Debt Consolidation Loan        440           368          352
##   Home Equity Loan               384           384          384
##   Mortgage Loan                  352           384          296
##   No Loan                          0             0            0
##   Not Specified                  360           376          400
##   Payday Loan                    424           440          384
##   Personal Loan                  368           392          288
##   Student Loan                   312           416          304
```

```
##
## $Last_Loan_1_Last_Loan_5$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    9328                    9800                    9968
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    9800                    9608                   11408
##           Not Specified             Payday Loan           Personal Loan
##                   10016                   10104                    9936
##            Student Loan
##                   10032
##
## $Last_Loan_1_Last_Loan_5$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    3424                    3496                    3528
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    3400                    3440                   69456
##           Not Specified             Payday Loan           Personal Loan
##                    3272                    3344                    3520
##            Student Loan
##                    3120
##
## $Last_Loan_1_Last_Loan_5$grand_total
## [1] 100000
##
## $Last_Loan_1_Last_Loan_5$cell_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                   0.288               0.376                   0.384
##   Credit-Builder Loan         0.528               0.400                   0.432
##   Debt Consolidation Loan     0.384               0.384                   0.368
##   Home Equity Loan            0.416               0.400                   0.344
##   Mortgage Loan               0.328               0.352                   0.352
##   No Loan                     0.000               0.000                   0.000
##   Not Specified               0.400               0.392                   0.408
##   Payday Loan                 0.328               0.392                   0.376
##   Personal Loan               0.384               0.392                   0.488
##   Student Loan                0.368               0.408                   0.376
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.232         0.384   6.224         0.296
##   Credit-Builder Loan                0.368         0.360   6.296         0.384
##   Debt Consolidation Loan            0.320         0.360   6.624         0.368
##   Home Equity Loan                   0.296         0.384   6.480         0.328
##   Mortgage Loan                      0.432         0.336   6.376         0.400
##   No Loan                            0.000         0.000  11.408         0.000
##   Not Specified                      0.336         0.392   6.648         0.304
##   Payday Loan                        0.464         0.400   6.536         0.360
##   Personal Loan                      0.496         0.424   6.288         0.416
##   Student Loan                       0.456         0.400   6.576         0.416
```

```
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.360         0.416        0.368
##   Credit-Builder Loan          0.344         0.344        0.344
##   Debt Consolidation Loan      0.440         0.368        0.352
##   Home Equity Loan             0.384         0.384        0.384
##   Mortgage Loan                0.352         0.384        0.296
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.360         0.376        0.400
##   Payday Loan                  0.424         0.440        0.384
##   Personal Loan                0.368         0.392        0.288
##   Student Loan                 0.312         0.416        0.304
##
## $Last_Loan_1_Last_Loan_5$row_percentage
##
##                           Auto Loan Credit-Builder Loan
##   Auto Loan                 3.087479            4.030875
##   Credit-Builder Loan       5.387755            4.081633
##   Debt Consolidation Loan   3.852327            3.852327
##   Home Equity Loan          4.244898            4.081633
##   Mortgage Loan             3.413822            3.663614
##   No Loan                   0.000000            0.000000
##   Not Specified             3.993610            3.913738
##   Payday Loan               3.246239            3.879652
##   Personal Loan             3.864734            3.945250
##   Student Loan              3.668262            4.066986
##
##                          Debt Consolidation Loan Home Equity Loan
##   Auto Loan                               4.116638         2.487136
##   Credit-Builder Loan                     4.408163         3.755102
##   Debt Consolidation Loan                 3.691814         3.210273
##   Home Equity Loan                        3.510204         3.020408
##   Mortgage Loan                           3.663614         4.496253
##   No Loan                                 0.000000         0.000000
##   Not Specified                           4.073482         3.354633
##   Payday Loan                             3.721298         4.592241
##   Personal Loan                           4.911433         4.991948
##   Student Loan                            3.748006         4.545455
##
##                          Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                    4.116638  66.723842      3.173242    3.859348
##   Credit-Builder Loan          3.673469  64.244898      3.918367    3.510204
##   Debt Consolidation Loan      3.611557  66.452648      3.691814    4.414125
##   Home Equity Loan             3.918367  66.122449      3.346939    3.918367
##   Mortgage Loan                3.497086  66.361366      4.163197    3.663614
##   No Loan                      0.000000 100.000000      0.000000    0.000000
##   Not Specified                3.913738  66.373802      3.035144    3.594249
##   Payday Loan                  3.958828  64.687253      3.562945    4.196358
##   Personal Loan                4.267311  63.285024      4.186795    3.703704
##   Student Loan                 3.987241  65.550239      4.146730    3.110048
##
```

```
##                          Personal Loan Student Loan
##   Auto Loan                   4.459691     3.945111
##   Credit-Builder Loan         3.510204     3.510204
##   Debt Consolidation Loan     3.691814     3.531300
##   Home Equity Loan            3.918367     3.918367
##   Mortgage Loan               3.996669     3.080766
##   No Loan                     0.000000     0.000000
##   Not Specified               3.753994     3.993610
##   Payday Loan                 4.354711     3.800475
##   Personal Loan               3.945250     2.898551
##   Student Loan                4.146730     3.030303
##
## $Last_Loan_1_Last_Loan_5$col_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                8.411215           10.755149               10.884354
##   Credit-Builder Loan     15.420561           11.441648               12.244898
##   Debt Consolidation Loan 11.214953           10.983982               10.430839
##   Home Equity Loan        12.149533           11.441648                9.750567
##   Mortgage Loan            9.579439           10.068650                9.977324
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           11.682243           11.212815               11.564626
##   Payday Loan              9.579439           11.212815               10.657596
##   Personal Loan           11.214953           11.212815               13.832200
##   Student Loan            10.747664           11.670481               10.657596
##
##                          Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                       6.823529     11.162791  8.961069
##   Credit-Builder Loan            10.823529     10.465116  9.064732
##   Debt Consolidation Loan         9.411765     10.465116  9.536973
##   Home Equity Loan                8.705882     11.162791  9.329648
##   Mortgage Loan                  12.705882      9.767442  9.179912
##   No Loan                         0.000000      0.000000 16.424787
##   Not Specified                   9.882353     11.395349  9.571527
##   Payday Loan                    13.647059     11.627907  9.410274
##   Personal Loan                  14.588235     12.325581  9.053214
##   Student Loan                   13.411765     11.627907  9.467865
##
##                          Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    9.046455   10.765550     11.818182    11.794872
##   Credit-Builder Loan         11.735941   10.287081      9.772727    11.025641
##   Debt Consolidation Loan     11.246944   13.157895     10.454545    11.282051
##   Home Equity Loan            10.024450   11.483254     10.909091    12.307692
##   Mortgage Loan               12.224939   10.526316     10.909091     9.487179
##   No Loan                      0.000000    0.000000      0.000000     0.000000
##   Not Specified                9.290954   10.765550     10.681818    12.820513
##   Payday Loan                 11.002445   12.679426     12.500000    12.307692
##   Personal Loan               12.713936   11.004785     11.136364     9.230769
##   Student Loan                12.713936    9.330144     11.818182     9.743590
##
##
```

```
## $Last_Loan_1_Last_Loan_6
## $Last_Loan_1_Last_Loan_6$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                    216                 296                     216
##   Credit-Builder Loan          312                 272                     264
##   Debt Consolidation Loan      272                 304                     328
##   Home Equity Loan             232                 312                     248
##   Mortgage Loan                224                 232                     232
##   No Loan                        0                   0                       0
##   Not Specified                320                 240                     264
##   Payday Loan                  224                 352                     272
##   Personal Loan                328                 360                     400
##   Student Loan                 344                 240                     240
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            160           264    7008           272
##   Credit-Builder Loan                  320           352    7120           312
##   Debt Consolidation Loan              224           312    7392           256
##   Home Equity Loan                     312           200    7288           304
##   Mortgage Loan                        312           200    7272           280
##   No Loan                                0             0   11408             0
##   Not Specified                        248           280    7432           296
##   Payday Loan                          280           200    7560           312
##   Personal Loan                        304           256    7104           296
##   Student Loan                         304           328    7400           296
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                      416           264          216
##   Credit-Builder Loan            256           336          256
##   Debt Consolidation Loan        272           296          312
##   Home Equity Loan               296           320          288
##   Mortgage Loan                  304           312          240
##   No Loan                          0             0            0
##   Not Specified                  336           336          264
##   Payday Loan                    256           344          304
##   Personal Loan                  256           272          360
##   Student Loan                   296           256          328
##
## $Last_Loan_1_Last_Loan_6$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    9328                    9800                    9968
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    9800                    9608                   11408
##           Not Specified             Payday Loan           Personal Loan
##                   10016                   10104                    9936
##            Student Loan
##                   10032
##
## $Last_Loan_1_Last_Loan_6$col_totals
```

```
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    2472                    2608                    2464
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    2464                    2392                   76984
##           Not Specified             Payday Loan           Personal Loan
##                    2624                    2688                    2736
##            Student Loan
##                    2568
##
## $Last_Loan_1_Last_Loan_6$grand_total
## [1] 100000
##
## $Last_Loan_1_Last_Loan_6$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.216               0.296                   0.216
##   Credit-Builder Loan        0.312               0.272                   0.264
##   Debt Consolidation Loan    0.272               0.304                   0.328
##   Home Equity Loan           0.232               0.312                   0.248
##   Mortgage Loan              0.224               0.232                   0.232
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.320               0.240                   0.264
##   Payday Loan                0.224               0.352                   0.272
##   Personal Loan              0.328               0.360                   0.400
##   Student Loan               0.344               0.240                   0.240
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.160         0.264   7.008         0.272
##   Credit-Builder Loan                0.320         0.352   7.120         0.312
##   Debt Consolidation Loan            0.224         0.312   7.392         0.256
##   Home Equity Loan                   0.312         0.200   7.288         0.304
##   Mortgage Loan                      0.312         0.200   7.272         0.280
##   No Loan                            0.000         0.000  11.408         0.000
##   Not Specified                      0.248         0.280   7.432         0.296
##   Payday Loan                        0.280         0.200   7.560         0.312
##   Personal Loan                      0.304         0.256   7.104         0.296
##   Student Loan                       0.304         0.328   7.400         0.296
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.416         0.264        0.216
##   Credit-Builder Loan          0.256         0.336        0.256
##   Debt Consolidation Loan      0.272         0.296        0.312
##   Home Equity Loan             0.296         0.320        0.288
##   Mortgage Loan                0.304         0.312        0.240
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.336         0.336        0.264
##   Payday Loan                  0.256         0.344        0.304
##   Personal Loan                0.256         0.272        0.360
##   Student Loan                 0.296         0.256        0.328
##
```

```
## $Last_Loan_1_Last_Loan_6$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 2.315609            3.173242
##   Credit-Builder Loan       3.183673            2.775510
##   Debt Consolidation Loan   2.728732            3.049759
##   Home Equity Loan          2.367347            3.183673
##   Mortgage Loan             2.331391            2.414654
##   No Loan                   0.000000            0.000000
##   Not Specified             3.194888            2.396166
##   Payday Loan               2.216944            3.483769
##   Personal Loan             3.301127            3.623188
##   Student Loan              3.429027            2.392344
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                2.315609         1.715266
##   Credit-Builder Loan                      2.693878         3.265306
##   Debt Consolidation Loan                  3.290530         2.247191
##   Home Equity Loan                         2.530612         3.183673
##   Mortgage Loan                            2.414654         3.247294
##   No Loan                                  0.000000         0.000000
##   Not Specified                            2.635783         2.476038
##   Payday Loan                              2.692003         2.771180
##   Personal Loan                            4.025765         3.059581
##   Student Loan                             2.392344         3.030303
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                     2.830189  75.128645      2.915952    4.459691
##   Credit-Builder Loan           3.591837  72.653061      3.183673    2.612245
##   Debt Consolidation Loan       3.130016  74.157303      2.568218    2.728732
##   Home Equity Loan              2.040816  74.367347      3.102041    3.020408
##   Mortgage Loan                 2.081599  75.686928      2.914238    3.164030
##   No Loan                       0.000000 100.000000      0.000000    0.000000
##   Not Specified                 2.795527  74.201278      2.955272    3.354633
##   Payday Loan                   1.979414  74.821853      3.087886    2.533650
##   Personal Loan                 2.576490  71.497585      2.979066    2.576490
##   Student Loan                  3.269537  73.763955      2.950558    2.950558
##
##                           Personal Loan Student Loan
##   Auto Loan                     2.830189     2.315609
##   Credit-Builder Loan           3.428571     2.612245
##   Debt Consolidation Loan       2.969502     3.130016
##   Home Equity Loan              3.265306     2.938776
##   Mortgage Loan                 3.247294     2.497918
##   No Loan                       0.000000     0.000000
##   Not Specified                 3.354633     2.635783
##   Payday Loan                   3.404592     3.008709
##   Personal Loan                 2.737520     3.623188
##   Student Loan                  2.551834     3.269537
##
## $Last_Loan_1_Last_Loan_6$col_percentage
```

```
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                8.737864           11.349693                8.766234
##   Credit-Builder Loan     12.621359           10.429448               10.714286
##   Debt Consolidation Loan 11.003236           11.656442               13.311688
##   Home Equity Loan         9.385113           11.963190               10.064935
##   Mortgage Loan            9.061489            8.895706                9.415584
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           12.944984            9.202454               10.714286
##   Payday Loan              9.061489           13.496933               11.038961
##   Personal Loan           13.268608           13.803681               16.233766
##   Student Loan            13.915858            9.202454                9.740260
##
##                          Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                        6.493506     11.036789  9.103190
##   Credit-Builder Loan             12.987013     14.715719  9.248675
##   Debt Consolidation Loan          9.090909     13.043478  9.601995
##   Home Equity Loan                12.662338      8.361204  9.466902
##   Mortgage Loan                   12.662338      8.361204  9.446119
##   No Loan                          0.000000      0.000000 14.818664
##   Not Specified                   10.064935     11.705686  9.653954
##   Payday Loan                     11.363636      8.361204  9.820222
##   Personal Loan                   12.337662     10.702341  9.227892
##   Student Loan                    12.337662     13.712375  9.612387
##
##                          Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    10.365854   15.476190      9.649123     8.411215
##   Credit-Builder Loan          11.890244    9.523810     12.280702     9.968847
##   Debt Consolidation Loan       9.756098   10.119048     10.818713    12.149533
##   Home Equity Loan             11.585366   11.011905     11.695906    11.214953
##   Mortgage Loan                10.670732   11.309524     11.403509     9.345794
##   No Loan                       0.000000    0.000000      0.000000     0.000000
##   Not Specified                11.280488   12.500000     12.280702    10.280374
##   Payday Loan                  11.890244    9.523810     12.573099    11.838006
##   Personal Loan                11.280488    9.523810      9.941520    14.018692
##   Student Loan                 11.280488   11.011905      9.356725    12.772586
##
##
## $Last_Loan_1_Last_Loan_7
## $Last_Loan_1_Last_Loan_7$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                    160                 208                     152
##   Credit-Builder Loan          168                 248                     160
##   Debt Consolidation Loan      184                 160                     144
##   Home Equity Loan             224                 200                     160
##   Mortgage Loan                152                 208                     120
##   No Loan                        0                   0                       0
##   Not Specified                184                 232                     168
##   Payday Loan                  128                 144                     168
##   Personal Loan                216                 208                     176
```

```
##   Student Loan                  184                 112                     184
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            112           200    7872           120
##   Credit-Builder Loan                  200           176    8136           144
##   Debt Consolidation Loan              184           208    8280           160
##   Home Equity Loan                     168           248    8192           136
##   Mortgage Loan                        200           160    8056           200
##   No Loan                                0             0   11408             0
##   Not Specified                        176           136    8248           232
##   Payday Loan                          200           232    8440           144
##   Personal Loan                        192           248    8016           192
##   Student Loan                          96           184    8480           280
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                       192           152          160
##   Credit-Builder Loan             264           176          128
##   Debt Consolidation Loan         144           256          248
##   Home Equity Loan                200           144          128
##   Mortgage Loan                   208           200          104
##   No Loan                           0             0            0
##   Not Specified                   200           200          240
##   Payday Loan                     232           248          168
##   Personal Loan                   272           200          216
##   Student Loan                    208           112          192
##
## $Last_Loan_1_Last_Loan_7$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    9328                    9800                    9968
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    9800                    9608                   11408
##           Not Specified             Payday Loan           Personal Loan
##                   10016                   10104                    9936
##            Student Loan
##                   10032
##
## $Last_Loan_1_Last_Loan_7$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    1600                    1720                    1432
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    1528                    1792                   85128
##           Not Specified             Payday Loan           Personal Loan
##                    1608                    1920                    1688
##            Student Loan
##                    1584
##
## $Last_Loan_1_Last_Loan_7$grand_total
## [1] 100000
##
```

```
## $Last_Loan_1_Last_Loan_7$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.160               0.208                   0.152
##   Credit-Builder Loan        0.168               0.248                   0.160
##   Debt Consolidation Loan    0.184               0.160                   0.144
##   Home Equity Loan           0.224               0.200                   0.160
##   Mortgage Loan              0.152               0.208                   0.120
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.184               0.232                   0.168
##   Payday Loan                0.128               0.144                   0.168
##   Personal Loan              0.216               0.208                   0.176
##   Student Loan               0.184               0.112                   0.184
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.112         0.200   7.872         0.120
##   Credit-Builder Loan                0.200         0.176   8.136         0.144
##   Debt Consolidation Loan            0.184         0.208   8.280         0.160
##   Home Equity Loan                   0.168         0.248   8.192         0.136
##   Mortgage Loan                      0.200         0.160   8.056         0.200
##   No Loan                            0.000         0.000  11.408         0.000
##   Not Specified                      0.176         0.136   8.248         0.232
##   Payday Loan                        0.200         0.232   8.440         0.144
##   Personal Loan                      0.192         0.248   8.016         0.192
##   Student Loan                       0.096         0.184   8.480         0.280
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.192         0.152        0.160
##   Credit-Builder Loan          0.264         0.176        0.128
##   Debt Consolidation Loan      0.144         0.256        0.248
##   Home Equity Loan             0.200         0.144        0.128
##   Mortgage Loan                0.208         0.200        0.104
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.200         0.200        0.240
##   Payday Loan                  0.232         0.248        0.168
##   Personal Loan                0.272         0.200        0.216
##   Student Loan                 0.208         0.112        0.192
##
## $Last_Loan_1_Last_Loan_7$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                1.7152659           2.2298456
##   Credit-Builder Loan      1.7142857           2.5306122
##   Debt Consolidation Loan  1.8459069           1.6051364
##   Home Equity Loan         2.2857143           2.0408163
##   Mortgage Loan            1.5820150           2.1648626
##   No Loan                  0.0000000           0.0000000
##   Not Specified            1.8370607           2.3162939
##   Payday Loan              1.2668250           1.4251781
##   Personal Loan            2.1739130           2.0933977
##   Student Loan             1.8341308           1.1164274
```

```
##
##                          Debt Consolidation Loan Home Equity Loan
##   Auto Loan                             1.6295026        1.2006861
##   Credit-Builder Loan                   1.6326531        2.0408163
##   Debt Consolidation Loan               1.4446228        1.8459069
##   Home Equity Loan                      1.6326531        1.7142857
##   Mortgage Loan                         1.2489592        2.0815987
##   No Loan                               0.0000000        0.0000000
##   Not Specified                         1.6773163        1.7571885
##   Payday Loan                           1.6627078        1.9794141
##   Personal Loan                         1.7713366        1.9323671
##   Student Loan                          1.8341308        0.9569378
##
##                          Mortgage Loan     No Loan Not Specified Payday Loan
##   Auto Loan                    2.1440823  84.3910806     1.2864494   2.0583190
##   Credit-Builder Loan          1.7959184  83.0204082     1.4693878   2.6938776
##   Debt Consolidation Loan      2.0866774  83.0658106     1.6051364   1.4446228
##   Home Equity Loan             2.5306122  83.5918367     1.3877551   2.0408163
##   Mortgage Loan                1.6652789  83.8467943     2.0815987   2.1648626
##   No Loan                      0.0000000 100.0000000     0.0000000   0.0000000
##   Not Specified                1.3578275  82.3482428     2.3162939   1.9968051
##   Payday Loan                  2.2961203  83.5312747     1.4251781   2.2961203
##   Personal Loan                2.4959742  80.6763285     1.9323671   2.7375201
##   Student Loan                 1.8341308  84.5295056     2.7910686   2.0733652
##
##                          Personal Loan Student Loan
##   Auto Loan                    1.6295026    1.7152659
##   Credit-Builder Loan          1.7959184    1.3061224
##   Debt Consolidation Loan      2.5682183    2.4879615
##   Home Equity Loan             1.4693878    1.3061224
##   Mortgage Loan                2.0815987    1.0824313
##   No Loan                      0.0000000    0.0000000
##   Not Specified                1.9968051    2.3961661
##   Payday Loan                  2.4544735    1.6627078
##   Personal Loan                2.0128824    2.1739130
##   Student Loan                 1.1164274    1.9138756
##
## $Last_Loan_1_Last_Loan_7$col_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               10.000000           12.093023               10.614525
##   Credit-Builder Loan     10.500000           14.418605               11.173184
##   Debt Consolidation Loan 11.500000            9.302326               10.055866
##   Home Equity Loan        14.000000           11.627907               11.173184
##   Mortgage Loan            9.500000           12.093023                8.379888
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           11.500000           13.488372               11.731844
##   Payday Loan              8.000000            8.372093               11.731844
##   Personal Loan           13.500000           12.093023               12.290503
##   Student Loan            11.500000            6.511628               12.849162
##
```

```
##                          Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                       7.329843     11.160714  9.247251
##   Credit-Builder Loan            13.089005      9.821429  9.557372
##   Debt Consolidation Loan        12.041885     11.607143  9.726529
##   Home Equity Loan               10.994764     13.839286  9.623156
##   Mortgage Loan                  13.089005      8.928571  9.463396
##   No Loan                         0.000000      0.000000 13.400996
##   Not Specified                  11.518325      7.589286  9.688939
##   Payday Loan                    13.089005     12.946429  9.914482
##   Personal Loan                  12.565445     13.839286  9.416408
##   Student Loan                    6.282723     10.267857  9.961470
##
##                          Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                     7.462687   10.000000      9.004739    10.101010
##   Credit-Builder Loan           8.955224   13.750000     10.426540     8.080808
##   Debt Consolidation Loan       9.950249    7.500000     15.165877    15.656566
##   Home Equity Loan              8.457711   10.416667      8.530806     8.080808
##   Mortgage Loan                12.437811   10.833333     11.848341     6.565657
##   No Loan                       0.000000    0.000000      0.000000     0.000000
##   Not Specified                14.427861   10.416667     11.848341    15.151515
##   Payday Loan                   8.955224   12.083333     14.691943    10.606061
##   Personal Loan                11.940299   14.166667     11.848341    13.636364
##   Student Loan                 17.412935   10.833333      6.635071    12.121212
##
##
## $Last_Loan_1_Last_Loan_8
## $Last_Loan_1_Last_Loan_8$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     80                  48                      40
##   Credit-Builder Loan           48                 112                      72
##   Debt Consolidation Loan       80                  88                      64
##   Home Equity Loan              56                 112                     120
##   Mortgage Loan                 64                 128                      64
##   No Loan                        0                   0                       0
##   Not Specified                 96                 104                      80
##   Payday Loan                  112                 112                      88
##   Personal Loan                 80                  80                     160
##   Student Loan                 104                  88                      72
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                           104            72    8624            88
##   Credit-Builder Loan                 104            80    9024            80
##   Debt Consolidation Loan              72            64    9256           104
##   Home Equity Loan                     72           112    9032            32
##   Mortgage Loan                        96            24    8976            88
##   No Loan                               0             0   11408             0
##   Not Specified                       128           120    9112            80
##   Payday Loan                          72            96    9248            72
##   Personal Loan                       152            72    8920           136
##   Student Loan                         88            96    9208            64
```

```
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                       96            96           80
##   Credit-Builder Loan            128            24          128
##   Debt Consolidation Loan         80            80           80
##   Home Equity Loan                96            64          104
##   Mortgage Loan                   64            16           88
##   No Loan                          0             0            0
##   Not Specified                  120            96           80
##   Payday Loan                     80           112          112
##   Personal Loan                   88           128          120
##   Student Loan                   136           104           72
##
## $Last_Loan_1_Last_Loan_8$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    9328                    9800                    9968
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    9800                    9608                   11408
##           Not Specified             Payday Loan           Personal Loan
##                   10016                   10104                    9936
##            Student Loan
##                   10032
##
## $Last_Loan_1_Last_Loan_8$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     720                     872                     760
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     888                     736                   92808
##           Not Specified             Payday Loan           Personal Loan
##                     744                     888                     720
##            Student Loan
##                     864
##
## $Last_Loan_1_Last_Loan_8$grand_total
## [1] 100000
##
## $Last_Loan_1_Last_Loan_8$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.080               0.048                   0.040
##   Credit-Builder Loan        0.048               0.112                   0.072
##   Debt Consolidation Loan    0.080               0.088                   0.064
##   Home Equity Loan           0.056               0.112                   0.120
##   Mortgage Loan              0.064               0.128                   0.064
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.096               0.104                   0.080
##   Payday Loan                0.112               0.112                   0.088
##   Personal Loan              0.080               0.080                   0.160
##   Student Loan               0.104               0.088                   0.072
```

```
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.104         0.072   8.624         0.088
##   Credit-Builder Loan                0.104         0.080   9.024         0.080
##   Debt Consolidation Loan            0.072         0.064   9.256         0.104
##   Home Equity Loan                   0.072         0.112   9.032         0.032
##   Mortgage Loan                      0.096         0.024   8.976         0.088
##   No Loan                            0.000         0.000  11.408         0.000
##   Not Specified                      0.128         0.120   9.112         0.080
##   Payday Loan                        0.072         0.096   9.248         0.072
##   Personal Loan                      0.152         0.072   8.920         0.136
##   Student Loan                       0.088         0.096   9.208         0.064
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.096         0.096        0.080
##   Credit-Builder Loan          0.128         0.024        0.128
##   Debt Consolidation Loan      0.080         0.080        0.080
##   Home Equity Loan             0.096         0.064        0.104
##   Mortgage Loan                0.064         0.016        0.088
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.120         0.096        0.080
##   Payday Loan                  0.080         0.112        0.112
##   Personal Loan                0.088         0.128        0.120
##   Student Loan                 0.136         0.104        0.072
##
## $Last_Loan_1_Last_Loan_8$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                0.8576329           0.5145798
##   Credit-Builder Loan      0.4897959           1.1428571
##   Debt Consolidation Loan  0.8025682           0.8828250
##   Home Equity Loan         0.5714286           1.1428571
##   Mortgage Loan            0.6661116           1.3322231
##   No Loan                  0.0000000           0.0000000
##   Not Specified            0.9584665           1.0383387
##   Payday Loan              1.1084719           1.1084719
##   Personal Loan            0.8051530           0.8051530
##   Student Loan             1.0366826           0.8771930
##
##                          Debt Consolidation Loan Home Equity Loan
##   Auto Loan                             0.4288165        1.1149228
##   Credit-Builder Loan                   0.7346939        1.0612245
##   Debt Consolidation Loan               0.6420546        0.7223114
##   Home Equity Loan                      1.2244898        0.7346939
##   Mortgage Loan                         0.6661116        0.9991674
##   No Loan                               0.0000000        0.0000000
##   Not Specified                         0.7987220        1.2779553
##   Payday Loan                           0.8709422        0.7125891
##   Personal Loan                         1.6103060        1.5297907
##   Student Loan                          0.7177033        0.8771930
##
```

```
##                          Mortgage Loan     No Loan Not Specified Payday Loan
##   Auto Loan                  0.7718696  92.4528302     0.9433962   1.0291595
##   Credit-Builder Loan        0.8163265  92.0816327     0.8163265   1.3061224
##   Debt Consolidation Loan    0.6420546  92.8571429     1.0433387   0.8025682
##   Home Equity Loan           1.1428571  92.1632653     0.3265306   0.9795918
##   Mortgage Loan              0.2497918  93.4221482     0.9159034   0.6661116
##   No Loan                    0.0000000 100.0000000     0.0000000   0.0000000
##   Not Specified              1.1980831  90.9744409     0.7987220   1.1980831
##   Payday Loan                0.9501188  91.5281077     0.7125891   0.7917656
##   Personal Loan              0.7246377  89.7745572     1.3687601   0.8856683
##   Student Loan               0.9569378  91.7862839     0.6379585   1.3556619
##
##                          Personal Loan Student Loan
##   Auto Loan                  1.0291595    0.8576329
##   Credit-Builder Loan        0.2448980    1.3061224
##   Debt Consolidation Loan    0.8025682    0.8025682
##   Home Equity Loan           0.6530612    1.0612245
##   Mortgage Loan              0.1665279    0.9159034
##   No Loan                    0.0000000    0.0000000
##   Not Specified              0.9584665    0.7987220
##   Payday Loan                1.1084719    1.1084719
##   Personal Loan              1.2882448    1.2077295
##   Student Loan               1.0366826    0.7177033
##
## $Last_Loan_1_Last_Loan_8$col_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan              11.111111            5.504587                5.263158
##   Credit-Builder Loan     6.666667           12.844037                9.473684
##   Debt Consolidation Loan 11.111111          10.091743                8.421053
##   Home Equity Loan        7.777778           12.844037               15.789474
##   Mortgage Loan           8.888889           14.678899                8.421053
##   No Loan                 0.000000            0.000000                0.000000
##   Not Specified          13.333333           11.926606               10.526316
##   Payday Loan            15.555556           12.844037               11.578947
##   Personal Loan          11.111111            9.174312               21.052632
##   Student Loan           14.444444           10.091743                9.473684
##
##                          Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                      11.711712      9.782609  9.292302
##   Credit-Builder Loan            11.711712     10.869565  9.723300
##   Debt Consolidation Loan         8.108108      8.695652  9.973278
##   Home Equity Loan                8.108108     15.217391  9.731920
##   Mortgage Loan                  10.810811      3.260870  9.671580
##   No Loan                         0.000000      0.000000 12.292044
##   Not Specified                  14.414414     16.304348  9.818119
##   Payday Loan                     8.108108     13.043478  9.964658
##   Personal Loan                  17.117117      9.782609  9.611240
##   Student Loan                    9.909910     13.043478  9.921558
##
##                          Not Specified Payday Loan Personal Loan Student Loan
```

```
##   Auto Loan                   11.827957   10.810811     13.333333     9.259259
##   Credit-Builder Loan         10.752688   14.414414      3.333333    14.814815
##   Debt Consolidation Loan     13.978495    9.009009     11.111111     9.259259
##   Home Equity Loan             4.301075   10.810811      8.888889    12.037037
##   Mortgage Loan               11.827957    7.207207      2.222222    10.185185
##   No Loan                      0.000000    0.000000      0.000000     0.000000
##   Not Specified               10.752688   13.513514     13.333333     9.259259
##   Payday Loan                  9.677419    9.009009     15.555556    12.962963
##   Personal Loan               18.279570    9.909910     17.777778    13.888889
##   Student Loan                 8.602151   15.315315     14.444444     8.333333
##
##
## $Last_Loan_1_Last_Loan_9
## $Last_Loan_1_Last_Loan_9$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     24                  40                      40
##   Credit-Builder Loan           48                  64                      80
##   Debt Consolidation Loan       56                  64                      40
##   Home Equity Loan              72                  80                      32
##   Mortgage Loan                 32                  24                      16
##   No Loan                        0                   0                       0
##   Not Specified                 32                  32                      64
##   Payday Loan                   24                  64                      32
##   Personal Loan                 64                  24                      56
##   Student Loan                  96                  40                      32
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            48            72    8896            72
##   Credit-Builder Loan                  24            80    9320            48
##   Debt Consolidation Loan              40            56    9552            40
##   Home Equity Loan                     64            40    9416            40
##   Mortgage Loan                        24            56    9264            56
##   No Loan                               0             0   11408             0
##   Not Specified                        48            80    9528            32
##   Payday Loan                          48            80    9688            40
##   Personal Loan                        72            40    9472            48
##   Student Loan                         40            16    9600            48
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                       72            16           48
##   Credit-Builder Loan             40            72           24
##   Debt Consolidation Loan         24            48           48
##   Home Equity Loan                24            16           16
##   Mortgage Loan                   80            24           32
##   No Loan                          0             0            0
##   Not Specified                   72            64           64
##   Payday Loan                     56            24           48
##   Personal Loan                   48            56           56
##   Student Loan                    48            64           48
##
```

```
## $Last_Loan_1_Last_Loan_9$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    9328                    9800                    9968
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    9800                    9608                   11408
##           Not Specified             Payday Loan           Personal Loan
##                   10016                   10104                    9936
##            Student Loan
##                   10032
##
## $Last_Loan_1_Last_Loan_9$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     448                     432                     392
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     408                     520                   96144
##           Not Specified             Payday Loan           Personal Loan
##                     424                     464                     384
##            Student Loan
##                     384
##
## $Last_Loan_1_Last_Loan_9$grand_total
## [1] 100000
##
## $Last_Loan_1_Last_Loan_9$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.024               0.040                   0.040
##   Credit-Builder Loan        0.048               0.064                   0.080
##   Debt Consolidation Loan    0.056               0.064                   0.040
##   Home Equity Loan           0.072               0.080                   0.032
##   Mortgage Loan              0.032               0.024                   0.016
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.032               0.032                   0.064
##   Payday Loan                0.024               0.064                   0.032
##   Personal Loan              0.064               0.024                   0.056
##   Student Loan               0.096               0.040                   0.032
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                         0.048         0.072   8.896         0.072
##   Credit-Builder Loan               0.024         0.080   9.320         0.048
##   Debt Consolidation Loan           0.040         0.056   9.552         0.040
##   Home Equity Loan                  0.064         0.040   9.416         0.040
##   Mortgage Loan                     0.024         0.056   9.264         0.056
##   No Loan                           0.000         0.000  11.408         0.000
##   Not Specified                     0.048         0.080   9.528         0.032
##   Payday Loan                       0.048         0.080   9.688         0.040
##   Personal Loan                     0.072         0.040   9.472         0.048
##   Student Loan                      0.040         0.016   9.600         0.048
##
```

```
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.072         0.016        0.048
##   Credit-Builder Loan          0.040         0.072        0.024
##   Debt Consolidation Loan      0.024         0.048        0.048
##   Home Equity Loan             0.024         0.016        0.016
##   Mortgage Loan                0.080         0.024        0.032
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.072         0.064        0.064
##   Payday Loan                  0.056         0.024        0.048
##   Personal Loan                0.048         0.056        0.056
##   Student Loan                 0.048         0.064        0.048
##
## $Last_Loan_1_Last_Loan_9$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                0.2572899           0.4288165
##   Credit-Builder Loan      0.4897959           0.6530612
##   Debt Consolidation Loan  0.5617978           0.6420546
##   Home Equity Loan         0.7346939           0.8163265
##   Mortgage Loan            0.3330558           0.2497918
##   No Loan                  0.0000000           0.0000000
##   Not Specified            0.3194888           0.3194888
##   Payday Loan              0.2375297           0.6334125
##   Personal Loan            0.6441224           0.2415459
##   Student Loan             0.9569378           0.3987241
##
##                          Debt Consolidation Loan Home Equity Loan
##   Auto Loan                             0.4288165        0.5145798
##   Credit-Builder Loan                   0.8163265        0.2448980
##   Debt Consolidation Loan               0.4012841        0.4012841
##   Home Equity Loan                      0.3265306        0.6530612
##   Mortgage Loan                         0.1665279        0.2497918
##   No Loan                               0.0000000        0.0000000
##   Not Specified                         0.6389776        0.4792332
##   Payday Loan                           0.3167063        0.4750594
##   Personal Loan                         0.5636071        0.7246377
##   Student Loan                          0.3189793        0.3987241
##
##                          Mortgage Loan     No Loan Not Specified Payday Loan
##   Auto Loan                   0.7718696  95.3687822     0.7718696   0.7718696
##   Credit-Builder Loan         0.8163265  95.1020408     0.4897959   0.4081633
##   Debt Consolidation Loan     0.5617978  95.8266453     0.4012841   0.2407705
##   Home Equity Loan            0.4081633  96.0816327     0.4081633   0.2448980
##   Mortgage Loan               0.5828476  96.4196503     0.5828476   0.8326395
##   No Loan                     0.0000000 100.0000000     0.0000000   0.0000000
##   Not Specified               0.7987220  95.1277955     0.3194888   0.7188498
##   Payday Loan                 0.7917656  95.8828187     0.3958828   0.5542359
##   Personal Loan               0.4025765  95.3301127     0.4830918   0.4830918
##   Student Loan                0.1594896  95.6937799     0.4784689   0.4784689
##
##                          Personal Loan Student Loan
```

```
##   Auto Loan                   0.1715266    0.5145798
##   Credit-Builder Loan         0.7346939    0.2448980
##   Debt Consolidation Loan     0.4815409    0.4815409
##   Home Equity Loan            0.1632653    0.1632653
##   Mortgage Loan               0.2497918    0.3330558
##   No Loan                     0.0000000    0.0000000
##   Not Specified               0.6389776    0.6389776
##   Payday Loan                 0.2375297    0.4750594
##   Personal Loan               0.5636071    0.5636071
##   Student Loan                0.6379585    0.4784689
##
## $Last_Loan_1_Last_Loan_9$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                5.357143            9.259259               10.204082
##   Credit-Builder Loan     10.714286           14.814815               20.408163
##   Debt Consolidation Loan 12.500000           14.814815               10.204082
##   Home Equity Loan        16.071429           18.518519                8.163265
##   Mortgage Loan            7.142857            5.555556                4.081633
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified            7.142857            7.407407               16.326531
##   Payday Loan              5.357143           14.814815                8.163265
##   Personal Loan           14.285714            5.555556               14.285714
##   Student Loan            21.428571            9.259259                8.163265
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                       11.764706     13.846154  9.252787
##   Credit-Builder Loan              5.882353     15.384615  9.693793
##   Debt Consolidation Loan          9.803922     10.769231  9.935097
##   Home Equity Loan                15.686275      7.692308  9.793643
##   Mortgage Loan                    5.882353     10.769231  9.635547
##   No Loan                          0.000000      0.000000 11.865535
##   Not Specified                   11.764706     15.384615  9.910135
##   Payday Loan                     11.764706     15.384615 10.076552
##   Personal Loan                   17.647059      7.692308  9.851889
##   Student Loan                     9.803922      3.076923  9.985022
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    16.981132   15.517241      4.166667    12.500000
##   Credit-Builder Loan          11.320755    8.620690     18.750000     6.250000
##   Debt Consolidation Loan       9.433962    5.172414     12.500000    12.500000
##   Home Equity Loan              9.433962    5.172414      4.166667     4.166667
##   Mortgage Loan                13.207547   17.241379      6.250000     8.333333
##   No Loan                       0.000000    0.000000      0.000000     0.000000
##   Not Specified                 7.547170   15.517241     16.666667    16.666667
##   Payday Loan                   9.433962   12.068966      6.250000    12.500000
##   Personal Loan                11.320755   10.344828     14.583333    14.583333
##   Student Loan                 11.320755   10.344828     16.666667    12.500000
##
##
## $Last_Loan_2_Last_Loan_3
```

```
## $Last_Loan_2_Last_Loan_3$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                    808                 808                     728
##   Credit-Builder Loan          784                1048                     720
##   Debt Consolidation Loan      720                 896                     760
##   Home Equity Loan             752                 880                     792
##   Mortgage Loan                600                 720                     832
##   No Loan                        0                   0                       0
##   Not Specified                688                 824                     872
##   Payday Loan                  936                 720                     688
##   Personal Loan                792                 720                     816
##   Student Loan                 728                 704                     720
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            800           688    1712           768
##   Credit-Builder Loan                  640           664    1784           736
##   Debt Consolidation Loan              768           672    1600           728
##   Home Equity Loan                     720           912    1656           776
##   Mortgage Loan                        592           648    1792           848
##   No Loan                                0             0   22536             0
##   Not Specified                        808           736    1664           832
##   Payday Loan                          840           888    1888           800
##   Personal Loan                        728           872    1840           744
##   Student Loan                         744           792    1776           848
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                      696           736          728
##   Credit-Builder Loan            648           736          768
##   Debt Consolidation Loan        768           648          712
##   Home Equity Loan               736           760          792
##   Mortgage Loan                  824           688          864
##   No Loan                          0             0            0
##   Not Specified                  792           784          728
##   Payday Loan                    832           768          744
##   Personal Loan                  648           656          824
##   Student Loan                   824           632          768
##
## $Last_Loan_2_Last_Loan_3$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    8472                    8528                    8272
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    8776                    8408                   22536
##           Not Specified             Payday Loan           Personal Loan
##                    8728                    9104                    8640
##            Student Loan
##                    8536
##
## $Last_Loan_2_Last_Loan_3$col_totals
##
```

```
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    6808                    7320                    6928
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    6640                    6872                   38248
##           Not Specified             Payday Loan           Personal Loan
##                    7080                    6768                    6408
##            Student Loan
##                    6928
##
## $Last_Loan_2_Last_Loan_3$grand_total
## [1] 100000
##
## $Last_Loan_2_Last_Loan_3$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.808               0.808                   0.728
##   Credit-Builder Loan        0.784               1.048                   0.720
##   Debt Consolidation Loan    0.720               0.896                   0.760
##   Home Equity Loan           0.752               0.880                   0.792
##   Mortgage Loan              0.600               0.720                   0.832
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.688               0.824                   0.872
##   Payday Loan                0.936               0.720                   0.688
##   Personal Loan              0.792               0.720                   0.816
##   Student Loan               0.728               0.704                   0.720
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.800         0.688   1.712         0.768
##   Credit-Builder Loan                0.640         0.664   1.784         0.736
##   Debt Consolidation Loan            0.768         0.672   1.600         0.728
##   Home Equity Loan                   0.720         0.912   1.656         0.776
##   Mortgage Loan                      0.592         0.648   1.792         0.848
##   No Loan                            0.000         0.000  22.536         0.000
##   Not Specified                      0.808         0.736   1.664         0.832
##   Payday Loan                        0.840         0.888   1.888         0.800
##   Personal Loan                      0.728         0.872   1.840         0.744
##   Student Loan                       0.744         0.792   1.776         0.848
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.696         0.736        0.728
##   Credit-Builder Loan          0.648         0.736        0.768
##   Debt Consolidation Loan      0.768         0.648        0.712
##   Home Equity Loan             0.736         0.760        0.792
##   Mortgage Loan                0.824         0.688        0.864
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.792         0.784        0.728
##   Payday Loan                  0.832         0.768        0.744
##   Personal Loan                0.648         0.656        0.824
##   Student Loan                 0.824         0.632        0.768
##
## $Last_Loan_2_Last_Loan_3$row_percentage
```

```
##
##                           Auto Loan Credit-Builder Loan
##   Auto Loan                 9.537299            9.537299
##   Credit-Builder Loan       9.193246           12.288931
##   Debt Consolidation Loan   8.704062           10.831721
##   Home Equity Loan          8.568824           10.027347
##   Mortgage Loan             7.136061            8.563273
##   No Loan                   0.000000            0.000000
##   Not Specified             7.882676            9.440880
##   Payday Loan              10.281195            7.908612
##   Personal Loan             9.166667            8.333333
##   Student Loan              8.528585            8.247423
##
##                          Debt Consolidation Loan Home Equity Loan
##   Auto Loan                               8.593012         9.442871
##   Credit-Builder Loan                     8.442777         7.504690
##   Debt Consolidation Loan                 9.187621         9.284333
##   Home Equity Loan                        9.024613         8.204193
##   Mortgage Loan                           9.895338         7.040913
##   No Loan                                 0.000000         0.000000
##   Not Specified                           9.990834         9.257562
##   Payday Loan                             7.557118         9.226714
##   Personal Loan                           9.444444         8.425926
##   Student Loan                            8.434864         8.716026
##
##                          Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                   8.120869  20.207743      9.065156    8.215297
##   Credit-Builder Loan         7.786116  20.919325      8.630394    7.598499
##   Debt Consolidation Loan     8.123791  19.342360      8.800774    9.284333
##   Home Equity Loan           10.391978  18.869644      8.842297    8.386509
##   Mortgage Loan               7.706946  21.313035     10.085633    9.800190
##   No Loan                     0.000000 100.000000      0.000000    0.000000
##   Not Specified               8.432631  19.065078      9.532539    9.074244
##   Payday Loan                 9.753954  20.738137      8.787346    9.138840
##   Personal Loan              10.092593  21.296296      8.611111    7.500000
##   Student Loan                9.278351  20.805998      9.934396    9.653233
##
##                          Personal Loan Student Loan
##   Auto Loan                   8.687441     8.593012
##   Credit-Builder Loan         8.630394     9.005629
##   Debt Consolidation Loan     7.833656     8.607350
##   Home Equity Loan            8.659982     9.024613
##   Mortgage Loan               8.182683    10.275928
##   No Loan                     0.000000     0.000000
##   Not Specified               8.982585     8.340972
##   Payday Loan                 8.435852     8.172232
##   Personal Loan               7.592593     9.537037
##   Student Loan                7.403936     8.997188
##
## $Last_Loan_2_Last_Loan_3$col_percentage
##
```

```
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               11.868390           11.038251               10.508083
##   Credit-Builder Loan     11.515864           14.316940               10.392610
##   Debt Consolidation Loan 10.575793           12.240437               10.969977
##   Home Equity Loan        11.045828           12.021858               11.431871
##   Mortgage Loan            8.813161            9.836066               12.009238
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           10.105758           11.256831               12.586605
##   Payday Loan             13.748531            9.836066                9.930716
##   Personal Loan           11.633373            9.836066               11.778291
##   Student Loan            10.693302            9.617486               10.392610
##
##                          Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                      12.048193     10.011641  4.476051
##   Credit-Builder Loan             9.638554      9.662398  4.664296
##   Debt Consolidation Loan        11.566265      9.778813  4.183225
##   Home Equity Loan               10.843373     13.271246  4.329638
##   Mortgage Loan                   8.915663      9.429569  4.685212
##   No Loan                         0.000000      0.000000 58.920728
##   Not Specified                  12.168675     10.710128  4.350554
##   Payday Loan                    12.650602     12.922002  4.936206
##   Personal Loan                  10.963855     12.689173  4.810709
##   Student Loan                   11.204819     11.525029  4.643380
##
##                          Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                   10.847458   10.283688     11.485643    10.508083
##   Credit-Builder Loan         10.395480    9.574468     11.485643    11.085450
##   Debt Consolidation Loan     10.282486   11.347518     10.112360    10.277136
##   Home Equity Loan            10.960452   10.874704     11.860175    11.431871
##   Mortgage Loan               11.977401   12.174941     10.736579    12.471132
##   No Loan                      0.000000    0.000000      0.000000     0.000000
##   Not Specified               11.751412   11.702128     12.234707    10.508083
##   Payday Loan                 11.299435   12.293144     11.985019    10.739030
##   Personal Loan               10.508475    9.574468     10.237203    11.893764
##   Student Loan                11.977401   12.174941      9.862672    11.085450
##
##
## $Last_Loan_2_Last_Loan_4
## $Last_Loan_2_Last_Loan_4$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                    648                 576                     552
##   Credit-Builder Loan          520                 680                     640
##   Debt Consolidation Loan      600                 560                     480
##   Home Equity Loan             488                 632                     680
##   Mortgage Loan                448                 648                     640
##   No Loan                        0                   0                       0
##   Not Specified                480                 576                     512
##   Payday Loan                  496                 672                     576
##   Personal Loan                536                 624                     440
##   Student Loan                 504                 696                     512
```

```
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                           552           592    3424           576
##   Credit-Builder Loan                 608           536    3424           464
##   Debt Consolidation Loan             560           480    3392           552
##   Home Equity Loan                    584           600    3528           664
##   Mortgage Loan                       536           560    3456           568
##   No Loan                               0             0   22536             0
##   Not Specified                       640           648    3432           544
##   Payday Loan                         600           584    3608           624
##   Personal Loan                       664           576    3592           544
##   Student Loan                        456           592    3608           584
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                      600           480          472
##   Credit-Builder Loan            488           608          560
##   Debt Consolidation Loan        616           528          504
##   Home Equity Loan               592           456          552
##   Mortgage Loan                  528           528          496
##   No Loan                          0             0            0
##   Not Specified                  720           616          560
##   Payday Loan                    688           608          648
##   Personal Loan                  472           544          648
##   Student Loan                   584           488          512
##
## $Last_Loan_2_Last_Loan_4$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    8472                    8528                    8272
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    8776                    8408                   22536
##           Not Specified             Payday Loan           Personal Loan
##                    8728                    9104                    8640
##            Student Loan
##                    8536
##
## $Last_Loan_2_Last_Loan_4$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    4720                    5664                    5032
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    5200                    5168                   54000
##           Not Specified             Payday Loan           Personal Loan
##                    5120                    5288                    4856
##            Student Loan
##                    4952
##
## $Last_Loan_2_Last_Loan_4$grand_total
## [1] 100000
##
## $Last_Loan_2_Last_Loan_4$cell_percentage
```

```
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.648               0.576                   0.552
##   Credit-Builder Loan        0.520               0.680                   0.640
##   Debt Consolidation Loan    0.600               0.560                   0.480
##   Home Equity Loan           0.488               0.632                   0.680
##   Mortgage Loan              0.448               0.648                   0.640
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.480               0.576                   0.512
##   Payday Loan                0.496               0.672                   0.576
##   Personal Loan              0.536               0.624                   0.440
##   Student Loan               0.504               0.696                   0.512
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.552         0.592   3.424         0.576
##   Credit-Builder Loan                0.608         0.536   3.424         0.464
##   Debt Consolidation Loan            0.560         0.480   3.392         0.552
##   Home Equity Loan                   0.584         0.600   3.528         0.664
##   Mortgage Loan                      0.536         0.560   3.456         0.568
##   No Loan                            0.000         0.000  22.536         0.000
##   Not Specified                      0.640         0.648   3.432         0.544
##   Payday Loan                        0.600         0.584   3.608         0.624
##   Personal Loan                      0.664         0.576   3.592         0.544
##   Student Loan                       0.456         0.592   3.608         0.584
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.600         0.480        0.472
##   Credit-Builder Loan          0.488         0.608        0.560
##   Debt Consolidation Loan      0.616         0.528        0.504
##   Home Equity Loan             0.592         0.456        0.552
##   Mortgage Loan                0.528         0.528        0.496
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.720         0.616        0.560
##   Payday Loan                  0.688         0.608        0.648
##   Personal Loan                0.472         0.544        0.648
##   Student Loan                 0.584         0.488        0.512
##
## $Last_Loan_2_Last_Loan_4$row_percentage
##
##                           Auto Loan Credit-Builder Loan
##   Auto Loan                 7.648725            6.798867
##   Credit-Builder Loan       6.097561            7.973734
##   Debt Consolidation Loan   7.253385            6.769826
##   Home Equity Loan          5.560620            7.201459
##   Mortgage Loan             5.328259            7.706946
##   No Loan                   0.000000            0.000000
##   Not Specified             5.499542            6.599450
##   Payday Loan               5.448155            7.381371
##   Personal Loan             6.203704            7.222222
##   Student Loan              5.904405            8.153702
##
```

```
##                          Debt Consolidation Loan Home Equity Loan
##   Auto Loan                               6.515581         6.515581
##   Credit-Builder Loan                     7.504690         7.129456
##   Debt Consolidation Loan                 5.802708         6.769826
##   Home Equity Loan                        7.748405         6.654512
##   Mortgage Loan                           7.611798         6.374881
##   No Loan                                 0.000000         0.000000
##   Not Specified                           5.866178         7.332722
##   Payday Loan                             6.326889         6.590510
##   Personal Loan                           5.092593         7.685185
##   Student Loan                            5.998126         5.342081
##
##                          Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                   6.987724  40.415486      6.798867    7.082153
##   Credit-Builder Loan         6.285178  40.150094      5.440901    5.722326
##   Debt Consolidation Loan     5.802708  41.005803      6.673114    7.446809
##   Home Equity Loan            6.836828  40.200547      7.566089    6.745670
##   Mortgage Loan               6.660324  41.103711      6.755471    6.279734
##   No Loan                     0.000000 100.000000      0.000000    0.000000
##   Not Specified               7.424381  39.321723      6.232814    8.249313
##   Payday Loan                 6.414763  39.630931      6.854130    7.557118
##   Personal Loan               6.666667  41.574074      6.296296    5.462963
##   Student Loan                6.935333  42.268041      6.841612    6.841612
##
##                          Personal Loan Student Loan
##   Auto Loan                   5.665722     5.571294
##   Credit-Builder Loan         7.129456     6.566604
##   Debt Consolidation Loan     6.382979     6.092843
##   Home Equity Loan            5.195989     6.289881
##   Mortgage Loan               6.279734     5.899144
##   No Loan                     0.000000     0.000000
##   Not Specified               7.057745     6.416132
##   Payday Loan                 6.678383     7.117750
##   Personal Loan               6.296296     7.500000
##   Student Loan                5.716963     5.998126
##
## $Last_Loan_2_Last_Loan_4$col_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               13.728814           10.169492               10.969793
##   Credit-Builder Loan     11.016949           12.005650               12.718601
##   Debt Consolidation Loan 12.711864            9.887006                9.538951
##   Home Equity Loan        10.338983           11.158192               13.513514
##   Mortgage Loan            9.491525           11.440678               12.718601
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           10.169492           10.169492               10.174881
##   Payday Loan             10.508475           11.864407               11.446741
##   Personal Loan           11.355932           11.016949                8.744038
##   Student Loan            10.677966           12.288136               10.174881
##
##                          Home Equity Loan Mortgage Loan   No Loan
```

```
##   Auto Loan                      10.615385     11.455108  6.340741
##   Credit-Builder Loan            11.692308     10.371517  6.340741
##   Debt Consolidation Loan        10.769231      9.287926  6.281481
##   Home Equity Loan               11.230769     11.609907  6.533333
##   Mortgage Loan                  10.307692     10.835913  6.400000
##   No Loan                         0.000000      0.000000 41.733333
##   Not Specified                  12.307692     12.538700  6.355556
##   Payday Loan                    11.538462     11.300310  6.681481
##   Personal Loan                  12.769231     11.145511  6.651852
##   Student Loan                    8.769231     11.455108  6.681481
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    11.250000   11.346445      9.884679     9.531502
##   Credit-Builder Loan           9.062500    9.228442     12.520593    11.308562
##   Debt Consolidation Loan      10.781250   11.649017     10.873147    10.177706
##   Home Equity Loan             12.968750   11.195159      9.390445    11.147011
##   Mortgage Loan                11.093750    9.984871     10.873147    10.016155
##   No Loan                       0.000000    0.000000      0.000000     0.000000
##   Not Specified                10.625000   13.615734     12.685338    11.308562
##   Payday Loan                  12.187500   13.010590     12.520593    13.085622
##   Personal Loan                10.625000    8.925870     11.202636    13.085622
##   Student Loan                 11.406250   11.043873     10.049423    10.339257
##
##
## $Last_Loan_2_Last_Loan_5
## $Last_Loan_2_Last_Loan_5$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     384                 416                     352
##   Credit-Builder Loan           448                 408                     432
##   Debt Consolidation Loan       352                 336                     392
##   Home Equity Loan              384                 360                     464
##   Mortgage Loan                 328                 384                     360
##   No Loan                         0                   0                       0
##   Not Specified                 344                 336                     400
##   Payday Loan                   384                 504                     416
##   Personal Loan                 400                 464                     376
##   Student Loan                  400                 288                     336
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            416           440    5152           216
##   Credit-Builder Loan                  304           352    5040           384
##   Debt Consolidation Loan              264           368    5024           408
##   Home Equity Loan                     440           432    5248           376
##   Mortgage Loan                        344           368    5088           392
##   No Loan                                0             0   22536             0
##   Not Specified                        520           360    5344           376
##   Payday Loan                          432           344    5408           400
##   Personal Loan                        312           336    5360           352
##   Student Loan                         368           440    5256           368
##
```

```
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                      384           368          344
##   Credit-Builder Loan            336           440          384
##   Debt Consolidation Loan        328           456          344
##   Home Equity Loan               384           328          360
##   Mortgage Loan                  432           352          360
##   No Loan                          0             0            0
##   Not Specified                  408           336          304
##   Payday Loan                    400           408          408
##   Personal Loan                  392           384          264
##   Student Loan                   280           448          352
##
## $Last_Loan_2_Last_Loan_5$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    8472                    8528                    8272
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    8776                    8408                   22536
##           Not Specified             Payday Loan           Personal Loan
##                    8728                    9104                    8640
##            Student Loan
##                    8536
##
## $Last_Loan_2_Last_Loan_5$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    3424                    3496                    3528
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    3400                    3440                   69456
##           Not Specified             Payday Loan           Personal Loan
##                    3272                    3344                    3520
##            Student Loan
##                    3120
##
## $Last_Loan_2_Last_Loan_5$grand_total
## [1] 100000
##
## $Last_Loan_2_Last_Loan_5$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.384               0.416                   0.352
##   Credit-Builder Loan        0.448               0.408                   0.432
##   Debt Consolidation Loan    0.352               0.336                   0.392
##   Home Equity Loan           0.384               0.360                   0.464
##   Mortgage Loan              0.328               0.384                   0.360
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.344               0.336                   0.400
##   Payday Loan                0.384               0.504                   0.416
##   Personal Loan              0.400               0.464                   0.376
##   Student Loan               0.400               0.288                   0.336
##
```

```
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.416         0.440   5.152         0.216
##   Credit-Builder Loan                0.304         0.352   5.040         0.384
##   Debt Consolidation Loan            0.264         0.368   5.024         0.408
##   Home Equity Loan                   0.440         0.432   5.248         0.376
##   Mortgage Loan                      0.344         0.368   5.088         0.392
##   No Loan                            0.000         0.000  22.536         0.000
##   Not Specified                      0.520         0.360   5.344         0.376
##   Payday Loan                        0.432         0.344   5.408         0.400
##   Personal Loan                      0.312         0.336   5.360         0.352
##   Student Loan                       0.368         0.440   5.256         0.368
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.384         0.368        0.344
##   Credit-Builder Loan          0.336         0.440        0.384
##   Debt Consolidation Loan      0.328         0.456        0.344
##   Home Equity Loan             0.384         0.328        0.360
##   Mortgage Loan                0.432         0.352        0.360
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.408         0.336        0.304
##   Payday Loan                  0.400         0.408        0.408
##   Personal Loan                0.392         0.384        0.264
##   Student Loan                 0.280         0.448        0.352
##
## $Last_Loan_2_Last_Loan_5$row_percentage
##
##                           Auto Loan Credit-Builder Loan
##   Auto Loan                4.532578            4.910293
##   Credit-Builder Loan      5.253283            4.784240
##   Debt Consolidation Loan  4.255319            4.061896
##   Home Equity Loan         4.375570            4.102097
##   Mortgage Loan            3.901047            4.567079
##   No Loan                  0.000000            0.000000
##   Not Specified            3.941338            3.849679
##   Payday Loan              4.217926            5.536028
##   Personal Loan            4.629630            5.370370
##   Student Loan             4.686036            3.373946
##
##                          Debt Consolidation Loan Home Equity Loan
##   Auto Loan                               4.154863         4.910293
##   Credit-Builder Loan                     5.065666         3.564728
##   Debt Consolidation Loan                 4.738878         3.191489
##   Home Equity Loan                        5.287147         5.013674
##   Mortgage Loan                           4.281637         4.091342
##   No Loan                                 0.000000         0.000000
##   Not Specified                           4.582951         5.957837
##   Payday Loan                             4.569420         4.745167
##   Personal Loan                           4.351852         3.611111
##   Student Loan                            3.936270         4.311153
##
##                          Mortgage Loan    No Loan Not Specified Payday Loan
```

```
##   Auto Loan                    5.193579  60.812087      2.549575    4.532578
##   Credit-Builder Loan          4.127580  59.099437      4.502814    3.939962
##   Debt Consolidation Loan      4.448743  60.735010      4.932302    3.965184
##   Home Equity Loan             4.922516  59.799453      4.284412    4.375570
##   Mortgage Loan                4.376784  60.513796      4.662226    5.137964
##   No Loan                      0.000000 100.000000      0.000000    0.000000
##   Not Specified                4.124656  61.228231      4.307974    4.674610
##   Payday Loan                  3.778559  59.402460      4.393673    4.393673
##   Personal Loan                3.888889  62.037037      4.074074    4.537037
##   Student Loan                 5.154639  61.574508      4.311153    3.280225
##
##                           Personal Loan Student Loan
##   Auto Loan                    4.343720     4.060434
##   Credit-Builder Loan          5.159475     4.502814
##   Debt Consolidation Loan      5.512573     4.158607
##   Home Equity Loan             3.737466     4.102097
##   Mortgage Loan                4.186489     4.281637
##   No Loan                      0.000000     0.000000
##   Not Specified                3.849679     3.483043
##   Payday Loan                  4.481547     4.481547
##   Personal Loan                4.444444     3.055556
##   Student Loan                 5.248360     4.123711
##
## $Last_Loan_2_Last_Loan_5$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               11.214953           11.899314                9.977324
##   Credit-Builder Loan     13.084112           11.670481               12.244898
##   Debt Consolidation Loan 10.280374            9.610984               11.111111
##   Home Equity Loan        11.214953           10.297483               13.151927
##   Mortgage Loan            9.579439           10.983982               10.204082
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           10.046729            9.610984               11.337868
##   Payday Loan             11.214953           14.416476               11.791383
##   Personal Loan           11.682243           13.272311               10.657596
##   Student Loan            11.682243            8.237986                9.523810
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                       12.235294     12.790698  7.417646
##   Credit-Builder Loan              8.941176     10.232558  7.256393
##   Debt Consolidation Loan          7.764706     10.697674  7.233356
##   Home Equity Loan                12.941176     12.558140  7.555863
##   Mortgage Loan                   10.117647     10.697674  7.325501
##   No Loan                          0.000000      0.000000 32.446441
##   Not Specified                   15.294118     10.465116  7.694080
##   Payday Loan                     12.705882     10.000000  7.786224
##   Personal Loan                    9.176471      9.767442  7.717116
##   Student Loan                    10.823529     12.790698  7.567381
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    6.601467   11.483254     10.454545    11.025641
```

```
##   Credit-Builder Loan         11.735941   10.047847     12.500000    12.307692
##   Debt Consolidation Loan     12.469438    9.808612     12.954545    11.025641
##   Home Equity Loan            11.491443   11.483254      9.318182    11.538462
##   Mortgage Loan               11.980440   12.918660     10.000000    11.538462
##   No Loan                      0.000000    0.000000      0.000000     0.000000
##   Not Specified               11.491443   12.200957      9.545455     9.743590
##   Payday Loan                 12.224939   11.961722     11.590909    13.076923
##   Personal Loan               10.757946   11.722488     10.909091     8.461538
##   Student Loan                11.246944    8.373206     12.727273    11.282051
##
##
## $Last_Loan_2_Last_Loan_6
## $Last_Loan_2_Last_Loan_6$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     184                 272                     256
##   Credit-Builder Loan           320                 304                     304
##   Debt Consolidation Loan       232                 256                     280
##   Home Equity Loan              376                 264                     280
##   Mortgage Loan                 232                 328                     256
##   No Loan                         0                   0                       0
##   Not Specified                 192                 224                     264
##   Payday Loan                   328                 320                     280
##   Personal Loan                 328                 280                     224
##   Student Loan                  280                 360                     320
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            216           256    5960           312
##   Credit-Builder Loan                  248           304    5968           208
##   Debt Consolidation Loan              312           280    5880           272
##   Home Equity Loan                     304           272    6072           232
##   Mortgage Loan                        320           240    5880           352
##   No Loan                                0             0   22536             0
##   Not Specified                        264           360    6136           336
##   Payday Loan                          248           248    6384           312
##   Personal Loan                        288           232    6128           304
##   Student Loan                         264           200    6040           296
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                       328           304          384
##   Credit-Builder Loan             296           272          304
##   Debt Consolidation Loan         248           240          272
##   Home Equity Loan                320           336          320
##   Mortgage Loan                   256           320          224
##   No Loan                           0             0            0
##   Not Specified                   320           320          312
##   Payday Loan                     368           304          312
##   Personal Loan                   272           344          240
##   Student Loan                    280           296          200
##
## $Last_Loan_2_Last_Loan_6$row_totals
```

```
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    8472                    8528                    8272
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    8776                    8408                   22536
##           Not Specified             Payday Loan           Personal Loan
##                    8728                    9104                    8640
##            Student Loan
##                    8536
##
## $Last_Loan_2_Last_Loan_6$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    2472                    2608                    2464
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    2464                    2392                   76984
##           Not Specified             Payday Loan           Personal Loan
##                    2624                    2688                    2736
##            Student Loan
##                    2568
##
## $Last_Loan_2_Last_Loan_6$grand_total
## [1] 100000
##
## $Last_Loan_2_Last_Loan_6$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.184               0.272                   0.256
##   Credit-Builder Loan        0.320               0.304                   0.304
##   Debt Consolidation Loan    0.232               0.256                   0.280
##   Home Equity Loan           0.376               0.264                   0.280
##   Mortgage Loan              0.232               0.328                   0.256
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.192               0.224                   0.264
##   Payday Loan                0.328               0.320                   0.280
##   Personal Loan              0.328               0.280                   0.224
##   Student Loan               0.280               0.360                   0.320
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.216         0.256   5.960         0.312
##   Credit-Builder Loan                0.248         0.304   5.968         0.208
##   Debt Consolidation Loan            0.312         0.280   5.880         0.272
##   Home Equity Loan                   0.304         0.272   6.072         0.232
##   Mortgage Loan                      0.320         0.240   5.880         0.352
##   No Loan                            0.000         0.000  22.536         0.000
##   Not Specified                      0.264         0.360   6.136         0.336
##   Payday Loan                        0.248         0.248   6.384         0.312
##   Personal Loan                      0.288         0.232   6.128         0.304
##   Student Loan                       0.264         0.200   6.040         0.296
##
##                          Payday Loan Personal Loan Student Loan
```

```
##   Auto Loan                     0.328         0.304        0.384
##   Credit-Builder Loan           0.296         0.272        0.304
##   Debt Consolidation Loan       0.248         0.240        0.272
##   Home Equity Loan              0.320         0.336        0.320
##   Mortgage Loan                 0.256         0.320        0.224
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.320         0.320        0.312
##   Payday Loan                   0.368         0.304        0.312
##   Personal Loan                 0.272         0.344        0.240
##   Student Loan                  0.280         0.296        0.200
##
## $Last_Loan_2_Last_Loan_6$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 2.171860            3.210576
##   Credit-Builder Loan       3.752345            3.564728
##   Debt Consolidation Loan   2.804642            3.094778
##   Home Equity Loan          4.284412            3.008204
##   Mortgage Loan             2.759277            3.901047
##   No Loan                   0.000000            0.000000
##   Not Specified             2.199817            2.566453
##   Payday Loan               3.602812            3.514938
##   Personal Loan             3.796296            3.240741
##   Student Loan              3.280225            4.217432
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                3.021719         2.549575
##   Credit-Builder Loan                      3.564728         2.908068
##   Debt Consolidation Loan                  3.384913         3.771760
##   Home Equity Loan                         3.190520         3.463993
##   Mortgage Loan                            3.044719         3.805899
##   No Loan                                  0.000000         0.000000
##   Not Specified                            3.024748         3.024748
##   Payday Loan                              3.075571         2.724077
##   Personal Loan                            2.592593         3.333333
##   Student Loan                             3.748828         3.092784
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                     3.021719  70.349386      3.682720    3.871577
##   Credit-Builder Loan           3.564728  69.981238      2.439024    3.470919
##   Debt Consolidation Loan       3.384913  71.083172      3.288201    2.998066
##   Home Equity Loan              3.099362  69.188696      2.643573    3.646308
##   Mortgage Loan                 2.854424  69.933397      4.186489    3.044719
##   No Loan                       0.000000 100.000000      0.000000    0.000000
##   Not Specified                 4.124656  70.302475      3.849679    3.666361
##   Payday Loan                   2.724077  70.123023      3.427065    4.042179
##   Personal Loan                 2.685185  70.925926      3.518519    3.148148
##   Student Loan                  2.343018  70.759138      3.467666    3.280225
##
##                           Personal Loan Student Loan
##   Auto Loan                     3.588291     4.532578
```

```
##   Credit-Builder Loan          3.189493     3.564728
##   Debt Consolidation Loan      2.901354     3.288201
##   Home Equity Loan             3.828624     3.646308
##   Mortgage Loan                3.805899     2.664129
##   No Loan                      0.000000     0.000000
##   Not Specified                3.666361     3.574702
##   Payday Loan                  3.339192     3.427065
##   Personal Loan                3.981481     2.777778
##   Student Loan                 3.467666     2.343018
##
## $Last_Loan_2_Last_Loan_6$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                7.443366           10.429448               10.389610
##   Credit-Builder Loan     12.944984           11.656442               12.337662
##   Debt Consolidation Loan  9.385113            9.815951               11.363636
##   Home Equity Loan        15.210356           10.122699               11.363636
##   Mortgage Loan            9.385113           12.576687               10.389610
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified            7.766990            8.588957               10.714286
##   Payday Loan             13.268608           12.269939               11.363636
##   Personal Loan           13.268608           10.736196                9.090909
##   Student Loan            11.326861           13.803681               12.987013
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                        8.766234     10.702341  7.741868
##   Credit-Builder Loan             10.064935     12.709030  7.752260
##   Debt Consolidation Loan         12.662338     11.705686  7.637951
##   Home Equity Loan                12.337662     11.371237  7.887353
##   Mortgage Loan                   12.987013     10.033445  7.637951
##   No Loan                          0.000000      0.000000 29.273615
##   Not Specified                   10.714286     15.050167  7.970487
##   Payday Loan                     10.064935     10.367893  8.292632
##   Personal Loan                   11.688312      9.698997  7.960096
##   Student Loan                    10.714286      8.361204  7.845786
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    11.890244   12.202381     11.111111    14.953271
##   Credit-Builder Loan           7.926829   11.011905      9.941520    11.838006
##   Debt Consolidation Loan      10.365854    9.226190      8.771930    10.591900
##   Home Equity Loan              8.841463   11.904762     12.280702    12.461059
##   Mortgage Loan                13.414634    9.523810     11.695906     8.722741
##   No Loan                       0.000000    0.000000      0.000000     0.000000
##   Not Specified                12.804878   11.904762     11.695906    12.149533
##   Payday Loan                  11.890244   13.690476     11.111111    12.149533
##   Personal Loan                11.585366   10.119048     12.573099     9.345794
##   Student Loan                 11.280488   10.416667     10.818713     7.788162
##
##
## $Last_Loan_2_Last_Loan_7
## $Last_Loan_2_Last_Loan_7$contingency_table
```

```
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                    240                 192                     112
##   Credit-Builder Loan          184                 152                     160
##   Debt Consolidation Loan      208                 168                     168
##   Home Equity Loan             104                 224                     192
##   Mortgage Loan                160                 216                     168
##   No Loan                        0                   0                       0
##   Not Specified                176                 192                     192
##   Payday Loan                  176                 216                     160
##   Personal Loan                208                 192                     160
##   Student Loan                 144                 168                     120
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            296           208    6784           120
##   Credit-Builder Loan                  184           200    6888           128
##   Debt Consolidation Loan              112           168    6664           216
##   Home Equity Loan                     160           232    7040           176
##   Mortgage Loan                        136           232    6752           176
##   No Loan                                0             0   22536             0
##   Not Specified                        104           216    7008           224
##   Payday Loan                          224           176    7432           192
##   Personal Loan                        160           192    7008           136
##   Student Loan                         152           168    7016           240
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                      232           152          136
##   Credit-Builder Loan            240           192          200
##   Debt Consolidation Loan        240           152          176
##   Home Equity Loan               232           216          200
##   Mortgage Loan                  240           192          136
##   No Loan                          0             0            0
##   Not Specified                  248           192          176
##   Payday Loan                    144           216          168
##   Personal Loan                  208           168          208
##   Student Loan                   136           208          184
##
## $Last_Loan_2_Last_Loan_7$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    8472                    8528                    8272
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    8776                    8408                   22536
##           Not Specified             Payday Loan           Personal Loan
##                    8728                    9104                    8640
##            Student Loan
##                    8536
##
## $Last_Loan_2_Last_Loan_7$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
```

```
##                    1600                    1720                    1432
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    1528                    1792                   85128
##           Not Specified             Payday Loan           Personal Loan
##                    1608                    1920                    1688
##            Student Loan
##                    1584
##
## $Last_Loan_2_Last_Loan_7$grand_total
## [1] 100000
##
## $Last_Loan_2_Last_Loan_7$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.240               0.192                   0.112
##   Credit-Builder Loan        0.184               0.152                   0.160
##   Debt Consolidation Loan    0.208               0.168                   0.168
##   Home Equity Loan           0.104               0.224                   0.192
##   Mortgage Loan              0.160               0.216                   0.168
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.176               0.192                   0.192
##   Payday Loan                0.176               0.216                   0.160
##   Personal Loan              0.208               0.192                   0.160
##   Student Loan               0.144               0.168                   0.120
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.296         0.208   6.784         0.120
##   Credit-Builder Loan                0.184         0.200   6.888         0.128
##   Debt Consolidation Loan            0.112         0.168   6.664         0.216
##   Home Equity Loan                   0.160         0.232   7.040         0.176
##   Mortgage Loan                      0.136         0.232   6.752         0.176
##   No Loan                            0.000         0.000  22.536         0.000
##   Not Specified                      0.104         0.216   7.008         0.224
##   Payday Loan                        0.224         0.176   7.432         0.192
##   Personal Loan                      0.160         0.192   7.008         0.136
##   Student Loan                       0.152         0.168   7.016         0.240
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.232         0.152        0.136
##   Credit-Builder Loan          0.240         0.192        0.200
##   Debt Consolidation Loan      0.240         0.152        0.176
##   Home Equity Loan             0.232         0.216        0.200
##   Mortgage Loan                0.240         0.192        0.136
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.248         0.192        0.176
##   Payday Loan                  0.144         0.216        0.168
##   Personal Loan                0.208         0.168        0.208
##   Student Loan                 0.136         0.208        0.184
##
## $Last_Loan_2_Last_Loan_7$row_percentage
##
```

```
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 2.832861            2.266289
##   Credit-Builder Loan       2.157598            1.782364
##   Debt Consolidation Loan   2.514507            2.030948
##   Home Equity Loan          1.185050            2.552416
##   Mortgage Loan             1.902950            2.568982
##   No Loan                   0.000000            0.000000
##   Not Specified             2.016499            2.199817
##   Payday Loan               1.933216            2.372583
##   Personal Loan             2.407407            2.222222
##   Student Loan              1.686973            1.968135
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                1.322002         3.493862
##   Credit-Builder Loan                      1.876173         2.157598
##   Debt Consolidation Loan                  2.030948         1.353965
##   Home Equity Loan                         2.187785         1.823154
##   Mortgage Loan                            1.998097         1.617507
##   No Loan                                  0.000000         0.000000
##   Not Specified                            2.199817         1.191567
##   Payday Loan                              1.757469         2.460457
##   Personal Loan                            1.851852         1.851852
##   Student Loan                             1.405811         1.780694
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                     2.455146  80.075543      1.416431    2.738432
##   Credit-Builder Loan           2.345216  80.769231      1.500938    2.814259
##   Debt Consolidation Loan       2.030948  80.560928      2.611219    2.901354
##   Home Equity Loan              2.643573  80.218778      2.005469    2.643573
##   Mortgage Loan                 2.759277  80.304472      2.093245    2.854424
##   No Loan                       0.000000 100.000000      0.000000    0.000000
##   Not Specified                 2.474794  80.293309      2.566453    2.841430
##   Payday Loan                   1.933216  81.634446      2.108963    1.581722
##   Personal Loan                 2.222222  81.111111      1.574074    2.407407
##   Student Loan                  1.968135  82.193065      2.811621    1.593252
##
##                           Personal Loan Student Loan
##   Auto Loan                     1.794145     1.605288
##   Credit-Builder Loan           2.251407     2.345216
##   Debt Consolidation Loan       1.837524     2.127660
##   Home Equity Loan              2.461258     2.278943
##   Mortgage Loan                 2.283539     1.617507
##   No Loan                       0.000000     0.000000
##   Not Specified                 2.199817     2.016499
##   Payday Loan                   2.372583     1.845343
##   Personal Loan                 1.944444     2.407407
##   Student Loan                  2.436739     2.155576
##
## $Last_Loan_2_Last_Loan_7$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
```

```
##   Auto Loan               15.000000           11.162791                7.821229
##   Credit-Builder Loan     11.500000            8.837209               11.173184
##   Debt Consolidation Loan 13.000000            9.767442               11.731844
##   Home Equity Loan         6.500000           13.023256               13.407821
##   Mortgage Loan           10.000000           12.558140               11.731844
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           11.000000           11.162791               13.407821
##   Payday Loan             11.000000           12.558140               11.173184
##   Personal Loan           13.000000           11.162791               11.173184
##   Student Loan             9.000000            9.767442                8.379888
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                      19.371728     11.607143  7.969176
##   Credit-Builder Loan            12.041885     11.160714  8.091345
##   Debt Consolidation Loan         7.329843      9.375000  7.828212
##   Home Equity Loan               10.471204     12.946429  8.269899
##   Mortgage Loan                   8.900524     12.946429  7.931585
##   No Loan                         0.000000      0.000000 26.473076
##   Not Specified                   6.806283     12.053571  8.232309
##   Payday Loan                    14.659686      9.821429  8.730382
##   Personal Loan                  10.471204     10.714286  8.232309
##   Student Loan                    9.947644      9.375000  8.241707
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                      7.462687   12.083333      9.004739     8.585859
##   Credit-Builder Loan            7.960199   12.500000     11.374408    12.626263
##   Debt Consolidation Loan       13.432836   12.500000      9.004739    11.111111
##   Home Equity Loan              10.945274   12.083333     12.796209    12.626263
##   Mortgage Loan                 10.945274   12.500000     11.374408     8.585859
##   No Loan                        0.000000    0.000000      0.000000     0.000000
##   Not Specified                 13.930348   12.916667     11.374408    11.111111
##   Payday Loan                   11.940299    7.500000     12.796209    10.606061
##   Personal Loan                  8.457711   10.833333      9.952607    13.131313
##   Student Loan                  14.925373    7.083333     12.322275    11.616162
##
##
## $Last_Loan_2_Last_Loan_8
## $Last_Loan_2_Last_Loan_8$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     120                 104                      80
##   Credit-Builder Loan            64                  40                      80
##   Debt Consolidation Loan        64                 136                      96
##   Home Equity Loan               64                 144                     104
##   Mortgage Loan                  88                  88                      80
##   No Loan                         0                   0                       0
##   Not Specified                  64                  80                      80
##   Payday Loan                    88                 136                      64
##   Personal Loan                  64                 104                     104
##   Student Loan                  104                  40                      72
##
```

```
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                           136            64    7624           136
##   Credit-Builder Loan                 120           104    7720            72
##   Debt Consolidation Loan             136            32    7472            88
##   Home Equity Loan                     88           136    7872            56
##   Mortgage Loan                       120            72    7576           104
##   No Loan                               0             0   22536             0
##   Not Specified                        64            48    8024           104
##   Payday Loan                          72           144    8280            64
##   Personal Loan                       104            64    7880            40
##   Student Loan                         48            72    7824            80
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                       64            48           96
##   Credit-Builder Loan            152            88           88
##   Debt Consolidation Loan        128            32           88
##   Home Equity Loan                88           136           88
##   Mortgage Loan                   96            88           96
##   No Loan                          0             0            0
##   Not Specified                   64           104           96
##   Payday Loan                     72            56          128
##   Personal Loan                  128            64           88
##   Student Loan                    96           104           96
##
## $Last_Loan_2_Last_Loan_8$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    8472                    8528                    8272
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    8776                    8408                   22536
##           Not Specified             Payday Loan           Personal Loan
##                    8728                    9104                    8640
##            Student Loan
##                    8536
##
## $Last_Loan_2_Last_Loan_8$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     720                     872                     760
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     888                     736                   92808
##           Not Specified             Payday Loan           Personal Loan
##                     744                     888                     720
##            Student Loan
##                     864
##
## $Last_Loan_2_Last_Loan_8$grand_total
## [1] 100000
##
## $Last_Loan_2_Last_Loan_8$cell_percentage
##
```

```
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.120               0.104                   0.080
##   Credit-Builder Loan        0.064               0.040                   0.080
##   Debt Consolidation Loan    0.064               0.136                   0.096
##   Home Equity Loan           0.064               0.144                   0.104
##   Mortgage Loan              0.088               0.088                   0.080
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.064               0.080                   0.080
##   Payday Loan                0.088               0.136                   0.064
##   Personal Loan              0.064               0.104                   0.104
##   Student Loan               0.104               0.040                   0.072
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.136         0.064   7.624         0.136
##   Credit-Builder Loan                0.120         0.104   7.720         0.072
##   Debt Consolidation Loan            0.136         0.032   7.472         0.088
##   Home Equity Loan                   0.088         0.136   7.872         0.056
##   Mortgage Loan                      0.120         0.072   7.576         0.104
##   No Loan                            0.000         0.000  22.536         0.000
##   Not Specified                      0.064         0.048   8.024         0.104
##   Payday Loan                        0.072         0.144   8.280         0.064
##   Personal Loan                      0.104         0.064   7.880         0.040
##   Student Loan                       0.048         0.072   7.824         0.080
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.064         0.048        0.096
##   Credit-Builder Loan          0.152         0.088        0.088
##   Debt Consolidation Loan      0.128         0.032        0.088
##   Home Equity Loan             0.088         0.136        0.088
##   Mortgage Loan                0.096         0.088        0.096
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.064         0.104        0.096
##   Payday Loan                  0.072         0.056        0.128
##   Personal Loan                0.128         0.064        0.088
##   Student Loan                 0.096         0.104        0.096
##
## $Last_Loan_2_Last_Loan_8$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                1.4164306           1.2275732
##   Credit-Builder Loan      0.7504690           0.4690432
##   Debt Consolidation Loan  0.7736944           1.6441006
##   Home Equity Loan         0.7292616           1.6408387
##   Mortgage Loan            1.0466223           1.0466223
##   No Loan                  0.0000000           0.0000000
##   Not Specified            0.7332722           0.9165903
##   Payday Loan              0.9666081           1.4938489
##   Personal Loan            0.7407407           1.2037037
##   Student Loan             1.2183693           0.4686036
##
##                          Debt Consolidation Loan Home Equity Loan
```

```
##   Auto Loan                              0.9442871        1.6052880
##   Credit-Builder Loan                    0.9380863        1.4071295
##   Debt Consolidation Loan                1.1605416        1.6441006
##   Home Equity Loan                       1.1850501        1.0027347
##   Mortgage Loan                          0.9514748        1.4272122
##   No Loan                                0.0000000        0.0000000
##   Not Specified                          0.9165903        0.7332722
##   Payday Loan                            0.7029877        0.7908612
##   Personal Loan                          1.2037037        1.2037037
##   Student Loan                           0.8434864        0.5623243
##
##                           Mortgage Loan     No Loan Not Specified Payday Loan
##   Auto Loan                   0.7554297  89.9905571     1.6052880   0.7554297
##   Credit-Builder Loan         1.2195122  90.5253283     0.8442777   1.7823640
##   Debt Consolidation Loan     0.3868472  90.3288201     1.0638298   1.5473888
##   Home Equity Loan            1.5496809  89.6991796     0.6381039   1.0027347
##   Mortgage Loan               0.8563273  90.1046622     1.2369172   1.1417697
##   No Loan                     0.0000000 100.0000000     0.0000000   0.0000000
##   Not Specified               0.5499542  91.9340055     1.1915674   0.7332722
##   Payday Loan                 1.5817223  90.9490334     0.7029877   0.7908612
##   Personal Loan               0.7407407  91.2037037     0.4629630   1.4814815
##   Student Loan                0.8434864  91.6588566     0.9372071   1.1246485
##
##                           Personal Loan Student Loan
##   Auto Loan                   0.5665722    1.1331445
##   Credit-Builder Loan         1.0318949    1.0318949
##   Debt Consolidation Loan     0.3868472    1.0638298
##   Home Equity Loan            1.5496809    1.0027347
##   Mortgage Loan               1.0466223    1.1417697
##   No Loan                     0.0000000    0.0000000
##   Not Specified               1.1915674    1.0999083
##   Payday Loan                 0.6151142    1.4059754
##   Personal Loan               0.7407407    1.0185185
##   Student Loan                1.2183693    1.1246485
##
## $Last_Loan_2_Last_Loan_8$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               16.666667           11.926606               10.526316
##   Credit-Builder Loan      8.888889            4.587156               10.526316
##   Debt Consolidation Loan  8.888889           15.596330               12.631579
##   Home Equity Loan         8.888889           16.513761               13.684211
##   Mortgage Loan           12.222222           10.091743               10.526316
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified            8.888889            9.174312               10.526316
##   Payday Loan             12.222222           15.596330                8.421053
##   Personal Loan            8.888889           11.926606               13.684211
##   Student Loan            14.444444            4.587156                9.473684
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                       15.315315      8.695652  8.214809
```

```
##   Credit-Builder Loan           13.513514     14.130435  8.318248
##   Debt Consolidation Loan       15.315315      4.347826  8.051030
##   Home Equity Loan               9.909910     18.478261  8.482027
##   Mortgage Loan                 13.513514      9.782609  8.163089
##   No Loan                        0.000000      0.000000 24.282389
##   Not Specified                  7.207207      6.521739  8.645806
##   Payday Loan                    8.108108     19.565217  8.921645
##   Personal Loan                 11.711712      8.695652  8.490647
##   Student Loan                   5.405405      9.782609  8.430308
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    18.279570    7.207207      6.666667    11.111111
##   Credit-Builder Loan           9.677419   17.117117     12.222222    10.185185
##   Debt Consolidation Loan      11.827957   14.414414      4.444444    10.185185
##   Home Equity Loan              7.526882    9.909910     18.888889    10.185185
##   Mortgage Loan                13.978495   10.810811     12.222222    11.111111
##   No Loan                       0.000000    0.000000      0.000000     0.000000
##   Not Specified                13.978495    7.207207     14.444444    11.111111
##   Payday Loan                   8.602151    8.108108      7.777778    14.814815
##   Personal Loan                 5.376344   14.414414      8.888889    10.185185
##   Student Loan                 10.752688   10.810811     14.444444    11.111111
##
##
## $Last_Loan_2_Last_Loan_9
## $Last_Loan_2_Last_Loan_9$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                      48                  24                      40
##   Credit-Builder Loan            40                  24                      32
##   Debt Consolidation Loan        32                  32                      72
##   Home Equity Loan               48                  72                      40
##   Mortgage Loan                  56                  72                      32
##   No Loan                         0                   0                       0
##   Not Specified                  24                  48                      40
##   Payday Loan                    64                  32                      80
##   Personal Loan                  88                  56                      24
##   Student Loan                   48                  72                      32
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                             48            80    7992            48
##   Credit-Builder Loan                   64            64    8112            64
##   Debt Consolidation Loan               32            48    7864            32
##   Home Equity Loan                      48            56    8280            72
##   Mortgage Loan                         56            56    7952            56
##   No Loan                                0             0   22536             0
##   Not Specified                         40            80    8376            24
##   Payday Loan                           32            64    8664            48
##   Personal Loan                         40            48    8216            32
##   Student Loan                          48            24    8152            48
##
##                           Payday Loan Personal Loan Student Loan
```

```
##   Auto Loan                        64            48           80
##   Credit-Builder Loan              80            32           16
##   Debt Consolidation Loan          48            48           64
##   Home Equity Loan                 56            56           48
##   Mortgage Loan                    40            48           40
##   No Loan                           0             0            0
##   Not Specified                    40            32           24
##   Payday Loan                      40            32           48
##   Personal Loan                    64            48           24
##   Student Loan                     32            40           40
##
## $Last_Loan_2_Last_Loan_9$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    8472                    8528                    8272
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    8776                    8408                   22536
##           Not Specified             Payday Loan           Personal Loan
##                    8728                    9104                    8640
##            Student Loan
##                    8536
##
## $Last_Loan_2_Last_Loan_9$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     448                     432                     392
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     408                     520                   96144
##           Not Specified             Payday Loan           Personal Loan
##                     424                     464                     384
##            Student Loan
##                     384
##
## $Last_Loan_2_Last_Loan_9$grand_total
## [1] 100000
##
## $Last_Loan_2_Last_Loan_9$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.048               0.024                   0.040
##   Credit-Builder Loan        0.040               0.024                   0.032
##   Debt Consolidation Loan    0.032               0.032                   0.072
##   Home Equity Loan           0.048               0.072                   0.040
##   Mortgage Loan              0.056               0.072                   0.032
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.024               0.048                   0.040
##   Payday Loan                0.064               0.032                   0.080
##   Personal Loan              0.088               0.056                   0.024
##   Student Loan               0.048               0.072                   0.032
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
```

```
##   Auto Loan                          0.048         0.080   7.992         0.048
##   Credit-Builder Loan                0.064         0.064   8.112         0.064
##   Debt Consolidation Loan            0.032         0.048   7.864         0.032
##   Home Equity Loan                   0.048         0.056   8.280         0.072
##   Mortgage Loan                      0.056         0.056   7.952         0.056
##   No Loan                            0.000         0.000  22.536         0.000
##   Not Specified                      0.040         0.080   8.376         0.024
##   Payday Loan                        0.032         0.064   8.664         0.048
##   Personal Loan                      0.040         0.048   8.216         0.032
##   Student Loan                       0.048         0.024   8.152         0.048
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                     0.064         0.048        0.080
##   Credit-Builder Loan           0.080         0.032        0.016
##   Debt Consolidation Loan       0.048         0.048        0.064
##   Home Equity Loan              0.056         0.056        0.048
##   Mortgage Loan                 0.040         0.048        0.040
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.040         0.032        0.024
##   Payday Loan                   0.040         0.032        0.048
##   Personal Loan                 0.064         0.048        0.024
##   Student Loan                  0.032         0.040        0.040
##
## $Last_Loan_2_Last_Loan_9$row_percentage
##
##                             Auto Loan Credit-Builder Loan
##   Auto Loan                 0.5665722           0.2832861
##   Credit-Builder Loan       0.4690432           0.2814259
##   Debt Consolidation Loan   0.3868472           0.3868472
##   Home Equity Loan          0.5469462           0.8204193
##   Mortgage Loan             0.6660324           0.8563273
##   No Loan                   0.0000000           0.0000000
##   Not Specified             0.2749771           0.5499542
##   Payday Loan               0.7029877           0.3514938
##   Personal Loan             1.0185185           0.6481481
##   Student Loan              0.5623243           0.8434864
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                              0.4721435        0.5665722
##   Credit-Builder Loan                    0.3752345        0.7504690
##   Debt Consolidation Loan                0.8704062        0.3868472
##   Home Equity Loan                       0.4557885        0.5469462
##   Mortgage Loan                          0.3805899        0.6660324
##   No Loan                                0.0000000        0.0000000
##   Not Specified                          0.4582951        0.4582951
##   Payday Loan                            0.8787346        0.3514938
##   Personal Loan                          0.2777778        0.4629630
##   Student Loan                           0.3748828        0.5623243
##
##                           Mortgage Loan     No Loan Not Specified Payday Loan
##   Auto Loan                    0.9442871  94.3342776     0.5665722   0.7554297
```

```
##   Credit-Builder Loan         0.7504690  95.1219512     0.7504690   0.9380863
##   Debt Consolidation Loan     0.5802708  95.0676983     0.3868472   0.5802708
##   Home Equity Loan            0.6381039  94.3482224     0.8204193   0.6381039
##   Mortgage Loan               0.6660324  94.5765937     0.6660324   0.4757374
##   No Loan                     0.0000000 100.0000000     0.0000000   0.0000000
##   Not Specified               0.9165903  95.9670027     0.2749771   0.4582951
##   Payday Loan                 0.7029877  95.1669596     0.5272408   0.4393673
##   Personal Loan               0.5555556  95.0925926     0.3703704   0.7407407
##   Student Loan                0.2811621  95.5014058     0.5623243   0.3748828
##
##                           Personal Loan Student Loan
##   Auto Loan                   0.5665722    0.9442871
##   Credit-Builder Loan         0.3752345    0.1876173
##   Debt Consolidation Loan     0.5802708    0.7736944
##   Home Equity Loan            0.6381039    0.5469462
##   Mortgage Loan               0.5708849    0.4757374
##   No Loan                     0.0000000    0.0000000
##   Not Specified               0.3666361    0.2749771
##   Payday Loan                 0.3514938    0.5272408
##   Personal Loan               0.5555556    0.2777778
##   Student Loan                0.4686036    0.4686036
##
## $Last_Loan_2_Last_Loan_9$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               10.714286            5.555556               10.204082
##   Credit-Builder Loan      8.928571            5.555556                8.163265
##   Debt Consolidation Loan  7.142857            7.407407               18.367347
##   Home Equity Loan        10.714286           16.666667               10.204082
##   Mortgage Loan           12.500000           16.666667                8.163265
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified            5.357143           11.111111               10.204082
##   Payday Loan             14.285714            7.407407               20.408163
##   Personal Loan           19.642857           12.962963                6.122449
##   Student Loan            10.714286           16.666667                8.163265
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                       11.764706     15.384615  8.312531
##   Credit-Builder Loan             15.686275     12.307692  8.437344
##   Debt Consolidation Loan          7.843137      9.230769  8.179398
##   Home Equity Loan                11.764706     10.769231  8.612082
##   Mortgage Loan                   13.725490     10.769231  8.270927
##   No Loan                          0.000000      0.000000 23.439840
##   Not Specified                    9.803922     15.384615  8.711932
##   Payday Loan                      7.843137     12.307692  9.011483
##   Personal Loan                    9.803922      9.230769  8.545515
##   Student Loan                    11.764706      4.615385  8.478948
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    11.320755   13.793103     12.500000    20.833333
##   Credit-Builder Loan          15.094340   17.241379      8.333333     4.166667
```

```
##   Debt Consolidation Loan      7.547170   10.344828     12.500000    16.666667
##   Home Equity Loan            16.981132   12.068966     14.583333    12.500000
##   Mortgage Loan               13.207547    8.620690     12.500000    10.416667
##   No Loan                      0.000000    0.000000      0.000000     0.000000
##   Not Specified                5.660377    8.620690      8.333333     6.250000
##   Payday Loan                 11.320755    8.620690      8.333333    12.500000
##   Personal Loan                7.547170   13.793103     12.500000     6.250000
##   Student Loan                11.320755    6.896552     10.416667    10.416667
##
##
## $Last_Loan_3_Last_Loan_4
## $Last_Loan_3_Last_Loan_4$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                    568                 616                     544
##   Credit-Builder Loan          616                 560                     624
##   Debt Consolidation Loan      600                 648                     736
##   Home Equity Loan             504                 544                     504
##   Mortgage Loan                432                 632                     584
##   No Loan                        0                   0                       0
##   Not Specified                512                 736                     560
##   Payday Loan                  448                 688                     576
##   Personal Loan                448                 504                     512
##   Student Loan                 592                 736                     392
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            520           720    1712           520
##   Credit-Builder Loan                  632           520    1952           624
##   Debt Consolidation Loan              608           464    1776           464
##   Home Equity Loan                     560           592    1616           576
##   Mortgage Loan                        592           648    1608           688
##   No Loan                                0             0   38248             0
##   Not Specified                        536           680    1800           608
##   Payday Loan                          528           536    1744           592
##   Personal Loan                        640           448    1816           560
##   Student Loan                         584           560    1728           488
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                      480           600          528
##   Credit-Builder Loan            664           544          584
##   Debt Consolidation Loan        536           536          560
##   Home Equity Loan               616           536          592
##   Mortgage Loan                  584           584          520
##   No Loan                          0             0            0
##   Not Specified                  616           544          488
##   Payday Loan                    592           464          600
##   Personal Loan                  488           504          488
##   Student Loan                   712           544          592
##
## $Last_Loan_3_Last_Loan_4$row_totals
##
```

```
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    6808                    7320                    6928
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    6640                    6872                   38248
##           Not Specified             Payday Loan           Personal Loan
##                    7080                    6768                    6408
##            Student Loan
##                    6928
##
## $Last_Loan_3_Last_Loan_4$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    4720                    5664                    5032
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    5200                    5168                   54000
##           Not Specified             Payday Loan           Personal Loan
##                    5120                    5288                    4856
##            Student Loan
##                    4952
##
## $Last_Loan_3_Last_Loan_4$grand_total
## [1] 100000
##
## $Last_Loan_3_Last_Loan_4$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.568               0.616                   0.544
##   Credit-Builder Loan        0.616               0.560                   0.624
##   Debt Consolidation Loan    0.600               0.648                   0.736
##   Home Equity Loan           0.504               0.544                   0.504
##   Mortgage Loan              0.432               0.632                   0.584
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.512               0.736                   0.560
##   Payday Loan                0.448               0.688                   0.576
##   Personal Loan              0.448               0.504                   0.512
##   Student Loan               0.592               0.736                   0.392
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.520         0.720   1.712         0.520
##   Credit-Builder Loan                0.632         0.520   1.952         0.624
##   Debt Consolidation Loan            0.608         0.464   1.776         0.464
##   Home Equity Loan                   0.560         0.592   1.616         0.576
##   Mortgage Loan                      0.592         0.648   1.608         0.688
##   No Loan                            0.000         0.000  38.248         0.000
##   Not Specified                      0.536         0.680   1.800         0.608
##   Payday Loan                        0.528         0.536   1.744         0.592
##   Personal Loan                      0.640         0.448   1.816         0.560
##   Student Loan                       0.584         0.560   1.728         0.488
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.480         0.600        0.528
```

```
##   Credit-Builder Loan           0.664         0.544        0.584
##   Debt Consolidation Loan       0.536         0.536        0.560
##   Home Equity Loan              0.616         0.536        0.592
##   Mortgage Loan                 0.584         0.584        0.520
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.616         0.544        0.488
##   Payday Loan                   0.592         0.464        0.600
##   Personal Loan                 0.488         0.504        0.488
##   Student Loan                  0.712         0.544        0.592
##
## $Last_Loan_3_Last_Loan_4$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 8.343126            9.048179
##   Credit-Builder Loan       8.415301            7.650273
##   Debt Consolidation Loan   8.660508            9.353349
##   Home Equity Loan          7.590361            8.192771
##   Mortgage Loan             6.286380            9.196740
##   No Loan                   0.000000            0.000000
##   Not Specified             7.231638           10.395480
##   Payday Loan               6.619385           10.165485
##   Personal Loan             6.991261            7.865169
##   Student Loan              8.545035           10.623557
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                7.990599         7.638073
##   Credit-Builder Loan                      8.524590         8.633880
##   Debt Consolidation Loan                 10.623557         8.775982
##   Home Equity Loan                         7.590361         8.433735
##   Mortgage Loan                            8.498254         8.614668
##   No Loan                                  0.000000         0.000000
##   Not Specified                            7.909605         7.570621
##   Payday Loan                              8.510638         7.801418
##   Personal Loan                            7.990012         9.987516
##   Student Loan                             5.658199         8.429561
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                    10.575793  25.146886      7.638073    7.050529
##   Credit-Builder Loan           7.103825  26.666667      8.524590    9.071038
##   Debt Consolidation Loan       6.697460  25.635104      6.697460    7.736721
##   Home Equity Loan              8.915663  24.337349      8.674699    9.277108
##   Mortgage Loan                 9.429569  23.399302     10.011641    8.498254
##   No Loan                       0.000000 100.000000      0.000000    0.000000
##   Not Specified                 9.604520  25.423729      8.587571    8.700565
##   Payday Loan                   7.919622  25.768322      8.747045    8.747045
##   Personal Loan                 6.991261  28.339576      8.739076    7.615481
##   Student Loan                  8.083141  24.942263      7.043880   10.277136
##
##                           Personal Loan Student Loan
##   Auto Loan                      8.813161     7.755582
##   Credit-Builder Loan            7.431694     7.978142
```

```
##   Debt Consolidation Loan      7.736721     8.083141
##   Home Equity Loan             8.072289     8.915663
##   Mortgage Loan                8.498254     7.566938
##   No Loan                      0.000000     0.000000
##   Not Specified                7.683616     6.892655
##   Payday Loan                  6.855792     8.865248
##   Personal Loan                7.865169     7.615481
##   Student Loan                 7.852194     8.545035
##
## $Last_Loan_3_Last_Loan_4$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               12.033898           10.875706               10.810811
##   Credit-Builder Loan     13.050847            9.887006               12.400636
##   Debt Consolidation Loan 12.711864           11.440678               14.626391
##   Home Equity Loan        10.677966            9.604520               10.015898
##   Mortgage Loan            9.152542           11.158192               11.605723
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           10.847458           12.994350               11.128776
##   Payday Loan              9.491525           12.146893               11.446741
##   Personal Loan            9.491525            8.898305               10.174881
##   Student Loan            12.542373           12.994350                7.790143
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                       10.000000     13.931889  3.170370
##   Credit-Builder Loan             12.153846     10.061920  3.614815
##   Debt Consolidation Loan         11.692308      8.978328  3.288889
##   Home Equity Loan                10.769231     11.455108  2.992593
##   Mortgage Loan                   11.384615     12.538700  2.977778
##   No Loan                          0.000000      0.000000 70.829630
##   Not Specified                   10.307692     13.157895  3.333333
##   Payday Loan                     10.153846     10.371517  3.229630
##   Personal Loan                   12.307692      8.668731  3.362963
##   Student Loan                    11.230769     10.835913  3.200000
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    10.156250    9.077156     12.355848    10.662359
##   Credit-Builder Loan          12.187500   12.556732     11.202636    11.793215
##   Debt Consolidation Loan       9.062500   10.136157     11.037891    11.308562
##   Home Equity Loan             11.250000   11.649017     11.037891    11.954766
##   Mortgage Loan                13.437500   11.043873     12.026359    10.500808
##   No Loan                       0.000000    0.000000      0.000000     0.000000
##   Not Specified                11.875000   11.649017     11.202636     9.854604
##   Payday Loan                  11.562500   11.195159      9.555189    12.116317
##   Personal Loan                10.937500    9.228442     10.378913     9.854604
##   Student Loan                  9.531250   13.464448     11.202636    11.954766
##
##
## $Last_Loan_3_Last_Loan_5
## $Last_Loan_3_Last_Loan_5$contingency_table
##
```

```
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                    368                 376                     312
##   Credit-Builder Loan          392                 384                     432
##   Debt Consolidation Loan      432                 448                     320
##   Home Equity Loan             288                 392                     408
##   Mortgage Loan                448                 344                     424
##   No Loan                        0                   0                       0
##   Not Specified                416                 392                     480
##   Payday Loan                  304                 328                     384
##   Personal Loan                408                 360                     344
##   Student Loan                 368                 472                     424
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            288           368    3536           320
##   Credit-Builder Loan                  464           408    3744           440
##   Debt Consolidation Loan              472           304    3440           416
##   Home Equity Loan                     280           320    3400           376
##   Mortgage Loan                        400           376    3368           384
##   No Loan                                0             0   38248             0
##   Not Specified                        424           512    3376           280
##   Payday Loan                          360           400    3472           448
##   Personal Loan                        320           312    3376           296
##   Student Loan                         392           440    3496           312
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                      424           480          336
##   Credit-Builder Loan            352           320          384
##   Debt Consolidation Loan        384           368          344
##   Home Equity Loan               352           440          384
##   Mortgage Loan                  432           392          304
##   No Loan                          0             0            0
##   Not Specified                  368           432          400
##   Payday Loan                    336           368          368
##   Personal Loan                  376           328          288
##   Student Loan                   320           392          312
##
## $Last_Loan_3_Last_Loan_5$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    6808                    7320                    6928
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    6640                    6872                   38248
##           Not Specified             Payday Loan           Personal Loan
##                    7080                    6768                    6408
##            Student Loan
##                    6928
##
## $Last_Loan_3_Last_Loan_5$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    3424                    3496                    3528
```

```
##        Home Equity Loan           Mortgage Loan                  No Loan
##                    3400                    3440                    69456
##           Not Specified             Payday Loan           Personal Loan
##                    3272                    3344                     3520
##            Student Loan
##                    3120
##
## $Last_Loan_3_Last_Loan_5$grand_total
## [1] 100000
##
## $Last_Loan_3_Last_Loan_5$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.368               0.376                   0.312
##   Credit-Builder Loan        0.392               0.384                   0.432
##   Debt Consolidation Loan    0.432               0.448                   0.320
##   Home Equity Loan           0.288               0.392                   0.408
##   Mortgage Loan              0.448               0.344                   0.424
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.416               0.392                   0.480
##   Payday Loan                0.304               0.328                   0.384
##   Personal Loan              0.408               0.360                   0.344
##   Student Loan               0.368               0.472                   0.424
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.288         0.368   3.536         0.320
##   Credit-Builder Loan                0.464         0.408   3.744         0.440
##   Debt Consolidation Loan            0.472         0.304   3.440         0.416
##   Home Equity Loan                   0.280         0.320   3.400         0.376
##   Mortgage Loan                      0.400         0.376   3.368         0.384
##   No Loan                            0.000         0.000  38.248         0.000
##   Not Specified                      0.424         0.512   3.376         0.280
##   Payday Loan                        0.360         0.400   3.472         0.448
##   Personal Loan                      0.320         0.312   3.376         0.296
##   Student Loan                       0.392         0.440   3.496         0.312
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.424         0.480        0.336
##   Credit-Builder Loan          0.352         0.320        0.384
##   Debt Consolidation Loan      0.384         0.368        0.344
##   Home Equity Loan             0.352         0.440        0.384
##   Mortgage Loan                0.432         0.392        0.304
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.368         0.432        0.400
##   Payday Loan                  0.336         0.368        0.368
##   Personal Loan                0.376         0.328        0.288
##   Student Loan                 0.320         0.392        0.312
##
## $Last_Loan_3_Last_Loan_5$row_percentage
##
##                           Auto Loan Credit-Builder Loan
```

```
##   Auto Loan                 5.405405            5.522914
##   Credit-Builder Loan       5.355191            5.245902
##   Debt Consolidation Loan   6.235566            6.466513
##   Home Equity Loan          4.337349            5.903614
##   Mortgage Loan             6.519208            5.005821
##   No Loan                   0.000000            0.000000
##   Not Specified             5.875706            5.536723
##   Payday Loan               4.491726            4.846336
##   Personal Loan             6.367041            5.617978
##   Student Loan              5.311778            6.812933
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                4.582844         4.230317
##   Credit-Builder Loan                      5.901639         6.338798
##   Debt Consolidation Loan                  4.618938         6.812933
##   Home Equity Loan                         6.144578         4.216867
##   Mortgage Loan                            6.169965         5.820722
##   No Loan                                  0.000000         0.000000
##   Not Specified                            6.779661         5.988701
##   Payday Loan                              5.673759         5.319149
##   Personal Loan                            5.368290         4.993758
##   Student Loan                             6.120092         5.658199
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                      5.405405  51.938895      4.700353    6.227967
##   Credit-Builder Loan            5.573770  51.147541      6.010929    4.808743
##   Debt Consolidation Loan        4.387991  49.653580      6.004619    5.542725
##   Home Equity Loan               4.819277  51.204819      5.662651    5.301205
##   Mortgage Loan                  5.471478  49.010477      5.587893    6.286380
##   No Loan                        0.000000 100.000000      0.000000    0.000000
##   Not Specified                  7.231638  47.683616      3.954802    5.197740
##   Payday Loan                    5.910165  51.300236      6.619385    4.964539
##   Personal Loan                  4.868914  52.684145      4.619226    5.867665
##   Student Loan                   6.351039  50.461894      4.503464    4.618938
##
##                           Personal Loan Student Loan
##   Auto Loan                      7.050529     4.935370
##   Credit-Builder Loan            4.371585     5.245902
##   Debt Consolidation Loan        5.311778     4.965358
##   Home Equity Loan               6.626506     5.783133
##   Mortgage Loan                  5.704307     4.423749
##   No Loan                        0.000000     0.000000
##   Not Specified                  6.101695     5.649718
##   Payday Loan                    5.437352     5.437352
##   Personal Loan                  5.118602     4.494382
##   Student Loan                   5.658199     4.503464
##
## $Last_Loan_3_Last_Loan_5$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               10.747664           10.755149                 8.843537
```

```
##   Credit-Builder Loan     11.448598           10.983982                12.244898
##   Debt Consolidation Loan 12.616822           12.814645                 9.070295
##   Home Equity Loan         8.411215           11.212815                11.564626
##   Mortgage Loan           13.084112            9.839817                12.018141
##   No Loan                  0.000000            0.000000                 0.000000
##   Not Specified           12.149533           11.212815                13.605442
##   Payday Loan              8.878505            9.382151                10.884354
##   Personal Loan           11.915888           10.297483                 9.750567
##   Student Loan            10.747664           13.501144                12.018141
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                        8.470588     10.697674  5.090993
##   Credit-Builder Loan             13.647059     11.860465  5.390463
##   Debt Consolidation Loan         13.882353      8.837209  4.952776
##   Home Equity Loan                 8.235294      9.302326  4.895185
##   Mortgage Loan                   11.764706     10.930233  4.849113
##   No Loan                          0.000000      0.000000 55.067957
##   Not Specified                   12.470588     14.883721  4.860631
##   Payday Loan                     10.588235     11.627907  4.998848
##   Personal Loan                    9.411765      9.069767  4.860631
##   Student Loan                    11.529412     12.790698  5.033402
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                      9.779951   12.679426     13.636364    10.769231
##   Credit-Builder Loan           13.447433   10.526316      9.090909    12.307692
##   Debt Consolidation Loan       12.713936   11.483254     10.454545    11.025641
##   Home Equity Loan              11.491443   10.526316     12.500000    12.307692
##   Mortgage Loan                 11.735941   12.918660     11.136364     9.743590
##   No Loan                        0.000000    0.000000      0.000000     0.000000
##   Not Specified                  8.557457   11.004785     12.272727    12.820513
##   Payday Loan                   13.691932   10.047847     10.454545    11.794872
##   Personal Loan                  9.046455   11.244019      9.318182     9.230769
##   Student Loan                   9.535452    9.569378     11.136364    10.000000
##
##
## $Last_Loan_3_Last_Loan_6
## $Last_Loan_3_Last_Loan_6$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     240                 296                     288
##   Credit-Builder Loan           264                 336                     312
##   Debt Consolidation Loan       336                 216                     248
##   Home Equity Loan              256                 264                     240
##   Mortgage Loan                 288                 304                     320
##   No Loan                         0                   0                       0
##   Not Specified                 288                 320                     248
##   Payday Loan                   256                 248                     304
##   Personal Loan                 256                 328                     280
##   Student Loan                  288                 296                     224
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
```

```
##   Auto Loan                          240          224    4440          224
##   Credit-Builder Loan                248          264    4656          344
##   Debt Consolidation Loan            280          320    4312          280
##   Home Equity Loan                   352          152    4168          344
##   Mortgage Loan                      200          288    4264          320
##   No Loan                              0            0   38248            0
##   Not Specified                      224          312    4344          392
##   Payday Loan                        368          256    4136          256
##   Personal Loan                      216          264    4072          224
##   Student Loan                       336          312    4344          240
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                      296           352          208
##   Credit-Builder Loan            280           272          344
##   Debt Consolidation Loan        344           272          320
##   Home Equity Loan               336           296          232
##   Mortgage Loan                  296           296          296
##   No Loan                          0             0            0
##   Not Specified                  312           352          288
##   Payday Loan                    288           304          352
##   Personal Loan                  240           256          272
##   Student Loan                   296           336          256
##
## $Last_Loan_3_Last_Loan_6$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    6808                    7320                    6928
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    6640                    6872                   38248
##           Not Specified             Payday Loan           Personal Loan
##                    7080                    6768                    6408
##            Student Loan
##                    6928
##
## $Last_Loan_3_Last_Loan_6$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    2472                    2608                    2464
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    2464                    2392                   76984
##           Not Specified             Payday Loan           Personal Loan
##                    2624                    2688                    2736
##            Student Loan
##                    2568
##
## $Last_Loan_3_Last_Loan_6$grand_total
## [1] 100000
##
## $Last_Loan_3_Last_Loan_6$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
```

```
##   Auto Loan                   0.240               0.296                    0.288
##   Credit-Builder Loan         0.264               0.336                    0.312
##   Debt Consolidation Loan     0.336               0.216                    0.248
##   Home Equity Loan            0.256               0.264                    0.240
##   Mortgage Loan               0.288               0.304                    0.320
##   No Loan                     0.000               0.000                    0.000
##   Not Specified               0.288               0.320                    0.248
##   Payday Loan                 0.256               0.248                    0.304
##   Personal Loan               0.256               0.328                    0.280
##   Student Loan                0.288               0.296                    0.224
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.240         0.224   4.440         0.224
##   Credit-Builder Loan                0.248         0.264   4.656         0.344
##   Debt Consolidation Loan            0.280         0.320   4.312         0.280
##   Home Equity Loan                   0.352         0.152   4.168         0.344
##   Mortgage Loan                      0.200         0.288   4.264         0.320
##   No Loan                            0.000         0.000  38.248         0.000
##   Not Specified                      0.224         0.312   4.344         0.392
##   Payday Loan                        0.368         0.256   4.136         0.256
##   Personal Loan                      0.216         0.264   4.072         0.224
##   Student Loan                       0.336         0.312   4.344         0.240
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                     0.296         0.352        0.208
##   Credit-Builder Loan           0.280         0.272        0.344
##   Debt Consolidation Loan       0.344         0.272        0.320
##   Home Equity Loan              0.336         0.296        0.232
##   Mortgage Loan                 0.296         0.296        0.296
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.312         0.352        0.288
##   Payday Loan                   0.288         0.304        0.352
##   Personal Loan                 0.240         0.256        0.272
##   Student Loan                  0.296         0.336        0.256
##
## $Last_Loan_3_Last_Loan_6$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 3.525264            4.347826
##   Credit-Builder Loan       3.606557            4.590164
##   Debt Consolidation Loan   4.849885            3.117783
##   Home Equity Loan          3.855422            3.975904
##   Mortgage Loan             4.190920            4.423749
##   No Loan                   0.000000            0.000000
##   Not Specified             4.067797            4.519774
##   Payday Loan               3.782506            3.664303
##   Personal Loan             3.995006            5.118602
##   Student Loan              4.157044            4.272517
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                4.230317         3.525264
```

```
##   Credit-Builder Loan                    4.262295         3.387978
##   Debt Consolidation Loan                3.579677         4.041570
##   Home Equity Loan                       3.614458         5.301205
##   Mortgage Loan                          4.656577         2.910361
##   No Loan                                0.000000         0.000000
##   Not Specified                          3.502825         3.163842
##   Payday Loan                            4.491726         5.437352
##   Personal Loan                          4.369538         3.370787
##   Student Loan                           3.233256         4.849885
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                    3.290247  65.217391      3.290247    4.347826
##   Credit-Builder Loan          3.606557  63.606557      4.699454    3.825137
##   Debt Consolidation Loan      4.618938  62.240185      4.041570    4.965358
##   Home Equity Loan             2.289157  62.771084      5.180723    5.060241
##   Mortgage Loan                4.190920  62.048894      4.656577    4.307334
##   No Loan                      0.000000 100.000000      0.000000    0.000000
##   Not Specified                4.406780  61.355932      5.536723    4.406780
##   Payday Loan                  3.782506  61.111111      3.782506    4.255319
##   Personal Loan                4.119850  63.545568      3.495630    3.745318
##   Student Loan                 4.503464  62.702079      3.464203    4.272517
##
##                           Personal Loan Student Loan
##   Auto Loan                    5.170388     3.055229
##   Credit-Builder Loan          3.715847     4.699454
##   Debt Consolidation Loan      3.926097     4.618938
##   Home Equity Loan             4.457831     3.493976
##   Mortgage Loan                4.307334     4.307334
##   No Loan                      0.000000     0.000000
##   Not Specified                4.971751     4.067797
##   Payday Loan                  4.491726     5.200946
##   Personal Loan                3.995006     4.244694
##   Student Loan                 4.849885     3.695150
##
## $Last_Loan_3_Last_Loan_6$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                 9.708738           11.349693                11.688312
##   Credit-Builder Loan      10.679612           12.883436                12.662338
##   Debt Consolidation Loan 13.592233            8.282209                10.064935
##   Home Equity Loan         10.355987           10.122699                 9.740260
##   Mortgage Loan            11.650485           11.656442                12.987013
##   No Loan                   0.000000            0.000000                 0.000000
##   Not Specified            11.650485           12.269939                10.064935
##   Payday Loan              10.355987            9.509202                12.337662
##   Personal Loan            10.355987           12.576687                11.363636
##   Student Loan             11.650485           11.349693                 9.090909
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                        9.740260      9.364548  5.767432
##   Credit-Builder Loan             10.064935     11.036789  6.048010
```

```
##   Debt Consolidation Loan        11.363636     13.377926  5.601164
##   Home Equity Loan               14.285714      6.354515  5.414112
##   Mortgage Loan                   8.116883     12.040134  5.538813
##   No Loan                         0.000000      0.000000 49.683051
##   Not Specified                   9.090909     13.043478  5.642731
##   Payday Loan                    14.935065     10.702341  5.372545
##   Personal Loan                   8.766234     11.036789  5.289411
##   Student Loan                   13.636364     13.043478  5.642731
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    8.536585   11.011905     12.865497     8.099688
##   Credit-Builder Loan         13.109756   10.416667      9.941520    13.395639
##   Debt Consolidation Loan     10.670732   12.797619      9.941520    12.461059
##   Home Equity Loan            13.109756   12.500000     10.818713     9.034268
##   Mortgage Loan               12.195122   11.011905     10.818713    11.526480
##   No Loan                      0.000000    0.000000      0.000000     0.000000
##   Not Specified               14.939024   11.607143     12.865497    11.214953
##   Payday Loan                  9.756098   10.714286     11.111111    13.707165
##   Personal Loan                8.536585    8.928571      9.356725    10.591900
##   Student Loan                 9.146341   11.011905     12.280702     9.968847
##
##
## $Last_Loan_3_Last_Loan_7
## $Last_Loan_3_Last_Loan_7$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     176                 120                      128
##   Credit-Builder Loan           136                 272                      176
##   Debt Consolidation Loan       208                 112                      152
##   Home Equity Loan              120                 240                      136
##   Mortgage Loan                 232                 232                      168
##   No Loan                         0                   0                        0
##   Not Specified                 232                 120                      152
##   Payday Loan                   192                 216                      208
##   Personal Loan                 168                 200                      144
##   Student Loan                  136                 208                      168
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            168           216    5248           192
##   Credit-Builder Loan                  176           208    5584           200
##   Debt Consolidation Loan              200           232    5080           240
##   Home Equity Loan                     192           208    5024           152
##   Mortgage Loan                        176           200    5208           168
##   No Loan                                0             0   38248             0
##   Not Specified                        176           224    5448           176
##   Payday Loan                          160           192    5096           160
##   Personal Loan                        104           184    4920           144
##   Student Loan                         176           128    5272           176
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                       192           184          184
```

```
##   Credit-Builder Loan             192           216          160
##   Debt Consolidation Loan         288           224          192
##   Home Equity Loan                200           216          152
##   Mortgage Loan                   192           160          136
##   No Loan                           0             0            0
##   Not Specified                   184           152          216
##   Payday Loan                     200           176          168
##   Personal Loan                   248           136          160
##   Student Loan                    224           224          216
##
## $Last_Loan_3_Last_Loan_7$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    6808                    7320                    6928
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    6640                    6872                   38248
##           Not Specified             Payday Loan           Personal Loan
##                    7080                    6768                    6408
##            Student Loan
##                    6928
##
## $Last_Loan_3_Last_Loan_7$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    1600                    1720                    1432
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    1528                    1792                   85128
##           Not Specified             Payday Loan           Personal Loan
##                    1608                    1920                    1688
##            Student Loan
##                    1584
##
## $Last_Loan_3_Last_Loan_7$grand_total
## [1] 100000
##
## $Last_Loan_3_Last_Loan_7$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.176               0.120                   0.128
##   Credit-Builder Loan        0.136               0.272                   0.176
##   Debt Consolidation Loan    0.208               0.112                   0.152
##   Home Equity Loan           0.120               0.240                   0.136
##   Mortgage Loan              0.232               0.232                   0.168
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.232               0.120                   0.152
##   Payday Loan                0.192               0.216                   0.208
##   Personal Loan              0.168               0.200                   0.144
##   Student Loan               0.136               0.208                   0.168
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.168         0.216   5.248         0.192
```

```
##   Credit-Builder Loan                0.176         0.208   5.584         0.200
##   Debt Consolidation Loan            0.200         0.232   5.080         0.240
##   Home Equity Loan                   0.192         0.208   5.024         0.152
##   Mortgage Loan                      0.176         0.200   5.208         0.168
##   No Loan                            0.000         0.000  38.248         0.000
##   Not Specified                      0.176         0.224   5.448         0.176
##   Payday Loan                        0.160         0.192   5.096         0.160
##   Personal Loan                      0.104         0.184   4.920         0.144
##   Student Loan                       0.176         0.128   5.272         0.176
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                     0.192         0.184        0.184
##   Credit-Builder Loan           0.192         0.216        0.160
##   Debt Consolidation Loan       0.288         0.224        0.192
##   Home Equity Loan              0.200         0.216        0.152
##   Mortgage Loan                 0.192         0.160        0.136
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.184         0.152        0.216
##   Payday Loan                   0.200         0.176        0.168
##   Personal Loan                 0.248         0.136        0.160
##   Student Loan                  0.224         0.224        0.216
##
## $Last_Loan_3_Last_Loan_7$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 2.585194            1.762632
##   Credit-Builder Loan       1.857923            3.715847
##   Debt Consolidation Loan   3.002309            1.616628
##   Home Equity Loan          1.807229            3.614458
##   Mortgage Loan             3.376019            3.376019
##   No Loan                   0.000000            0.000000
##   Not Specified             3.276836            1.694915
##   Payday Loan               2.836879            3.191489
##   Personal Loan             2.621723            3.121099
##   Student Loan              1.963048            3.002309
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                1.880141         2.467685
##   Credit-Builder Loan                      2.404372         2.404372
##   Debt Consolidation Loan                  2.193995         2.886836
##   Home Equity Loan                         2.048193         2.891566
##   Mortgage Loan                            2.444703         2.561118
##   No Loan                                  0.000000         0.000000
##   Not Specified                            2.146893         2.485876
##   Payday Loan                              3.073286         2.364066
##   Personal Loan                            2.247191         1.622971
##   Student Loan                             2.424942         2.540416
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                     3.172738  77.085781      2.820212    2.820212
##   Credit-Builder Loan           2.841530  76.284153      2.732240    2.622951
```

```
##   Debt Consolidation Loan      3.348730  73.325635      3.464203    4.157044
##   Home Equity Loan             3.132530  75.662651      2.289157    3.012048
##   Mortgage Loan                2.910361  75.785797      2.444703    2.793946
##   No Loan                      0.000000 100.000000      0.000000    0.000000
##   Not Specified                3.163842  76.949153      2.485876    2.598870
##   Payday Loan                  2.836879  75.295508      2.364066    2.955083
##   Personal Loan                2.871411  76.779026      2.247191    3.870162
##   Student Loan                 1.847575  76.096998      2.540416    3.233256
##
##                           Personal Loan Student Loan
##   Auto Loan                    2.702703     2.702703
##   Credit-Builder Loan          2.950820     2.185792
##   Debt Consolidation Loan      3.233256     2.771363
##   Home Equity Loan             3.253012     2.289157
##   Mortgage Loan                2.328289     1.979045
##   No Loan                      0.000000     0.000000
##   Not Specified                2.146893     3.050847
##   Payday Loan                  2.600473     2.482270
##   Personal Loan                2.122347     2.496879
##   Student Loan                 3.233256     3.117783
##
## $Last_Loan_3_Last_Loan_7$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               11.000000            6.976744                8.938547
##   Credit-Builder Loan      8.500000           15.813953               12.290503
##   Debt Consolidation Loan 13.000000            6.511628               10.614525
##   Home Equity Loan         7.500000           13.953488                9.497207
##   Mortgage Loan           14.500000           13.488372               11.731844
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           14.500000            6.976744               10.614525
##   Payday Loan             12.000000           12.558140               14.525140
##   Personal Loan           10.500000           11.627907               10.055866
##   Student Loan             8.500000           12.093023               11.731844
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                       10.994764     12.053571  6.164834
##   Credit-Builder Loan             11.518325     11.607143  6.559534
##   Debt Consolidation Loan         13.089005     12.946429  5.967484
##   Home Equity Loan                12.565445     11.607143  5.901701
##   Mortgage Loan                   11.518325     11.160714  6.117846
##   No Loan                          0.000000      0.000000 44.929988
##   Not Specified                   11.518325     12.500000  6.399774
##   Payday Loan                     10.471204     10.714286  5.986279
##   Personal Loan                    6.806283     10.267857  5.779532
##   Student Loan                    11.518325      7.142857  6.193027
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    11.940299   10.000000     10.900474    11.616162
##   Credit-Builder Loan          12.437811   10.000000     12.796209    10.101010
##   Debt Consolidation Loan      14.925373   15.000000     13.270142    12.121212
```

```
##   Home Equity Loan             9.452736   10.416667     12.796209     9.595960
##   Mortgage Loan               10.447761   10.000000      9.478673     8.585859
##   No Loan                      0.000000    0.000000      0.000000     0.000000
##   Not Specified               10.945274    9.583333      9.004739    13.636364
##   Payday Loan                  9.950249   10.416667     10.426540    10.606061
##   Personal Loan                8.955224   12.916667      8.056872    10.101010
##   Student Loan                10.945274   11.666667     13.270142    13.636364
##
##
## $Last_Loan_3_Last_Loan_8
## $Last_Loan_3_Last_Loan_8$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     80                  80                      72
##   Credit-Builder Loan           96                 104                      72
##   Debt Consolidation Loan      104                 104                      96
##   Home Equity Loan              80                 136                      72
##   Mortgage Loan                 88                 112                      88
##   No Loan                        0                   0                       0
##   Not Specified                120                  80                      64
##   Payday Loan                   56                  64                     112
##   Personal Loan                 24                 128                     104
##   Student Loan                  72                  64                      80
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                           128            96    6016            72
##   Credit-Builder Loan                  96            88    6488            56
##   Debt Consolidation Loan             112            56    6112            72
##   Home Equity Loan                     64            80    5880            96
##   Mortgage Loan                       136            72    6000            64
##   No Loan                               0             0   38248             0
##   Not Specified                       104            88    6272            72
##   Payday Loan                          72            72    5952           144
##   Personal Loan                        64            96    5712            56
##   Student Loan                        112            88    6128           112
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                       80            72          112
##   Credit-Builder Loan            144            96           80
##   Debt Consolidation Loan         72           112           88
##   Home Equity Loan                56            80           96
##   Mortgage Loan                  120            72          120
##   No Loan                          0             0            0
##   Not Specified                  112            56          112
##   Payday Loan                    128            88           80
##   Personal Loan                   80            64           80
##   Student Loan                    96            80           96
##
## $Last_Loan_3_Last_Loan_8$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
```

```
##                    6808                    7320                    6928
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    6640                    6872                   38248
##           Not Specified             Payday Loan           Personal Loan
##                    7080                    6768                    6408
##            Student Loan
##                    6928
##
## $Last_Loan_3_Last_Loan_8$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     720                     872                     760
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     888                     736                   92808
##           Not Specified             Payday Loan           Personal Loan
##                     744                     888                     720
##            Student Loan
##                     864
##
## $Last_Loan_3_Last_Loan_8$grand_total
## [1] 100000
##
## $Last_Loan_3_Last_Loan_8$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.080               0.080                   0.072
##   Credit-Builder Loan        0.096               0.104                   0.072
##   Debt Consolidation Loan    0.104               0.104                   0.096
##   Home Equity Loan           0.080               0.136                   0.072
##   Mortgage Loan              0.088               0.112                   0.088
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.120               0.080                   0.064
##   Payday Loan                0.056               0.064                   0.112
##   Personal Loan              0.024               0.128                   0.104
##   Student Loan               0.072               0.064                   0.080
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                         0.128         0.096   6.016         0.072
##   Credit-Builder Loan               0.096         0.088   6.488         0.056
##   Debt Consolidation Loan           0.112         0.056   6.112         0.072
##   Home Equity Loan                  0.064         0.080   5.880         0.096
##   Mortgage Loan                     0.136         0.072   6.000         0.064
##   No Loan                           0.000         0.000  38.248         0.000
##   Not Specified                     0.104         0.088   6.272         0.072
##   Payday Loan                       0.072         0.072   5.952         0.144
##   Personal Loan                     0.064         0.096   5.712         0.056
##   Student Loan                      0.112         0.088   6.128         0.112
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.080         0.072        0.112
##   Credit-Builder Loan          0.144         0.096        0.080
```

```
##   Debt Consolidation Loan       0.072         0.112        0.088
##   Home Equity Loan              0.056         0.080        0.096
##   Mortgage Loan                 0.120         0.072        0.120
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.112         0.056        0.112
##   Payday Loan                   0.128         0.088        0.080
##   Personal Loan                 0.080         0.064        0.080
##   Student Loan                  0.096         0.080        0.096
##
## $Last_Loan_3_Last_Loan_8$row_percentage
##
##                             Auto Loan Credit-Builder Loan
##   Auto Loan                 1.1750881           1.1750881
##   Credit-Builder Loan       1.3114754           1.4207650
##   Debt Consolidation Loan   1.5011547           1.5011547
##   Home Equity Loan          1.2048193           2.0481928
##   Mortgage Loan             1.2805588           1.6298021
##   No Loan                   0.0000000           0.0000000
##   Not Specified             1.6949153           1.1299435
##   Payday Loan               0.8274232           0.9456265
##   Personal Loan             0.3745318           1.9975031
##   Student Loan              1.0392610           0.9237875
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                             1.0575793        1.8801410
##   Credit-Builder Loan                   0.9836066        1.3114754
##   Debt Consolidation Loan               1.3856813        1.6166282
##   Home Equity Loan                      1.0843373        0.9638554
##   Mortgage Loan                         1.2805588        1.9790454
##   No Loan                               0.0000000        0.0000000
##   Not Specified                         0.9039548        1.4689266
##   Payday Loan                           1.6548463        1.0638298
##   Personal Loan                         1.6229713        0.9987516
##   Student Loan                          1.1547344        1.6166282
##
##                           Mortgage Loan     No Loan Not Specified Payday Loan
##   Auto Loan                    1.4101058  88.3666275     1.0575793   1.1750881
##   Credit-Builder Loan          1.2021858  88.6338798     0.7650273   1.9672131
##   Debt Consolidation Loan      0.8083141  88.2217090     1.0392610   1.0392610
##   Home Equity Loan             1.2048193  88.5542169     1.4457831   0.8433735
##   Mortgage Loan                1.0477299  87.3108265     0.9313155   1.7462165
##   No Loan                      0.0000000 100.0000000     0.0000000   0.0000000
##   Not Specified                1.2429379  88.5875706     1.0169492   1.5819209
##   Payday Loan                  1.0638298  87.9432624     2.1276596   1.8912530
##   Personal Loan                1.4981273  89.1385768     0.8739076   1.2484395
##   Student Loan                 1.2702079  88.4526559     1.6166282   1.3856813
##
##                           Personal Loan Student Loan
##   Auto Loan                    1.0575793    1.6451234
##   Credit-Builder Loan          1.3114754    1.0928962
##   Debt Consolidation Loan      1.6166282    1.2702079
```

```
##   Home Equity Loan            1.2048193    1.4457831
##   Mortgage Loan               1.0477299    1.7462165
##   No Loan                     0.0000000    0.0000000
##   Not Specified               0.7909605    1.5819209
##   Payday Loan                 1.3002364    1.1820331
##   Personal Loan               0.9987516    1.2484395
##   Student Loan                1.1547344    1.3856813
##
## $Last_Loan_3_Last_Loan_8$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               11.111111            9.174312                9.473684
##   Credit-Builder Loan     13.333333           11.926606                9.473684
##   Debt Consolidation Loan 14.444444           11.926606               12.631579
##   Home Equity Loan        11.111111           15.596330                9.473684
##   Mortgage Loan           12.222222           12.844037               11.578947
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           16.666667            9.174312                8.421053
##   Payday Loan              7.777778            7.339450               14.736842
##   Personal Loan            3.333333           14.678899               13.684211
##   Student Loan            10.000000            7.339450               10.526316
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                       14.414414     13.043478  6.482200
##   Credit-Builder Loan             10.810811     11.956522  6.990777
##   Debt Consolidation Loan         12.612613      7.608696  6.585639
##   Home Equity Loan                 7.207207     10.869565  6.335661
##   Mortgage Loan                   15.315315      9.782609  6.464960
##   No Loan                          0.000000      0.000000 41.211964
##   Not Specified                   11.711712     11.956522  6.758038
##   Payday Loan                      8.108108      9.782609  6.413240
##   Personal Loan                    7.207207     13.043478  6.154642
##   Student Loan                    12.612613     11.956522  6.602879
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                      9.677419    9.009009     10.000000    12.962963
##   Credit-Builder Loan            7.526882   16.216216     13.333333     9.259259
##   Debt Consolidation Loan        9.677419    8.108108     15.555556    10.185185
##   Home Equity Loan              12.903226    6.306306     11.111111    11.111111
##   Mortgage Loan                  8.602151   13.513514     10.000000    13.888889
##   No Loan                        0.000000    0.000000      0.000000     0.000000
##   Not Specified                  9.677419   12.612613      7.777778    12.962963
##   Payday Loan                   19.354839   14.414414     12.222222     9.259259
##   Personal Loan                  7.526882    9.009009      8.888889     9.259259
##   Student Loan                  15.053763   10.810811     11.111111    11.111111
##
##
## $Last_Loan_3_Last_Loan_9
## $Last_Loan_3_Last_Loan_9$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
```

```
##   Auto Loan                      40                  32                      88
##   Credit-Builder Loan            72                  64                      40
##   Debt Consolidation Loan        32                  48                      48
##   Home Equity Loan               48                  48                       8
##   Mortgage Loan                  80                  24                      96
##   No Loan                         0                   0                       0
##   Not Specified                  56                  64                       8
##   Payday Loan                    32                  40                      24
##   Personal Loan                  40                  56                      40
##   Student Loan                   48                  56                      40
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                             32            48    6344            64
##   Credit-Builder Loan                   64            40    6832            48
##   Debt Consolidation Loan               40           120    6544            32
##   Home Equity Loan                      32            56    6240            40
##   Mortgage Loan                         56            32    6440            40
##   No Loan                                0             0   38248             0
##   Not Specified                         88            72    6632             8
##   Payday Loan                           16            72    6328            56
##   Personal Loan                         64            40    6016            64
##   Student Loan                          16            40    6520            72
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                        32            80           48
##   Credit-Builder Loan              72            48           40
##   Debt Consolidation Loan          24            24           16
##   Home Equity Loan                 64            32           72
##   Mortgage Loan                    32            48           24
##   No Loan                           0             0            0
##   Not Specified                    64            32           56
##   Payday Loan                      88            40           72
##   Personal Loan                    48            16           24
##   Student Loan                     40            64           32
##
## $Last_Loan_3_Last_Loan_9$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    6808                    7320                    6928
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    6640                    6872                   38248
##           Not Specified             Payday Loan           Personal Loan
##                    7080                    6768                    6408
##            Student Loan
##                    6928
##
## $Last_Loan_3_Last_Loan_9$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     448                     432                     392
##        Home Equity Loan           Mortgage Loan                 No Loan
```

```
##                     408                     520                   96144
##           Not Specified             Payday Loan           Personal Loan
##                     424                     464                     384
##            Student Loan
##                     384
##
## $Last_Loan_3_Last_Loan_9$grand_total
## [1] 100000
##
## $Last_Loan_3_Last_Loan_9$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.040               0.032                   0.088
##   Credit-Builder Loan        0.072               0.064                   0.040
##   Debt Consolidation Loan    0.032               0.048                   0.048
##   Home Equity Loan           0.048               0.048                   0.008
##   Mortgage Loan              0.080               0.024                   0.096
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.056               0.064                   0.008
##   Payday Loan                0.032               0.040                   0.024
##   Personal Loan              0.040               0.056                   0.040
##   Student Loan               0.048               0.056                   0.040
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.032         0.048   6.344         0.064
##   Credit-Builder Loan                0.064         0.040   6.832         0.048
##   Debt Consolidation Loan            0.040         0.120   6.544         0.032
##   Home Equity Loan                   0.032         0.056   6.240         0.040
##   Mortgage Loan                      0.056         0.032   6.440         0.040
##   No Loan                            0.000         0.000  38.248         0.000
##   Not Specified                      0.088         0.072   6.632         0.008
##   Payday Loan                        0.016         0.072   6.328         0.056
##   Personal Loan                      0.064         0.040   6.016         0.064
##   Student Loan                       0.016         0.040   6.520         0.072
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.032         0.080        0.048
##   Credit-Builder Loan          0.072         0.048        0.040
##   Debt Consolidation Loan      0.024         0.024        0.016
##   Home Equity Loan             0.064         0.032        0.072
##   Mortgage Loan                0.032         0.048        0.024
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.064         0.032        0.056
##   Payday Loan                  0.088         0.040        0.072
##   Personal Loan                0.048         0.016        0.024
##   Student Loan                 0.040         0.064        0.032
##
## $Last_Loan_3_Last_Loan_9$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                0.5875441           0.4700353
```

```
##   Credit-Builder Loan       0.9836066           0.8743169
##   Debt Consolidation Loan   0.4618938           0.6928406
##   Home Equity Loan          0.7228916           0.7228916
##   Mortgage Loan             1.1641444           0.3492433
##   No Loan                   0.0000000           0.0000000
##   Not Specified             0.7909605           0.9039548
##   Payday Loan               0.4728132           0.5910165
##   Personal Loan             0.6242197           0.8739076
##   Student Loan              0.6928406           0.8083141
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                               1.2925969        0.4700353
##   Credit-Builder Loan                     0.5464481        0.8743169
##   Debt Consolidation Loan                 0.6928406        0.5773672
##   Home Equity Loan                        0.1204819        0.4819277
##   Mortgage Loan                           1.3969732        0.8149010
##   No Loan                                 0.0000000        0.0000000
##   Not Specified                           0.1129944        1.2429379
##   Payday Loan                             0.3546099        0.2364066
##   Personal Loan                           0.6242197        0.9987516
##   Student Loan                            0.5773672        0.2309469
##
##                           Mortgage Loan     No Loan Not Specified Payday Loan
##   Auto Loan                     0.7050529  93.1844888     0.9400705   0.4700353
##   Credit-Builder Loan           0.5464481  93.3333333     0.6557377   0.9836066
##   Debt Consolidation Loan       1.7321016  94.4572748     0.4618938   0.3464203
##   Home Equity Loan              0.8433735  93.9759036     0.6024096   0.9638554
##   Mortgage Loan                 0.4656577  93.7136205     0.5820722   0.4656577
##   No Loan                       0.0000000 100.0000000     0.0000000   0.0000000
##   Not Specified                 1.0169492  93.6723164     0.1129944   0.9039548
##   Payday Loan                   1.0638298  93.4988180     0.8274232   1.3002364
##   Personal Loan                 0.6242197  93.8826467     0.9987516   0.7490637
##   Student Loan                  0.5773672  94.1108545     1.0392610   0.5773672
##
##                           Personal Loan Student Loan
##   Auto Loan                     1.1750881    0.7050529
##   Credit-Builder Loan           0.6557377    0.5464481
##   Debt Consolidation Loan       0.3464203    0.2309469
##   Home Equity Loan              0.4819277    1.0843373
##   Mortgage Loan                 0.6984866    0.3492433
##   No Loan                       0.0000000    0.0000000
##   Not Specified                 0.4519774    0.7909605
##   Payday Loan                   0.5910165    1.0638298
##   Personal Loan                 0.2496879    0.3745318
##   Student Loan                  0.9237875    0.4618938
##
## $Last_Loan_3_Last_Loan_9$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  8.928571            7.407407               22.448980
##   Credit-Builder Loan       16.071429           14.814815               10.204082
```

```
##   Debt Consolidation Loan  7.142857           11.111111                12.244898
##   Home Equity Loan        10.714286           11.111111                 2.040816
##   Mortgage Loan           17.857143            5.555556                24.489796
##   No Loan                  0.000000            0.000000                 0.000000
##   Not Specified           12.500000           14.814815                 2.040816
##   Payday Loan              7.142857            9.259259                 6.122449
##   Personal Loan            8.928571           12.962963                10.204082
##   Student Loan            10.714286           12.962963                10.204082
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                        7.843137      9.230769  6.598436
##   Credit-Builder Loan             15.686275      7.692308  7.106008
##   Debt Consolidation Loan          9.803922     23.076923  6.806457
##   Home Equity Loan                 7.843137     10.769231  6.490265
##   Mortgage Loan                   13.725490      6.153846  6.698286
##   No Loan                          0.000000      0.000000 39.781994
##   Not Specified                   21.568627     13.846154  6.897986
##   Payday Loan                      3.921569     13.846154  6.581794
##   Personal Loan                   15.686275      7.692308  6.257281
##   Student Loan                     3.921569      7.692308  6.781494
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    15.094340    6.896552     20.833333    12.500000
##   Credit-Builder Loan          11.320755   15.517241     12.500000    10.416667
##   Debt Consolidation Loan       7.547170    5.172414      6.250000     4.166667
##   Home Equity Loan              9.433962   13.793103      8.333333    18.750000
##   Mortgage Loan                 9.433962    6.896552     12.500000     6.250000
##   No Loan                       0.000000    0.000000      0.000000     0.000000
##   Not Specified                 1.886792   13.793103      8.333333    14.583333
##   Payday Loan                  13.207547   18.965517     10.416667    18.750000
##   Personal Loan                15.094340   10.344828      4.166667     6.250000
##   Student Loan                 16.981132    8.620690     16.666667     8.333333
##
##
## $Last_Loan_4_Last_Loan_5
## $Last_Loan_4_Last_Loan_5$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     320                 392                     400
##   Credit-Builder Loan           464                 520                     344
##   Debt Consolidation Loan       320                 352                     416
##   Home Equity Loan              344                 440                     344
##   Mortgage Loan                 512                 328                     336
##   No Loan                         0                   0                       0
##   Not Specified                 368                 360                     448
##   Payday Loan                   424                 392                     408
##   Personal Loan                 320                 408                     408
##   Student Loan                  352                 304                     424
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            320           432    1592           232
```

```
##   Credit-Builder Loan                 448           392    1920           336
##   Debt Consolidation Loan             432           408    1664           384
##   Home Equity Loan                    416           432    1640           392
##   Mortgage Loan                       336           360    1760           440
##   No Loan                               0             0   54000             0
##   Not Specified                       368           336    1640           424
##   Payday Loan                         368           384    1824           408
##   Personal Loan                       320           344    1712           320
##   Student Loan                        392           352    1704           336
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                       360           360          312
##   Credit-Builder Loan             384           464          392
##   Debt Consolidation Loan         368           384          304
##   Home Equity Loan                496           360          336
##   Mortgage Loan                   344           360          392
##   No Loan                           0             0            0
##   Not Specified                   440           384          352
##   Payday Loan                     344           448          288
##   Personal Loan                   280           336          408
##   Student Loan                    328           424          336
##
## $Last_Loan_4_Last_Loan_5$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    4720                    5664                    5032
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    5200                    5168                   54000
##           Not Specified             Payday Loan           Personal Loan
##                    5120                    5288                    4856
##            Student Loan
##                    4952
##
## $Last_Loan_4_Last_Loan_5$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    3424                    3496                    3528
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    3400                    3440                   69456
##           Not Specified             Payday Loan           Personal Loan
##                    3272                    3344                    3520
##            Student Loan
##                    3120
##
## $Last_Loan_4_Last_Loan_5$grand_total
## [1] 100000
##
## $Last_Loan_4_Last_Loan_5$cell_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                   0.320               0.392                    0.400
```

```
##   Credit-Builder Loan         0.464               0.520                    0.344
##   Debt Consolidation Loan     0.320               0.352                    0.416
##   Home Equity Loan            0.344               0.440                    0.344
##   Mortgage Loan               0.512               0.328                    0.336
##   No Loan                     0.000               0.000                    0.000
##   Not Specified               0.368               0.360                    0.448
##   Payday Loan                 0.424               0.392                    0.408
##   Personal Loan               0.320               0.408                    0.408
##   Student Loan                0.352               0.304                    0.424
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.320         0.432   1.592         0.232
##   Credit-Builder Loan                0.448         0.392   1.920         0.336
##   Debt Consolidation Loan            0.432         0.408   1.664         0.384
##   Home Equity Loan                   0.416         0.432   1.640         0.392
##   Mortgage Loan                      0.336         0.360   1.760         0.440
##   No Loan                            0.000         0.000  54.000         0.000
##   Not Specified                      0.368         0.336   1.640         0.424
##   Payday Loan                        0.368         0.384   1.824         0.408
##   Personal Loan                      0.320         0.344   1.712         0.320
##   Student Loan                       0.392         0.352   1.704         0.336
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                     0.360         0.360        0.312
##   Credit-Builder Loan           0.384         0.464        0.392
##   Debt Consolidation Loan       0.368         0.384        0.304
##   Home Equity Loan              0.496         0.360        0.336
##   Mortgage Loan                 0.344         0.360        0.392
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.440         0.384        0.352
##   Payday Loan                   0.344         0.448        0.288
##   Personal Loan                 0.280         0.336        0.408
##   Student Loan                  0.328         0.424        0.336
##
## $Last_Loan_4_Last_Loan_5$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 6.779661            8.305085
##   Credit-Builder Loan       8.192090            9.180791
##   Debt Consolidation Loan   6.359300            6.995231
##   Home Equity Loan          6.615385            8.461538
##   Mortgage Loan             9.907121            6.346749
##   No Loan                   0.000000            0.000000
##   Not Specified             7.187500            7.031250
##   Payday Loan               8.018154            7.413011
##   Personal Loan             6.589786            8.401977
##   Student Loan              7.108239            6.138934
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                8.474576         6.779661
##   Credit-Builder Loan                      6.073446         7.909605
```

```
##   Debt Consolidation Loan                8.267091         8.585056
##   Home Equity Loan                       6.615385         8.000000
##   Mortgage Loan                          6.501548         6.501548
##   No Loan                                0.000000         0.000000
##   Not Specified                          8.750000         7.187500
##   Payday Loan                            7.715582         6.959153
##   Personal Loan                          8.401977         6.589786
##   Student Loan                           8.562197         7.915994
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                    9.152542  33.728814      4.915254    7.627119
##   Credit-Builder Loan          6.920904  33.898305      5.932203    6.779661
##   Debt Consolidation Loan      8.108108  33.068362      7.631161    7.313196
##   Home Equity Loan             8.307692  31.538462      7.538462    9.538462
##   Mortgage Loan                6.965944  34.055728      8.513932    6.656347
##   No Loan                      0.000000 100.000000      0.000000    0.000000
##   Not Specified                6.562500  32.031250      8.281250    8.593750
##   Payday Loan                  7.261725  34.493192      7.715582    6.505295
##   Personal Loan                7.084020  35.255354      6.589786    5.766063
##   Student Loan                 7.108239  34.410339      6.785137    6.623586
##
##                           Personal Loan Student Loan
##   Auto Loan                    7.627119     6.610169
##   Credit-Builder Loan          8.192090     6.920904
##   Debt Consolidation Loan      7.631161     6.041335
##   Home Equity Loan             6.923077     6.461538
##   Mortgage Loan                6.965944     7.585139
##   No Loan                      0.000000     0.000000
##   Not Specified                7.500000     6.875000
##   Payday Loan                  8.472012     5.446293
##   Personal Loan                6.919275     8.401977
##   Student Loan                 8.562197     6.785137
##
## $Last_Loan_4_Last_Loan_5$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                 9.345794           11.212815               11.337868
##   Credit-Builder Loan      13.551402           14.874142                9.750567
##   Debt Consolidation Loan   9.345794           10.068650               11.791383
##   Home Equity Loan         10.046729           12.585812                9.750567
##   Mortgage Loan            14.953271            9.382151                9.523810
##   No Loan                   0.000000            0.000000                0.000000
##   Not Specified            10.747664           10.297483               12.698413
##   Payday Loan              12.383178           11.212815               11.564626
##   Personal Loan             9.345794           11.670481               11.564626
##   Student Loan             10.280374            8.695652               12.018141
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                         9.411765     12.558140  2.292099
##   Credit-Builder Loan              13.176471     11.395349  2.764340
##   Debt Consolidation Loan          12.705882     11.860465  2.395761
```

```
##   Home Equity Loan               12.235294     12.558140  2.361207
##   Mortgage Loan                   9.882353     10.465116  2.533978
##   No Loan                         0.000000      0.000000 77.747063
##   Not Specified                  10.823529      9.767442  2.361207
##   Payday Loan                    10.823529     11.162791  2.626123
##   Personal Loan                   9.411765     10.000000  2.464870
##   Student Loan                   11.529412     10.232558  2.453352
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                      7.090465   10.765550     10.227273    10.000000
##   Credit-Builder Loan           10.268949   11.483254     13.181818    12.564103
##   Debt Consolidation Loan       11.735941   11.004785     10.909091     9.743590
##   Home Equity Loan              11.980440   14.832536     10.227273    10.769231
##   Mortgage Loan                 13.447433   10.287081     10.227273    12.564103
##   No Loan                        0.000000    0.000000      0.000000     0.000000
##   Not Specified                 12.958435   13.157895     10.909091    11.282051
##   Payday Loan                   12.469438   10.287081     12.727273     9.230769
##   Personal Loan                  9.779951    8.373206      9.545455    13.076923
##   Student Loan                  10.268949    9.808612     12.045455    10.769231
##
##
## $Last_Loan_4_Last_Loan_6
## $Last_Loan_4_Last_Loan_6$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     256                 320                      280
##   Credit-Builder Loan           264                 320                      232
##   Debt Consolidation Loan       328                 296                      296
##   Home Equity Loan              328                 296                      216
##   Mortgage Loan                 256                 264                      328
##   No Loan                         0                   0                        0
##   Not Specified                 208                 376                      296
##   Payday Loan                   296                 280                      360
##   Personal Loan                 248                 248                      248
##   Student Loan                  288                 208                      208
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            240           208    2208           360
##   Credit-Builder Loan                  264           240    2864           336
##   Debt Consolidation Loan              264           256    2528           288
##   Home Equity Loan                     296           344    2464           248
##   Mortgage Loan                        272           224    2704           288
##   No Loan                                0             0   54000             0
##   Not Specified                        288           256    2392           304
##   Payday Loan                          312           264    2720           304
##   Personal Loan                        288           312    2472           256
##   Student Loan                         240           288    2632           240
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                       320           272          256
##   Credit-Builder Loan             472           376          296
```

```
##   Debt Consolidation Loan         208           288          280
##   Home Equity Loan                320           352          336
##   Mortgage Loan                   280           312          240
##   No Loan                           0             0            0
##   Not Specified                   344           336          320
##   Payday Loan                     288           216          248
##   Personal Loan                   192           312          280
##   Student Loan                    264           272          312
##
## $Last_Loan_4_Last_Loan_6$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    4720                    5664                    5032
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    5200                    5168                   54000
##           Not Specified             Payday Loan           Personal Loan
##                    5120                    5288                    4856
##            Student Loan
##                    4952
##
## $Last_Loan_4_Last_Loan_6$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    2472                    2608                    2464
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    2464                    2392                   76984
##           Not Specified             Payday Loan           Personal Loan
##                    2624                    2688                    2736
##            Student Loan
##                    2568
##
## $Last_Loan_4_Last_Loan_6$grand_total
## [1] 100000
##
## $Last_Loan_4_Last_Loan_6$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.256               0.320                   0.280
##   Credit-Builder Loan        0.264               0.320                   0.232
##   Debt Consolidation Loan    0.328               0.296                   0.296
##   Home Equity Loan           0.328               0.296                   0.216
##   Mortgage Loan              0.256               0.264                   0.328
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.208               0.376                   0.296
##   Payday Loan                0.296               0.280                   0.360
##   Personal Loan              0.248               0.248                   0.248
##   Student Loan               0.288               0.208                   0.208
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.240         0.208   2.208         0.360
##   Credit-Builder Loan                0.264         0.240   2.864         0.336
```

```
##   Debt Consolidation Loan            0.264         0.256   2.528         0.288
##   Home Equity Loan                   0.296         0.344   2.464         0.248
##   Mortgage Loan                      0.272         0.224   2.704         0.288
##   No Loan                            0.000         0.000  54.000         0.000
##   Not Specified                      0.288         0.256   2.392         0.304
##   Payday Loan                        0.312         0.264   2.720         0.304
##   Personal Loan                      0.288         0.312   2.472         0.256
##   Student Loan                       0.240         0.288   2.632         0.240
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                     0.320         0.272        0.256
##   Credit-Builder Loan           0.472         0.376        0.296
##   Debt Consolidation Loan       0.208         0.288        0.280
##   Home Equity Loan              0.320         0.352        0.336
##   Mortgage Loan                 0.280         0.312        0.240
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.344         0.336        0.320
##   Payday Loan                   0.288         0.216        0.248
##   Personal Loan                 0.192         0.312        0.280
##   Student Loan                  0.264         0.272        0.312
##
## $Last_Loan_4_Last_Loan_6$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 5.423729            6.779661
##   Credit-Builder Loan       4.661017            5.649718
##   Debt Consolidation Loan   6.518283            5.882353
##   Home Equity Loan          6.307692            5.692308
##   Mortgage Loan             4.953560            5.108359
##   No Loan                   0.000000            0.000000
##   Not Specified             4.062500            7.343750
##   Payday Loan               5.597579            5.295008
##   Personal Loan             5.107084            5.107084
##   Student Loan              5.815832            4.200323
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                5.932203         5.084746
##   Credit-Builder Loan                      4.096045         4.661017
##   Debt Consolidation Loan                  5.882353         5.246423
##   Home Equity Loan                         4.153846         5.692308
##   Mortgage Loan                            6.346749         5.263158
##   No Loan                                  0.000000         0.000000
##   Not Specified                            5.781250         5.625000
##   Payday Loan                              6.807867         5.900151
##   Personal Loan                            5.107084         5.930807
##   Student Loan                             4.200323         4.846527
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                     4.406780  46.779661      7.627119    6.779661
##   Credit-Builder Loan           4.237288  50.564972      5.932203    8.333333
##   Debt Consolidation Loan       5.087440  50.238474      5.723370    4.133545
```

```
##   Home Equity Loan             6.615385  47.384615      4.769231    6.153846
##   Mortgage Loan                4.334365  52.321981      5.572755    5.417957
##   No Loan                      0.000000 100.000000      0.000000    0.000000
##   Not Specified                5.000000  46.718750      5.937500    6.718750
##   Payday Loan                  4.992436  51.437216      5.748865    5.446293
##   Personal Loan                6.425041  50.906096      5.271829    3.953871
##   Student Loan                 5.815832  53.150242      4.846527    5.331179
##
##                           Personal Loan Student Loan
##   Auto Loan                    5.762712     5.423729
##   Credit-Builder Loan          6.638418     5.225989
##   Debt Consolidation Loan      5.723370     5.564388
##   Home Equity Loan             6.769231     6.461538
##   Mortgage Loan                6.037152     4.643963
##   No Loan                      0.000000     0.000000
##   Not Specified                6.562500     6.250000
##   Payday Loan                  4.084720     4.689864
##   Personal Loan                6.425041     5.766063
##   Student Loan                 5.492730     6.300485
##
## $Last_Loan_4_Last_Loan_6$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               10.355987           12.269939               11.363636
##   Credit-Builder Loan     10.679612           12.269939                9.415584
##   Debt Consolidation Loan 13.268608           11.349693               12.012987
##   Home Equity Loan        13.268608           11.349693                8.766234
##   Mortgage Loan           10.355987           10.122699               13.311688
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified            8.414239           14.417178               12.012987
##   Payday Loan             11.974110           10.736196               14.610390
##   Personal Loan           10.032362            9.509202               10.064935
##   Student Loan            11.650485            7.975460                8.441558
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                        9.740260      8.695652  2.868128
##   Credit-Builder Loan             10.714286     10.033445  3.720254
##   Debt Consolidation Loan         10.714286     10.702341  3.283799
##   Home Equity Loan                12.012987     14.381271  3.200665
##   Mortgage Loan                   11.038961      9.364548  3.512418
##   No Loan                          0.000000      0.000000 70.144446
##   Not Specified                   11.688312     10.702341  3.107139
##   Payday Loan                     12.662338     11.036789  3.533202
##   Personal Loan                   11.688312     13.043478  3.211057
##   Student Loan                     9.740260     12.040134  3.418892
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    13.719512   11.904762      9.941520     9.968847
##   Credit-Builder Loan          12.804878   17.559524     13.742690    11.526480
##   Debt Consolidation Loan      10.975610    7.738095     10.526316    10.903427
##   Home Equity Loan              9.451220   11.904762     12.865497    13.084112
```

```
##   Mortgage Loan               10.975610   10.416667     11.403509     9.345794
##   No Loan                      0.000000    0.000000      0.000000     0.000000
##   Not Specified               11.585366   12.797619     12.280702    12.461059
##   Payday Loan                 11.585366   10.714286      7.894737     9.657321
##   Personal Loan                9.756098    7.142857     11.403509    10.903427
##   Student Loan                 9.146341    9.821429      9.941520    12.149533
##
##
## $Last_Loan_4_Last_Loan_7
## $Last_Loan_4_Last_Loan_7$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                    176                 224                     112
##   Credit-Builder Loan          304                 168                     128
##   Debt Consolidation Loan      160                 192                     168
##   Home Equity Loan             192                 232                     176
##   Mortgage Loan                200                 192                     208
##   No Loan                        0                   0                       0
##   Not Specified                160                 160                     200
##   Payday Loan                  160                 136                     184
##   Personal Loan                168                 184                     144
##   Student Loan                  80                 232                     112
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                           160           232    3080           120
##   Credit-Builder Loan                 184           200    3808           200
##   Debt Consolidation Loan             224           192    3376           136
##   Home Equity Loan                    152           208    3392           200
##   Mortgage Loan                       160           160    3464           184
##   No Loan                               0             0   54000             0
##   Not Specified                       184           240    3424           240
##   Payday Loan                         192           192    3632           232
##   Personal Loan                       128           208    3328           168
##   Student Loan                        144           160    3624           128
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                      208           240          168
##   Credit-Builder Loan            272           224          176
##   Debt Consolidation Loan        208           200          176
##   Home Equity Loan               272           176          200
##   Mortgage Loan                  184           168          248
##   No Loan                          0             0            0
##   Not Specified                  192           176          144
##   Payday Loan                    208           160          192
##   Personal Loan                  176           192          160
##   Student Loan                   200           152          120
##
## $Last_Loan_4_Last_Loan_7$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    4720                    5664                    5032
```

```
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    5200                    5168                   54000
##           Not Specified             Payday Loan           Personal Loan
##                    5120                    5288                    4856
##            Student Loan
##                    4952
##
## $Last_Loan_4_Last_Loan_7$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    1600                    1720                    1432
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    1528                    1792                   85128
##           Not Specified             Payday Loan           Personal Loan
##                    1608                    1920                    1688
##            Student Loan
##                    1584
##
## $Last_Loan_4_Last_Loan_7$grand_total
## [1] 100000
##
## $Last_Loan_4_Last_Loan_7$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.176               0.224                   0.112
##   Credit-Builder Loan        0.304               0.168                   0.128
##   Debt Consolidation Loan    0.160               0.192                   0.168
##   Home Equity Loan           0.192               0.232                   0.176
##   Mortgage Loan              0.200               0.192                   0.208
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.160               0.160                   0.200
##   Payday Loan                0.160               0.136                   0.184
##   Personal Loan              0.168               0.184                   0.144
##   Student Loan               0.080               0.232                   0.112
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.160         0.232   3.080         0.120
##   Credit-Builder Loan                0.184         0.200   3.808         0.200
##   Debt Consolidation Loan            0.224         0.192   3.376         0.136
##   Home Equity Loan                   0.152         0.208   3.392         0.200
##   Mortgage Loan                      0.160         0.160   3.464         0.184
##   No Loan                            0.000         0.000  54.000         0.000
##   Not Specified                      0.184         0.240   3.424         0.240
##   Payday Loan                        0.192         0.192   3.632         0.232
##   Personal Loan                      0.128         0.208   3.328         0.168
##   Student Loan                       0.144         0.160   3.624         0.128
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.208         0.240        0.168
##   Credit-Builder Loan          0.272         0.224        0.176
##   Debt Consolidation Loan      0.208         0.200        0.176
```

```
##   Home Equity Loan              0.272         0.176        0.200
##   Mortgage Loan                 0.184         0.168        0.248
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.192         0.176        0.144
##   Payday Loan                   0.208         0.160        0.192
##   Personal Loan                 0.176         0.192        0.160
##   Student Loan                  0.200         0.152        0.120
##
## $Last_Loan_4_Last_Loan_7$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 3.728814            4.745763
##   Credit-Builder Loan       5.367232            2.966102
##   Debt Consolidation Loan   3.179650            3.815580
##   Home Equity Loan          3.692308            4.461538
##   Mortgage Loan             3.869969            3.715170
##   No Loan                   0.000000            0.000000
##   Not Specified             3.125000            3.125000
##   Payday Loan               3.025719            2.571861
##   Personal Loan             3.459638            3.789127
##   Student Loan              1.615509            4.684976
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                2.372881         3.389831
##   Credit-Builder Loan                      2.259887         3.248588
##   Debt Consolidation Loan                  3.338633         4.451510
##   Home Equity Loan                         3.384615         2.923077
##   Mortgage Loan                            4.024768         3.095975
##   No Loan                                  0.000000         0.000000
##   Not Specified                            3.906250         3.593750
##   Payday Loan                              3.479576         3.630862
##   Personal Loan                            2.965404         2.635914
##   Student Loan                             2.261712         2.907916
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                     4.915254  65.254237      2.542373    4.406780
##   Credit-Builder Loan           3.531073  67.231638      3.531073    4.802260
##   Debt Consolidation Loan       3.815580  67.090620      2.702703    4.133545
##   Home Equity Loan              4.000000  65.230769      3.846154    5.230769
##   Mortgage Loan                 3.095975  67.027864      3.560372    3.560372
##   No Loan                       0.000000 100.000000      0.000000    0.000000
##   Not Specified                 4.687500  66.875000      4.687500    3.750000
##   Payday Loan                   3.630862  68.683812      4.387292    3.933434
##   Personal Loan                 4.283361  68.533773      3.459638    3.624382
##   Student Loan                  3.231018  73.182553      2.584814    4.038772
##
##                           Personal Loan Student Loan
##   Auto Loan                     5.084746     3.559322
##   Credit-Builder Loan           3.954802     3.107345
##   Debt Consolidation Loan       3.974563     3.497615
##   Home Equity Loan              3.384615     3.846154
```

```
##   Mortgage Loan                3.250774     4.798762
##   No Loan                      0.000000     0.000000
##   Not Specified                3.437500     2.812500
##   Payday Loan                  3.025719     3.630862
##   Personal Loan                3.953871     3.294893
##   Student Loan                 3.069467     2.423263
##
## $Last_Loan_4_Last_Loan_7$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               11.000000           13.023256                7.821229
##   Credit-Builder Loan     19.000000            9.767442                8.938547
##   Debt Consolidation Loan 10.000000           11.162791               11.731844
##   Home Equity Loan        12.000000           13.488372               12.290503
##   Mortgage Loan           12.500000           11.162791               14.525140
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           10.000000            9.302326               13.966480
##   Payday Loan             10.000000            7.906977               12.849162
##   Personal Loan           10.500000           10.697674               10.055866
##   Student Loan             5.000000           13.488372                7.821229
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                       10.471204     12.946429  3.618081
##   Credit-Builder Loan             12.041885     11.160714  4.473264
##   Debt Consolidation Loan         14.659686     10.714286  3.965793
##   Home Equity Loan                 9.947644     11.607143  3.984588
##   Mortgage Loan                   10.471204      8.928571  4.069166
##   No Loan                          0.000000      0.000000 63.433888
##   Not Specified                   12.041885     13.392857  4.022178
##   Payday Loan                     12.565445     10.714286  4.266516
##   Personal Loan                    8.376963     11.607143  3.909407
##   Student Loan                     9.424084      8.928571  4.257119
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    7.462687   10.833333     14.218009    10.606061
##   Credit-Builder Loan         12.437811   14.166667     13.270142    11.111111
##   Debt Consolidation Loan      8.457711   10.833333     11.848341    11.111111
##   Home Equity Loan            12.437811   14.166667     10.426540    12.626263
##   Mortgage Loan               11.442786    9.583333      9.952607    15.656566
##   No Loan                      0.000000    0.000000      0.000000     0.000000
##   Not Specified               14.925373   10.000000     10.426540     9.090909
##   Payday Loan                 14.427861   10.833333      9.478673    12.121212
##   Personal Loan               10.447761    9.166667     11.374408    10.101010
##   Student Loan                 7.960199   10.416667      9.004739     7.575758
##
##
## $Last_Loan_4_Last_Loan_8
## $Last_Loan_4_Last_Loan_8$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                      96                 136                      72
```

```
##   Credit-Builder Loan          104                  88                      80
##   Debt Consolidation Loan       48                  72                      80
##   Home Equity Loan              48                  96                      64
##   Mortgage Loan                 32                 136                     104
##   No Loan                        0                   0                       0
##   Not Specified                120                  72                      32
##   Payday Loan                  128                  72                     112
##   Personal Loan                 72                  96                      88
##   Student Loan                  72                 104                     128
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                             96            72    3832            88
##   Credit-Builder Loan                  192            96    4704           104
##   Debt Consolidation Loan              104           104    4288            32
##   Home Equity Loan                      96           104    4400           104
##   Mortgage Loan                         72           104    4328           128
##   No Loan                                0             0   54000             0
##   Not Specified                         56            96    4400            72
##   Payday Loan                          128            64    4432            72
##   Personal Loan                         96            56    4128            64
##   Student Loan                          48            40    4296            80
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                       96           104          128
##   Credit-Builder Loan            128            96           72
##   Debt Consolidation Loan         88            64          152
##   Home Equity Loan               120            88           80
##   Mortgage Loan                   96            72           96
##   No Loan                          0             0            0
##   Not Specified                   80            96           96
##   Payday Loan                    136            80           64
##   Personal Loan                   72            88           96
##   Student Loan                    72            32           80
##
## $Last_Loan_4_Last_Loan_8$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    4720                    5664                    5032
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    5200                    5168                   54000
##           Not Specified             Payday Loan           Personal Loan
##                    5120                    5288                    4856
##            Student Loan
##                    4952
##
## $Last_Loan_4_Last_Loan_8$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     720                     872                     760
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     888                     736                   92808
```

```
##           Not Specified             Payday Loan           Personal Loan
##                     744                     888                     720
##            Student Loan
##                     864
##
## $Last_Loan_4_Last_Loan_8$grand_total
## [1] 100000
##
## $Last_Loan_4_Last_Loan_8$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.096               0.136                   0.072
##   Credit-Builder Loan        0.104               0.088                   0.080
##   Debt Consolidation Loan    0.048               0.072                   0.080
##   Home Equity Loan           0.048               0.096                   0.064
##   Mortgage Loan              0.032               0.136                   0.104
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.120               0.072                   0.032
##   Payday Loan                0.128               0.072                   0.112
##   Personal Loan              0.072               0.096                   0.088
##   Student Loan               0.072               0.104                   0.128
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.096         0.072   3.832         0.088
##   Credit-Builder Loan                0.192         0.096   4.704         0.104
##   Debt Consolidation Loan            0.104         0.104   4.288         0.032
##   Home Equity Loan                   0.096         0.104   4.400         0.104
##   Mortgage Loan                      0.072         0.104   4.328         0.128
##   No Loan                            0.000         0.000  54.000         0.000
##   Not Specified                      0.056         0.096   4.400         0.072
##   Payday Loan                        0.128         0.064   4.432         0.072
##   Personal Loan                      0.096         0.056   4.128         0.064
##   Student Loan                       0.048         0.040   4.296         0.080
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.096         0.104        0.128
##   Credit-Builder Loan          0.128         0.096        0.072
##   Debt Consolidation Loan      0.088         0.064        0.152
##   Home Equity Loan             0.120         0.088        0.080
##   Mortgage Loan                0.096         0.072        0.096
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.080         0.096        0.096
##   Payday Loan                  0.136         0.080        0.064
##   Personal Loan                0.072         0.088        0.096
##   Student Loan                 0.072         0.032        0.080
##
## $Last_Loan_4_Last_Loan_8$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                2.0338983           2.8813559
##   Credit-Builder Loan      1.8361582           1.5536723
```

```
##   Debt Consolidation Loan   0.9538951           1.4308426
##   Home Equity Loan          0.9230769           1.8461538
##   Mortgage Loan             0.6191950           2.6315789
##   No Loan                   0.0000000           0.0000000
##   Not Specified             2.3437500           1.4062500
##   Payday Loan               2.4205749           1.3615734
##   Personal Loan             1.4827018           1.9769357
##   Student Loan              1.4539580           2.1001616
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                             1.5254237        2.0338983
##   Credit-Builder Loan                   1.4124294        3.3898305
##   Debt Consolidation Loan               1.5898251        2.0667727
##   Home Equity Loan                      1.2307692        1.8461538
##   Mortgage Loan                         2.0123839        1.3931889
##   No Loan                               0.0000000        0.0000000
##   Not Specified                         0.6250000        1.0937500
##   Payday Loan                           2.1180030        2.4205749
##   Personal Loan                         1.8121911        1.9769357
##   Student Loan                          2.5848142        0.9693053
##
##                           Mortgage Loan     No Loan Not Specified Payday Loan
##   Auto Loan                    1.5254237  81.1864407     1.8644068   2.0338983
##   Credit-Builder Loan          1.6949153  83.0508475     1.8361582   2.2598870
##   Debt Consolidation Loan      2.0667727  85.2146264     0.6359300   1.7488076
##   Home Equity Loan             2.0000000  84.6153846     2.0000000   2.3076923
##   Mortgage Loan                2.0123839  83.7461300     2.4767802   1.8575851
##   No Loan                      0.0000000 100.0000000     0.0000000   0.0000000
##   Not Specified                1.8750000  85.9375000     1.4062500   1.5625000
##   Payday Loan                  1.2102874  83.8124054     1.3615734   2.5718608
##   Personal Loan                1.1532125  85.0082372     1.3179572   1.4827018
##   Student Loan                 0.8077544  86.7528271     1.6155089   1.4539580
##
##                           Personal Loan Student Loan
##   Auto Loan                    2.2033898    2.7118644
##   Credit-Builder Loan          1.6949153    1.2711864
##   Debt Consolidation Loan      1.2718601    3.0206677
##   Home Equity Loan             1.6923077    1.5384615
##   Mortgage Loan                1.3931889    1.8575851
##   No Loan                      0.0000000    0.0000000
##   Not Specified                1.8750000    1.8750000
##   Payday Loan                  1.5128593    1.2102874
##   Personal Loan                1.8121911    1.9769357
##   Student Loan                 0.6462036    1.6155089
##
## $Last_Loan_4_Last_Loan_8$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               13.333333           15.596330                9.473684
##   Credit-Builder Loan     14.444444           10.091743               10.526316
##   Debt Consolidation Loan  6.666667            8.256881               10.526316
```

```
##   Home Equity Loan         6.666667           11.009174                8.421053
##   Mortgage Loan            4.444444           15.596330               13.684211
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           16.666667            8.256881                4.210526
##   Payday Loan             17.777778            8.256881               14.736842
##   Personal Loan           10.000000           11.009174               11.578947
##   Student Loan            10.000000           11.926606               16.842105
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                       10.810811      9.782609  4.128954
##   Credit-Builder Loan             21.621622     13.043478  5.068529
##   Debt Consolidation Loan         11.711712     14.130435  4.620291
##   Home Equity Loan                10.810811     14.130435  4.740971
##   Mortgage Loan                    8.108108     14.130435  4.663391
##   No Loan                          0.000000      0.000000 58.184639
##   Not Specified                    6.306306     13.043478  4.740971
##   Payday Loan                     14.414414      8.695652  4.775450
##   Personal Loan                   10.810811      7.608696  4.447892
##   Student Loan                     5.405405      5.434783  4.628911
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    11.827957   10.810811     14.444444    14.814815
##   Credit-Builder Loan          13.978495   14.414414     13.333333     8.333333
##   Debt Consolidation Loan       4.301075    9.909910      8.888889    17.592593
##   Home Equity Loan             13.978495   13.513514     12.222222     9.259259
##   Mortgage Loan                17.204301   10.810811     10.000000    11.111111
##   No Loan                       0.000000    0.000000      0.000000     0.000000
##   Not Specified                 9.677419    9.009009     13.333333    11.111111
##   Payday Loan                   9.677419   15.315315     11.111111     7.407407
##   Personal Loan                 8.602151    8.108108     12.222222    11.111111
##   Student Loan                 10.752688    8.108108      4.444444     9.259259
##
##
## $Last_Loan_4_Last_Loan_9
## $Last_Loan_4_Last_Loan_9$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                      80                  40                      48
##   Credit-Builder Loan            48                  40                      32
##   Debt Consolidation Loan        48                  40                      24
##   Home Equity Loan               48                  80                      48
##   Mortgage Loan                  64                  40                      48
##   No Loan                         0                   0                       0
##   Not Specified                  48                  72                      64
##   Payday Loan                    56                  56                      32
##   Personal Loan                  24                  32                      40
##   Student Loan                   32                  32                      56
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                             24            48    4240            72
##   Credit-Builder Loan                   48            72    5160            88
```

```
##   Debt Consolidation Loan               56            32    4672            32
##   Home Equity Loan                      64            40    4784            32
##   Mortgage Loan                         40            80    4704            24
##   No Loan                                0             0   54000             0
##   Not Specified                         40            80    4728            24
##   Payday Loan                           16            40    4808            64
##   Personal Loan                         56            88    4448            48
##   Student Loan                          64            40    4600            40
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                        88            32           48
##   Credit-Builder Loan              80            40           56
##   Debt Consolidation Loan          32            48           48
##   Home Equity Loan                 48            40           16
##   Mortgage Loan                    72            64           32
##   No Loan                           0             0            0
##   Not Specified                    16            32           16
##   Payday Loan                      64            72           80
##   Personal Loan                    32            40           48
##   Student Loan                     32            16           40
##
## $Last_Loan_4_Last_Loan_9$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    4720                    5664                    5032
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    5200                    5168                   54000
##           Not Specified             Payday Loan           Personal Loan
##                    5120                    5288                    4856
##            Student Loan
##                    4952
##
## $Last_Loan_4_Last_Loan_9$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     448                     432                     392
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     408                     520                   96144
##           Not Specified             Payday Loan           Personal Loan
##                     424                     464                     384
##            Student Loan
##                     384
##
## $Last_Loan_4_Last_Loan_9$grand_total
## [1] 100000
##
## $Last_Loan_4_Last_Loan_9$cell_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                   0.080               0.040                   0.048
##   Credit-Builder Loan         0.048               0.040                   0.032
```

```
##   Debt Consolidation Loan     0.048               0.040                    0.024
##   Home Equity Loan            0.048               0.080                    0.048
##   Mortgage Loan               0.064               0.040                    0.048
##   No Loan                     0.000               0.000                    0.000
##   Not Specified               0.048               0.072                    0.064
##   Payday Loan                 0.056               0.056                    0.032
##   Personal Loan               0.024               0.032                    0.040
##   Student Loan                0.032               0.032                    0.056
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.024         0.048   4.240         0.072
##   Credit-Builder Loan                0.048         0.072   5.160         0.088
##   Debt Consolidation Loan            0.056         0.032   4.672         0.032
##   Home Equity Loan                   0.064         0.040   4.784         0.032
##   Mortgage Loan                      0.040         0.080   4.704         0.024
##   No Loan                            0.000         0.000  54.000         0.000
##   Not Specified                      0.040         0.080   4.728         0.024
##   Payday Loan                        0.016         0.040   4.808         0.064
##   Personal Loan                      0.056         0.088   4.448         0.048
##   Student Loan                       0.064         0.040   4.600         0.040
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                     0.088         0.032        0.048
##   Credit-Builder Loan           0.080         0.040        0.056
##   Debt Consolidation Loan       0.032         0.048        0.048
##   Home Equity Loan              0.048         0.040        0.016
##   Mortgage Loan                 0.072         0.064        0.032
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.016         0.032        0.016
##   Payday Loan                   0.064         0.072        0.080
##   Personal Loan                 0.032         0.040        0.048
##   Student Loan                  0.032         0.016        0.040
##
## $Last_Loan_4_Last_Loan_9$row_percentage
##
##                             Auto Loan Credit-Builder Loan
##   Auto Loan                 1.6949153           0.8474576
##   Credit-Builder Loan       0.8474576           0.7062147
##   Debt Consolidation Loan   0.9538951           0.7949126
##   Home Equity Loan          0.9230769           1.5384615
##   Mortgage Loan             1.2383901           0.7739938
##   No Loan                   0.0000000           0.0000000
##   Not Specified             0.9375000           1.4062500
##   Payday Loan               1.0590015           1.0590015
##   Personal Loan             0.4942339           0.6589786
##   Student Loan              0.6462036           0.6462036
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                               1.0169492        0.5084746
##   Credit-Builder Loan                     0.5649718        0.8474576
##   Debt Consolidation Loan                 0.4769475        1.1128776
```

```
##   Home Equity Loan                      0.9230769        1.2307692
##   Mortgage Loan                         0.9287926        0.7739938
##   No Loan                               0.0000000        0.0000000
##   Not Specified                         1.2500000        0.7812500
##   Payday Loan                           0.6051437        0.3025719
##   Personal Loan                         0.8237232        1.1532125
##   Student Loan                          1.1308562        1.2924071
##
##                           Mortgage Loan     No Loan Not Specified Payday Loan
##   Auto Loan                   1.0169492  89.8305085     1.5254237   1.8644068
##   Credit-Builder Loan         1.2711864  91.1016949     1.5536723   1.4124294
##   Debt Consolidation Loan     0.6359300  92.8457870     0.6359300   0.6359300
##   Home Equity Loan            0.7692308  92.0000000     0.6153846   0.9230769
##   Mortgage Loan               1.5479876  91.0216718     0.4643963   1.3931889
##   No Loan                     0.0000000 100.0000000     0.0000000   0.0000000
##   Not Specified               1.5625000  92.3437500     0.4687500   0.3125000
##   Payday Loan                 0.7564297  90.9228442     1.2102874   1.2102874
##   Personal Loan               1.8121911  91.5980231     0.9884679   0.6589786
##   Student Loan                0.8077544  92.8917609     0.8077544   0.6462036
##
##                           Personal Loan Student Loan
##   Auto Loan                   0.6779661    1.0169492
##   Credit-Builder Loan         0.7062147    0.9887006
##   Debt Consolidation Loan     0.9538951    0.9538951
##   Home Equity Loan            0.7692308    0.3076923
##   Mortgage Loan               1.2383901    0.6191950
##   No Loan                     0.0000000    0.0000000
##   Not Specified               0.6250000    0.3125000
##   Payday Loan                 1.3615734    1.5128593
##   Personal Loan               0.8237232    0.9884679
##   Student Loan                0.3231018    0.8077544
##
## $Last_Loan_4_Last_Loan_9$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               17.857143            9.259259               12.244898
##   Credit-Builder Loan     10.714286            9.259259                8.163265
##   Debt Consolidation Loan 10.714286            9.259259                6.122449
##   Home Equity Loan        10.714286           18.518519               12.244898
##   Mortgage Loan           14.285714            9.259259               12.244898
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           10.714286           16.666667               16.326531
##   Payday Loan             12.500000           12.962963                8.163265
##   Personal Loan            5.357143            7.407407               10.204082
##   Student Loan             7.142857            7.407407               14.285714
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                        5.882353      9.230769  4.410052
##   Credit-Builder Loan             11.764706     13.846154  5.366950
##   Debt Consolidation Loan         13.725490      6.153846  4.859378
##   Home Equity Loan                15.686275      7.692308  4.975870
```

```
##   Mortgage Loan                   9.803922     15.384615  4.892661
##   No Loan                         0.000000      0.000000 56.165751
##   Not Specified                   9.803922     15.384615  4.917624
##   Payday Loan                     3.921569      7.692308  5.000832
##   Personal Loan                  13.725490     16.923077  4.626394
##   Student Loan                   15.686275      7.692308  4.784490
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    16.981132   18.965517      8.333333    12.500000
##   Credit-Builder Loan          20.754717   17.241379     10.416667    14.583333
##   Debt Consolidation Loan       7.547170    6.896552     12.500000    12.500000
##   Home Equity Loan              7.547170   10.344828     10.416667     4.166667
##   Mortgage Loan                 5.660377   15.517241     16.666667     8.333333
##   No Loan                       0.000000    0.000000      0.000000     0.000000
##   Not Specified                 5.660377    3.448276      8.333333     4.166667
##   Payday Loan                  15.094340   13.793103     18.750000    20.833333
##   Personal Loan                11.320755    6.896552     10.416667    12.500000
##   Student Loan                  9.433962    6.896552      4.166667    10.416667
##
##
## $Last_Loan_5_Last_Loan_6
## $Last_Loan_5_Last_Loan_6$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     240                 304                     280
##   Credit-Builder Loan           280                 336                     272
##   Debt Consolidation Loan       216                 336                     304
##   Home Equity Loan              304                 288                     256
##   Mortgage Loan                 272                 272                     184
##   No Loan                         0                   0                       0
##   Not Specified                 200                 232                     368
##   Payday Loan                   440                 248                     280
##   Personal Loan                 240                 216                     224
##   Student Loan                  280                 376                     296
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            328           312     880           216
##   Credit-Builder Loan                  296           232     720           336
##   Debt Consolidation Loan              368           256     888           264
##   Home Equity Loan                     208           288     856           352
##   Mortgage Loan                        280           344     872           312
##   No Loan                                0             0   69456             0
##   Not Specified                        272           240     832           320
##   Payday Loan                          272           224     816           192
##   Personal Loan                        232           296     960           320
##   Student Loan                         208           200     704           312
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                       296           288          280
##   Credit-Builder Loan             328           408          288
##   Debt Consolidation Loan         296           256          344
```

```
##   Home Equity Loan                272           336          240
##   Mortgage Loan                   264           304          336
##   No Loan                           0             0            0
##   Not Specified                   328           296          184
##   Payday Loan                     304           264          304
##   Personal Loan                   376           328          328
##   Student Loan                    224           256          264
##
## $Last_Loan_5_Last_Loan_6$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    3424                    3496                    3528
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    3400                    3440                   69456
##           Not Specified             Payday Loan           Personal Loan
##                    3272                    3344                    3520
##            Student Loan
##                    3120
##
## $Last_Loan_5_Last_Loan_6$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    2472                    2608                    2464
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    2464                    2392                   76984
##           Not Specified             Payday Loan           Personal Loan
##                    2624                    2688                    2736
##            Student Loan
##                    2568
##
## $Last_Loan_5_Last_Loan_6$grand_total
## [1] 100000
##
## $Last_Loan_5_Last_Loan_6$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.240               0.304                   0.280
##   Credit-Builder Loan        0.280               0.336                   0.272
##   Debt Consolidation Loan    0.216               0.336                   0.304
##   Home Equity Loan           0.304               0.288                   0.256
##   Mortgage Loan              0.272               0.272                   0.184
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.200               0.232                   0.368
##   Payday Loan                0.440               0.248                   0.280
##   Personal Loan              0.240               0.216                   0.224
##   Student Loan               0.280               0.376                   0.296
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.328         0.312   0.880         0.216
##   Credit-Builder Loan                0.296         0.232   0.720         0.336
##   Debt Consolidation Loan            0.368         0.256   0.888         0.264
```

```
##   Home Equity Loan                  0.208         0.288   0.856         0.352
##   Mortgage Loan                     0.280         0.344   0.872         0.312
##   No Loan                           0.000         0.000  69.456         0.000
##   Not Specified                     0.272         0.240   0.832         0.320
##   Payday Loan                       0.272         0.224   0.816         0.192
##   Personal Loan                     0.232         0.296   0.960         0.320
##   Student Loan                      0.208         0.200   0.704         0.312
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                     0.296         0.288        0.280
##   Credit-Builder Loan           0.328         0.408        0.288
##   Debt Consolidation Loan       0.296         0.256        0.344
##   Home Equity Loan              0.272         0.336        0.240
##   Mortgage Loan                 0.264         0.304        0.336
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.328         0.296        0.184
##   Payday Loan                   0.304         0.264        0.304
##   Personal Loan                 0.376         0.328        0.328
##   Student Loan                  0.224         0.256        0.264
##
## $Last_Loan_5_Last_Loan_6$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 7.009346            8.878505
##   Credit-Builder Loan       8.009153            9.610984
##   Debt Consolidation Loan   6.122449            9.523810
##   Home Equity Loan          8.941176            8.470588
##   Mortgage Loan             7.906977            7.906977
##   No Loan                   0.000000            0.000000
##   Not Specified             6.112469            7.090465
##   Payday Loan              13.157895            7.416268
##   Personal Loan             6.818182            6.136364
##   Student Loan              8.974359           12.051282
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                8.177570         9.579439
##   Credit-Builder Loan                      7.780320         8.466819
##   Debt Consolidation Loan                  8.616780        10.430839
##   Home Equity Loan                         7.529412         6.117647
##   Mortgage Loan                            5.348837         8.139535
##   No Loan                                  0.000000         0.000000
##   Not Specified                           11.246944         8.312958
##   Payday Loan                              8.373206         8.133971
##   Personal Loan                            6.363636         6.590909
##   Student Loan                             9.487179         6.666667
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                     9.112150  25.700935      6.308411    8.644860
##   Credit-Builder Loan           6.636156  20.594966      9.610984    9.382151
##   Debt Consolidation Loan       7.256236  25.170068      7.482993    8.390023
##   Home Equity Loan              8.470588  25.176471     10.352941    8.000000
```

```
##   Mortgage Loan               10.000000  25.348837      9.069767    7.674419
##   No Loan                      0.000000 100.000000      0.000000    0.000000
##   Not Specified                7.334963  25.427873      9.779951   10.024450
##   Payday Loan                  6.698565  24.401914      5.741627    9.090909
##   Personal Loan                8.409091  27.272727      9.090909   10.681818
##   Student Loan                 6.410256  22.564103     10.000000    7.179487
##
##                           Personal Loan Student Loan
##   Auto Loan                    8.411215     8.177570
##   Credit-Builder Loan         11.670481     8.237986
##   Debt Consolidation Loan      7.256236     9.750567
##   Home Equity Loan             9.882353     7.058824
##   Mortgage Loan                8.837209     9.767442
##   No Loan                      0.000000     0.000000
##   Not Specified                9.046455     5.623472
##   Payday Loan                  7.894737     9.090909
##   Personal Loan                9.318182     9.318182
##   Student Loan                 8.205128     8.461538
##
## $Last_Loan_5_Last_Loan_6$col_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                9.7087379          11.6564417
##   Credit-Builder Loan     11.3268608          12.8834356
##   Debt Consolidation Loan  8.7378641          12.8834356
##   Home Equity Loan        12.2977346          11.0429448
##   Mortgage Loan           11.0032362          10.4294479
##   No Loan                  0.0000000           0.0000000
##   Not Specified            8.0906149           8.8957055
##   Payday Loan             17.7993528           9.5092025
##   Personal Loan            9.7087379           8.2822086
##   Student Loan            11.3268608          14.4171779
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                           11.3636364       13.3116883
##   Credit-Builder Loan                 11.0389610       12.0129870
##   Debt Consolidation Loan             12.3376623       14.9350649
##   Home Equity Loan                    10.3896104        8.4415584
##   Mortgage Loan                        7.4675325       11.3636364
##   No Loan                              0.0000000        0.0000000
##   Not Specified                       14.9350649       11.0389610
##   Payday Loan                         11.3636364       11.0389610
##   Personal Loan                        9.0909091        9.4155844
##   Student Loan                        12.0129870        8.4415584
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                  13.0434783  1.1430947     8.2317073  11.0119048
##   Credit-Builder Loan         9.6989967  0.9352593    12.8048780  12.2023810
##   Debt Consolidation Loan    10.7023411  1.1534864    10.0609756  11.0119048
##   Home Equity Loan           12.0401338  1.1119194    13.4146341  10.1190476
##   Mortgage Loan              14.3812709  1.1327029    11.8902439   9.8214286
```

```
##   No Loan                     0.0000000 90.2213447     0.0000000   0.0000000
##   Not Specified              10.0334448  1.0807441    12.1951220  12.2023810
##   Payday Loan                 9.3645485  1.0599605     7.3170732  11.3095238
##   Personal Loan              12.3745819  1.2470124    12.1951220  13.9880952
##   Student Loan                8.3612040  0.9144757    11.8902439   8.3333333
##
##                           Personal Loan Student Loan
##   Auto Loan                  10.5263158   10.9034268
##   Credit-Builder Loan        14.9122807   11.2149533
##   Debt Consolidation Loan     9.3567251   13.3956386
##   Home Equity Loan           12.2807018    9.3457944
##   Mortgage Loan              11.1111111   13.0841121
##   No Loan                     0.0000000    0.0000000
##   Not Specified              10.8187135    7.1651090
##   Payday Loan                 9.6491228   11.8380062
##   Personal Loan              11.9883041   12.7725857
##   Student Loan                9.3567251   10.2803738
##
##
## $Last_Loan_5_Last_Loan_7
## $Last_Loan_5_Last_Loan_7$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     160                 176                     128
##   Credit-Builder Loan           168                 200                     216
##   Debt Consolidation Loan       144                 216                     152
##   Home Equity Loan              208                 216                     160
##   Mortgage Loan                 176                 184                     176
##   No Loan                         0                   0                       0
##   Not Specified                 168                 216                     160
##   Payday Loan                   208                 120                     152
##   Personal Loan                 144                 256                     160
##   Student Loan                  224                 136                     128
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            152           176    1728           264
##   Credit-Builder Loan                  112           240    1768           128
##   Debt Consolidation Loan              216           168    1888           184
##   Home Equity Loan                     216           200    1584           184
##   Mortgage Loan                        144           200    1824           184
##   No Loan                                0             0   69456             0
##   Not Specified                        184           152    1664           176
##   Payday Loan                          184           208    1728           144
##   Personal Loan                        152           216    1944           184
##   Student Loan                         168           232    1544           160
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                       216           208          216
##   Credit-Builder Loan             272           208          184
##   Debt Consolidation Loan         208           176          176
##   Home Equity Loan                264           208          160
```

```
##   Mortgage Loan                   176           184          192
##   No Loan                           0             0            0
##   Not Specified                   184           184          184
##   Payday Loan                     240           192          168
##   Personal Loan                   184           136          144
##   Student Loan                    176           192          160
##
## $Last_Loan_5_Last_Loan_7$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    3424                    3496                    3528
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    3400                    3440                   69456
##           Not Specified             Payday Loan           Personal Loan
##                    3272                    3344                    3520
##            Student Loan
##                    3120
##
## $Last_Loan_5_Last_Loan_7$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    1600                    1720                    1432
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    1528                    1792                   85128
##           Not Specified             Payday Loan           Personal Loan
##                    1608                    1920                    1688
##            Student Loan
##                    1584
##
## $Last_Loan_5_Last_Loan_7$grand_total
## [1] 100000
##
## $Last_Loan_5_Last_Loan_7$cell_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                   0.160               0.176                   0.128
##   Credit-Builder Loan         0.168               0.200                   0.216
##   Debt Consolidation Loan     0.144               0.216                   0.152
##   Home Equity Loan            0.208               0.216                   0.160
##   Mortgage Loan               0.176               0.184                   0.176
##   No Loan                     0.000               0.000                   0.000
##   Not Specified               0.168               0.216                   0.160
##   Payday Loan                 0.208               0.120                   0.152
##   Personal Loan               0.144               0.256                   0.160
##   Student Loan                0.224               0.136                   0.128
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.152         0.176   1.728         0.264
##   Credit-Builder Loan                0.112         0.240   1.768         0.128
##   Debt Consolidation Loan            0.216         0.168   1.888         0.184
##   Home Equity Loan                   0.216         0.200   1.584         0.184
```

```
##   Mortgage Loan                      0.144         0.200   1.824         0.184
##   No Loan                            0.000         0.000  69.456         0.000
##   Not Specified                      0.184         0.152   1.664         0.176
##   Payday Loan                        0.184         0.208   1.728         0.144
##   Personal Loan                      0.152         0.216   1.944         0.184
##   Student Loan                       0.168         0.232   1.544         0.160
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                     0.216         0.208        0.216
##   Credit-Builder Loan           0.272         0.208        0.184
##   Debt Consolidation Loan       0.208         0.176        0.176
##   Home Equity Loan              0.264         0.208        0.160
##   Mortgage Loan                 0.176         0.184        0.192
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.184         0.184        0.184
##   Payday Loan                   0.240         0.192        0.168
##   Personal Loan                 0.184         0.136        0.144
##   Student Loan                  0.176         0.192        0.160
##
## $Last_Loan_5_Last_Loan_7$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 4.672897            5.140187
##   Credit-Builder Loan       4.805492            5.720824
##   Debt Consolidation Loan   4.081633            6.122449
##   Home Equity Loan          6.117647            6.352941
##   Mortgage Loan             5.116279            5.348837
##   No Loan                   0.000000            0.000000
##   Not Specified             5.134474            6.601467
##   Payday Loan               6.220096            3.588517
##   Personal Loan             4.090909            7.272727
##   Student Loan              7.179487            4.358974
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                3.738318         4.439252
##   Credit-Builder Loan                      6.178490         3.203661
##   Debt Consolidation Loan                  4.308390         6.122449
##   Home Equity Loan                         4.705882         6.352941
##   Mortgage Loan                            5.116279         4.186047
##   No Loan                                  0.000000         0.000000
##   Not Specified                            4.889976         5.623472
##   Payday Loan                              4.545455         5.502392
##   Personal Loan                            4.545455         4.318182
##   Student Loan                             4.102564         5.384615
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                  5.140187  50.467290      7.710280    6.308411
##   Credit-Builder Loan        6.864989  50.572082      3.661327    7.780320
##   Debt Consolidation Loan    4.761905  53.514739      5.215420    5.895692
##   Home Equity Loan           5.882353  46.588235      5.411765    7.764706
##   Mortgage Loan              5.813953  53.023256      5.348837    5.116279
```

```
##   No Loan                      0.000000 100.000000      0.000000    0.000000
##   Not Specified                4.645477  50.855746      5.378973    5.623472
##   Payday Loan                  6.220096  51.674641      4.306220    7.177033
##   Personal Loan                6.136364  55.227273      5.227273    5.227273
##   Student Loan                 7.435897  49.487179      5.128205    5.641026
##
##                           Personal Loan Student Loan
##   Auto Loan                    6.074766     6.308411
##   Credit-Builder Loan          5.949657     5.263158
##   Debt Consolidation Loan      4.988662     4.988662
##   Home Equity Loan             6.117647     4.705882
##   Mortgage Loan                5.348837     5.581395
##   No Loan                      0.000000     0.000000
##   Not Specified                5.623472     5.623472
##   Payday Loan                  5.741627     5.023923
##   Personal Loan                3.863636     4.090909
##   Student Loan                 6.153846     5.128205
##
## $Last_Loan_5_Last_Loan_7$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               10.000000           10.232558                8.938547
##   Credit-Builder Loan     10.500000           11.627907               15.083799
##   Debt Consolidation Loan  9.000000           12.558140               10.614525
##   Home Equity Loan        13.000000           12.558140               11.173184
##   Mortgage Loan           11.000000           10.697674               12.290503
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           10.500000           12.558140               11.173184
##   Payday Loan             13.000000            6.976744               10.614525
##   Personal Loan            9.000000           14.883721               11.173184
##   Student Loan            14.000000            7.906977                8.938547
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                        9.947644      9.821429  2.029884
##   Credit-Builder Loan              7.329843     13.392857  2.076872
##   Debt Consolidation Loan         14.136126      9.375000  2.217837
##   Home Equity Loan                14.136126     11.160714  1.860727
##   Mortgage Loan                    9.424084     11.160714  2.142656
##   No Loan                          0.000000      0.000000 81.590076
##   Not Specified                   12.041885      8.482143  1.954704
##   Payday Loan                     12.041885     11.607143  2.029884
##   Personal Loan                    9.947644     12.053571  2.283620
##   Student Loan                    10.994764     12.946429  1.813739
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    16.417910   11.250000     12.322275    13.636364
##   Credit-Builder Loan           7.960199   14.166667     12.322275    11.616162
##   Debt Consolidation Loan      11.442786   10.833333     10.426540    11.111111
##   Home Equity Loan             11.442786   13.750000     12.322275    10.101010
##   Mortgage Loan                11.442786    9.166667     10.900474    12.121212
##   No Loan                       0.000000    0.000000      0.000000     0.000000
```

```
##   Not Specified               10.945274    9.583333     10.900474    11.616162
##   Payday Loan                  8.955224   12.500000     11.374408    10.606061
##   Personal Loan               11.442786    9.583333      8.056872     9.090909
##   Student Loan                 9.950249    9.166667     11.374408    10.101010
##
##
## $Last_Loan_5_Last_Loan_8
## $Last_Loan_5_Last_Loan_8$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                    104                 144                     128
##   Credit-Builder Loan           72                  72                      64
##   Debt Consolidation Loan       96                  80                      56
##   Home Equity Loan              64                  88                      80
##   Mortgage Loan                 88                  64                      96
##   No Loan                        0                   0                       0
##   Not Specified                104                 128                      64
##   Payday Loan                   64                 112                      80
##   Personal Loan                 80                  64                     128
##   Student Loan                  48                 120                      64
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                           104            88    2512           112
##   Credit-Builder Loan                 152            72    2704            72
##   Debt Consolidation Loan              80            80    2744            88
##   Home Equity Loan                    104            80    2600            72
##   Mortgage Loan                        80           128    2672            56
##   No Loan                               0             0   69456             0
##   Not Specified                       104            72    2504            64
##   Payday Loan                          72            88    2560           112
##   Personal Loan                       120            72    2688            96
##   Student Loan                         72            56    2368            72
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                       96            64           72
##   Credit-Builder Loan             88            72          128
##   Debt Consolidation Loan         88            80          136
##   Home Equity Loan               104            96          112
##   Mortgage Loan                   72            88           96
##   No Loan                          0             0            0
##   Not Specified                   88            56           88
##   Payday Loan                     88            80           88
##   Personal Loan                  120            88           64
##   Student Loan                   144            96           80
##
## $Last_Loan_5_Last_Loan_8$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    3424                    3496                    3528
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    3400                    3440                   69456
```

```
##           Not Specified             Payday Loan           Personal Loan
##                    3272                    3344                    3520
##            Student Loan
##                    3120
##
## $Last_Loan_5_Last_Loan_8$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     720                     872                     760
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     888                     736                   92808
##           Not Specified             Payday Loan           Personal Loan
##                     744                     888                     720
##            Student Loan
##                     864
##
## $Last_Loan_5_Last_Loan_8$grand_total
## [1] 100000
##
## $Last_Loan_5_Last_Loan_8$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.104               0.144                   0.128
##   Credit-Builder Loan        0.072               0.072                   0.064
##   Debt Consolidation Loan    0.096               0.080                   0.056
##   Home Equity Loan           0.064               0.088                   0.080
##   Mortgage Loan              0.088               0.064                   0.096
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.104               0.128                   0.064
##   Payday Loan                0.064               0.112                   0.080
##   Personal Loan              0.080               0.064                   0.128
##   Student Loan               0.048               0.120                   0.064
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.104         0.088   2.512         0.112
##   Credit-Builder Loan                0.152         0.072   2.704         0.072
##   Debt Consolidation Loan            0.080         0.080   2.744         0.088
##   Home Equity Loan                   0.104         0.080   2.600         0.072
##   Mortgage Loan                      0.080         0.128   2.672         0.056
##   No Loan                            0.000         0.000  69.456         0.000
##   Not Specified                      0.104         0.072   2.504         0.064
##   Payday Loan                        0.072         0.088   2.560         0.112
##   Personal Loan                      0.120         0.072   2.688         0.096
##   Student Loan                       0.072         0.056   2.368         0.072
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.096         0.064        0.072
##   Credit-Builder Loan          0.088         0.072        0.128
##   Debt Consolidation Loan      0.088         0.080        0.136
##   Home Equity Loan             0.104         0.096        0.112
##   Mortgage Loan                0.072         0.088        0.096
```

```
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.088         0.056        0.088
##   Payday Loan                   0.088         0.080        0.088
##   Personal Loan                 0.120         0.088        0.064
##   Student Loan                  0.144         0.096        0.080
##
## $Last_Loan_5_Last_Loan_8$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 3.037383            4.205607
##   Credit-Builder Loan       2.059497            2.059497
##   Debt Consolidation Loan   2.721088            2.267574
##   Home Equity Loan          1.882353            2.588235
##   Mortgage Loan             2.558140            1.860465
##   No Loan                   0.000000            0.000000
##   Not Specified             3.178484            3.911980
##   Payday Loan               1.913876            3.349282
##   Personal Loan             2.272727            1.818182
##   Student Loan              1.538462            3.846154
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                3.738318         3.037383
##   Credit-Builder Loan                      1.830664         4.347826
##   Debt Consolidation Loan                  1.587302         2.267574
##   Home Equity Loan                         2.352941         3.058824
##   Mortgage Loan                            2.790698         2.325581
##   No Loan                                  0.000000         0.000000
##   Not Specified                            1.955990         3.178484
##   Payday Loan                              2.392344         2.153110
##   Personal Loan                            3.636364         3.409091
##   Student Loan                             2.051282         2.307692
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                     2.570093  73.364486      3.271028    2.803738
##   Credit-Builder Loan           2.059497  77.345538      2.059497    2.517162
##   Debt Consolidation Loan       2.267574  77.777778      2.494331    2.494331
##   Home Equity Loan              2.352941  76.470588      2.117647    3.058824
##   Mortgage Loan                 3.720930  77.674419      1.627907    2.093023
##   No Loan                       0.000000 100.000000      0.000000    0.000000
##   Not Specified                 2.200489  76.528117      1.955990    2.689487
##   Payday Loan                   2.631579  76.555024      3.349282    2.631579
##   Personal Loan                 2.045455  76.363636      2.727273    3.409091
##   Student Loan                  1.794872  75.897436      2.307692    4.615385
##
##                           Personal Loan Student Loan
##   Auto Loan                     1.869159     2.102804
##   Credit-Builder Loan           2.059497     3.661327
##   Debt Consolidation Loan       2.267574     3.854875
##   Home Equity Loan              2.823529     3.294118
##   Mortgage Loan                 2.558140     2.790698
##   No Loan                       0.000000     0.000000
```

```
##   Not Specified                1.711491     2.689487
##   Payday Loan                  2.392344     2.631579
##   Personal Loan                2.500000     1.818182
##   Student Loan                 3.076923     2.564103
##
## $Last_Loan_5_Last_Loan_8$col_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               14.444444           16.513761               16.842105
##   Credit-Builder Loan     10.000000            8.256881                8.421053
##   Debt Consolidation Loan 13.333333            9.174312                7.368421
##   Home Equity Loan         8.888889           10.091743               10.526316
##   Mortgage Loan           12.222222            7.339450               12.631579
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           14.444444           14.678899                8.421053
##   Payday Loan              8.888889           12.844037               10.526316
##   Personal Loan           11.111111            7.339450               16.842105
##   Student Loan             6.666667           13.761468                8.421053
##
##                          Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                       11.711712     11.956522  2.706663
##   Credit-Builder Loan             17.117117      9.782609  2.913542
##   Debt Consolidation Loan          9.009009     10.869565  2.956642
##   Home Equity Loan                11.711712     10.869565  2.801483
##   Mortgage Loan                    9.009009     17.391304  2.879062
##   No Loan                          0.000000      0.000000 74.838376
##   Not Specified                   11.711712      9.782609  2.698043
##   Payday Loan                      8.108108     11.956522  2.758383
##   Personal Loan                   13.513514      9.782609  2.896302
##   Student Loan                     8.108108      7.608696  2.551504
##
##                          Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    15.053763   10.810811      8.888889     8.333333
##   Credit-Builder Loan           9.677419    9.909910     10.000000    14.814815
##   Debt Consolidation Loan      11.827957    9.909910     11.111111    15.740741
##   Home Equity Loan              9.677419   11.711712     13.333333    12.962963
##   Mortgage Loan                 7.526882    8.108108     12.222222    11.111111
##   No Loan                       0.000000    0.000000      0.000000     0.000000
##   Not Specified                 8.602151    9.909910      7.777778    10.185185
##   Payday Loan                  15.053763    9.909910     11.111111    10.185185
##   Personal Loan                12.903226   13.513514     12.222222     7.407407
##   Student Loan                  9.677419   16.216216     13.333333     9.259259
##
##
## $Last_Loan_5_Last_Loan_9
## $Last_Loan_5_Last_Loan_9$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                      24                  32                      96
##   Credit-Builder Loan            56                  64                      64
##   Debt Consolidation Loan        72                  48                      24
```

```
##   Home Equity Loan              32                  32                      24
##   Mortgage Loan                 64                  64                      48
##   No Loan                        0                   0                       0
##   Not Specified                 40                  48                      16
##   Payday Loan                   56                  40                      48
##   Personal Loan                 40                  56                      48
##   Student Loan                  64                  48                      24
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                             32            48    2984            72
##   Credit-Builder Loan                   24            40    3048            64
##   Debt Consolidation Loan               32            72    3096            64
##   Home Equity Loan                      24            80    3008            40
##   Mortgage Loan                         56            24    3000            56
##   No Loan                                0             0   69456             0
##   Not Specified                         64            48    2832            40
##   Payday Loan                           88            56    2952            32
##   Personal Loan                         64           104    3064             8
##   Student Loan                          24            48    2704            48
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                       40            40           56
##   Credit-Builder Loan             72            32           32
##   Debt Consolidation Loan         32            16           72
##   Home Equity Loan                64            40           56
##   Mortgage Loan                   48            32           48
##   No Loan                          0             0            0
##   Not Specified                   80            56           48
##   Payday Loan                      8            48           16
##   Personal Loan                   56            64           16
##   Student Loan                    64            56           40
##
## $Last_Loan_5_Last_Loan_9$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    3424                    3496                    3528
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    3400                    3440                   69456
##           Not Specified             Payday Loan           Personal Loan
##                    3272                    3344                    3520
##            Student Loan
##                    3120
##
## $Last_Loan_5_Last_Loan_9$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     448                     432                     392
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     408                     520                   96144
##           Not Specified             Payday Loan           Personal Loan
##                     424                     464                     384
```

```
##            Student Loan
##                     384
##
## $Last_Loan_5_Last_Loan_9$grand_total
## [1] 100000
##
## $Last_Loan_5_Last_Loan_9$cell_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                   0.024               0.032                   0.096
##   Credit-Builder Loan         0.056               0.064                   0.064
##   Debt Consolidation Loan     0.072               0.048                   0.024
##   Home Equity Loan            0.032               0.032                   0.024
##   Mortgage Loan               0.064               0.064                   0.048
##   No Loan                     0.000               0.000                   0.000
##   Not Specified               0.040               0.048                   0.016
##   Payday Loan                 0.056               0.040                   0.048
##   Personal Loan               0.040               0.056                   0.048
##   Student Loan                0.064               0.048                   0.024
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.032         0.048   2.984         0.072
##   Credit-Builder Loan                0.024         0.040   3.048         0.064
##   Debt Consolidation Loan            0.032         0.072   3.096         0.064
##   Home Equity Loan                   0.024         0.080   3.008         0.040
##   Mortgage Loan                      0.056         0.024   3.000         0.056
##   No Loan                            0.000         0.000  69.456         0.000
##   Not Specified                      0.064         0.048   2.832         0.040
##   Payday Loan                        0.088         0.056   2.952         0.032
##   Personal Loan                      0.064         0.104   3.064         0.008
##   Student Loan                       0.024         0.048   2.704         0.048
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                     0.040         0.040        0.056
##   Credit-Builder Loan           0.072         0.032        0.032
##   Debt Consolidation Loan       0.032         0.016        0.072
##   Home Equity Loan              0.064         0.040        0.056
##   Mortgage Loan                 0.048         0.032        0.048
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.080         0.056        0.048
##   Payday Loan                   0.008         0.048        0.016
##   Personal Loan                 0.056         0.064        0.016
##   Student Loan                  0.064         0.056        0.040
##
## $Last_Loan_5_Last_Loan_9$row_percentage
##
##                             Auto Loan Credit-Builder Loan
##   Auto Loan                 0.7009346           0.9345794
##   Credit-Builder Loan       1.6018307           1.8306636
##   Debt Consolidation Loan   2.0408163           1.3605442
##   Home Equity Loan          0.9411765           0.9411765
```

```
##   Mortgage Loan             1.8604651           1.8604651
##   No Loan                   0.0000000           0.0000000
##   Not Specified             1.2224939           1.4669927
##   Payday Loan               1.6746411           1.1961722
##   Personal Loan             1.1363636           1.5909091
##   Student Loan              2.0512821           1.5384615
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                               2.8037383        0.9345794
##   Credit-Builder Loan                     1.8306636        0.6864989
##   Debt Consolidation Loan                 0.6802721        0.9070295
##   Home Equity Loan                        0.7058824        0.7058824
##   Mortgage Loan                           1.3953488        1.6279070
##   No Loan                                 0.0000000        0.0000000
##   Not Specified                           0.4889976        1.9559902
##   Payday Loan                             1.4354067        2.6315789
##   Personal Loan                           1.3636364        1.8181818
##   Student Loan                            0.7692308        0.7692308
##
##                           Mortgage Loan     No Loan Not Specified Payday Loan
##   Auto Loan                   1.4018692  87.1495327     2.1028037   1.1682243
##   Credit-Builder Loan         1.1441648  87.1853547     1.8306636   2.0594966
##   Debt Consolidation Loan     2.0408163  87.7551020     1.8140590   0.9070295
##   Home Equity Loan            2.3529412  88.4705882     1.1764706   1.8823529
##   Mortgage Loan               0.6976744  87.2093023     1.6279070   1.3953488
##   No Loan                     0.0000000 100.0000000     0.0000000   0.0000000
##   Not Specified               1.4669927  86.5525672     1.2224939   2.4449878
##   Payday Loan                 1.6746411  88.2775120     0.9569378   0.2392344
##   Personal Loan               2.9545455  87.0454545     0.2272727   1.5909091
##   Student Loan                1.5384615  86.6666667     1.5384615   2.0512821
##
##                           Personal Loan Student Loan
##   Auto Loan                   1.1682243    1.6355140
##   Credit-Builder Loan         0.9153318    0.9153318
##   Debt Consolidation Loan     0.4535147    2.0408163
##   Home Equity Loan            1.1764706    1.6470588
##   Mortgage Loan               0.9302326    1.3953488
##   No Loan                     0.0000000    0.0000000
##   Not Specified               1.7114914    1.4669927
##   Payday Loan                 1.4354067    0.4784689
##   Personal Loan               1.8181818    0.4545455
##   Student Loan                1.7948718    1.2820513
##
## $Last_Loan_5_Last_Loan_9$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                5.357143            7.407407               24.489796
##   Credit-Builder Loan     12.500000           14.814815               16.326531
##   Debt Consolidation Loan 16.071429           11.111111                6.122449
##   Home Equity Loan         7.142857            7.407407                6.122449
##   Mortgage Loan           14.285714           14.814815               12.244898
```

```
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified            8.928571           11.111111                4.081633
##   Payday Loan             12.500000            9.259259               12.244898
##   Personal Loan            8.928571           12.962963               12.244898
##   Student Loan            14.285714           11.111111                6.122449
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                        7.843137      9.230769  3.103678
##   Credit-Builder Loan              5.882353      7.692308  3.170245
##   Debt Consolidation Loan          7.843137     13.846154  3.220170
##   Home Equity Loan                 5.882353     15.384615  3.128640
##   Mortgage Loan                   13.725490      4.615385  3.120320
##   No Loan                          0.000000      0.000000 72.241638
##   Not Specified                   15.686275      9.230769  2.945582
##   Payday Loan                     21.568627     10.769231  3.070394
##   Personal Loan                   15.686275     20.000000  3.186886
##   Student Loan                     5.882353      9.230769  2.812448
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    16.981132    8.620690     10.416667    14.583333
##   Credit-Builder Loan          15.094340   15.517241      8.333333     8.333333
##   Debt Consolidation Loan      15.094340    6.896552      4.166667    18.750000
##   Home Equity Loan              9.433962   13.793103     10.416667    14.583333
##   Mortgage Loan                13.207547   10.344828      8.333333    12.500000
##   No Loan                       0.000000    0.000000      0.000000     0.000000
##   Not Specified                 9.433962   17.241379     14.583333    12.500000
##   Payday Loan                   7.547170    1.724138     12.500000     4.166667
##   Personal Loan                 1.886792   12.068966     16.666667     4.166667
##   Student Loan                 11.320755   13.793103     14.583333    10.416667
##
##
## $Last_Loan_6_Last_Loan_7
## $Last_Loan_6_Last_Loan_7$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     160                 232                     104
##   Credit-Builder Loan           152                 200                     176
##   Debt Consolidation Loan       160                 192                     184
##   Home Equity Loan              184                 160                     160
##   Mortgage Loan                 192                 144                     120
##   No Loan                         0                   0                       0
##   Not Specified                 152                 208                     184
##   Payday Loan                   208                 176                     200
##   Personal Loan                 208                 200                     200
##   Student Loan                  184                 208                     104
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                            112           152     936           176
##   Credit-Builder Loan                  176           144     976           216
##   Debt Consolidation Loan              144           216     864           160
##   Home Equity Loan                     184           232     880           208
```

```
##   Mortgage Loan                        192           152     768           120
##   No Loan                                0             0   76984             0
##   Not Specified                        192           264     848           232
##   Payday Loan                          168           184     968           176
##   Personal Loan                        200           224     944           168
##   Student Loan                         160           224     960           152
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                       216           176          208
##   Credit-Builder Loan             208           160          200
##   Debt Consolidation Loan         184           144          216
##   Home Equity Loan                240           152           64
##   Mortgage Loan                   280           240          184
##   No Loan                           0             0            0
##   Not Specified                   152           216          176
##   Payday Loan                     232           208          168
##   Personal Loan                   232           144          216
##   Student Loan                    176           248          152
##
## $Last_Loan_6_Last_Loan_7$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    2472                    2608                    2464
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    2464                    2392                   76984
##           Not Specified             Payday Loan           Personal Loan
##                    2624                    2688                    2736
##            Student Loan
##                    2568
##
## $Last_Loan_6_Last_Loan_7$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    1600                    1720                    1432
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    1528                    1792                   85128
##           Not Specified             Payday Loan           Personal Loan
##                    1608                    1920                    1688
##            Student Loan
##                    1584
##
## $Last_Loan_6_Last_Loan_7$grand_total
## [1] 100000
##
## $Last_Loan_6_Last_Loan_7$cell_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                   0.160               0.232                   0.104
##   Credit-Builder Loan         0.152               0.200                   0.176
##   Debt Consolidation Loan     0.160               0.192                   0.184
##   Home Equity Loan            0.184               0.160                   0.160
```

```
##   Mortgage Loan               0.192               0.144                   0.120
##   No Loan                     0.000               0.000                   0.000
##   Not Specified               0.152               0.208                   0.184
##   Payday Loan                 0.208               0.176                   0.200
##   Personal Loan               0.208               0.200                   0.200
##   Student Loan                0.184               0.208                   0.104
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.112         0.152   0.936         0.176
##   Credit-Builder Loan                0.176         0.144   0.976         0.216
##   Debt Consolidation Loan            0.144         0.216   0.864         0.160
##   Home Equity Loan                   0.184         0.232   0.880         0.208
##   Mortgage Loan                      0.192         0.152   0.768         0.120
##   No Loan                            0.000         0.000  76.984         0.000
##   Not Specified                      0.192         0.264   0.848         0.232
##   Payday Loan                        0.168         0.184   0.968         0.176
##   Personal Loan                      0.200         0.224   0.944         0.168
##   Student Loan                       0.160         0.224   0.960         0.152
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                     0.216         0.176        0.208
##   Credit-Builder Loan           0.208         0.160        0.200
##   Debt Consolidation Loan       0.184         0.144        0.216
##   Home Equity Loan              0.240         0.152        0.064
##   Mortgage Loan                 0.280         0.240        0.184
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.152         0.216        0.176
##   Payday Loan                   0.232         0.208        0.168
##   Personal Loan                 0.232         0.144        0.216
##   Student Loan                  0.176         0.248        0.152
##
## $Last_Loan_6_Last_Loan_7$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 6.472492            9.385113
##   Credit-Builder Loan       5.828221            7.668712
##   Debt Consolidation Loan   6.493506            7.792208
##   Home Equity Loan          7.467532            6.493506
##   Mortgage Loan             8.026756            6.020067
##   No Loan                   0.000000            0.000000
##   Not Specified             5.792683            7.926829
##   Payday Loan               7.738095            6.547619
##   Personal Loan             7.602339            7.309942
##   Student Loan              7.165109            8.099688
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                4.207120         4.530744
##   Credit-Builder Loan                      6.748466         6.748466
##   Debt Consolidation Loan                  7.467532         5.844156
##   Home Equity Loan                         6.493506         7.467532
##   Mortgage Loan                            5.016722         8.026756
```

```
##   No Loan                                0.000000         0.000000
##   Not Specified                          7.012195         7.317073
##   Payday Loan                            7.440476         6.250000
##   Personal Loan                          7.309942         7.309942
##   Student Loan                           4.049844         6.230530
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                    6.148867  37.864078      7.119741    8.737864
##   Credit-Builder Loan          5.521472  37.423313      8.282209    7.975460
##   Debt Consolidation Loan      8.766234  35.064935      6.493506    7.467532
##   Home Equity Loan             9.415584  35.714286      8.441558    9.740260
##   Mortgage Loan                6.354515  32.107023      5.016722   11.705686
##   No Loan                      0.000000 100.000000      0.000000    0.000000
##   Not Specified               10.060976  32.317073      8.841463    5.792683
##   Payday Loan                  6.845238  36.011905      6.547619    8.630952
##   Personal Loan                8.187135  34.502924      6.140351    8.479532
##   Student Loan                 8.722741  37.383178      5.919003    6.853583
##
##                           Personal Loan Student Loan
##   Auto Loan                    7.119741     8.414239
##   Credit-Builder Loan          6.134969     7.668712
##   Debt Consolidation Loan      5.844156     8.766234
##   Home Equity Loan             6.168831     2.597403
##   Mortgage Loan               10.033445     7.692308
##   No Loan                      0.000000     0.000000
##   Not Specified                8.231707     6.707317
##   Payday Loan                  7.738095     6.250000
##   Personal Loan                5.263158     7.894737
##   Student Loan                 9.657321     5.919003
##
## $Last_Loan_6_Last_Loan_7$col_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan               10.0000000          13.4883721
##   Credit-Builder Loan      9.5000000          11.6279070
##   Debt Consolidation Loan 10.0000000          11.1627907
##   Home Equity Loan        11.5000000           9.3023256
##   Mortgage Loan           12.0000000           8.3720930
##   No Loan                  0.0000000           0.0000000
##   Not Specified            9.5000000          12.0930233
##   Payday Loan             13.0000000          10.2325581
##   Personal Loan           13.0000000          11.6279070
##   Student Loan            11.5000000          12.0930233
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                               7.2625698        7.3298429
##   Credit-Builder Loan                    12.2905028       11.5183246
##   Debt Consolidation Loan                12.8491620        9.4240838
##   Home Equity Loan                       11.1731844       12.0418848
##   Mortgage Loan                           8.3798883       12.5654450
##   No Loan                                 0.0000000        0.0000000
```

```
##   Not Specified                        12.8491620       12.5654450
##   Payday Loan                          13.9664804       10.9947644
##   Personal Loan                        13.9664804       13.0890052
##   Student Loan                          7.2625698       10.4712042
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                   8.4821429  1.0995207    10.9452736  11.2500000
##   Credit-Builder Loan         8.0357143  1.1465088    13.4328358  10.8333333
##   Debt Consolidation Loan    12.0535714  1.0149422     9.9502488   9.5833333
##   Home Equity Loan           12.9464286  1.0337374    12.9353234  12.5000000
##   Mortgage Loan               8.4821429  0.9021708     7.4626866  14.5833333
##   No Loan                     0.0000000 90.4332300     0.0000000   0.0000000
##   Not Specified              14.7321429  0.9961470    14.4278607   7.9166667
##   Payday Loan                10.2678571  1.1371112    10.9452736  12.0833333
##   Personal Loan              12.5000000  1.1089183    10.4477612  12.0833333
##   Student Loan               12.5000000  1.1277136     9.4527363   9.1666667
##
##                           Personal Loan Student Loan
##   Auto Loan                  10.4265403   13.1313131
##   Credit-Builder Loan         9.4786730   12.6262626
##   Debt Consolidation Loan     8.5308057   13.6363636
##   Home Equity Loan            9.0047393    4.0404040
##   Mortgage Loan              14.2180095   11.6161616
##   No Loan                     0.0000000    0.0000000
##   Not Specified              12.7962085   11.1111111
##   Payday Loan                12.3222749   10.6060606
##   Personal Loan               8.5308057   13.6363636
##   Student Loan               14.6919431    9.5959596
##
##
## $Last_Loan_6_Last_Loan_8
## $Last_Loan_6_Last_Loan_8$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                      64                  88                     104
##   Credit-Builder Loan            56                  88                      40
##   Debt Consolidation Loan        72                  64                      80
##   Home Equity Loan               64                 128                     112
##   Mortgage Loan                  96                 128                      88
##   No Loan                         0                   0                       0
##   Not Specified                 104                 120                      64
##   Payday Loan                   112                  80                      88
##   Personal Loan                  88                  80                      72
##   Student Loan                   64                  96                     112
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                             64            96    1776            48
##   Credit-Builder Loan                   96            56    1904            72
##   Debt Consolidation Loan               96           104    1696           104
##   Home Equity Loan                      88            72    1552           120
##   Mortgage Loan                        120            88    1552            96
```

```
##   No Loan                                0             0   76984             0
##   Not Specified                         96            48    1832            64
##   Payday Loan                          120            88    1848            96
##   Personal Loan                        128            88    1912            64
##   Student Loan                          80            96    1752            80
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                        88            64           80
##   Credit-Builder Loan              80           112          104
##   Debt Consolidation Loan         104            24          120
##   Home Equity Loan                 88           112          128
##   Mortgage Loan                    80            56           88
##   No Loan                           0             0            0
##   Not Specified                   128            88           80
##   Payday Loan                     112            96           48
##   Personal Loan                    72           112          120
##   Student Loan                    136            56           96
##
## $Last_Loan_6_Last_Loan_8$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    2472                    2608                    2464
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    2464                    2392                   76984
##           Not Specified             Payday Loan           Personal Loan
##                    2624                    2688                    2736
##            Student Loan
##                    2568
##
## $Last_Loan_6_Last_Loan_8$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     720                     872                     760
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     888                     736                   92808
##           Not Specified             Payday Loan           Personal Loan
##                     744                     888                     720
##            Student Loan
##                     864
##
## $Last_Loan_6_Last_Loan_8$grand_total
## [1] 100000
##
## $Last_Loan_6_Last_Loan_8$cell_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                   0.064               0.088                   0.104
##   Credit-Builder Loan         0.056               0.088                   0.040
##   Debt Consolidation Loan     0.072               0.064                   0.080
##   Home Equity Loan            0.064               0.128                   0.112
##   Mortgage Loan               0.096               0.128                   0.088
```

```
##   No Loan                     0.000              0.000                   0.000
##   Not Specified               0.104              0.120                   0.064
##   Payday Loan                 0.112              0.080                   0.088
##   Personal Loan               0.088              0.080                   0.072
##   Student Loan                0.064              0.096                   0.112
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.064         0.096   1.776         0.048
##   Credit-Builder Loan                0.096         0.056   1.904         0.072
##   Debt Consolidation Loan            0.096         0.104   1.696         0.104
##   Home Equity Loan                   0.088         0.072   1.552         0.120
##   Mortgage Loan                      0.120         0.088   1.552         0.096
##   No Loan                            0.000         0.000  76.984         0.000
##   Not Specified                      0.096         0.048   1.832         0.064
##   Payday Loan                        0.120         0.088   1.848         0.096
##   Personal Loan                      0.128         0.088   1.912         0.064
##   Student Loan                       0.080         0.096   1.752         0.080
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                     0.088         0.064        0.080
##   Credit-Builder Loan           0.080         0.112        0.104
##   Debt Consolidation Loan       0.104         0.024        0.120
##   Home Equity Loan              0.088         0.112        0.128
##   Mortgage Loan                 0.080         0.056        0.088
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.128         0.088        0.080
##   Payday Loan                   0.112         0.096        0.048
##   Personal Loan                 0.072         0.112        0.120
##   Student Loan                  0.136         0.056        0.096
##
## $Last_Loan_6_Last_Loan_8$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 2.588997            3.559871
##   Credit-Builder Loan       2.147239            3.374233
##   Debt Consolidation Loan   2.922078            2.597403
##   Home Equity Loan          2.597403            5.194805
##   Mortgage Loan             4.013378            5.351171
##   No Loan                   0.000000            0.000000
##   Not Specified             3.963415            4.573171
##   Payday Loan               4.166667            2.976190
##   Personal Loan             3.216374            2.923977
##   Student Loan              2.492212            3.738318
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                4.207120         2.588997
##   Credit-Builder Loan                      1.533742         3.680982
##   Debt Consolidation Loan                  3.246753         3.896104
##   Home Equity Loan                         4.545455         3.571429
##   Mortgage Loan                            3.678930         5.016722
##   No Loan                                  0.000000         0.000000
```

```
##   Not Specified                          2.439024         3.658537
##   Payday Loan                            3.273810         4.464286
##   Personal Loan                          2.631579         4.678363
##   Student Loan                           4.361371         3.115265
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                    3.883495  71.844660      1.941748    3.559871
##   Credit-Builder Loan          2.147239  73.006135      2.760736    3.067485
##   Debt Consolidation Loan      4.220779  68.831169      4.220779    4.220779
##   Home Equity Loan             2.922078  62.987013      4.870130    3.571429
##   Mortgage Loan                3.678930  64.882943      4.013378    3.344482
##   No Loan                      0.000000 100.000000      0.000000    0.000000
##   Not Specified                1.829268  69.817073      2.439024    4.878049
##   Payday Loan                  3.273810  68.750000      3.571429    4.166667
##   Personal Loan                3.216374  69.883041      2.339181    2.631579
##   Student Loan                 3.738318  68.224299      3.115265    5.295950
##
##                           Personal Loan Student Loan
##   Auto Loan                    2.588997     3.236246
##   Credit-Builder Loan          4.294479     3.987730
##   Debt Consolidation Loan      0.974026     4.870130
##   Home Equity Loan             4.545455     5.194805
##   Mortgage Loan                2.341137     3.678930
##   No Loan                      0.000000     0.000000
##   Not Specified                3.353659     3.048780
##   Payday Loan                  3.571429     1.785714
##   Personal Loan                4.093567     4.385965
##   Student Loan                 2.180685     3.738318
##
## $Last_Loan_6_Last_Loan_8$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                 8.888889           10.091743               13.684211
##   Credit-Builder Loan       7.777778           10.091743                5.263158
##   Debt Consolidation Loan  10.000000            7.339450               10.526316
##   Home Equity Loan          8.888889           14.678899               14.736842
##   Mortgage Loan            13.333333           14.678899               11.578947
##   No Loan                   0.000000            0.000000                0.000000
##   Not Specified            14.444444           13.761468                8.421053
##   Payday Loan              15.555556            9.174312               11.578947
##   Personal Loan            12.222222            9.174312                9.473684
##   Student Loan              8.888889           11.009174               14.736842
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                        7.207207     13.043478  1.913628
##   Credit-Builder Loan             10.810811      7.608696  2.051547
##   Debt Consolidation Loan         10.810811     14.130435  1.827429
##   Home Equity Loan                 9.909910      9.782609  1.672270
##   Mortgage Loan                   13.513514     11.956522  1.672270
##   No Loan                          0.000000      0.000000 82.949746
##   Not Specified                   10.810811      6.521739  1.973968
```

```
##   Payday Loan                    13.513514     11.956522  1.991208
##   Personal Loan                  14.414414     11.956522  2.060167
##   Student Loan                    9.009009     13.043478  1.887768
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    6.451613    9.909910      8.888889     9.259259
##   Credit-Builder Loan          9.677419    9.009009     15.555556    12.037037
##   Debt Consolidation Loan     13.978495   11.711712      3.333333    13.888889
##   Home Equity Loan            16.129032    9.909910     15.555556    14.814815
##   Mortgage Loan               12.903226    9.009009      7.777778    10.185185
##   No Loan                      0.000000    0.000000      0.000000     0.000000
##   Not Specified                8.602151   14.414414     12.222222     9.259259
##   Payday Loan                 12.903226   12.612613     13.333333     5.555556
##   Personal Loan                8.602151    8.108108     15.555556    13.888889
##   Student Loan                10.752688   15.315315      7.777778    11.111111
##
##
## $Last_Loan_6_Last_Loan_9
## $Last_Loan_6_Last_Loan_9$contingency_table
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                      64                  72                      48
##   Credit-Builder Loan            72                  56                      80
##   Debt Consolidation Loan        40                  24                      24
##   Home Equity Loan               32                  48                      24
##   Mortgage Loan                  48                  56                      56
##   No Loan                         0                   0                       0
##   Not Specified                  72                  32                      48
##   Payday Loan                    24                  64                      32
##   Personal Loan                  48                  32                      48
##   Student Loan                   48                  48                      32
##
##                           Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                             56            24    2080            32
##   Credit-Builder Loan                   40            48    2168            40
##   Debt Consolidation Loan               48            48    2112            56
##   Home Equity Loan                      40            56    2032            40
##   Mortgage Loan                         40            80    1904            48
##   No Loan                                0             0   76984             0
##   Not Specified                         40            40    2200            40
##   Payday Loan                           40            80    2264            56
##   Personal Loan                         32            56    2304            48
##   Student Loan                          72            88    2096            64
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                        24            48           24
##   Credit-Builder Loan              64            24           16
##   Debt Consolidation Loan          24            32           56
##   Home Equity Loan                 56            64           72
##   Mortgage Loan                    72            48           40
##   No Loan                           0             0            0
```

```
##   Not Specified                    40            48           64
##   Payday Loan                      72            32           24
##   Personal Loan                    64            56           48
##   Student Loan                     48            32           40
##
## $Last_Loan_6_Last_Loan_9$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    2472                    2608                    2464
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    2464                    2392                   76984
##           Not Specified             Payday Loan           Personal Loan
##                    2624                    2688                    2736
##            Student Loan
##                    2568
##
## $Last_Loan_6_Last_Loan_9$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     448                     432                     392
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     408                     520                   96144
##           Not Specified             Payday Loan           Personal Loan
##                     424                     464                     384
##            Student Loan
##                     384
##
## $Last_Loan_6_Last_Loan_9$grand_total
## [1] 100000
##
## $Last_Loan_6_Last_Loan_9$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.064               0.072                   0.048
##   Credit-Builder Loan        0.072               0.056                   0.080
##   Debt Consolidation Loan    0.040               0.024                   0.024
##   Home Equity Loan           0.032               0.048                   0.024
##   Mortgage Loan              0.048               0.056                   0.056
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.072               0.032                   0.048
##   Payday Loan                0.024               0.064                   0.032
##   Personal Loan              0.048               0.032                   0.048
##   Student Loan               0.048               0.048                   0.032
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.056         0.024   2.080         0.032
##   Credit-Builder Loan                0.040         0.048   2.168         0.040
##   Debt Consolidation Loan            0.048         0.048   2.112         0.056
##   Home Equity Loan                   0.040         0.056   2.032         0.040
##   Mortgage Loan                      0.040         0.080   1.904         0.048
##   No Loan                            0.000         0.000  76.984         0.000
```

```
##   Not Specified                      0.040         0.040   2.200         0.040
##   Payday Loan                        0.040         0.080   2.264         0.056
##   Personal Loan                      0.032         0.056   2.304         0.048
##   Student Loan                       0.072         0.088   2.096         0.064
##
##                           Payday Loan Personal Loan Student Loan
##   Auto Loan                     0.024         0.048        0.024
##   Credit-Builder Loan           0.064         0.024        0.016
##   Debt Consolidation Loan       0.024         0.032        0.056
##   Home Equity Loan              0.056         0.064        0.072
##   Mortgage Loan                 0.072         0.048        0.040
##   No Loan                       0.000         0.000        0.000
##   Not Specified                 0.040         0.048        0.064
##   Payday Loan                   0.072         0.032        0.024
##   Personal Loan                 0.064         0.056        0.048
##   Student Loan                  0.048         0.032        0.040
##
## $Last_Loan_6_Last_Loan_9$row_percentage
##
##                             Auto Loan Credit-Builder Loan
##   Auto Loan                 2.5889968           2.9126214
##   Credit-Builder Loan       2.7607362           2.1472393
##   Debt Consolidation Loan   1.6233766           0.9740260
##   Home Equity Loan          1.2987013           1.9480519
##   Mortgage Loan             2.0066890           2.3411371
##   No Loan                   0.0000000           0.0000000
##   Not Specified             2.7439024           1.2195122
##   Payday Loan               0.8928571           2.3809524
##   Personal Loan             1.7543860           1.1695906
##   Student Loan              1.8691589           1.8691589
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                               1.9417476        2.2653722
##   Credit-Builder Loan                     3.0674847        1.5337423
##   Debt Consolidation Loan                 0.9740260        1.9480519
##   Home Equity Loan                        0.9740260        1.6233766
##   Mortgage Loan                           2.3411371        1.6722408
##   No Loan                                 0.0000000        0.0000000
##   Not Specified                           1.8292683        1.5243902
##   Payday Loan                             1.1904762        1.4880952
##   Personal Loan                           1.7543860        1.1695906
##   Student Loan                            1.2461059        2.8037383
##
##                           Mortgage Loan     No Loan Not Specified Payday Loan
##   Auto Loan                    0.9708738  84.1423948     1.2944984   0.9708738
##   Credit-Builder Loan          1.8404908  83.1288344     1.5337423   2.4539877
##   Debt Consolidation Loan      1.9480519  85.7142857     2.2727273   0.9740260
##   Home Equity Loan             2.2727273  82.4675325     1.6233766   2.2727273
##   Mortgage Loan                3.3444816  79.5986622     2.0066890   3.0100334
##   No Loan                      0.0000000 100.0000000     0.0000000   0.0000000
##   Not Specified                1.5243902  83.8414634     1.5243902   1.5243902
```

```
##   Payday Loan                 2.9761905  84.2261905     2.0833333   2.6785714
##   Personal Loan               2.0467836  84.2105263     1.7543860   2.3391813
##   Student Loan                3.4267913  81.6199377     2.4922118   1.8691589
##
##                           Personal Loan Student Loan
##   Auto Loan                   1.9417476    0.9708738
##   Credit-Builder Loan         0.9202454    0.6134969
##   Debt Consolidation Loan     1.2987013    2.2727273
##   Home Equity Loan            2.5974026    2.9220779
##   Mortgage Loan               2.0066890    1.6722408
##   No Loan                     0.0000000    0.0000000
##   Not Specified               1.8292683    2.4390244
##   Payday Loan                 1.1904762    0.8928571
##   Personal Loan               2.0467836    1.7543860
##   Student Loan                1.2461059    1.5576324
##
## $Last_Loan_6_Last_Loan_9$col_percentage
##
##                           Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               14.285714           16.666667               12.244898
##   Credit-Builder Loan     16.071429           12.962963               20.408163
##   Debt Consolidation Loan  8.928571            5.555556                6.122449
##   Home Equity Loan         7.142857           11.111111                6.122449
##   Mortgage Loan           10.714286           12.962963               14.285714
##   No Loan                  0.000000            0.000000                0.000000
##   Not Specified           16.071429            7.407407               12.244898
##   Payday Loan              5.357143           14.814815                8.163265
##   Personal Loan           10.714286            7.407407               12.244898
##   Student Loan            10.714286           11.111111                8.163265
##
##                           Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                       13.725490      4.615385  2.163422
##   Credit-Builder Loan              9.803922      9.230769  2.254951
##   Debt Consolidation Loan         11.764706      9.230769  2.196705
##   Home Equity Loan                 9.803922     10.769231  2.113496
##   Mortgage Loan                    9.803922     15.384615  1.980363
##   No Loan                          0.000000      0.000000 80.071559
##   Not Specified                    9.803922      7.692308  2.288234
##   Payday Loan                      9.803922     15.384615  2.354801
##   Personal Loan                    7.843137     10.769231  2.396405
##   Student Loan                    17.647059     16.923077  2.180063
##
##                           Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                    7.547170    5.172414     12.500000     6.250000
##   Credit-Builder Loan          9.433962   13.793103      6.250000     4.166667
##   Debt Consolidation Loan     13.207547    5.172414      8.333333    14.583333
##   Home Equity Loan             9.433962   12.068966     16.666667    18.750000
##   Mortgage Loan               11.320755   15.517241     12.500000    10.416667
##   No Loan                      0.000000    0.000000      0.000000     0.000000
##   Not Specified                9.433962    8.620690     12.500000    16.666667
##   Payday Loan                 13.207547   15.517241      8.333333     6.250000
```

```
##   Personal Loan               11.320755   13.793103     14.583333    12.500000
##   Student Loan                15.094340   10.344828      8.333333    10.416667
##
##
## $Last_Loan_7_Last_Loan_8
## $Last_Loan_7_Last_Loan_8$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     96                  64                      64
##   Credit-Builder Loan           80                  72                      80
##   Debt Consolidation Loan       80                 104                     104
##   Home Equity Loan              56                  80                      80
##   Mortgage Loan                128                 120                      88
##   No Loan                        0                   0                       0
##   Not Specified                 64                  56                      72
##   Payday Loan                   88                 152                     136
##   Personal Loan                 72                  96                      56
##   Student Loan                  56                 128                      80
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                           160           104     824            80
##   Credit-Builder Loan                  56            80     928            80
##   Debt Consolidation Loan              80            48     816            24
##   Home Equity Loan                     72            56     824           112
##   Mortgage Loan                       112            72     840            96
##   No Loan                               0             0   85128             0
##   Not Specified                       104            88     888            64
##   Payday Loan                         112            96     904            72
##   Personal Loan                        96           104     936           120
##   Student Loan                         96            88     720            96
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                       88            56           64
##   Credit-Builder Loan            144            80          120
##   Debt Consolidation Loan         56            64           56
##   Home Equity Loan                56            72          120
##   Mortgage Loan                   88           112          136
##   No Loan                          0             0            0
##   Not Specified                  120            72           80
##   Payday Loan                    160            72          128
##   Personal Loan                   56            88           64
##   Student Loan                   120           104           96
##
## $Last_Loan_7_Last_Loan_8$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    1600                    1720                    1432
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    1528                    1792                   85128
##           Not Specified             Payday Loan           Personal Loan
##                    1608                    1920                    1688
```

```
##            Student Loan
##                    1584
##
## $Last_Loan_7_Last_Loan_8$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     720                     872                     760
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     888                     736                   92808
##           Not Specified             Payday Loan           Personal Loan
##                     744                     888                     720
##            Student Loan
##                     864
##
## $Last_Loan_7_Last_Loan_8$grand_total
## [1] 100000
##
## $Last_Loan_7_Last_Loan_8$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.096               0.064                   0.064
##   Credit-Builder Loan        0.080               0.072                   0.080
##   Debt Consolidation Loan    0.080               0.104                   0.104
##   Home Equity Loan           0.056               0.080                   0.080
##   Mortgage Loan              0.128               0.120                   0.088
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.064               0.056                   0.072
##   Payday Loan                0.088               0.152                   0.136
##   Personal Loan              0.072               0.096                   0.056
##   Student Loan               0.056               0.128                   0.080
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.160         0.104   0.824         0.080
##   Credit-Builder Loan                0.056         0.080   0.928         0.080
##   Debt Consolidation Loan            0.080         0.048   0.816         0.024
##   Home Equity Loan                   0.072         0.056   0.824         0.112
##   Mortgage Loan                      0.112         0.072   0.840         0.096
##   No Loan                            0.000         0.000  85.128         0.000
##   Not Specified                      0.104         0.088   0.888         0.064
##   Payday Loan                        0.112         0.096   0.904         0.072
##   Personal Loan                      0.096         0.104   0.936         0.120
##   Student Loan                       0.096         0.088   0.720         0.096
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.088         0.056        0.064
##   Credit-Builder Loan          0.144         0.080        0.120
##   Debt Consolidation Loan      0.056         0.064        0.056
##   Home Equity Loan             0.056         0.072        0.120
##   Mortgage Loan                0.088         0.112        0.136
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.120         0.072        0.080
```

```
##   Payday Loan                   0.160         0.072        0.128
##   Personal Loan                 0.056         0.088        0.064
##   Student Loan                  0.120         0.104        0.096
##
## $Last_Loan_7_Last_Loan_8$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 6.000000            4.000000
##   Credit-Builder Loan       4.651163            4.186047
##   Debt Consolidation Loan   5.586592            7.262570
##   Home Equity Loan          3.664921            5.235602
##   Mortgage Loan             7.142857            6.696429
##   No Loan                   0.000000            0.000000
##   Not Specified             3.980100            3.482587
##   Payday Loan               4.583333            7.916667
##   Personal Loan             4.265403            5.687204
##   Student Loan              3.535354            8.080808
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                4.000000        10.000000
##   Credit-Builder Loan                      4.651163         3.255814
##   Debt Consolidation Loan                  7.262570         5.586592
##   Home Equity Loan                         5.235602         4.712042
##   Mortgage Loan                            4.910714         6.250000
##   No Loan                                  0.000000         0.000000
##   Not Specified                            4.477612         6.467662
##   Payday Loan                              7.083333         5.833333
##   Personal Loan                            3.317536         5.687204
##   Student Loan                             5.050505         6.060606
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                     6.500000  51.500000      5.000000    5.500000
##   Credit-Builder Loan           4.651163  53.953488      4.651163    8.372093
##   Debt Consolidation Loan       3.351955  56.983240      1.675978    3.910615
##   Home Equity Loan              3.664921  53.926702      7.329843    3.664921
##   Mortgage Loan                 4.017857  46.875000      5.357143    4.910714
##   No Loan                       0.000000 100.000000      0.000000    0.000000
##   Not Specified                 5.472637  55.223881      3.980100    7.462687
##   Payday Loan                   5.000000  47.083333      3.750000    8.333333
##   Personal Loan                 6.161137  55.450237      7.109005    3.317536
##   Student Loan                  5.555556  45.454545      6.060606    7.575758
##
##                           Personal Loan Student Loan
##   Auto Loan                     3.500000     4.000000
##   Credit-Builder Loan           4.651163     6.976744
##   Debt Consolidation Loan       4.469274     3.910615
##   Home Equity Loan              4.712042     7.853403
##   Mortgage Loan                 6.250000     7.589286
##   No Loan                       0.000000     0.000000
##   Not Specified                 4.477612     4.975124
##   Payday Loan                   3.750000     6.666667
```

```
##   Personal Loan                5.213270     3.791469
##   Student Loan                 6.565657     6.060606
##
## $Last_Loan_7_Last_Loan_8$col_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan               13.3333333           7.3394495
##   Credit-Builder Loan     11.1111111           8.2568807
##   Debt Consolidation Loan 11.1111111          11.9266055
##   Home Equity Loan         7.7777778           9.1743119
##   Mortgage Loan           17.7777778          13.7614679
##   No Loan                  0.0000000           0.0000000
##   Not Specified            8.8888889           6.4220183
##   Payday Loan             12.2222222          17.4311927
##   Personal Loan           10.0000000          11.0091743
##   Student Loan             7.7777778          14.6788991
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                               8.4210526       18.0180180
##   Credit-Builder Loan                    10.5263158        6.3063063
##   Debt Consolidation Loan                13.6842105        9.0090090
##   Home Equity Loan                       10.5263158        8.1081081
##   Mortgage Loan                          11.5789474       12.6126126
##   No Loan                                 0.0000000        0.0000000
##   Not Specified                           9.4736842       11.7117117
##   Payday Loan                            17.8947368       12.6126126
##   Personal Loan                           7.3684211       10.8108108
##   Student Loan                           10.5263158       10.8108108
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                    14.1304348  0.8878545    10.7526882   9.9099099
##   Credit-Builder Loan          10.8695652  0.9999138    10.7526882  16.2162162
##   Debt Consolidation Loan       6.5217391  0.8792345     3.2258065   6.3063063
##   Home Equity Loan              7.6086957  0.8878545    15.0537634   6.3063063
##   Mortgage Loan                 9.7826087  0.9050944    12.9032258   9.9099099
##   No Loan                       0.0000000 91.7248513     0.0000000   0.0000000
##   Not Specified                11.9565217  0.9568141     8.6021505  13.5135135
##   Payday Loan                  13.0434783  0.9740540     9.6774194  18.0180180
##   Personal Loan                14.1304348  1.0085337    16.1290323   6.3063063
##   Student Loan                 11.9565217  0.7757952    12.9032258  13.5135135
##
##                           Personal Loan Student Loan
##   Auto Loan                    7.7777778    7.4074074
##   Credit-Builder Loan         11.1111111   13.8888889
##   Debt Consolidation Loan      8.8888889    6.4814815
##   Home Equity Loan            10.0000000   13.8888889
##   Mortgage Loan               15.5555556   15.7407407
##   No Loan                      0.0000000    0.0000000
##   Not Specified               10.0000000    9.2592593
##   Payday Loan                 10.0000000   14.8148148
##   Personal Loan               12.2222222    7.4074074
```

```
##   Student Loan               14.4444444   11.1111111
##
##
## $Last_Loan_7_Last_Loan_9
## $Last_Loan_7_Last_Loan_9$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     32                  40                      24
##   Credit-Builder Loan           32                  40                      96
##   Debt Consolidation Loan       40                  40                      56
##   Home Equity Loan              56                  56                      16
##   Mortgage Loan                 48                  80                      40
##   No Loan                        0                   0                       0
##   Not Specified                 80                  24                      24
##   Payday Loan                   56                  56                      80
##   Personal Loan                 48                  40                      32
##   Student Loan                  56                  56                      24
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                             56            56    1176            88
##   Credit-Builder Loan                   32            56    1320            40
##   Debt Consolidation Loan               48            48    1112            16
##   Home Equity Loan                      64            56    1152            16
##   Mortgage Loan                         32            48    1280            80
##   No Loan                                0             0   85128             0
##   Not Specified                         40            64    1200            32
##   Payday Loan                           40            48    1408            32
##   Personal Loan                         56            64    1256            48
##   Student Loan                          40            80    1112            72
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                       56            40           32
##   Credit-Builder Loan             24            48           32
##   Debt Consolidation Loan         32            24           16
##   Home Equity Loan                48            32           32
##   Mortgage Loan                   72            72           40
##   No Loan                          0             0            0
##   Not Specified                   32            40           72
##   Payday Loan                     88            56           56
##   Personal Loan                   72            24           48
##   Student Loan                    40            48           56
##
## $Last_Loan_7_Last_Loan_9$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                    1600                    1720                    1432
##        Home Equity Loan           Mortgage Loan                 No Loan
##                    1528                    1792                   85128
##           Not Specified             Payday Loan           Personal Loan
##                    1608                    1920                    1688
##            Student Loan
```

```
##                    1584
##
## $Last_Loan_7_Last_Loan_9$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     448                     432                     392
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     408                     520                   96144
##           Not Specified             Payday Loan           Personal Loan
##                     424                     464                     384
##            Student Loan
##                     384
##
## $Last_Loan_7_Last_Loan_9$grand_total
## [1] 100000
##
## $Last_Loan_7_Last_Loan_9$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.032               0.040                   0.024
##   Credit-Builder Loan        0.032               0.040                   0.096
##   Debt Consolidation Loan    0.040               0.040                   0.056
##   Home Equity Loan           0.056               0.056                   0.016
##   Mortgage Loan              0.048               0.080                   0.040
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.080               0.024                   0.024
##   Payday Loan                0.056               0.056                   0.080
##   Personal Loan              0.048               0.040                   0.032
##   Student Loan               0.056               0.056                   0.024
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.056         0.056   1.176         0.088
##   Credit-Builder Loan                0.032         0.056   1.320         0.040
##   Debt Consolidation Loan            0.048         0.048   1.112         0.016
##   Home Equity Loan                   0.064         0.056   1.152         0.016
##   Mortgage Loan                      0.032         0.048   1.280         0.080
##   No Loan                            0.000         0.000  85.128         0.000
##   Not Specified                      0.040         0.064   1.200         0.032
##   Payday Loan                        0.040         0.048   1.408         0.032
##   Personal Loan                      0.056         0.064   1.256         0.048
##   Student Loan                       0.040         0.080   1.112         0.072
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.056         0.040        0.032
##   Credit-Builder Loan          0.024         0.048        0.032
##   Debt Consolidation Loan      0.032         0.024        0.016
##   Home Equity Loan             0.048         0.032        0.032
##   Mortgage Loan                0.072         0.072        0.040
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.032         0.040        0.072
##   Payday Loan                  0.088         0.056        0.056
```

```
##   Personal Loan                 0.072         0.024        0.048
##   Student Loan                  0.040         0.048        0.056
##
## $Last_Loan_7_Last_Loan_9$row_percentage
##
##                            Auto Loan Credit-Builder Loan
##   Auto Loan                 2.000000            2.500000
##   Credit-Builder Loan       1.860465            2.325581
##   Debt Consolidation Loan   2.793296            2.793296
##   Home Equity Loan          3.664921            3.664921
##   Mortgage Loan             2.678571            4.464286
##   No Loan                   0.000000            0.000000
##   Not Specified             4.975124            1.492537
##   Payday Loan               2.916667            2.916667
##   Personal Loan             2.843602            2.369668
##   Student Loan              3.535354            3.535354
##
##                           Debt Consolidation Loan Home Equity Loan
##   Auto Loan                                1.500000         3.500000
##   Credit-Builder Loan                      5.581395         1.860465
##   Debt Consolidation Loan                  3.910615         3.351955
##   Home Equity Loan                         1.047120         4.188482
##   Mortgage Loan                            2.232143         1.785714
##   No Loan                                  0.000000         0.000000
##   Not Specified                            1.492537         2.487562
##   Payday Loan                              4.166667         2.083333
##   Personal Loan                            1.895735         3.317536
##   Student Loan                             1.515152         2.525253
##
##                           Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                     3.500000  73.500000      5.500000    3.500000
##   Credit-Builder Loan           3.255814  76.744186      2.325581    1.395349
##   Debt Consolidation Loan       3.351955  77.653631      1.117318    2.234637
##   Home Equity Loan              3.664921  75.392670      1.047120    3.141361
##   Mortgage Loan                 2.678571  71.428571      4.464286    4.017857
##   No Loan                       0.000000 100.000000      0.000000    0.000000
##   Not Specified                 3.980100  74.626866      1.990050    1.990050
##   Payday Loan                   2.500000  73.333333      1.666667    4.583333
##   Personal Loan                 3.791469  74.407583      2.843602    4.265403
##   Student Loan                  5.050505  70.202020      4.545455    2.525253
##
##                           Personal Loan Student Loan
##   Auto Loan                     2.500000     2.000000
##   Credit-Builder Loan           2.790698     1.860465
##   Debt Consolidation Loan       1.675978     1.117318
##   Home Equity Loan              2.094241     2.094241
##   Mortgage Loan                 4.017857     2.232143
##   No Loan                       0.000000     0.000000
##   Not Specified                 2.487562     4.477612
##   Payday Loan                   2.916667     2.916667
##   Personal Loan                 1.421801     2.843602
```

```
##   Student Loan                 3.030303     3.535354
##
## $Last_Loan_7_Last_Loan_9$col_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan               7.142857            9.259259                6.122449
##   Credit-Builder Loan     7.142857            9.259259               24.489796
##   Debt Consolidation Loan 8.928571            9.259259               14.285714
##   Home Equity Loan       12.500000           12.962963                4.081633
##   Mortgage Loan          10.714286           18.518519               10.204082
##   No Loan                 0.000000            0.000000                0.000000
##   Not Specified          17.857143            5.555556                6.122449
##   Payday Loan            12.500000           12.962963               20.408163
##   Personal Loan          10.714286            9.259259                8.163265
##   Student Loan           12.500000           12.962963                6.122449
##
##                          Home Equity Loan Mortgage Loan   No Loan
##   Auto Loan                      13.725490     10.769231  1.223165
##   Credit-Builder Loan             7.843137     10.769231  1.372941
##   Debt Consolidation Loan        11.764706      9.230769  1.156598
##   Home Equity Loan               15.686275     10.769231  1.198203
##   Mortgage Loan                   7.843137      9.230769  1.331336
##   No Loan                         0.000000      0.000000 88.542187
##   Not Specified                   9.803922     12.307692  1.248128
##   Payday Loan                     9.803922      9.230769  1.464470
##   Personal Loan                  13.725490     12.307692  1.306374
##   Student Loan                    9.803922     15.384615  1.156598
##
##                          Not Specified Payday Loan Personal Loan Student Loan
##   Auto Loan                   20.754717   12.068966     10.416667     8.333333
##   Credit-Builder Loan          9.433962    5.172414     12.500000     8.333333
##   Debt Consolidation Loan      3.773585    6.896552      6.250000     4.166667
##   Home Equity Loan             3.773585   10.344828      8.333333     8.333333
##   Mortgage Loan               18.867925   15.517241     18.750000    10.416667
##   No Loan                      0.000000    0.000000      0.000000     0.000000
##   Not Specified                7.547170    6.896552     10.416667    18.750000
##   Payday Loan                  7.547170   18.965517     14.583333    14.583333
##   Personal Loan               11.320755   15.517241      6.250000    12.500000
##   Student Loan                16.981132    8.620690     12.500000    14.583333
##
##
## $Last_Loan_8_Last_Loan_9
## $Last_Loan_8_Last_Loan_9$contingency_table
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                     16                  32                      56
##   Credit-Builder Loan           88                  48                      56
##   Debt Consolidation Loan       88                  32                      32
##   Home Equity Loan              48                  56                      40
##   Mortgage Loan                 56                  56                      48
##   No Loan                        0                   0                       0
```

```
##   Not Specified                  16                  48                      24
##   Payday Loan                    80                  32                      24
##   Personal Loan                  24                  24                      64
##   Student Loan                   32                 104                      48
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                             40            48     312            56
##   Credit-Builder Loan                   16            80     424            64
##   Debt Consolidation Loan               80            64     384            24
##   Home Equity Loan                      72            40     344            64
##   Mortgage Loan                         48            40     296            40
##   No Loan                                0             0   92808             0
##   Not Specified                         40            80     344            32
##   Payday Loan                           56            48     440            64
##   Personal Loan                         32            64     336            64
##   Student Loan                          24            56     456            16
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                       56            48           56
##   Credit-Builder Loan             32            48           16
##   Debt Consolidation Loan         32             0           24
##   Home Equity Loan                80            80           64
##   Mortgage Loan                   80            24           48
##   No Loan                          0             0            0
##   Not Specified                   48            56           56
##   Payday Loan                     56            48           40
##   Personal Loan                   40            32           40
##   Student Loan                    40            48           40
##
## $Last_Loan_8_Last_Loan_9$row_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     720                     872                     760
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     888                     736                   92808
##           Not Specified             Payday Loan           Personal Loan
##                     744                     888                     720
##            Student Loan
##                     864
##
## $Last_Loan_8_Last_Loan_9$col_totals
##
##               Auto Loan     Credit-Builder Loan Debt Consolidation Loan
##                     448                     432                     392
##        Home Equity Loan           Mortgage Loan                 No Loan
##                     408                     520                   96144
##           Not Specified             Payday Loan           Personal Loan
##                     424                     464                     384
##            Student Loan
##                     384
##
```

```
## $Last_Loan_8_Last_Loan_9$grand_total
## [1] 100000
##
## $Last_Loan_8_Last_Loan_9$cell_percentage
##
##                          Auto Loan Credit-Builder Loan Debt Consolidation Loan
##   Auto Loan                  0.016               0.032                   0.056
##   Credit-Builder Loan        0.088               0.048                   0.056
##   Debt Consolidation Loan    0.088               0.032                   0.032
##   Home Equity Loan           0.048               0.056                   0.040
##   Mortgage Loan              0.056               0.056                   0.048
##   No Loan                    0.000               0.000                   0.000
##   Not Specified              0.016               0.048                   0.024
##   Payday Loan                0.080               0.032                   0.024
##   Personal Loan              0.024               0.024                   0.064
##   Student Loan               0.032               0.104                   0.048
##
##                          Home Equity Loan Mortgage Loan No Loan Not Specified
##   Auto Loan                          0.040         0.048   0.312         0.056
##   Credit-Builder Loan                0.016         0.080   0.424         0.064
##   Debt Consolidation Loan            0.080         0.064   0.384         0.024
##   Home Equity Loan                   0.072         0.040   0.344         0.064
##   Mortgage Loan                      0.048         0.040   0.296         0.040
##   No Loan                            0.000         0.000  92.808         0.000
##   Not Specified                      0.040         0.080   0.344         0.032
##   Payday Loan                        0.056         0.048   0.440         0.064
##   Personal Loan                      0.032         0.064   0.336         0.064
##   Student Loan                       0.024         0.056   0.456         0.016
##
##                          Payday Loan Personal Loan Student Loan
##   Auto Loan                    0.056         0.048        0.056
##   Credit-Builder Loan          0.032         0.048        0.016
##   Debt Consolidation Loan      0.032         0.000        0.024
##   Home Equity Loan             0.080         0.080        0.064
##   Mortgage Loan                0.080         0.024        0.048
##   No Loan                      0.000         0.000        0.000
##   Not Specified                0.048         0.056        0.056
##   Payday Loan                  0.056         0.048        0.040
##   Personal Loan                0.040         0.032        0.040
##   Student Loan                 0.040         0.048        0.040
##
## $Last_Loan_8_Last_Loan_9$row_percentage
##
##                           Auto Loan Credit-Builder Loan
##   Auto Loan                2.222222            4.444444
##   Credit-Builder Loan     10.091743            5.504587
##   Debt Consolidation Loan 11.578947            4.210526
##   Home Equity Loan         5.405405            6.306306
##   Mortgage Loan            7.608696            7.608696
##   No Loan                  0.000000            0.000000
##   Not Specified            2.150538            6.451613
```

```
##   Payday Loan               9.009009            3.603604
##   Personal Loan             3.333333            3.333333
##   Student Loan              3.703704           12.037037
##
##                          Debt Consolidation Loan Home Equity Loan
##   Auto Loan                               7.777778         5.555556
##   Credit-Builder Loan                     6.422018         1.834862
##   Debt Consolidation Loan                 4.210526        10.526316
##   Home Equity Loan                        4.504505         8.108108
##   Mortgage Loan                           6.521739         6.521739
##   No Loan                                 0.000000         0.000000
##   Not Specified                           3.225806         5.376344
##   Payday Loan                             2.702703         6.306306
##   Personal Loan                           8.888889         4.444444
##   Student Loan                            5.555556         2.777778
##
##                          Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                   6.666667  43.333333      7.777778    7.777778
##   Credit-Builder Loan         9.174312  48.623853      7.339450    3.669725
##   Debt Consolidation Loan     8.421053  50.526316      3.157895    4.210526
##   Home Equity Loan            4.504505  38.738739      7.207207    9.009009
##   Mortgage Loan               5.434783  40.217391      5.434783   10.869565
##   No Loan                     0.000000 100.000000      0.000000    0.000000
##   Not Specified              10.752688  46.236559      4.301075    6.451613
##   Payday Loan                 5.405405  49.549550      7.207207    6.306306
##   Personal Loan               8.888889  46.666667      8.888889    5.555556
##   Student Loan                6.481481  52.777778      1.851852    4.629630
##
##                          Personal Loan Student Loan
##   Auto Loan                   6.666667     7.777778
##   Credit-Builder Loan         5.504587     1.834862
##   Debt Consolidation Loan     0.000000     3.157895
##   Home Equity Loan            9.009009     7.207207
##   Mortgage Loan               3.260870     6.521739
##   No Loan                     0.000000     0.000000
##   Not Specified               7.526882     7.526882
##   Payday Loan                 5.405405     4.504505
##   Personal Loan               4.444444     5.555556
##   Student Loan                5.555556     4.629630
##
## $Last_Loan_8_Last_Loan_9$col_percentage
##
##                           Auto Loan Credit-Builder Loan
##   Auto Loan                3.5714286           7.4074074
##   Credit-Builder Loan     19.6428571          11.1111111
##   Debt Consolidation Loan 19.6428571           7.4074074
##   Home Equity Loan        10.7142857          12.9629630
##   Mortgage Loan           12.5000000          12.9629630
##   No Loan                  0.0000000           0.0000000
##   Not Specified            3.5714286          11.1111111
##   Payday Loan             17.8571429           7.4074074
```

```
##   Personal Loan            5.3571429           5.5555556
##   Student Loan             7.1428571          24.0740741
##
##                          Debt Consolidation Loan Home Equity Loan
##   Auto Loan                             14.2857143        9.8039216
##   Credit-Builder Loan                   14.2857143        3.9215686
##   Debt Consolidation Loan                8.1632653       19.6078431
##   Home Equity Loan                      10.2040816       17.6470588
##   Mortgage Loan                         12.2448980       11.7647059
##   No Loan                                0.0000000        0.0000000
##   Not Specified                          6.1224490        9.8039216
##   Payday Loan                            6.1224490       13.7254902
##   Personal Loan                         16.3265306        7.8431373
##   Student Loan                          12.2448980        5.8823529
##
##                          Mortgage Loan    No Loan Not Specified Payday Loan
##   Auto Loan                   9.2307692  0.3245132    13.2075472  12.0689655
##   Credit-Builder Loan        15.3846154  0.4410052    15.0943396   6.8965517
##   Debt Consolidation Loan    12.3076923  0.3994009     5.6603774   6.8965517
##   Home Equity Loan            7.6923077  0.3577966    15.0943396  17.2413793
##   Mortgage Loan               7.6923077  0.3078715     9.4339623  17.2413793
##   No Loan                     0.0000000 96.5302047     0.0000000   0.0000000
##   Not Specified              15.3846154  0.3577966     7.5471698  10.3448276
##   Payday Loan                 9.2307692  0.4576469    15.0943396  12.0689655
##   Personal Loan              12.3076923  0.3494758    15.0943396   8.6206897
##   Student Loan               10.7692308  0.4742886     3.7735849   8.6206897
##
##                          Personal Loan Student Loan
##   Auto Loan                   12.5000000   14.5833333
##   Credit-Builder Loan         12.5000000    4.1666667
##   Debt Consolidation Loan      0.0000000    6.2500000
##   Home Equity Loan            20.8333333   16.6666667
##   Mortgage Loan                6.2500000   12.5000000
##   No Loan                      0.0000000    0.0000000
##   Not Specified               14.5833333   14.5833333
##   Payday Loan                 12.5000000   10.4166667
##   Personal Loan                8.3333333   10.4166667
##   Student Loan                12.5000000   10.4166667
```

Create dataframe, with replica to excel spreadhseet

```
# Load necessary libraries
if (!require("progress")) install.packages("progress")
if (!require("lsr")) install.packages("lsr")  # lsr package for etaSquared function
if (!require("Hmisc")) install.packages("Hmisc")  # Hmisc package for rcorr function
if (!require("vcd")) install.packages("vcd")  # vcd package for Cramer's V
if (!require("stats")) install.packages("stats")  # For Chi-squared test

library(progress)
library(lsr)
library(Hmisc)
library(vcd)
library(stats)

# Define function for ANOVA and Eta-squared
calc_anova_eta_squared <- function(data, categorical_vars, numeric_vars) {
  total_combinations <- length(categorical_vars) * length(numeric_vars)
  pb <- progress_bar$new(total = total_combinations, format = "Processing [:bar] :percent ETA
: :eta", clear = FALSE, width = 60)

  results <- data.frame()

  for (cat_var in categorical_vars) {
    for (num_var in numeric_vars) {
      data[[cat_var]] <- as.factor(data[[cat_var]])
      anova_model <- aov(data[[num_var]] ~ data[[cat_var]], data = data)
      p_value <- summary(anova_model)[[1]][["Pr(>F)"]][1]
      formatted_p_value <- if (p_value < 0.0001) "<.0001" else format.pval(p_value, digits =
4)
      eta_squared <- etaSquared(anova_model)[1, "eta.sq"]
      significance <- ifelse(p_value < 0.05, "Significant", "Not Significant")
      interpretation <- ifelse(eta_squared < 0.06, "Weak correlation", ifelse(eta_squared <
0.14, "Moderate correlation", "Strong correlation"))

      # Add to the results dataframe
      results <- rbind(results, data.frame(
        Attribute1 = cat_var,
        Attribute2 = num_var,
        p_value = formatted_p_value,
        Significance = significance,
        Interpretation = interpretation,
        Eta_Squared = eta_squared,
        correlation_coefficient = NA,
        cramers_v = NA
      ))
      pb$tick()
    }
  }
  return(results)
}

# Define function for Pearson Correlation
```

```
calc_pearson_correlation <- function(data, numeric_vars) {
  cor_results <- rcorr(as.matrix(data[, numeric_vars]), type = "pearson")
  cor_matrix <- cor_results$r
  p_matrix <- cor_results$P
  results <- data.frame()

  for (i in 1:(ncol(cor_matrix) - 1)) {
    for (j in (i + 1):ncol(cor_matrix)) {
      cor_val <- cor_matrix[i, j]
      p_val <- p_matrix[i, j]
      formatted_p_val <- if (p_val < 0.0001) "<.0001" else formatC(p_val, format = "f", digit
s = 4)
      significance <- ifelse(p_val < 0.05, "Significant", "Not Significant")
      interpretation <- if (cor_val == 1) {
        "Perfect positive correlation"
      } else if (cor_val == -1) {
        "Perfect negative correlation"
      } else if (cor_val >= 0.7) {
        "Strong positive correlation"
      } else if (cor_val >= 0.3) {
        "Moderate positive correlation"
      } else if (cor_val > 0) {
        "Weak positive correlation"
      } else if (cor_val > -0.3) {
        "Weak negative correlation"
      } else if (cor_val >= -0.7) {
        "Moderate negative correlation"
      } else {
        "Strong negative correlation"
      }

      # Add to the results dataframe
      results <- rbind(results, data.frame(
        Attribute1 = names(data)[i],
        Attribute2 = names(data)[j],
        p_value = formatted_p_val,
        Significance = significance,
        Interpretation = interpretation,
        Eta_Squared = NA,
        correlation_coefficient = cor_val,
        cramers_v = NA
      ))
    }
  }
  return(results)
}

# Define function for Chi-squared and Cramer's V
calc_chisq_cramers_v <- function(data, categorical_vars) {
  total_combinations <- length(categorical_vars) * (length(categorical_vars) - 1) / 2
  pb <- progress_bar$new(total = total_combinations, format = "Processing [:bar] :percent ETA
```

```
: :eta", clear = FALSE, width = 60)

  results <- data.frame()

  for (i in 1:(length(categorical_vars) - 1)) {
    for (j in (i + 1):length(categorical_vars)) {
      cat_var1 <- categorical_vars[i]
      cat_var2 <- categorical_vars[j]
      contingency_table <- table(data[[cat_var1]], data[[cat_var2]])
      chi_sq_test <- chisq.test(contingency_table)
      p_value <- chi_sq_test$p.value
      formatted_p_value <- if (p_value < 0.0001) "<.0001" else format.pval(p_value, digits =
4)
      cramers_v <- assocstats(contingency_table)$cramer
      association_strength <- if (cramers_v < 0.1) {
        "Weak association"
      } else if (cramers_v >= 0.1 && cramers_v < 0.3) {
        "Moderate association"
      } else if (cramers_v >= 0.3 && cramers_v < 0.5) {
        "Strong association"
      } else {
        "Very strong association"
      }

      # Add to the results dataframe
      results <- rbind(results, data.frame(
        Attribute1 = cat_var1,
        Attribute2 = cat_var2,
        p_value = formatted_p_value,
        Significance = ifelse(p_value < 0.05, "Significant", "Not Significant"),
        Interpretation = association_strength,
        Eta_Squared = NA,
        correlation_coefficient = NA,
        cramers_v = cramers_v
      ))
      pb$tick()
    }
  }
  return(results)
}

# Example usage
# Assuming 'credit_score' is your dataframe

categorical_vars <- c("Credit_Mix", "Payment_of_Min_Amount", "Payment_Behaviour", "Occupatio
n",
                      "Credit_Score", "Last_Loan_1", "Last_Loan_2", "Last_Loan_3", "Last_Loan
_4",
                      "Last_Loan_5", "Last_Loan_6", "Last_Loan_7", "Last_Loan_8", "Last_Loan_
9")
numeric_vars <- c("Age", "Annual_Income", "Monthly_Inhand_Salary", "Num_Bank_Accounts", "Num_
```

```
Credit_Card",
                  "Interest_Rate", "Num_of_Loan", "Delay_from_due_date", "Num_of_Delayed_Paym
ent",
                  "Changed_Credit_Limit", "Num_Credit_Inquiries", "Outstanding_Debt", "Credit
_Utilization_Ratio",
                  "Credit_History_Age", "Total_EMI_per_month", "Amount_invested_monthly", "Mo
nthly_Balance")

# Call the functions
result1 <- calc_anova_eta_squared(credit_score, categorical_vars, numeric_vars)
result2 <- calc_pearson_correlation(credit_score, numeric_vars)
result3 <- calc_chisq_cramers_v(credit_score, categorical_vars)
```

```
## Warning in chisq.test(contingency_table): Chi-squared approximation may be
## incorrect
```

```
# Combine all results
final_results <- rbind(result1, result2, result3)

# Print the final results
print(final_results)
```

```
##                  Attribute1               Attribute2 p_value    Significance
## 1                 Credit_Mix                      Age  <.0001     Significant
## 2                 Credit_Mix            Annual_Income  <.0001     Significant
## 3                 Credit_Mix    Monthly_Inhand_Salary  <.0001     Significant
## 4                 Credit_Mix        Num_Bank_Accounts  <.0001     Significant
## 5                 Credit_Mix          Num_Credit_Card  <.0001     Significant
## 6                 Credit_Mix            Interest_Rate  <.0001     Significant
## 7                 Credit_Mix              Num_of_Loan  <.0001     Significant
## 8                 Credit_Mix      Delay_from_due_date  <.0001     Significant
## 9                 Credit_Mix   Num_of_Delayed_Payment  <.0001     Significant
## 10                Credit_Mix     Changed_Credit_Limit  <.0001     Significant
## 11                Credit_Mix     Num_Credit_Inquiries  <.0001     Significant
## 12                Credit_Mix         Outstanding_Debt  <.0001     Significant
## 13                Credit_Mix Credit_Utilization_Ratio  <.0001     Significant
## 14                Credit_Mix       Credit_History_Age  <.0001     Significant
## 15                Credit_Mix      Total_EMI_per_month  <.0001     Significant
## 16                Credit_Mix  Amount_invested_monthly  <.0001     Significant
## 17                Credit_Mix          Monthly_Balance  <.0001     Significant
## 18     Payment_of_Min_Amount                      Age  <.0001     Significant
## 19     Payment_of_Min_Amount            Annual_Income  <.0001     Significant
## 20     Payment_of_Min_Amount    Monthly_Inhand_Salary  <.0001     Significant
## 21     Payment_of_Min_Amount        Num_Bank_Accounts  <.0001     Significant
## 22     Payment_of_Min_Amount          Num_Credit_Card  <.0001     Significant
## 23     Payment_of_Min_Amount            Interest_Rate  <.0001     Significant
## 24     Payment_of_Min_Amount              Num_of_Loan  <.0001     Significant
## 25     Payment_of_Min_Amount      Delay_from_due_date  <.0001     Significant
## 26     Payment_of_Min_Amount   Num_of_Delayed_Payment  <.0001     Significant
## 27     Payment_of_Min_Amount     Changed_Credit_Limit  <.0001     Significant
## 28     Payment_of_Min_Amount     Num_Credit_Inquiries  <.0001     Significant
## 29     Payment_of_Min_Amount         Outstanding_Debt  <.0001     Significant
## 30     Payment_of_Min_Amount Credit_Utilization_Ratio  <.0001     Significant
## 31     Payment_of_Min_Amount       Credit_History_Age  <.0001     Significant
## 32     Payment_of_Min_Amount      Total_EMI_per_month  <.0001     Significant
## 33     Payment_of_Min_Amount  Amount_invested_monthly  <.0001     Significant
## 34     Payment_of_Min_Amount          Monthly_Balance  <.0001     Significant
## 35         Payment_Behaviour                      Age  <.0001     Significant
## 36         Payment_Behaviour            Annual_Income  <.0001     Significant
## 37         Payment_Behaviour    Monthly_Inhand_Salary  <.0001     Significant
## 38         Payment_Behaviour        Num_Bank_Accounts  <.0001     Significant
## 39         Payment_Behaviour          Num_Credit_Card  <.0001     Significant
## 40         Payment_Behaviour            Interest_Rate  <.0001     Significant
## 41         Payment_Behaviour              Num_of_Loan  <.0001     Significant
## 42         Payment_Behaviour      Delay_from_due_date  <.0001     Significant
## 43         Payment_Behaviour   Num_of_Delayed_Payment  <.0001     Significant
## 44         Payment_Behaviour     Changed_Credit_Limit  <.0001     Significant
## 45         Payment_Behaviour     Num_Credit_Inquiries  <.0001     Significant
## 46         Payment_Behaviour         Outstanding_Debt  <.0001     Significant
## 47         Payment_Behaviour Credit_Utilization_Ratio  <.0001     Significant
## 48         Payment_Behaviour       Credit_History_Age  <.0001     Significant
## 49         Payment_Behaviour      Total_EMI_per_month  <.0001     Significant
## 50         Payment_Behaviour  Amount_invested_monthly  <.0001     Significant
```

```
## 51          Payment_Behaviour          Monthly_Balance  <.0001     Significant
## 52                 Occupation                      Age  <.0001     Significant
## 53                 Occupation            Annual_Income  <.0001     Significant
## 54                 Occupation    Monthly_Inhand_Salary  <.0001     Significant
## 55                 Occupation        Num_Bank_Accounts  <.0001     Significant
## 56                 Occupation          Num_Credit_Card  <.0001     Significant
## 57                 Occupation            Interest_Rate  <.0001     Significant
## 58                 Occupation              Num_of_Loan  <.0001     Significant
## 59                 Occupation      Delay_from_due_date  <.0001     Significant
## 60                 Occupation   Num_of_Delayed_Payment  <.0001     Significant
## 61                 Occupation     Changed_Credit_Limit  <.0001     Significant
## 62                 Occupation     Num_Credit_Inquiries  <.0001     Significant
## 63                 Occupation         Outstanding_Debt  <.0001     Significant
## 64                 Occupation Credit_Utilization_Ratio    0.08 Not Significant
## 65                 Occupation       Credit_History_Age  <.0001     Significant
## 66                 Occupation      Total_EMI_per_month  <.0001     Significant
## 67                 Occupation  Amount_invested_monthly   5e-04     Significant
## 68                 Occupation          Monthly_Balance   4e-04     Significant
## 69               Credit_Score                      Age  <.0001     Significant
## 70               Credit_Score            Annual_Income  <.0001     Significant
## 71               Credit_Score    Monthly_Inhand_Salary  <.0001     Significant
## 72               Credit_Score        Num_Bank_Accounts  <.0001     Significant
## 73               Credit_Score          Num_Credit_Card  <.0001     Significant
## 74               Credit_Score            Interest_Rate  <.0001     Significant
## 75               Credit_Score              Num_of_Loan  <.0001     Significant
## 76               Credit_Score      Delay_from_due_date  <.0001     Significant
## 77               Credit_Score   Num_of_Delayed_Payment  <.0001     Significant
## 78               Credit_Score     Changed_Credit_Limit  <.0001     Significant
## 79               Credit_Score     Num_Credit_Inquiries  <.0001     Significant
## 80               Credit_Score         Outstanding_Debt  <.0001     Significant
## 81               Credit_Score Credit_Utilization_Ratio  <.0001     Significant
## 82               Credit_Score       Credit_History_Age  <.0001     Significant
## 83               Credit_Score      Total_EMI_per_month  <.0001     Significant
## 84               Credit_Score  Amount_invested_monthly  <.0001     Significant
## 85               Credit_Score          Monthly_Balance  <.0001     Significant
## 86                Last_Loan_1                      Age  <.0001     Significant
## 87                Last_Loan_1            Annual_Income  <.0001     Significant
## 88                Last_Loan_1    Monthly_Inhand_Salary  <.0001     Significant
## 89                Last_Loan_1        Num_Bank_Accounts  <.0001     Significant
## 90                Last_Loan_1          Num_Credit_Card  <.0001     Significant
## 91                Last_Loan_1            Interest_Rate  <.0001     Significant
## 92                Last_Loan_1              Num_of_Loan  <.0001     Significant
## 93                Last_Loan_1      Delay_from_due_date  <.0001     Significant
## 94                Last_Loan_1   Num_of_Delayed_Payment  <.0001     Significant
## 95                Last_Loan_1     Changed_Credit_Limit  <.0001     Significant
## 96                Last_Loan_1     Num_Credit_Inquiries  <.0001     Significant
## 97                Last_Loan_1         Outstanding_Debt  <.0001     Significant
## 98                Last_Loan_1 Credit_Utilization_Ratio  <.0001     Significant
## 99                Last_Loan_1       Credit_History_Age  <.0001     Significant
## 100               Last_Loan_1      Total_EMI_per_month  <.0001     Significant
## 101               Last_Loan_1  Amount_invested_monthly  <.0001     Significant
```

```
## 102              Last_Loan_1          Monthly_Balance  <.0001     Significant
## 103              Last_Loan_2                      Age  <.0001     Significant
## 104              Last_Loan_2            Annual_Income  <.0001     Significant
## 105              Last_Loan_2    Monthly_Inhand_Salary  <.0001     Significant
## 106              Last_Loan_2        Num_Bank_Accounts  <.0001     Significant
## 107              Last_Loan_2          Num_Credit_Card  <.0001     Significant
## 108              Last_Loan_2            Interest_Rate  <.0001     Significant
## 109              Last_Loan_2              Num_of_Loan  <.0001     Significant
## 110              Last_Loan_2      Delay_from_due_date  <.0001     Significant
## 111              Last_Loan_2   Num_of_Delayed_Payment  <.0001     Significant
## 112              Last_Loan_2     Changed_Credit_Limit  <.0001     Significant
## 113              Last_Loan_2     Num_Credit_Inquiries  <.0001     Significant
## 114              Last_Loan_2         Outstanding_Debt  <.0001     Significant
## 115              Last_Loan_2 Credit_Utilization_Ratio  <.0001     Significant
## 116              Last_Loan_2       Credit_History_Age  <.0001     Significant
## 117              Last_Loan_2      Total_EMI_per_month  <.0001     Significant
## 118              Last_Loan_2  Amount_invested_monthly  <.0001     Significant
## 119              Last_Loan_2          Monthly_Balance  <.0001     Significant
## 120              Last_Loan_3                      Age  <.0001     Significant
## 121              Last_Loan_3            Annual_Income  <.0001     Significant
## 122              Last_Loan_3    Monthly_Inhand_Salary  <.0001     Significant
## 123              Last_Loan_3        Num_Bank_Accounts  <.0001     Significant
## 124              Last_Loan_3          Num_Credit_Card  <.0001     Significant
## 125              Last_Loan_3            Interest_Rate  <.0001     Significant
## 126              Last_Loan_3              Num_of_Loan  <.0001     Significant
## 127              Last_Loan_3      Delay_from_due_date  <.0001     Significant
## 128              Last_Loan_3   Num_of_Delayed_Payment  <.0001     Significant
## 129              Last_Loan_3     Changed_Credit_Limit  <.0001     Significant
## 130              Last_Loan_3     Num_Credit_Inquiries  <.0001     Significant
## 131              Last_Loan_3         Outstanding_Debt  <.0001     Significant
## 132              Last_Loan_3 Credit_Utilization_Ratio  <.0001     Significant
## 133              Last_Loan_3       Credit_History_Age  <.0001     Significant
## 134              Last_Loan_3      Total_EMI_per_month  <.0001     Significant
## 135              Last_Loan_3  Amount_invested_monthly  <.0001     Significant
## 136              Last_Loan_3          Monthly_Balance  <.0001     Significant
## 137              Last_Loan_4                      Age  <.0001     Significant
## 138              Last_Loan_4            Annual_Income  <.0001     Significant
## 139              Last_Loan_4    Monthly_Inhand_Salary  <.0001     Significant
## 140              Last_Loan_4        Num_Bank_Accounts  <.0001     Significant
## 141              Last_Loan_4          Num_Credit_Card  <.0001     Significant
## 142              Last_Loan_4            Interest_Rate  <.0001     Significant
## 143              Last_Loan_4              Num_of_Loan  <.0001     Significant
## 144              Last_Loan_4      Delay_from_due_date  <.0001     Significant
## 145              Last_Loan_4   Num_of_Delayed_Payment  <.0001     Significant
## 146              Last_Loan_4     Changed_Credit_Limit  <.0001     Significant
## 147              Last_Loan_4     Num_Credit_Inquiries  <.0001     Significant
## 148              Last_Loan_4         Outstanding_Debt  <.0001     Significant
## 149              Last_Loan_4 Credit_Utilization_Ratio  <.0001     Significant
## 150              Last_Loan_4       Credit_History_Age  <.0001     Significant
## 151              Last_Loan_4      Total_EMI_per_month  <.0001     Significant
## 152              Last_Loan_4  Amount_invested_monthly  <.0001     Significant
```

```
## 153              Last_Loan_4          Monthly_Balance  <.0001     Significant
## 154              Last_Loan_5                      Age  <.0001     Significant
## 155              Last_Loan_5            Annual_Income  <.0001     Significant
## 156              Last_Loan_5    Monthly_Inhand_Salary  <.0001     Significant
## 157              Last_Loan_5        Num_Bank_Accounts  <.0001     Significant
## 158              Last_Loan_5          Num_Credit_Card  <.0001     Significant
## 159              Last_Loan_5            Interest_Rate  <.0001     Significant
## 160              Last_Loan_5              Num_of_Loan  <.0001     Significant
## 161              Last_Loan_5      Delay_from_due_date  <.0001     Significant
## 162              Last_Loan_5   Num_of_Delayed_Payment  <.0001     Significant
## 163              Last_Loan_5     Changed_Credit_Limit  <.0001     Significant
## 164              Last_Loan_5     Num_Credit_Inquiries  <.0001     Significant
## 165              Last_Loan_5         Outstanding_Debt  <.0001     Significant
## 166              Last_Loan_5 Credit_Utilization_Ratio  <.0001     Significant
## 167              Last_Loan_5       Credit_History_Age  <.0001     Significant
## 168              Last_Loan_5      Total_EMI_per_month  <.0001     Significant
## 169              Last_Loan_5  Amount_invested_monthly  <.0001     Significant
## 170              Last_Loan_5          Monthly_Balance  <.0001     Significant
## 171              Last_Loan_6                      Age  <.0001     Significant
## 172              Last_Loan_6            Annual_Income  <.0001     Significant
## 173              Last_Loan_6    Monthly_Inhand_Salary  <.0001     Significant
## 174              Last_Loan_6        Num_Bank_Accounts  <.0001     Significant
## 175              Last_Loan_6          Num_Credit_Card  <.0001     Significant
## 176              Last_Loan_6            Interest_Rate  <.0001     Significant
## 177              Last_Loan_6              Num_of_Loan  <.0001     Significant
## 178              Last_Loan_6      Delay_from_due_date  <.0001     Significant
## 179              Last_Loan_6   Num_of_Delayed_Payment  <.0001     Significant
## 180              Last_Loan_6     Changed_Credit_Limit  <.0001     Significant
## 181              Last_Loan_6     Num_Credit_Inquiries  <.0001     Significant
## 182              Last_Loan_6         Outstanding_Debt  <.0001     Significant
## 183              Last_Loan_6 Credit_Utilization_Ratio  <.0001     Significant
## 184              Last_Loan_6       Credit_History_Age  <.0001     Significant
## 185              Last_Loan_6      Total_EMI_per_month  <.0001     Significant
## 186              Last_Loan_6  Amount_invested_monthly  <.0001     Significant
## 187              Last_Loan_6          Monthly_Balance  <.0001     Significant
## 188              Last_Loan_7                      Age  <.0001     Significant
## 189              Last_Loan_7            Annual_Income  <.0001     Significant
## 190              Last_Loan_7    Monthly_Inhand_Salary  <.0001     Significant
## 191              Last_Loan_7        Num_Bank_Accounts  <.0001     Significant
## 192              Last_Loan_7          Num_Credit_Card  <.0001     Significant
## 193              Last_Loan_7            Interest_Rate  <.0001     Significant
## 194              Last_Loan_7              Num_of_Loan  <.0001     Significant
## 195              Last_Loan_7      Delay_from_due_date  <.0001     Significant
## 196              Last_Loan_7   Num_of_Delayed_Payment  <.0001     Significant
## 197              Last_Loan_7     Changed_Credit_Limit  <.0001     Significant
## 198              Last_Loan_7     Num_Credit_Inquiries  <.0001     Significant
## 199              Last_Loan_7         Outstanding_Debt  <.0001     Significant
## 200              Last_Loan_7 Credit_Utilization_Ratio  <.0001     Significant
## 201              Last_Loan_7       Credit_History_Age  <.0001     Significant
## 202              Last_Loan_7      Total_EMI_per_month  <.0001     Significant
## 203              Last_Loan_7  Amount_invested_monthly  <.0001     Significant
```

```
## 204              Last_Loan_7          Monthly_Balance  <.0001     Significant
## 205              Last_Loan_8                      Age  <.0001     Significant
## 206              Last_Loan_8            Annual_Income  <.0001     Significant
## 207              Last_Loan_8    Monthly_Inhand_Salary  <.0001     Significant
## 208              Last_Loan_8        Num_Bank_Accounts  <.0001     Significant
## 209              Last_Loan_8          Num_Credit_Card  <.0001     Significant
## 210              Last_Loan_8            Interest_Rate  <.0001     Significant
## 211              Last_Loan_8              Num_of_Loan  <.0001     Significant
## 212              Last_Loan_8      Delay_from_due_date  <.0001     Significant
## 213              Last_Loan_8   Num_of_Delayed_Payment  <.0001     Significant
## 214              Last_Loan_8     Changed_Credit_Limit  <.0001     Significant
## 215              Last_Loan_8     Num_Credit_Inquiries  <.0001     Significant
## 216              Last_Loan_8         Outstanding_Debt  <.0001     Significant
## 217              Last_Loan_8 Credit_Utilization_Ratio  <.0001     Significant
## 218              Last_Loan_8       Credit_History_Age  <.0001     Significant
## 219              Last_Loan_8      Total_EMI_per_month  <.0001     Significant
## 220              Last_Loan_8  Amount_invested_monthly  <.0001     Significant
## 221              Last_Loan_8          Monthly_Balance  <.0001     Significant
## 222              Last_Loan_9                      Age  <.0001     Significant
## 223              Last_Loan_9            Annual_Income  <.0001     Significant
## 224              Last_Loan_9    Monthly_Inhand_Salary  <.0001     Significant
## 225              Last_Loan_9        Num_Bank_Accounts  <.0001     Significant
## 226              Last_Loan_9          Num_Credit_Card  <.0001     Significant
## 227              Last_Loan_9            Interest_Rate  <.0001     Significant
## 228              Last_Loan_9              Num_of_Loan  <.0001     Significant
## 229              Last_Loan_9      Delay_from_due_date  <.0001     Significant
## 230              Last_Loan_9   Num_of_Delayed_Payment  <.0001     Significant
## 231              Last_Loan_9     Changed_Credit_Limit  <.0001     Significant
## 232              Last_Loan_9     Num_Credit_Inquiries  <.0001     Significant
## 233              Last_Loan_9         Outstanding_Debt  <.0001     Significant
## 234              Last_Loan_9 Credit_Utilization_Ratio  <.0001     Significant
## 235              Last_Loan_9       Credit_History_Age  <.0001     Significant
## 236              Last_Loan_9      Total_EMI_per_month  <.0001     Significant
## 237              Last_Loan_9  Amount_invested_monthly  <.0001     Significant
## 238              Last_Loan_9          Monthly_Balance  <.0001     Significant
## 239              Customer_ID                      Age  <.0001     Significant
## 240              Customer_ID               Occupation  <.0001     Significant
## 241              Customer_ID            Annual_Income  <.0001     Significant
## 242              Customer_ID    Monthly_Inhand_Salary  <.0001     Significant
## 243              Customer_ID        Num_Bank_Accounts  <.0001     Significant
## 244              Customer_ID          Num_Credit_Card  <.0001     Significant
## 245              Customer_ID            Interest_Rate  <.0001     Significant
## 246              Customer_ID              Num_of_Loan  <.0001     Significant
## 247              Customer_ID      Delay_from_due_date  <.0001     Significant
## 248              Customer_ID   Num_of_Delayed_Payment  <.0001     Significant
## 249              Customer_ID     Changed_Credit_Limit  <.0001     Significant
## 250              Customer_ID     Num_Credit_Inquiries  <.0001     Significant
## 251              Customer_ID               Credit_Mix  <.0001     Significant
## 252              Customer_ID         Outstanding_Debt  <.0001     Significant
## 253              Customer_ID Credit_Utilization_Ratio  <.0001     Significant
## 254              Customer_ID       Credit_History_Age  <.0001     Significant
```

```
## 255                    Age              Occupation  <.0001     Significant
## 256                    Age           Annual_Income  <.0001     Significant
## 257                    Age    Monthly_Inhand_Salary  <.0001     Significant
## 258                    Age        Num_Bank_Accounts  <.0001     Significant
## 259                    Age          Num_Credit_Card  <.0001     Significant
## 260                    Age            Interest_Rate  <.0001     Significant
## 261                    Age              Num_of_Loan  <.0001     Significant
## 262                    Age      Delay_from_due_date  <.0001     Significant
## 263                    Age   Num_of_Delayed_Payment  <.0001     Significant
## 264                    Age     Changed_Credit_Limit  <.0001     Significant
## 265                    Age     Num_Credit_Inquiries  <.0001     Significant
## 266                    Age               Credit_Mix  <.0001     Significant
## 267                    Age         Outstanding_Debt  <.0001     Significant
## 268                    Age Credit_Utilization_Ratio  <.0001     Significant
## 269                    Age       Credit_History_Age  <.0001     Significant
## 270             Occupation           Annual_Income  <.0001     Significant
## 271             Occupation    Monthly_Inhand_Salary  <.0001     Significant
## 272             Occupation        Num_Bank_Accounts  <.0001     Significant
## 273             Occupation          Num_Credit_Card  <.0001     Significant
## 274             Occupation            Interest_Rate  <.0001     Significant
## 275             Occupation              Num_of_Loan  <.0001     Significant
## 276             Occupation      Delay_from_due_date  <.0001     Significant
## 277             Occupation   Num_of_Delayed_Payment  <.0001     Significant
## 278             Occupation     Changed_Credit_Limit  <.0001     Significant
## 279             Occupation     Num_Credit_Inquiries  <.0001     Significant
## 280             Occupation               Credit_Mix  <.0001     Significant
## 281             Occupation         Outstanding_Debt  <.0001     Significant
## 282             Occupation Credit_Utilization_Ratio  <.0001     Significant
## 283             Occupation       Credit_History_Age  <.0001     Significant
## 284           Annual_Income    Monthly_Inhand_Salary  <.0001     Significant
## 285           Annual_Income        Num_Bank_Accounts  <.0001     Significant
## 286           Annual_Income          Num_Credit_Card  <.0001     Significant
## 287           Annual_Income            Interest_Rate  <.0001     Significant
## 288           Annual_Income              Num_of_Loan  <.0001     Significant
## 289           Annual_Income      Delay_from_due_date  <.0001     Significant
## 290           Annual_Income   Num_of_Delayed_Payment  <.0001     Significant
## 291           Annual_Income     Changed_Credit_Limit  <.0001     Significant
## 292           Annual_Income     Num_Credit_Inquiries  <.0001     Significant
## 293           Annual_Income               Credit_Mix  <.0001     Significant
## 294           Annual_Income         Outstanding_Debt  <.0001     Significant
## 295           Annual_Income Credit_Utilization_Ratio  <.0001     Significant
## 296           Annual_Income       Credit_History_Age  <.0001     Significant
## 297   Monthly_Inhand_Salary        Num_Bank_Accounts  <.0001     Significant
## 298   Monthly_Inhand_Salary          Num_Credit_Card  <.0001     Significant
## 299   Monthly_Inhand_Salary            Interest_Rate  <.0001     Significant
## 300   Monthly_Inhand_Salary              Num_of_Loan  <.0001     Significant
## 301   Monthly_Inhand_Salary      Delay_from_due_date  <.0001     Significant
## 302   Monthly_Inhand_Salary   Num_of_Delayed_Payment  <.0001     Significant
## 303   Monthly_Inhand_Salary     Changed_Credit_Limit  <.0001     Significant
## 304   Monthly_Inhand_Salary     Num_Credit_Inquiries  <.0001     Significant
## 305   Monthly_Inhand_Salary               Credit_Mix  <.0001     Significant
```

```
## 306    Monthly_Inhand_Salary         Outstanding_Debt  <.0001     Significant
## 307    Monthly_Inhand_Salary Credit_Utilization_Ratio  <.0001     Significant
## 308    Monthly_Inhand_Salary       Credit_History_Age  <.0001     Significant
## 309        Num_Bank_Accounts          Num_Credit_Card  <.0001     Significant
## 310        Num_Bank_Accounts            Interest_Rate  <.0001     Significant
## 311        Num_Bank_Accounts              Num_of_Loan  <.0001     Significant
## 312        Num_Bank_Accounts      Delay_from_due_date  <.0001     Significant
## 313        Num_Bank_Accounts   Num_of_Delayed_Payment  <.0001     Significant
## 314        Num_Bank_Accounts     Changed_Credit_Limit  <.0001     Significant
## 315        Num_Bank_Accounts     Num_Credit_Inquiries  <.0001     Significant
## 316        Num_Bank_Accounts               Credit_Mix  <.0001     Significant
## 317        Num_Bank_Accounts         Outstanding_Debt  <.0001     Significant
## 318        Num_Bank_Accounts Credit_Utilization_Ratio  <.0001     Significant
## 319        Num_Bank_Accounts       Credit_History_Age  <.0001     Significant
## 320          Num_Credit_Card            Interest_Rate  <.0001     Significant
## 321          Num_Credit_Card              Num_of_Loan  <.0001     Significant
## 322          Num_Credit_Card      Delay_from_due_date  <.0001     Significant
## 323          Num_Credit_Card   Num_of_Delayed_Payment  <.0001     Significant
## 324          Num_Credit_Card     Changed_Credit_Limit  <.0001     Significant
## 325          Num_Credit_Card     Num_Credit_Inquiries  <.0001     Significant
## 326          Num_Credit_Card               Credit_Mix  <.0001     Significant
## 327          Num_Credit_Card         Outstanding_Debt  <.0001     Significant
## 328          Num_Credit_Card Credit_Utilization_Ratio  <.0001     Significant
## 329          Num_Credit_Card       Credit_History_Age  <.0001     Significant
## 330            Interest_Rate              Num_of_Loan  <.0001     Significant
## 331            Interest_Rate      Delay_from_due_date  <.0001     Significant
## 332            Interest_Rate   Num_of_Delayed_Payment  <.0001     Significant
## 333            Interest_Rate     Changed_Credit_Limit  <.0001     Significant
## 334            Interest_Rate     Num_Credit_Inquiries  <.0001     Significant
## 335            Interest_Rate               Credit_Mix  <.0001     Significant
## 336            Interest_Rate         Outstanding_Debt  <.0001     Significant
## 337            Interest_Rate Credit_Utilization_Ratio  <.0001     Significant
## 338            Interest_Rate       Credit_History_Age  <.0001     Significant
## 339              Num_of_Loan      Delay_from_due_date  <.0001     Significant
## 340              Num_of_Loan   Num_of_Delayed_Payment  <.0001     Significant
## 341              Num_of_Loan     Changed_Credit_Limit  <.0001     Significant
## 342              Num_of_Loan     Num_Credit_Inquiries  <.0001     Significant
## 343              Num_of_Loan               Credit_Mix  <.0001     Significant
## 344              Num_of_Loan         Outstanding_Debt  <.0001     Significant
## 345              Num_of_Loan Credit_Utilization_Ratio  <.0001     Significant
## 346              Num_of_Loan       Credit_History_Age  <.0001     Significant
## 347      Delay_from_due_date   Num_of_Delayed_Payment  <.0001     Significant
## 348      Delay_from_due_date     Changed_Credit_Limit  <.0001     Significant
## 349      Delay_from_due_date     Num_Credit_Inquiries  <.0001     Significant
## 350      Delay_from_due_date               Credit_Mix  <.0001     Significant
## 351      Delay_from_due_date         Outstanding_Debt  <.0001     Significant
## 352      Delay_from_due_date Credit_Utilization_Ratio  <.0001     Significant
## 353      Delay_from_due_date       Credit_History_Age  <.0001     Significant
## 354   Num_of_Delayed_Payment     Changed_Credit_Limit  <.0001     Significant
## 355   Num_of_Delayed_Payment     Num_Credit_Inquiries  <.0001     Significant
## 356   Num_of_Delayed_Payment               Credit_Mix  <.0001     Significant
```

```
## 357   Num_of_Delayed_Payment         Outstanding_Debt  <.0001     Significant
## 358   Num_of_Delayed_Payment Credit_Utilization_Ratio  <.0001     Significant
## 359   Num_of_Delayed_Payment       Credit_History_Age  <.0001     Significant
## 360     Changed_Credit_Limit     Num_Credit_Inquiries  <.0001     Significant
## 361     Changed_Credit_Limit               Credit_Mix  <.0001     Significant
## 362     Changed_Credit_Limit         Outstanding_Debt  <.0001     Significant
## 363     Changed_Credit_Limit Credit_Utilization_Ratio  <.0001     Significant
## 364     Changed_Credit_Limit       Credit_History_Age  <.0001     Significant
## 365     Num_Credit_Inquiries               Credit_Mix  <.0001     Significant
## 366     Num_Credit_Inquiries         Outstanding_Debt  <.0001     Significant
## 367     Num_Credit_Inquiries Credit_Utilization_Ratio  0.0671 Not Significant
## 368     Num_Credit_Inquiries       Credit_History_Age  <.0001     Significant
## 369               Credit_Mix         Outstanding_Debt  <.0001     Significant
## 370               Credit_Mix Credit_Utilization_Ratio  <.0001     Significant
## 371               Credit_Mix       Credit_History_Age  <.0001     Significant
## 372         Outstanding_Debt Credit_Utilization_Ratio  <.0001     Significant
## 373         Outstanding_Debt       Credit_History_Age  <.0001     Significant
## 374 Credit_Utilization_Ratio       Credit_History_Age  <.0001     Significant
## 375               Credit_Mix    Payment_of_Min_Amount  <.0001     Significant
## 376               Credit_Mix        Payment_Behaviour  <.0001     Significant
## 377               Credit_Mix               Occupation  <.0001     Significant
## 378               Credit_Mix             Credit_Score  <.0001     Significant
## 379               Credit_Mix              Last_Loan_1  <.0001     Significant
## 380               Credit_Mix              Last_Loan_2  <.0001     Significant
## 381               Credit_Mix              Last_Loan_3  <.0001     Significant
## 382               Credit_Mix              Last_Loan_4  <.0001     Significant
## 383               Credit_Mix              Last_Loan_5  <.0001     Significant
## 384               Credit_Mix              Last_Loan_6  <.0001     Significant
## 385               Credit_Mix              Last_Loan_7  <.0001     Significant
## 386               Credit_Mix              Last_Loan_8  <.0001     Significant
## 387               Credit_Mix              Last_Loan_9  <.0001     Significant
## 388    Payment_of_Min_Amount        Payment_Behaviour  <.0001     Significant
## 389    Payment_of_Min_Amount               Occupation  <.0001     Significant
## 390    Payment_of_Min_Amount             Credit_Score  <.0001     Significant
## 391    Payment_of_Min_Amount              Last_Loan_1  <.0001     Significant
## 392    Payment_of_Min_Amount              Last_Loan_2  <.0001     Significant
## 393    Payment_of_Min_Amount              Last_Loan_3  <.0001     Significant
## 394    Payment_of_Min_Amount              Last_Loan_4  <.0001     Significant
## 395    Payment_of_Min_Amount              Last_Loan_5  <.0001     Significant
## 396    Payment_of_Min_Amount              Last_Loan_6  <.0001     Significant
## 397    Payment_of_Min_Amount              Last_Loan_7  <.0001     Significant
## 398    Payment_of_Min_Amount              Last_Loan_8  <.0001     Significant
## 399    Payment_of_Min_Amount              Last_Loan_9  <.0001     Significant
## 400        Payment_Behaviour               Occupation  <.0001     Significant
## 401        Payment_Behaviour             Credit_Score  <.0001     Significant
## 402        Payment_Behaviour              Last_Loan_1  <.0001     Significant
## 403        Payment_Behaviour              Last_Loan_2  <.0001     Significant
## 404        Payment_Behaviour              Last_Loan_3  <.0001     Significant
## 405        Payment_Behaviour              Last_Loan_4  <.0001     Significant
## 406        Payment_Behaviour              Last_Loan_5  <.0001     Significant
## 407        Payment_Behaviour              Last_Loan_6  <.0001     Significant
```

```
## 408        Payment_Behaviour              Last_Loan_7  <.0001     Significant
## 409        Payment_Behaviour              Last_Loan_8  <.0001     Significant
## 410        Payment_Behaviour              Last_Loan_9  <.0001     Significant
## 411               Occupation             Credit_Score  <.0001     Significant
## 412               Occupation              Last_Loan_1  <.0001     Significant
## 413               Occupation              Last_Loan_2  <.0001     Significant
## 414               Occupation              Last_Loan_3  <.0001     Significant
## 415               Occupation              Last_Loan_4  <.0001     Significant
## 416               Occupation              Last_Loan_5  <.0001     Significant
## 417               Occupation              Last_Loan_6  <.0001     Significant
## 418               Occupation              Last_Loan_7  <.0001     Significant
## 419               Occupation              Last_Loan_8  <.0001     Significant
## 420               Occupation              Last_Loan_9  <.0001     Significant
## 421             Credit_Score              Last_Loan_1  <.0001     Significant
## 422             Credit_Score              Last_Loan_2  <.0001     Significant
## 423             Credit_Score              Last_Loan_3  <.0001     Significant
## 424             Credit_Score              Last_Loan_4  <.0001     Significant
## 425             Credit_Score              Last_Loan_5  <.0001     Significant
## 426             Credit_Score              Last_Loan_6  <.0001     Significant
## 427             Credit_Score              Last_Loan_7  <.0001     Significant
## 428             Credit_Score              Last_Loan_8  <.0001     Significant
## 429             Credit_Score              Last_Loan_9  <.0001     Significant
## 430              Last_Loan_1              Last_Loan_2  <.0001     Significant
## 431              Last_Loan_1              Last_Loan_3  <.0001     Significant
## 432              Last_Loan_1              Last_Loan_4  <.0001     Significant
## 433              Last_Loan_1              Last_Loan_5  <.0001     Significant
## 434              Last_Loan_1              Last_Loan_6  <.0001     Significant
## 435              Last_Loan_1              Last_Loan_7  <.0001     Significant
## 436              Last_Loan_1              Last_Loan_8  <.0001     Significant
## 437              Last_Loan_1              Last_Loan_9  <.0001     Significant
## 438              Last_Loan_2              Last_Loan_3  <.0001     Significant
## 439              Last_Loan_2              Last_Loan_4  <.0001     Significant
## 440              Last_Loan_2              Last_Loan_5  <.0001     Significant
## 441              Last_Loan_2              Last_Loan_6  <.0001     Significant
## 442              Last_Loan_2              Last_Loan_7  <.0001     Significant
## 443              Last_Loan_2              Last_Loan_8  <.0001     Significant
## 444              Last_Loan_2              Last_Loan_9  <.0001     Significant
## 445              Last_Loan_3              Last_Loan_4  <.0001     Significant
## 446              Last_Loan_3              Last_Loan_5  <.0001     Significant
## 447              Last_Loan_3              Last_Loan_6  <.0001     Significant
## 448              Last_Loan_3              Last_Loan_7  <.0001     Significant
## 449              Last_Loan_3              Last_Loan_8  <.0001     Significant
## 450              Last_Loan_3              Last_Loan_9  <.0001     Significant
## 451              Last_Loan_4              Last_Loan_5  <.0001     Significant
## 452              Last_Loan_4              Last_Loan_6  <.0001     Significant
## 453              Last_Loan_4              Last_Loan_7  <.0001     Significant
## 454              Last_Loan_4              Last_Loan_8  <.0001     Significant
## 455              Last_Loan_4              Last_Loan_9  <.0001     Significant
## 456              Last_Loan_5              Last_Loan_6  <.0001     Significant
## 457              Last_Loan_5              Last_Loan_7  <.0001     Significant
## 458              Last_Loan_5              Last_Loan_8  <.0001     Significant
```

```
## 459              Last_Loan_5              Last_Loan_9  <.0001     Significant
## 460              Last_Loan_6              Last_Loan_7  <.0001     Significant
## 461              Last_Loan_6              Last_Loan_8  <.0001     Significant
## 462              Last_Loan_6              Last_Loan_9  <.0001     Significant
## 463              Last_Loan_7              Last_Loan_8  <.0001     Significant
## 464              Last_Loan_7              Last_Loan_9  <.0001     Significant
## 465              Last_Loan_8              Last_Loan_9  <.0001     Significant
##                    Interpretation  Eta_Squared correlation_coefficient
## 1                Weak correlation 0.0594116373                      NA
## 2            Moderate correlation 0.1185959340                      NA
## 3            Moderate correlation 0.1177830572                      NA
## 4              Strong correlation 0.5260646583                      NA
## 5              Strong correlation 0.3161195856                      NA
## 6              Strong correlation 0.5687012194                      NA
## 7              Strong correlation 0.4548029415                      NA
## 8              Strong correlation 0.5024109375                      NA
## 9              Strong correlation 0.5753075941                      NA
## 10             Strong correlation 0.1923268119                      NA
## 11             Strong correlation 0.4490802302                      NA
## 12             Strong correlation 0.5932736087                      NA
## 13               Weak correlation 0.0082000341                      NA
## 14             Strong correlation 0.4238448239                      NA
## 15               Weak correlation 0.0311966651                      NA
## 16               Weak correlation 0.0483887410                      NA
## 17             Strong correlation 0.1427186426                      NA
## 18           Moderate correlation 0.0727174418                      NA
## 19           Moderate correlation 0.0958696467                      NA
## 20           Moderate correlation 0.0953752526                      NA
## 21             Strong correlation 0.3423564139                      NA
## 22             Strong correlation 0.1726277130                      NA
## 23             Strong correlation 0.4019252281                      NA
## 24             Strong correlation 0.2682129096                      NA
## 25             Strong correlation 0.2591889584                      NA
## 26             Strong correlation 0.3536356078                      NA
## 27             Strong correlation 0.2576029162                      NA
## 28             Strong correlation 0.4014399703                      NA
## 29             Strong correlation 0.2430213446                      NA
## 30               Weak correlation 0.0069526841                      NA
## 31             Strong correlation 0.3505139130                      NA
## 32               Weak correlation 0.0156868376                      NA
## 33               Weak correlation 0.0388921424                      NA
## 34           Moderate correlation 0.1139093622                      NA
## 35               Weak correlation 0.0027757882                      NA
## 36             Strong correlation 0.2885956390                      NA
## 37             Strong correlation 0.2901667943                      NA
## 38               Weak correlation 0.0234812005                      NA
## 39               Weak correlation 0.0133897924                      NA
## 40               Weak correlation 0.0295762077                      NA
## 41               Weak correlation 0.0233304574                      NA
## 42               Weak correlation 0.0205732412                      NA
## 43               Weak correlation 0.0230931055                      NA
```

```
## 44              Weak correlation 0.0089325646                      NA
## 45              Weak correlation 0.0245486374                      NA
## 46              Weak correlation 0.0265518247                      NA
## 47              Weak correlation 0.0144130656                      NA
## 48              Weak correlation 0.0250184278                      NA
## 49          Moderate correlation 0.0722065613                      NA
## 50              Weak correlation 0.0515176534                      NA
## 51            Strong correlation 0.2441823102                      NA
## 52              Weak correlation 0.0013174103                      NA
## 53              Weak correlation 0.0005189060                      NA
## 54              Weak correlation 0.0004953275                      NA
## 55              Weak correlation 0.0007162473                      NA
## 56              Weak correlation 0.0007129517                      NA
## 57              Weak correlation 0.0005931851                      NA
## 58              Weak correlation 0.0011163095                      NA
## 59              Weak correlation 0.0007065380                      NA
## 60              Weak correlation 0.0005594529                      NA
## 61              Weak correlation 0.0010000679                      NA
## 62              Weak correlation 0.0009477425                      NA
## 63              Weak correlation 0.0011460221                      NA
## 64              Weak correlation 0.0002193123                      NA
## 65              Weak correlation 0.0008017997                      NA
## 66              Weak correlation 0.0009488726                      NA
## 67              Weak correlation 0.0003840157                      NA
## 68              Weak correlation 0.0003905053                      NA
## 69              Weak correlation 0.0259835412                      NA
## 70              Weak correlation 0.0457993179                      NA
## 71              Weak correlation 0.0446120689                      NA
## 72            Strong correlation 0.1564286766                      NA
## 73            Strong correlation 0.1635849085                      NA
## 74            Strong correlation 0.2357163658                      NA
## 75          Moderate correlation 0.1295394112                      NA
## 76            Strong correlation 0.1866246328                      NA
## 77            Strong correlation 0.1502466506                      NA
## 78              Weak correlation 0.0509347124                      NA
## 79            Strong correlation 0.1924927336                      NA
## 80            Strong correlation 0.1542020031                      NA
## 81              Weak correlation 0.0021015717                      NA
## 82            Strong correlation 0.1511611048                      NA
## 83              Weak correlation 0.0057662903                      NA
## 84              Weak correlation 0.0246714987                      NA
## 85              Weak correlation 0.0435032239                      NA
## 86              Weak correlation 0.0105031409                      NA
## 87              Weak correlation 0.0142727564                      NA
## 88              Weak correlation 0.0139725247                      NA
## 89              Weak correlation 0.0313293671                      NA
## 90              Weak correlation 0.0219655842                      NA
## 91              Weak correlation 0.0493285914                      NA
## 92            Strong correlation 0.2690683350                      NA
## 93              Weak correlation 0.0326994975                      NA
## 94              Weak correlation 0.0329488244                      NA
```

```
## 95              Weak correlation 0.0215617182                      NA
## 96              Weak correlation 0.0538428921                      NA
## 97              Weak correlation 0.0427782125                      NA
## 98              Weak correlation 0.0043255182                      NA
## 99              Weak correlation 0.0568177590                      NA
## 100         Moderate correlation 0.1335849441                      NA
## 101             Weak correlation 0.0052526935                      NA
## 102         Moderate correlation 0.0653145175                      NA
## 103             Weak correlation 0.0208442509                      NA
## 104             Weak correlation 0.0273705370                      NA
## 105             Weak correlation 0.0270001603                      NA
## 106         Moderate correlation 0.0725831407                      NA
## 107             Weak correlation 0.0548873968                      NA
## 108         Moderate correlation 0.1101513580                      NA
## 109           Strong correlation 0.4491415259                      NA
## 110         Moderate correlation 0.0749973579                      NA
## 111         Moderate correlation 0.0711440348                      NA
## 112             Weak correlation 0.0451343853                      NA
## 113         Moderate correlation 0.1241024911                      NA
## 114         Moderate correlation 0.0978899255                      NA
## 115             Weak correlation 0.0069623766                      NA
## 116         Moderate correlation 0.1327681933                      NA
## 117           Strong correlation 0.1869027260                      NA
## 118             Weak correlation 0.0106402368                      NA
## 119         Moderate correlation 0.1081438320                      NA
## 120             Weak correlation 0.0224865370                      NA
## 121             Weak correlation 0.0290840671                      NA
## 122             Weak correlation 0.0290470442                      NA
## 123         Moderate correlation 0.0937145529                      NA
## 124         Moderate correlation 0.0713440263                      NA
## 125         Moderate correlation 0.1334366915                      NA
## 126           Strong correlation 0.6064427587                      NA
## 127         Moderate correlation 0.1024611951                      NA
## 128         Moderate correlation 0.0958064758                      NA
## 129         Moderate correlation 0.0607209607                      NA
## 130           Strong correlation 0.1475958267                      NA
## 131           Strong correlation 0.1485867624                      NA
## 132             Weak correlation 0.0072053048                      NA
## 133           Strong correlation 0.1663425655                      NA
## 134           Strong correlation 0.1981489285                      NA
## 135             Weak correlation 0.0126961118                      NA
## 136         Moderate correlation 0.1192307032                      NA
## 137             Weak correlation 0.0353611657                      NA
## 138             Weak correlation 0.0448939404                      NA
## 139             Weak correlation 0.0447396378                      NA
## 140           Strong correlation 0.1509458500                      NA
## 141         Moderate correlation 0.1158311106                      NA
## 142           Strong correlation 0.2076277621                      NA
## 143           Strong correlation 0.6860636166                      NA
## 144           Strong correlation 0.1563954214                      NA
## 145           Strong correlation 0.1515613084                      NA
```

```
## 146          Moderate correlation 0.0921522389                      NA
## 147            Strong correlation 0.2198657339                      NA
## 148            Strong correlation 0.2498041625                      NA
## 149              Weak correlation 0.0074018738                      NA
## 150            Strong correlation 0.2513111916                      NA
## 151            Strong correlation 0.1402952996                      NA
## 152              Weak correlation 0.0186041802                      NA
## 153          Moderate correlation 0.1338757086                      NA
## 154              Weak correlation 0.0506086939                      NA
## 155          Moderate correlation 0.0714601095                      NA
## 156          Moderate correlation 0.0709090269                      NA
## 157            Strong correlation 0.2465806313                      NA
## 158            Strong correlation 0.1991372933                      NA
## 159            Strong correlation 0.3506369311                      NA
## 160            Strong correlation 0.6923063069                      NA
## 161            Strong correlation 0.2752314832                      NA
## 162            Strong correlation 0.2443191708                      NA
## 163            Strong correlation 0.1582061232                      NA
## 164            Strong correlation 0.3624485895                      NA
## 165            Strong correlation 0.4570059428                      NA
## 166              Weak correlation 0.0068017962                      NA
## 167            Strong correlation 0.4105915470                      NA
## 168          Moderate correlation 0.0753197926                      NA
## 169              Weak correlation 0.0301582489                      NA
## 170          Moderate correlation 0.1374797433                      NA
## 171              Weak correlation 0.0327335993                      NA
## 172              Weak correlation 0.0505306702                      NA
## 173              Weak correlation 0.0500466711                      NA
## 174            Strong correlation 0.1810462712                      NA
## 175            Strong correlation 0.1451678050                      NA
## 176            Strong correlation 0.2543723288                      NA
## 177            Strong correlation 0.6451306757                      NA
## 178            Strong correlation 0.2070275831                      NA
## 179            Strong correlation 0.1782680292                      NA
## 180          Moderate correlation 0.1123997462                      NA
## 181            Strong correlation 0.2527083228                      NA
## 182            Strong correlation 0.3570039399                      NA
## 183              Weak correlation 0.0053423239                      NA
## 184            Strong correlation 0.2943054085                      NA
## 185          Moderate correlation 0.0620329682                      NA
## 186              Weak correlation 0.0210998777                      NA
## 187          Moderate correlation 0.1101719265                      NA
## 188              Weak correlation 0.0193773660                      NA
## 189              Weak correlation 0.0329037545                      NA
## 190              Weak correlation 0.0325288237                      NA
## 191          Moderate correlation 0.1258934342                      NA
## 192          Moderate correlation 0.0995108496                      NA
## 193            Strong correlation 0.1671163377                      NA
## 194            Strong correlation 0.5174890652                      NA
## 195            Strong correlation 0.1551894045                      NA
## 196          Moderate correlation 0.1255077652                      NA
```

```
## 197          Moderate correlation 0.0714501275                      NA
## 198            Strong correlation 0.1560560230                      NA
## 199            Strong correlation 0.2741405203                      NA
## 200              Weak correlation 0.0036099671                      NA
## 201            Strong correlation 0.1921504063                      NA
## 202              Weak correlation 0.0418144921                      NA
## 203              Weak correlation 0.0139465998                      NA
## 204          Moderate correlation 0.0778285267                      NA
## 205              Weak correlation 0.0089547939                      NA
## 206              Weak correlation 0.0172715802                      NA
## 207              Weak correlation 0.0170297782                      NA
## 208          Moderate correlation 0.0816803439                      NA
## 209          Moderate correlation 0.0684297053                      NA
## 210          Moderate correlation 0.0999446073                      NA
## 211            Strong correlation 0.3241682812                      NA
## 212          Moderate correlation 0.1111827357                      NA
## 213          Moderate correlation 0.0880009321                      NA
## 214              Weak correlation 0.0480370016                      NA
## 215          Moderate correlation 0.0859471406                      NA
## 216            Strong correlation 0.2135933510                      NA
## 217              Weak correlation 0.0019170606                      NA
## 218          Moderate correlation 0.1035773390                      NA
## 219              Weak correlation 0.0214984164                      NA
## 220              Weak correlation 0.0074364462                      NA
## 221              Weak correlation 0.0424060506                      NA
## 222              Weak correlation 0.0056668967                      NA
## 223              Weak correlation 0.0088081317                      NA
## 224              Weak correlation 0.0086723503                      NA
## 225              Weak correlation 0.0415347011                      NA
## 226              Weak correlation 0.0342679005                      NA
## 227              Weak correlation 0.0521582972                      NA
## 228            Strong correlation 0.2003074041                      NA
## 229              Weak correlation 0.0578583335                      NA
## 230              Weak correlation 0.0477520559                      NA
## 231              Weak correlation 0.0224761748                      NA
## 232              Weak correlation 0.0430731802                      NA
## 233          Moderate correlation 0.1078362204                      NA
## 234              Weak correlation 0.0010760497                      NA
## 235              Weak correlation 0.0535404413                      NA
## 236              Weak correlation 0.0120468745                      NA
## 237              Weak correlation 0.0035843094                      NA
## 238              Weak correlation 0.0243221716                      NA
## 239     Weak positive correlation           NA             0.091498098
## 240     Weak positive correlation           NA             0.090599841
## 241     Weak negative correlation           NA            -0.190376861
## 242     Weak negative correlation           NA            -0.148510743
## 243     Weak negative correlation           NA            -0.217519363
## 244     Weak negative correlation           NA            -0.213515895
## 245     Weak negative correlation           NA            -0.174132024
## 246     Weak negative correlation           NA            -0.184291579
## 247     Weak negative correlation           NA            -0.156620624
```

```
## 248     Weak negative correlation           NA            -0.250929883
## 249     Weak negative correlation           NA            -0.202366718
## 250     Weak positive correlation           NA             0.025498718
## 251     Weak positive correlation           NA             0.234622382
## 252     Weak negative correlation           NA            -0.069253439
## 253     Weak positive correlation           NA             0.057360393
## 254     Weak positive correlation           NA             0.116591576
## 255   Strong positive correlation           NA             0.998201298
## 256     Weak negative correlation           NA            -0.284047617
## 257     Weak negative correlation           NA            -0.217876942
## 258 Moderate negative correlation           NA            -0.302352436
## 259     Weak negative correlation           NA            -0.255559600
## 260     Weak negative correlation           NA            -0.250162613
## 261     Weak negative correlation           NA            -0.286056400
## 262     Weak negative correlation           NA            -0.175354471
## 263     Weak negative correlation           NA            -0.280824917
## 264     Weak negative correlation           NA            -0.269319340
## 265     Weak positive correlation           NA             0.175930085
## 266     Weak positive correlation           NA             0.272377442
## 267 Moderate positive correlation           NA             0.389863173
## 268 Moderate positive correlation           NA             0.627441939
## 269   Strong positive correlation           NA             0.705527756
## 270     Weak negative correlation           NA            -0.283254291
## 271     Weak negative correlation           NA            -0.216945089
## 272 Moderate negative correlation           NA            -0.301906999
## 273     Weak negative correlation           NA            -0.254405633
## 274     Weak negative correlation           NA            -0.249554591
## 275     Weak negative correlation           NA            -0.285143370
## 276     Weak negative correlation           NA            -0.175308035
## 277     Weak negative correlation           NA            -0.280429622
## 278     Weak negative correlation           NA            -0.269078383
## 279     Weak positive correlation           NA             0.176080563
## 280     Weak positive correlation           NA             0.271509238
## 281 Moderate positive correlation           NA             0.388376946
## 282 Moderate positive correlation           NA             0.628107251
## 283   Strong positive correlation           NA             0.706741379
## 284 Moderate positive correlation           NA             0.442682385
## 285 Moderate positive correlation           NA             0.584296791
## 286 Moderate positive correlation           NA             0.472483374
## 287 Moderate positive correlation           NA             0.560276399
## 288 Moderate positive correlation           NA             0.601374916
## 289 Moderate positive correlation           NA             0.331353387
## 290 Moderate positive correlation           NA             0.519331276
## 291 Moderate positive correlation           NA             0.507045026
## 292     Weak negative correlation           NA            -0.071778031
## 293 Moderate negative correlation           NA            -0.485332180
## 294     Weak positive correlation           NA             0.097519772
## 295     Weak negative correlation           NA            -0.182209302
## 296 Moderate negative correlation           NA            -0.300448088
## 297 Moderate positive correlation           NA             0.497912254
## 298 Moderate positive correlation           NA             0.417965172
```

```
## 299 Moderate positive correlation           NA             0.479574545
## 300 Moderate positive correlation           NA             0.423560528
## 301     Weak positive correlation           NA             0.253215392
## 302 Moderate positive correlation           NA             0.460208373
## 303 Moderate positive correlation           NA             0.490361538
## 304     Weak negative correlation           NA            -0.055247815
## 305 Moderate negative correlation           NA            -0.417223413
## 306     Weak positive correlation           NA             0.105110826
## 307     Weak negative correlation           NA            -0.142999950
## 308     Weak negative correlation           NA            -0.238717951
## 309 Moderate positive correlation           NA             0.559156230
## 310 Moderate positive correlation           NA             0.589087833
## 311 Moderate positive correlation           NA             0.571416284
## 312 Moderate positive correlation           NA             0.366942876
## 313 Moderate positive correlation           NA             0.633519364
## 314 Moderate positive correlation           NA             0.629414446
## 315     Weak negative correlation           NA            -0.075699778
## 316 Moderate negative correlation           NA            -0.576226084
## 317     Weak positive correlation           NA             0.134071707
## 318     Weak negative correlation           NA            -0.194956671
## 319 Moderate negative correlation           NA            -0.328844625
## 320 Moderate positive correlation           NA             0.501243265
## 321 Moderate positive correlation           NA             0.474618018
## 322 Moderate positive correlation           NA             0.370380275
## 323 Moderate positive correlation           NA             0.566149032
## 324 Moderate positive correlation           NA             0.638712643
## 325     Weak negative correlation           NA            -0.100469173
## 326 Moderate negative correlation           NA            -0.605724924
## 327 Moderate positive correlation           NA             0.432064514
## 328     Weak negative correlation           NA            -0.164422747
## 329 Moderate negative correlation           NA            -0.433743891
## 330 Moderate positive correlation           NA             0.542237578
## 331     Weak positive correlation           NA             0.293718951
## 332 Moderate positive correlation           NA             0.540785530
## 333 Moderate positive correlation           NA             0.571712834
## 334     Weak negative correlation           NA            -0.063796125
## 335 Moderate negative correlation           NA            -0.491183210
## 336     Weak positive correlation           NA             0.138504451
## 337     Weak negative correlation           NA            -0.162542565
## 338     Weak negative correlation           NA            -0.281932965
## 339 Moderate positive correlation           NA             0.326109289
## 340 Moderate positive correlation           NA             0.500726935
## 341 Moderate positive correlation           NA             0.504874335
## 342     Weak negative correlation           NA            -0.073752469
## 343 Moderate negative correlation           NA            -0.482197077
## 344     Weak positive correlation           NA             0.110521870
## 345     Weak negative correlation           NA            -0.182650235
## 346 Moderate negative correlation           NA            -0.305263200
## 347 Moderate positive correlation           NA             0.380253899
## 348 Moderate positive correlation           NA             0.458871974
## 349     Weak negative correlation           NA            -0.047999797
```

```
## 350 Moderate negative correlation           NA            -0.425164261
## 351     Weak positive correlation           NA             0.101970941
## 352     Weak negative correlation           NA            -0.115154553
## 353     Weak negative correlation           NA            -0.205366154
## 354 Moderate positive correlation           NA             0.598013939
## 355     Weak negative correlation           NA            -0.077280135
## 356 Moderate negative correlation           NA            -0.610655016
## 357     Weak positive correlation           NA             0.154512706
## 358     Weak negative correlation           NA            -0.176470588
## 359 Moderate negative correlation           NA            -0.322271754
## 360     Weak negative correlation           NA            -0.071164782
## 361 Moderate negative correlation           NA            -0.629317326
## 362     Weak positive correlation           NA             0.165506168
## 363     Weak negative correlation           NA            -0.173813039
## 364 Moderate negative correlation           NA            -0.317173954
## 365     Weak positive correlation           NA             0.073235326
## 366     Weak positive correlation           NA             0.013262830
## 367     Weak negative correlation           NA            -0.005790128
## 368     Weak positive correlation           NA             0.245164194
## 369     Weak negative correlation           NA            -0.171810368
## 370     Weak positive correlation           NA             0.175744603
## 371 Moderate positive correlation           NA             0.325762749
## 372     Weak positive correlation           NA             0.241913127
## 373     Weak positive correlation           NA             0.051229242
## 374     Weak positive correlation           NA             0.046782448
## 375       Very strong association           NA                      NA
## 376          Moderate association           NA                      NA
## 377              Weak association           NA                      NA
## 378            Strong association           NA                      NA
## 379          Moderate association           NA                      NA
## 380          Moderate association           NA                      NA
## 381          Moderate association           NA                      NA
## 382            Strong association           NA                      NA
## 383       Very strong association           NA                      NA
## 384            Strong association           NA                      NA
## 385            Strong association           NA                      NA
## 386            Strong association           NA                      NA
## 387          Moderate association           NA                      NA
## 388          Moderate association           NA                      NA
## 389              Weak association           NA                      NA
## 390            Strong association           NA                      NA
## 391          Moderate association           NA                      NA
## 392            Strong association           NA                      NA
## 393            Strong association           NA                      NA
## 394            Strong association           NA                      NA
## 395       Very strong association           NA                      NA
## 396            Strong association           NA                      NA
## 397            Strong association           NA                      NA
## 398          Moderate association           NA                      NA
## 399          Moderate association           NA                      NA
## 400              Weak association           NA                      NA
```

```
## 401             Weak association          NA                      NA
## 402             Weak association          NA                      NA
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## 404             Weak association          NA                      NA
## 405             Weak association          NA                      NA
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## 420             Weak association          NA                      NA
## 421         Moderate association          NA                      NA
## 422         Moderate association          NA                      NA
## 423         Moderate association          NA                      NA
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## 425         Moderate association          NA                      NA
## 426         Moderate association          NA                      NA
## 427         Moderate association          NA                      NA
## 428         Moderate association          NA                      NA
## 429             Weak association          NA                      NA
## 430         Moderate association          NA                      NA
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## 432         Moderate association          NA                      NA
## 433             Weak association          NA                      NA
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## 435             Weak association          NA                      NA
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## 437             Weak association          NA                      NA
## 438         Moderate association          NA                      NA
## 439         Moderate association          NA                      NA
## 440         Moderate association          NA                      NA
## 441         Moderate association          NA                      NA
## 442             Weak association          NA                      NA
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## 445         Moderate association          NA                      NA
## 446         Moderate association          NA                      NA
## 447         Moderate association          NA                      NA
## 448         Moderate association          NA                      NA
## 449             Weak association          NA                      NA
## 450             Weak association          NA                      NA
## 451         Moderate association          NA                      NA
```

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## 452          Moderate association           NA                        NA
## 453          Moderate association           NA                        NA
## 454          Moderate association           NA                        NA
## 455              Weak association           NA                        NA
## 456          Moderate association           NA                        NA
## 457          Moderate association           NA                        NA
## 458          Moderate association           NA                        NA
## 459          Moderate association           NA                        NA
## 460          Moderate association           NA                        NA
## 461          Moderate association           NA                        NA
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## 1           NA
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## 31          NA
## 32          NA
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## 34          NA
## 35          NA
## 36          NA
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## 37          NA
## 38          NA
## 39          NA
## 40          NA
## 41          NA
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## 45          NA
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## 85          NA
## 86          NA
## 87          NA
```

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## 88          NA
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## 90          NA
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## 132         NA
## 133         NA
## 134         NA
## 135         NA
## 136         NA
## 137         NA
## 138         NA
```

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## 139          NA
## 140          NA
## 141          NA
## 142          NA
## 143          NA
## 144          NA
## 145          NA
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## 147          NA
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## 185          NA
## 186          NA
## 187          NA
## 188          NA
## 189          NA
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## 190          NA
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## 238          NA
## 239          NA
## 240          NA
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## 241          NA
## 242          NA
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## 276          NA
## 277          NA
## 278          NA
## 279          NA
## 280          NA
## 281          NA
## 282          NA
## 283          NA
## 284          NA
## 285          NA
## 286          NA
## 287          NA
## 288          NA
## 289          NA
## 290          NA
## 291          NA
```

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## 292          NA
## 293          NA
## 294          NA
## 295          NA
## 296          NA
## 297          NA
## 298          NA
## 299          NA
## 300          NA
## 301          NA
## 302          NA
## 303          NA
## 304          NA
## 305          NA
## 306          NA
## 307          NA
## 308          NA
## 309          NA
## 310          NA
## 311          NA
## 312          NA
## 313          NA
## 314          NA
## 315          NA
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## 317          NA
## 318          NA
## 319          NA
## 320          NA
## 321          NA
## 322          NA
## 323          NA
## 324          NA
## 325          NA
## 326          NA
## 327          NA
## 328          NA
## 329          NA
## 330          NA
## 331          NA
## 332          NA
## 333          NA
## 334          NA
## 335          NA
## 336          NA
## 337          NA
## 338          NA
## 339          NA
## 340          NA
## 341          NA
## 342          NA
```

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## 343         NA
## 344         NA
## 345         NA
## 346         NA
## 347         NA
## 348         NA
## 349         NA
## 350         NA
## 351         NA
## 352         NA
## 353         NA
## 354         NA
## 355         NA
## 356         NA
## 357         NA
## 358         NA
## 359         NA
## 360         NA
## 361         NA
## 362         NA
## 363         NA
## 364         NA
## 365         NA
## 366         NA
## 367         NA
## 368         NA
## 369         NA
## 370         NA
## 371         NA
## 372         NA
## 373         NA
## 374         NA
## 375 0.81941008
## 376 0.13582694
## 377 0.03347124
## 378 0.44994121
## 379 0.17054007
## 380 0.24949430
## 381 0.29443835
## 382 0.37806288
## 383 0.50165682
## 384 0.44117027
## 385 0.38274565
## 386 0.35252464
## 387 0.25360906
## 388 0.16351605
## 389 0.02870354
## 390 0.47350231
## 391 0.23182371
## 392 0.34456695
## 393 0.36255122
```

```
## 394 0.43567449
## 395 0.54788536
## 396 0.45174808
## 397 0.34532665
## 398 0.22999254
## 399 0.16545849
## 400 0.02595969
## 401 0.09071711
## 402 0.04031992
## 403 0.04936506
## 404 0.04747032
## 405 0.05909902
## 406 0.07067990
## 407 0.06080149
## 408 0.05280862
## 409 0.04182599
## 410 0.03421204
## 411 0.03005710
## 412 0.03119128
## 413 0.03375707
## 414 0.03860316
## 415 0.03372037
## 416 0.03159487
## 417 0.03027574
## 418 0.03040452
## 419 0.03368530
## 420 0.03254446
## 421 0.11316613
## 422 0.16664049
## 423 0.17406019
## 424 0.21025318
## 425 0.27128322
## 426 0.22503579
## 427 0.17677884
## 428 0.12228283
## 429 0.08719994
## 430 0.22301955
## 431 0.15363956
## 432 0.11419480
## 433 0.08278096
## 434 0.07072193
## 435 0.05757405
## 436 0.04386797
## 437 0.03608927
## 438 0.23017891
## 439 0.16798007
## 440 0.12230868
## 441 0.10190062
## 442 0.08017032
## 443 0.05928495
## 444 0.04378910
```

```
## 445 0.24429951
## 446 0.17666840
## 447 0.14658261
## 448 0.11349395
## 449 0.07815494
## 450 0.06408399
## 451 0.24189752
## 452 0.20079097
## 453 0.15513232
## 454 0.10818740
## 455 0.08098319
## 456 0.27949548
## 457 0.21359970
## 458 0.14534739
## 459 0.11110726
## 460 0.25961410
## 461 0.17633474
## 462 0.13009604
## 463 0.23055106
## 464 0.17121990
## 465 0.25906234
```

```
write.csv(final_results, "C:\\Users\\jivko\\Documents\\Data Analytics, Big Data, and Predicti
ve Analytics\\Personal Project\\Credit_Score_Multiclass.csv", row.names = FALSE)
```